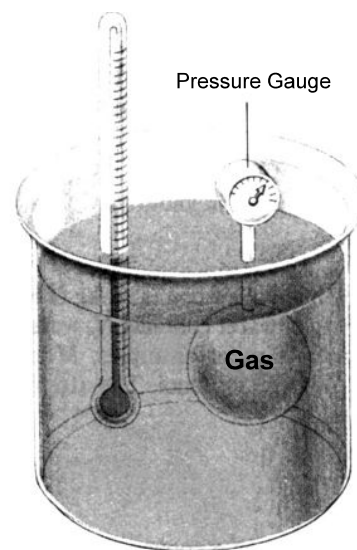


Heat and Thermal Expansion

FEW WORDS FOR STUDENTS

Thermal Physics is the branch of Physics which deals with the theory and application of thermal energy. In this chapter we'll mainly deal with the measurement of temperature and the different effects which takes place on changing the temperature of body or due to supply of thermal energy. In next chapters we'll discuss some fundamental laws of thermal physics related to heat exchange between bodies, called thermodynamic laws. This chapter gives you a thorough understanding of basic ideas needed in understanding thermodynamic laws in depth.



CHAPTER CONTENTS

- 1.1 *Heat and Temperature*
- 1.2 *Temperature and States of Matter*
- 1.3 *Thermometry*
- 1.4 *Thermal Equilibrium*
- 1.5 *Thermal Expansion*
- 1.6 *Calorimetry*

COVER APPLICATION



Figure (a)



Figure (b)

To avoid damage in over-bridges caused by thermal expansion as shown in figure-(b), during construction steel girders with slots are fitted leaving gap between these as shown in figure-(a). In summers due to expansion this gap decreases and keeps the bridge safe.

In previous chapters till now we have discussed only three fundamental quantities i.e. length, mass and time. All other mechanical quantities such as force, energy, momentum can be expressed in terms of these quantities. We now start another phenomena, the heat phenomenon which requires a fourth fundamental quantity called temperature to express a physical quantity of heat phenomenon in terms of fundamental quantities.

When we touch a body, we sense its coldness or hotness relatively with respect to our body and we say it is cool or warm. This property of the object as well as our body by which we are revising whether it is hot or cold, called temperature. The hotter it feels, the higher is the temperature.

1.1 Heat and Temperature

To define temperature quantitatively some systems are used in which a measurable property of system varies with hotness or coldness of the system. A simple example is a liquid (like mercury or alcohol) in a bulb attached to a thin tube as shown in figure-1.1. When the system becomes hot due to thermal expansion the volume of liquid increases and the length of liquid in the tube increases. The scale used to measure length of liquid column can be calibrated from an arbitrarily fixed point for measurement of temperature. Another simple system is quantity of gas in a constant volume container as shown in figure-1.2. the pressure of the gas measured by a pressure gauge or by a manometer increases or decreases as the gas becomes hot or colder. Another example of such a system is an electrical resistance R of a wire, which also varies with hotness or coldness of surrounding. In each of these examples, the quantity describing the varying state of the system, such as the length, the pressure or the resistance R is called a state coordinate of the system. Such systems which are used to measure temperature are called thermometers.

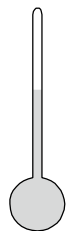


Figure 1.1

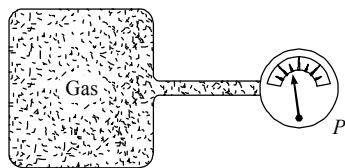


Figure 1.2

1.1.1 Thermal Partitions

When two bodies A and B with thermometers T_A and T_B attached to them as shown in figure-1.3(a) at different hotness level are placed in contact as shown in figure-1.3(b). Their state coordinates changes due to different energy content of the

bodies. The one which is hotter than other is said to have more energy content and will start losing this extra energy to the colder body, hence state coordinates of the bodies start changing as they come in contact depending on the relative hotness or coldness level. Qualitatively, we say that initially one is hotter than the other and each system changes the state of the other.

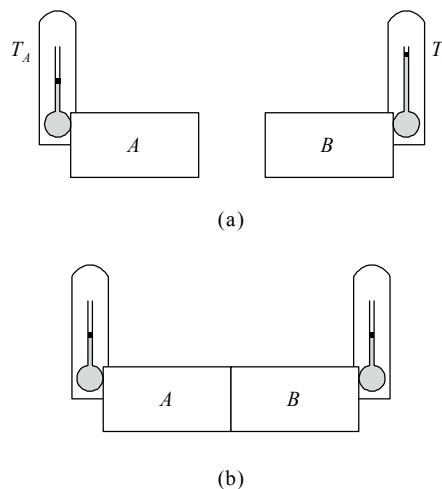


Figure 1.3

If the two system (bodies A and B) are separated by an insulating material such as wood, plastic foam or fibre glass, they influence each other much more slowly. Such a separator or an insulating wall through which the energy content of a body, which we call thermal energy or heat, can not flow is called an adiabatic separator or an adiabatic wall. Due to such a separator the state coordinates of both the system vary independently of each other. Such a partition is an idealized model which can be only approximately realized in real world.

The opposite of an adiabatic wall is a partition that does allow the two systems on opposite sides to influence each other or which allows the heat to flow through it, such a wall is called a diathermic wall or a diathermic separator.

1.2 Temperature and States of Matter

If we talk about water, we know it can exist in form of ice (solid), water (liquid) or steam (gas). Many other material can exist as solid, liquid or gases. Such distinct forms or states of matter are called phases. The change from one state or phase to another, such as melting of ice, is usually caused by a transfer of thermal energy.

The molecules of a gas move about freely, except when they collide with other gas molecules or the walls of the container. The average separation between molecules is large compared with their own size, and as a result, a gas has no definite volume.

Consequently a gas may be compressed or expanded and will fill a container of any shape or size. In a liquid, the average separation between molecules is comparable to their own diameters. Individual molecules are free to move about, but because of the forces between them they move so that the average separation between the neighbouring molecules remains same. As a result, a liquid is virtually incompressible and has a definite volume, although its shape can change to match the shape of its container. In solids the separation between molecules are comparable to that in liquids but binding forces are so strong that the atoms in a solid are not free to move about fixed positions. Thus a solid has not only a definite volume but a definite shape as well.

In every state of matter with the motion of molecules of the matter some energy is associated. What we've discussed in previous article about temperature is that it gives us a feeling for how hot or cold the object is depending on its temperature and temperature of an object can be measured by a device called thermometer. Changes in temperature, changes the energy associated with the different types of motion (linear, rotation and vibrational) depending on the state of matter (solid, liquid or gas). Changes in one state to another are also the changes in energy of the atoms and molecules that compose the material. The energy of molecules of a body can be measured quantitatively as dependent on temperature. But it doesn't mean the two different bodies at same temperature have same energy content in their molecules. What we can say for one body is, if we increase its temperature, total energy associated with its molecule will increase. If kinetic energy of molecules increases then it means the temperature of body increases. If the phase of substance is changing, it is due to change in potential energy of the molecules, which exist due to inter molecular forces of the body.

As we've discussed that thermometers are used to measure temperature of a body quantitatively what a human body can feel qualitatively. But range of measurement of thermometers can be extended for beyond the sense of touch. The range of thermometers extends from temperature low enough to freeze the gases of air to the enormous temperature at the interior of sun.

1.3 Thermometry

The science and measurement of temperature is thermometry. The basis of thermometry is that some physical properties vary with temperature in a quantitative and repeatable fashion. Some of these thermometric properties are the volume of a gas or liquid, the length of a metallic strip, the electric resistance of a conductor and the light transmitting properties of a crystal. Some crystal has their transmission coefficient or light dependent on their temperature thus as temperature of crystal

changes, the amount of light transmitted through it changes. Any physical system whose properties change with temperature can be used as a thermometer. The choice of thermometer depends primarily on range of temperature to be measured. We measure the change in some property, say the length of a column of liquid, and there associate the change in temperature with our measurement of the change in length of liquid column.

The most popular thermometer is liquid in glass thermometer. In this thermometer a liquid is sealed into a glass capillary tube having a glass bulb at one end. When temperature increases, both the volume of the glass bulb and the volume of liquid increases. If the liquid expanded at the same rate as that of glass, we would observe no change. But as liquid expands at a greater rate than glass does, the liquid will rise in the capillary connected to bulb as the temperature increases. By using a large bulb and a narrow tube, it is possible to make a thermometer that we can read easily from a scale scribed on the glass. The common fever thermometer is made this way.

As discussed, some thermometers determine temperature by measuring the amount of light transmitted by a crystal as shown in figure-1.4. The amount of light transmitted by the crystal depends on the temperature and is reproducible. This type of thermometer is used in medical applications that require a small, remotely readable thermometer.

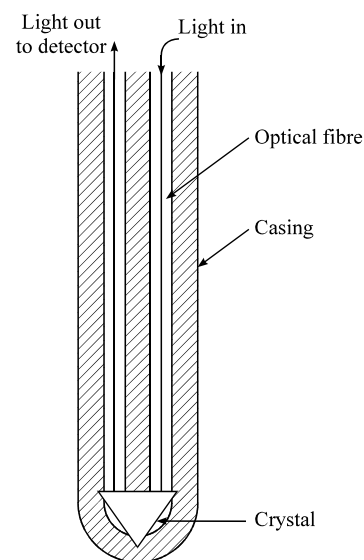


Figure 1.4

Any thermometer, whether liquid in glass or one that depends on other thermal properties, must be calibrated to make it a useful instrument. While calibrating the scale marked on thermometer it must be having the range of temperatures to measure temperature between standard or reference temperatures.

For example in case of liquid in glass thermometer temperature between reference marks are interpreted as proportional to the length of liquid column. There are so many temperature scales are defined for measurement of temperatures for thermometers. First we discuss the most common one-

Celsius temperature scale. The two fixed points lower and upper

for references of measurement are chosen as ice point and steam point of water. The ice point is defined as the equilibrium temperature of a mixture of ice and water at a pressure of one atmosphere and the steam point is defined as the equilibrium temperature of water and steam at a pressure of one atmosphere. The numbers assigned to these point in the Celsius scale are arbitrarily chosen as 0 for the ice point and 100 for the steam point.

Assuming the cross-section of the thermometer capillary is uniform and rate of expansion of the liquid with change in temperature is constant then we can mark the distance between ice point and steam point into 100 equal parts and each part we call degree. At any temperature we can easily compare the level of the liquid to the nearest mark. This scale was originally known as the centigrade scale because it has one hundred divisions between the principal reference marks.

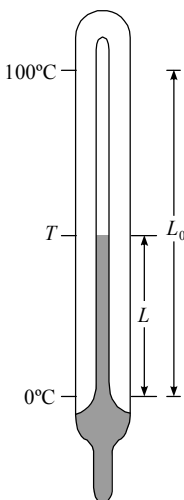


Figure 1.5

Figure-1.5 shows a Celsius scale thermometer at temperature T , at which the liquid is extended to a distance L beyond the zero position. Thus here temperature can be calculated by

$$T = \left(\frac{L}{L_0} \right) \times 100 \quad \dots (1.1)$$

Although the Celsius temperature scale is widely used, there is nothing fundamental about choosing the ice point to be 0° and the steam point as 100° . The Fahrenheit temperature scale assigns a value of 32° to the ice point and 212° to the steam point, a difference of exact 180° . Similarly there can be so many temperature scales having different numerical values assigned for ice point and steam point respectively. It is easy to transform temperature in one system into temperatures in the other system. The general transformation formula used for this is

$$\frac{\text{Temperature in X Scale} - \text{ice point in X Scale}}{\text{Steam point in X Scale} - \text{ice point in X Scale}} = \text{constant for all temperature Scales}$$

The above formula, if we use to transform a temperature T_C in Celsius scale to its value T_F in Fahrenheit scale, we have

$$\frac{T_C - 0}{100 - 0} = \frac{T_F - 32}{212 - 32} \quad \dots (1.2)$$

$$\text{or} \quad T_F = \frac{9}{5} T_C + 32 \quad \dots (1.3)$$

In both the Fahrenheit and Celsius temperature scale, the assignment of the zero point is arbitrary. We can readily achieve temperatures below these zero points. However, one temperature scale has more fundamental choice of zero point. This was given by Lord Kelvin from the study of gases. Kelvin scale uses intervals equal to those of the Celsius degree but with zero set at the lowest theoretical temperature that a gas can reach. The scale is based on the fact that a gas at 0°C will lose $\frac{1}{273.15}$ of its volume for a 1°C drop in temperature. If this reduction in volume were to continue with decreasing temperature and if the gas did not liquefy, the volume would become zero at -273.15°C . This is a temperature called absolute zero. The temperature scale based on this zero is the Kelvin temperature scale. If we use equation-(1.2) to establish a relation in temperatures measured in Celsius scale T_C and Kelvin Scale

$$T_K, \text{ we have } \frac{T_C - 0}{100 - 0} = \frac{T_K - 273.15}{373.15 - 273.15}$$

$$\text{or} \quad T_K = T_C + 273.15 \quad \dots (1.4)$$

This temperature in Kelvin scale is used with a unit of Kelvin (K) and it is not written with a degree sign. The temperature in Kelvin is called absolute temperature.

Illustrative Example 1.1

What are the following temperature on the Kelvin scale :

(a) 37°C , (b) 80°F , (c) -196°C ?

Solution

(a) Temperature on Kelvin scale T_K is related to temperature T_C on Celsius scale as

$$T_K = T_C + 273$$

$$\text{or} \quad T_K = 37 + 273$$

$$= 310 \text{ K}$$

(b) Temperature T_K on Kelvin scale and T_F on Fahrenheit scale are related as

$$\frac{T_K - 273}{373 - 273} = \frac{T_F - 32}{212 - 32}$$

$$\text{or} \quad \frac{T_K - 273}{100} = \frac{T_F - 32}{180}$$

$$\text{or} \quad T_K = \frac{5}{9} (T_F - 32) + 273$$

Here $T_F = 80^\circ\text{F}$ thus

$$T_K = \frac{5}{9} (80 - 32) + 273$$

$$= 299.66 \text{ K}$$

(c) Again from relation used in part (a)

$$\begin{aligned} T_K &= T_c + 273 \\ &= -196 + 273 \\ &= 77 \text{ K} \end{aligned}$$

Illustrative Example 1.2

Typical temperatures in the interior of the earth and sun are about $4 \times 10^3^\circ\text{C}$ and $1.5 \times 10^7^\circ\text{C}$, respectively. (a) What are these temperature in Kelvins? (b) What percentage error is made in each case if a person forgets to change $^\circ\text{C}$ to K?

Solution

(a) Temperature on Kelvin scale are

$$T_K = T_c + 273$$

For $T_c = 4000^\circ\text{C}$, we have $T_K = 4000 + 273 = 4273 \text{ K}$ and for $T_c = 1.5 \times 10^7^\circ\text{C}$ we have

$$T_K = (1.5 \times 10^7 + 273) \text{ K}$$

(b) Percentage error can be given as

$$\% \text{ error} = \frac{T_K - T_c}{T_c} \times 100 = \frac{273}{T_c} \times 100$$

For $T_c = 4000^\circ\text{C}$, we have

$$\% \text{ error} = \frac{273}{4000} \times 100 = 6.825\%$$

For $T_c = 1.5 \times 10^7^\circ\text{C}$ we have

$$\begin{aligned} \% \text{ error} &= \frac{273}{1.5 \times 10^7} \times 100 \\ &= 1.82 \times 10^{-3} \% \end{aligned}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Thermal Expansion & Thermometry

Module Number - 2

Practice Exercise 1.1

(i) (a) "Room temperature" is often taken to be 68°F ; what will it be on Celsius scale? (b) The temperature of the filament in a light bulb is about 1800°C ; what will it be on Fahrenheit scale?

[(a) 20°C , (b) 3272°F]

(ii) If a temperature scale defined the freezing point of water as 100° and the boiling point as 0° . What temperature on this scale corresponds to 25°C ?

[75°]

(iii) In an alcohol-in-glass thermometer, the alcohol column has length 12.45 cm at 0.0°C and length 21.30 cm at 100.0°C . What is the temperature if the column has length (a) 15.10 cm, and (b) 22.95 cm.

[(a) 29.94°C , (b) 118.64°C]

(iv) At what temperature will the Fahrenheit and Centigrade scales yield the same numerical value?

[-40°]

1.4 Thermal Equilibrium

When two different system at different temperatures are kept in contact for a long duration or connected through a diathermic wall for a long duration till all the changes in their temperature are ceased, this situation is called thermal equilibrium.

The condition of thermal equilibrium can be simply related by the temperature of respective bodies. Earlier we've discussed that temperature of a body determines the total kinetic energy of all the molecules of the body. When two bodies at different temperature are connected by a heat conducting material, thermal energy starts flowing from a high temperature body to the body at lower temperature. This energy which flows from one body to another due to its thermal properties, is called heat. Remember the word "heat" is only used for the energy which is being transferred or energy in flow from or to a body due to its thermal properties. The word "heat" can never be used for energy stored in a body whether or not due to its thermal properties.

As stated above when two bodies at different temperatures are connected by a diathermic material, heat starts flowing between them from high temperature body to the lower temperature body till their temperature becomes equal and after that no net heat flow takes place between them. They are now said to be in thermal equilibrium. Thus we can state that when two bodies in thermal equilibrium (at same temperatures) are connected, no net heat flow takes place between them. On the basis of this concept a law is defined called zeroth law of thermodynamics and is stated as

"When two different bodies are in thermal equilibrium with a third body independently then both of these are also in thermal equilibrium."

1.5 Thermal Expansion

When a body is heated, its temperature rises and as we've discussed that on increasing temperature total kinetic energy of its molecules increases. In a solid, molecules can only have thermal agitation (random vibrations). As temperature of a body increases, the vibrations of molecules will become fast and due to this the rate of collision among neighbouring molecules increases. As the collisions between neighbouring molecules increases, it develops a thermal stress in the body and due to this the intermolecular separation increases which results in thermal expansion of body. In the similar way the block diagram shown in figure-1.6 explains the way how thermal expansion takes place.

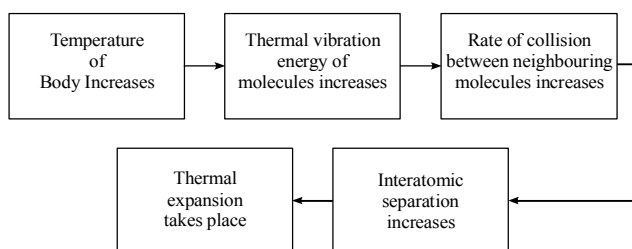


Figure 1.6

Thermal expansion of a substance can be classified in three broad categories, these are (i) Linear Expansion (ii) Superficial Expansion and (iii) Cubical Expansion or Volume Expansion

1.5.1 Linear Expansion

This is expansion in dimension of length of a solid body. It takes place only in solids. To explain linear expansion we take a simple example of a ruler scale. Figure-1.7 shows a ruler scale of length L at temperature T . If the temperature of scale increases, due to thermal expansion, its length also increase. It is observed that on increasing temperature by ΔT , the expansion in its length ΔL is directly proportional to the rise in temperature ΔT as

$$\Delta L \propto \Delta T \quad \dots (1.5)$$

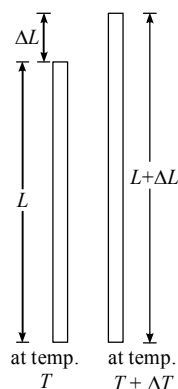


Figure 1.7

It is also observed that expansion of large object is more due to more number of molecule and equal increase in all intermolecular spacing increases the dimensions proportionally, thus we can say that the expansion in its length ΔL is also proportional to the initial length of the scale as

$$\Delta L \propto L \quad \dots (1.6)$$

From equation-(1.5) and (1.6), we have

$$\Delta L \propto L \Delta T$$

or

$$\Delta L = \alpha L \Delta T \quad \dots (1.7)$$

Thus the final length at higher temperature can be given as

$$L_f = L + \Delta L$$

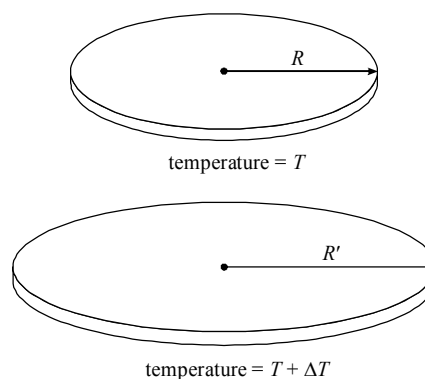
or

$$L_f = L (1 + \alpha \Delta T) \quad \dots (1.8)$$

Here α is the proportionality constant which depends on the material of the body in simple cases, and is called coefficient of linear expansion. Theoretically α is supposed to be a constant but practically it is seen that the value of α varies with temperature and its relation also changes with temperature depending on the range of temperatures. But for a short range of temperature α can be taken as a constant.

The above relation given by equation-(1.8) is not only valid for the length of an object but it can also be used to obtain the enclosed length of an object or distance between two points on an object. For example consider the disc of radius R shown in figure-1.8. If temperature of this disc increases, its size will also increase, as we have discussed, any change in length due to temperature variation can be given by equation-(1.8), we can use the same result to find the final radius of the disc as

$$R' = R (1 + \alpha \Delta T) \quad \dots (1.9)$$



Similarly this expression can also be used to find the distance between any two points on a body. For example figure-1.9 shows a copper rod bent in the form of a semicircle. We consider

two points on it marked as A and B . Let the separation between the two is l at a temperature T . If the temperature of rod is increased by ΔT . It will expand and hence the length l will also increase and become l' . As here A & B are the points on a body made of similar molecules and one lattice structure, the separation between A and B follows the similar law of linear expansion. Thus l' can be given as

$$l' = l(1 + \alpha \Delta T) \quad \dots (1.10)$$

Where α is the coefficient of linear expansion of material of rod.

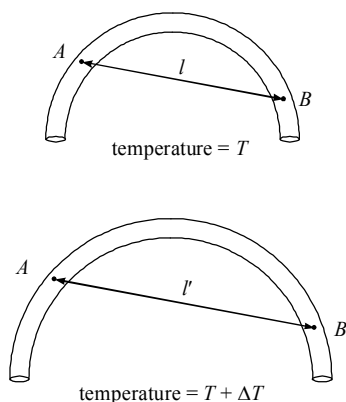


Figure 1.9

Similarly the same result can also be used for lengths and dimensions of an object enclosed by a material like radius of a hole in a disc or radius of a spherical cavity in a solid object. While applying the concept of linear expansion in such cases actually the length is that of air section but we use the coefficient of linear expansion for the material enclosing this air section. Now we take some examples for understanding the linear expansion in detail.

1.5.2 Stress in Objects Due to Thermal Expansion

In thermal expansion the inter molecular separation between molecules of an object increases. It results several applications. For example consider a metal wire of length l , just taut between two rigid clamps at same separation l as shown in figure-1.10(a). If the temperature of surrounding increases, the length of wire will increase due to thermal expansion and there will be a sag in the wire as shown in figure-1.10(b). If the temperature of wire will decrease, it will tend to contract but due to clamps at the ends, it is not allowed to contract. If due to fall in temperature ΔT , its contraction would be $\Delta L = \alpha L \Delta T$ then this change in length will be balanced by the elastic strain in the wire. As temperature starts decreasing, intermolecular spacing between molecules of wire tend to decrease but due to clamps, a tension starts developing in the wire which stretches the wire by the same amount to keep its length constant. If total drop in

temperature is ΔT , change in length due to thermal expansion can be written as

$$\Delta L = \alpha L \Delta T \quad \dots (1.11)$$

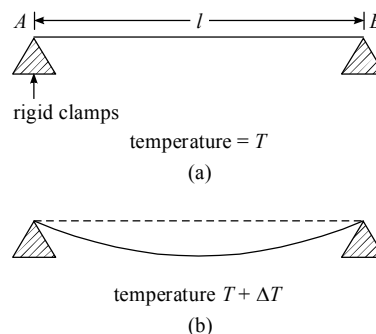


Figure 1.10

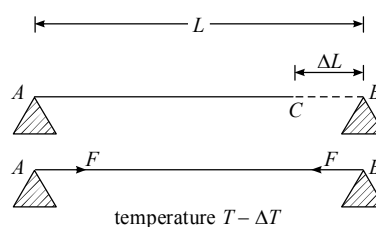


Figure 1.11

As shown in figure-1.11 if wire is not clamped at end B , its new length should become AC which is $L - \Delta L$. If it is externally stretched and the end C is tied to clamp B , this ΔL will be the extension in string length due to its elasticity. If cross-sectional area of wire is S and F be the tension developed in wire due to stretching. Thus the stress developed in the wire due to this tension F is given as

$$\text{stress} = \frac{F}{S} \quad \dots (1.12)$$

Strain produced in wire due to its elastic properties is

$$\text{strain} = \frac{\Delta L}{L} = \alpha \Delta T \quad \dots (1.13)$$

If Young's modulus of the material of wire is Y , we have

$$Y = \frac{\text{stress}}{\text{strain}}$$

or

$$Y = \frac{F/S}{\alpha \Delta T}$$

or

$$F = Y S \alpha \Delta T \quad \dots (1.14)$$

Equation-(1.14) gives the expression for tension in the wire due to decrease in its temperature by ΔT . This result gives the tension in wire if initially wire is just taut between clamps. If it already has some tension in it then this expression will give the increment in tension in the wire.

Figure-1.12(a) shows another similar example. A rod of cross sectional area S and length L is supported between two rigid walls at a separation exactly equal to that of length of rod so that rod just fits between the two walls.

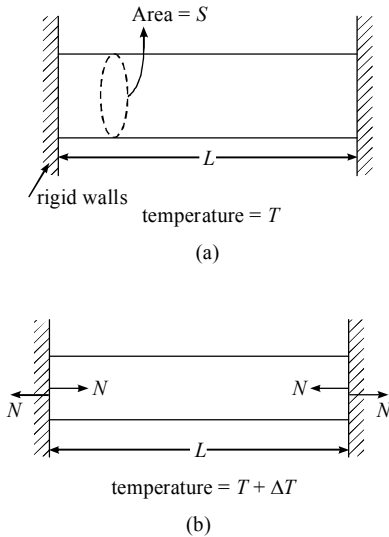


Figure 1.12

If surrounding temperature decreases, the rod contracts and it will fall down as the separation between the wall does not change. But if temperature of surrounding increase, the rod will tend to expand and start pushing the two walls with a normal force as shown in figure-1.12(b) and as temperature of system increases, the magnitude of normal force between walls and rod is also increased. As rigid walls do not allow rod to expand from its original length L , as temperature increases, N will increase to produce an elastic strain in it due to which at any higher temperature the increment in length of rod due to thermal expansion is exactly compensated by the elastic compression in the rod due to normal reaction acting on it by the walls.

If temperature of surrounding increases by ΔT , the natural increase in length due to thermal expansion is given as

$$\Delta L = \alpha L \Delta T$$

If N is the normal reaction on rod's cross-section then due to walls, elastic stress on rod is

$$\text{stress} = \frac{N}{S} \quad \dots (1.15)$$

As length of rod remains constant, this implies that due to the normal reaction on rod, it is elastically compressed by exactly the same amount by which it would thermally expand. Thus elastic strain in rod is given as

$$\text{strain} = \frac{\Delta L}{L} = \alpha \Delta T \quad \dots (1.16)$$

If Y is the Young's modulus of the material of rod then from equation-(1.15) & (1.16), we have

$$Y = \frac{\text{stress}}{\text{strain}} = \frac{N/S}{\Delta L/L}$$

or

$$N = Y S \alpha \Delta T \quad \dots (1.17)$$

Similar to the previous case the expression in equation-(1.17) gives us increment in normal reaction between walls and rod if initially some normal force between the two exists.

1.5.3 Variation in Time Period of Pendulum Clocks by Thermal Expansion

In pendulum clocks, their oscillation period depends on length of pendulum and the acceleration due to gravity and is given as

$$t = 2\pi \sqrt{\frac{l}{g}} \quad \dots (1.18)$$

If a pendulum clock is taken to a region where temperature is ΔT above this temperature. The new time period of oscillation of pendulum at higher temperature is given as

$$t' = 2\pi \sqrt{\frac{l'}{g}}$$

or

$$t' = 2\pi \sqrt{\frac{l(1 + \alpha_0 \Delta T)}{g}}$$

[If α_0 is the coefficient of linear expansion of material of pendulum rod]

or

$$t' = t(1 + \alpha_0 \Delta T)^{1/2}$$

or

$$t' = t \left(1 + \frac{1}{2} \alpha_0 \Delta T \right)$$

[As α_0 is very small $(1 + \alpha_0 \Delta T)^{1/2} \simeq (1 + \frac{1}{2} \alpha_0 \Delta T)$]

Change in period of oscillation can be given as

$$t' - t = \frac{1}{2} \alpha_0 t \Delta T \quad \dots (1.19)$$

Equation-(1.19) gives the increase in period of oscillation of the pendulum and we can see that the increase in time period depends on the rise in temperature as well as the initial time period of oscillations.

Here if we find the change in time per second, we get

$$\frac{t' - t}{t} = \frac{1}{2} \alpha_0 \Delta T \quad \dots (1.20)$$

Equation-(1.20) gives the time lost per second by the pendulum clock. If ΔT is the fall in temperature, same equation will give the time gained by the clock per second as its oscillation will become faster due to reduction in its length.

Now we take some numerical examples to understand the applications of thermal expansion in a better way.

Illustrative Example 1.3

In an aluminium sheet there is a hole of diameter 2 m and is horizontally mounted on a stand. Onto this hole an iron sphere of diameter 2.004 m is resting. Initial temperature of this system is 25°C. Find at what temperature, the iron sphere will fall down through the hole in sheet. The coefficients of linear expansion for aluminium and iron are 2.4×10^{-5} and 1.2×10^{-5} respectively.

Solution

As value of coefficient of linear expansion for aluminium is more than that for iron, it expands faster than iron. So at some higher temperature when diameter of hole will exactly become equal to that of iron sphere, the sphere will pass through the hole. Let it happen at some higher temperature T . Thus we have at this temperature T ,

$$(\text{diameter of hole})_{Al} = (\text{diameter of sphere})_{iron}$$

$$2[1 + \alpha_{Al}(T - 25)] = 2.004 [1 + \alpha_{iron}(T - 25)]$$

$$2\alpha_{Al}(T - 25) = 0.004 + 2.004 \alpha_{iron}(T - 25)$$

$$\text{or } T = \left(\frac{0.004}{2\alpha_{Al} - 2.004\alpha_{iron}} + 25 \right) ^\circ\text{C}$$

$$\text{or } T = \frac{0.004}{2 \times 2.4 \times 10^{-5} - 2.004 \times 1.2 \times 10^{-5}} + 25$$

$$\text{or } T = 191.7^\circ\text{C}$$

Illustrative Example 1.4

A metal rod A of 25 cm length expands by 0.05 cm when its temperature is raised from 0°C to 100°C. Another rod B of a different metal of length 40 cm expands by 0.04 cm for the same rise in temperature. A third rod C of 50 cm length is made up of pieces of rods A and B placed end to end expands by 0.03 cm on heating from 0°C to 50°C. Find the length of each portion of composite rod C .

Solution

From the given data for rod A , we have

$$\Delta L = \alpha_A L \Delta T$$

or

$$\begin{aligned} \alpha_A &= \frac{\Delta L}{L \Delta T} = \frac{0.05}{25 \times 100} \\ &= 2 \times 10^{-5} ^\circ\text{C}^{-1} \end{aligned}$$

For rod B , we have $\Delta L = \alpha_B L \Delta T$

or

$$\begin{aligned} \alpha_B &= \frac{\Delta L}{L \Delta T} = \frac{0.04}{40 \times 100} \\ &= 10^{-5} ^\circ\text{C}^{-1} \end{aligned}$$

If rod C is made of segments of rod A and B of lengths l_1 and l_2 respectively then we have at $^\circ\text{C}$

$$l_1 + l_2 = 50 \text{ cm} \quad \dots (1.21)$$

At

$$T = 50^\circ\text{C}$$

$$l'_1 + l'_2 = 50.03 \text{ cm}$$

Thus $\alpha_A l_1 \Delta T + \alpha_B l_2 \Delta T = 0.03 \text{ cm}$

$$\text{or } 2 \times 10^{-5} \times l_1 \times 50 + 10^{-5} \times l_2 \times 50 = 0.03 \text{ cm}$$

$$\text{or } 2l_1 + l_2 = \frac{0.03}{50} \times 10^5 = 60 \text{ cm} \quad \dots (1.22)$$

Solving-(1.21) and (1.22) we get $l_1 = 10 \text{ cm}$ and $l_2 = 40 \text{ cm}$

Illustrative Example 1.5

Two straight thin bars, one of brass and the other of steel are joined together side by side by short steel cross-pieces at 0°C, one cm long, the centre lines of the bars being one cm apart. When heated to 100°C, the composite bar becomes bent into the arc of a circle. Calculate the radius of this circle.

$$\alpha \text{ for brass} = 19 \times 10^{-6} \text{ per } ^\circ\text{C},$$

and

$$\alpha \text{ for steel} = 11 \times 10^{-6} \text{ per } ^\circ\text{C}.$$

Solution

As the expansion of brass is greater and hence the combination will bend with brass rod outside as shown in figure-1.13.

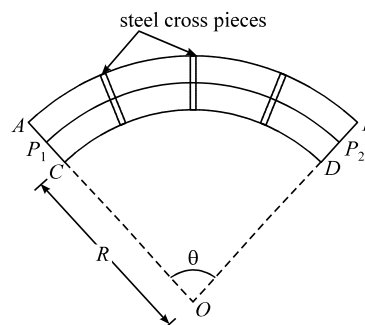


Figure 1.13

In figure, AB is brass bar and CD is steel bar.

Let $OC = R$ and $\angle COD = \theta$.

Length of steel cross pieces at 100°C

$$= 1(1 + 11 \times 10^{-6} \times 100) = 1.0011 \text{ cm}$$

Now $OA = (R + 1.0011) \text{ cm}$

and $CD = R\theta$... (1.23)

and $AB = (R + 1.0011)\theta$... (1.24)

Also, $AB = L_0(1 + 19 \times 10^{-6} \times 100)$... (1.25)

and $CD = L_0(1 + 11 \times 10^{-6} \times 100)$... (1.26)

From equation-(1.23), (1.24), (1.25) and (1.26), we have

$$\frac{AB}{CD} = \frac{R + 1.0011}{R} = \frac{(1 + 19 \times 10^{-6} \times 100)}{(1 + 11 \times 10^{-6} \times 100)}$$

$$\text{or } 1 + \frac{1.001}{R} = \frac{1 + 19 \times 10^{-4}}{1 + 11 \times 10^{-4}} \cdot \frac{1.0011}{R} = \frac{8 \times 10^{-4}}{1 + 11 \times 10^{-4}}$$

$$\begin{aligned} \text{or } R &= (1.0011) \frac{(1 + 11 \times 10^{-4})}{8 \times 10^{-4}} \\ &= 1252.8 \text{ cm} \end{aligned}$$

Illustrative Example 1.6

A clock with a metallic pendulum is 5 seconds fast each day at a temperature of 15°C and 10 seconds slow each day at a temperature of 30°C . Find coefficient of linear expansion for the metal.

Solution

We've discussed that time lost or gained per second by a pendulum clock is given by

$$\delta t = \frac{1}{2} \alpha \Delta T$$

Here temperature is higher than graduation temperature thus clock will lose time and if it is lower than graduation temperature will gain time.

Thus time lost or gained per day is

$$\delta t = \frac{1}{2} \alpha \Delta T \times 86400 \quad [\text{As 1 day} = 86400 \text{ s.}]$$

If graduation temperature of clock is T_0 then we have

At 15°C , clock is gaining time, thus

$$5 = \frac{1}{2} \cdot \alpha \cdot (T_0 - 15) \times 86400 \quad \dots (1.27)$$

At 30°C clock is losing time, thus

$$10 = \frac{1}{2} \alpha (30 - T_0) \times 86400 \quad \dots (1.28)$$

Dividing equation-(1.28) by (1.27), we get

$$2(T_0 - 15) = (30 - T_0)$$

or $T_0 = 20^\circ\text{C}$

Thus from equation-(1.27)

$$\begin{aligned} 5 &= \frac{1}{2} \times \alpha \times [20 - 15] \times 86400 \alpha \\ &= 2.31 \times 10^{-5} ^\circ\text{C}^{-1} \end{aligned}$$

Illustrative Example 1.7

What should be the lengths of steel and copper rod so that the length of steel rod is 5 cm longer than the copper rod at all the temperatures. Coefficients of linear expansion for copper and steel are

$$\alpha_{Cu} = 1.7 \times 10^{-5} ^\circ\text{C}^{-1} \quad \text{and} \quad \alpha_{steel} = 1.1 \times 10^{-5} ^\circ\text{C}^{-1}$$

Solution

It is given that the difference in length of the two rods is always 5 cm. Thus the expansion in both the rods must be same for all temperature differences. Thus we can say that at all temperature differences, we have

$$\Delta L_{Cu} = \Delta L_{steel}$$

$$\text{or } \alpha_{Cu} l_1 \Delta T = \alpha_{st} l_2 \Delta T$$

[If l_1 and l_2 are the initial lengths of Cu and steel rods]

$$\text{or } \alpha_{Cu} l_1 = \alpha_{st} l_2$$

$$\text{or } 1.7 l_1 = 1.1 l_2 \quad \dots (1.29)$$

$$\text{It is given that } l_2 - l_1 = 5 \text{ cm} \quad \dots (1.30)$$

From equations-(1.29) and (1.30) we have

$$\left(\frac{1.7}{1.1} - 1 \right) l_1 = 5 \text{ cm}$$

$$\text{or } l_1 = \frac{5 \times 1.1}{0.6} = 9.17 \text{ cm}$$

Now from equation-(1.30)

$$l_2 = 14.17 \text{ cm}$$

Illustrative Example 1.8

A steel wire of cross-sectional area 0.5 mm^2 is held between two rigid clamps so that it is just taut at 20°C . Find the tension in the wire at 0°C . Given that Young's Modulus of steel is $Y_{st} = 2.1 \times 10^{12} \text{ dynes/cm}^2$ and coefficient of linear expansion of steel is $\alpha_{st} = 1.1 \times 10^{-5} ^\circ\text{C}^{-1}$.

Solution

We know that due to drop in temperature, the tension increment in a clamped wire is

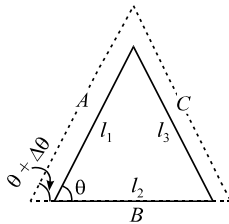
$$\begin{aligned} T &= YA \alpha \Delta T \\ &= 2.1 \times 10^{12} \times 0.5 \times 10^{-2} \times 1.1 \times 10^{-5} \times 20 \\ &= 2.31 \times 10^{-6} \text{ dynes.} \end{aligned}$$

Illustrative Example 1.9

Three rods A , B and C , having identical shape and size, are hinged together at ends to form an equilateral triangle. Rods A and B are made of same material having coefficient of linear thermal expansion α_1 while that of material of rod C is α_2 . By how many kelvin should the system of rods be heated to increase the angle opposite to rod C by $\Delta\theta$.

Solution

When the system is heated, the rods expand and the triangle does not remain equilateral. Let lengths of rods A , B and C be l_1 , l_2 and l_3 respectively. Then

**Figure 1.14**

$$\cos \theta = \frac{l_1^2 + l_2^2 - l_3^2}{2 l_1 l_2}$$

$$\text{or } 2 l_1 l_2 \cos \theta = l_1^2 + l_2^2 - l_3^2 \quad \dots (1.31)$$

Differentiating equation-(1.30), we get

$$\begin{aligned} 2 l_1 \cos \theta dl_2 + 2 l_1 \cos \theta dl_1 - 2 l_1 l_2 \sin \theta d\theta \\ = 2 l_1 dl_1 + 2 l_2 dl_2 - 2 l_3 dl_3 \quad \dots (1.32) \end{aligned}$$

Let the temperature of the system be increased by ΔT . Then

$$dl_1 = l_1 \alpha_1 \Delta T$$

$$dl_2 = l_2 \alpha_1 \Delta T \quad \dots (1.33)$$

and

$$dl_3 = l_3 \alpha_2 \Delta T$$

In case of equilateral triangle $l_1 = l_2 = l_3 = l$ (say) and $\theta = 60^\circ$ and for small change in angle $d\theta = \Delta\theta$

From equation-(1.32) and (1.33), we get

$$\begin{aligned} 2 l^2 \cos (60^\circ) \alpha_1 \Delta T + 2 l^2 \cos (60^\circ) \alpha_1 \Delta T - 2 l^2 \sin (60^\circ) \Delta\theta \\ = 2 l^2 \alpha_1 \Delta T + 2 l^2 \alpha_1 \Delta T - 2 l^2 \alpha_2 \Delta T \\ \frac{\alpha_1}{2} \Delta T + \frac{\alpha_1}{2} \Delta T - \frac{\sqrt{3}}{2} \Delta\theta \\ = \alpha_1 \Delta T + \alpha_1 \Delta T - \alpha_2 \Delta T \\ \alpha_1 \Delta T - \frac{\sqrt{3}}{2} \Delta\theta = 2 \alpha_1 \Delta T - \alpha_2 \Delta T \\ (\alpha_2 - \alpha_1) \Delta T = \frac{\sqrt{3}}{2} \Delta\theta \\ \Delta T = \frac{\sqrt{3} \Delta\theta}{2(\alpha_2 - \alpha_1)} \end{aligned}$$

Illustrative Example 1.10

A pendulum clock and a digital clock both are synchronized to keep correct time at temperature 20°C in the morning on 1st March, 2003. At 12:00 noon temperature increases to 40°C and remains constant for three months. Now on 1st June, 2003, at 12:00 noon temperature drops to 10°C and remains constant for a very long duration. Find the date and time on which both the clocks will again be synchronized for a moment.

Solution

As a digital clock (if ideal) always keeps correct time. But on increasing temperature on 1st March 12:00 noon, the pendulum clock slows down and start losing time. We know that time lost by a pendulum clock per second is given as

$$\delta t = \frac{1}{2} \alpha \Delta T$$

[If α is the coefficient of linear expansion for the material of pendulum]

$$= \frac{1}{2} \alpha (20)$$

In three months (March + April + May = 92 days) it loses time

$$\delta t_{92 \text{ days}} = \frac{1}{2} \times \alpha \times 20 \times 92 \times 86400$$

On 1st June, 12:00 noon, temperature drops to 10°C which is 10° less than the temperature at which clock keeps correct time,

thus now clock starts gaining time and if after N days it gains exactly the time lost during previous three months, it shows right time again for a moment. Thus time gained by the clock in N days is

$$\delta t_{N \text{ days}} = \frac{1}{2} \alpha (10) \times N \times 86400$$

We have $\delta t_{92 \text{ days}} (\text{lost}) = \delta t_{N \text{ days}} (\text{gained})$

$$\text{or } \frac{1}{2} \times \alpha \times 20 \times 92 \times 86400 = \frac{1}{2} \times \alpha \times 10 \times N \times 86400$$

$$\text{or } N = 184 \text{ days.}$$

Thus after 184 days from 1st June 2003, pendulum clock will show correct time and both the clocks will be in synchronization for a moment and after 184 days means the date is 2nd Dec. 2003 and time is 12:00 noon.

Illustrative Example 1.11

At room temperature (25°C) the length of a steel rod is measured using a brass centimeter scale. The measured length is 20 cm. If the scale is calibrated to read accurately at temperature 0°C. Find the actual length of steel rod at room temperature

Solution

The brass scale is calibrated to read accurately at 0°C, this means at 0°C, each division of scale has exact 1 cm length. Thus at higher temperature the division length of scale will be more than 1 cm due to thermal expansion. Thus at higher temperatures the scale reading for length measurement is not appropriate and as at higher temperature the division length is more, the length this scale reads will be lesser than the actual length to be measured. For example in this case the length of each division on brass scale at 25°C is

$$l_{1 \text{ div}} = (1 \text{ cm}) [1 + \alpha_{br} (25 - 0)] \\ = 1 + \alpha_{br} (25)$$

It is given that at 25°C the length of steel rod measured is 20 cm. Actually it is not 20 cm, it is 20 divisions on the brass scale. Now we can find the actual length of the steel rod at 25°C as

$$l_{25^\circ\text{C}} = (20 \text{ cm}) \times l_{1 \text{ div}} \\ \text{or } l_{\text{actual}} = 20 [1 + \alpha_{br} (25)] \quad \dots (1.34)$$

The above expression is a general relation using which you can find the actual lengths of the objects of which lengths are measured by a metallic scale at some temperature other than the graduation temperature of the scale.

Illustrative Example 1.12

A rod AB of length l is pivoted at an end A and freely rotated in a horizontal plane at an angular speed ω about a vertical axis passing through A . If coefficient of linear expansion of material of rod is α , find the percentage change in its angular velocity if temperature of system is increased by ΔT .

Solution

If temperature of surrounding increases by ΔT , the new length of rod becomes

$$l' = l (1 + \alpha \Delta T)$$

Due to change in length, moment of inertia of rod also changes

and it is about an end A and is given as $I'_A = \frac{Ml'^2}{3}$

As no external force or torque is acting on rod thus its angular momentum remains constant during heating thus we have

$$I_A \omega = I'_A \omega'$$

[If ω' is the final angular velocity of rod after heating]

$$\text{or } \frac{Ml^2}{3} \omega = \frac{Ml'^2 (1 + \alpha \Delta T)^2}{3} \omega'$$

$$\text{or } \omega' = \omega (1 - 2\alpha \Delta T)$$

[Using binomial expansion for small α]

Thus percentage change in angular velocity of rod due to heating can be given as

$$\Delta \omega = \frac{\omega - \omega'}{\omega} \times 100\% \\ = 2\alpha \Delta T \times 100\%$$

Illustrative Example 1.13

A compensated pendulum shown in figure-1.15 is in the form of an isosceles triangle of base length $l_1 = 5 \text{ cm}$ and coefficient of linear expansion $\alpha_1 = 18 \times 10^{-6}$ and side length l_2 and coefficient of linear expansion $\alpha_2 = 12 \times 10^{-6}$. Find l_2 so that the distance of centre of mass of the bob from suspension centre O may remain the same at all the temperature.

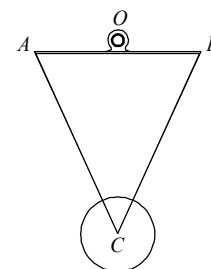


Figure 1.15

Solution

First we must know what is a compensated pendulum. We've discussed that due to change in temperature the time period of

a pendulum clock changes. Due to this in pendulum clocks, to make a pendulum some specific metals are used which have very low coefficient of expansion so that the error introduced in their time is very small. The other alternative of minimizing the error in time measurement, is to use compensated pendulum. This is a pendulum made up of two or more metals of such an arrangement in which the effective length of centre of mass of the whole body remains unchanged. Such a situation is given in figure-1.15.

In this case the distance of centre of mass from suspension point O is

$$h = \sqrt{l_2^2 - \frac{l_1^2}{4}}$$

At some higher temperature Δt , new value of h will become

$$h' = \sqrt{l_2^2 (1 + \alpha_2 \Delta t)^2 + \frac{l_1^2}{4} (1 + \alpha_1 \Delta t)^2}$$

As we require $h = h'$, we have

$$l_2^2 \frac{l_1^2}{4} = l_2^2 (1 + 2\alpha_2 \Delta t) - \frac{l_1^2}{4} (1 + 2\alpha_1 \Delta t)$$

or $2\alpha_2 l_2^2 = \frac{l_1^2}{4} \alpha_1$

or $l_2 = \frac{l_1}{2} \sqrt{\frac{\alpha_1}{\alpha_2}}$

Illustrative Example 1.14

Two same length rods of brass and steel of equal cross-sectional area are joined end to end as shown in figure-1.16 and supported between two rigid vertical walls. Initially the rods are unstrained. If the temperature of system is raised by Δt . Find the displacement of the junction of two rods. Given that the coefficients of linear expansion and young's modulus of brass and steel are α_b , α_s ($\alpha_b > \alpha_s$), Y_b and Y_s respectively.

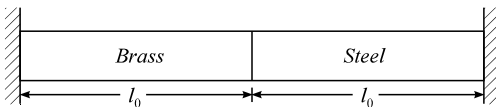


Figure 1.16

Solution

Here the initial lengths of the two rods are l_0 and as $\alpha_b > \alpha_s$, we can directly state that the junction of the two rods is displaced toward right. Due to this brass rod is some what expanded but less as compared to free expansion and steel is overall compressed due to the stress developed between the two rods.

Figure-1.17 shows the final situation of rods at higher temperature Δt . In figure, A is the initial position of junction and B is its final position.

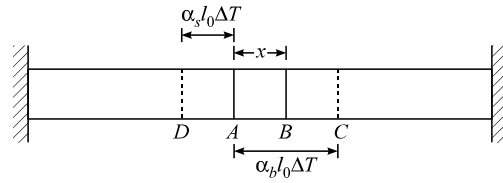


Figure 1.17

If only brass rod is present, expansion in it due to increase in temperature by ΔT is

$$\Delta l_b = \alpha_b l_0 \Delta T$$

Which is equal to AC in figure-1.17. But finally the rod is expanded by x , thus the elastic compression in the rod due to stress between the two rods is BC , which is given as

$$BC = \alpha_b l_0 \Delta T - x$$

Thus elastic strain in brass rod is

$$(\text{Strain})_{\text{brass}} = \frac{\alpha_b l_0 \Delta T - x}{l_0} \quad [\text{As } \alpha_b l_0 \Delta T \text{ is small}]$$

Similarly if steel rod alone is there, it would have been expanded by

$$\Delta l_s = \alpha_s l_0 \Delta T$$

Which is equal to AD in figure-1.17. But finally steel rod is compressed by x , thus the elastic compression in steel rod due to stress between the two rods is BD , which is given as

$$BD = \alpha_s l_0 \Delta T + x$$

Thus elastic strain in steel rod is

$$(\text{Strain})_{\text{steel}} = \frac{\alpha_s l_0 \Delta T + x}{l_0} \quad [\text{As } \alpha_s l_0 \Delta T \text{ is small}]$$

As the two rods are in contact, stress developed in the two rods must be equal thus we have

$$(\text{Stress})_{\text{brass}} = (\text{Stress})_{\text{steel}}$$

or $Y_b \times (\text{Strain})_{\text{brass}} = Y_s \times (\text{Strain})_{\text{steel}}$

or $Y_b \left[\frac{\alpha_b l_0 \Delta T - x}{l_0} \right] = Y_s \left[\frac{\alpha_s l_0 \Delta T + x}{l_0} \right]$

or $x = \frac{(Y_b \alpha_b - Y_s \alpha_s) l_0 \Delta T}{(Y_b + Y_s)}$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Thermal Expansion & Thermometry

Module Numbers - 1 to 10

Practice Exercise 1.2

(i) A steel tape measures the length of a copper rod as 90.0 cm and when both are at 10°C, which is the graduation temperature of the tape. What would the tape read for the length of the rod when both are at 30°C. Given that coefficient of linear expansion of steel is $1.2 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$ and that of copper is $1.7 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.

[90.01 cm]

(ii) In the construction of a flyover, steel girders of length 12 m are used to place one after another. How much gap should be left between the two at their junction so that in summers when peak temperature is about 48°C, there should not be any compression. Given that the construction is done in peak winters when temperature is 18°C. Given that the coefficient of linear expansion of steel is $1.1 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.

[3.96 × 10⁻³ m]

(iii) A steel rod is clamped between two rigid supports at 20°C. If the temperature of rod increases to 50°C, find the strain developed in the rod due to this. Given that the coefficient of linear expansion of steel is $1.2 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.

[− 3.6 × 10⁻⁴]

(iv) Figure-1.18 shows a steel rod of cross sectional area $2 \times 10^{-6} \text{ m}^2$ is fixed between two vertical walls. Initially at 20°C, there is no force between the ends of the rod and the walls. Find the force which the rod will exert on walls at 100°C. Given that the coefficient of linear expansion of steel is $1.2 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$ and its Young's modulus is $2 \times 10^{11} \text{ N/m}^2$.

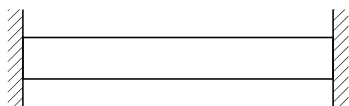


Figure 1.18

[384 N]

(v) A glass window is to be fitted in an aluminium frame. The temperature on the working day is 40°C and the size of the glass piece is a rectangle of sides 20 cm and 30 cm. What should be the dimensions of the aluminium frame so that in winters glass does not experience any stress when temperature drops to 0°C. Given that the coefficients of linear expansion for glass is $9 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ and that for aluminium is $2.4 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.

[20.012 cm and 30.018 cm]

(vi) A cylindrical steel component machined on an engine lathe is heated to temperature of 80°C. The diameter of the component should be 5 cm at temperature of 10°C and the permissible error should not exceed 10 microns from the specified dimension. Do we need corrections for the thermal expansion component be introduced during the process of machining? If so, what diameter should be prepared? Take : $\alpha_{\text{steel}} = 12 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$.

[Yes, 5.0052 cm]

(vii) At a certain temperature the pendulum of a clock keeps correct time. The coefficient of linear expansion for the pendulum material is $\alpha = 1.85 \times 10^{-5} \text{ K}^{-1}$. How much will the clock gain or lose in 24 hours if the ambient temperature is 10°C higher?

[7.992 s]

(viii) A clock pendulum made of invar has a period of 0.5 s at 20°C. If the clock is used in a climate where average temperature is 30°C, what correction may be necessary at the end of 30 days. $\alpha_{\text{invar}} = 7 \times 10^{-7} (\text{ }^\circ\text{C})^{-1}$.

[9 s (to be added)]

(ix) The time period of a physical pendulum is given by

$T = 2\pi \sqrt{\frac{I}{mgl}}$ where m = mass of the pendulum, I = moment of inertia about the axis of suspension, l = distance of centre of mass of bob from the centre of suspension. Calculate the change in time period when temperature changes by ΔT . The coefficient of linear expansion of the material of pendulum is α .

$[\pi \alpha \Delta T \sqrt{\frac{I}{mgl}}]$

(x) Two rods of material X sandwich another rod of material Y as shown in figure-1.19. At temperature T , the three rods are in a state of zero strain and of length L and riveted to each other in this state. If the temperature of the system increases to $T + \Delta T$, find the final length of the system of the three rods. Given that the coefficients of linear expansions of the rods and their Young's modulus for material X and Y are α_x , α_y , Y_x and Y_y respectively. Consider $\alpha_y > \alpha_x$.

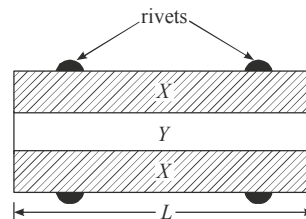


Figure 1.19

$\left[L \left[1 + \frac{\alpha_y Y_y + 2\alpha_x Y_x}{Y_y + 2Y_x} \Delta T \right] \right]$

1.5.4 Superficial Expansion

This is expansion in surface of a body or two dimensional expansion of an object. Figure-1.20 shows a rectangular plate of size $l \times b$, when its temperature increases from T to $T + \Delta T$, both of its length and width increases as thermal expansion takes place uniformly in all direction in a material. If α be the coefficient of linear expansion for the material of plate, its final length and width at higher temperature can be given as

$$l' = l(1 + \alpha\Delta T) \quad \dots (1.35)$$

$$b' = b(1 + \alpha\Delta T) \quad \dots (1.36)$$

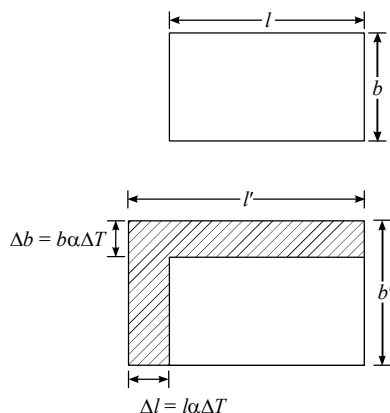


Figure 1.20

Final area of plate is

$$A' = l' b' = l b (1 + \alpha\Delta T)^2$$

$$\text{or} \quad A' = A (1 + 1 \alpha\Delta T)^2 \quad [\text{Initial area } A = l b]$$

$$\text{or} \quad A' = A (1 + 2 \alpha\Delta T) \quad [\text{As } \alpha\Delta T \ll 1, \text{ using binomial expansion}]$$

$$\text{or} \quad A' = A (1 + \beta\Delta T) \quad \dots (1.37)$$

Here $\beta = 2\alpha$ is called coefficient of superficial expansion for the material of object. In some objects if linear expansion coefficient α is not very small (which is very rare), we can not use $\beta = 2\alpha$ for that material as binomial expansion can not be used and value of β can only be experimentally evaluated.

From equation-(1.37) it can be seen that temperature of a body if decreases, its final area at lower temperature can be given as

$$A_f = \frac{A}{(1 + \beta\Delta T)}$$

For small β we can write this expression as

$$A_f = A (1 - \beta\Delta T)^{-1} = A (1 + \beta\Delta T) \quad \dots (1.38)$$

1.5.5 Cubical or Volume Expansion

As we know when temperature of a body increases, thermal expansion takes place uniformly in all directions. Due to this

volume of body also increases. Consider a box of length, width and height $l \times b \times h$. If its temperature increases by ΔT , and the coefficient of linear expansion for the material of box is α , the final volume of box will be given as

$$V_f = l b h (1 + \alpha\Delta T)^3$$

$$\text{or} \quad V_f = V_i (1 + 3\alpha\Delta T) \quad [\text{Where } V_i = l b h \text{ and for } \alpha\Delta T \ll 1]$$

$$\text{or} \quad V_f = V_i (1 + \gamma\Delta T) \quad \dots (1.39)$$

Here $\gamma = 3\alpha$ is called coefficient of volume or cubical expansion of the material of body. In case of expansion of liquids, length and area do not have any significance and so for a liquid we can only define coefficient of volume expansion. In liquids molecules are more mobile as compared to solid thus expansion of liquids is always more than that of solids, thus the value of γ is also higher for liquids than solids.

1.5.6 Variation in Density of a Substance

We know volume of a substance increases with rise in temperature. Thus we can obviously state that the density of object decreases with rise in temperature as its mass is a constant. For example if a body of mass m has density ρ_1 and volume V_1 at temperature T , then at some higher temperature $T + \Delta T$ the volume of body V_2 can be given as

$$V_2 = V_1 (1 + \gamma\Delta T) \quad \dots (1.40)$$

If the density of body now becomes ρ_2 then it can be given as

$$\rho_2 = \frac{m}{V_2} = \frac{\rho_1 V_1}{V_1 (1 + \gamma\Delta T)}$$

$$\text{or} \quad \rho_2 = \frac{\rho_1}{1 + \gamma\Delta T} \quad \dots (1.41)$$

$$\text{or} \quad \rho_2 = \rho_1 (1 + \gamma\Delta T)^{-1} = \rho_1 (1 - \gamma\Delta T) \quad \dots (1.42)$$

[As for $\gamma\Delta T \ll 1$]

We have discussed that with temperature expansion rate of liquids is faster than solids thus some times in case of liquids when coefficient of expansion γ is not a very small value, above equation-(1.42) will not be valid, as we can not use binomial approximation as $\gamma\Delta T$ is not very less than 1. Then variation of density of such liquids is given by equation-(1.41) only.

1.5.7 Weight Thermometer

This is a very good application of thermal expansion. In ancient times this device was used to measure surrounding temperature of a region. In it a liquid is taken in a container as shown in figure-1.21. Initially the liquid is filled to completely fill up the container and both have initial volume V_0 . If temperature of

region [container] is increased by ΔT , final volume of liquid and container will become V_l & V_c respectively. If γ_l and γ_c are the coefficient of volume expansion of liquid and container then V_l and V_c are given as

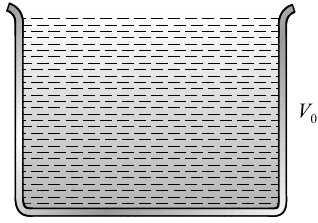


Figure 1.21

$$V_l = V_0 (1 + \gamma_l \Delta T) \quad \dots (1.43)$$

and
$$V_c = V_0 (1 + \gamma_c \Delta T) \quad \dots (1.44)$$

As $\gamma_l > \gamma_c$, liquid will expand more than container and some liquid will expel from the container. The volume of overflow liquid can be given as

$$\begin{aligned} \Delta V &= V_l - V_c \\ &= V_0 (\gamma_l - \gamma_c) \Delta T \end{aligned} \quad \dots (1.45)$$

Here this volume ΔV can be taken as apparent expansion of liquid relative to container and the term $(\gamma_l - \gamma_c)$ is called as coefficient of apparent expansion of liquid with respect to glass and can also be denoted by γ_{lc} , thus we can write the volume of liquid overflow is

$$\Delta V = V_0 \gamma_{lc} \Delta T \quad \dots (1.46)$$

The density of liquid at this higher temperature is given as

$$\rho_l' = \frac{\rho_l}{(1 + \gamma_l \Delta T)}$$

Mass of liquid overflow is $\Delta m = \Delta V \times \rho_l'$

$$\Delta m = \frac{V_0 (\gamma_l - \gamma_c) \Delta T \rho_l}{1 + \gamma_l \Delta T}$$

or
$$\Delta m + \gamma_l \Delta m \Delta T = V_0 (\gamma_l - \gamma_c) \Delta T \rho_l$$

or
$$\Delta T = \left[\frac{\Delta m}{V_0 (\gamma_l - \gamma_c) - \Delta m \gamma_l} \right] \frac{1}{\rho_l} \quad \dots (1.47)$$

If we weight the mass of liquid overflow, then using this in expression of equation-(1.47) we can find the rise in temperature.

1.5.8 Isotropic and Anisotropic Expansion

We've discussed that when a substance is heated, its temperature rises and thermal expansion takes place uniformly

in all direction. But this is not always true, the expansion of a body in different direction depends on the lattice structure of the material of body. In most of the solids, atoms are arranged at uniform interatomic distances thus expansion takes place uniformly in all directions. But in some solids interatomic separations in different dimensions are different. For example in graphite, the separation of carbon atoms in layers is different then the separation between layers as shown in figure-1.22. Thus when graphite is heated and its temperature increases, the expansion in the plane of layers of carbon atoms is uniform but only in two dimensional plane but the expansion along z direction in figure-1.22 i.e. in the direction normal to layers the expansion is different. If a cube made of graphite is heated, we can conclusively state that at some higher temperature two opposite faces of the cube out of six remains expanded squares, other four faces become rectangular.

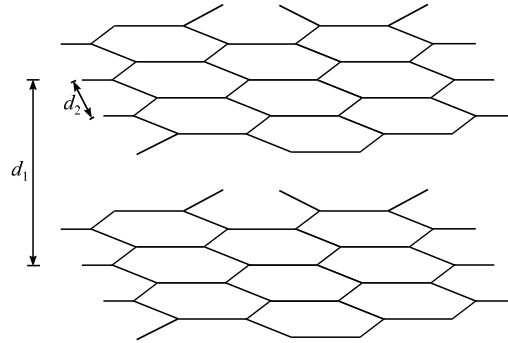


Figure 1.22

Thus when a solid expands uniformly in all three dimensions, this is called isotropic expansion of solid. In isotropic expansion the coefficient of linear expansion α remains same in all three dimension and thus the volume coefficient of thermal expansion can be written as

$$\gamma = 3\alpha \quad \dots (1.48)$$

When expansion of solid is different in the three dimensions due to atomic arrangement in solid lattice structure, this is termed as anisotropic expansion of solids. In anisotropic expansion the coefficient of linear expansion of solid is different in different dimensions of solid. For example in a piece of graphite if α_1 is the linear expansion coefficient in x and y direction (along the plane of carbon layers) and α_2 is the linear expansion coefficient in z direction (normal to graphite layers) then the coefficient of cubical expansion of solid is given as

$$\gamma = 2\alpha_1 + \alpha_2 \quad \dots (1.49)$$

Some times the value of α is different in all three dimensions in a solid say α_1 , α_2 and α_3 then in that case the value of γ is given as

$$\gamma = \alpha_1 + \alpha_2 + \alpha_3 \quad \dots (1.50)$$

As we've observed the anisotropic expansion is due to the properties of a lattice structure in atomic arrangement, it can be seen only in solids not in liquids or gases.

Now we take some examples to discuss thermal expansion in detail.

Illustrative Example 1.15

A one litre closed flask contains some mercury. It is found that at different temperatures the volume of air inside the flask remains the same. What is the volume of mercury in flask? Given coefficient of linear expansion of glass = $9 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$. Coefficient of volume expansion of Hg = $1.8 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$.

Solution

Here the volume of air in the flask remains the same. This is only possible when the expansion of flask is exactly the same as the expansion of mercury in the flask.

Here coefficient of cubical expansion of glass is

$$\gamma_g = 3\alpha = 3 \times 9 \times 10^{-6} = 27 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$$

Here to keep the volume of air in flask constant, we have

Expansion of flask = Expansion of mercury

$$(1000)\gamma_g \Delta T = x \cdot \gamma_m \Delta T$$

[Let x be the volume of mercury in the flask]

$$\begin{aligned} \text{or} \quad x &= V \cdot \frac{\gamma_g}{\gamma_m} \\ &= 1000 \times \frac{27 \times 10^{-6}}{1.8 \times 10^{-4}} = 150 \text{ cm}^3. \end{aligned}$$

Illustrative Example 1.16

A sphere of radius 3.5 cm and mass 250 gm floats in a liquid at 20°C . As the temperature is raised, the sphere just begin to sink at a temperature of 35°C . If the density of liquid at 0°C is 1.527 gm/cm^3 , find the coefficient of cubical expansion of the liquid. Neglect expansion of the sphere.

Solution

It is given that at 20°C , sphere is floating in liquid but at 35°C it begins to sink as on increasing temperature liquid expands and its density decreases. At 35°C , its density becomes equal to that of sphere and then the buoyancy force on sphere is just equal to the weight of sphere and it is fully submerged in the liquid.

Given that density of sphere is

$$\begin{aligned} \rho_s &= \frac{m}{\frac{4}{3}\pi r^3} \\ &= \frac{250}{\frac{4}{3} \times 3.14 \times (3.5)^3} = 1.396 \text{ gm/cm}^3 \end{aligned}$$

It is given that density of liquid at 0°C is $\rho_0 = 1.527 \text{ gm/cm}^3$ if γ is the coefficient of cubical expansion of liquid then its density at 35°C is given as

$$\rho_{35} = \frac{\rho_0}{1 + \gamma(35)}$$

At 35°C , as sphere begins to sink, we have

density of sphere = density of liquid

$$1.396 = \frac{1.527}{1 + \gamma(35)}$$

[As sphere does not expand, its density remains constant.]

$$\begin{aligned} \text{or} \quad \gamma &= \left(\frac{1.527}{1.396} - 1 \right) \times \frac{1}{35} \\ &= 2.681 \times 10^{-3} \text{ }^\circ\text{C}^{-1} \end{aligned}$$

Illustrative Example 1.17

A piece of metal weighs 46 gm in air. When it is immersed in a liquid of specific gravity 1.24 at 27°C , it weighs 30 gm. When the temperature of the liquid is raised to 42°C , the metal piece weighs 30.5 gm. The specific gravity of the liquid at 42°C is 1.20. Calculate the coefficient of linear expansion of the metal.

Solution

Here we first calculate the volume of the metal piece at 27°C as well as the volume of metal piece at 42°C . It is given that

Weight of the piece of metal in air = 46 gm

Weight of the piece of metal in liquid at 27°C = 30 gm

Thus loss of the weight of the piece of metal in liquid = $46 - 30 = 16 \text{ gm}$ = weight of liquid displaced

Thus the volume of metal piece at 27°C is

$$\begin{aligned} &= \frac{\text{weight of liquid displaced}}{\text{density}} = \frac{16}{1.24} \text{ cm}^3 \\ V_1 &= \frac{16}{1.24} \text{ cm}^3 \end{aligned}$$

Similarly, the volume of the piece of metal at 42 °C can be calculated in the following way :

Weight of the piece of metal in air = 46 gm

Weight of the piece of metal in liquid at 42 °C = 30.5 gm

Loss of the weight of the piece of metal = 46 – 30.5 = 15.5 gm

Weight of the volume displaced of liquid at 42 °C = 15.5 gm

Now the volume of the liquid displaced

$$= \frac{\text{weight of the liquid displaced}}{\text{density}}$$

$$= \frac{15.5}{1.20} \text{ cm}^3$$

Thus the volume of piece of metal at 42 °C

$$V_2 = \frac{15.5}{1.20} \text{ cm}^3$$

Applying the formula $V_2 = V_1 (1 + \gamma \Delta T)$, we have

$$\frac{15.5}{1.20} = \frac{16}{1.24} (1 + \gamma \cdot 15) \quad [\text{As } \Delta T = 42 - 27 = 15]$$

$$\Rightarrow 1 + 15\gamma = \frac{15.5}{1.20} \times \frac{1.24}{16}$$

$$\text{or } \gamma = \frac{1}{15} \left[\frac{15.5}{1.20} \times \frac{1.24}{16} - 1 \right] = \frac{1}{15} \times \frac{1}{960}$$

$$\Rightarrow \alpha = \frac{1}{3 \times 15 \times 960} = 2.315 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}.$$

Illustrative Example 1.18

A sinker of weight W_0 has an apparent weight W_1 when weighed in a liquid at a temperature T_1 and W_2 when weighed in the same liquid at a temperature T_2 . The coefficient of cubical expansion of the material of the sinker is β . What is the coefficient of volume expansion of the liquid ?

Solution

Here, weight of sinker in air = W_0

Weight of sinker in liquid at temperature $T_1 = W_1$

Weight of sinker in liquid at temperature $T_2 = W_2$

(i) At temperature T_1

Loss of weight of the sinker in liquid = $W_0 - W_1$

Weight of the liquid displaced = $W_0 - W_1$

Volume of liquid displaced

$$V_{T_1} = \frac{\text{weight of liquid displaced}}{\text{density}}$$

$$V_{T_1} = \frac{(W_0 - W_1)}{d_1} \quad \dots (1.51)$$

Where d_1 is the density of the liquid at temperature T_1 .

(ii) At temperature T_2

Similarly, we can calculate the volume of liquid displaced V_{T_2} at temperature T_2 .

$$V_{T_2} = \frac{(W_0 - W_2)}{d_2} \quad \dots (1.52)$$

Where d_2 is the density of the liquid at temperature T_2

We know that

$$V_{T_2} = V_{T_1} [1 + \gamma (T_2 - T_1)]$$

Where γ is the coefficient of cubical expansion of sinker

According to the given problem $\gamma = \beta$, thus

$$V_{T_2} = V_{T_1} [1 + \beta (T_2 - T_1)] \quad \dots (1.53)$$

Substituting the value of V_{T_2} and V_{T_1} , we have

$$\left(\frac{(W_0 - W_2)}{d_2} \right) = \left(\frac{(W_0 - W_1)}{d_1} \right) [1 + \beta (T_2 - T_1)] \quad \dots (1.54)$$

$$\text{But } \frac{d_1}{d_2} = \frac{M / V_{T_1}}{M / V_{T_2}} = \frac{V_{T_2}}{V_{T_1}} = \frac{V_{T_1} [1 + \gamma_L (T_2 - T_1)]}{V_{T_1}}$$

$$\text{or } \frac{d_1}{d_2} = [1 + \gamma_L (T_2 - T_1)] \quad \dots (1.55)$$

Where γ_L is the coefficient of cubical expansion of the liquid.

From equations-(1.54) and (1.55), we have

$$[1 + \gamma_L (T_2 - T_1)] = \left(\frac{W_0 - W_1}{W_0 - W_2} \right) [1 + \beta (T_2 - T_1)]$$

Solving we have

$$\gamma_L = \frac{(W_2 - W_1) + \beta (W_0 - W_1)(T_2 - T_1)}{(W_0 - W_2)(T_2 - T_1)}$$

Illustrative Example 1.19

A barometer reads 75 cm on a steel scale. The room temperature is 30°C. The scale is correctly graduated for 0°C. Find the correct atmospheric pressure. Given that coefficient of linear expansion of steel is $1.1 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$ and that of cubical expansion of mercury is $1.8 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$. Neglect expansion of glass tube.

Solution

For steel scale at 30°C, length of 1 cm division will become

$$\begin{aligned} l_{1 \text{ cm at } 30^\circ\text{C}} &= (1 \text{ cm}) [1 + \alpha_{st} \times (30)] \\ &= 1 + 1.1 \times 10^{-5} \times 30 \\ &= 1.00033 \text{ cm} \end{aligned}$$

Thus actual length of mercury column at 30°C will be

$$l_{Hg \text{ at } 30^\circ\text{C}} = 1.00033 \times 75 = 75.02 \text{ cm}$$

If area of cross section of capillary tube is A length of mercury column at 0°C can be given as

$$l_{Hg \text{ at } 30^\circ\text{C}} \times A = l_{Hg \text{ at } 0^\circ\text{C}} \times A \times [1 + \gamma_{Hg} (30)]$$

[Given that A does not change with temperature]

$$\text{or } l_{Hg \text{ at } 0^\circ\text{C}} = \frac{75.024}{(1 + 1.8 \times 10^{-4} \times 30)} = 74.62 \text{ cm.}$$

Atmospheric pressure is measured in terms of length of mercury column at 0°C thus the correct barometric pressure is 74.62 cm.

Illustrative Example 1.20

A cylindrical Can can made up of aluminium contains 500 cm³ of beer. The area of cross-section of Can is 125 cm² at 10°C. Find the rise in level of beer if temperature of Can increases to 80°C. Given that coefficients of linear expansion of aluminium is $\alpha_{Al} = 2.3 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$ and that for cubical expansion of beer is $\gamma_{Beer} = 3.2 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$.

Solution

It is given that at 10 °C, volume of beer is 500 cm³ and the area of cross section of Can is 125 cm². Thus height of beer level is

$$h = \frac{500}{125} = 4 \text{ cm}$$

Now at 80°, volume of beer becomes

$$\begin{aligned} V_{80^\circ\text{C}} &= 500 (1 + \gamma_{Beer} \times 70) \\ &= 500 (1 + 3.2 \times 10^{-4} \times 70) \\ &= 511.2 \text{ cm}^3 \end{aligned}$$

At 80°, area of cross-section of Can becomes

$$\begin{aligned} A_{80^\circ\text{C}} &= 125 [1 + 2\alpha_{Al} \times 70] \\ &= 125 [1 + 2 \times 2.3 \times 10^{-5} \times 70] \\ &= 125.402 \text{ cm}^2 \end{aligned}$$

Thus new height of beer level at 80°C is

$$h' = \frac{V_{80^\circ\text{C}}}{A_{80^\circ\text{C}}} = \frac{511.2}{125.402} = 4.076 \text{ cm}$$

Thus rise in level of beer is

$$\begin{aligned} \Delta h &= h' - h \\ &= 4.076 - 4.0 \\ &= 0.076 \text{ cm} \end{aligned}$$

Illustrative Example 1.21

A steel ball initially at a pressure of 10^5 Pa is heated from 20°C to 120°C keeping its volume constant. Find the final pressure inside the ball. Given that coefficient of linear expansion of steel is $1.1 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$ and Bulk modulus of steel is $1.6 \times 10^{11} \text{ N/m}^2$.

Solution

On increasing temperature of ball by 100°C (from 20°C to 120°C), the thermal expansion in its volume can be given as

$$\begin{aligned} \Delta V &= \gamma_{st} V \Delta T \\ &= 3 \alpha_{st} V \Delta T \end{aligned} \quad \dots (1.56)$$

Here it is given that no change in volume is allowed. This implies that the volume increment by thermal expansion is compressed elastically by external pressure. Thus elastic compression in the sphere must be equal to that given in equation-(1.56). Bulk modulus of a material is defined as

$$B = \frac{\text{increase in pressure}}{\text{volume strain}} = \frac{\Delta P}{\Delta V/V}$$

Here the externally applied pressure to keep the volume of ball constant is given as

$$\begin{aligned} \Delta P &= B \times \frac{\Delta V}{V} = B (3\alpha_{st} \Delta T) \\ &= 1.6 \times 10^{11} \times 3 \times 1.1 \times 10^{-5} \times 100 \\ &= 5.28 \times 10^8 \text{ N/m}^2 \\ &= 5.28 \times 10^8 \text{ Pa} \end{aligned}$$

Thus this must be the excess pressure inside the ball at 120°C to keep its volume constant during heating.

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Thermal Expansion & Thermometry

Module Numbers - 11 to 19

Practice Exercise 1.3

(i) In a mercury in glass thermometer the cross-section of the capillary is A_0 and volume of the bulb is V_0 at 0°C . If mercury just fills the bulb at 0°C , find that the length of mercury in the capillary at temperature $t^\circ\text{C}$. Given that β = coefficient of cubical expansion of mercury and α = coefficient of linear expansion of glass.

$$\left[\frac{V_0(\beta - 3\alpha)t}{A_0(1 + 2\alpha t)} \right]$$

(ii) A one litre flask contains some mercury. It is found that at different temperatures the volume of air inside the flask remains the same. What is the volume of mercury in flask? Given that the coefficient of linear expansion of glass = $9 \times 10^{-6}^\circ\text{C}^{-1}$ and coefficient of volume expansion of mercury = $1.8 \times 10^{-4}^\circ\text{C}^{-1}$.

$$[150 \text{ cm}^3]$$

(iii) Some steel balls are fixed at the bottom of a silica bulb of negligible expansivity. The bulb holds 340 gm of mercury at 0°C when filled, in the absence of the steel balls and 255 gm of mercury, when the steel balls are inside. On heating the bulb and its contents (steel balls and mercury) to 100°C , 4.8 gm of mercury overflows. Find the coefficient of linear expansion of steel. $\gamma_{\text{Hg}} = 180 \times 10^{-6}^\circ\text{C}^{-1}$.

$$[11.76 \times 10^{-6}^\circ\text{C}^{-1}]$$

(iv) If a liquid is contained in a long narrow rigid vessel so it can expand in essentially one direction only, show that the effective coefficient of linear expansion α of liquid is approximately equal to the coefficient of volume expansion β of liquid.

(v) In observing the real expansion of a liquid a certain part of glass vessel is to be filled with a liquid of volume expansion $\gamma = 18 \times 10^{-5}$ in order to exclude the effect of the change in remaining volume of the vessel during heating. What fraction of the vessel should be filled with the liquid? The coefficient of linear expansion of glass is $\alpha = 9.0 \times 10^{-6}$.

$$\left[\frac{3}{20} \right]$$

(vi) A clock with a metallic pendulum gains 6 seconds each day when the temperature is 20°C and loses 6 second when the temperature is 40°C . Find the coefficient of linear expansion of the metal.

$$[\alpha = 1.4 \times 10^{-5}^\circ\text{C}^{-1}]$$

(vii) The coefficient of apparent expansion of a liquid when determined using two different vessels A and B are γ_1 and γ_2 respectively. If the coefficient of linear expansion of the vessel A is α_1 , Find the coefficient of linear expansion of the vessel B .

$$\left[\frac{\gamma_1 - \gamma_2 + 3\alpha_1}{3} \right]$$

(viii) A solid whose volume does not change with temperature floats in a liquid. For two different temperatures T_1 and T_2 of the liquid, fractions f_1 and f_2 of the volume of the solid remain submerged in the liquid. Find the coefficient of volume expansion of the liquid.

$$\left[\frac{f_2 - f_1}{f_1 T_2 - f_2 T_1} \right]$$

(ix) A small quantity of a liquid which does not mix with water sinks to the bottom at 20°C , the densities of the liquid and water at that temperature being 1021 and 998 kg m^{-3} , respectively. To what temperature must the mixture be uniformly heated in order that the liquid form globules which just float on water? The coefficient of cubical expansion of the liquid and water over the temperature ranges are $85 \times 10^{-5}^\circ\text{C}^{-1}$ and $45 \times 10^{-5}^\circ\text{C}^{-1}$, respectively.

$$[79.15^\circ\text{C}]$$

1.6 Calorimetry

The process of measurements of heat exchanges between bodies and system is known as calorimetry in early ages some chemists of the time found that when a hot object such as an iron ball, was immersed in a water bath, the resulting change in temperature of the water depend on both the mass and initial temperature of the ball. The temperature change was interpreted as a measure of the heat contained in the object. Further experiments showed that when two similar iron blocks at the same initial temperature were immersed in identical water baths, the more massive block caused a greater temperature change. Similarly for two identical blocks at different temperatures, the hotter block gave rise to a greater temperature change in the bath. Finally, for blocks of same mass and same initial temperature, the change in temperature of both was different for different materials. Thus the amount of heat supplied by a body depends on its mass, initial temperature and its material. Lets discuss the things in details and in scientific terms.

1.6.1 Specific Heat and Absorption of Heat

When heat is supplied to a substance it increases the energy of its molecule and hence its temperature increases. The rise in temperature of substance is proportional to heat supplied to it and also depends on the mass and material of the substance. If ΔQ is the amount of heat supplied to a body of mass m and due to this if its temperature increases by ΔT , then this heat supplied ΔQ and the rise in temperature ΔT are related as

$$\Delta Q = ms\Delta T \quad \dots (1.57)$$

Where s is a constant for material of body known as specific heat of the material of body, from above expression in equation-(1.57), if $m = 1 \text{ gm}$ and $\Delta T = 1^\circ\text{C}$ then $s = \Delta Q$ thus specific heat of a material is defined as “the amount of heat for unit mass of a body to raise its temperature by 1°C .” Thus material with a high specific heat requires a lot of heat to change its temperature, while a material with a low specific heat requires little heat to change its temperature. If an object cools, then the temperature change is taken negative and heat ΔQ is given by the object. The units of specific heat are $\text{cal/gm } ^\circ\text{C}$ or $\text{J/kg } ^\circ\text{C}$. Specific heats of some common materials used in general life are given in the lists shown in table-1.1

Table -1.1

Substance	Specific heat $\text{J/kg } ^\circ\text{C}$	Specific heat $\text{Cal/gm } ^\circ\text{C}$
Water	4187	1.00
Ice	2090	0.50
Steam	2010	0.48
Wood	1700	0.40
Aluminium	900	0.215
Glass	840	0.200
Iron	448	0.107
Copper	390	0.092
Lead	128	0.0305

In the expression shown in equation-(1.57), the term ms , the product of mass of body and specific heat of its material is described as the heat capacity of the body which is the *amount of heat required to change an objects temperature by 1°C* . Thus bodies of the same material but of different masses have heat capacities proportional to their mass.

In most of the materials, with temperature their specific heat varies slightly but for short temperature ranges we can take it to a constant. All the specific heats given in Table -1.1 are at temperature 25°C .

1.6.2 Mechanical Equivalent of Heat

At the time of discovery of heat and temperature, the distinction between the two was not clear and the two were often confused. At that time it was generally taught that heat was some kind of fluid, which could be added to or taken away from a substance to make it hot or cold. We know that heat is a form of “Energy Transfer” that occurs when there is a temperature difference between objects. A common example of the distinction between heat and temperature is taken by comparing a candle flame and a warm hot water radiator in a cool room. The candle flame is at a much higher temperature than the radiator but we can not expect it to appreciably warm the room. Although the radiator is at much lower temperature than the candle but enough heat flows from it to keep the whole room warm. In both cases energy (Heat) is transferred from an object at a higher temperature to surrounding at a lower temperature.

In year 1842 Julius Mayer suggested that heat and mechanical work were equivalent and that one could be transformed into the other, he also showed that the temperature of water could be raised by 1°C or more by mechanical agitation alone. But he failed to determine the amount of work required for such a change.

The conclusive relation between heat flow and work was demonstrated by James Prescott Joule in 1843.

He demonstrated by his experiments that mechanical energy can be directly converted into thermal energy and he also measured the numerical factor relating mechanical units to thermal units. A common unit of energy was devised and given name in the honor of Joule’s contribution to science. This unit joule is equal to roughly one fourth the size of calorie as

$$1 \text{ calorie} = 4.187 \text{ joules}$$

This relation is called “*Mechanical equivalent of heat*” and denoted by ‘ J ’. The calorie is not accepted unit of energy as SI units. The appropriated SI unit for energy is joule. But calorie is still used in several practical applications. Therefore we use both units in our examples.

1.6.3 Water Equivalent of a Substance

Water equivalent of a substance is defined as the quantity of water which requires the same amount of heat which the substance requires for same rise in temperature. For example if a body of mass m and specific heat s is heated from temperature T_1 to T_2 then the amount of heat required for the purpose is

$$Q = ms(T_2 - T_1) \quad \dots (1.58)$$

If m_w is the mass of water which requires the same heat given in equation-(1.58), to range its temperature from T_1 to T_2 then we have

$$Q = ms(T_2 - T_1) = m_w s_w (T_2 - T_1)$$

or
$$m_w = \frac{ms}{s_w} \quad \dots (1.59)$$

Equation-(1.59) gives the water equivalent of the body. Now we take some examples to understand the heat flow and variation in temperature of different bodies.

Illustrative Example 1.22

A lead ball at 25°C is dropped from a height of 2 km. It is heated due to air resistance and it is assumed that all of its kinetic energy is used in increasing the temperature of ball. If specific heat of lead is $s = 126$ Joule/kg °C, find the final temperature of ball.

Solution

If there is no air resistance, velocity of ball on reaching ground is

$$V = \sqrt{2gh} = \sqrt{2 \times 10 \times 2000} = 200 \text{ m/s}$$

Kinetic energy of ball is

$$\begin{aligned} K &= \frac{1}{2} mv^2 \\ &= \frac{1}{2} \times m \times (200)^2 = 2 \times 10^4 \times m \text{ Joule} \end{aligned}$$

[If mass m is in kg]

If all the kinetic energy is absorbed by the ball as heat and due to this temperature of ball is increased by ΔT , we have

$$K = ms\Delta T$$

or
$$2 \times 10^4 \times m = m \times 126 \times \Delta T$$

or
$$\Delta T = \frac{2 \times 10^4}{126} = 158.73^\circ\text{C}$$

Illustrative Example 1.23

The temperature of equal masses of three different liquids A , B and C are 15°C, 20°C and 30°C respectively. When A and B are mixed, their equilibrium temperature is 18°C. When B and C are mixed, it is 22°C. What will be the equilibrium temperature when liquids A and C are mixed.

Solution

Let specific heats of liquids are s_A , s_B and s_C respectively. Then for mixing of liquids A and B , we have

heat lost by B = heat gained by A

or
$$m \times s_B \times (20 - 18) = m \times s_A \times (18 - 15)$$

$$2s_B = 3s_A \quad \dots (1.60)$$

For mixing of liquids B and C , we have

heat lost by C = heat gained by B

$$m \times s_C \times (30 - 22) = m \times s_B \times (22 - 20)$$

$$8s_C = 2s_B$$

or
$$s_B = 4s_C \quad \dots (1.61)$$

From equation-(1.60) and (1.61), we get

$$8s_C = 3s_A \quad \dots (1.62)$$

Now if liquids A and C are mixed, let their equilibrium temperature in T_e then we have

Heat lost by C = heat gained by A

$$m \times s_C \times (30 - T_e) = m \times s_A \times (T_e - 15)$$

or
$$\frac{3s_A}{8} \times (30 - T_e) = s_A \times (T_e - 15)$$

or
$$90 - 3T_e = 8T_e - 120$$

or
$$11T_e = 210$$

or
$$T_e = \frac{210}{11} = 19.09^\circ\text{C}$$

Illustrative Example 1.24

A metal block of density 5000 kg/m³ and mass 2 kg is suspended by a spring of force constant 200 Nt/m. The spring block system is submerged in a water vessel. Total mass of water in it is 300 g and in equilibrium the block is at a height 40 cm above the bottom of vessel. If the support is broken. Find the rise in temperature of water. Specific heat of the material of block is 250 J/kg K and that of water is 4200 J/kg K. Neglect the heat capacities of the vessel and the spring.

Solution

When the block is in equilibrium in the water and spring is stretched by a distance x and the spring force balances the effective weight of block i.e. weight of block minus the buoyant force on block. Thus for equilibrium of block in water, we have

$kx + \text{weight of liquid displaced} = \text{weight of block}$

$$200x + \frac{2}{5000} \times 1000 \times 10 = 2 \times 10$$

$$\text{or } x = \frac{20-4}{200} = \frac{16}{200} = 0.08 \text{ m} = 8 \text{ cm}$$

Thus in equilibrium, energy stored in spring is

$$U = \frac{1}{2} kx^2$$

$$= \frac{1}{2} \times 200 \times (0.08)^2 = 0.64 \text{ Joule}$$

When the support is broken, the mass fall down to the bottom of vessel and the potential energy stored in spring and the gravitational potential energy of block is released and used in heating the water and block. Thus we have

$$0.64 + mg(0.4) = m_w \times s_w \times \Delta T + m_B \times s_B \times \Delta T$$

$$\text{or } 0.64 + 2 \times 10 \times 0.4 = [0.3 \times 4200 + 2 \times 250] \Delta T$$

$$\text{or } \Delta T = \frac{8.64}{1760} = 0.0049^\circ\text{C}$$

Illustrative Example 1.25

A copper cube of mass 200 gm slides down on a rough inclined plane of inclination 37° at a constant speed. Assume that any loss in mechanical energy goes into the block as thermal energy. Find the increase in temperature of block as it slides down through 60 cm. Given that specific heat of copper is 420 J/kg.K .

Solution

As block slides at constant speed, friction on block exactly balances the gravitational pull on it, $mg \sin \theta$. Thus

$$f = mg \sin \theta = 0.2 \times 10 \times \sin 37^\circ$$

$$= 2 \times \frac{3}{5} = 1.2 \text{ N}$$

As block slides 60 cm distance, work done by it against friction is

$$W = f \cdot l$$

$$= 1.2 \times 0.6 = 0.72 \text{ J}$$

This energy is only used in increasing the temperature of copper block, thus

$$0.72 = m.s. \Delta T$$

$$\text{or } \Delta T = \frac{0.72}{ms}$$

$$= \frac{0.72}{0.2 \times 420} = 0.00857^\circ\text{C}$$

Illustrative Example 1.26

A metal container of mass 500 gm contains 200 gm of water at 20°C . A block of iron also of mass 200 gm at 100°C is dropped into water. Find the equilibrium temperature of the water. Given that specific heats of metal of container and iron and that of water are 910 J/kg K , 470 J/kg K and 4200 J/kg K respectively.

Solution

Here container and water are at 20°C , thus when iron block is dropped into water it will loose energy to it and its temperature will fall. If equilibrium temperature is T_0 , then we have

heat lost by iron block = heat gained by water plus container

$$m_i \times s_i (100 - T_0) = m_w \times s_w (T_0 - 20) + m_c \times s_c (T_0 - 20)$$

$$\text{or } T_0 = \frac{m_i s_i (100) + m_w s_w (20) + m_c s_c (20)}{m_i s_i + m_w s_w + m_c s_c}$$

$$= \frac{0.2 \times 470 \times 100 + 0.2 \times 4200 \times 20 + 0.5 \times 910 \times 20}{0.2 \times 470 + 0.2 \times 4200 + 0.5 \times 910}$$

$$= \frac{9400 + 16800 + 9100}{94 + 840 + 455}$$

$$= \frac{35300}{1329} = 26.56^\circ\text{C}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Calorimetry

Module Numbers - 1 to 6

Practice Exercise 1.4

(i) A can of water of volume 0.5 m^3 at a temperature 30°C is cooled to 15°C . Neglect heat capacity of the can. If the amount of heat released by water is used to lift a box of weight 10 kg, find the height to which it can be lifted. Given that specific heat of water is $4200 \text{ J/kg}^\circ\text{C}$. Take $g = 10 \text{ m/s}^2$.

$$[3.15 \times 10^5 \text{ m}]$$

(ii) Equal masses of three liquids have temperatures 10°C , 25°C and 40°C respectively. If First two liquids are mixed, the mixture has a temperature of 15°C . If second and third are mixed, the equilibrium temperature is 30°C . Find the final temperature if first and third liquids are mixed.

$$[16^\circ\text{C}]$$

(iii) A water heater can generate 8500 kcal/hr. How much water can it heat from 10°C to 60°C per hour ? Use specific heat of water 4200 J/kg°C.

[170 kg/hr]

(iv) When a 290 gm piece of iron at 180°C is placed in a 100 gm aluminium calorimeter cup containing 250 gm of glycerin at 10°C, the final temperature is observed to be 38°C. What is the specific heat of glycerin ? Use specific heat of iron 470 J/kg K and that of aluminium is 900 J/kg K.

[2405 J/kg K]

(v) The specific heat of many solids at low temperatures varies with absolute temperature T according to the relation $s = aT^3$, where a is a constant. Find the heat energy required to raise the temperature of a mass m of such a solid from 0 to 20K.

[$4 \times 10^4 ma$]

(vi) A certain calorimeter has a water equivalent of 4.9 gm. That is, in heat exchanges, the calorimeter behaves like 4.9 gm of water. It contains 40 gm of oil at 50.0°C. When 100 gm of lead at 30.0°C is added, the final temperature is 48.0°C. What is the specific heat capacity of the oil ?

[Take $S_{\text{water}} = 1 \text{ cal/gm}^\circ\text{C}$, $S_{\text{lead}} = 0.305 \text{ cal/gm}^\circ\text{C}$]

[0.563 cal/gm°C]

1.6.4 Phase Transformation

When heat is supplied to a substance, its temperature increases and simultaneously the agitation (vibration energy) of its molecules also increases. If the substance is a solid, we have discussed that in a solid intermolecular forces are strong enough to make the position of molecule fixed in the lattice and the substance has a definite shape unlike to liquid or gases. When thermal agitation of molecules increases with rise in temperature, molecules start exerting an additional force onto each other and with temperature this force also increases. When temperature of a substance attains a value at which this force due to thermal agitation exceeds the cohesive force due to which molecules are bounded to each other, the molecules start becoming free and the heat energy due to which temperature of body was increasing will be now used in breaking the molecules from the lattice and further supply of heat will break more molecules from the lattice and this phenomenon is called melting of solid and during melting temperature of substance do not increase with further supply of heat. During melting supplied energy is used to break the bonds between molecules to melt the substance. Thus no temperature rise take place until the whole substance is melt. The amount of heat required to melt a given mass of a solid

substance is measured in terms of Latent Heat which is characteristic property of a solid substance. It is defined as *the amount of heat required to melt a unit mass of solid*, denoted by L_f and termed as latent heat of fusion. If Q is the heat, which melt a mass m of a solid substance with latent heat of fusion L_f , then we have

$$Q = mL_f \quad \dots (1.63)$$

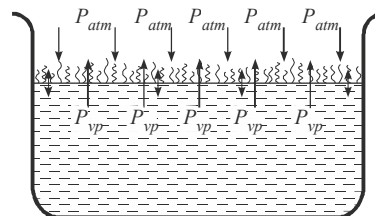


Figure 1.23

A similar phenomenon vaporization can also be defined, for transformation of a liquid to vapour. Figure-1.23 shows some liquid in a container, we have already discussed that at a temperature every molecule has some thermal agitation energy which is proportional to the absolute temperature of substance. Due to this, molecules on the surface of liquid are in continuous random oscillations as shown in figure-1.23. Some surface molecules randomly jumps in atmosphere from the surface but due to atmospheric pressure comes back to liquid surface. Due to thermal agitation the surface molecules repeatedly bounce from the surface and exert a pressure on atmosphere above the liquid surface, this pressure is called vapour pressure of the liquid which is exerted by those molecules which leave the surface of liquid for a short time randomly and repeatedly which may be treated as the vapour of liquid for the time these are in air. This vapour just above the liquid surface exerts an upward pressure on atmospheric air molecules as shown in figure-1.23. As temperature of liquid increases, due to increase in thermal agitation its vapour pressure will also increase. When vapour pressure of liquid exceeds atmospheric pressure, liquid molecules start escaping from surface to atmospheric air. This is called vaporization of liquid. At this stage if heat is supplied to the liquid, its temperature would not increase and the amount of energy supplied is taken by molecules on liquid surface in vapourization. *The amount of heat required to vapourize unit mass of a liquid is called as latent heat of vapourization of the liquid* and is denoted by L_v . The amount of heat required to vapourize a liquid of mass m is

$$Q = mL_v \quad \dots (1.64)$$

To understand the application of latent heat of fusion and vapourization, we take few examples.

Illustrative Example 1.27

Find the result of mixing 0.5 kg ice at 0°C with 2 kg water at 30°C. Given that latent heat of ice is $L = 3.36 \times 10^5 \text{ J kg}^{-1}$ and specific heat of water is 4200 Joule $\text{kg}^{-1} \text{ K}^{-1}$.

Solution

In such mixing problems, it is advisable to first convert all the phases of mixing substance into a single phase at a common temperature and keep the excess or required heat for this aside and finally supply or extract that amount of heat to get the final equilibrium temperature of the mixture.

As in this problem 0.5 kg of ice is given at 0°C. To convert it into water at 0°C if we require Q_1 amount of heat, then we have

$$Q_1 = -mL_f = -0.5 \times 3.36 \times 10^5 \\ = -1.68 \times 10^5 \text{ Joule} [- \text{ve sign for heat required}]$$

The water 2 kg is available at 30°C, to convert it into 0°C, it release some heat say Q_2 , then we have

$$Q_2 = ms\Delta T \\ = 2 \times 4200 \times 30 = 2.52 \times 10^5 \text{ Joule}$$

Thus we have the final mixture as

$$\text{Final mixture} = 2.5 \text{ kg water at } 0^\circ + 2.52 \times 10^5 \text{ J} - 1.68 \times 10^5 \text{ J} \\ = 2.5 \text{ kg water at } 0^\circ + 8.4 \times 10^4 \text{ J}$$

Thus finally we can supply the available heat to the 2.5 kg water at 0°C to get the final temperature of mixture as

$$2.5 \times 4200 \times (T_g - 0) = 8.4 \times 10^4$$

$$\text{or } T_g = 8^\circ\text{C}$$

Thus final result is 8°C of 2.5 kg water after mixing.

Illustrative Example 1.28

When 2 kg block of copper at 100°C is put in an ice container with 0.75 kg of ice at 0°C, find the equilibrium temperature and final composition of the mixture. Given that specific heat of copper is 378 J/kg K and that of water is 4200 J/kg K and the latent heat of fusion of ice is $3.36 \times 10^5 \text{ J/kg}$.

Solution

Here in the given mixture initially we have

$$\text{Mixture} = 2 \text{ kg copper at } 100^\circ\text{C} + 0.75 \text{ kg of ice at } 0^\circ\text{C}$$

Now heat released by copper to attain 0°C is

$$Q_1 = 2 \times 378 \times 100 = 7.56 \times 10^4 \text{ Joule}$$

Heat gained by ice to melt is

$$Q_2 = -0.75 \times 3.36 \times 10^5 = -25.2 \times 10^4 \text{ Joule}$$

Thus mixture contains

$$\begin{aligned} \text{Mixture} &= 2 \text{ kg copper at } 0^\circ\text{C} + 0.75 \text{ kg} \\ &\text{water at } 0^\circ\text{C} + 7.56 \times 10^4 \text{ J} - 25.2 \times 10^4 \text{ J} \\ &= 2 \text{ kg copper at } 0^\circ\text{C} + 0.75 \text{ kg water at } 0^\circ\text{C} - 17.64 \times 10^4 \text{ J} \end{aligned}$$

Here the available heat is negative which implies that whole ice will not melt here we can freeze the available water at 0°C upto the extent when the available negative heat is balanced. $17.64 \times 10^4 \text{ J}$ heat is released when m mass of water is frozen thus m is given as

$$m = \frac{17.64 \times 10^4}{3.36 \times 10^5} = 0.525 \text{ kg}$$

Thus final mixture contains

$$\begin{aligned} \text{Mixture} &= 2 \text{ kg copper at } 0^\circ\text{C} + 0.525 \text{ kg} \\ &\text{ice at } 0^\circ\text{C} + 0.225 \text{ kg water at } 0^\circ\text{C} \end{aligned}$$

Thus it is obvious that the equilibrium temperature is 0°C.

Illustrative Example 1.29

How should 1 kg of water at 50°C is divided in two parts so that if one part is turned into ice at 0°C, it would release sufficient amount of heat to vaporize the other part. Given that latent heat of fusion of ice is $3.36 \times 10^5 \text{ J/kg}$, latent heat of vapourization of water is $22.5 \times 10^5 \text{ J/kg}$ and specific heat of water is 4200 J/kg K.

Solution

Let x kg of water is frozen then the amount of heat it releases is

$$\begin{aligned} Q_1 &= x \times 4200 \times 50 + x \times 3.36 \times 10^5 \text{ Joule} \\ &= x \times 5.46 \times 10^5 \text{ Joule} \end{aligned}$$

The heat required to vapourize the $(1-x)$ kg of water from 50°C is

$$Q_2 = (1-x) \times 22.5 \times 10^5$$

Here we've taken heat required to vapourize the water as only mass $\times$ latent heat of vapourization and not the heat required to first raise the temperature of $(1-x)$ kg of water from 50° to 100° plus the mass $\times$ latent heat similar as when heat is supplied to water from an external source, it first reaches 100°C then its vaporization starts but when heat is taken by water itself it vapourize (evaporation) at 50°C as in this case. The similar

case we see in our general life in cooling of water in a pitcher, drying of cloths hanging in open air etc.

Thus heat Q_2 must be provided by first part of water, we have

$$Q_2 = Q_1$$

$$(1-x) \times 22.5 \times 10^5 = x \times 5.46 \times 10^5$$

$$\text{or } 22.5 - 22.5x = 5.46x$$

$$\text{or } x = \frac{22.5}{27.96} = 0.805 \text{ kg}$$

Illustrative Example 1.30

In a pitcher when water is filled some water comes to outer surface slowly through its porous walls and gets evaporated. Most of the latent heat needed for evaporation is taken from water inside and hence this water is cooled down. If 10 kg water is taken in the pitcher and 12 gm water comes out and evaporated per minute. Neglect heat transfer by convection and radiation to surrounding, find the time in which the temperature of water in pitcher decreases by 5°C .

Solution

It is given that 12 gm water is evaporated per minute, thus heat required per minute for it is

$$\begin{aligned} Q &= mL_v \\ &= 12 \times 540 = 6480 \text{ cal/min} \end{aligned}$$

After time t , mass of inside water is

$$m = 10000 - 12t$$

If in further time dt , $dm = 12 dt$, mass is vapourized, and the temperature of inside water falls by dT , we have

$$(10000 - 12t) \times 1 \times dT = 12 dt \times 540$$

$$\text{or } dT = 12 \times 540 \frac{dt}{10000 - 12t}$$

Integrating the above expression in proper limits, we get,

$$\int_{T_0}^{T_0-5} dT = 12 \times 540 \times \int_0^t \frac{dt}{10000 - 12t}$$

$$5 = 540 \ln \left(\frac{10000}{10000 - 12t} \right)$$

$$\text{or } e^{5/540} = \frac{10000}{10000 - 12t}$$

$$\begin{aligned} \text{or } t &= \frac{10000}{12} [e^{1/108} - 1] \\ &= 7.737 \text{ minutes} \end{aligned}$$

Alternative method

As here the amount of water vaporized is very small, we can assume that the quantity of water inside remains constant. Thus to reduce the temperature of inside water, amount of heat to be rejected is

$$\begin{aligned} Q &= m^3 \Delta T \\ &= 10000 \times 1 \times 5 \\ &= 50,000 \text{ calories} \end{aligned}$$

As obtained from the given data that 6480 calories of heat is required per minute to vapourize 12 gm water thus if in time t minutes, temperature of inside water falls by 5°C , we have

$$50000 = 6480 t$$

$$\begin{aligned} \text{or } t &= \frac{50000}{6480} \\ &= 7.716 \text{ minutes} \end{aligned}$$

Which is approximately same as obtained earlier. But students should note that this method will give the correct answer if the mass of inside water actually remains constant *i.e.* when the rate of evaporation is very very low.

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Calorimetry

Module Numbers - 7 to 11

Practice Exercise 1.5

(i) 5g ice at 0°C is mixed with 5g of steam at 100°C . What is the final temperature?

[100°C]

(ii) A block of ice of mass $M = 50 \text{ kg}$ slides on a horizontal surface. It starts with a speed $v = 5.0 \text{ m/s}$ and finally stops after moving a distance $s = 30 \text{ m}$. What is the mass of ice that has melted due to friction between block and surface? Latent heat of fusion of ice is $L_f = 80 \text{ cal/g}$.

[1.86 g]

(iii) The temperature of a body rises by 44°C when a certain amount of heat is given to it. The same heat when supplied to 22 g of ice at -8°C , raises its temperature to 16°C . Find the water equivalent of the body.

[Given : $s_{\text{water}} = 1 \text{ cal/g}^{\circ}\text{C}$ & $L_f = 80 \text{ cal/g}$, $s_{\text{ice}} = 0.5 \text{ cal/g}^{\circ}\text{C}$]

[50g]

(iv) In a mixture of 35 g of ice and 35 g of water in equilibrium, 4 gm steam is passed. The whole mixture is in a copper calorimeter of mass 50 g. Find the equilibrium temperature of the mixture. Given that specific heat of water is 4200 J/kgK and that of copper is 420 J/kgK and latent heat of fusion of ice is $3.36 \times 10^5 \text{ J/kg}$ and latent heat of vaporization of water is $2.25 \times 10^6 \text{ J/kg}$.

[0°C , mixture = 3.22 g of ice + 70.78 g water]

(v) A lead bullet melts when stopped by a wall. Assuming that 25% of the energy is absorbed and distributed in the wall. Find the velocity of the bullet if its temperature first rises by 300K and then it melts. Given that the specific heat of lead is $0.03 \text{ cal/g}^{\circ}\text{C}$ and latent heat of fusion of lead is 6000 cal/kg .

[409.87 m/s]

(vi) An aluminium container of mass 100 g contains 200 g of ice at -20°C . Heat is added to the system at the rate 100 calories per second. What is temperature of the system after four minutes ? Specific heat of ice = $0.5 \text{ cal/g}^{\circ}\text{C}^{-1}$ and latent heat of fusion of ice = 80 cal/gm and specific heat of aluminium is $0.215 \text{ cal/g}^{\circ}\text{C}^{-1}$.

[27.08°C]

(vii) 1 kg of ice at 0°C is mixed with 1 kg of steam at 100°C . Find the equilibrium temperature and the final composition of the

mixture. Given that latent heat of fusion of ice is $3.36 \times 10^5 \text{ J/kg}$ and latent heat of vaporization of water is $2.27 \times 10^6 \text{ J/kg}$. Specific heat of water is $4200 \text{ J/kg}^{\circ}\text{C}^{-1}$.

[100°C , mixture = 1335 g water + 665 g steam]

(viii) When a block of metal of specific heat $0.1 \text{ cal/g}^{\circ}\text{C}$ and weighing 110 gm is heated to 100°C and then quickly transferred to a calorimeter containing 200 g of liquid at 10°C , the resulting temperature is 18°C . On repeating the experiment with 400 g of same liquid in the same calorimeter at the same initial temperature, the resulting temperature is 14.5°C . Find :

(i) specific heat of the liquid

(ii) the water equivalent of the calorimeter.

[$S_L = 0.481 \text{ cal g}^{-1}^{\circ}\text{C}^{-1}$; $W_c = 16.55 \text{ g}$]

(ix) A pitcher contains 10 kg of water at 20°C . Water comes to its outer surface through its porous walls and gets evaporated. The latent heat required for evaporation is taken from the water inside the pitcher, thus the inside water is cooled down. It is given that the rate of evaporation is 0.2 g/s . Calculate the time in which the temperature of the water inside drops to 15°C . Given that the specific heat of water is $4200 \text{ J/kg}^{\circ}\text{C}$ and latent heat of vaporization of water is $2.27 \times 10^6 \text{ J/kg}$.

[Approx. 462 s]

(x) A pitcher contains 1 kg water at 40°C . It is given that the rate of evaporation of water from the surface of pitcher is 50 gm/s . Find the time it will take to cool down the water inside to 30°C . Given that latent heat of vaporization of water is 540 cal/g and specific heat of water is $1 \text{ cal/g}^{\circ}\text{C}$.

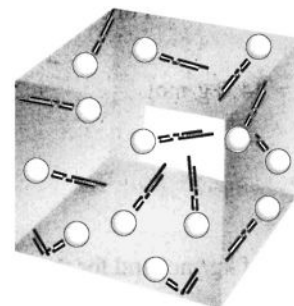
[$20 (1 - e^{-1/54}) \text{ s}$]

* * * * *

Kinetic Theory of Gases and Gas Laws

FEW WORDS FOR STUDENTS

In this chapter we'll discuss about the general behaviour of gases. We use the basic ideas that we've discussed in previous chapter and some concept of mechanics to understand the gas behaviour. Mainly how a gas exerts pressure and its relation with volume and temperature of gas and the total energy possessed by gas and its relation to the temperature of gas.



CHAPTER CONTENTS

- 2.1 *Postulates of Kinetic Theory of Gases*
- 2.2 *Gas Laws*
- 2.3 *Pressure of Air*
- 2.4 *Distribution of Molecular Speeds*
- 2.5 *Pressure Exerted by a Gas*
- 2.6 *Kinetic Energy of Gas Molecules*
- 2.7 *Microscopic Interpretation of Temperature*
- 2.8 *Degrees of Freedom and Equipartition of Energy*
- 2.9 *Internal Energy of a Gas*
- 2.10 *Path of Molecules and Mean Free Path*
- 2.11 *Vander Walls Gas Equation*
- 2.12 *Diffusion*

COVER APPLICATION



Figure (a)

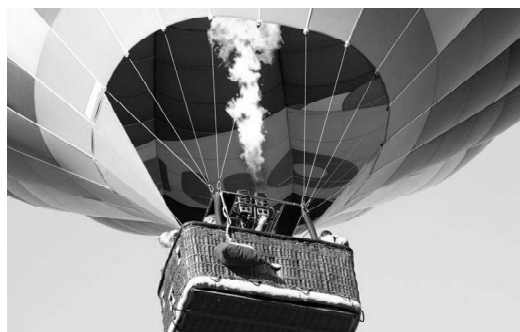


Figure (b)

Warmer air rises in cooler air is the basis of how hot air balloon works. This is because hot air is lighter than cool air as it has lower density. The actual balloon (called an envelope) has to be so large as it can take a large amount of heated air to lift it off the ground. For example, to lift 1000 pounds worth of weight you would need almost 65,000 cubic feet of heated air! To help keep the balloon in the air and rising, hot air needs to be propelled upwards into the envelope using the burner (see separate section on burners for more information.)

In this chapter, mainly we'll deal with the behaviour of gases. In fact a gas is composed of large number of molecules in random motion. Characteristics of a gas depends on the properties of the motion of these molecules. The analysis of a gas at such a microscopic level is called kinetic theory of gases.

2.1 Postulates of Kinetic Theory of Gases

Before proceeding for investigations on properties of a gas from the point of view of kinetic theory, we first define an idealized model of a gas. For such a model of an ideal gas we make some assumptions for molecules of a gas. These assumptions are called basic postulates of kinetic theory of gases. These are :

(i) In a gas size of molecules is negligible or the average separation between them is large compared with their dimensions. This means that the volume of the molecules is negligible when compared with the volume of the container.

(ii) Each molecule is considered to be a hard sphere and collide elastically with the other molecules and container walls. The pressure that a gas exerts on the walls of its container is a consequence of the repeated collisions of the gas molecules with the walls.

(iii) It is assumed that molecules obey Newton's Laws of motion but during motion they do not interact each other except during collisions and these are not deformed during collisions thus forces between molecules are short range forces which only acts during collisions.

(iv) As during motion molecules do not interact, the total energy of a gas can be considered as the sum of kinetic energies of all of its molecules and hence a gas contain zero potential energy.

(v) During collision the time of contact is negligible compared to the time between two successive collision of a gas molecule, which is called relaxation time.

(vi) As a whole molecules move randomly. By "randomly" we mean that any molecule can move in any direction at any speed. The direction of motion of gas molecules changes only when it collides with the other gas molecule or the container wall. This results a random zig-zag path of a molecule shown in figure-2.1.

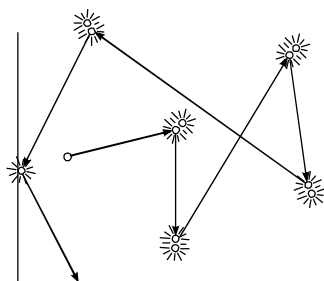


Figure 2.1

This type of random motion is termed "Brownian Motion". We also assume that the distribution of speeds does not change in time, despite the collisions between molecules. That is at any given moment, a certain percentage of molecules move at high speeds, a certain percentage move at low speeds and so on.

(vii) The gas under consideration is assumed a pure substance. That is, all of its molecules are identical.

(viii) Effect of gravity on gas molecules in a container is neglected. That is in a container we assume throughout its volume density of gas is same.

(ix) At all temperature and pressures, all gases obey ideal gas law which relates the microscope characteristics of a gas i.e. pressure, volume and temperature. The ideal gas law is stated as

$$PV = nRT \quad \dots (2.1)$$

Here n are the number of moles of a gas and R is universal gas constant. In next sections of the chapter we'll discuss gas law in details.

Generally real gases obey ideal gas law $PV = nRT$ only at very high temperature and very low pressure as at very high temperature kinetic energy of the gas molecules is so high that even if some interaction between molecules is present, it will not contribute any energy and at very low pressure, separation between molecule is very large thus reduces their interaction. So we can say that at high temperature and low pressure, real gases behave like ideal gases.

2.2 Gas Laws

In mathematical analysis and behaviour of gases, three parameters are important in describing the properties of a gas. These are pressure, volume and temperature. Several experiments are done for analysis of a gas and to relate these parameters. Some laws are established on the basis of these experiments for relating these parameters. These laws are known as gas laws. Now we'll discuss these laws in details with some examples.

2.2.1 Boyle's Law

This law is an experimental result obtained by Boyle between the pressure and the volume of an enclosed gas at a constant temperature. Boyle's Law states that the pressure exerted by a gas at constant temperature is inversely proportional to the volume in which it is enclosed. Thus Boyle's law is written as

$$P \propto \frac{1}{V}$$

or $PV = \text{constant} \quad \dots (2.2)$

Where P is the gas pressure, V its volume and the value of constant depends on the gas temperature and its quantity. The complete statement of Boyle's law includes the condition that both the temperature and the amount of gas must be held constant.

Alternatively Boyle's law may be written as

$$P_1 V_1 = P_2 V_2 \quad \dots (2.3)$$

Where the subscripts 1 and 2 refer to different physical states of the same sample of gas with the temperature held constant. The variation of pressure and volume of gas at constant temperature is shown in graph in figure-2.2.

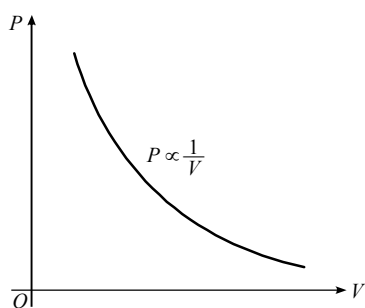


Figure 2.2

Another important point about Boyle's law is to be noted that while Boyle's law is applicable over a wide range of pressure but it does not always apply. For example, if the temperature is low enough, a sample of gas will be condensed to a liquid at sufficiently high pressure. In further sections we'll discuss about liquification of gases in detail.

If we find the volume of one mole molecules of gas using gas law, at standard pressure 1 atm and standard temperature 273 K, we get

$$PV = nRT$$

$$\begin{aligned} \text{or } V &= \frac{1 \times 8.314 \times 273}{1.013 \times 10^5} \\ &= 2.2406 \times 10^{-2} \text{ m}^3 \\ &= 22406 \text{ cm}^3 \\ &= 22.4 \text{ litre} \end{aligned}$$

Thus we can state that one mole of all gases occupy a volume of 22.4 litre at standard pressure and temperature.

2.2.2 Charles and Gay-Lussac Law

Boyle's law relates the pressure and the volume of a gas at constant temperature. However, we can also investigate the effect of temperature change on the volume of a gas at constant

pressure. The apparatus for such an experiment is shown in figure-2.3.

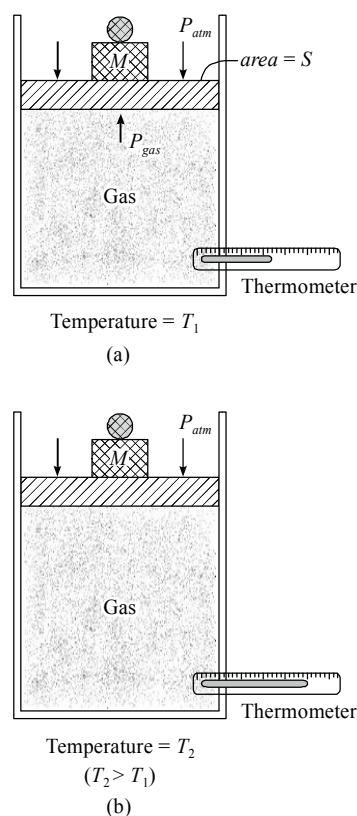


Figure 2.3

We introduced a gas into a cylinder and place a weight on the piston of mass M . This arrangement if in equilibrium at a temperature T_1 as shown in figure-2.3(a) develops a constant pressure in the gas. For equilibrium of piston, the gas pressure can be given as

$$P_{\text{gas}} = P_{\text{atm}} + \frac{Mg}{S} \quad \dots (2.4)$$

If gas is heated this pressure remains constant as for equilibrium the gas expands to maintain the pressure. Now after heating the gas to different temperatures and recording its data for temperature and corresponding volume, we plot a graph. The respective graph is shown in figure-2.4.

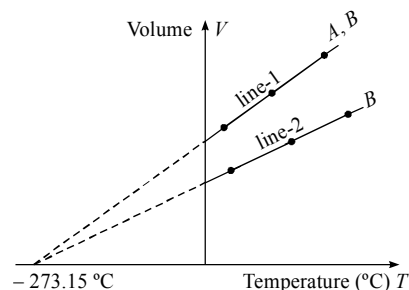


Figure 2.4

If the data is recorded for a gas A we get line-1 shown in figure-2.4 and if we repeat the experiment with another gas B , with the same initial volume and temperature, we get the same results and obtain line-1. But if we use gas B with different initial volume, we obtain data producing a new line, line-2. Whatever gas we use the behaviour is same. The graph will always be a straight line.

Here we can see that these straight lines if extended intersect with temperature axis at -273.15°C . Theoretically it shows that the volume of gas become zero at this temperature. The reason already we've assumed that the size of molecules is negligible and at -273.15°C or 0 K temperature, the kinetic energy of molecules become zero or all motions are frozen at 0 K temperature thus no movement is there in gas molecules of negligible size at this temperature. If the graph shown in figure-2.4 is again plotted with Kelvin scale we get one as shown in figure-2.5.

This scale is absolute scale with the sense that its zero (-273.15°C) is the lower limit for temperature and in practical nature to attain a temperature below this is not possible due to the reason discussed above. In further analysis of gases we will use Kelvin scale.

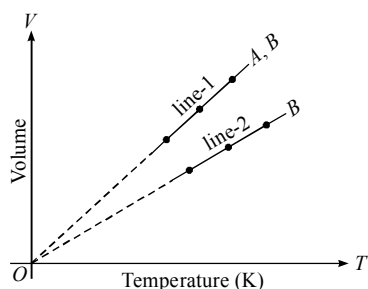


Figure 2.5

Thus if temperature is expressed in Kelvins, we find that when the pressure is held constant, the volume is proportional to the temperature. This statement is the law of Charles and Gay-Lussac. This can be expressed mathematically as

$$\frac{V_1}{V_2} = \frac{T_1}{T_2} \quad \dots (2.5)$$

or
$$\frac{V}{T} = \text{constant} \quad \dots (2.6)$$

As with Boyle's law, the amount of gas also must be held constant for equation-(2.5) and (2.6) to be valid.

2.2.3 Ideal Gas Law

Boyle's law and the law of Charles and Gay-Lussac are the particular cases of a more general expression called the "ideal gas law". It can be written as

$$PV = nRT \quad \dots (2.7)$$

Here, P , V and T stand for pressure, volume and temperature respectively and R is a constant that is same for all gases and so is called the universal gas constant and n are the number of moles of gas. If pressure is measured in SI units of units of pascal, volume in cubic meters, and temperature in kelvins, then R has a value $8.314\text{ Joule/mole-K}$. If in a state gas pressure, volume and temperature are P_1 , V_1 and T_1 and after some experiment on same sample of gas if its pressure, volume and temperature are P_2 , V_2 and T_2 , then from equation-(2.7) we have

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \quad \dots (2.8)$$

As number of moles in initial and final state are equal.

The amount of a gas is generally measured in moles, given as n . A mole (mol) is the amount of material whose mass in grams is numerically equal to the molecular mass of substance. For example molecular mass of O_2 is 32, Thus a mole of oxygen is 32 grams.

2.2.4 Avogadro's Number and Avogadro Hypothesis

We've discussed that the gas constant R , has the same value for all gases. This fact was first recognised in a slightly different form, by an Italian Scientist Amedeo Avogadro. Avogadro stated that equal volumes of gas at the same pressure and temperature contain equal numbers of molecules. This statement is called Avogadro's hypothesis. We can see that this statement is consistent with R being the same for all gases.

The number of molecule in a mole is known as Avogadro's number, N_A . Although Avogadro was not able to actually determine the value of N_A , several methods have been devised to measure N_A and the acceptable value found is

$$N_A = 6.023 \times 10^{23} \text{ molecules/mole}$$

Thus in n moles of a gas total number of molecules of the gas are

$$N = n N_A \quad \dots (2.9)$$

or number of moles of a gas can be given as

$$n = \frac{N}{N_A} = \frac{m' N}{m' N_A} = \frac{m}{M} \quad \dots (2.10)$$

Here m' is the mass of each molecule, m is the total mass of gas and M is the mass of 1 mole of molecules of gas *i.e.* its molecular mass.

From gas law, we have

$$PV = nRT$$

or

$$PV = \frac{N}{N_A} RT = \frac{m}{M} RT$$

or

$$PV = NkT \quad \dots (2.11)$$

Where $k = \frac{R}{N_{AV}}$ is the Boltzmann's constant and has the value
 $k = 1.38 \times 10^{-23} \text{ J/K}$

This equation-(2.11) is another form of ideal gas law which is in terms of total number of molecules of gas N instead of moles n .

2.2.5 Dalton's Law of Partial Pressures

This law states that the total pressure of a mixture of gases is equal to the sum of the partial pressures of the gases in the mixture. In a container of volume V if n different gases are taken at a common temperature independently at pressure $P_1, P_2, \dots, P_n$, then on mixing all these gases in the same container at the same temperature, the total pressure of the mixture is given by

$$P_T = P_1 + P_2 + \dots + P_n \quad \dots (2.12)$$

2.2.6 Different Forms of Ideal Gas Law

As discussed the pressure volume and temperature of a gas in a state related by ideal gas law given as

$$PV = nRT \quad \dots (2.13)$$

Where n are the number of moles of gas in the enclosed volume V . If m mass of a gas (molecular mass = M) is taken in the container of volume V , we have number of moles of gas as

$$n = \frac{m}{M} \quad \dots (2.14)$$

Now from equation-(2.13)

$$PV = \frac{m}{M} RT$$

$$\text{or} \quad P = \frac{m}{VM} RT$$

$$\text{or} \quad P = \frac{\rho RT}{M} \quad \dots (2.15)$$

Equation-(2.15) is a modified form of ideal gas law which relates gas pressure, its density and temperature in a physical state. Equation-(2.13) is termed as container form of Gas Law as it is generally used when a gas is enclosed in a container and equation-(2.15) is termed as atmospheric form of gas law as it relates density of gas with the pressure and temperature and is used widely for open atmosphere or in a region where gas is not enclosed.

2.2.7 Real Gases and Change of Phase

The ideal gas law $PV = nRT$ gives the behaviour of a gas as long as its pressure is not too high, as discussed earlier. To discuss this in depth look at PV graph shown in figure-2.6 for a given

amount of gas. These curves show how pressure of gas varies with the change in volume at constant temperatures. Here the dotted curves represent the behaviour of an ideal gas at $PV = \text{Constant}$. The solid curve represents the behaviour of a real gas at the same temperature. Notice that at high pressure the volume in a real gas is less than that predicted by the ideal gas law and the deviation is more when gas is about to liquefy.

Students should also note that at high pressure, the molecules of the gas are close together, particularly at lower temperatures, the potential energy associated with the attractive forces between molecules, which we ignore (assumptions of KTG) are now no longer negligible compared to the low kinetic energy of molecules. These attractive forces tend to pull the molecules close together so that at a given pressure the volume of gas is less than expected from ideal gas law. If we see at further lower temperatures, these forces cause liquification of the gas.

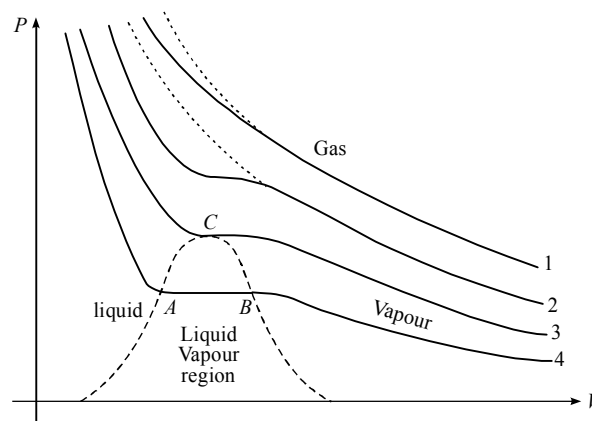


Figure 2.6

In figure-2.6, curve-4 represents the situation when liquification occurs. At low pressure on curve-4, the substance is a gas and occupies large volume. As the pressure is increased, the volume decreases till point B . Beyond point B , the volume decreases without change in pressure. In this region the gas is gradually changing to liquid up to point A . At point A gas is completely liquefied. Further if we increase the pressure, volume changes slightly as liquids are almost incompressible so here curve is very steep. The region, where gas and liquid phase exist together in equilibrium.

If we carefully look on curve-3, we can see that at point C , curve is almost horizontal and it happens only at this point. This point C is called critical point and the gas temperature of curve-3 is called critical temperature of the gas. Critical temperature of a gas is defined as - "A gas will change to liquid phase if its temperature is less than critical temperature if sufficient pressure is applied on the gas". Above critical temperature, no matter whatever be the amount of pressure applied, the gas

can not be liquefied. As pressure is increased, gas becomes denser and denser and acquires the properties resembling to a liquid but does not condense. A distinction is made between “gas” and “vapour”, students should keep in mind that a substance below its critical temperature in gaseous state is called “vapour” (which can be liquefied on applying sufficient pressure on it) and a substance above its critical temperature in gaseous state is called “gas” (which can not be liquefied).

Illustrative Example 2.1

A cylindrical container is shown in figure-2.7 in which a gas is enclosed. Its initial volume is V and temperature is T . As no external pressure is applied on the light piston shown, gas pressure must be equal to the atmospheric pressure. If gas temperature is doubled, find its final volume. In its final state if piston is clamped and temperature is again doubled, find the final pressure of the gas.

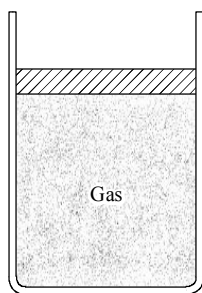


Figure 2.7

Solution

In the initial state the pressure, volume and temperature of gas P_0 , V and T respectively if P_0 is the atmospheric pressure.

It is given that temperature of gas is increased to double its value i.e. up to $2T$. As in initial and final state pressure of gas remains constant as initially as well as finally the piston is exposed to atmospheric pressure. Thus we have from charls and Gay Lussac Law

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Here
$$V_2 = \frac{V_1 T_2}{T_1} = 2V$$
 [As $V_1 = V$, $T_1 = T$ and $T_2 = 2T$]

Thus after increasing the gas temperature to $2T$, its volume becomes $2V$. Now the piston is clamped that means, now the volume of gas remains constant. Again its temperature is doubled from $2T$ to $4T$, let the pressure changes from P_0 to P' . Thus we have

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

or

$$P_2 = P' = \frac{P_1 T_2}{T_1} = 2P_0$$

[As $P_1 = P_0$, $T_1 = 2T$ and $T_2 = 4T$]

Thus in final state gas pressure becomes $2P_0$.

Illustrative Example 2.2

Two glass bulbs of equal volume are connected by a narrow tube and are filled with a gas at 0°C and a pressure of 76 cm of mercury. One of the bulbs is then placed in melting ice and the other is placed in water bath maintained at 62°C . What is the new value of the pressure inside the bulbs? The volume of the connecting tube is negligible.

Solution

If we assume the volume of each bulbs is V , the number of moles present in each bulb initially is

$$n = \frac{PV}{RT} = \frac{(76\text{cm})V}{R(273)}$$

Thus total number of moles in the two connected bulbs is

$2n = \frac{2 \times (76\text{cm})V}{R(273)}$ when one bulbs is placed in melting ice (273 K) and other is placed in water at 62°C ($62 + 273 = 335$ K) still total number of moles in the two bulbs will be equal to the initial moles i.e. $2n$. If P_f be the final pressure in the two bulbs we have

$$\begin{aligned} \frac{P_i(2V)}{T_i} &= \frac{P_f V}{T_{f1}} + \frac{P_f V}{T_{f2}} \\ \frac{(76\text{cm})(2V)}{273} &= \frac{P_f V}{273} + \frac{P_f V}{335} \\ \text{or } P_f &= \frac{2 \times 76}{273} \times \frac{273 \times 335}{608} \\ &= 83.75 \text{ cm of Hg} \end{aligned}$$

Illustrative Example 2.3

Two closed containers of equal volume of air are initially at 1.05×10^5 Pa, and 300 K temperature. If the containers are connected by a narrow tube and one container is maintained at 300 K temperature and other at 400 K temperature. Find the final pressure in the containers.

Solution

Similar to previous example final pressure in both the containers remain same as they are connected. If initial pressure volume and temperatures of the two containers are denoted by P_0 , V_0 and T_0 and P_f , V_0 , T_1 and P_f , V_0 , T_2 are the final pressure volume

and temperature in the two containers then according to gas law and constant number of moles, we have

$$\frac{P_0(2V_0)}{T_0} = \frac{P_f V_0}{T_1} + \frac{P_f V_0}{T_2}$$

or
$$P_f = \frac{2P_0}{T_0} \left(\frac{T_1 T_2}{T_1 + T_2} \right)$$

Here it is given that $P_0 = 105 \text{ kPa}$, $T_0 = 300 \text{ K}$, $T_1 = 300 \text{ K}$, $T_2 = 400 \text{ K}$, then we get

$$P_f = \frac{2 \times 105 \times 10^3 \times 400}{300 + 400} = 120 \text{ kPa}$$

Illustrative Example 2.4

Equal masses of a gas are sealed in two vessels, one of volume V_0 and other of volume $2V_0$. If the first vessel is at temperature 300 K and the other is at 600 K . Find the ratio of pressures in the two vessels.

Solution

As number of moles in the two vessels are equal, we have

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

It is given that $V_1 = V_0$, $V_2 = 2V_0$ and $T_1 = 300 \text{ K}$, $T_2 = 600 \text{ K}$. Thus we have

$$\frac{P_1 V_0}{300} = \frac{P_2 (2V_0)}{600}$$

or
$$P_1 = P_2$$

or
$$\frac{P_1}{P_2} = 1$$

Illustrative Example 2.5

A glass container encloses a gas at a pressure of $8 \times 10^5 \text{ Pa}$ and 300 K temperature. The container walls can bear a maximum pressure of 10^6 Pa . If the temperature of container is gradually increased, find the temperature at which container will break.

Solution

From gas law, we have for a constant volume container

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

or
$$T_2 = \left(\frac{P_2}{P_1} \right) T_1$$

Given that

$$P_1 = 8 \times 10^5 \text{ Pa}, T_1 = 300 \text{ K},$$

$$P_2 = 10^6 \text{ Pa}, \text{ then we have}$$

$$T_2 = \frac{10^6}{8 \times 10^5} \times 300 = 375 \text{ K}$$

Illustrative Example 2.6

Find the minimum attainable pressure of one mole of an ideal gas if during its expansion its temperature and volume are related as $T = T_0 + \alpha V^2$ where T_0 and α are positive constants.

Solution

Given that one mole of gas is used, thus from gas law, we have

$$PV = RT$$

or
$$P = \frac{RT}{V} = \frac{R}{V} (T_0 + \alpha V^2)$$

$$[\text{As } T = T_0 + \alpha V^2]$$

Here pressure P will be minimum when

$$\frac{dP}{dV} = 0$$

or
$$\frac{dP}{dV} = -\frac{RT_0}{V^2} + \alpha R = 0$$

or
$$V = \sqrt{\frac{T_0}{\alpha}}$$

Thus pressure of gas is minimum when its volume is $V = \sqrt{\frac{T_0}{\alpha}}$

and at this volume its temperature is given as

$$T = T_0 + \alpha V^2$$

$$T = T_0 + \alpha \left(\sqrt{\frac{T_0}{\alpha}} \right)^2 = 2T_0$$

Thus minimum value of pressure is

$$P_{\min} = \frac{RT}{V} = \frac{R(2T_0)}{\sqrt{\frac{T_0}{\alpha}}} = 2R\sqrt{T_0\alpha}$$

Illustrative Example 2.7

A smooth vertical tube having two different cross section is open from both the ends but closed by two sliding pistons as shown in figure-2.8 and tied with an inextensible string. One mole of an ideal gas is enclosed between the pistons. The difference in cross-sectional areas of the two pistons is given ΔS . The masses of piston are m_1 and m_2 for larger and smaller one respectively. Find the temperature by which tube is raised

so that the pistons will be displaced by a distance l . Take atmospheric pressure equal to P_0 .

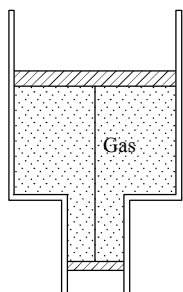


Figure 2.8

Solution

If initial pressure of gas is P and let S_1 and S_2 are the cross-sectional areas of the larger and smaller piston then for equilibrium of the two pistons we have

For larger piston

$$P_0 S_1 + m_1 g + T = P S_1$$

[If T is the tension in string] ... (2.16)

For smaller piston

$$P S_2 + m_2 g = T + P_0 S_2$$

... (2.17)

Adding equation-(2.16) and-(2.17), we get

$$P_0 (S_1 - S_2) + m_1 g + m_2 g = P (S_1 - S_2)$$

$$\text{or } P_0 + \left(\frac{m_1 + m_2}{\Delta S} \right) g = P$$

... (2.18)

If gas temperature is increased from T_1 to T_2 the volume of gas increases from V to $V + l\Delta S$ as l is the displacement of pistons, then from gas law we must have

$$P \cdot V = R T_1 \quad [\text{For initial state}] \quad \dots (2.19)$$

$$P (V + l\Delta S) = R T_2 \quad [\text{For final state}] \quad \dots (2.20)$$

According to equation-(2.18)

Pressure of gas does not change as it does not depend on temperature

From equation-(2.19) and (2.20), if we subtract these equation, we get

$$P \cdot l\Delta S = R (T_2 - T_1)$$

$$\text{or } T_2 - T_1 = \frac{P l \Delta S}{R}$$

$$= \left(P_0 + \frac{(m_1 + m_2)}{\Delta S} g \right) \frac{l \Delta S}{R}$$

$$= [P_0 \Delta S + (m_1 + m_2) g] \frac{l}{R}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Kinetic Theory of Gases

Module Number - 1 to 13

Practice Exercise 2.1

(i) A rubber balloon is filled with air at $2 \times 10^5 \text{ N/m}^2$ pressure at a temperature 20°C . When its temperature is increased to 40°C , the volume of balloon is increased by 2%. Find the final air pressure inside the balloon at 40°C .

$$[2.095 \times 10^5 \text{ N/m}^2]$$

(ii) There are two containers, each of volume V containing ideal gases. The pressure and temperature of the gases in the two vessels are p_1, T_1 and p_2, T_2 respectively. If the vessels are now connected by a thin long tube of negligible volume, the final temperature of the two after mixing is T . Find the final pressure of the gas.

$$\left[\frac{T}{2} \left(\frac{p_1}{T_1} + \frac{p_2}{T_2} \right) \right]$$

(iii) A vessel of volume $V = 20$ litre contains a mixture of H_2 and He at 20°C and pressure $P = 2$ atm. The mass of the mixture is $m = 5$ g. Find the ratio of masses m_1/m_2 where $m_1 =$ mass of H_2 and $m_2 =$ mass of He.

$$[1/2]$$

(iv) The temperature of a gas contained in a closed vessel increases by 1°C when pressure of the gas is increased by 1%. Find the initial temperature of the gas.

$$[100 \text{ K}]$$

(v) A vessel contains a mixture of nitrogen ($m_1 = 7$ g) and carbon dioxide ($m_2 = 11$ g) at a temperature $T = 290$ K and pressure $p_0 = 1$ atm. Find the density of this mixture, assuming the gases to be ideal.

$$[1.494 \text{ kg m}^{-3}]$$

(vi) A vessel of volume $V = 30$ litre is separated into 3 equal parts by stationary semipermeable partitions. The left, middle and right parts are filled with $m_1 = 30$ g of hydrogen (H_2), $m_2 = 160$ g of oxygen (O_2), and $m_3 = 70$ g of nitrogen (N_2) respectively. The left partition lets only hydrogen, while the right partition lets through hydrogen and nitrogen. What will be the pressure in each part of the vessel after the equilibrium has been set in if the vessel is kept at a constant temperature $T = 300$ K ?

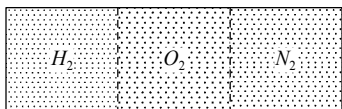


Figure 2.9

$[P_1 = 12.471 \times 10^5 \text{ Pa}, P_2 = 28.06 \times 10^5 \text{ Pa}, P_3 = 15.589 \times 10^5 \text{ Pa}]$

(vii) A vessel of volume 30 litres contains ideal gas at a temperature 63°C . After a portion of the gas has been let out, the pressure in the vessel decreased by 0.415 bar. If density of gas at STP is 1.3 gm/ltr, find the mass of gas released from the vessel.

[13.14 g]

(viii) A vessel of volume 8.3 litres contains a mixture of ideal gases at temperature 300 K, 0.1 mol of nitrogen, 0.2 mol of oxygen and 0.3 mol of carbon dioxide. Find the pressure and molecular weight of the mixture.

$[1.8 \times 10^5 \text{ Pa}; 37.33]$

(ix) Two glass bulbs of volume 3 litres and 1 litre, respectively, are connected by a capillary tube. Air at a pressure of 76 cm of mercury at 30°C is contained in the apparatus which is then hermetically sealed. If the 3litre bulb is immersed in steam at 100°C , the other remaining at 30°C , what would be the pressure of the air in the bulbs ? Neglect expansion of the 3-litre bulb.

[88.45 cm of Hg]

(x) A freely moving piston divides a vertical cylinder, closed at both ends, into two parts each containing 1 mole of air. In equilibrium, at $T = 300$ K, volume of the upper part is $\eta = 4$ times greater than the lower part. At what temperature will the ratio of these volumes be equal to $\eta' = 2$?

[750 K]

2.3 Pressure of Air

Figure-2.10 shows a Toricellian tube, the principle of today's barometer. If a glass tube, closed at one end, is completely filled with mercury and then inverted into a bowl of mercury, the

column of mercury in the tube drops until it reaches a height of about 76 cm above the lower surface as shown in figure-2.10 this is what Torricelli predicted before experiment. According to Pascal's principle, the pressure of the atmosphere on the surface of the mercury in the bowl at point A is equal to the pressure due to the weight of mercury in the tube at point B if this were not so, the mercury would flow because it would not be in static equilibrium. The space between the top of liquid and the end of the tube contains no air and was named the Torricellian vacuum.

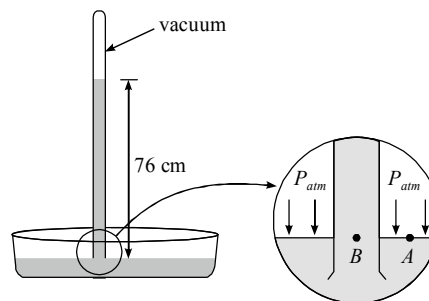


Figure 2.10

This is how barometer works and the height of mercury column in the tube gives the atmospheric pressure of air. If cross-sectional area of tube is A then the weight of mercury column is

$$W = h A \rho g$$

Thus pressure at its bottom is given as

$$p = \frac{W}{A} = h \rho g \quad \dots (2.21)$$

In general the height of mercury column in open atmosphere in standard conditions is about 76 cm, thus atmospheric pressure of air is given as

$$\begin{aligned} p &= h \rho g \\ &= (0.76) \times (13.6 \times 10^3) \times (9.81) \text{ W/m}^2 \\ &= 1.01 \times 10^5 \text{ N/m}^2 \end{aligned}$$

$$p_{\text{air}} = 1.01 \times 10^5 \text{ Pa} = 1 \text{ atm}$$

This pressure is atmospheric pressure. An atmosphere (1 atm) is a unit of pressure equal to the pressure of the earth's atmosphere at sea level.

2.3.1 Barometric Relation

After Toricelli's experiments, pascal suggested that the atmosphere is like a big ocean of air, in which pressure is greater at the bottom than at higher altitudes. This assumption was verified and confirmed when a Torricellian tube was carried from sea level to a mountain top and the mercury column was observed to be shorter at higher elevations.

Maxwell developed a relation to find atmospheric pressure as a function of height from ground level (more precisely sea level). The relation is named Barometric Relation. We first derive the relation mathematically and then we'll discuss some uses and applications of this relation.

We consider an atmospheric layer of width dx at a height x above the earth surface as shown in figure-2.11. According to Pascal's assumption as we go up in air, pressure decreases thus if just below this elemental layer if pressure is P and just above it is $P-dP$, then dP is the pressure difference due to this small layer of width dx . Thus we can write

$$dP = -dx \rho g \quad \dots (2.22)$$

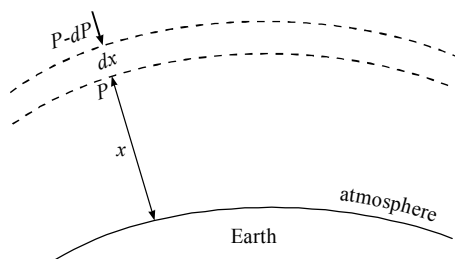


Figure 2.11

Here minus sign indicates that the pressure decreases as altitude increases where ρ is the air density in the elemental layer at a height dx . According to gas law the pressure at this layer P is given as

$$P = \rho \frac{RT}{M} \quad \dots (2.23)$$

From equation-(2.22) and (2.23)

$$dP = -dx \frac{PM}{RT} g$$

or

$$\frac{dP}{P} = -\frac{Mg}{RT} dx$$

Integrating this expression within proper limits, gives

$$\int_{P_0}^P \frac{dP}{P} = - \int_0^h \frac{Mg}{RT} dx \quad \dots (2.24)$$

or

$$\ln \frac{P}{P_0} = -\frac{Mgh}{RT}$$

[Assuming throughout the atmosphere temperature $T = \text{constant}$]

or

$$P = P_0 e^{-\frac{Mgh}{RT}} \quad \dots (2.25)$$

Here P_0 is the atmospheric pressure on ground-level or sea-level and equation-(2.25) gives the pressure of atmospheric air

at a height h above the earth surface and is called as Barometric formula. Using this we can also find the atmospheric air density at a height h above the ground level.

Using equation-(2.23) we have

$$\rho = \frac{PM}{RT}$$

or

$$\rho = \frac{P_0 M}{RT} e^{-\frac{Mgh}{RT}}$$

or

$$\rho = \rho_0 e^{-\frac{Mgh}{RT}} \quad \dots (2.26)$$

Here $\rho_0 = \frac{P_0 M}{RT}$ can be taken as atmosphere air density at the ground level. Another form of Barometric formula can be written in terms of number of molecules per unit volume or the molecular density in atmosphere and can be given as

$$n = n_0 e^{-\frac{Mgh}{RT}} \quad \dots (2.27)$$

Above equation-(2.25), (2.26) and (2.27) are used with the assumption that the atmospheric temperature T and the acceleration due to gravity g remains constant within the limits of integration of expression in equation-(2.24). If in some region temperature gradient (dT/dh) exist then the above formula can be modified while integrating the expression in equation-(2.24).

Illustrative Example 2.8

A vertical cylinder of height 100 cm contains air at a constant temperature and its top is closed by a frictionless piston at atmospheric pressure (76 cm of Hg) as shown in figure-2.12. If mercury is slowly poured on the piston, due to its weight air is compressed. Find the maximum height of the mercury column which can be put on the piston.

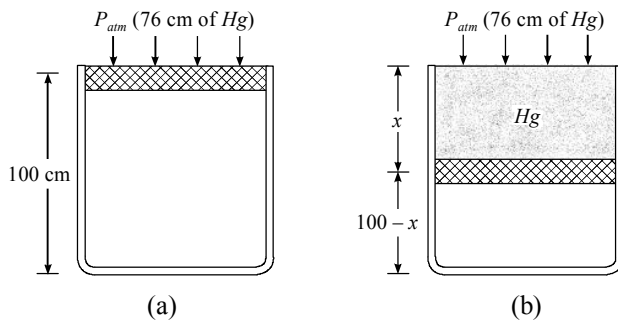


Figure 2.12

Solution

When mercury is poured on the top of the piston, due to increase in pressure, the volume of air will decrease according to Boyle's

Law. If final mercury column of height x is poured on the piston then gas pressure in equilibrium can be given as

$$P_f = (76 + x) \text{ cm of Hg}$$

As atmospheric pressure is equivalent to the pressure due to a mercury column of height 76 cm. If A be the area of cross section of cylinder then we have according to Boyle's Law

$$P_1 V_1 = P_2 V_2$$

$$\text{or } (76 \text{ cm})(100 A) = (76 + x)(100 - x) A$$

$$\text{or } 7600 = 7600 + 24x - x^2$$

$$\text{or } x = 24 \text{ cm}$$

Illustrative Example 2.9

A vertical hollow cylinder of height 1.52 m is fitted with a movable piston of negligible mass and thickness. The lower part of cylinder contains an ideal gas and the upper part is filled with mercury as shown in figure-2.13. Initially the temperature of system is 300 K and the lengths of gas and mercury columns are equal. Find the temperature to which system is raised so that half of mercury overflows. Take atmospheric pressure is 76 cm of Hg and neglect thermal expansion of mercury.

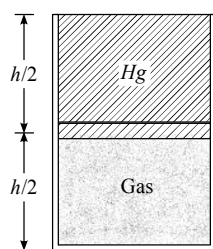


Figure 2.13

Solution

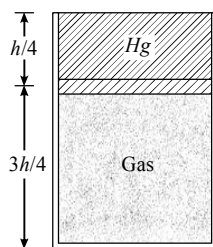


Figure 2.14

Final state of gas is shown in figure-2.14 after half of mercury overflows. Let this temperature be T . Then from gas law for initial and final state of gas we have

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Here initial pressure, volume and temperature of gas are

$$P_1 = 76 + \frac{h}{2} = 76 + 76 = 152 \text{ cm of Hg}$$

$$V_1 = \frac{h}{2} A = 76 A$$

$$T_1 = 300 \text{ K}$$

Similarly after heating pressure, volume and temperature of gas are

$$P_2 = 76 + \frac{h}{4} = 76 + 38 = 114 \text{ cm of Hg}$$

$$V_2 = \frac{3h}{4} A = 114 A$$

Thus from gas law, we have

$$\frac{152 \times 76 A}{300} = \frac{114 \times 114 A}{T_2}$$

$$\text{or } T_2 = 337.5 \text{ K}$$

Illustrative Example 2.10

A very tall cylindrical vessel with an ideal gas of molar mass M , filled in it is placed in a uniform gravitational field. It is given that the temperature of gas varies with height in such a manner that its density remains same throughout the container. Find the temperature gradient in the container dT/dh .

Solution

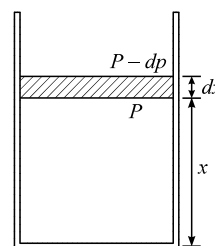


Figure 2.15

In the container shown in figure-2.15, consider a layer of width dx at a height x above its bottom. If P is the pressure just below this layer then just above this layer, we can consider pressure to be $P - dP$. Here dP is the pressure due to the gas layer of width dx which can be given as

$$dP = - dx \rho g$$

$$[\text{If } \rho \text{ is the density of gas}] \quad \dots (2.28)$$

From open atmospheric form of gas law, we have

$$P = \frac{\rho R T}{M}$$

As it is given that density of gas is constant, we have

$$dP = \frac{\rho R}{M} dT \quad \dots (2.29)$$

From (2.28) and (2.29) we have

$$\frac{\rho R}{M} dT = -dx \rho g$$

$$\text{or} \quad \frac{dT}{dx} = -\frac{Mg}{R}$$

Illustrative Example 2.11

An open glass tube is immersed in mercury in such a way that the length of 8 cm extends above the mercury level. Now the open end of the tube is closed by a finger and raised by further 44 cm. What will be the length of air column above mercury in the tube. Take atmospheric pressure to be 76 cm of mercury. Neglect capillary effect.

Solution

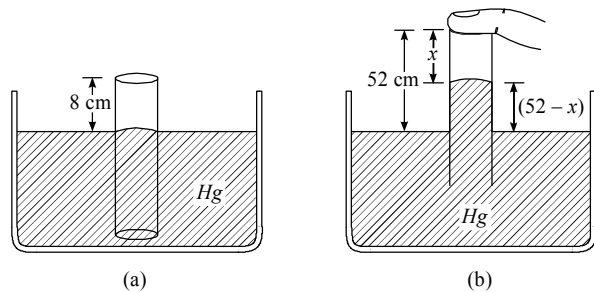


Figure 2.16

Figure-2.16(a) shows the tube in initial state then it is closed at top and raised up so that its length above mercury becomes $8 + 44 = 52$ cm as shown in figure-2.16(b). As initially tube is open to atmosphere, pressure of air inside is 76 cm of Hg and its volume is $8A$ if A is the cross-sectional area of the tube.

When its upper end is closed and raised up, situation is shown in figure-2.16(b). Let the air column be of length x then its final volume will be xA . Pressure at the mercury level in the container is equal to atmospheric pressure 76 cm of Hg and the air column is separated by another mercury column of height $52-x$ above this level which will oppose the atmospheric pressure thus pressure inside the column will be $76 - (52 - x) = (24 + x)$ cm of mercury.

As during the process temperature of system remains constant, thus we can use Boyle's law for the initial and final states as

$$P_1 V_1 = P_2 V_2$$

$$\text{or} \quad 76 \times 8A = (24 + x) \times xA$$

$$\text{or} \quad 608 = 24x + x^2$$

$$\text{or} \quad x^2 + 24x - 608 = 0$$

On solving, we get $x = 15.4$ cm or $x = -39.4$ cm

Thus the acceptable value of final air column is 15.4 cm.

Illustrative Example 2.12

A uniform tube closed at one end, contains a pallet of mercury 10 cm long. When the tube is kept vertically with the closed end up, the length of air column trapped by mercury and the closed end of the tube is 20 cm. If the tube is inverted so that its open end becomes upward. Find the final length of the air column trapped. Take atmospheric pressure to be 76 cm of Hg.

Solution

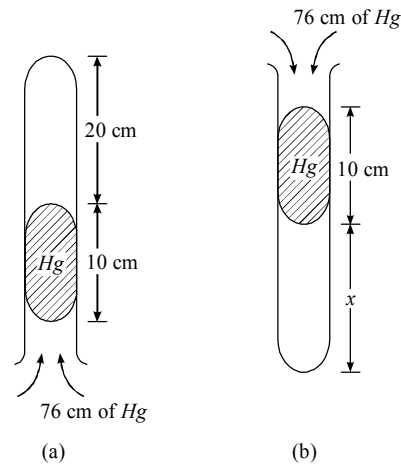


Figure 2.17

Figure-2.17(a) shows the initial state of the tube. Here pressure below the mercury pallet is the atmospheric pressure. 76 cm of Hg and due to opposite of pallet's weight the pressure of air column is $76 - 10 = 66$ cm of Hg.

When the tube is inverted with its open end upward, situation is shown in figure-2.17(b). If the length of air column is x and the pressure on this air column is atmospheric pressure planes the weight of mercury pallet which becomes $76 + 10 = 86$ cm of Hg, then according to Boyle's Law we have

$$P_1 V_1 = P_2 V_2$$

$$66 \times 20A = 86 \times xA$$

[If A is the cross sectional area of tube]

$$\text{or} \quad x = 15.35 \text{ cm}$$

Illustrative Example 2.13

Figure-2.18 shows a horizontal cylindrical container of length 30 cm, which is partitioned by a tight fitting separator. The separator is diathermic but conducts heat very slowly. Initially the separator is in the state shown in figure-2.18. The temperature of left part of cylinder is 100 K and that on right part is 400 K. Initially the separator is in equilibrium. As heat is conducted from right to left part, separator displaces to the right. Find the displacement of separator after a long time when gases on the parts of cylinder are in thermal equilibrium.

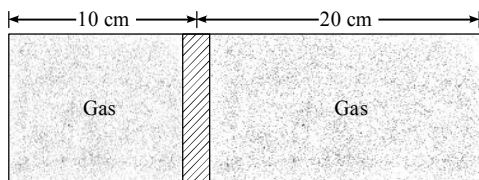


Figure 2.18

Solution

It is given that initially the separator is in equilibrium thus pressure on both sides of gas are equal say, it is P_i . If A be the area of cross-section of cylinder, number of moles of gas in left and right part n_1 and n_2 can be given as

$$n_1 = \frac{P_i(10A)}{R(100)} \quad \text{and} \quad n_2 = \frac{P_i(20A)}{R(400)}$$

Finally if separator is displaced to right by a distance x , we have

$$n_1 = \frac{P_f(10+x)A}{RT_f} \quad \text{and} \quad n_2 = \frac{P_f(20-x)A}{RT_f}$$

If P_f and T_f be the final pressure and temperature on both sides after a long time.

Now if we equate the ratio of moles n_1/n_2 in initial and final state we get

$$\frac{n_1}{n_2} = \frac{\left(\frac{10A}{100}\right)}{\left(\frac{20A}{400}\right)} = \frac{(10+x)A}{(20-x)A}$$

$$\text{or} \quad 2(20-x) = 10+x$$

$$\text{or} \quad x = 10 \text{ cm}$$

Thus in final state when gases in both parts are in thermal equilibrium, the piston is displaced to 10 cm right from its initial position.

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Kinetic Theory of Gases

Module Number - 14 to 17

Practice Exercise 2.2

(i) A glass tube, which is closed at one end, is completely submerged with open end downward in a vessel of mercury, so that air column of length $l = 10$ cm is entrapped inside the tube. To what height must the upper end of the tube be raised above the level of mercury in the vessel so that the level of mercury in the tube coincides with that in the vessel? Also calculate the mass of the air if the temperature remains constant at 27°C . The area of cross-section of the tube is $\alpha = 1.0 \text{ cm}^2$. Given that atmospheric pressure $p_0 = 1.013 \times 10^5 \text{ Pa} = 76 \text{ cm of Hg}$, molecular weight of air = 29.

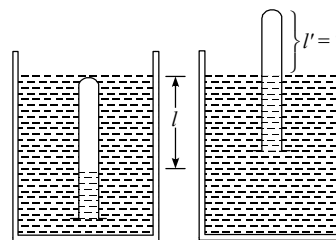


Figure 2.19

$$[l' = 11.315 \text{ cm}, m = 13.32 \text{ mg}]$$

(ii) An ideal gas is enclosed in a tube and is held in the vertical position with the closed end upward. The length of the pellet of mercury entrapping the gas is $h = 10$ cm and the length of the tube occupied by gas is $l = 40$ cm. Calculate the length occupied by the gas when it is turned through $\theta = 60^\circ$ and 90° . Atmospheric pressure, $H = 76$ cm of mercury.

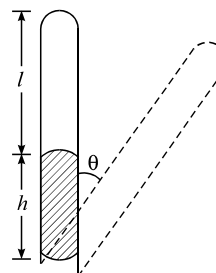


Figure 2.20

$$[37.18 \text{ cm}, 34.73 \text{ cm}]$$

(iii) A glass tube of length $l = 50$ cm and cross-section $A = 0.5 \text{ cm}^2$ is sealed at one end and submerged into water as shown in the figure-2.21. What force should be applied to hold the tube under the water if the distance from the surface of the

water to the closed end is $h = 20$ cm and the atmospheric pressure is 10^5 Pa ? The weight of the tube is 15 gm.

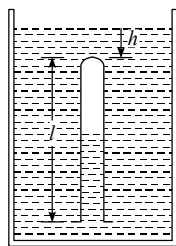


Figure 2.21

[0.084 N]

(iv) An ideal gas of molar mass M is contained in a tall vertical cylindrical vessel whose base area is S and height h . The temperature of the gas is T , its pressure on the bottom base is p_0 . Assuming the temperature and the free-fall acceleration g to be independent of the height, find the mass of gas in the vessel.

$[m = (1 - e^{-Mgh/RT}) p_0 S/g]$

(v) An ideal gas is trapped between a mercury column and the closed end of a narrow vertical tube of uniform bore. The upper end of the tube is open to the atmosphere (atmospheric pressure = 76 cm of mercury). The lengths of the mercury and the trapped gas columns are 20 cm and 43 cm respectively. What will be the length of the gas column when the tube is turned slowly in a vertical plane through an angle of 60° ? Assume the temperature to be constant

[48 cm]

(vi) An ideal gas of molar mass M is contained in a very tall vertical cylindrical vessel in the uniform gravitational field in which the free-fall acceleration equals g . Assuming the gas temperature to be the same and equal to T , find the height at which the centre of gravity of the gas is located.

$[RT/Mg]$

(vii) A barometer gives wrong readings because of some air in the space above the mercury column. At a pressure 75.5 cm Hg the barometer shows 74.8 cm, and at 74 cm it shows 73.6 cm. Find the length l of the barometer tube above the surface of mercury.

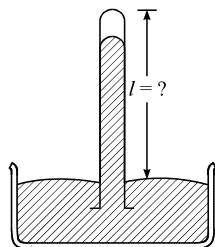


Figure 2.22

[94.1 cm]

(viii) A horizontal cylinder closed from one end is rotated with a constant angular velocity ω about a vertical axis passing through the open end of the cylinder. The outside air pressure is equal to p_0 , the temperature to T , and the molar mass of air to M . Find the air pressure as a function of the distance r from the rotation axis. The molar mass is assumed to be independent of r .

$[p = p_0 e^{M\omega^2 r^2 / 2RT}]$

(ix) An ideal gas of molar mass M is located in the uniform gravitational field in which the free-fall acceleration is equal to g . Find the gas pressure as a function of height h , if $p = p_0$ at $h = 0$, and the temperature varies with height as

(a) $T = T_0 (1 - ah)$; (b) $T = T_0 (1 + ah)$

[(a) $p = p_0 (1 - ah)^n$, $h < 1/a$; (b) $p = p_0 / (1 + ah)^n$. Here $n = Mg/aRT_0$]

2.4 Distribution of Molecular Speeds

The molecules in a gas are assumed to be in random motion, which means that many molecules have speeds less than the average speed and others have speeds greater than the average. On the basis of kinetic theory and experiments Maxwell plotted a graph of variation of speeds with relative number of molecules as shown in figure-2.23. The graph shows how the speeds of molecules in a gas are distributed. This is known as Maxwell distribution of speeds. The speeds vary from zero up to many times the average speed, but as can be seen from the graph, most molecules have speeds that are not far from the average.

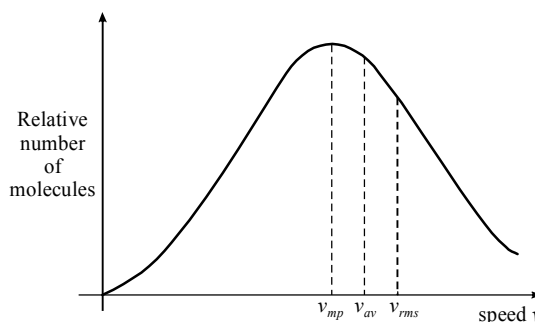


Figure 2.23

The exact distribution of speeds is described by a distribution function by Maxwell, who defined a function named probability distribution function $f(v)$, so that, of the N molecules in a gaseous system, the number dN molecules which have range of speeds between v and $v + dv$ or in the very short neighbourhood of v is given by

$$dN = f(v) dv \quad \dots (2.30)$$

Thus $f(v)$ is the number of molecules per unit range of speeds. As we have discussed that the speeds of molecules vary in a

wide range from zero to infinity, from equation-(2.30), we can write for total number of molecules as

$$N = \int_0^{\infty} f(v) dv \quad \dots (2.31)$$

The distribution function $f(v)$ was obtained by statistical analysis and found out as

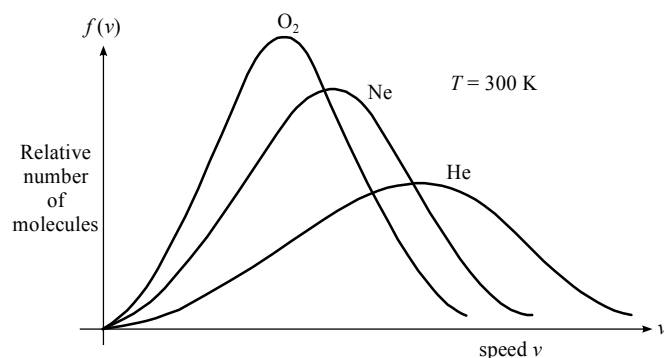
$$f(v) = 4\pi N v^2 \left[\frac{M}{2\pi RT} \right]^{3/2} e^{-\frac{Mv^2}{2RT}} \quad \dots (2.32)$$

Where R is universal gas constant and M is the molar mass, using equation-(2.32) we can write

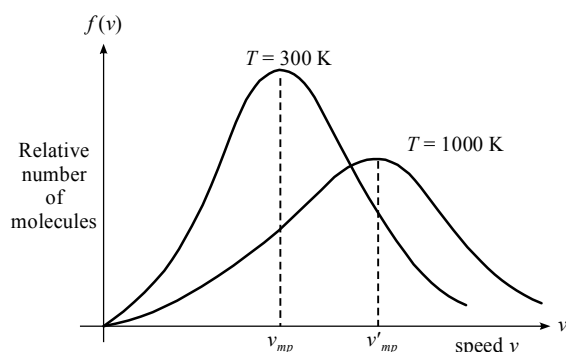
$$f(v) dv = 4\pi N v^2 \left[\frac{M}{2\pi RT} \right]^{3/2} e^{-\frac{Mv^2}{2RT}} dv \quad \dots (2.33)$$

Here $f(v) dv$ is the number of molecules that have speeds between v and $v + dv$ and this equation-(2.33) is called as analytical Maxwell Boltzmann distribution function. Figure-2.24(a) shows the function computed for oxygen, neon and helium gases at 300 K temperature.

Here we can observe that the peak of each curve represents the speed, the maximum number of molecules have and it is called the most probable speed for that gas at particular temperature. This most probable speed is exhibited by large number of molecules of a gas at a given temperature.



(a)



(b)

Figure 2.24

It can also be seen from figure-2.24(a) that more massive molecules have lower most probable speed. For a given gas, the most probable molecular speed becomes greater with increase in temperature as shown in figure-2.24(b). It can also be seen that more molecules have high speeds and fewer molecule have lower speeds throughout the range of speeds.

2.4.1 Different Speeds For Molecules of a Gas

(i) Average velocity of Gas Molecules

Initially we've discussed that all molecules of a gas in a container are in brownian motion, thus due to randomness the directions of motion of different molecules are random and continuously changing due to repeated collisions randomly. If $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_N$ are the instantaneous velocity vectors of all N molecules of a gas, the average velocity vector of these N molecules can be written as

$$\langle \vec{v} \rangle = \frac{\vec{v}_1 + \vec{v}_2 + \dots + \vec{v}_N}{N} = 0 \quad \dots (2.34)$$

It is obvious that as all vectors are randomly scattered in space thus their sum i.e. the numerator of expression in equation-(2.34) can be approximated to zero. Thus the average velocity vector of molecules in a gas is always taken as zero.

(ii) Root-Mean-Square Velocity of Gas Molecules

As the name implies this is the square root of mean of squares of velocities of all the molecules of a gas. Mean of squares can be simply written as

$$\langle v^2 \rangle = \left(\frac{|\vec{v}_1|^2 + |\vec{v}_2|^2 + \dots + |\vec{v}_N|^2}{N} \right)$$

and $v_{rms} = \sqrt{\langle v^2 \rangle}$

$$= \sqrt{\left(\frac{|\vec{v}_1|^2 + |\vec{v}_2|^2 + \dots + |\vec{v}_N|^2}{N} \right)} \quad \dots (2.35)$$

Mathematically its value can be calculated by using the distribution function and taking the average of the square of the speed. Since $f(v) dv$ is the number of molecules with speed v in the range dv , the mean square speed can be given as

$$\langle v^2 \rangle = \frac{\int_0^{\infty} v^2 f(v) dv}{N} \quad \dots (2.36)$$

Integrating the expression in equation-(2.36), we finally get

$$\langle v^2 \rangle = \frac{3RT}{M}$$

So that rms speed can be given as

$$v_{rms} = \sqrt{\frac{3RT}{M}} \quad \dots (2.37)$$

In most of numeric calculation and analysis of kinetic theory we use rms velocity for gas molecules.

(iii) Mean Speed of Gas Molecules

We have discussed that due to randomness average velocity vector of all the molecules of a gas comes out zero. But it is not same in case of mean or average speed. We obtain the mean speed by averaging the speed of the molecules. It can be simply defined as

$$\langle v \rangle = \frac{|\vec{v}_1| + |\vec{v}_2| + \dots + |\vec{v}_N|}{N} \quad \dots (2.38)$$

Using distribution function we can calculate the mean speed as

$$\langle v \rangle = \frac{\int_0^\infty v f(v) dv}{N} \quad \dots (2.39)$$

Integrating expression in equation-(2.39) results

$$v_{mean} = \sqrt{\frac{8RT}{\pi M}} \quad \dots (2.40)$$

We can see on comparing equation-(2.37) and equation-(2.40)

$$v_{mean} < v_{rms}$$

(iv) Most Probable Speed of Gas Molecules

As discussed this is the speed maximum number of molecules have, and this is the speed corresponding to which distribution function has maximum or peak value. Its value can be obtained by differentiating the distribution function $f(v)$ and setting the derivative equal to zero. The final result obtained is

$$v_{mp} = \sqrt{\frac{2RT}{M}} \quad \dots (2.41)$$

Comparing equation-(2.37), (2.40) and (2.41) we can see

$$v_{mp} < v_{mean} < v_{rms}$$

2.5 Pressure Exerted by a Gas

When a gas is enclosed in a rigid container, we have discussed that the molecules of gas are in Brownian motion and randomly collide with each other and container walls. All these collisions are assumed to be perfectly elastic. Due to these continuous collisions of large number of gas molecules with container walls, a pressure is exerted on the walls. Now we calculate this pressure analytically. Consider N molecules of a gas enclosed in a

container as shown in figure-2.25. The length, width and height of container are l , w and h respectively along x , y and z direction. Out of several molecules, we consider one gas molecule of mass m' which is moving at an instant with a velocity v . This velocity has three components v_x , v_y and v_z along x , y and z directions respectively.

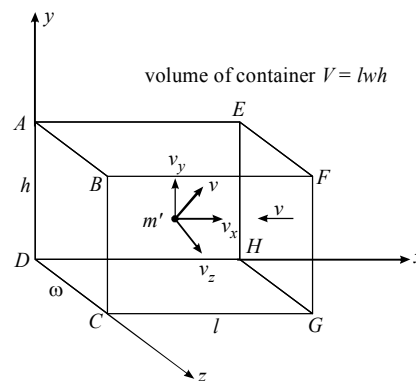


Figure 2.25

Let us first consider the motion of this molecule in x direction only. If it is moving with a velocity v_x towards the wall $EFGH$. It collides with the wall elastically and rebounds with the same speed v_x . Now it is travelling towards the wall $ABCD$. During a collision the momentum imparted to the wall by the molecule is

$$\Delta p_x = 2 m' v_x \quad \dots (2.42)$$

Now the molecule strikes the wall $ABCD$ after travelling a distance l and strike again onto the wall $EFGH$ with same speed v_x . The average time between two collisions on same wall by the molecule is

$$\Delta t = \frac{2l}{v_x} \quad \dots (2.43)$$

Thus the frequency of collision by the molecule on same wall is

$$\Delta n = \frac{1}{\Delta t} = \frac{v_x}{2l} \quad \dots (2.44)$$

Thus the momentum imparted to the same wall by this molecule per second or the force exerted by this molecule in x -direction is

$$F_x = \frac{\Delta p_x}{\Delta t} = \frac{2 m' v_x^2}{2l} = \frac{m' v_x^2}{l} \quad \dots (2.45)$$

The pressure exerted by this molecule on wall $ABCD$ is

$$P_x = \frac{F_x}{wh} = \frac{m' v_x^2}{lwh} = \frac{m' v_x^2}{V} \quad \dots (2.46)$$

Similarly we can write the pressure exerted by the molecule on walls in y and z direction can be given as

$$P_y = \frac{m' v_y^2}{V} \quad \dots (2.47)$$

$$\text{and } P_z = \frac{m'v_z^2}{V} \quad \dots (2.48)$$

The average pressure by the molecule on container walls is

$$P = \frac{1}{3} (P_x + P_y + P_z)$$

$$\text{or } P = \frac{1}{3} \frac{m'}{v} (v_x^2 + v_y^2 + v_z^2)$$

$$\text{or } P = \frac{1}{3} \frac{m'}{v} v^2 \quad \dots (2.49)$$

Where $v^2 = v_x^2 + v_y^2 + v_z^2$ is the root mean square (rms) velocity of a gas molecule we've already discussed. Equation-(2.49) gives the average pressure on container walls due to motion of only one molecule. There are total N molecules of gas enclosed in the container. Thus total average pressure exerted by a gas on its container walls is

$$P = \frac{1}{3} \frac{m'}{v} v^2 \times N$$

$$\text{or } P = \frac{1}{3} \frac{m}{v} v^2$$

[Where $m = m' N$ is the total mass of gas]

$$\text{or } P = \frac{1}{3} \rho v^2$$

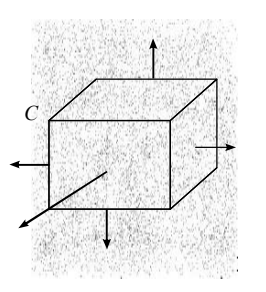
$$\text{[Where } \rho = \frac{m}{V} \text{ is the density of gas]} \quad \dots (2.50)$$

Here we can also write

$$P = \frac{2}{3} \left(\frac{1}{2} \rho v^2 \right) = \frac{2}{3} e \quad \dots (2.51)$$

Here e represents the average kinetic energy of gas per unit volume or average kinetic energy density of gas pressure of a gas in a container can also be derived in another way as shown in figure-2.26. Figure shows a container wall and the gas molecules in its neighbourhood. If total number of molecules in the container are N , then the molecular density n_0 can be given as

$$n_0 = \frac{N}{V} \quad \dots (2.52)$$



Container wall

Figure 2.26

We consider a cube C of unit volume 1 m^3 near the wall shown. Total number of molecules in this cube are n_0 (molecular density) and each molecule is assumed to be moving in a random direction with the rms speed v . Due to randomness it can be assumed that toward every face of the cube $\frac{n_0}{6}$ molecules are moving with this speed. Thus number of collisions with a container wall per second per square meter of its surface can be written as

$$N_c = \frac{n_0}{6} v \quad \dots (2.53)$$

Equation-(2.53) gives the number of collision per unit area of container wall in contact with a gas. As collisions are elastic, the momentum transferred to wall in each collision by a molecule is

$$\Delta p = 2m'v \quad [\text{If } m' \text{ is the mass of each molecule}]$$

Thus momentum transferred to a container wall per second per unit of its surface area or average pressure can be given as

$$P = \frac{n_0 v}{6} \times 2m'v = \frac{1}{3} \rho v^2 \quad \dots (2.54)$$

[As $n_0 \cdot m' = \rho$ density of gas]

Thus we can see that equation-(2.54) is identical with equation-(2.50). Equation-(2.50) was derived for a box shaped container but in derivation of equation-(2.54) we haven't taken any specific shape of container. Thus, this relation of average pressure always remains same irrespective of the shape of container.

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Kinetic Theory of Gases

Module Number - 18 to 20

2.6 Kinetic Energy of Gas Molecules

We've discussed that all gas molecules in a container are in continuous random motion repeatedly colliding with neighbouring molecules and container walls elastically. As all collisions are considered to be perfectly elastic we can assume that the total kinetic energy of all the gas molecules remains constant and all molecules are assumed to be always moving with their rms velocity. In this situation kinetic energy of a gas molecule of mass m' can be given as

$$E_K = \frac{1}{2} m' v_{rms}^2$$

or

$$E_K = \frac{1}{2} m' \left(\frac{3RT}{M} \right) \quad \dots (2.55)$$

By definition, the molecular mass of a gas molecule can be given as

$$M = m' N_{AV} \quad [N_{AV} = \text{Avogadro Number}] \quad \dots (2.56)$$

From equation-(2.55) and (2.56)

$$E_K = \frac{1}{2} m' \left(\frac{3RT}{m' N_{AV}} \right) = \frac{3}{2} \frac{R}{N_{AV}} T = \frac{3}{2} kT \quad \dots (2.57)$$

Here $k = \frac{R}{N_{AV}} = 1.38 \times 10^{-23}$ Joule/molecule, k is a universal constant called Boltzmann's constant. This equation-(2.57) gives the kinetic energy of a moving gas molecule in a gas at absolute temperature T . If we find the total kinetic energy of all the molecules of a gas, it is given as

$$E_T = \frac{3}{2} kT \times N$$

[If N are the total no. of gas molecules in a container]

If n moles of gas are there in a container, we have

$$N = n \times N_{AV}$$

Thus
$$E_T = \frac{3}{2} \left(\frac{R}{N_{AV}} \right) T \times n \times N_{AV}$$

or
$$E_T = \frac{3}{2} n R T \quad \dots (2.58)$$

As we've discussed that total energy of a gas is the sum of kinetic energy of all of its molecules. Equation-(2.58) thus gives the total energy of a gas in its molecular motion or the expression in equation-(2.58) gives the total translational energy of all the gas molecules at absolute temperature T .

2.7 Microscopic Interpretation of Temperature

In previous section we've discussed that the average molecular translational kinetic energy and absolute temperature are proportional. The higher the temperature of a system. The greater, proportionally, is the average translational kinetic energy of the molecules of that system. Thus in terms of average kinetic energy of a gas molecule E_T , the temperature can be given by equation-(2.57) as

$$T = \frac{2}{3} \frac{E_T}{k} \quad \dots (2.59)$$

Thus the average translational kinetic energy of gas molecules in an ideal gas depends only on the temperature, not on the pressure or type of gas. Thus equation-(2.58) shows that temperature is a measure of kinetic energies of molecules or temperature is a large-scale manifestation of motion at the molecular level of gases, liquids and solids.

Now we take few example to understand the molecular speeds and their kinetic energy in detail.

Illustrative Example 2.14

- (a) Calculate (i) root mean square speed and (ii) the mean kinetic energy of one mole of hydrogen at S.T.P. (given that density of hydrogen is 0.09 kg/m^3).
- (b) Given that the mass of a molecule of hydrogen is $3.34 \times 10^{-27} \text{ kg}$. Calculate Avogadro's number.
- (c) Calculate Boltzmann's constant.

Solution

- (a) (i) We know that the pressure of a gas is given as

$$P = \frac{1}{3} \rho v_{rms}^2$$

$$v_{rms} = \sqrt{\left(\frac{3P}{\rho} \right)}$$

$$v_{rms} = \sqrt{\left(\frac{3 \times 0.76 \times 13.6 \times 10^3 \times 9.8}{0.09} \right)}$$

$$= 1837 \text{ m/sec} = 1.837 \text{ km/sec}$$

(ii) Kinetic energy $= \frac{1}{2} m v_{rms}^2$

Here $M = 2 \text{ gm} = 2 \times 10^{-3} \text{ kg}$.

or
$$\text{K.E.} = \frac{1}{2} \times 2 \times 10^{-3} \times (1837)^2$$

$$= 3374.56 \text{ joules}$$

(b) Mass of one molecule of $\text{H}_2 = 3.34 \times 10^{-27} \text{ kg}$

Molecular mass of hydrogen $= 2 \times 10^{-3} \text{ kg}$

The Avogadro's number N_A , which is the number of molecules in one gram molecule of hydrogen is given by

$$N_A = \frac{2 \times 10^{-3}}{3.34 \times 10^{-27}} = 5.988 \times 10^{23} \text{ molecules.}$$

(c) We know that $k = \frac{R}{N_A} = \frac{8.3}{5.988 \times 10^{23}}$

$$= 1.37 \times 10^{-23} \text{ J/mol.K}$$

Illustrative Example 2.15

Calculate the number of molecules in 1 cm^3 of an ideal gas at 27°C and a pressure of 10 mm of mercury. Mean kinetic energy of a molecule at 27°C is 4×10^{-14} erg; the density of mercury is 13.6 gm/cc .

Solution

The pressure exerted by a gas is given by

$$P = \frac{2}{3} (\text{K.E. per unit volume})$$

or
$$\text{K.E.} = \frac{3}{2} P$$

Here $P = 10 \text{ mm of mercury} = 1 \text{ cm of mercury}$,

$$= 1 \times 13.6 \times 980 = 1.33 \times 10^4 \text{ dynes/cm}^2$$

Thus, we have
$$\text{K.E.} = \frac{3}{2} \times 1.33 \times 10^4 \times 1 = 1.99 \times 10^4 \text{ ergs}$$

As mean kinetic energy per molecule is 4×10^{-14} ergs, the number of molecule will be

$$\frac{1.99 \times 10^4}{4 \times 10^{-14}} = 4.9 \times 10^{17} \approx 5 \times 10^{17} \text{ molecules.}$$

Illustrative Example 2.16

A parallel beam of nitrogen molecules moving with velocity $v = 400 \text{ m/s}$ impinges on a wall at an angle $\theta = 30^\circ$ to its normal. The concentration of molecules in the beam $n_0 = 0.9 \times 10^{19} \text{ cm}^{-3}$. Find the pressure exerted by the beam on the wall assuming the collisions of molecules with the wall are perfectly elastic.

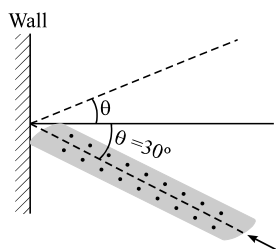
Solution

Figure 2.27

Figure-2.27 shows a beam of molecules in cylindrical form striking the wall. According to elastic collision, each molecule is reflected from the wall at the same angle of incidence as shown with same speed. If m_0 is the mass of each nitrogen molecule, momentum of each molecule is

$$P_0 = m_0 v$$

During collision change in momentum of each molecule or momentum imparted to wall by each molecule is

$$\Delta P_0 = 2m_0 v \cos \theta$$

If S is the cross-sectional area of beam and v be the velocity of molecules in the beam then number of molecules incident on wall per second are

$$N = n_0 v S \quad [n_0 = \text{molecular density}]$$

Thus total momentum imparted to the wall per second by the beam or the force exerted on wall is

$$F = \frac{\Delta P_0}{\Delta t} = 2m_0 v \cos \theta \times n_0 v S \quad \dots (2.60)$$

As equation-(2.60) gives that total momentum imparted to the wall per second, this is the net force exerted by beam on wall.

Thus the pressure on wall by beam is

$$\begin{aligned} P &= \frac{F}{S} = 2 m_0 v^2 \cos \theta \cdot n_0 \\ &= 2 \times \frac{28 \times 10^{-3}}{6.023 \times 10^{23}} \times 9 \times 10^{24} \times (400)^2 \times \cos^2(30^\circ) \\ &= 10^5 \text{ Pa} \approx 1 \text{ atm} \end{aligned}$$

Illustrative Example 2.17

Calculate the temperature at which rms velocity of a gas molecule is same as that of a molecule of another gas at 47°C . Molecular weight of first and second gases are 64 and 32 respectively.

Solution

We know rms speed of a gas molecule is given as

$$v = \sqrt{\frac{3RT}{M}}$$

For first and second gas if temperature are T_1 and T_2 respectively, we have

$$v_{\text{rms1}} = v_{\text{rms2}}$$

or
$$\frac{T_1}{M_1} = \frac{T_2}{M_2}$$

or
$$\frac{T_1}{64 \times 10^{-3}} = \frac{320}{32} \quad [\text{As } T_2 = 47 + 273 = 320 \text{ K}]$$

or
$$T_1 = 640 \text{ K}$$

Illustrative Example 2.18

Figure-2.28 shows a cylindrical container which is divided in two equal parts by a clamped diathermic piston. Different ideal

gases are filled in the two parts. It is found that the rms speed of molecules in the lower part is equal to the mean speed of molecules in the upper part. Find the ratio of mass of molecule of gas in lower part to that of the gas in upper part.

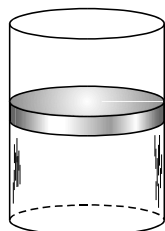


Figure 2.28

Solution

As piston is diathermic the two gases must be in thermal equilibrium. If M_1 and M_2 be the molecular masses of the gases filled in lower and upper part of the container, the rms speed of gas molecules in lower part is

$$v_{rms} = \sqrt{\frac{3RT}{M_1}} \quad \dots (2.61)$$

The mean speed of gas molecules in upper part is

$$v_{mean} = \sqrt{\frac{8RT}{\pi M_2}} \quad \dots (2.62)$$

From equation-(2.61) and (2.62), according to the given situation, we have

$$\begin{aligned} v_{rms} &= v_{mean} \\ \text{or } \sqrt{\frac{3RT}{M_1}} &= \sqrt{\frac{8RT}{\pi M_2}} \\ \text{or } \frac{M_1}{M_2} &= \frac{3\pi}{8} = 1.178 \end{aligned}$$

As ratio of masses of molecules and that of molecular masses is same, it is 1.178.

Illustrative Example 2.19

In a closed container of volume 10^{-3} m^3 , O_2 gas is filled at temperature 400 K and pressure 1.5 atm. A small hole is made in the container from which gas leaks out to open atmosphere. After some time the temperature and pressure of container become equals to that of surrounding. Find the mass of gas that leaks out from the container. (Atmospheric temperature = 300 K)

Solution

If initially there is m_1 mass of O_2 gas in the container, from gas law we have

$$PV = \frac{m_1}{32} RT$$

or

$$\begin{aligned} m_1 &= \frac{32PV}{RT} = \frac{32 \times 1.5 \times 10^5 \times 10^{-3}}{8.314 \times 400} \\ &= 1.443 \text{ gm} \end{aligned}$$

Finally if m_2 mass of O_2 is left in the container, from gas law we have

$$\begin{aligned} PV &= \frac{m_2}{32} RT \\ \text{or } m_2 &= \frac{32PV}{RT} \\ &= \frac{32 \times 10^5 \times 10^{-3}}{8.314 \times 300} = 1.283 \text{ gm} \end{aligned}$$

Thus mass of gas leaked out is

$$\begin{aligned} \Delta m &= m_1 - m_2 \\ &= 1.443 - 1.283 = 0.16 \text{ gm} \end{aligned}$$

Illustrative Example 2.20

In a certain region of outer space there are only 100 molecules per cm^3 on an average. The temperature there is about 3 K. What is the average pressure of this very dilute gas.

Solution

We know that the gas pressure is given as

$$P = \frac{1}{3} \rho v_{rms}^2$$

Now we can write the density of gas as $\rho = m_0 N$, where m_0 is the mass of the gas molecule and N is the number of molecules in 1 m^3 of the gas.

In the present case

$$N = 100 \times 10^6$$

or

$$P = \frac{1}{3} \times (100 \times 10^6) m_0 v_{rms}^2$$

As we know rms velocity of gas molecules is given as

$$\begin{aligned} v_{rms} &= \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3kT}{m_0}} \\ [\text{As } M &= m_0 N_{AV} \text{ and } \frac{R}{N_{AV}} = k] \end{aligned}$$

$$\text{So, we have } P = \frac{1}{3} \times (100 \times 10^6) \times 3 k T$$

$$\begin{aligned} &= (100 \times 10^6) k T \\ &= (100 \times 10^6) (1.38 \times 10^{-23}) (3) \quad [\text{As } T = 3 \text{ K}] \\ &= 41.4 \times 10^{-16} \\ &= 4.14 \times 10^{-15} \text{ Pa} \end{aligned}$$

Illustrative Example 2.21

One gram-mole of oxygen at 27°C and one atmospheric pressure is enclosed in a vessel. Assuming the molecules to be moving with v_{rms} , find the number of collisions per second which the molecules makes with one square meter area of the vessel wall.

Solution

From gas law we have

$$PV = \frac{m}{M} RT$$

or
$$PV = \frac{m'N}{m'N_{AV}} RT$$

[If m' = Mass of one molecule and N = total number of molecules]

or
$$PV = NkT \quad \left[\text{As } \frac{R}{N_{AV}} = k \right]$$

or
$$P = n_0 k T \quad \dots (2.63)$$

Here $n_0 = \frac{N}{V}$ the number of molecules per unit volume of gas. Students should note that equation (2.63) relates the gas pressure, temperature and molecular density. This expression can be used as a standard equation in several numerical cases as an alternative form of gas law.

Thus here molecular density is given as

$$\begin{aligned} n_0 &= \frac{P}{kT} \\ &= \frac{(1 \text{ atm})}{(1.38 \times 10^{-23}) \times (300)} \\ &= \frac{10^5}{1.38 \times 10^{-23} \times 300} = 2.44 \times 10^{25} \text{ m}^{-3} \end{aligned}$$

The rms speed of gas molecules is given as

$$\begin{aligned} v_{rms} &= \sqrt{\frac{3RT}{M}} \\ &= \sqrt{\frac{3 \times 8.314 \times 300}{32 \times 10^{-3}}} = 483.4 \text{ m/s} \end{aligned}$$

Now number of collision per square metre of vessel wall is given as

$$\begin{aligned} N &= \frac{n_0 v_{rms}}{6} = \frac{2.44 \times 10^{25} \times 483.4}{6} \\ &= 1.97 \times 10^{27} \text{ m}^{-2} \end{aligned}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Kinetic Theory of Gases

Module Number - 21 to 25

Practice Exercise 2.3

(i) The rms velocity of hydrogen molecules at a certain temperature is 300 m/s. If the temperature is doubled and hydrogen gas dissociates into atomic hydrogen. Find the rms speed of the gas molecules now.

[600 m/s]

(ii) In a container, the molecular density of an enclosed gas is 10^{26} molecules/m³, each of mass 3×10^{-27} kg. If rms velocity of the gas molecules is 2000 m/s, find the number of molecules hitting per square meter of the container wall every second and the pressure exerted on the walls of container by the molecules.

[$3.33 \times 10^{28} \text{ s}^{-1}$, $4 \times 10^5 \text{ N/m}^2$]

(iii) An ideal gas is enclosed between the closed end of a uniform cross-sectional tube and a pellet of mercury of length $h = 10$ cm. The length of the tube occupied by the gas when held vertical with the closed end upward is $l_1 = 40$ cm. When it is turned through $\theta = 60^\circ$ the length occupied by the gas is only $l_2 = 38$ cm. Calculate the frictional force between the wall of the tube and mercury pellet. The radius of the tube is $r = 2$ mm. Take $g = 10 \text{ m/s}^2$ and density of mercury 13600 kg/m^3 .

[0.026 N]

(iv) The average translational kinetic energy of O_2 at a particular temperature is 0.048 eV. Find the average translational kinetic energy of N_2 molecules in eV at the same temperature.

[0.048 eV]

(v) Find the number of molecules per unit volume of air at temperature 0°C and a pressure of $p = 1.013 \times 10^5 \text{ Pa}$ and the average distance between molecules. Boltzmann constant is $k = 1.38 \times 10^{-23} \text{ J/K}$.

[$2.688 \times 10^{25} \text{ m}^{-3}$, $3.31 \times 10^{-9} \text{ m}$]

(vi) Evaluate the speeds v_{mp} , $\langle v \rangle$, v_{rms} for (a) H_2 at 300 K and (b) O_2 at 300 K.

[(a) 1579.3 m/s, 1782.5 m/s, 1934.25 m/s (b) 394.83 m/s, 445.63 m/s, 483.56 m/s]

(vii) Twelve molecules have the following speeds, given in arbitrary units : 5, 2, 2, 6, 0, 4, 1, 3, 5, 1, 7, 3,. Calculate (a) the mean speed and (b) the rms speed.

[(a) 3.25 m/s, (b) 3.86 m/s]

(viii) What is the rms speed of nitrogen molecules contained in an 8.0 m^3 volume at 2.1 atm if the total amount of nitrogen is 1300 mol ?

[374.68 m/s]

2.8 Degrees of Freedom and Equipartition of Energy

In previous section we've derived a relation among temperature and average translational kinetic energy of a gas molecule as

$$E_T = \frac{1}{2} m' v_{rms}^2 = \frac{3}{2} kT$$

Initially we have discussed that if a molecule has v_x , v_y and v_z , average velocity components in three dimensional co-ordinate system,

We have

$$v_{rms}^2 = v_x^2 + v_y^2 + v_z^2$$

As for a single molecule in space the three directions x , y and z are identical in all respect, we can assume that the average values of v_x , v_y and v_z must be same, hence

$$v_x^2 = v_y^2 = v_z^2 = \frac{v_{rms}^2}{3}$$

Thus, the average kinetic energies of a gas molecule in the three directions respectively can be given as

$$\frac{1}{2} m v_x^2 = \frac{1}{2} m v_y^2 = \frac{1}{2} m v_z^2 = \frac{1}{2} kT \quad \dots (2.64)$$

The three kinetic energies in expression of equation-(2.64) are the three parts of total kinetic energy of a gas molecule used corresponding to the three velocity components in the three directions in space.

Thus a gas molecule uses its total translational kinetic energy in three ways of translations. These ways are called translational degrees of freedom. Here each degree of freedom corresponds to the ability of a molecule to participate in a one dimensional motion which contributes to the total mechanical energy of that molecule. Since there are three spatial directions in which the molecule can move, a gas molecule has three translational degrees of freedom and from equation-(2.64) it can be seen that energy associated with each degree of freedom is $\frac{1}{2} kT$. It is further observed that energy associated with each degree of freedom whether it is due to translational motion, rotational motion or any type of motion is always $\frac{1}{2} kT$, irrespective of type of motion. This is called as law of equipartition of energy, stated as

“The total energy of an ideal gas molecule is distributed equally among all of its degrees of freedom and energy associated with each degree of freedom is $\frac{1}{2} kT$.”

2.9 Internal Energy of a Gas

Law of equipartition of energy states that the total energy of a gas molecule is equally divided among all of its degrees of freedom or the number of ways in which the molecule can contribute to its mechanical energy, and energy in each contribution type is $\frac{1}{2} kT$.

Thus if a gas molecule has f degrees of freedom or it can move in f number of ways, the total energy of that molecule can be given as

$$E = f \times \frac{1}{2} kT = \frac{f}{2} kT \quad \dots (2.65)$$

If an ideal gas having f degrees of freedom and have n moles in a container than total energy of this gas is given as

$$U = \frac{f}{2} kT \times n \times N_{AV}$$

$$\text{or} \quad U = \frac{f}{2} n R T \quad [\text{As } k N_{AV} = R] \quad \dots (2.66)$$

The expression in equation-(2.66) gives the sum of total kinetic energies of all types of motion of all the molecules of a gas, called as *total internal energy* of a gas. Our initial assumption for an ideal gas was, its molecules do not interact thus no potential energy exist for gas molecules thus energy given by equation-(2.66) is the net amount of energy a gas can contain i.e. the internal energy of gas and it depends only on absolute temperature of the gas.

2.9.1 Degrees of Freedom and Internal Energy For Different Types of Gases

Monoatomic Gases

The molecule of a monoatomic gas is just a single atom. It can have several types of motion shown in figure-2.29.

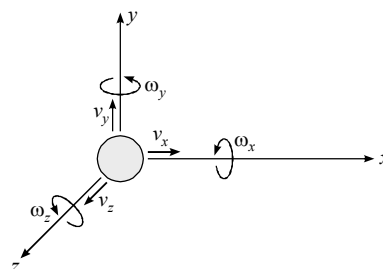


Figure 2.29

A single atom, can translate in x , y and z direction with speeds v_x , v_y , and v_z respectively as discussed earlier and hence has three translational degrees of freedom and energies in these three degrees of freedom can be written as $\frac{1}{2}mv_x^2$, $\frac{1}{2}mv_y^2$ and

$\frac{1}{2}mv_z^2$ respectively. Similar to this translational motion the atom can also spin about the three co-ordinate axes as shown with respective angular speeds ω_x , ω_y and ω_z . Thus this gas molecule must also has three rotational degrees of freedom and energy in these three degrees of freedom can be given as

$\frac{1}{2}I\omega_x^2$, $\frac{1}{2}I\omega_y^2$ and $\frac{1}{2}I\omega_z^2$. Here I is the moment of inertia of the gas molecule. According to our initial assumption for an ideal gas, we neglect the size of a gas molecule. If size or radius of an atom is neglected, we must take its moment of inertia negligible which is proportional to the square of its radius. Thus in a monoatomic gas molecule we can consider that no energy is contributed by its rotational motion hence a monoatomic gas molecule has only three degrees of freedom and all these are called translational degrees of freedom.

Thus total internal energy of n moles of a monoatomic gas at a temperature T is given as

$$U = \frac{f}{2} n R T = \frac{3}{2} n R T \quad \dots (2.67)$$

Diatomic Gases

The molecule of a diatomic gas consists of two atoms connected by a bond or interatomic forces between the two atoms. Figure-2.30 shows a rough sketch of such an atom. In free space this atom is free to move in the three dimensional co-ordinate system, hence has three velocity components along the three axes shown in figure-2.30 thus it has three translational degrees of freedom. In fact all type of molecules, irrespective of its atomic configuration has three translational degrees of freedom in space. Here also if we consider the rotational motion of the molecule about x , y and z axis from figure-2.30 we can see that the rotation of this molecule about y and z axis in this situation is significant as the bond length between the two atoms can not be neglected however, we are ignoring the radii of individual atoms. Masses of individual atoms are concentrated at their position thus moment of inertia of this molecule about y and z axis is a significant value but when we consider the rotation of this molecule about x -axis in the coordinate system shown in figure-2.30, its moment of inertia is again negligible thus rotation of the molecule about x -axis does not contribute any energy to the total energy of molecule so here we can state that a diatomic gas molecule has two rotational degrees of freedom in addition to three translational degrees of freedom. Thus in general, a diatomic gas molecule has five degrees of freedom.

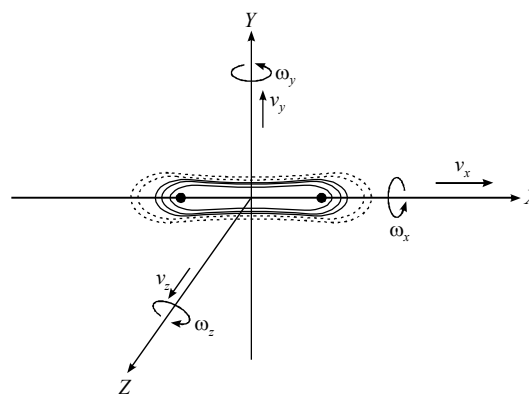


Figure 2.30

Some times vibrational motion of molecules can also contribute to the total mechanical energy of a gas molecule as the molecular bonds are not rigid, they can stretch bend, may behave like a spring as shown in figure-2.30. This result lead to additional degrees of freedom and as well as energy. For most diatomic gases, however the vibrational motion does not contribute appreciably to the total energy of a gas molecule. The reason for this involves the concept of quantum mechanics and beyond the scope of this book. Briefly, vibrational energy of an atom can change only in some finite steps in a quantized manner. If the energy change of the first step is much larger then the energy possessed by most molecules, then nearly all the molecules remain in the minimum energy state of motion and so at low temperature, changing the temperature does not change their average vibrational energy appreciable and do not contribute any energy change to the total mechanical energy of a gas molecule. At low temperature the vibrational degrees of freedom are said to be frozen out and not included in total energy calculation and changes of energy. But at very high temperatures like 5000 K and above the atoms of the diatomic gas starts vibrating at their local position appreciably and bond behaves like a spring as shown in figure-2.31 and then after we include two additional vibrational degrees of freedom for a diatomic gas.

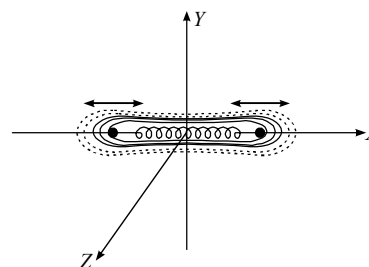


Figure 2.31

According to quantum mechanical properties, rotational energy of a molecule also changes in finite steps, but these steps are usually very small. Thus the freezing out of rotational degrees of freedom is very rare for example in hydrogen gas it is observed

practically that at very low temperature about 50 K or below only translational energy is at a significant value and rotational energies are almost negligible and so in laboratory calculations only three translational degrees of freedom are used for H_2 gas at temperatures below 50 K temperature. Thus total mechanical energy or internal energy of an ideal diatomic gas at a temperature T is given as

$$U = \frac{f}{2} n R T$$

At low temperatures $f = 5$, thus

$$U = \frac{5}{2} n R T \quad \dots (2.68)$$

At very high temperatures $f = 7$, thus

$$U = \frac{7}{2} n R T \quad \dots (2.69)$$

Polyatomic Gases

A polyatomic gas molecule should also have three translational degrees of freedom and the number of rotational degrees of freedom depending on the geometry of the molecule. If the molecule is a linear one like CO_2 then it has only two rotational degrees of freedom due to the obvious reason discussed in previous section, and if it is a non linear molecule like H_2O , CH_4 etc. then it has three rotational degrees of freedom as about all three co-ordinate axes, some significant moment of inertia exists. Number of vibrational degrees of freedom in complex polyatomic molecules varies in different ways. There is no simple theoretical way to calculate the exact number of active vibrational degrees of freedom in a polyatomic molecule.

Thus the total internal energy of n moles of a polyatomic gas at a temperature T is given as

For a linear molecule

$$U = \frac{f}{2} n R T = \left(\frac{5+x}{2} \right) n R T \quad \dots (2.70)$$

Where x are the number of vibrational degrees of freedoms

For a non linear molecule which has x number of vibrational degrees of freedom

$$U = \frac{f}{2} n R T = \left(\frac{6+x}{2} \right) n R T \quad \dots (2.71)$$

2.9.2 Equivalent Degrees of Freedom For a Gaseous Mixture

We know if two substances at same temperature are connected or mixed, they do not exchange any thermal energy and the temperature of mixture remains same.

If N gases with degrees of freedom $f_1, f_2, f_3 \dots f_N$ are mixed with $n_1, n_2, n_3 \dots n_N$ moles at same temperature T then their total internal energy before mixing can be given as

$$U_i = \frac{f_1}{2} n_1 R T + \frac{f_2}{2} n_2 R T + \dots + \frac{f_N}{2} n_N R T \quad \dots (2.72)$$

If after homogeneous mixing, for analytical purpose we assume f_{eq} are the number of degrees of freedom for the mixture then after mixing the internal energy of this mixture can be given as

$$U_f = \frac{f_{eq}}{2} (n_1 + n_2 + \dots n_N) R T \quad \dots (2.73)$$

As no energy loss is taking place during mixing of gases, we have from equation-(2.72) and (2.73)

$$U_i = U_f$$

$$\text{or} \quad \frac{f_1}{2} n_1 R T + \frac{f_2}{2} n_2 R T + \dots + \frac{f_N}{2} n_N R T$$

$$= \frac{f_{eq}}{2} (n_1 + n_2 + \dots n_N) R T$$

$$\text{or} \quad f_{eq} = \frac{f_1 n_1 + f_2 n_2 + \dots f_N n_N}{n_1 + n_2 + \dots n_N} \quad \dots (2.74)$$

2.9.3 Mixing of Gases at Constant Volume

When some gases at different temperature are mixed at constant volume in a thermally insulated vessel, the total internal energy of all the gases remains constant. For example if N gases with $n_1, n_2, \dots n_N$ moles at temperature $T_1, T_2 \dots T_N$ are mixed in a container and if the gases have degrees of freedom $f_1, f_2, \dots f_N$, then the total internal energy of gases before mixing is

$$U_i = \frac{f_1}{2} n_1 R T_1 + \frac{f_2}{2} n_2 R T_2 + \dots + \frac{f_N}{2} n_N R T_N \quad \dots (2.75)$$

If after mixing temperature of mixture become T_f then the total internal energy of gas after mixing of gases is

$$U_f = \frac{f_1}{2} n_1 R T_f + \frac{f_2}{2} n_2 R T_f + \dots + \frac{f_N}{2} n_N R T_f \quad \dots (2.76)$$

As total internal energy of the gaseous mixture must remains constant thus we have from equation-(2.75) and (2.76),

$$U_i = U_f$$

$$\text{or} \quad \frac{f_1}{2} n_1 R T_f + \frac{f_2}{2} n_2 R T_f + \dots + \frac{f_N}{2} n_N R T_f$$

$$= \frac{f_1}{2} n_1 R T_1 + \frac{f_2}{2} n_2 R T_2 + \dots + \frac{f_N}{2} n_N R T_N$$

$$\text{or} \quad T_f = \frac{f_1 n_1 T_1 + f_2 n_2 T_2 + \dots f_N n_N T_N}{f_1 n_1 + f_2 n_2 + \dots f_N n_N} \quad \dots (2.77)$$

Here equation-(2.77) can be used to find the final temperature of a gaseous mixture of different gases after mixing in a thermally insulated container.

Illustrative Example 2.22

Calculate (a) the average kinetic energy of translation of an oxygen molecule at 27°C, (b) the total kinetic energy of an oxygen molecule at 27°C, (c) the total kinetic energy in joule of one mole of oxygen at 27°C. Given Avogadro number = 6.02×10^{23} and Boltzmann's constant = 1.38×10^{-23} J/molecule-K.

Solution

(a) An oxygen molecule has three translational degrees of freedom, thus the average translational kinetic energy of an oxygen molecule at 27°C is given as

$$E_T = \frac{3}{2} kT = \frac{3}{2} \times 1.38 \times 10^{-23} \times 300$$

$$= 6.21 \times 10^{-21} \text{ J/mole}$$

(b) An oxygen molecule has total five degrees of freedom, hence its total kinetic energy is given as

$$E_T = \frac{5}{2} kT = \frac{5}{2} \times 1.38 \times 10^{-23} \times 300$$

$$= 10.35 \times 10^{-21} \text{ J/mole}$$

(c) Total kinetic energy of one mole of oxygen is its internal energy, which can be given as

$$U = \frac{5}{2} nRT$$

$$= \frac{5}{2} \times 1 \times 8.314 \times 300$$

$$= 6235.5 \text{ J/mole}$$

Illustrative Example 2.23

15 gm of nitrogen is enclosed in a vessel at temperature $T = 300$ K. Find the amount of heat required to double the root mean square velocity of these molecules.

Solution

We know *rms* speed of gas molecules is given as

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

As v_{rms} is directly proportional to $\sqrt{T}$, to double *rms* speed, temperature must be raised four times to $T_f = 1200$ K.

As the gas is enclosed in a container, it can not expand or it can not lose any energy to external system. Its initial internal energy is given as

$$U_i = \frac{f}{2} nRT = \frac{5}{2} nRT \text{ [As for a diatomic gas } f=5]$$

$$= \frac{5}{2} \times \frac{15}{28} \times 8.314 \times 300$$

$$= 3340.45 \text{ joule}$$

Finally when gas temperature is raised to 1200 K, its internal energy is given as

$$U_f = \frac{5}{2} nRT_f$$

$$= \frac{5}{2} \times \frac{15}{28} \times 8.314 \times 1200$$

$$= 13361.6 \text{ joule}$$

Increase in internal energy or amount of heat supplied is

$$\Delta Q = U_f - U_i = 10021.35 \text{ joule}$$

Illustrative Example 2.24

1 gm of helium having *rms* velocity of molecules 1000 m/s and 4 gm of oxygen having *rms* velocity of molecules 1000 m/s are mixed in a container which is thermally isolated. What are the *rms* velocities of helium and oxygen molecules after equilibrium is attained?

Solution

Given that *rms* velocity of He molecules is 1000 m/s. If gas is at temperature T , we have

$$v_{rms} = \sqrt{\frac{3RT}{M_1}}$$

or

$$1000 = \sqrt{\frac{3 \times 8.314 \times T_1}{2 \times 10^{-3}}}$$

or

$$T_1 = \frac{2 \times 10^{-3} \times 10^6}{3 \times 8.314} = 80.186 \text{ K}$$

It is given that *rms* velocity of oxygen molecules is also 1000 m/s. If temperature of this gas is T_2 , we have

$$v_{rms} = \sqrt{\frac{3RT_2}{M_2}}$$

or
$$1000 = \sqrt{\frac{3 \times 8.314 \times T_2}{32 \times 10^{-3}}}$$

or
$$T_2 = \frac{32 \times 10^{-3} \times 10^6}{3 \times 8.314} = 1282.976 \text{ K}$$

It is given that 1 gm of He and 4 gm of O_2 is mixed. If their number of moles are n_1 and n_2 then

$$n_1 = \frac{1}{2} \text{ and } n_2 = \frac{4}{32} = \frac{1}{8}$$

We know that gases at different temperature are mixed at constant volume (or in a container), the total internal energy of system remains constant before and after mixing. If in this case final temperature of mixture is T_f then we have

$$T_f = \frac{f_1 n_1 T_1 + f_2 n_2 T_2}{f_1 n_1 + f_2 n_2}$$

Here we have $f_1 = 3$, $f_2 = 5$, $n_1 = \frac{1}{2}$, $n_2 = \frac{1}{8}$, $T_1 = 80.186$ and $T_2 = 1282.76$

Thus final temperature of mixture is given as

$$\begin{aligned} T_f &= \frac{3 \times \frac{1}{2} \times 80.186 + 5 \times \frac{1}{8} \times 1282.76}{3 \times \frac{1}{2} + 5 \times \frac{1}{8}} \\ &= \frac{120.28 \times 801.725}{2.125} = 433.89 \text{ K} \end{aligned}$$

Thus final *rms* velocity of He gas molecules is

$$\begin{aligned} v_{rms} &= \sqrt{\frac{3RT_f}{M_1}} \\ &= \sqrt{\frac{3 \times 8.314 \times 433.89}{2 \times 10^{-3}}} = 2326.17 \text{ m/s} \end{aligned}$$

Final *rms* velocity of O_2 gas molecules is

$$\begin{aligned} v_{rms} &= \sqrt{\frac{3RT_f}{M_2}} \\ &= \sqrt{\frac{3 \times 8.314 \times 433.89}{32 \times 10^{-3}}} = 581.54 \text{ m/s} \end{aligned}$$

Illustrative Example 2.25

A cubical vessel of side 1 m contains one mole of nitrogen at a temperature of 300 K. If the molecules are assuming to move with the *rms* velocity (a) find the number of collisions per second which the molecules may make with the wall of the vessel, (b) further if the vessel now thermally insulated moved with a constant speed v and then suddenly stopped and this results in rise of temperature by 2°C , find v .

Solution

Given that volume of container = 1 m^3

Temperature of gas = 300 K

Amount of gas = 1 mole

(a) If n_0 be the number of molecules per unit volume, we have

$$\begin{aligned} n_0 &= 1 \text{ mole} \\ &= 6.023 \times 10^{23} \text{ m}^{-3} \end{aligned}$$

The *rms* velocity of molecules at 300 K temperature is

$$\begin{aligned} v_{rms} &= \sqrt{\frac{3RT}{M}} \\ &= \sqrt{\frac{3 \times 8.314 \times 300}{28 \times 10^{-3}}} \\ &= 516.75 \text{ m/s} \end{aligned}$$

We know that number of collisions per second per square meter of wall is given by

$$\begin{aligned} N &= \frac{n_0 v}{6} \\ &= \frac{6.023 \times 10^{23} \times 516.75}{6} \\ &= 5.187 \times 10^{25} \text{ s}^{-1} \text{ m}^{-2} \end{aligned}$$

(b) When container is moving at speed v , the kinetic energy in the nitrogen molecules is

$$\begin{aligned} E_k &= \frac{1}{2} M v^2 \\ &= \frac{1}{2} \times 28 \times 10^{-3} \times v^2 \\ &[\text{Mass of 1 mole of gas} = 28 \times 10^{-3} \text{ kg}] \end{aligned}$$

When the container is suddenly stopped, this kinetic energy is transformed into thermal energy and increases the internal energy of gas as container is insulated. If temperature increment of gas is ΔT , rise in its internal energy is

$$\Delta U = \frac{f}{2} n R \Delta T$$

or

$$\Delta U = \frac{5}{2} n R \Delta T$$

or

$$= \frac{5}{2} \times 1 \times 8.314 \times 2$$

or

$$= 41.57 \text{ joule}$$

From energy conservation, we have

$$\frac{1}{2} \times 28 \times 10^{-3} \times v^2 = 41.57$$

$$\text{or } v = \sqrt{\frac{41.57 \times 2}{28 \times 10^{-3}}} \\ = 54.5 \text{ m/s}$$

Illustrative Example 2.26

An adiabatic vessel contains $n_1 = 3$ mole of a diatomic gas. Moment of inertia of each molecule is $I = 2.76 \times 10^{-46} \text{ kg m}^2$ and root mean square angular velocity is $\omega_0 = 5 \times 10^{12} \text{ rad/sec}$. Another adiabatic vessel contains $n_2 = 5$ mole of a monatomic gas at a temperature 470 K . Assume gases to be ideal, calculate root mean square angular velocity of diatomic molecules when the two vessels are connected by a thin tube of negligible volume. Boltzmann constant $k = 1.38 \times 10^{-23} \text{ J/molecule}$.

Solution

We know according to law of equipartition of energy, each gas molecule has $\frac{1}{2} kT$ energy associated with each of its degrees of freedom. As a diatomic gas molecule has two rotational degrees of freedom, its total rotational energy must be $2 \times \frac{1}{2} kT = kT$.

If initial temperature of diatomic gas is T_1 , we have

$$\frac{1}{2} I \omega_{rms}^2 = kT_1$$

$$\text{or } T_1 = \frac{I \omega_{rms}^2}{2k} \\ = \frac{(2.76 \times 10^{-46})(5 \times 10^{12})^2}{2 \times 1.38 \times 10^{-23}} \\ T_1 = 250 \text{ K}$$

When the two vessels containing diatomic and monoatomic gases are connected, these gases exchange this thermal energy but no energy is lost to surrounding as vessels are adiabatic. Thus this mixing of gases takes place at constant volume, total internal energy of system remains constant. If T_f be the final temperature of the system, we have

$$T_f = \frac{f_1 n_1 T_1 + f_2 n_2 T_2}{f_1 n_1 + f_2 n_2} \quad \dots (2.78)$$

Here for diatomic gas

$$f_1 = 5, n = 3 \text{ and } T_1 = 250 \text{ K}$$

For monoatomic gas

$$f_2 = 3, n_2 = 5 \text{ and } T_2 = 470 \text{ K}$$

Thus from equation-(2.78)

$$\text{or } T_f = \frac{5 \times 3 \times 250 + 3 \times 5 \times 470}{5 \times 3 + 3 \times 5} \\ T_f = 360 \text{ K}$$

Thus final mixture of the two gases is at temperature 360 K . If final *rms* angular velocity of diatomic gas molecules is ω_{rmsf} , according to law of equipartition of energy, we have

$$\frac{1}{2} I \omega_{rmsf}^2 = kT \\ \text{or } \omega_{rmsf} = \sqrt{\frac{2kT}{I}} \\ = \sqrt{\frac{2 \times 1.38 \times 10^{-23} \times 360}{2.76 \times 10^{-46}}} \\ \omega_{rmsf} = 6 \times 10^{12} \text{ rad/s}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Kinetic Theory of Gases

Module Number - 26 to 32

Practice Exercise 2.4

(i) The ratio of translational and rotational kinetic energy at 100 K temperature is $3 : 2$. Find the internal energy of one mole gas at this temperature.

[2075 J]

(ii) The temperature of a gas consisting of rigid diatomic molecules is $T = 300 \text{ K}$. Calculate the angular root mean square velocity of a rotating molecule if its moment of inertia is equal to $I = 2.1 \times 10^{-39} \text{ kg m}^2$.

[$1.985 \times 10^9 \text{ rad/s}$]

(iii) A nonlinear polyatomic gas at 650 K has molecules with 12 active degrees of freedom. (a) Evaluate the average molecular mechanical energy. (b) What is the value of v_{rms} ? Take mass of one molecule of gas $m = 1.3 \times 10^{-25} \text{ kg}$. (c) How many vibrational degrees of freedom are active?

[(a) $5.382 \times 10^{-20} \text{ J}$ (b) 455.03 m/s (c) 6]

(iv) The first allowed excited state of hydrogen atom is 10.2 eV above its lowest energy (ground) state. To what temperature should hydrogen gas be raised so that inelastic collisions may excite an appreciable number of atoms to their first excited state?

[$7.88 \times 10^4 \text{ K}$]

(v) A cylindrical container of volume V is divided in two equal parts by a fixed diathermic partition. Identical gas is filled in the two parts at initial pressure and temperature p_1, T_1 and p_2, T_2 ($T_2 > T_1$) respectively. After a long time when the two gases are in thermal equilibrium, find the final temperature and pressure of the gases in the two parts.

$$\left[\frac{p_1 T_2 (p_1 + p_2)}{p_1 T_2 + p_2 T_1}, \frac{p_2 T_1 (p_1 + p_2)}{p_1 T_2 + p_2 T_1}, \frac{T_1 T_2 (p_1 + p_2)}{p_1 T_2 + p_2 T_1} \right]$$

(vi) Find the amount of heat supplied by gas in one part to that in other part in previous question if gas taken is monoatomic.

$$\left[\frac{3 p_1 p_2 (T_2 - T_1) V}{2 (p_1 T_2 + p_2 T_1)} \right]$$

2.10 Path of Molecules and Mean Free Path

If we find the mean speed of a molecules of air at room temperature 25°C , we get

$$\begin{aligned} v_{av} &= \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8 \times 8.314 \times 298}{3.14 \times 29 \times 10^{-3}}} \\ &= 466.54 \text{ m/s} \end{aligned}$$

It shows that in general average speed of a common gas molecule or air molecules is comparable or more than the speed of sound. In this situation a question arises, why the smell of a scent appears to spread in a room at much slower than this. If the speed of the molecules of the scent were of the order of about 500 m/s then the molecules of the scent should have taken almost no time to reach any corner of the room. The reason behind this is the explanation of pressure by kinetic theory assumptions. In assumptions of kinetic theory for an ideal gas we've ignored the size of gas molecules. If all the molecules of a gas are assumed point sized, they will never collide during their motion but in fact molecules are finite in size so they repeatedly collide with the neighbouring molecules randomly.

A molecule in its path undergoes a number of collisions so the path traversed by it is not a straight line but somewhat zigzag as shown in figure-2.32. As a result of this zigzag motion, its net displacement per unit time is comparatively small. So the scent takes more time to reach a place than the time predicted by kinetic theory. Between two successive collisions a molecule however travels in a straight line with uniform velocity. As shown in figure-2.32, the motion is random thus the distance travelled by the molecule between two successive collisions are not always equal. Due to this we use a parameter to describe this random motion. This is mean free path of the molecules, denoted by λ . As the name implies it is the average distance traversed by a molecule between two successive collisions. We expect λ to

vary inversely with the molecular density N/V , the number of molecules per unit volume. The larger the molecular density is, the more collisions there should be in the dense gas and the smaller the mean free path. We also expect λ to vary inversely with the size of molecules. If the molecules were point sized, they would never collide and the mean free path would be infinite. Thus the larger the molecules are, the smaller the mean free path.

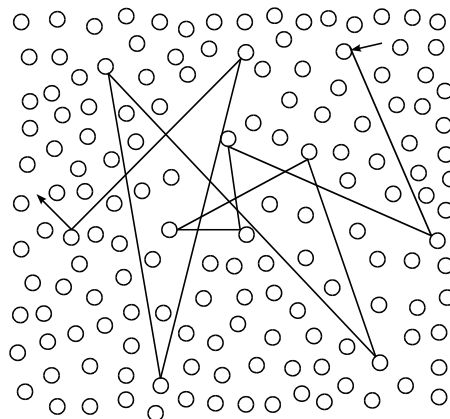


Figure 2.32

The exact expression for mean free path is turned out to be

$$\lambda = \frac{1}{\sqrt{2} m d^2 (N/V)} \quad \dots (2.79)$$

Here m is the mass of molecules and d is the molecular diameter.

From gas law we have $PV = nRT$ or $PV = NkT$ as $\frac{R}{N_{AV}} = k$ if N are the total number of molecules in a gas, thus equation-(2.79) now become

$$\lambda = \frac{kT}{\sqrt{2} m d^2 P} \quad \dots (2.80)$$

2.11 Vander Walls Gas Equation

In the beginning of chapter we've discussed the assumptions of kinetic theory for ideal gases. In these assumptions the volume of gas molecules as well as the interaction between them are neglected. But in practical or real gases this is not true. Due to the size of the gas molecules, actual volume available for motion of molecules is less than the container in which gas is enclosed. Similarly the pressure which the molecules exert on container walls, we've derived in equation-(2.50) should also be different due to the interaction between the neighbouring molecules during their collisions.

The ideal gas equation in equation-(2.7) can be used for different ideal gases but not for real gases due to the above reason. This ideal gas law can be modified for real gases and by introducing

some appropriate corrections, we can develop another equation of state for a real gas. This equation was developed by Dutch physicist J.D. Vander Walls. The equation is named vander walls gas equation after him and is given as

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT \quad \dots (2.81)$$

Here a and b are empirical constants, different for different gases. Roughly b is the volume of one mole of molecules, thus the total volume of molecules in n moles of a gas is then nb , and the total volume available for the molecules to move around in the gas is $(V - nb)$. The constant a depends on the attractive intermolecular forces, which reduces the pressure of gas for given values of n , V and T by pulling the molecules together as they push the walls of container during collisions. The decrease in pressure is proportional to the number of molecules per unit volume in a layer near the wall which are colliding and exerting pressure on walls and is also proportional to the number per unit volume in the next layer beyond the wall which are attracting the first layer molecules. Hence the decrease in pressure due to intermolecular forces is proportional to $\frac{n^2}{V^2}$.

When a gas is at very low pressure, ($\frac{n}{V}$) for the gas is very small or we can say that the gas is dilute, the separation between the gas molecules is large, the correlations in the vander walls equation become insignificant and equation-(2.81) reduces to ideal gas equation. Some times at very high temperature when kinetic energy of gas molecules is very high, the effect of molecular interaction is negligible. Thus we can say that at very low pressure or at very high temperature real gases may behave like ideal gases.

2.12 Diffusion

If we care fully place a drop of ink in a glass of water, we'll find that the ink colour spreads throughout the water. After a long time eventually the colour will become uniform. This mixing occurs because of the random movement of the molecules and is called diffusion. Diffusion also occurs in gases. Common examples of diffusion is the spreading of smell of a perfume in a room or smoke diffusing in air, although convection often plays a greater role in this spreading than does diffusion. In all cases the diffusing substance moves from a region where its concentration is high to one where its concentration is low.

The simplest explanation of diffusion is on the basis of kinetic theory and the random motion of molecules. We take a simple example to understand this. Consider a long tube of cross-sectional area A as shown in figure-2.33. Containing molecules in a higher concentration on the left than on the right. According to the kinetic theory molecules are in brownian motion. If we consider a small elemental length Δx of the tube then molecules from both regions left and right cross into this section of length Δx due to their random motion. Since there is higher concentration on left side, more molecules cross into this elemental section from left side then to the right side. Thus there is a net flow of molecules from left to right, from high concentration toward low concentration. When after sometime the two side concentration become equal, the flow still continues but equal from both the sides thus there is no net flow of molecules from either side when concentration on both sides become equal.

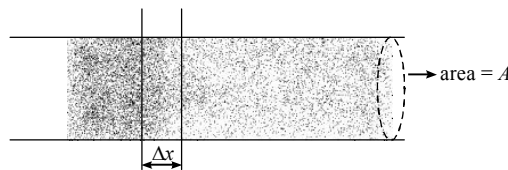


Figure 2.33

Detailed mathematical analysis of diffusion is beyond the scope of this book. At elementary level we discuss the Adolf formula which gives the average rate of diffusion from one region of concentration C_1 to another region of concentration C_2 separated by a central section of width Δx , as discussed in figure-2.33, is given as

$$J = DA \frac{C_1 - C_2}{\Delta x} \quad \dots (2.82)$$

Here $(C_1 - C_2)/\Delta x$ is the change in concentration per unit distance and some times written as dC/dx for mathematical analysis, and is called as concentration gradient.

Here D is the proportionality constant called diffusion constant and the above equation-(2.82) is called as "Adolf formula" or "Adolf Fick's Law of Diffusion". Above equation is not only valid for a gas diffusing in itself called self diffusion but also applicable for a gas diffusing in a second gas. If the concentrations C_1 and C_2 here are in mole/m³ then J is the number of moles passing a given area per second. The diffusion constant D will depend on the properties of the substances involved, and also on the temperature and the external pressure.

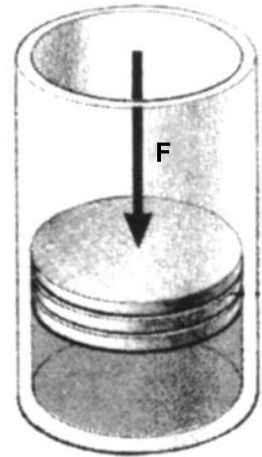
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Thermodynamics Laws & Specific Heats of Gases

FEW WORDS FOR STUDENTS

In this chapter we'll discuss the fundamental laws of thermodynamics. These laws are the basics for us to understand flow of heat and energy as in mechanics the fundamentals to understand motion are energy and momentum conservation laws. These thermodynamics laws are very useful in explaining the physical behaviour of a system when it responds to the flow of heat or thermal energy.

The two laws, first and second laws of thermodynamics completely govern the flow of thermal energy in nature.



CHAPTER CONTENTS

- 3.1 Thermal Equilibrium and Zeroth Law of Thermodynamics
- 3.2 First Law of Thermodynamics
- 3.3 State of a Gas and Indicator Diagram
- 3.4 Molar Specific Heats of a Gas
- 3.5 Different Type of Thermodynamic Processes
- 3.6 Free Expansion of a Gas
- 3.7 Polytropic Process
- 3.8 Second Law of Thermodynamics

COVER APPLICATION



Figure-(a)

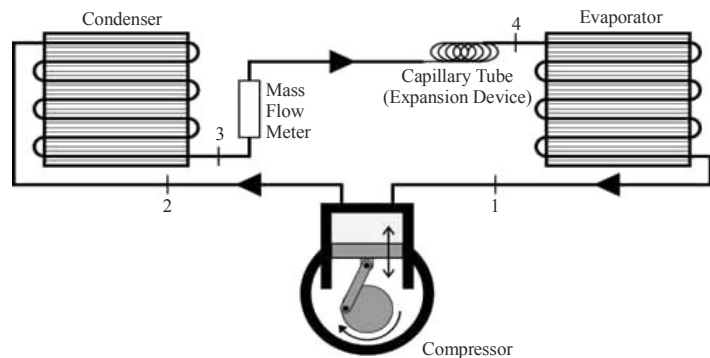


Figure-(b)

Refrigeration, or cooling process, is the removal of unwanted heat from a selected object, substance, or space and its transfer to another object, substance, or space. The main components of the refrigeration system are condenser, evaporator and compressor. Removal of heat lowers the temperature and may be accomplished by use of ice, snow, chilled water or mechanical refrigeration. Figure-(a) is the industrial setup of a refrigeration system and Figure-(b) is the block diagram of different units and assembly of a refrigeration system.

The branch of physics thermodynamics is concerned with the relationships between heat and work. In previous chapters of calorimetry and kinetic theory of gases, we have discussed the measurements of heat or thermal energy in the form of kinetic energy of gas molecules. Thermodynamics is the study of thermal energy in different forms and it is based on two basic laws. First is that you can not get more energy out of a system than you put into it in all forms. That is the basic law of conservation of energy. Second law says that the transfer of energy by heat flow has a direction generally from high temperature region to a low temperature region or in other words we can say that processes in nature are not reversible.

The principles of thermodynamics were developed in eighteenth and nineteenth century and these are the most useful relations established between heat and work which are used very widely in different experiments and researches.

3.1 Thermal Equilibrium and Zeroth Law of Thermodynamics

In previous chapters we've discussed that two objects in thermal contact can exchange heat as long as they are at different temperatures. The hot object cools and colder one gets warm until they reach a common temperature at which no further change takes place. These two objects in this state are said to be in thermal equilibrium.

Based on thermal equilibrium earlier we've defined the zeroth law of thermodynamics as if two bodies are independently in thermal equilibrium with a third body then those two bodies are also in thermal equilibrium with each other. This is called zeroth law of thermodynamics because this is the logical basis of first and second laws of thermodynamics.

In further sections of this chapter we'll discuss about the concept of a thermodynamics system and about energy into or out of the system in the form of heat or work. A thermodynamic system is a collection of objects considered together and rest of the environment is called surrounding of system. A thermodynamic system interacts with its surrounding by exchange of energy in the form of heat transfer or work. As a result of this exchange of energy, the system's internal energy may change. We've discussed that by internal energy we mean the total kinetic energy and potential energy associated with the internal state of atoms composing the system. In addition to internal energy a system may have kinetic and potential energies due to the outside forces acting on system.

3.2 First Law of Thermodynamics

The first law of thermodynamics is based on the idea that energy is neither created nor destroyed in any thermodynamic

system. In the usual formulation of first law, we consider the transfer of heat into a system, the work performed by the system and the change in the system's internal energy.

This law simply states that the total amount of heat supplied to a gaseous system is used in two parts. A part of supplied heat increases the kinetic energy of gas molecule or increases the temperature of system and the other part of supplied heat is used to do work against surrounding.

If dQ is the heat supplied to a gas during a heating process and due to this internal energy of a gas increases by dU and gas does a work dW against surrounding then according to first law of thermodynamics, we have

$$dQ = dU + dW \quad \dots (3.1)$$

This equation-(3.1) is called differential form of first law of thermodynamics. If a gas is heated from an initial state to a final state by a process and if total heat supplied in the heating is ΔQ , total work done by the gas is ΔW and total change in internal energy of the gas is ΔU then we have

$$\Delta Q = \Delta W + \Delta U \quad \dots (3.2)$$

Equation-(3.2) is state form of first law of thermodynamics and is used to relate the heat, work and change in internal energy of gas between initial and final states of a thermodynamics heating process.

3.2.1 Specific Heat Capacities of Gases

In solids and liquids we define specific heat as the amount of heat required for per degree rise in temperature of unit mass of substance. In the same sense we can define specific heat for a gas. Specific heat of a gas can be defined as the amount of heat required for a unit mass of a gas to raise its temperature by one degree. If instead of unit mass we take one mole of gas then it is termed as molar specific heat or molar heat capacity of the gas. It is denoted by C and if n moles of a gas is heated on dQ supply of heat, and if gas temperature increases by dT , then we have

$$dQ = n C dT \quad \dots (3.3)$$

If temperature of gas changes from T_1 to T_2 . Then total heat supplied is

$$Q = \int_{T_1}^{T_2} dQ = \int_{T_1}^{T_2} n C dT = n C (T_2 - T_1) \quad \dots (3.4)$$

In solids and liquids generally specific heat remains constant for a material or varies slightly with temperature. But the case is not same for gases. In gases their specific heat also depends on the way of heating. If the process changes by which heat is supplied to a gas, it will also change the amount of heat required

for a given amount of gas to change its temperature by a specific value. Before discussing the specific heats of a gas, we first discuss the phenomenon of heating a gas or how heat supplied to a gas is used by it.

3.2.2 Heating of a Gaseous System and Work done by a Gas

Figure-3.1(a) shows a cylindrical container in which a gas is enclosed and top of the container is closed by a light piston. Initially piston is in equilibrium as the pressure exerted by gas due to collisions of molecules is balanced by the atmospheric pressure due to air molecules outside. In this state if some heat is supplied to the gas by the burner as shown in figure-3.1 say a small amount dQ heat is supplied to the gas as shown in figure-3.1(b). If due to this dQ amount of heat, the temperature of gas is also slightly increased, say by amount dT . Thus this increase in temperature increases the kinetic energy of moving gas molecules due to which pressure exerted by gas on piston increases and piston will start displacing upward till again the pressure of gas becomes equal to atmosphere pressure. If the piston is displaced up by a distance dx , and this is due to the pressure of gas. We can say that in displacement of piston, work done by the gas is dW , which is given as

$$dW = P_{\text{gas}} S \cdot dx \quad [S = \text{area of piston}]$$

or
$$dW = P_{\text{gas}} dV \quad \dots (3.5)$$

[$dV = Sdx$ is the increase in volume of gas]

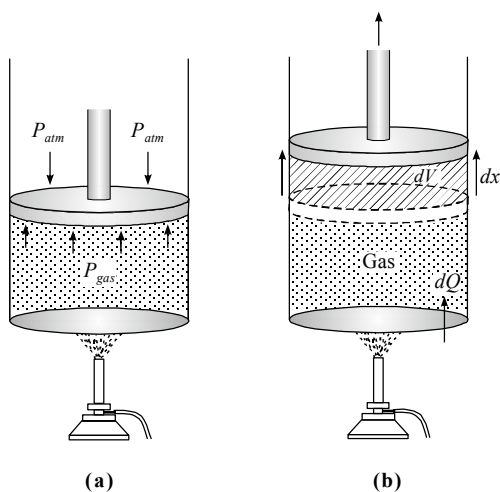


Figure 3.1

If during heating total volume of gas changes from V_1 to V_2 then the net work done in the heating process is

$$W = \int_{V_1}^{V_2} P_{\text{gas}} dV \quad \dots (3.6)$$

Equation-(3.5) gives the net work done in heating a gaseous system. In different cases during heating, pressure of gas may

vary in different ways which results in different total work from integral in equation-(3.6) however the volume of initial and final state remain same.

As discussed, whenever volume of gas increases, we can say the work is done by the gas and when a gas is compressed, we say work is done on the gas by atmosphere or by some external agent responsible for the compression of gas.

3.2.3 Change in Internal Energy of a Gas on Heating

In previous section we've discussed that when heat is supplied to a system, it may be possible that gas expands. If it expands, we say gas does work against surrounding. It means some energy out of supplied heat goes to the surrounding in this work done. Similarly it may be possible that a part of supplied heat increases the kinetic energy of gas molecules. This means the temperature of gas increases. If temperature of gas increases by a small amount dT , we can say that the total internal energy of gas is also increased by a small amount, say dU , which can be given as

$$dU = \frac{f}{2} n R dT \quad \dots (3.7)$$

Similarly if in a case gas temperature decreases, then equation-(3.7) gives the decrement in internal energy and we write this equation with a negative sign on either side of equality.

If total change in temperature of gas is from T_1 to T_2 then total change in its internal energy from initial to final state is

$$\Delta U = \int dU = \int_{T_1}^{T_2} \frac{f}{2} n R dT$$

or
$$\Delta U = \frac{f}{2} n R (T_2 - T_1) \quad \dots (3.8)$$

Illustrative Example 3.1

One mole of an ideal gas is heated from 0°C to 100°C at a constant pressure of 1 atmosphere. Calculate the work done in the process.

Solution

One mole of gas occupies a volume of 22400 cm^3 at 0°C and at 1 atmosphere pressure. Thus, the initial volume of the gas is $V_1 = 22400\text{ cm}^3 = 22400 \times 10^{-6}\text{ m}^3 = 0.0224\text{ m}^3$. The final volume V_2 can be calculated by using gas laws as

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Here

$$T_1 = 0^\circ\text{C} = 273\text{ K}$$

$$T_2 = 100^\circ\text{C} = 373\text{ K}$$

$$\begin{aligned} P &= P_2 = 1 \text{ atmosphere} = 0.76 \text{ m of Hg} \\ &= 0.76 \times 9.8 \times 13600 \text{ Nm}^{-2} \\ &= 1.013 \times 10^5 \text{ Nm}^{-2} \end{aligned}$$

or

$$V_2 = V_1 \frac{T_2}{T_1}$$

$$\text{Thus we have} = 0.0224 \times \frac{373}{273} \text{ m}^3 = 0.0306 \text{ m}^3$$

We know work done in the process can be calculated as

$$\begin{aligned} W &= \int_{V_1}^{V_2} P dV = P \int_{V_1}^{V_2} dV \quad [\text{As } P \text{ is constant}] \\ &= P (V_2 - V_1) \\ &= 1.013 \times 10^5 (0.0306 - 0.0224) \\ &= 830.7 \text{ J} \end{aligned}$$

Here as gas expands, work is done by the gas.

Illustrative Example 3.2

In thermodynamic system internal energy decreases by 400 J while it is doing 250 J of work. What net heat is taken in by the system in the process.

Solution

Here it is given that

$$\Delta U = -400 \text{ J}$$

and

$$W = +250 \text{ J}$$

Now from first law of thermodynamics, we have

$$\begin{aligned} Q &= \Delta U + W \\ &= (-400) + (+250) = -150 \text{ J} \end{aligned}$$

The amount of heat Q is negative thus in the process this thermodynamic system rejects 150 J of heat to the surroundings.

Illustrative Example 3.3

An ideal gas in a cylindrical vessel is confined by a piston at a constant pressure of 10^5 Pa . When $2 \times 10^4 \text{ J}$ of heat is added to it, the volume of gas expands from 0.15 m^3 to 0.25 m^3 . (a) What is the work done by the system in this process. (b) What is the change in internal energy of the system.

Solution

(a) We know that work done by a gas in a process is given as

$$\begin{aligned} W &= \int_{V_1}^{V_2} P dV = P (V_2 - V_1) \\ &\quad [\text{As } P = \text{constant here}] \\ &= 10^5 [0.25 - 0.15] = 10^4 \text{ J} \end{aligned}$$

(b) From first law of thermodynamics, in the process we have

$$Q = \Delta U + W$$

$$\text{Here} \quad 2 \times 10^4 = \Delta U + 10^4$$

$$\text{or} \quad \Delta U = 10^4 \text{ J.}$$

Illustrative Example 3.4

A vertical hollow cylinder contains an ideal gas. The gas is enclosed by a 5 kg movable piston having a cross-sectional area of $5 \times 10^{-3} \text{ m}^2$. Now the gas is heated from 300 K to 350 K and the piston rises by 0.1 m. The piston is now clamped in this position and the gas is cooled back to 300 K. Find the difference between the heat energy added during heating and the heat energy lost during cooling. (1 atmospheric pressure = 10^5 Nm^{-2} and $g = 10 \text{ ms}^{-2}$).

Solution

Given, mass of piston $M = 5 \text{ kg}$, cross-sectional area of piston $A = 5 \times 10^{-3} \text{ m}^2$, $g = 10 \text{ ms}^{-2}$ and atmospheric pressure $P_0 = 10^5 \text{ Nm}^{-2}$.

The initial pressure of the gas in the cylinder is

$P = \text{atmospheric pressure} + \text{pressure due to weight } Mg \text{ of piston}$

$$= P_0 + \frac{Mg}{A} \quad \text{or} \quad P = 10^5 + \frac{5 \times 10}{5 \times 10^{-6}} = 1.1 \times 10^5 \text{ Nm}^{-2}$$

When the piston rises by $x = 0.1 \text{ m}$, the increase in the volume of the gas is

$$\Delta V = Ax = (5 \times 10^{-3}) \times 0.1 = 5 \times 10^{-4} \text{ m}^3$$

Thus work done by the gas is

$$\begin{aligned} W &= \int_{V_1}^{V_2} P dV \\ &= P (V_2 - V_1) \\ &= 1.1 \times 10^5 \times 5 \times 10^{-4} = 55 \text{ J} \end{aligned}$$

If ΔU is the increase in the internal energy during heating, then from the first law of thermodynamics, the heat energy supplied to the gas is given as

$$Q = \Delta U + W = (\Delta U + 55) \text{ joule}$$

Since the piston is clamped, the volume of the gas remains constant during cooling. Hence work done during cooling is zero.

As the gas is cooled back to the initial temperature, the change in the internal energy during cooling is

$$\Delta U' = -\Delta U$$

Thus heat energy lost by the gas during cooling is

$$Q' = \Delta U' + W' = -\Delta U + 0 = -\Delta U$$

The difference between heat energy added during heating and heat energy lost during cooling is

$$\Delta Q = Q - Q' = (\Delta U + 55) - \Delta U = 55 \text{ joule.}$$

Illustrative Example 3.5

Gaseous hydrogen initially at STP in a container of volume $5 \times 10^{-5} \text{ m}^3$ is cooled by 55 K. Find the change in internal energy and amount of heat lost by the gas.

Solution

Initially in standard conditions the gas pressure is

$$P = 10^5 \text{ Pa}$$

Gas temperature is $T = 273 \text{ K}$

Gas volume is $V = 5 \times 10^{-5} \text{ m}^3$

If n moles of gas are there, then from gas law, we have

$$PV = nRT$$

$$\text{or } n = \frac{PV}{RT} = \frac{10^5 \times 5 \times 10^{-5}}{8.314 \times 273} = 0.22 \text{ mole.}$$

Thus change in internal energy in a process is given as

$$\begin{aligned} \Delta U &= \frac{f}{2} nR\Delta T \\ &= \frac{5}{2} \times 0.22 \times 8.314 \times 55 = 251.5 \text{ J.} \end{aligned}$$

As gas is enclosed in a container its volume remains constant during the process thus work done in this process is zero and according to first law of thermodynamics, we have

$$Q = \Delta U = -251.5 \text{ J}$$

Thus the decrease in internal energy is lost by the gas in the form of heat to the surrounding.

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Laws of Thermodynamics

Module Number - 1 to 6

Practice Exercise 3.1

(i) A gas undergoes a change of state during which 100 J of heat is supplied to it and it does 20 J of work. The system is brought back to its original state through a process during which 20 J of heat is rejected by the gas. Find the work done by the gas in the second process.

[60 J]

(ii) A sample of an ideal gas is taken through a process $ABCD$. It absorbs 50 J of energy during the process AB which is an isometric process, no heat during BC and it rejects 70 J of heat during isobaric process CD . It is also given that 40 J of work is done on the gas during the process BC . Internal energy of gas in state A is 1500 J. Find the internal energy of gas in state C .

[1590 J]

(iii) In a gaseous system, a gas expands from 10^{-4} m^3 to $2 \times 10^{-4} \text{ m}^3$ while its pressure remains constant at 10^5 Nt/m^2 . Calculate the amount of heat absorbed by the gas in the expansion. $[\gamma = 1.67]$

[24.92 J]

(iv) At atmospheric pressure, $m = 1 \text{ g}$ of water boils into steam which occupies a volume of $V_g = 1671 \text{ cm}^3$. The latent heat of vapourisation of $L = 539 \text{ cal/g}$ and specific volume of water $V_L = 1 \text{ cm}^3$. Find the increase in internal energy of the system. Atmospheric pressure is $p_0 = 1.013 \times 10^5 \text{ Pa}$. Take $1 \text{ cal} = 4.2 \text{ J}$

[2094.63 J]

(v) N_2 is confined in a cylindrical vessel with a movable piston exposed to open atmosphere. If 25 kcal of heat is added to it and the internal energy of the gas increases by 8 kcal, find the work done by the gas.

[17 kcal]

(vi) A sample of an ideal diatomic gas is heated at constant pressure. If an amount of 100 J of heat is supplied to the gas, find the work done by the gas.

[28.57 J]

3.3 State of a Gas and Indicator Diagram

When a gas is heated its thermodynamic parameters pressure, volume and temperature changes. These three parameters of a gas at an instant, pressure P , volume V and temperature T , combinely define the state of a gas. Whenever the state of a gas (P , V , T) is changed, we say the gaseous system is undergone a thermodynamic process. The graphical representation of the change in state of a gas by a thermodynamic process is called indicator diagram. Indicator diagram is plotted generally in pressure and volume of a gas. Figure-3.2 shows a general PV indicator diagram. This indicator diagram representing the change of state of a gaseous system from state-1 (P_1 V_1 T_1) to state-2 (P_2 V_2 T_2). Each point on indicator diagram represents a unique state of gas. In the figure-3.2 shown when gas is in its state-1 its pressure, volume and temperature are P_1 , V_1 and T_1 respectively. Now the gas state is to be changed to state-2 with pressure, volume and temperature P_2 , V_2 and T_2 respectively. Figure-3.2 shows three different paths along which the gas can be taken from state-1 to state-2. In path-I, initially volume of gas is kept constant at V_1 and its pressure is increased from P_1 to P_2 then pressure P_2 is kept constant and volume is increased to V_2 . In path-III first pressure is kept constant and volume is increased to V_2 then pressure is kept constant at P_2 and pressure is increased to P_2 . In path-II both pressure and volume are changed simultaneously along the curve shown to attain the desired values of pressure and volume of state-2. Here each path from which a gas is carried from initial to final state is called a thermodynamic process. There are infinite ways or thermodynamic processes by which state of a gas can be changed from given initial to final state. It can also be stated that when a gas is carried from one state to another, there are several adjacent intermediate states exist between initial and final states of the gas. Thus an indicator diagram curve represents the locus of all intermediate states between the two terminal states.

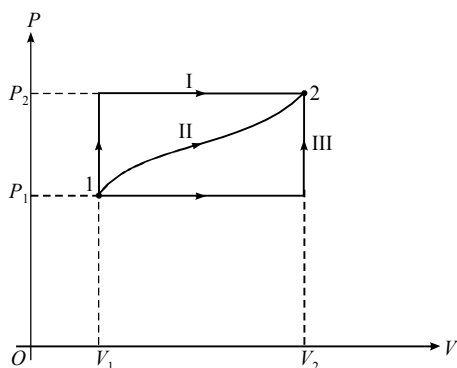


Figure 3.2

3.3.1 Properties of an Indicator Diagram

Figure-3.3 shows a general indicator diagram for a thermodynamic process for heating of a gas from state-1 to state-2. The path of this process is shown in figure-3.3 from initial to final state. Some standard characteristic of a P - V curve plotted for a thermodynamic process are

(i) Each point on an indicator diagram represents a unique state of a gas. In a P - V diagram, each points gives a specific value of pressure and volume of a gas and if number of moles are known, we can get its temperature using gas law. Hence a PV curve is the locus of the states of a gas during a thermodynamic process between initial and final states.

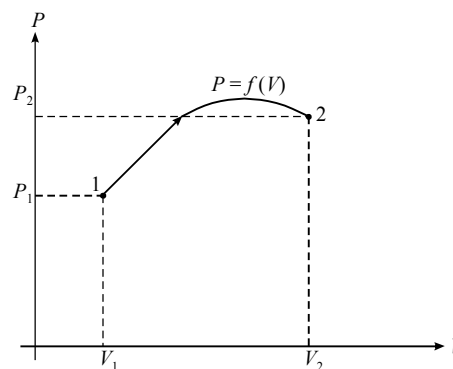


Figure 3.3

(ii) The path of a thermodynamic process on a P - V diagram relates pressure and volume of gas during the heating process. Each curve on P - V diagram has a characteristic function between P and V as $P = f(V)$. This function between pressure and volume is called as process equation or equation of the thermodynamic process between state-1 and state-2.

(iii) In a thermodynamic process, the work done by the gas, we've already discussed, is given as

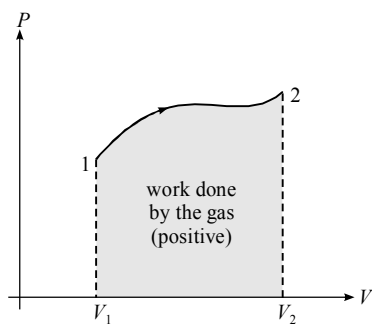
$$W = \int_{V_1}^{V_2} P dV \quad \dots (3.9)$$

If the process equation is known we can solve the integral in equation-(3.9) as

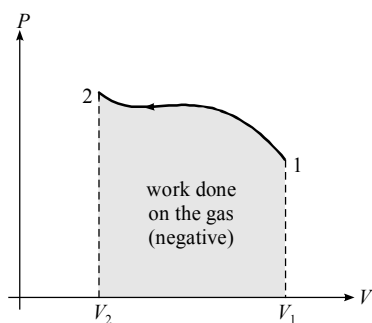
$$W = \int_{V_1}^{V_2} f(V) dV \quad \dots (3.10)$$

This equation-(3.10) gives the work done by the gas during the process between state-1 and state-2 and mathematically this equation gives the area under the P - V curve on indicator diagram. As shown in figure-3.4(a) and 3.4(b), when final volume of gas is more then its initial volume, the area below the P - V curve gives the work done by the gas as it expands and when final volume of gas is lesser then its initial volume the curve is

drawn from right to left as shown in figure and the area below the P - V curve gives the work done on the gas and is taken negative in numerical calculations.



(a)



(b)

Figure 3.4

3.3.2 State Variable and Path Variables

When a gas is taken from its initial state (P_1, V_1, T_1) to a final state (P_2, V_2, T_2), out of the three thermodynamic variables, heat supplied Q and work done by the gas W , depend on the path along which the gas is taken from initial to final state. Dependency on path means it depends on the function $P = f(V)$ or the way how pressure and volume of gas are related during the process. Thus both Q and W are called path variables. Unlike to these internal energy of a gas only depends on its temperature or only on the initial and final state temperatures (T_1 and T_2) of the gas irrespective of how gas is heated to change its temperature from T_1 to T_2 . Thus internal energy of a gas is called state variable.

If a gas is taken from a state-1 to another state-2 by several different paths and the heat supplied and work done in the respective paths are (Q_1, W_1) , (Q_2, W_2) , (Q_3, W_3) ... and so on then according to first law of thermodynamics, we have

$$\Delta U_{12} = Q_1 - W_1 = Q_2 - W_2 = \dots \quad \dots (3.11)$$

3.3.3 Cyclic Processes

A thermodynamic process in which initial and final state are same are called cyclic processes. Figure-3.5 shows an indicator

diagram of a cyclic process. A gas in state-1 is heated and is expand to change its state to state-2 shown in figure along path-I. Now it is compressed along path-II and taken back to its initial state to restore its pressure, volume and temperature to its initial values.

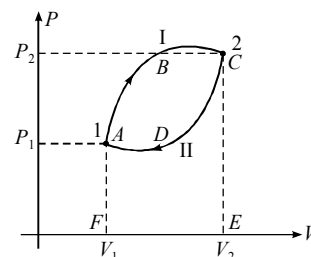
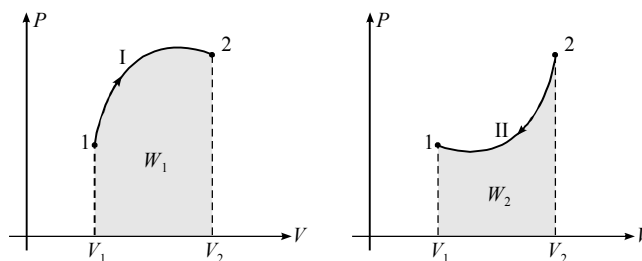


Figure 3.5

In a cyclic process we can say that no change in internal energy takes place as it is a state function and here also $\Delta T = 0$, thus $\Delta U = 0$. But as heat supplied and work done are path functions, we can numerically obtain these variables easily using first law of thermodynamics for paths I and II respectively.

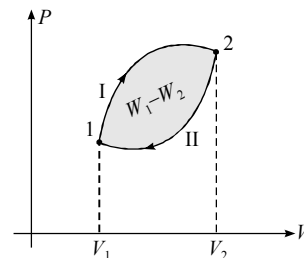


$W_1 =$ work done by the gas

(a)

$W_2 =$ work done on the gas

(b)



$W_1 - W_2 =$ net work done by the gas in cycle

(c)

Figure 3.6

Figure-3.6(a) shows the expansion of gas from state-1 to state-2 along path-I, here during expansion work is done by the gas and is given by the area below the curve as shown in this figure-3.6(a) say it is W_1 . When the gas is taken back to its initial state from state-2, during compression work is done on the gas and it is given by the area below the PV curve of path-II as shown in figure-3.6(b) say it is W_2 . From figure-3.5 it is clear that W_1 is more than W_2 hence in the complete cycle net work is

done by the gas and it is given by the area enclosed by the PV -curve of the complete cycle as shown in figure-3.6(c). Thus net work done by the gas in a cyclic process is

$$W_1 - W_2 = \text{Area enclosed by the } PV\text{-curve of the cyclic process}$$

As discussed initially that in a cyclic process net change in internal energy of gas is zero thus the total work done is equal to the total amount of heat supplied to the gas. In fact during first part of the cycle i.e. during expansion of gas, heat is supplied and work is done by the gas, say heat supplied to the gas is Q_1 . In second part of the cycle i.e. during compression of gas, work is done on the gas and some heat is rejected by the gas to its surrounding say this amount is Q_2 . As total work is done by the gas thus we generally have $Q_1 > Q_2$ so total heat supplied to the gas can be written as

$$\Delta Q = Q_1 - Q_2 \quad \dots (3.12)$$

This must be equal to the total work done by gas as $\Delta U = 0$ in the complete cycle. Thus

$$\Delta Q = Q_1 - Q_2 = \Delta W = W_1 - W_2 \quad \dots (3.13)$$

3.3.4 Positive and Negative Cycles

In previous section we have discussed that net work done by a gas in a cyclic process is equal to the area enclosed by its PV -curve on indicator diagram. We've taken example shown by the curve in figure-3.5 and figure-3.6. If we carefully look on the path of process, it is clockwise. Another cyclic process can also be realized of which PV curve can be anticlockwise. Such a process is shown in figure-3.7. There it can be clearly seen that during expansion the work done by the gas along path-I is less than the work done on the gas during its compression along path-II. Thus in cyclic processes having anticlockwise cycles, net work is done on the gas during the complete cycle and is again given by the area enclosed by the cyclic curve as shown by figure-3.7. Again as no net change in internal energy takes place in a cyclic process, in anticlockwise cycles net heat is rejected by the gas to the surrounding and is equal to the net work done on the gas.

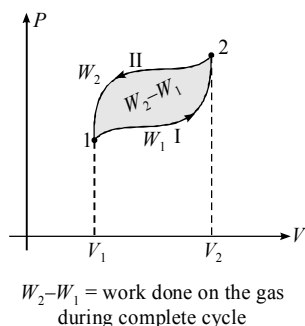


Figure 3.7

Cyclic processes in which PV curve is clockwise, net work is done by the gas, such cycles are called heat engine cycles. And those processes in which PV curve is anticlockwise, net work is done on the gas and heat is rejected by the gas to the surrounding, such cycles are called refrigeration cycles. In further sections we'll discuss these cycles in more details.

Illustrative Example 3.6

When a thermodynamic system is taken from an initial state I to a final state F along the path IAF , as shown in figure-3.8, the heat energy absorbed by the system is $Q = 55 \text{ J}$ and the work done by the system is $W = 25 \text{ J}$. If the same system is taken along the path IBF , the value of $Q = 35 \text{ J}$.

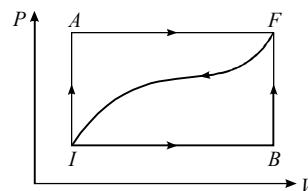


Figure 3.8

- Find the work done along the path IBF .
- If $W = -15 \text{ J}$ for the curved path FI , how much heat energy is lost by the system along this path?
- If $U_I = 10 \text{ J}$, what is U_F ?
- If $U_B = 20 \text{ J}$, what is Q for the processes BF and IB ?

Solution

The first law of thermodynamics states that

$$\Delta Q = \Delta U + \Delta W \text{ or } Q = (U_F - U_I) + W$$

Here U_I and U_F are the internal energies in the initial and the final state. Given that, for path IAF , $Q = 55 \text{ J}$ and $W = 25 \text{ J}$. Therefore,

$$\Delta U = U_F - U_I = Q - W = 55 - 25 = 30 \text{ J}$$

The internal energy is independent of the path; it depends only on the initial and final states of the system. Thus the internal energy between I and F states is 30 J irrespective of the path followed by the system.

- For path IBF , $Q = 35 \text{ J}$ and $\Delta U = 30 \text{ J}$. Therefore,

$$W = Q - \Delta U = 35 - 30 = 5 \text{ J}$$

- For path FI , $W = -15 \text{ J}$, but $\Delta U = -30 \text{ J}$

Therefore $Q = W + \Delta U = -15 - 30 = -45 \text{ J}$

(c) Given $U_I = 10 \text{ J}$. Therefore, $U_F = \Delta U + U_I = 30 + 10 = 40 \text{ J}$

(d) The process BF is isochoric, i.e. the volume is constant. Hence $W = 0$. Therefore

$$Q = (\Delta U)_{BF} = U_F - U_B = 40 - 20 = 20 \text{ J}$$

The process IB is isobaric (constant pressure). Therefore,

$$Q = (Q)_{IBF} - (Q)_{BF} = 35 - 20 = 15 \text{ J}.$$

Illustrative Example 3.7

Figure-3.9 shows an ideal gas changing its state from state A to state C by two different paths ABC and AC .

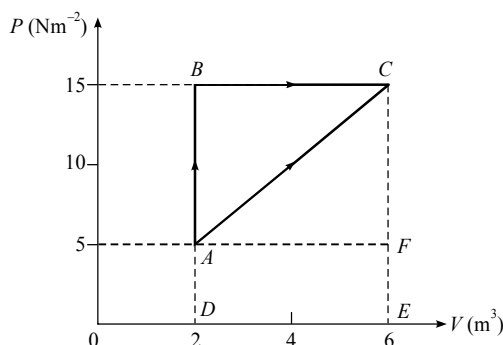


Figure 3.9

- Find the path along which the work done is the least.
- The internal energy of the gas at A is 10 J and the amount of heat supplied to change its state to C through the path AC is 200 J . Find the internal energy at C .
- The internal energy of the gas at state B is 20 J . Find the amount of heat supplied to the gas to go from state A to state B .

Solution

(a) We know work done is given by the area below PV curve thus by observing the two PV curves AC and ABC , we can say that work done in path AC is less than that in path ABC .

(b) Work done in path AC by the gas is

$$\begin{aligned} W_{AC} &= \text{area of } ACFEDA \\ &= \text{area of } ACF + \text{Area of } AFED \\ &= \frac{1}{2} \times (15 - 5) \times (6 - 2) + (6 - 2) \times 5 \\ &= 20 + 20 = 40 \text{ J} \end{aligned}$$

It is given that heat supplied in process AC is $Q_{AC} = 200 \text{ J}$

Thus change in internal energy of gas in path AC is from first law of thermodynamics, given as

$$Q_{AC} = W_{AC} + \Delta U_{AC}$$

$$\text{or } \Delta U_{AC} = Q_{AC} - W_{AC} = 200 - 40 = 160 \text{ J}$$

As it is given that at state A , internal energy of gas is 10 J thus at state C , its internal energy is

$$\Delta U_{AC} = U_C - U_A$$

$$\text{or } U_C = \Delta U_{AC} + U_A = 160 + 10 = 170 \text{ J}$$

(c) As in process AB , no volume change takes place thus no work is done by or on the gas during path AB . Thus according to first law of thermodynamics, we have

$$Q_{AB} = \Delta U_{AB} + W_{AB}$$

$$\text{or } Q_{AB} = U_B - U_A + 0$$

$$\text{or } Q_{AB} = 20 - 10 = 10 \text{ J} \quad [\text{As it is given that } U_B = 20 \text{ J}]$$

Illustrative Example 3.8

A gas is taken from state-1 to state-2 along the path shown in figure-3.10. If 70 cal of heat is extracted from the gas in the process, calculate the change in internal energy of the system.

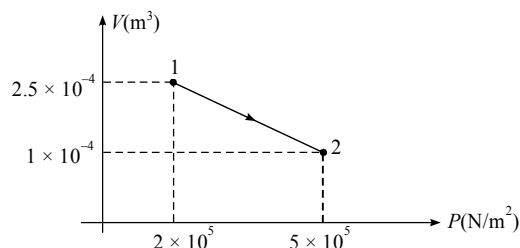


Figure 3.10

Solution

We know work done by a gas is given by the area under PV -curve or the area between PV -curve and the volume axis. Generally we take volume on x -axis while plotting PV -curve but in figure-3.10 it is taken on y -axis thus the work done is given by the shaded area shown in figure-3.11. In this process volume of gas decreases thus work is done on the gas and it is given as

$$\begin{aligned} W &= -\frac{1}{2} \times 1.5 \times 10^{-4} \times (2 + 5) \times 10^5 \\ &= -52.5 \text{ J} \end{aligned}$$

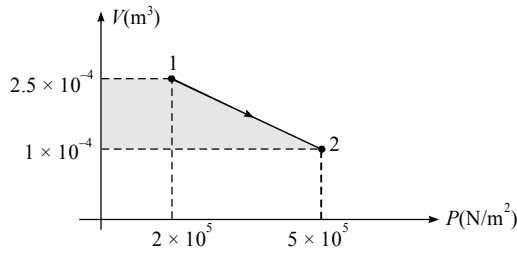


Figure 3.11

It is given that heat extracted in the process is 70 cal thus

$$\begin{aligned} Q &= -70 \text{ cal} = 70 \times 4.2 \text{ J} \\ &= -294 \text{ J} \end{aligned}$$

Now from first law of thermodynamics, we have

$$Q = W + \Delta U$$

or

$$\begin{aligned} \Delta U &= Q - W \\ &= (-294) - (-52.5) \\ &= -241.5 \text{ J} \end{aligned}$$

Thus in the process internal energy of gas decreases by 241.5 J

Illustrative Example 3.9

Figure-3.12 shows a process $ABCA$ performed on one mole of an ideal gas. Find the net heat supplied to the gaseous system during the process.

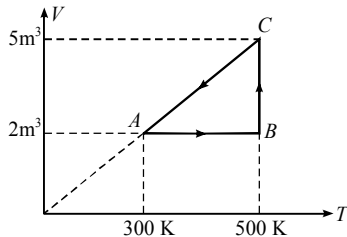


Figure 3.12

Solution

As we know in a cyclic process gas finally returns to its initial state hence total change in internal energy of gas is zero thus the total heat supplied to the gas is equal to the work done by the gas. Now we find the work done by the gas in different paths of the cycle.

In Process AB

As shown in graph during process AB , volume of gas remains constant thus work done by gas is zero

$$W_{AB} = 0$$

In Process BC

In process BC , volume of gas changes from $V_1 = 2\text{m}^3$ to $V_2 = 5\text{m}^3$ thus work done can be obtained as

$$W_{BC} = \int_2^5 PdV$$

As in process BC , temperature of gas remains constant at 500 K, thus we can write pressure of gas from gas law as

$$P = \frac{RT}{V} = \frac{500 R}{V} \quad [\text{As } n = 1 \text{ mole}]$$

Now work done is

$$\begin{aligned} W_{BC} &= \int_2^5 \frac{500 R}{V} dV \\ &= 500 R \ln \left(\frac{5}{2} \right) \end{aligned}$$

In Process CA

As in this process path is a straight line passing through origin thus $V \propto T$ or pressure of gas remains constant and we know if gas pressure is constant work done is given as

$$W_{CA} = nR(T_2 - T_1) = nR(300 - 500) = -200 R$$

This is negative as gas is being compressed from volume 5m^3 to 2m^3 or work is done on the gas.

Now we can find the total work done by the gas in the complete cycle $ABCA$ as

$$\begin{aligned} W_{ABCA} &= W_{AB} + W_{BC} + W_{CA} \\ &= 0 + 500 R \ln \left(\frac{5}{2} \right) - 200 R \\ &= R(500 \ln \frac{5}{2} - 200) \\ &= 2146.22 \text{ J} = \text{heat supplied to the gas} \end{aligned}$$

Illustrative Example 3.10

An ideal gas is taken round a cyclic thermodynamic process $ABCA$ as shown in figure-3.13. If the internal energy of the gas at point A is assumed zero while at B it is 50 J. The heat absorbed by the gas in the process BC is 90 J.

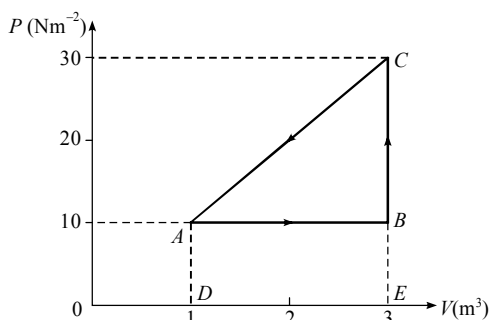


Figure 3.13

- (a) What is the internal energy of the gas at point C ?
 (b) How much heat energy is absorbed by the gas in the process AB ?
 (c) Find the heat energy rejected or absorbed by the gas in the process CA.
 (d) What is the net work done by the gas in the complete cycle ABCA ?

Solution

Given that $U_A = 0$, $U_B = 50$ J and $Q_{BC} = 90$ J. Also
 $P_A = P_B = 10$ Nm⁻², $P_C = 30$ Nm⁻²,
 $V_A = 1$ m³ and $V_B = V_C = 3$ m³.

- (a) In process BC as volume of gas remains constant, work done by gas in this process is zero, thus

$$W_{BC} = 0.$$

Heat absorbed by the gas is $Q_{BC} = 90$ J. From the first law of thermodynamics.

$$(\Delta U)_{BC} = U_C - U_B = Q_{BC} - W_{BC} = 90 \text{ J} - 0 = 90 \text{ J}.$$

or $U_C = (\Delta U)_{BC} + U_B = 90 \text{ J} + 50 \text{ J} = 140 \text{ J}$

- (b) In process AB, we have

$$\begin{aligned} (\Delta U)_{AB} &= U_B - U_A \\ &= 50 - 0 = 50 \text{ J} \end{aligned}$$

Work done is given as

$$\begin{aligned} W_{AB} &= \text{area under AB in } P-V \text{ diagram} \\ &= \text{area of rectangle ABED} \\ &= AB \times AD = (3 \text{ m}^3 - 1 \text{ m}^3) \times 10 \text{ Nm}^{-2} \\ &= 20 \text{ J} \end{aligned}$$

Thus heat absorbed by the system is

$$Q_{AB} = (\Delta U)_{AB} + W_{AB} = 50 + 20 = 70 \text{ J}$$

- (c) For process CA

$$(\Delta U)_{CA} = U_A - U_C = 0 - 140 = -140 \text{ J}$$

Work done is given as

$$\begin{aligned} W_{CA} &= \text{area ACED} \\ &= \text{area of triangle ACB} + \text{area of rectangle ABED} \\ &= \frac{1}{2} \times AB \times BC + AB \times AD \\ &= \frac{1}{2} \times (3 - 1) \text{ m}^3 \times (30 - 10) \text{ Nm}^{-2} + 20 \\ &= 20 + 20 = 40 \text{ J} \end{aligned}$$

In this process, the volume decreases, the work is done on the gas. Hence, the work done is negative. Thus

$$W_{CA} = -40 \text{ J}$$

Thus heat rejected by the gas is

$$Q_{CA} = (\Delta U)_{CA} + W_{CA} = -140 - 40 = -180 \text{ J}.$$

- (d) Net work done in the complete cyclic process ABCA is

$$W = \text{area of triangle ABC} = \frac{1}{2} \times 2 \times 20 = 20 \text{ J}$$

As the cycle is anticlockwise, net work is done on the gas.

Illustrative Example 3.11

A sample of 2 kg of monoatomic helium (assumed ideal) is taken through the process ABC and another sample of 2 kg of the same gas is taken through the process ADC as shown in figure-3.14. Given, molecular mass of helium = 4.

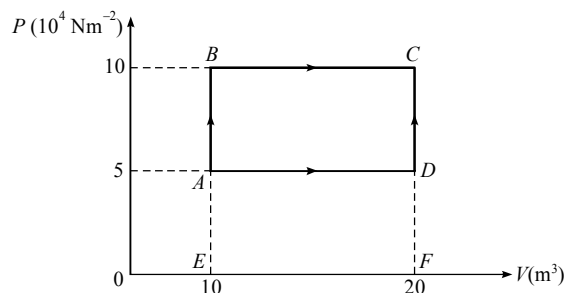


Figure 3.14

- (a) What is the temperature of helium in each of the states A, B, C and D ?
 (b) Is there any way of telling afterwards which sample of helium went through the process ABC and which went through the process ADC ? Write yes or no.
 (c) How much is the heat involved in each of the process ABC and ADC ?

Solution

Given that mass of helium used in the process is $m = 2$ kg thus number of moles can be given as

$$n = \frac{m}{M} = \frac{2}{4 \times 10^{-3}} = 500$$

At different states, the pressure and volume of gas are also given, from figure-3.14

$$P_A = P_D = 5 \times 10^4 \text{ N/m}^2$$

$$P_B = P_C = 10^5 \text{ N/m}^2$$

$$V_A = V_B = 10 \text{ m}^3$$

$$V_C = V_D = 20 \text{ m}^3$$

(a) From gas law, we have

$$T_A = \frac{P_A V_A}{nR} = \frac{5 \times 10^4 \times 10}{500 \times 8.314} = 120.3 \text{ K}$$

$$T_B = \frac{P_B V_B}{nR} = \frac{10^5 \times 10}{500 \times 8.314} = 240.6 \text{ K}$$

$$T_C = \frac{P_C V_C}{nR} = \frac{10^5 \times 20}{500 \times 8.314} = 481.11 \text{ K}$$

and $T_D = \frac{P_D V_D}{nR} = \frac{5 \times 10^4 \times 20}{500 \times 8.314} = 120.3 \text{ K}$

(b) Since the gas is taken from same initial state to same final state C no matters whatever be the path, the answer is No.

(c) In process ABC , the change in internal energy is

$$\begin{aligned} \Delta U_{ABC} &= U_C - U_A = \frac{f}{2} nR (T_C - T_A) \\ &= \frac{3}{2} \times 500 \times 8.314 (481.11 - 120.3) \\ &= 2.25 \times 10^6 \text{ J} \end{aligned}$$

Net work done in process ABC is

$$\begin{aligned} W_{ABC} &= W_{AB} + W_{BC} \\ &= 0 + \text{area below curve } BC \\ &= 0 + 10^5 \times 10 \\ &= 10^6 \text{ J} \end{aligned}$$

Thus from first law of thermodynamics, heat supplied in process ABC is

$$Q = W + \Delta U$$

or

$$\begin{aligned} Q &= 10^6 + 2.25 \times 10^6 \\ &= 3.25 \times 10^6 \text{ J} \end{aligned}$$

Similarly in process ADC as being a state function change in internal energy remains same as initial and final states are same. Thus

$$\Delta U_{ADC} = 2.25 \times 10^6 \text{ J}$$

Thus work done by the gas in process ADC is

$$\begin{aligned} W_{ADC} &= W_{AD} + W_{DC} \\ &= \text{area below curve } AD + 0 \\ &= 5 \times 10^4 \times 10 + 0 \\ &= 0.5 \times 10^6 \text{ J} \end{aligned}$$

Thus from first law of thermodynamics, heat supplied in the process ADC is given as

$$\begin{aligned} Q &= W + \Delta U \\ \text{or} \quad Q &= 0.5 \times 10^6 + 2.25 \times 10^6 \\ &= 2.75 \times 10^6 \text{ J} \end{aligned}$$

Illustrative Example 3.12

One mole of an ideal monatomic gas is taken round the cyclic process $ABCA$ as shown in figure-3.15. Calculate

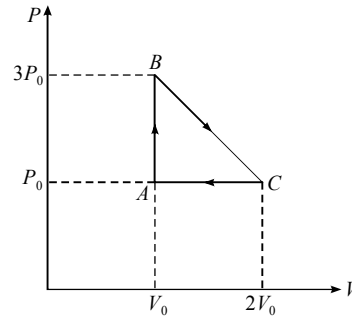


Figure 3.15

- the workdone by the gas
- the heat rejected by the gas in the path CA and the heat absorbed by the gas in the path AB
- the net heat absorbed by the gas in the path BC
- the maximum temperature attained by the gas during the cycle.

Solution

- The workdone by the gas is equal to the area under the closed curve. Thus workdone in cycle is

$$\begin{aligned}
 W &= \frac{1}{2} (2 V_0 - V_0) (3 P_0) (3 P_0 - P_0) \\
 &= \frac{1}{2} V_0 \times 2 P_0 \\
 &= P_0 V_0
 \end{aligned}$$

(b) Heat rejected in path CA is given as

$$\begin{aligned}
 Q_{CA} &= \eta C_p \Delta T = \eta C_p (T_C - T_A) \\
 &= 1 \times 5/2 \cdot R \left[\frac{P_0 2V_0}{R} - \frac{P_0 V_0}{R} \right] \\
 &= \frac{5}{2} P_0 V_0 \quad [\text{As } n = 1 \text{ mole}]
 \end{aligned}$$

Heat absorbed in path AC

$$\begin{aligned}
 Q_{AC} &= \eta C_p (T_B - T_A) \\
 &= 1 \times \frac{3}{2} R \times \left[\frac{3 P_0 V_0}{R} - \frac{P_0 V_0}{R} \right] \\
 &= 3 P_0 V_0
 \end{aligned}$$

(c) For cycle ABC , we have

Heat supplied = workdone by the gas

$$\text{or} \quad = -\frac{5}{2} P_0 V_0 + 3 P_0 V_0 + Q_{BC} = P_0 V_0$$

Heat supplied in path BC is given by

$$\begin{aligned}
 Q_{BC} &= P_0 V_0 + \frac{5}{2} P_0 V_0 - 3 P_0 V_0 \\
 &= P_0 V_0/2
 \end{aligned}$$

(d) We know that $PV/T = \text{constant}$. So, when PV is maximum, T is also maximum. PV is maximum for part BC . Hence temperature will be maximum between B and C .

Let equation of BC be as

$$P = k V + k' \text{ satisfying both the point } B \text{ and } C$$

$$\text{For point } B, \quad 3 P_0 = k V_0 + k'$$

$$\text{For point } C, \quad P_0 = k (2 V_0) + k'$$

Solving these equations, we get

$$k = -2 (P_0/V_0) \text{ and } k' = 5 P_0$$

So, the equation for time BC is

$$P = -2 \frac{P_0}{V_0} \times V + 5 P_0$$

$$\text{or} \quad \frac{RT}{V} = -\frac{2 P_0 V}{V_0} + 5 P_0$$

$$\text{or} \quad T = \frac{P_0}{R} \left[5V - 2 \frac{V^2}{V_0} \right] \quad \dots (3.14)$$

For maximum, $dT/dV = 0$

$$\text{So,} \quad \frac{dT}{dV} = \frac{P_0}{R} \left[5 - \frac{4V}{V_0} \right] = 0$$

$$\text{Hence} \quad 5 - \frac{4V}{V_0} = 0 \quad \text{or} \quad 5 V_0 - 4 V = 0$$

$$\text{or} \quad V = \frac{5}{4} V_0 \quad \dots (3.15)$$

Substituting the value of V from equation-(3.15) in equation-(3.14), we get

$$\begin{aligned}
 T_{\max} &= \frac{P_0}{R} \left[5 \times \left(\frac{5}{4} V_0 \right) - 2 \left(\frac{5 V_0}{4} \right)^2 \frac{1}{V_0} \right] \\
 &= \frac{P_0}{R} \left[\frac{25 V_0}{4} - \frac{25 V_0}{8} \right] = \frac{P_0}{8} \times \frac{25 V_0}{8} \\
 &= \frac{25 P_0 V_0}{8 R}
 \end{aligned}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Laws of Thermodynamics

Module Number - 9 to 16

Practice Exercise 3.2

(i) A thermodynamic system undergoes cyclic process 1423451 as shown in figure-3.16. Find the work done by the system.

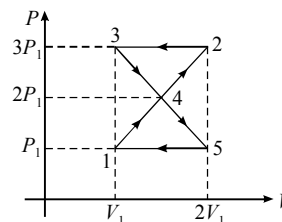


Figure 3.16

[0]

(ii) n moles of an ideal gas undergoes a thermodynamic process in which pressure of gas varies linearly with the volume of gas as shown in figure-3.17. Find the maximum temperature of the gas during the process.

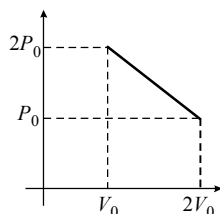


Figure 3.17

$$\left[\frac{9P_0V_0}{4nR} \right]$$

(iii) The PV diagram shown in figure-3.18 for a thermodynamic process is a semicircle. Find the work done on the gas in the process ABC .

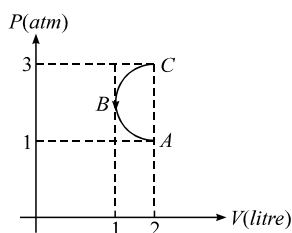


Figure 3.18

$$[\pi/2 \text{ atm-ltr}]$$

(iv) The figure-3.19 shows p - V diagram of the thermodynamic process of an ideal gas. Compute from this graph : (a) work done in the processes $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$, and $D \rightarrow A$. (b) work done in the complete cycle. $1 \text{ atm} = 1.0 \times 10^5 \text{ N/m}^2$.

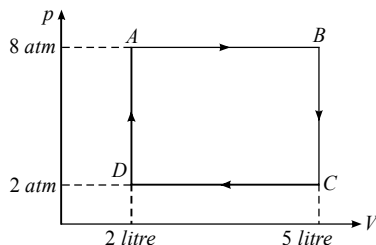


Figure 3.19

$$[2400 \text{ J}, 0, -600 \text{ J}, 0; 1800 \text{ J}]$$

(v) A cyclic process $ABCA$ shown in V - T diagram (figure-3.20) is performed with a constant mass of an ideal gas. Show the same process on a P - V diagram.

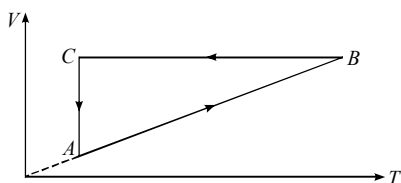


Figure 3.20

In the figure-3.20, CA is parallel to the V -axis and BC is parallel to the T -axis.

(vi) Figure-3.21 shows the PV diagram for a gas confined to a cylinder by a piston. How much work does the gas do as it expands from A to C along the curve?

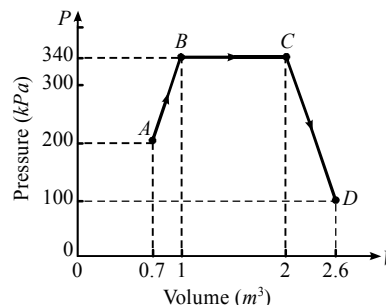


Figure 3.21

$$[421 \text{ kJ}]$$

(vii) The volume of a monoatomic ideal gas increases linearly with pressure, as shown in the figure-3.22. Calculate (a) increase in internal energy, (b) work done by the gas, and (c) heat supplied to the gas.

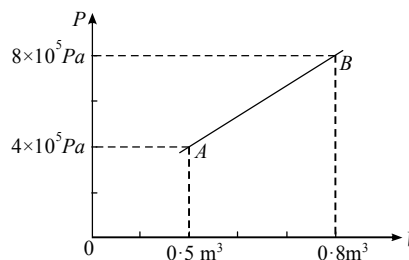


Figure 3.22

$$[6.6 \times 10^5, 1.8 \times 10^5, 8.4 \times 10^5]$$

3.4 Molar Specific Heats of a Gas

As we have already discussed that depending on ways of supplying heat to a gas, it may have infinite different molar specific heats. But as a characteristic property a gas has two standard molar specific heats or molar heat capacities. These are

- Molar heat capacity at constant volume (C_V)
- Molar heat capacity at constant pressure (C_P)

3.4.1 Molar Heat Capacity at Constant Volume (C_V)

This is the amount of heat required for one mole of a gas to raise its temperature by one degree at constant volume. This is denoted by C_V .

If a gas is heated at constant volume, we can simply state that no work is done by the gas or on the gas during heating. Thus if we apply first law of thermodynamics to such type of a heating process, we get

$$dQ = dU + dW$$

or $dQ = dU$ [As $v = \text{constant}$, $dW = 0$]

or $n C_V dT = \frac{f}{2} n R dT$ [As $dQ = n C_V dT$]

or $C_V = \frac{f}{2} R$... (3.16)

Above relation states that when volume of a gas is constant during heating, no work is done. This implies that the total amount of heat supplied to the gas is used in increasing internal energy of gas, that is to raise the temperature.

Here equation-(3.16) gives the characteristic property of a gas, the molar specific heat at constant volume. For different types of gases values of C_V are listed in table-3.1

Table-3.1

Type of Gas	No. of degrees of freedom f	Molar heat capacity at constant volume C_V
Monoatomic gas	3	$\frac{3}{2} R$
Diatomic gas at lower temperature	5	$\frac{5}{2} R$
Diatomic gas at higher temperature	7	$\frac{7}{2} R$
Polyatomic gas linear molecule	$5 + x$ (vib.)	$\left(\frac{5+x}{2}\right) R$
Polyatomic gas non linear molecule	$6 + x$ (vib.)	$\frac{6+x}{2} R$

3.4.2 Molar Heat Capacity at Constant Pressure (C_P)

It is the amount of heat required for one mole of a gas to raise its temperature by one degree at constant pressure. It is denoted by C_P .

When a gas is heated at constant pressure, with rise in temperature volume of gas must increase by gas law as

$$PV = n R T$$

Differentiating gas law gives

$$PdV + VdP = n R dT \quad \dots (3.17)$$

This equation-(3.17) is called “*differential form of gas law*”. If during heating pressure of gas is a constant, second term on left hand side of equality will be zero thus we have

$$P dV = n R dT$$

[As $dP = 0$, for constant pressure]

Which is also the elemental work done by the gas. This is a process when gas pressure does not change during heating, if volume of gas changes from V_1 to V_2 and temperature changes from T_1 to T_2 , the total work done is given as

$$W = \int_{V_1}^{V_2} P dV = \int_{T_1}^{T_2} n R dT$$

or $W = P (V_2 - V_1) = n R (T_2 - T_1) \quad \dots (3.18)$

According to first law of thermodynamics, if dQ is the heat supplied to a gaseous system at constant pressure, dU is the increase in internal energy and if dW is the work done by the gas, then we have

$$dQ = dU + dW$$

If C_P is the molar heat capacity for this gas then we have

$$n C_P dT = \frac{f}{2} n R dT + PdV \quad [\text{As } dQ = n C_P dT]$$

or $n C_P dT = \frac{f}{2} n R dT + n R dT$

[as $PdV = nRT$ for constant pressure]

or $C_P = \frac{f}{2} R + R = \left(\frac{f+2}{2}\right) R \quad \dots (3.19)$

or $C_P = C_V + R \quad \dots (3.20)$

Equation-(3.19) gives another characteristic property of a gas, the molar specific heat at constant pressure. Equation-(3.20) gives a relation among the molar specific heats of a gas. This relation is called Mayor's Relation. It shows that always values of C_P are higher than C_V for a gas because when a gas is heated at constant pressure, more heat is required to raise the temperature of gas as compared to the case when it is heated at constant volume because at constant pressure some amount of supplied heat is used to do work against surrounding where in heating at constant volume total amount of supplied heat is used in increasing internal energy i.e. the temperature of gas.

Table-3.2 gives the values of C_P for different types of gases.

Table 3.2

Type of Gas	No. of degrees of freedom f	Molar heat capacity at constant Pressure C_p
Monoatomic gas	3	$\frac{5}{2} R$
Diatomic gas at lower temperature	5	$\frac{7}{2} R$
Diatomic gas at higher temperature	7	$\frac{9}{2} R$
Polyatomic gas linear molecule	$5 + x$ (vib.)	$\left(\frac{7+x}{2}\right) R$
Polyatomic gas non linear molecule	$6 + x$ (vib.)	$\left(\frac{8+x}{2}\right) R$

3.4.3 Ratio of Heat Capacities of a Gas

In previous section we've obtained C_V and C_p as

$$C_V = \frac{f}{2} R$$

and

$$C_p = \frac{f+2}{2} R$$

The values of C_p and C_V are the characteristic properties of a gas. One more characteristic property is widely used in analyzing the behaviour of gas. This is ratio of the two specific heats of a gas and is termed as adiabatic exponent of a gas. It is denoted by γ as

$$\gamma = \frac{C_p}{C_V} = \frac{f+2}{f} \quad \dots (3.21)$$

Later we'll see that γ is very useful property of a gas in calculating different thermodynamic parameters of a gas when it undergoes a thermodynamic process. From equation-(3.21), we can find the values of γ for different types of gases.

Table-3.3 gives the values of γ for different types of gases

Table-3.3

Type of Gas	No. of degrees of freedom f	Ratio of specific heat γ
Monoatomic gas	3	$\frac{5}{3} = 1.67$
Diatomic gas at lower temperature	5	$\frac{7}{5} = 1.4$
Diatomic gas at higher temperature	7	$\frac{9}{7} = 1.28$
Polyatomic gas linear molecule	$5 + x$ (vib.)	$\frac{7+x}{5+x}$
Polyatomic gas non linear molecule	$6 + x$ (vib.)	$\frac{8+x}{6+x}$

From equation-(3.21), we can represent number of degrees of freedom in terms of γ as

$$f = \left(\frac{2}{\gamma - 1} \right) \quad \dots (3.22)$$

Using equation-(3.22) we can represent C_p and C_V in terms of γ as

$$C_V = \frac{f}{2} R = \frac{R}{\gamma - 1} \quad \dots (3.23)$$

and

$$C_p = \gamma C_V = \frac{\gamma R}{\gamma - 1} \quad \dots (3.24)$$

Equation-(3.23) and (3.24) are commonly used in wide range of numerical problems so students are advised to keep these results on their tips.

3.4.4 Ratio of Specific Heats for a Mixture of Gases

In previous chapter we've discussed that if N gases of $n_1, n_2, \dots, n_N$ moles are mixed at a common temperature T , the internal energy of the mixture remains constant. If these N gases have values of C_V as $C_{V1}, C_{V2}, \dots, C_{VN}$ then we have

$$\begin{aligned} n_1 C_{V1} T + \dots + n_N C_{VN} T \\ = (n_1 + n_2 + \dots + n_N) C_{Veq} T \end{aligned} \quad \dots (3.25)$$

$$\text{or} \quad C_{Veq} = \frac{n_1 C_{V1} + n_2 C_{V2} + \dots + n_N C_{VN}}{n_1 + n_2 + \dots + n_N} \quad \dots (3.26)$$

Where C_{Veq} is the equivalent molar specific heat for the mixture of gases at constant volume similarly C_{peq} for the mixture of gases can be given as

$$\text{or} \quad C_{peq} = \frac{n_1 C_{p1} + n_2 C_{p2} + \dots + n_N C_{pN}}{n_1 + n_2 + \dots + n_N} \quad \dots (3.27)$$

If $\gamma_1, \gamma_2, \dots, \gamma_N$ are the ratio of specific heats for these independent gas then from equation-(3.25) we have

$$\begin{aligned} \frac{n_1 RT}{\gamma_1 - 1} + \frac{n_2 RT}{\gamma_2 - 1} + \dots + \frac{n_N RT}{\gamma_N - 1} \\ = (n_1 + n_2 + \dots + n_N) \left(\frac{R}{\gamma_{eq} - 1} \right) T \quad \left[\text{As } C_V = \frac{R}{\gamma - 1} \right] \end{aligned}$$

Here γ_{eq} is the equivalent adiabatic exponent for the mixture of gases and given by the equation

$$\frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1} + \dots + \frac{n_N}{\gamma_N - 1} = \frac{n_1 + n_2 + \dots + n_N}{\gamma_{eq} - 1} \quad \dots (3.28)$$

Illustrative Example 3.13

Calculate the heat absorbed by a system in going through the cyclic process shown in figure-3.23.

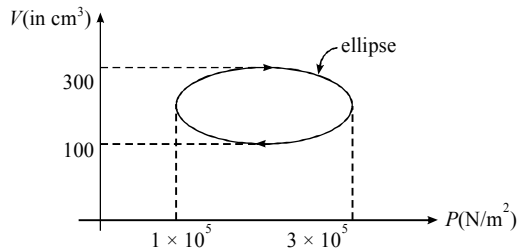


Figure 3.23

Solution

In the figure-3.23 shown, the cycle is clockwise, thus net work is done by the gas. As in a cyclic process no change in internal energy takes place thus heat supplied is equal to the work done by the gas in one complete cycle so in this case heat supplied to the gas is given as

$$\begin{aligned} Q &= \text{Work done by the gas } W \\ &= \text{Area of ellipse} \\ &= \pi ab \end{aligned}$$

Where a and b are semi major and semi-minor axis of the ellipse which are given from figure-3.23 as

$$a = 1.0 \times 10^5 \text{ N/m}$$

$$\text{and } b = 100 \times 10^{-6} \text{ m}^3$$

Thus area of ellipse is

$$\begin{aligned} Q = W &= \pi \times 1.0 \times 10^5 \times 100 \times 10^{-6} \\ &= 3.14 \times 10 \\ &= 31.4 \text{ J.} \end{aligned}$$

Illustrative Example 3.14

An ideal gas has a specific heat at constant pressure $C_p = (5 R/2)$. The gas is kept in a closed vessel of volume 0.0083 m^3 , at a temperature of 300 K and a pressure of $1.6 \times 10^6 \text{ N/m}^2$. An amount of $2.49 \times 10^4 \text{ J}$ of heat energy is supplied to the gas. Calculate the final temperature and pressure of the gas.

Solution

Given that initial pressure, volume and temperature of gas is

$$P_1 = 1.6 \times 10^6 \text{ N/m}^2$$

$$V_1 = 0.0083 \text{ m}^3$$

$$T_1 = 300 \text{ K}$$

From gas law we can find the number of moles of gas in the container as

$$\begin{aligned} n &= \frac{PV}{RT} = \frac{1.6 \times 10^6 \times 0.0083}{8.314 \times 300} \\ &= \frac{16}{3} \text{ mole} \end{aligned}$$

It is given that molar specific heat of gas at constant pressure is

$$C_p = \frac{5R}{2}$$

Thus gas is monoatomic, hence its molar specific heat at constant volume is given as

$$C_v = \frac{3R}{2}$$

As gas is heated in a closed vessel i.e. at constant volume, if its temperature is raised from T_1 to T_2 then, we have heat supplied to the gas is

$$Q = n C_v (T_2 - T_1)$$

$$\text{or } 2.49 \times 10^4 = \frac{16}{3} \times \frac{3}{2} R (T_2 - 300)$$

$$\begin{aligned} \text{or } T_2 &= 300 + \frac{3 \times 2 \times 2.49 \times 10^4}{16 \times 3 \times 8.314} \\ &= 300 + 374.36 \\ &= 674.36 \text{ K} \end{aligned}$$

For constant volume process, we have

$$\frac{P_2}{P_1} = \frac{T_2}{T_1}$$

$$\text{Thus } P_2 = \frac{T_2}{T_1} \times P_1$$

$$\begin{aligned} &= \frac{674.36}{300} \times 1.6 \times 10^6 \\ &= 3.6 \times 10^6 \text{ N/m}^2 \end{aligned}$$

Illustrative Example 3.15

Figure-3.24 shows a cylindrical container containing oxygen gas and closed by a piston of mass 50 kg . Piston can slide smoothly in the cylinder. Its cross sectional area is 100 cm^2 and atmospheric pressure is 10^5 Pa . Some heat is supplied to the cylinder so that the piston is slowly displaced up by 20 cm .

Find the amount of heat supplied to the gas.

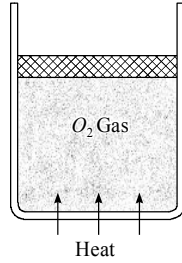


Figure 3.24

Solution

As here piston is open to atmosphere, the gas is under a constant pressure given as

$$\begin{aligned} P_{\text{gas}} &= P_{\text{atm}} + \frac{Mg}{S} \\ &= 10^5 + \frac{50 \times 10}{100 \times 10^{-4}} \text{ Pa} \\ &= 10^5 + 5 \times 10^4 = 1.5 \times 10^5 \text{ Pa} \end{aligned}$$

It is given that the piston moves out by 20 cm, thus increment in its volume is given as

$$\begin{aligned} \Delta V &= S \Delta x \\ &= 100 \times 10^{-4} \times 20 \times 10^{-2} \text{ m}^3 \\ &= 2 \times 10^{-3} \text{ m}^3 \end{aligned}$$

Thus in the process work done by gas is

$$\begin{aligned} W &= P_{\text{gas}} \Delta V \quad [\text{As } P_{\text{gas}} = \text{constant}] \\ &= 1.5 \times 10^5 \times 2 \times 10^{-3} \\ &= 300 \text{ J} \end{aligned}$$

For a process in which gas pressure is constant, work done can also be given as

$$W = nR\Delta T = P_{\text{gas}} \Delta V$$

or $nR\Delta T = 300 \text{ J}$

As in the process gas pressure is constant thus heat supplied in raising the temperature by ΔT is given as

$$\begin{aligned} Q &= n C_p \Delta T \\ &= n \left(\frac{7}{2} R \right) \Delta T \end{aligned}$$

[As for O_2 as being diatomic gas $C_p = \frac{7R}{2}$]

$$\begin{aligned} Q &= \frac{7}{2} \times 300 \\ &= 1050 \text{ J} \end{aligned}$$

Illustrative Example 3.16

Two moles of an ideal monoatomic gas are confined within a cylinder by a massless spring loaded with a frictionless piston of negligible mass and of cross-sectional area $4 \times 10^{-3} \text{ m}^2$. The spring is initially in its relaxed state. Now the gas is heated by a heater for some time. During this time the gas expands and does 50 J of work in moving the piston through a distance of 0.1 m. The temperature of the gas increases by 50 K. Calculate the spring constant and the heat supplied by the heater.

Solution

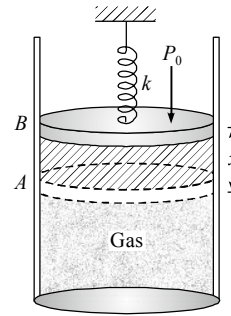


Figure 3.25

Refer to figure-3.25. A is the initial (equilibrium) position of the piston when the spring is relaxed. When the gas is heated, it expands and pushes the piston up by a distance, say, x . The spring is compressed, if k is the force constant of the spring and A the area of cross-section of the piston (which is equal to the cross-sectional area of the cylinder), the force in the spring is $F = kx$ and the pressure exerted on the gas by the spring is

$$P_s = \frac{F}{A} = \frac{kx}{A}$$

If P_0 is the atmospheric pressure, at equilibrium of piston the pressure of the gas in the cylinder is

$$P = P_0 + P_s = P_0 + \frac{kx}{A}$$

The increase in the volume of the gas by infinitesimal movement dx of the piston is

$$dV = A dx$$

Thus work done is given as

$$W = \int P dV = \int_0^x \left(P_0 + \frac{kx}{A} \right) A dx$$

$$= P_0 A \int_0^x dx + k \int_0^x x dx$$

or $W = P_0 A x + \frac{1}{2} k x^2$

Given $A = 4 \times 10^{-3} \text{ m}^2$, $x = 0.1 \text{ m}$, $W = 50 \text{ J}$. The atmospheric pressure $P_0 = 0.76 \text{ m of Hg} = 0.76 \times 9.8 \times 13600 = 1.013 \times 10^5 \text{ N/m}^2$. Using these values in above expression of work and solving for k , we get

$$k = 1896 \text{ N/m}$$

To find heat energy Q supplied by the heater, we use the first law of thermodynamics.

$$Q = \Delta U + W$$

Now
$$\Delta U = \frac{f}{2} nR \Delta T = nC_V \Delta T$$

$$\begin{aligned} \Delta U &= \frac{3}{2} nR \Delta T \\ &= \frac{3}{2} \times 2 \times 8.31 \times 50 = 1246.5 \text{ J} \end{aligned}$$

[For a monoatomic gas we have $C_V = \frac{3}{2} R$]

Thus we have, heat supplied given as

$$Q = 1246.5 + 50 = 1296.5 \text{ J.}$$

Illustrative Example 3.17

Consider the cyclic process $ABCA$ shown in figure-3.26. An ideal gas of 2 moles is undergone this process. A total of 1200 J heat is rejected by the gas in the complete cycle. Find the work done by the gas during the process BC .

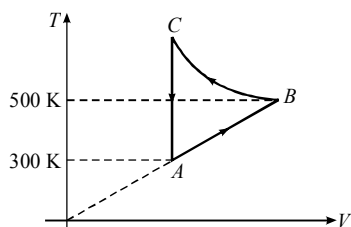


Figure 3.26

Solution

As we can see that in figure-3.26 CA is a process where gas volume remains constant thus no work is done in this process and in process AB gas pressure remains constant as $T \propto V$. Work done by the gas in this process can be given as

$$W_{AB} = nR(T_B - T_A)$$

[As if $P = \text{constant}$ $W = P(V_2 - V_1) = nR(T_2 - T_1)$]

$$\begin{aligned} &= 2 \times 8.314 \times (500 - 300) \\ &= 2 \times 8.314 \times 200 \\ &= 3325.6 \text{ J} \end{aligned}$$

As the cycle is anticlockwise, net work is done on the gas and the equal amount of heat must be rejected by the gas as in complete cycle no change in internal energy takes place. Thus heat rejected by the gas in complete cycle is given as

$$Q = \text{Net Work done on gas}$$

or
$$-1200 = W_{AB} + W_{BC} + W_{CA}$$

$$-1200 = 3325.6 + W_{BC} + 0$$

or
$$W_{BC} = -4525.6 \text{ J}$$

Illustrative Example 3.18

An ideal monoatomic gas is confined in a cylinder by a spring loaded piston of cross-section $8 \times 10^{-3} \text{ m}^2$. Initially the gas is at 300 K and occupies a volume of $2.4 \times 10^{-3} \text{ m}^3$ and the spring is in its relaxed (unstretched, uncompressed) state (see figure-3.27). The gas is heated by a small electric heater until the piston moves out slowly by 0.1 m. Calculate the final temperature of the gas and the heat supplied (in joule) by the heater. The force constant of the spring is 8000 N/m, atmospheric pressure is $1 \times 10^5 \text{ N/m}^2$. The cylinder and the piston are thermally insulated. The piston is massless and there is no friction between the piston and the cylinder. Neglect heat-loss through the lead wires of the heater. The heat-capacity of the heater coil is negligible. Assume the spring to be massless.

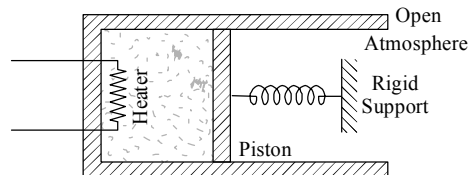


Figure 3.27

Solution

Initially, the pressure of the gas in the cylinder is atmospheric pressure as spring is in relaxed state. Therefore

$$P_1 = \text{atmospheric pressure} = 1.0 \times 10^5 \text{ N/m}^2$$

$$V_1 = \text{initial volume} = 2.4 \times 10^{-3} \text{ m}^3$$

$$T_1 = \text{initial temperature} = 300 \text{ K}$$

When heat is supplied by the heater, the piston is compressed by 0.1 m. The reaction force of compression of spring is equal to kx which acts on the piston or on the gas as

$$F = kx = 8000 \times 0.1 = 800 \text{ Nt}$$

Pressure exerted on the piston by the spring

$$\Delta P = \frac{F}{A} = \frac{800}{8 \times 10^{-3}} = 1 \times 10^5 \text{ N/m}^2$$

The total pressure P_2 of the gas inside cylinder is

$$\begin{aligned} P_2 &= P_{atm} + \Delta P = 1 \times 10^5 + 1 \times 10^5 \\ &= 2 \times 10^5 \text{ N/m}^2 \end{aligned}$$

Since the piston has moved outwards, there has been an increase ΔV in the volume of the gas

$$\begin{aligned} \Delta V &= A \times x = (8 \times 10^{-3}) \times (0.1) \\ &= 8 \times 10^{-4} \text{ m}^3 \end{aligned}$$

The final volume of the gas

$$\begin{aligned} V_2 &= V_1 + \Delta V = 2.4 \times 10^{-3} + 8 \times 10^{-4} \\ &= 3.2 \times 10^{-3} \text{ m}^3 \end{aligned}$$

Let T_2 be the final temperature of gas. Then

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\begin{aligned} \text{or } T_2 &= T_1 \frac{P_2 V_2}{P_1 V_1} \\ &= 300 \times \frac{2 \times 10^5 \times 3.2 \times 10^{-3}}{10^5 \times 2.4 \times 10^{-3}} \\ &= 800 \text{ K} \end{aligned}$$

Let the heat supplied by the heater be Q . This is used in two parts : a part is used in doing external work W due to expansion of the gas and the other part is used in increasing the internal energy of the gas. Hence

$$Q = W + \Delta U$$

$$\text{Now } W = \int_{V_1}^{V_2} P dV = \int_0^x \left(P_{atm} + \frac{kx}{A} \right) A dx$$

[As pressure is $(P_{atm} + kx/A)$ and $dV = A dx$]

$$\begin{aligned} \text{or } W &= P_{atm} A x + \frac{kx^2}{2} \\ &= \left[10^5 \times (8 \times 10^{-3}) (0.1) + \frac{8000 \times (0.1)^2}{2} \right] \\ &= 120 \text{ joule} \end{aligned}$$

$$\text{Further } U = n C_v \Delta T$$

Number of moles of gas can be obtained from initial conditions and gas law as

$$\begin{aligned} n &= \frac{PV}{RT} = \frac{1 \times 10^5 \times 2.4 \times 10^{-3}}{8.314 \times 300} \\ &= 0.096 \text{ mole} \end{aligned}$$

Thus change in internal energy of gas is given as

$$\Delta U = n \left(\frac{3}{2} R \right) \Delta T$$

$$[\text{As for monoatomic gas } C_v = \frac{3}{2} R]$$

$$\text{or } U = 0.096 \times \frac{3}{2} \times 8.314 \times 500 = 598.6 \text{ Joule}$$

Heat supplied by heater

$$\begin{aligned} &= (120 + 598.6) \\ &= 718.6 \text{ joule} \end{aligned}$$

Illustrative Example 3.19

A monoatomic ideal gas is taken through the process ABC as shown in figure-3.28. The temperature at the point A is 300 K. Find the temperatures at points B and C . Also find the work done and heat supplied to the gas in paths AB and BC .

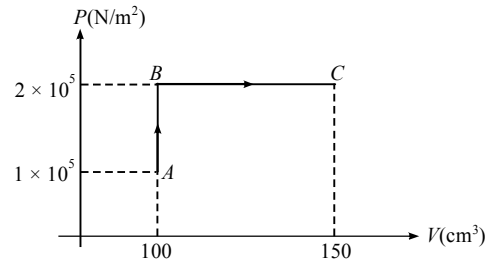


Figure 3.28

Solution

At initial state pressure, volume and temperature of gas is given as

$$P_A = 1 \times 10^5 \text{ Pa}$$

$$V_A = 100 \times 10^{-6} \text{ m}^3$$

$$T_A = 300 \text{ K}$$

Thus from gas law, we can find number of moles of gas, as

$$n = \frac{P_A V_A}{RT_A} = \frac{10^5 \times 10^{-4}}{8.314 \times 300} = 0.004 \text{ mole}$$

Thus at point B , gas temperature is given by

$$\frac{P_A}{T_A} = \frac{P_B}{T_B}$$

$$\text{or } T_B = \frac{P_B}{P_A} \times T_A$$

$$= \frac{2 \times 10^5}{10^5} \times 300$$

$$= 600 \text{ K}$$

Similarly at point C, gas temperature is given by

$$\frac{T_C}{V_C} = \frac{T_B}{V_B}$$

or

$$T_C = \frac{V_C}{V_B} \times T_B$$

$$= \frac{150 \times 10^{-6}}{100 \times 10^{-6}} \times 600$$

$$= 900 \text{ K}$$

Now in process AB, gas volume is constant thus no work is done by the gas and heat supplied to the gas can be given as

$$Q = n C_V (T_B - T_A)$$

$$= n \left[\frac{3}{2} R \right] (T_B - T_A)$$

$$[\text{As for a monoatomic gas } C_V = \frac{3R}{2}]$$

or

$$Q = 0.004 \times \frac{3}{2} \times 8.314 \times [600 - 300]$$

$$= 14.96 \text{ J}$$

In the process BC, gas pressure is constant thus work done by the gas can be given as

$$W = P_B (V_C - V_B)$$

$$= 2 \times 10^5 [150 - 100] \times 10^{-6}$$

$$= 10 \text{ J}$$

The heat supplied to the gas in this constant pressure process BC can be given as

$$Q = n C_P (T_C - T_B)$$

$$= n \left[\frac{5}{2} R \right] (T_C - T_B)$$

$$[\text{As for monoatomic gas the value of } C_P = \frac{5}{2} R]$$

$$= 0.004 \times \frac{5}{2} \times 8.314 \times (900 - 600)$$

$$= 24.94 \text{ J}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Laws of Thermodynamics

Module Number - 7 to 8

3.5 Different Type of Thermodynamic Processes

We've already discussed that a thermodynamic process is a way of heating a gas or doing work on it by which state of a gas can be changed from one to another by a specific path. Each thermodynamic process can be represented by a specific $P = f(V)$ curve on a P - V indicator diagram. In each thermodynamic process, there are three variables related. Heat supplied to a gas, change in its internal energy and work done by a gas. According to first law of thermodynamics a part of supplied heat is used as rise in internal energy of gas and rest is used to do work against surrounding. As we change the process of heating, the fraction of heat supplied which is used to increase the internal energy of gas and the fraction of heat supplied consumed in doing work will change. These three variables heat supplied Q , change in internal energy of a gas ΔU and work done W are called thermodynamic variables for a process.

There are some standard thermodynamic processes, commonly used in practice, these are

- (1) Isochoric or Isometric Process
- (2) Isobaric Process
- (3) Isothermal Process
- (4) Adiabatic Process

We first discuss the above processes in detail, then we'll discuss about, some other thermodynamic processes as a special case.

3.5.1 Isochoric or Isometric Process

In this process volume of gas remains constant during heating or state changing process on a gas. Thus during the heating process, we have

$$dV = 0 \quad [\text{During the process } V = \text{constant}]$$

$$\text{and} \quad \Delta V = 0 \quad [\text{As } V_1 = V_2]$$

As no change in volume is taking place, work done by the gas in the process is zero. Some time in few processes $\Delta V = 0$ but $dV \neq 0$. We can not treat these processes as isochoric because

in such processes initial and final volume of the gas are same but during the process, volume change takes place. We can differentiate these processes easily by looking carefully on their PV diagrams as shown in figure-3.29.

Figure-3.29(a) shows a PV diagram of an isochoric process in which, when a gas is carried from state-1 to state-2, its volume V remains constant during heating, only increase in pressure takes place as shown and the effective area below the PV -curve is also zero which shows no work is done during the process.

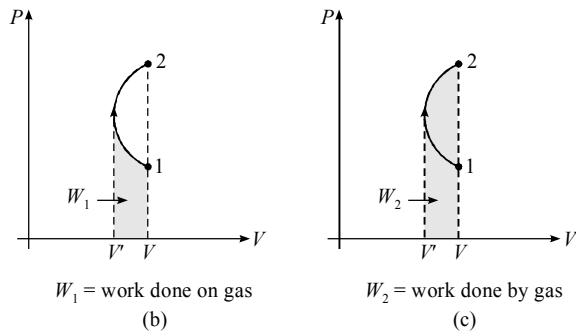
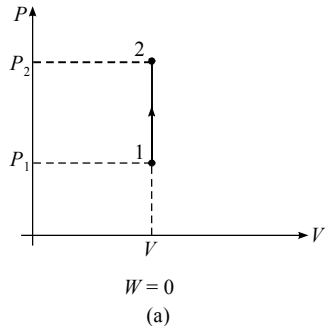


Figure 3.29

Figure-3.29(b) and (c) shows a process in which the state of gas is changed from same state-1 to state-2 and the volume of gas at both the states is same V . But we can see that during the process first the volume of gas is compressed from V to V' then it is expanded from V' to V . Figure-3.29(b) shows the area below the graph during the process when gas is compressed from volume V to V' . This area represents the work W_1 done on the gas (-ve) upto this intermediate state with volume V' from initial state-1. Figure-3.29(c) shows the area below the graph during the process when gas is expanded from this intermediate state to the final state to restore the volume V of the gas. This area during expansion of gas represents the work W_2 done by the gas from this intermediate state with volume V' to the final state-2. It is clear that $W_2 > W_1$ hence the total work in this process is done by the gas and is given as

$$W_{\text{gas}} = W_2 - W_1$$

Although in the process $\Delta V = 0$ but the process is not isochoric.

Heat supplied in Isochoric Process

If n moles of a gas is heated from temperature T_1 to T_2 keeping its volume constant during heating, the total amount of heat supplied to the gas can be obtained by using molar heat capacity of gas at constant volume as

$$dQ = n C_V dT$$

and

$$Q = \int dQ = \int_{T_1}^{T_2} n C_V dT$$

$$Q = n C_V (T_2 - T_1)$$

$$= \frac{n R}{\gamma - 1} (T_2 - T_1) \quad \dots (3.29)$$

Work done in Isochoric Process

As discussed earlier, no change in volume implies no work done, thus

$$W = 0 \quad \dots (3.30)$$

Change in Internal Energy in Isochoric Process

As discussed earlier, internal energy is a state variable so in all process if a gas is heated from temperature T_1 to T_2 , change in internal energy is given by

$$\Delta U = \frac{f}{2} n R \Delta T = n C_V \Delta T$$

or

$$\Delta U = \frac{n R}{\gamma - 1} (T_2 - T_1) \quad \dots (3.31)$$

This is same as equation-(3.29) because according to first law of thermodynamics if no work is done in a process, whole of supplied heat to a gas will appear as increase in its internal energy.

As we know internal energy of a gas is a state function, in all the process the change in internal energy is always given by equation-(3.31). As in isochoric process, volume of gas during the process remains constant from gas law the pressure and temperature of gas between its initial and final state can be related as

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \quad \dots (3.32)$$

3.5.2 Isobaric Process

In this process during change of state of a gas, its pressure remains constant. Thus

$$dP = 0$$

and

$$\Delta P = 0$$

Again some process can be there in which $\Delta P = 0$ but $dP \neq 0$. Such process can not be taken as isobaric process. We can discuss these by their PV - diagrams shown in figure-3.30. Figure-3.30(a) represents an isobaric process in which a gas is taken from state-1 to state-2 and during the process only volume of gas is being changed from V_1 to V_2 , pressure remains constant at P . Thus the area below the curve $P(V_2 - V_1)$ gives the work done in the process. But in figure-3.30(b) and (c), the gas is taken from same state-1 to state-2 by different paths in which during the process, the pressure of gas is not constant thus in these processes, even $\Delta P = 0$ but $dP \neq 0$ and due to this work done or area below these PV curves are either less or more than the case when the pressure remains constant i.e. in an isobaric process.

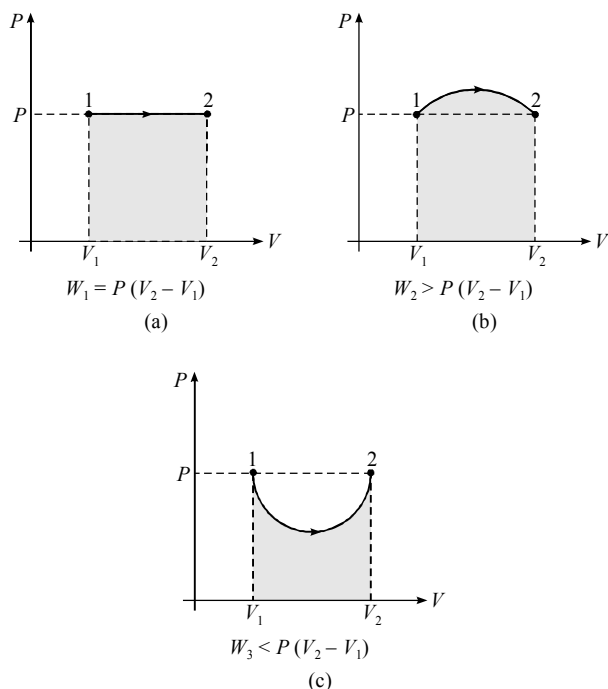


Figure 3.30

Heat Supplied in Isobaric Process

If n moles of a gas is heated at constant pressure from temperature T_1 to T_2 the amount of heat supplied can be easily calculated by using molar heat capacity of gas at constant pressure, thus

$$dQ = n C_p dT$$

$$Q = \int dQ = \int_{T_1}^{T_2} n C_p dT$$

or $Q = n C_p (T_2 - T_1)$

or $Q = \frac{n \gamma R}{\gamma - 1} (T_2 - T_1) \quad \dots (3.33)$

Work Done by gas in Isobaric Process

At the beginning of chapter we've discussed that when a gas is heated from volume V_1 to V_2 , work done can be calculated as

$$W = \int dW = \int_{V_1}^{V_2} P dV$$

As pressure of gas is constant, we have

$$W = \int_{V_1}^{V_2} P dV = \int_{T_1}^{T_2} n R dT$$

[As $PdV = nRdT$ for constant pressure]

or $W = P(V_2 - V_1)$
 $= n R (T_2 - T_1) \quad \dots (3.34)$

Change in Internal Energy in Isobaric Process

If n moles of a gas is heated from temperature T_1 to T_2 , as being a state variable, change in internal energy can be given as

$$\Delta U = n C_v (T_2 - T_1)$$

or $\Delta U = \frac{n R}{\gamma - 1} (T_2 - T_1) \quad \dots (3.35)$

From equation-(3.33), (3.34) and (3.35) we can also verify that these three thermodynamic variables are satisfying first law of thermodynamics.

3.5.3 Isothermal Process

In this process temperature of gas remains constant during heating. Thus during the process

$$dT = 0$$

and $\Delta T = 0$

If temperature of gas does not change then it implies that there is no change in internal energy of gas during the process thus in an isothermal process, we also have

$$dU = 0$$

and $\Delta U = 0$

As no change is there in internal energy of a gas, then according to first law of thermodynamics we can say that the total amount of heat supplied to a gas is used in doing work against surrounding i.e. in expending the gas.

Indicator diagrams for an Isothermal Process

To plot an indicator diagram we must require a process equation which can be given by gas law, in case of an isothermal process. According to gas law

$$PV = nRT$$

As in isothermal process gas temperature remains constant, thus, we have

$$P \propto \frac{1}{V}$$

or
$$P = \frac{k}{V} \quad \dots (3.36)$$

Thus according to equation-(3.35), it is clear that PV curve on an indicator diagram is a rectangular hyperbola as shown in figure-3.31. Figure-3.32 shows a gas undergoes several isothermal processes at different temperatures. Here we can see that all curves for these processes are almost parallel to each other and no two curves can intersect as these are at different temperatures these curves are called “*Isotherms*”. If a gas undergone a process in which $\Delta T = 0$ but $dT \neq 0$ then it is obvious that during the process gas temperature is changing. The PV curve for such a process must intersect with the series of isotherms shown in figure-3.32 such a process is shown in figure-3.33 this type of a process can never be taken as isothermal process, however in this process also internal energy of gas in its initial and final state are same.

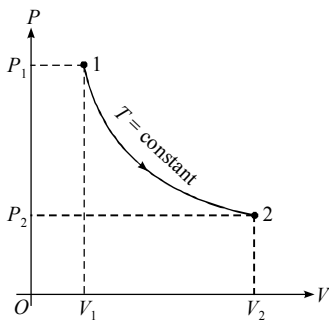


Figure 3.31

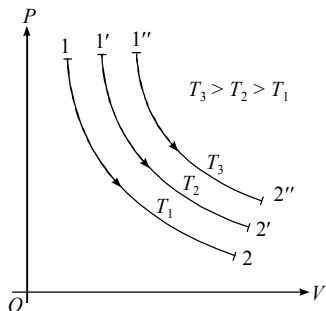


Figure 3.32

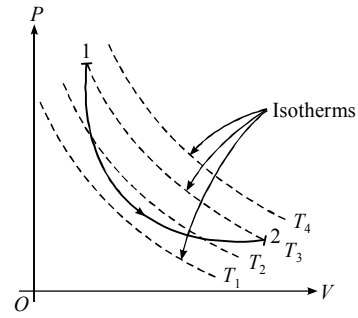


Figure 3.33

Heat Supplied and Work done in Isothermal Process

As discussed, in isothermal process temperature of gas does not change thus the total heat supplied to the gas is used in doing work against surrounding. If during the process volume of gas increases from V_1 to V_2 then pressure also very simultaneously with it and as temperature remains constant we can use Boyle's law to relate pressure and volume of the gas as

$$P_1 V_1 = P_2 V_2 \quad \dots (3.37)$$

[If P_1 and P_2 are the initial and final pressures of gas]

Work done by the gas can be calculated as

$$W = \int_{V_1}^{V_2} P dV$$

$$W = \int_{V_1}^{V_2} \frac{nRT}{V} dV$$

[As $P = \frac{nRT}{V}$ at every intermediate state of gas during heating]

or
$$W = nRT \ln \left(\frac{V_2}{V_1} \right)$$

$$= nRT \ln \left(\frac{P_1}{P_2} \right) \quad \dots (3.38)$$

As change in internal energy in the process is zero, we have heat supplied

$$Q = W = nRT \ln \left(\frac{V_2}{V_1} \right)$$

$$= nRT \ln \left(\frac{P_1}{P_2} \right)$$

Here we can also write

$$P_1 V_1 = P_2 V_2 = n R T \quad [\text{As } T = \text{constant}]$$

$$\begin{aligned} Q = W &= P_1 V_1 \ln \left(\frac{V_2}{V_1} \right) \\ &= P_2 V_2 \ln \left(\frac{V_2}{V_1} \right) \quad \dots (3.39) \end{aligned}$$

3.5.4 Adiabatic Process

In this process no heat is supplied to the gaseous system during the process. More precisely the system remains thermally insulated so that no heat can be given to it or can be taken out from it during the process. Thus here $\Delta Q = 0$ as well as $dQ = 0$. If no heat is supplied to a gas then according to first law of thermodynamics, during the process we have

$$dW + dU = 0$$

$$\text{or} \quad dW = -dU \quad \dots (3.40)$$

Equation-(3.40) shows that if a gas expands in such a process or work is done by the gas, then by the same amount internal energy of gas decreases. It is obvious that if gas is doing work against surrounding then the amount of energy required in this work is extracted from the internal energy of gas as there is no external heat supply. This is called adiabatic expansion and in adiabatic expansion always the gas is cooled. Similarly when some external work is done on the gas, by the same amount its internal energy must increase as there is no rejection of heat in adiabatic process. Thus in adiabatic compression always gas temperature increases.

Indicator Diagram for an adiabatic process

To plot an indicator diagram we must have the relation in pressure and volume of a gas undergoing the process at one of its intermediate state during the process or the process equation for the process. In previous three standard thermodynamic processes, the process equations are directly given by the gas laws. Now we first discuss how to obtain the process equation in a general thermodynamic process.

How to derive the process equation for a thermodynamic process

In a general thermodynamic process to derive its process equation, we generally try to develop a relation in heat, work and change in internal energy for the process and then solve this relation with differential form of gas law to obtain the desired relation in any two of the thermodynamic parameters

of a gas pressure, volume and its temperature. For example first we derive the process equation for an adiabatic process.

For adiabatic process the differential form of first law of thermodynamics can be written as

$$dW = -dU$$

$$PdV = -nC_v dT$$

$$\text{or} \quad PdV = -\frac{nR}{\gamma-1} dT \quad \dots (3.41)$$

Equation-(3.41) is the differential form of first law of thermodynamics in terms of gas parameters pressure, volume and temperature.

Differential form of gas law can be written as

$$PdV + VdP = nRdT \quad \dots (3.42)$$

Now from equation-(3.41) and (3.42), we get

$$PdV + VdP = (1-\gamma)PdV$$

$$\text{or} \quad VdP = -\gamma PdV \quad \dots (3.43)$$

$$\text{or} \quad \int \frac{dP}{P} = -\gamma \int \frac{dV}{V}$$

$$\text{or} \quad \ln P = -\gamma \ln V + C$$

[C $\rightarrow$ constant of integration]

$$\text{or} \quad \ln PV^\gamma = C$$

$$\text{or} \quad PV^\gamma = \text{constant} \quad \dots (3.44)$$

$$\text{or} \quad P \propto \frac{1}{V^\gamma}$$

Equation-(3.44) shows how pressure of gas vary with its volume during the process. From equation-(3.43) we can get the slope of PV curve plotted according to equation-(3.44) as

$$\frac{dP}{dV} = -\gamma \left(\frac{P}{V} \right) \quad \dots (3.45)$$

Equation-(3.45) gives the slope of the PV -curve plotted for a gas undergone an adiabatic process. Similarly we can find the slope of an isothermal curve by differential form of gas law as

$$PdV + VdP = 0$$

[As $T = \text{constant}$ in an isothermal process]

$$\text{or} \quad \frac{dP}{dV} = -\left(\frac{P}{V} \right) \quad \dots (3.46)$$

Comparing this with equation-(3.45). We can see that in magnitude slope of an adiabatic curve is γ times more than the slope of an isothermal curve as

$$\left| \frac{dP}{dV} \right|_{\text{adiabatic process}} = \gamma \left| \frac{dP}{dV} \right|_{\text{isothermal process}} \quad \dots (3.47)$$

From equation-(3.47) we can state qualitatively that an adiabatic curve is always steeper than an isothermal curve. If we note the values of γ for a gas then it is maximum for a monoatomic gas. If we compare the isothermal and adiabatic curves for different types of gases starting with same initial state of a gas, we can see that the curve of a monoatomic gas in adiabatic process is the steepest and the curve for an isothermal process (independent from γ) for a gas is having least slope. Figure-3.34 shows the comparison. We can also note that the work done, the area below the $P-V$ curve is minimum for the steepest graph.

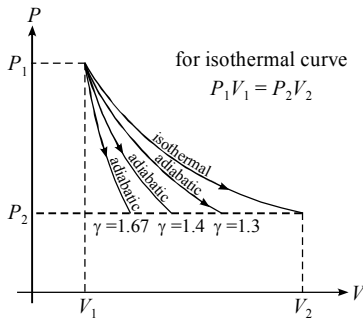


Figure 3.34

We have derived that the process equation for an adiabatic process, which can be written as

$$PV^\gamma = \text{constant} \quad \dots (3.48)$$

From gas law we can write

$$P = \frac{nRT}{V}$$

Thus $\frac{nRT}{V} V^\gamma = \text{constant}$

or $TV^{\gamma-1} = \text{constant} \quad \dots (3.49)$

Similarly volume of gas can be written as

$$V = \frac{nRT}{P}$$

or $TP^{1-\gamma} = \text{constant} \quad \dots (3.50)$

Equation-(3.48), (3.49) and (3.50) represent the same process equation for an adiabatic process in terms of different gas parameters.

One more important point students should keep in mind that in nature if a process occurs very fast, like bursting of a tyre,

sudden compression or expansion of a gas, these are regarded as approximately an adiabatic process as flow of heat is a slow and gradual process. It is assumed that in sudden thermodynamic changes, there is no sufficient time available so hardly any heat flow can take place. Similarly if a thermodynamic process is very slow and it is given that the system is in good thermal contact with the surrounding then in such processes it is assumed that during the process gas remains in thermal equilibrium with the surrounding and thus it can be regarded as an isothermal process.

Heat Supplied in Adiabatic Process

We've discussed that in an adiabatic process no heat flow takes place between system and surrounding during the process (expansion or compression) thus

$$Q=0$$

Work done and change in internal energy in adiabatic process

As already discussed, according to first law of thermodynamics the amount of work done by gas in an adiabatic process is equal to the negative of change in internal energy between final and initial state of the process. We know that internal energy is a state function thus does not depend as to how the state of a gas is changed. If gas temperatures are T_1 and T_2 before and after the process or in its initial and final states then the change (increase) in internal energy can be given as

$$\begin{aligned} \Delta U &= \int dU = \int_{T_1}^{T_2} \frac{nR}{\gamma-1} dT \\ &= \frac{nR}{\gamma-1} (T_2 - T_1) \end{aligned}$$

Thus is adiabatic process work done by the gas can be written as

$$W = -\Delta U = -\frac{nR}{\gamma-1} (T_2 - T_1)$$

or

$$W = \left(\frac{P_1 V_1 - P_2 V_2}{\gamma-1} \right) \quad \dots (3.51)$$

$$[\text{As } P_1 V_1 = nRT_1 \text{ and } P_2 V_2 = nRT_2]$$

These four processes we've discussed are the most common processes realized in practice. But initially we've discussed that there can be infinite ways of heating a gas are there thus there is no limitation on number of thermodynamic processes for a gas. Now we first take few examples to understand the above discussed processes in details then we will discuss a

general thermodynamic process named polytropic process, there after we'll see that the above four processes are also the special cases of polytropic process.

Illustrative Example 3.20

The volume of an ideal diatomic gas with $\gamma = 1.5$ is changed adiabatically from 16 litre to 12 litre. Find the ratio of the final and initial pressures and temperatures.

Solution

We know in an adiabatic process pressure and volume of different states of a gas are related as

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

Thus here
$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^\gamma$$

or
$$= \left(\frac{16}{12} \right)^{1.5} = \left(\frac{4}{3} \right)^{1.5} = 1.533$$

Similarly in an adiabatic process gas volume and temperature of its different states are related as

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

or
$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{\gamma-1}$$

$$= \left(\frac{16}{12} \right)^{0.5} = \left(\frac{4}{3} \right)^{0.5} = 1.153$$

Illustrative Example 3.21

A sample of diatomic gas with $\gamma = 1.5$ is compressed from a volume of 1600 cc to 400 cc adiabatically. The initial pressure of gas was 1.5×10^5 Pa. Find the final pressure and work done by the gas in the process.

Solution

We know in an adiabatic process, pressure and volume of gas of its different states are related as

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

or
$$P_2 = \left(\frac{V_1}{V_2} \right)^\gamma \cdot P_1$$

$$= \left(\frac{1600}{400} \right)^{1.5} \times 1.5 \times 10^5$$

$$= (4)^{1.5} \times 1.5 \times 10^5$$

$$= 1.2 \times 10^6 \text{ Pa}$$

For an adiabatic process work done by a gas is given as

$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

$$= \frac{1.5 \times 10^5 \times 1600 \times 10^{-6} - 1.2 \times 10^6 \times 400 \times 10^{-6}}{1.5 - 1}$$

$$= \frac{240 - 480}{0.5} = -480 \text{ J}$$

Here work done by gas comes out a negative value thus we can state that as gas is being compressed, work is done on the gas and so work done by gas is -480 J .

Illustrative Example 3.22

Two moles of a certain ideal gas at temperature $T_0 = 300 \text{ K}$ were cooled isochorically so that the gas pressure reduced $\eta = 2.0$ times. Then, as a result of the isobaric process, the gas expanded till its temperature get back to the initial value. Find the total amount of heat absorbed by the gas in this process.

Solution

In the first process, under isochoric process $W = 0$ (as $\Delta V = 0$). From gas law, if the pressure is reduced to η times, then the temperature is also reduced to η times i.e., the new temperature becomes T_0/η .

Thus from first law of thermodynamics, we have

$$Q_1 = \Delta U_1 = n C_V \Delta T = \frac{n R}{\gamma - 1} \Delta T$$

or
$$Q_1 = \frac{n R}{\gamma - 1} \left[\frac{T_0}{\eta} - T_0 \right] = \frac{n R T_0 (1 - \eta)}{\eta (\gamma - 1)}$$

During second process (under isobaric process), workdone is equal to $P \Delta V = n R \Delta T$. And from first law of thermodynamics, we have

$$Q_2 = \Delta U_2 + W_2 = \frac{n R \Delta T}{\gamma - 1} + n R \Delta T$$

$$= n R \Delta T \left[\frac{1}{\gamma - 1} + 1 \right] = n R \Delta T \left[\frac{\gamma}{\gamma - 1} \right]$$

$$= \frac{n R \gamma}{\gamma - 1} \left[T_0 - \frac{T_0}{\eta} \right] = \frac{n R \gamma (\eta - 1)}{\eta (\gamma - 1)}$$

Now total amount of heat supplied is

$$\begin{aligned} Q &= Q_1 + Q_2 \\ &= \frac{n R T_0 (1 - \eta)}{\eta(\gamma - 1)} + \frac{n R \gamma (\eta - 1)}{\eta(\gamma - 1)} \\ &= n R T_0 \left(1 - \frac{1}{\eta} \right) \end{aligned}$$

Here we have $n = 2, R = 8.3, T_0 = 300 \text{ K}$ and $\eta = 2$

$$\text{Thus } Q = 2 \times 8.3 \times 300 \left(1 - \frac{1}{2} \right) \text{ J}$$

$$\text{or } = 2490 \text{ J} = 2.5 \text{ kJ.}$$

Illustrative Example 3.23

One mole of a gas is isothermally expanded at 27°C till the volume is doubled. Then it is adiabatically compressed to its original volume. Find the total work done.

($\gamma = 1.4$ and $R = 8.4 \text{ joule/mole/}^\circ\text{K}$).

Solution

In case of isothermal expansion, the workdone is given by

$$dW_1 = R T \cdot \ln(V_f/V_i)$$

Here $R = 8.314, T = 27^\circ\text{C} = 273 + 27 = 300^\circ\text{K}$,

$$(V_f/V_i) = 2$$

$$\begin{aligned} \text{or } dW_1 &= 8.314 \times 300 \times \ln(2) \\ &= 8.314 \times 300 \times 0.693 \\ &= 1728.48 \text{ J.} \end{aligned}$$

Now the gas is adiabatically compressed to its original volume. Initially at the beginning of adiabatic compression, the temperature of the gas is 300 K and at the end of adiabatic compression, the temperature becomes T_2 because the temperature is changed. The initial volume of the gas is $2V_1$ and after compression it again becomes the original volume i.e., V_1 . For an adiabatic process, we know temperature and volume are related as

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\text{or } 300 \times (2 V_1)^{\gamma-1} = T_2 \times (V_1)^{\gamma-1}$$

$$\text{or } 300 \times (2)^{1.4-1} = T_2 (1)^{1.4-1}$$

$$\text{or } T_2 = 300 \times (2)^{0.4} = 395.85 \text{ K}$$

Workdone during adiabatic process is

$$W_2 = \frac{nR}{(\gamma-1)} (T_2 - T_1)$$

$$= \frac{8.314}{-(1.4-1)} (395.85 - 300)$$

$$= -1992.24 \text{ J}$$

$$\text{Total workdone} = -1728.47 - 1992.24$$

$$= -263.77 \text{ J.}$$

Illustrative Example 3.24

Calculate the work done when one mole of an ideal monoatomic gas is compressed adiabatically. The initial pressure and volume of the gas are 10^5 N/m^2 and 6 litre respectively. The final volume of the gas is 2 litres . Molar specific heat of the gas at constant volume is $3R/2$.

Solution

For an adiabatic change

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$\text{or } P_2 = P_1 \left(\frac{V_1}{V_2} \right)^\gamma$$

Here $P_1 = 10^5 \text{ N/m}^2$, $V_1 = 6 \text{ litre}$, $V_2 = 2 \text{ litre}$ and for a monoatomic gas, we have

$$C_V = \frac{3R}{2}$$

$$\text{and } C_P = \frac{3R}{2} + R = \frac{5R}{2}$$

$$\text{or } \gamma = \frac{C_P}{C_V} = \frac{5R/2}{3R/2} = \frac{5}{3}$$

$$\text{Thus } P_2 = 10^5 \times \left(\frac{6}{2} \right)^{5/3} = 10^5 \times (3)^{5/3}$$

We know that work done on gas ΔW in adiabatic change is given by

$$\begin{aligned} \Delta W &= \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} \\ &= \frac{10^5 \times (3)^{5/3} \times (2 \times 10^{-3}) - 10^5 \times (6 \times 10^{-3})}{(5/3) - 1} \end{aligned}$$

$$[\text{Here } V_2 = 2 \text{ litre} = 2 \times 10^{-3} \text{ m}^3 \text{ and } V_1 = 6 \text{ litre} = 6 \times 10^{-3} \text{ m}^3]$$

$$= \frac{2 \times 10^2 [(3)^{5/3} - 3]}{(2/3)}$$

$$= \frac{2 \times 10^2 [6.19 - 3]}{(2/3)} = 3 \times 10^2 \times (3.19)$$

$$= 957 \text{ J.}$$

Illustrative Example 3.25

An ideal gas at 75 cm mercury pressure is compressed isothermally until its volume is reduced to three quarters of its original volume. It is then allowed to expand adiabatically to a volume 20% greater than its original volume. If the initial temperature of the gas is 17 °C, calculate the final pressure and temperature ($\gamma = 1.5$).

Solution

First of all the gas is compressed isothermally. Using Boyle's law

$$P_1 V_1 = P_2 V_2$$

or
$$P_2 = (P_1 V_1 / V_2)$$

Here $P_1 = 75$ cm of mercury and $V_2 = \frac{3}{4} V_1$

Thus
$$P_2 = \frac{75 V_1}{(3/4) V_1} = 100 \text{ cm of mercury}$$

The gas is now expanded adiabatically to 20% greater of its original value. Under adiabatic change the pressure and volume of gas are related as

$$P_2 V_2^\gamma = P_3 V_3^\gamma$$

or
$$P_3 = P_2 \left(\frac{V_2}{V_3} \right)^\gamma$$

Here $V_2 = \frac{3}{4} V_1$ and $V_3 = \frac{120}{100} V_1$

Thus
$$\begin{aligned} P_3 &= 100 \times \left(\frac{3 V_1}{4} \right)^{1.5} \times \left(\frac{100}{120 V_1} \right)^{1.5} \\ &= 100 \times \left(\frac{3}{4} \right)^{1.5} \times \left(\frac{5}{6} \right)^{1.5} \\ &= 100 \times \left(\frac{5}{8} \right)^{1.5} \\ &= 100 \times 0.494 = 49.4 \text{ cm of mercury} \end{aligned}$$

Let the final temperature after adiabatic change be T_3 then from the relation of temperature and volume in an adiabatic process, we have

Now
$$\begin{aligned} T_2 V_2^{\gamma-1} &= T_3 V_3^{\gamma-1} \\ T_2 = 17^\circ\text{C} &= (273 + 27) = 290 \text{ K} \end{aligned}$$

Now

$$\begin{aligned} T_3 &= T_2 \left(\frac{V_2}{V_3} \right)^{\gamma-1} \\ &= 290 \times \left(\frac{3 V_1}{4} \right)^{1.5-1} \times \left(\frac{100}{120 V_1} \right)^{1.5-1} \\ &= 290 \times \left(\frac{5}{8} \right)^{0.5} = 229.3 \text{ K} \end{aligned}$$

Hence the final temperature will be -43.7°C

Illustrative Example 3.26

One mole of a certain ideal gas is contained under a weightless piston of a vertical cylinder at a temperature T . The space over the piston opens into the atmosphere. What work has to be performed in order to increase isothermally the gas volume under the piston η times by slowly raising the piston? The friction of the piston against the cylinder walls is negligibly small.

Solution

As gas expands from its initial volume (say V) to η time of initial volume (ηV) we can say that work is done by the gas in this expansion. As this expansion is isothermal, work done by the gas is simply given as

$$W = nRT \ln \left(\frac{V_2}{V_1} \right)$$

Here $n = 1$ mole, $V_2 = \eta V_1$ thus

$$W = RT \ln \eta \quad \dots (3.52)$$

As gas is expanding, it is doing work and some external agent is pulling the piston up to increase the volume of gas we can say that gas is supporting the external agent and atmosphere is opposing this expansion thus work is done on atmosphere. As atmospheric pressure to be constant, and change in volume of gas is from V to ηV , the work done on atmosphere is

$$W_{\text{atm}} = -P(\Delta V)$$

$$= PV(1 - \eta)$$

Initially the piston was in equilibrium thus gas pressure was equal to atmospheric pressure so, we have

$$PV = RT \quad [\text{As } n = 1 \text{ mole}]$$

or
$$W_{\text{atm}} = -RT(\eta - 1) \quad \dots (3.53)$$

If W_{ext} is the work done by external agent in pulling the gas here we must have

$$|W_{ext}| + |W_{gas}| = |W_{atm}| \quad \dots(3.54)$$

$$W_{ext} = W_{atm} - W_{gas}$$

$$W_{ext} = RT(\eta - 1) - RT \ln \eta$$

Here equation-(3.54) the basic equation we've encountered several times in mechanics as if some work is being done then at least two objects are to be involved in the system, one who is doing the work (energy supplier) and the other on which work is done (energy acceptor). Only one body can never do any work. This we'll discuss in next section and then we'll take some more examples on such concepts.

3.6 Free Expansion of a Gas

When heat is added to a thermodynamics system, it undergoes a change of state which depends on the path from the initial state to final state. We take an example of it. Figure-3.35 shows an ideal gas contained in a cylinder with a piston having an initial volume of 5 litre at temperature 300 K. We want to increase its volume from 5 litre to 8 litre. We put this cylinder on to a heater at same temperature 300 K. Heater supplies heat to the gas and the gas expands slowly and after expanding in the slow and isobaric manner, the gas reaches its final volume of 8 litre. In this process gas absorbs a definite amount of heat.

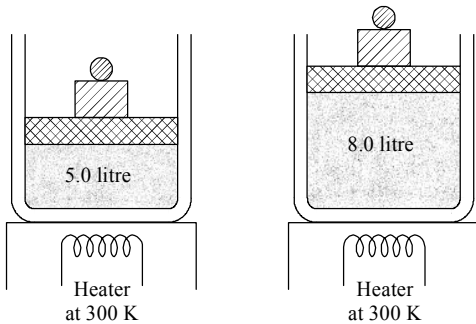


Figure 3.35

Figure-3.36 shows a different process leading to similar final state. In this case an insulated cylinder is divided in two parts by a thin massless fixed piston. The volume of lower compartment is 5.0 litre and that of upper is 3.0 litre. In the lower part we place the same amount of the same gas used in previous case at same temperature 300 K. The initial state is same as before. In the upper part there is vacuum or no pressure region. If we release the piston. It undergoes a sudden expansion due to vacuum on other side and the gas fills the whole space of 8.0 litre volume of cylinder rapidly. In this expansion we can see that no heat is supplied to the gas as

walls are insulated. During such an expansion gas does no work because it is not pushing anything during expansion or in this case there is nothing (vacuum) on which work can be done. Thus if no heat is supplied and no work is done by the gas, its internal energy will also remain constant. Such an expansion of gas is called “free expansion” of gas. As during free expansion temperature of gas remains constant, Boyle’s law will remain valid in such cases; thus

$$P_1 V_1 = P_2 V_2$$

Here subscript 1 and 2 are used for the state of gas before and after the free expansion.

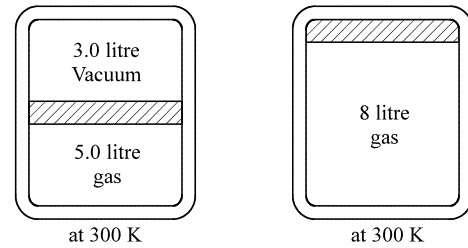


Figure 3.36

Illustrative Example 3.27

Figure-3.37 shows a cylindrical container of volume V . Whose walls are adiabatic. Initially a light adiabatic piston divides the container in two equal parts as shown. In left part there is n moles of an ideal gas with adiabatic exponent γ is filled at temperature T_A and in other part there is vacuum. If the piston is released, the gas fills the whole container uniformly. Find the final pressure and temperature of gas. Now if the piston is slowly displaced externally back to its initial position. Find the final pressure and temperature of gas.

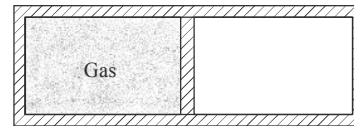


Figure 3.37

Solution

When the piston is released the gas expands to fill the complete volume of container. As there is nothing in the other part of cylinder, this is the free expansion of gas hence no work is done by the gas and as container is thermally insulated from surrounding, the gas temperature remains constant and thus according to Boyle’s Law as the volume of gas is doubled so its final pressure is reduced to half. Thus

$$P_f = \frac{P_i}{2}$$

Initial pressure P_i of gas can be given by gas law as

$$P_i = \frac{nRT}{V/2}$$

Thus, we have $P_f = \frac{P_i}{2} = \frac{nRT}{V}$

Now if piston is displaced back to its initial position, this is adiabatic compression of gas in which gas volume decreases to half i.e. $V/2$. As we know in an adiabatic process, pressure and volume of gas in different states are related as

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

Here $P_1 = \frac{nRT}{V}$, $V_1 = V$

and $V_2 = \frac{V}{2}$ thus

$$\begin{aligned} P_2 &= P_1 \left(\frac{V_1}{V_2} \right)^\gamma \\ &= P_1 (2)^\gamma \\ &= (2)^\gamma \frac{nRT}{V} \end{aligned}$$

Similarly in adiabatic process, gas volume and temperature in different states are related as

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

or $T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1}$

Here $T_1 = T$, $V_1 = V$

and $V_2 = \frac{V}{2}$ thus

$$T_2 = T(2)^{\gamma-1}$$

Illustrative Example 3.28

There are two thermally insulated vessel. One with 0.025 moles of helium and other with n moles of hydrogen. Initially both the gases are at room temperature. Now equal amount of heat is supplied to both the vessels. It is found that in both the gases temperature rises by same amount. Find the number of moles of hydrogen in second vessel

Solution

As the gases are enclosed in closed vessels, the heating can be taken as isochoric heating and as heat supplied to both

vessels are same, we have

$$Q = n_1 C_{V1} \Delta T = n_2 C_{V2} \Delta T$$

or $0.025 \times \frac{3}{2} R \Delta T = n \frac{5}{2} R \Delta T$

[As for He, $C_{V1} = \frac{3}{2} R$ and for H_2 , $C_{V2} = \frac{5}{2} R$]

or $n = \frac{0.025 \times 3}{5} = 0.015 \text{ mole}$

Illustrative Example 3.29

There are two vessels. Each of them contains one mole of a monatomic ideal gas. Initial volume of the gas in each vessel is $8.3 \times 10^{-3} \text{ m}^3$ at 27°C . Equal amount of heat is supplied to each vessel. In one of the vessels, the volume of the gas is doubled without change in its internal energy, whereas the volume of the gas is held constant in the second vessel. The vessels are now connected to allow free mixing of the gas. Find the final temperature and pressure of the combined gas system.

Solution

According to first law of thermodynamics

$$\Delta Q = \Delta U + \Delta W$$

For vessel I, $\Delta U = 0$ [As no change in temperature]

or $\Delta Q = \Delta W$

or $Q = \int_{V_1}^{V_2} P dV$

From gas law, we have

$$\begin{aligned} Q &= \int_{V_1}^{V_2} \frac{nRT}{V} dV = nRT \ln \left(\frac{V_2}{V_1} \right) \\ &= nRT \ln(2) \quad [\text{As } V_2 = 2 V_1] \quad \dots (3.55) \end{aligned}$$

For second vessel, $\Delta W = 0$ (as volume is constant)

Thus heat supplied is given as

$$\begin{aligned} Q &= n C_V \Delta T = n \left(\frac{3}{2} R \right) \Delta T \quad \dots (3.56) \\ &[\text{As for monoatomic gas } C_V = \frac{3}{2} R] \end{aligned}$$

From equation-(3.55) and (3.56), we get

$$nRT \ln(2) = 2n \left(\frac{3}{2} R \right) \Delta T$$

or
$$\Delta T = \frac{2}{3} \times 300 \times 0.693 = 138.6 \text{ K}$$

This is the change in the temperature of the gas of second vessel.

Now the final temperature of the gas in second vessel is

$$T' = T + \Delta T = 300 + 138.6 = 438.6 \text{ K}$$

Let after mixing, T_f and P_f be the final temperature and pressure respectively. Then as moles are equal, we have

$$T_f = \frac{T + (T + \Delta T)}{2} = \frac{300 + 438.6}{2} = 369.3 \text{ K}$$

For a gas system from gas law, we have

$$P_f V_f = n R T_f$$

or
$$P_f = \frac{n R T_f}{V_f} = \frac{2 \times 8.3 \times 369.3}{2 \times 8.3 \times 10^{-3} + 8.3 \times 10^{-3}} = 2.46 \times 10^5 \text{ N/m}^2$$

Illustrative Example 3.30

Two moles of helium gas undergo a cyclic process as shown in figure-3.38. Assuming the gas to be ideal, calculate the following quantities in this process.

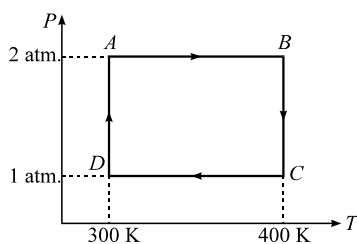


Figure 3.38

- The net change in the heat energy.
- The net work done.
- The net change in internal energy.

Solution

As we know in a cyclic process, the change in heat energy or heat supplied to the gas is equal to the net work done by the gas.

Here AB is isobaric process. Hence work done during this process from A to B is

$$W_{AB} = P(V_2 - V_1) = n R (T_2 - T_1)$$

or
$$W_{AB} = 2 \times 8.314 \times (400 - 300) = 1662.8 \text{ joule}$$

Work done during isothermal process from B to C

$$\begin{aligned} W_{BC} &= n R T_C \ln(V_2/V_1) = n R T_C \ln(P_1/P_2) \\ &= 2 \times 8.314 \times 400 \times \ln(2) \\ &= 2 \times 8.314 \times 400 \times 0.693 \\ &= 4610.2 \text{ joule} \end{aligned}$$

Workdone during isobaric process from C to D

$$\begin{aligned} W_{CD} &= n R (T_D - T_C) = 2 \times 8.314 \times (300 - 400) \\ &= -1662.8 \text{ joule} \end{aligned}$$

Workdone during isothermal process from D to A

$$\begin{aligned} W_{DA} &= n R T_D \ln(P_D/P_A) \\ &= n R T_D \ln(2) \\ &= -2 \times 8.314 \times 300 \times 0.693 \\ &= -3457.7 \text{ joule} \end{aligned}$$

Net workdone

$$\begin{aligned} &= W_{AB} + W_{BC} + W_{CD} + W_{DA} \\ &= 1662.8 + 4610.2 - 1662.5 - 3457.7 \\ &= 1152.5 \text{ joule} \end{aligned}$$

Now from first law of thermodynamics

$$\Delta Q = \Delta U + \Delta W$$

Here $\Delta U = 0$, thus we have

$$\Delta Q = \Delta W = 1152.5 \text{ joule}$$

So the heat given to the system is 1152.5 joule

As the gas returns to its original state, hence there is no change in internal energy.

Illustrative Example 3.31

Two moles of helium gas ($\gamma = 5/3$) are initially at temperature 27°C and occupy a volume of 20 litres. The gas is first expanded at constant pressure until the volume is doubled. Then it undergoes an adiabatic change until the temperature returns to its initial value.

- Sketch the process on a P - V diagram.
- What are the final volume and pressure of the gas ?
- What is the work done by the gas ?

Solution

For a perfect gas

$$P V = n R T$$

Given that $V = 20$ litres
 $= 20 \times 10^{-3} \text{ m}^3$
 $T = 27^\circ\text{C} = 300 \text{ K}$ and number of molecules
 $n = 2$

Thus initial pressure is given as

$$P = \frac{n R T}{V} = \frac{2 \times 8.3 \times 300}{20 \times 10^{-3}} \\ = 2.5 \times 10^5 \text{ N/m}^2$$

(i) Figure-3.39 shows the indicator diagram of the complete process.

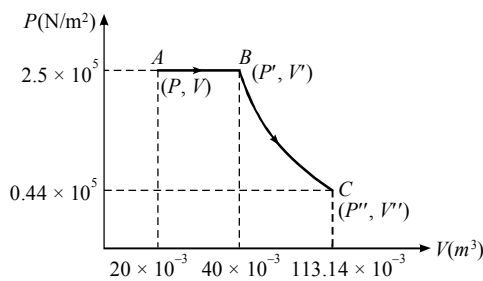


Figure 3.39

(ii) At point B,

Pressure $P' = P = 2.5 \times 10^5 \text{ N/m}^2$,
 $V' = 2 V = 40 \times 10^{-3} \text{ m}^3$

As pressure is constant in the process AB, making its volume doubled, its temperature will also be doubled.

Thus temperature at point B is $T' = 600 \text{ K}$

The gas now undergoes adiabatic expansion to cool down to

$$T'' = T = 300 \text{ K}$$

We know for an adiabatic process $T V^{\gamma-1} = \text{constant}$

$$T' (V')^{\gamma-1} = T'' (V'')^{\gamma-1} \\ \left(\frac{V''}{V'} \right) = \left(\frac{T'}{T''} \right)^{1/(\gamma-1)} = \left(\frac{600}{300} \right)^{1/(\frac{5}{3}-1)} \\ = (2)^{3/2} = 2\sqrt{2}$$

Thus final volume is

$$V'' = (2\sqrt{2}) V' \\ = 2 \times 1.414 \times 40 \times 10^{-3} \\ = 113.14 \times 10^{-3} \text{ m}^3$$

Similarly final pressure is given by process equation as

$$P' V'^{\gamma} = P'' V''^{\gamma}$$

$$\text{or } P'' = P' \left(\frac{V'}{V''} \right)^{\gamma} \\ = 2.5 \times 10^5 \times \left(\frac{40 \times 10^{-3}}{113.14 \times 10^{-3}} \right)^{5/3} \\ = 4.42 \times 10^4 \text{ Pa}$$

(iii) Work done under isobaric process AB is

$$\text{or } W_1 = P \Delta V \\ \text{or } W_1 = 2.5 \times 10^5 \times (40 - 20) \times 10^{-3} \\ \text{or } = 4980 \text{ J}$$

Workdone during adiabatic process BC is given as

$$\text{or } W_2 = \frac{n R}{\gamma - 1} [T_1 - T_2] \\ \text{or } = \frac{2 \times 8.3}{[1 - (5/3)]} [300 - 600] = 7470 \text{ J}$$

$$\text{Total work done} = W_1 + W_2 = 4980 + 7470 \\ = 12450 \text{ J}$$

Illustrative Example 3.32

Figure-3.40 shows on adiabatic cylindrical container of volume V_0 divided by an adiabatic smooth piston in two equal parts. An ideal gas ($C_P/C_V = \gamma$) is at a pressure P_1 and temperature T_1 in left part and gas at pressure P_2 and temperature T_2 in right part. The piston is slowly displaced and released at a position where it can stay in equilibrium. Find the final pressure, volume and temperature of the two parts.

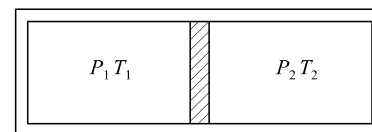


Figure 3.40

Solution

As finally the piston is in equilibrium, both the gases must be at same pressure P_f . Let the displacement of piston be in final state x and if A is the area of cross-section of the piston the final volumes of the left and right part finally can be given by figure-3.41 as

$$V_L = \frac{V_0}{2} + Ax$$

$$\text{and } V_R = \frac{V_0}{2} - Ax$$

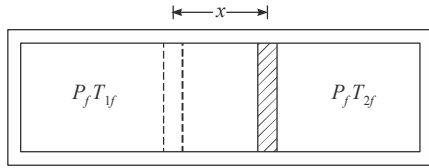


Figure 3.41

As it is given that the container walls and piston are adiabatic in left side the gas undergoes adiabatic expansion and on right side the gas undergoes adiabatic compressive. Thus we have for initial and final state of gas on left side

$$P_1 \left(\frac{V_0}{2} \right)^\gamma = P_f \left(\frac{V_0}{2} + Ax \right)^\gamma \quad \dots (3.57)$$

Similarly for gas on right side, we have

$$P_2 \left(\frac{V_0}{2} \right)^\gamma = P_f \left(\frac{V_0}{2} - Ax \right)^\gamma \quad \dots (3.58)$$

Dividing equation-(3.57) by (3.58), we get

$$\frac{P_1}{P_2} = \frac{\left(\frac{V_0}{2} + Ax \right)^\gamma}{\left(\frac{V_0}{2} - Ax \right)^\gamma}$$

$$\text{or} \quad \left(\frac{V_0}{2} - Ax \right) P_1^{1/\gamma} = \left(\frac{V_0}{2} + Ax \right) P_2^{1/\gamma}$$

$$\text{or} \quad Ax = \frac{V_0}{2} \frac{\left[P_1^{1/\gamma} - P_2^{1/\gamma} \right]}{\left[P_1^{1/\gamma} + P_2^{1/\gamma} \right]}$$

Now from equation-(3.57)

$$P_f = \frac{P_1 \left(\frac{V_0}{2} \right)^\gamma}{\left[\frac{V_0}{2} + Ax \right]^\gamma}$$

$$\text{or} \quad P_f = \frac{P_1 \left(\frac{V_0}{2} \right)^\gamma}{\left[\frac{V_0}{2} + \frac{V_0}{2} \frac{\left(P_1^{1/\gamma} - P_2^{1/\gamma} \right)}{\left(P_1^{1/\gamma} + P_2^{1/\gamma} \right)} \right]^\gamma}$$

$$\text{or} \quad P_f = \frac{P_1 \left(P_1^{1/\gamma} + P_2^{1/\gamma} \right)^\gamma}{(2)^\gamma P_1}$$

$$\text{or} \quad P_f = \left[\frac{P_1^{1/\gamma} + P_2^{1/\gamma}}{2} \right]^\gamma$$

Similarly the final temperatures of the two parts are T_{1f} and T_{2f} respectively then for left part, we have

$$T_1 \left(\frac{V_0}{2} \right)^{\gamma-1} = T_{1f} \left(\frac{V_0}{2} + Ax \right)^{\gamma-1} \quad \dots (3.59)$$

or

$$T_{1f} = \frac{T_1 \left(\frac{V_0}{2} \right)^{\gamma-1}}{\left[\frac{V_0}{2} + \frac{V_0}{2} \frac{\left(P_1^{1/\gamma} - P_2^{1/\gamma} \right)}{\left(P_1^{1/\gamma} + P_2^{1/\gamma} \right)} \right]^{\gamma-1}}$$

or

$$T_f = \frac{T_1}{P_1} \left[\frac{\left(P_1^{1/\gamma} + P_2^{1/\gamma} \right)}{2} \right]^{\gamma-1}$$

Similarly for right part of gas, we have

$$T_2 \left(\frac{V_0}{2} \right)^{\gamma-1} = T_{2f} \left(\frac{V_0}{2} - Ax \right)^{\gamma-1} \quad \dots (3.60)$$

or

$$T_{2f} = \frac{T_2 \left(\frac{V_0}{2} \right)^{\gamma-1}}{\left[\frac{V_0}{2} - \frac{V_0}{2} \frac{\left(P_1^{1/\gamma} - P_2^{1/\gamma} \right)}{\left(P_1^{1/\gamma} + P_2^{1/\gamma} \right)} \right]^{\gamma-1}}$$

or

$$T_f = \frac{T_2}{P_2} \left[\frac{\left(P_1^{1/\gamma} + P_2^{1/\gamma} \right)}{2} \right]^{\gamma-1}$$

Illustrative Example 3.33

A piston can freely move inside a horizontal cylinder closed from both ends. Initially, the piston separates the inside space of the cylinder into two equal parts each of volume V_0 , in which an ideal gas is contained under the same pressure P_0 and at the same temperature. What work has to be performed in order to increase isothermally the volume of one part of gas η times compared to that of the other by slowly moving the piston ?

Solution

As the piston is displaced externally some external work is done in the process. If piston is displaced toward right the gas on left side expands and does some work. Similarly gas on right is compressed and on it work is done. As discussed earlier, here we have

$$|W_{\text{gas in left part}}| + |W_{\text{ext}}| = |W_{\text{gas in right part}}| \quad \dots (3.61)$$

Here $|W_{\text{gas on left}}|$ = magnitude of work done by gas in left part

$|W_{\text{ext}}|$ = magnitude of work done by external agent

$|W_{\text{gas in right part}}|$ = magnitude of work done on gas in right part

It is given that initial volume of both the parts is V_0 and in the process final volume of one part is η times that of the other part. If the final volume of right part is V then that of left part will become ηV . As total volume of container is $2V_0$, then we have

$$V + \eta V = 2V_0$$

$$\text{or} \quad V = \frac{2V_0}{\eta + 1}$$

For gas in left part work done by gas in isothermal expansion is

$$W_{\text{by gas}} = nRT \ln \frac{V_2}{V_1}$$

$$\Rightarrow \quad = P_0 V_0 \ln \frac{\eta V}{V_0}$$

[As from gas law for gas on left part $P_0 V_0 = nRT$]

$$\Rightarrow \quad = P_0 V_0 \ln \left(\frac{2\eta}{\eta + 1} \right) \quad \dots (3.62)$$

Similarly for gas in right part, work done on the gas in isothermal compression is

$$W_{\text{on gas}} = nRT \ln \frac{V_2}{V_1}$$

$$\Rightarrow \quad = P_0 V_0 \ln \frac{V}{V_0}$$

$$\Rightarrow \quad = P_0 V_0 \ln \left(\frac{2}{\eta + 1} \right)$$

$$\Rightarrow \quad = -P_0 V_0 \ln \left(\frac{\eta + 1}{2} \right)$$

Now from equation-(3.61), (3.62) and (3.63), we have

$$\begin{aligned} |W_{\text{ext}}| &= |W_{\text{by gas in left part}}| + |W_{\text{on gas in right part}}| \\ &= -P_0 V_0 \ln \frac{2\eta}{\eta + 1} + P_0 V_0 \ln \left(\frac{\eta + 1}{2} \right) \\ &= P_0 V_0 \ln \left[\frac{(\eta + 1)^2}{4\eta} \right] \end{aligned}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Laws of Thermodynamics

Module Number - 17 to 28

Practice Exercise 3.3

(i) One mole of an ideal gas undergoes a process whose process equation is given as

$$P = \frac{P_0}{1 + \left(\frac{V_0}{V} \right)^2}$$

Here P_0 and V_0 are constant. When the volume of gas is changed from V_0 to $2V_0$, find the change in temperature of the gas.

$$\left[\frac{11 P_0 V_0}{10 R} \right]$$

(ii) For a thermodynamic system the pressure, volume and temperature are related as new gas law given as

$$P = \frac{\alpha T^2}{V}$$

Here α is a constant. Find the work done by the system in this process when pressure remains constant and its temperature changes from T_0 to $2T_0$.

$$[3 \alpha T_0^2]$$

(iii) Temperature of 1 mole of an ideal gas is increased from 300 K to 310 K under isochoric process. Heat supplied to the gas in this process is $25R$. What amount of work has to be done by the gas if temperature of the gas decreases from 310 K to 300 K adiabatically.

$$[25 R]$$

(iv) During the adiabatic expansion of 2 moles of a gas, the increase in internal energy was found to be equal to -100 J . Find the work done by the gas in the process.

$$[100 \text{ J}]$$

(v) One litre of an ideal gas ($\gamma = 1.5$) at 300 K temperature and 10^5 Pa pressure, is suddenly compressed to half of its original volume. Find the final temperature of the gas. It is then cooled to 300 K by isobaric process and then it is expanded isothermally to achieve its original volume of 1 litre. Calculate the work done by the gas in each process and also calculate the total work done in the cycle.

$$[424.26 \text{ K}, -82.84 \text{ J}, -41.4 \text{ J}, 103.95 \text{ J}, -20.29 \text{ J}]$$

(vi) Three identical diatomic gases ($\gamma = 1.5$) are enclosed in three identical containers but at different pressures and same temperatures. These gases are expanded to double their volumes. In first container the process is isothermal, in second container the process is adiabatic and in third container process is isobaric. If the final pressures are equal in the three containers, find the ratio of the initial pressures in the three containers.

$$[2 : 2\sqrt{2} : 1]$$

(vii) Calculate the work done when 1 mole of a perfect gas is compressed adiabatically. The initial pressure and volume of the gas are 10^5 N/m^2 and 6 litres respectively. The final volume is 2 litres. Molar specific heat of the gas at constant volume is $\frac{3R}{2}$.

$$[-974.07 \text{ J}]$$

(viii) An ideal di-atomic gas is heated at constant pressure such that it performs a work $W = 2.0 \text{ J}$. Find the amount of heat supplied.

$$[7 \text{ J}]$$

(ix) Two identical gases whose adiabatic exponent is γ are filled in two identical containers at equal pressures. In both the containers the volume of gas is doubled. In first container it is done by an isothermal process and in second container it is done by adiabatic process. Find the condition for which the work done by the gas in the two expansion process is same.

$$[1 - 2^{1-\gamma} = (\gamma - 1) \ln 2]$$

(x) A rectangular box shown in the figure-3.42 has a partition which can slide without friction along the length of the box. Initially each of the two chambers of the box has one mole of a monatomic gas $\left(\gamma = \frac{5}{3}\right)$ at a pressure P_0 , volume V_0 , and temperature T_0 . The chamber on the left is slowly heated by an electric heater. The walls of the box and the partition are thermally insulated. Heat loss through the lead wires is negligible. The gas in the left chamber expands, pushing the partition until the final pressure in both chambers becomes $\frac{243p_0}{32}$. Determine : (a) the final temperature of the gas in each chamber, (b) the work done by the gas in the right chamber.

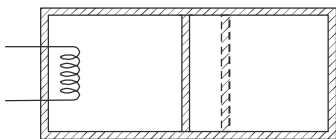


Figure 3.42

$$[(a) \frac{9}{4} T_0, \frac{207}{16} T_0 \text{ (b) } -\frac{15}{8} p_0 V_0]$$

(xi) One mole of oxygen, initially at temperature $T = 290 \text{ K}$ is compressed adiabatically so that its pressure increases $\eta = 10$ times. Find the final temperature and work done on it.

$$[559.90 \text{ K}, 5609.87 \text{ J}]$$

3.7 Polytropic Process

A polytropic process is one in which the molar specific heat of a gas during the process (heating) remains constant or does not change with the gas parameters pressure, volume or temperature. If for a polytropic process molar specific heat is taken C then according to first law of thermodynamics, we have

$$dQ = dU + dW$$

$$\text{or } n C dT = \frac{n R dT}{\gamma - 1} + P dV$$

$$\text{or } C = \frac{R}{\gamma - 1} + \frac{P dV}{n dT} \quad \dots (3.63)$$

From gas law

$$P dV + V dP = n R dT \quad \dots (3.64)$$

From (3.63) and (3.64)

$$C = \frac{R}{\gamma - 1} + \frac{R P dV}{P dV + V dP} \quad \dots (3.65)$$

$$\text{or } P dV \left(C - \frac{R}{\gamma - 1} - R \right) = - \left(C - \frac{R}{\gamma - 1} \right) V dP$$

$$\text{or } \frac{dV}{V} \left(\frac{m - R}{m} \right) = - \frac{dP}{P}$$

$$[\text{Where } m = C - \frac{R}{\gamma - 1} = \text{constant}]$$

Integrating the expression, we get

$$\left(\frac{m - R}{m} \right) \int \frac{dV}{V} = - \int \frac{dP}{P}$$

$$\text{or } \left(\frac{m - R}{m} \right) \ln V = - \ln P + C_1$$

$$[C_1 = \text{integration constant}]$$

$$\text{or } \ln P V^{\left(\frac{m - R}{m} \right)} = C_1$$

$$\text{or } P V^n = \text{constant} \quad \dots (3.66)$$

Where $n = \frac{m - R}{m}$ is another constant which is called *polytropic constant* whose value depends on γ of gas, gas

constant R and the molar specific heat C of the process which is taken a constant equation-(3.66) shows that if the process equation of a process is represented in the form of $PV^n = \text{constant}$ then it can be regarded as a polytropic process and the molar specific heat for a polytropic process can be defined by using first law of thermodynamics by equation-(3.65) as

$$C = \frac{R}{\gamma-1} + \frac{R PdV}{PdV + VdP} \quad \dots(3.67)$$

As $PV^n = \text{const}$ we have

$$dPV^n + nPV^{n-1} dV = 0$$

or
$$dP = -\frac{nP}{V} dV \quad \dots(3.68)$$

From (3.67) and (3.68) we have

$$C = \frac{R}{\gamma-1} + \frac{R PdV}{PdV + V\left(-\frac{nP}{V} dV\right)}$$

or
$$C = \frac{R}{\gamma-1} + \frac{R}{1-n} \quad \dots(3.69)$$

or
$$C = \frac{R(\gamma-n)}{(\gamma-1)(1-n)} \quad \dots(3.70)$$

Equation-(3.69) and (3.70) gives the molar heat capacity for a general polytropic process. If the polytropic constant n is known we can directly find the molar heat capacity C for the respective process using equation-(3.69) or (3.70) which directly gives the total amount of heat supplied to a gas as if n moles of gas are heated from a temperature T_1 to T_2 then the total heat supplied for the purpose is given as

$$Q = \int_{T_1}^{T_2} nC dT$$

$$Q = nC(T_2 - T_1) \quad \dots(3.71)$$

As internal energy change for any process can be given as

$$\Delta U = n C_V (T_2 - T_1) \quad \dots(3.72)$$

Thus work done in a polytropic process can be written from equation-(3.72) and (3.73) as

$$W = Q - \Delta U$$

$$W = n(C - C_V)(T_2 - T_1) \quad \dots(3.73)$$

3.7.1 Standard Processes as a Special Case of Polytropic Process

As discussed earlier the four standard processes are the special cases of general polytropic processes. Now we discuss these processes again in terms of polytropic process

(i) Isochoric Process

We know in an isochoric process, the process equation for the process is

$$V = \text{constant}$$

The general equation for polytropic process $PV^n = \text{constant}$, approaches this result when $n \rightarrow \infty$ and in this case the molar specific heat of the process is given by equation-(3.69) as

$$C = \frac{R}{\gamma-1} + \frac{R}{1-n}$$

As $n \rightarrow \infty$
$$C = \frac{R}{\gamma-1} = C_V$$

Which is in favour of the process

(ii) Isobaric process

Here in isobaric process the process equation is given as

$$P = \text{constant}$$

From the general equation of polytropic process $PV^n = \text{constant}$, above equation is obtained by substituting polytropic constant $n = 0$. Thus molar specific heat of this process can be given by equation-(3.69) as

$$C = \frac{R}{\gamma-1} + \frac{R}{1-n}$$

or
$$C = \frac{R}{\gamma-1} + R = C_V + R = C_P$$

Hence it is also same as that used for constant pressure heating.

(iii) Isothermal process

Here during the process temperature of system remains constant thus the process equation is given as

$$PV = \text{constant}$$

From general equation of polytropic process here we can see that for an isothermal process the polytropic constant $n = 1$. Thus if we find molar specific heat for isothermal process from equation-(3.65), we get

$$C = \frac{R}{\gamma-1} + \frac{R}{1-n}$$

or
$$C \longrightarrow \infty$$

This is also obvious that in isothermal process temperature of gas never changes and molar heat capacity is the amount of heat required to change the temperature for one mole of gas by one degree. So if we continuously supply heat to a gas infinitely then also its temperature will not change then it is undergoing an isothermal process.

(iv) Adiabatic process

In previous section we've derived the process equation of an adiabatic process, given as

$$P V^\gamma = \text{constant}$$

Thus for an adiabatic process, the polytropic constant is equal to the adiabatic exponent of the gas, $n = \gamma$. Now from equation-(3.69) we can see that the molar heat capacity for a gas in adiabatic process is

$$C = \frac{R}{\gamma - 1} + \frac{R}{1 - n}$$

or $C = 0$

Again here it is obvious that in an adiabatic expansion or compression, temperature of gas changes without any supply of heat. Thus in this process no heat is required to raise the temperature of gas.

3.7.2 Indicator Diagram For a Polytropic Process

The general process equation is

$$P V^n = \text{constant}$$

or $P \propto \frac{1}{V^n}$

We know if $n = 1$, it becomes an isothermal process, PV -curve of which is a rectangular hyperbola and the slope of the curve is given as

$$\frac{dP}{dV} = -n \left(\frac{P}{V} \right)$$

or $\left| \frac{dP}{dV} \right|_{\text{Polytropic Process}} = n \left| \frac{dP}{dV} \right|_{\text{Isothermal Process}} \quad \dots (3.74)$

Thus for a process if polytropic constant n is more than unity then this curve will be steeper than isothermal curve and if $n < 1$ then its slope will be less than isothermal curve. Figure-3.43 shows various PV -curves for different value of n for some standard and general processes.

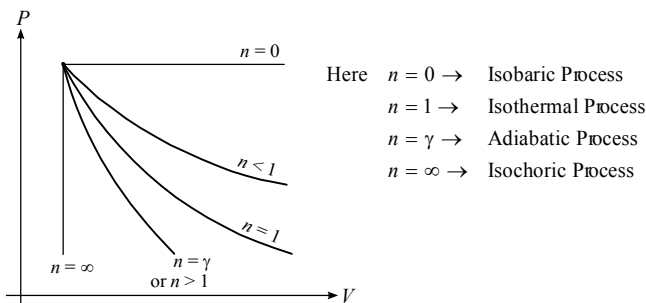


Figure 3.43

Illustrative Example 3.34

An amount Q of heat is added to a monoatomic ideal gas in a thermodynamic process. In the process gas expands and does a work $\frac{Q}{2}$ on its surrounding. Find the molar specific heat of gas in this process.

Solution

From first law of thermodynamics we have

$$Q = \Delta U + W$$

It is given that $W = \frac{Q}{2}$

Thus $Q = \Delta U + \frac{Q}{2}$

or $\Delta U = \frac{Q}{2} \quad \dots (3.75)$

If C is the molar specific heat of the gas in this process then the amount of heat supplied can be given as

$$Q = nC\Delta T \quad \dots (3.76)$$

[If ΔT is rise in temperature of gas]

For change in internal energy of gas, we can write

$$\Delta U = nC_V\Delta T$$

or $= n \left(\frac{3R}{2} \right) \Delta T \quad \dots (3.77)$

[As for a monoatomic gas $C_V = \left(\frac{3R}{2} \right)$]

Thus from equation-(3.75), (3.76) and (3.77) we have

$$nC_V\Delta T = \frac{1}{2} (nC\Delta T)$$

or $C = 2 C_V = 3R$

Illustrative Example 3.35

An ideal gas has a molar heat capacity C_V at constant volume. Find the molar heat capacity of this gas as a function of its volume V , if the gas undergoes the following process :

(a) $T = T_0 e^{\alpha V}$ and (b) $P = P_0 e^{\alpha V}$.

Solution

(a) The process equation for the thermodynamic process is given as

$$T = T_0 e^{\alpha V} \quad \dots (3.78)$$

For a general thermodynamic process we know the molar heat capacity is given by

$$C = C_V + \frac{R P dV}{P dV + V dP} \quad \dots (3.79)$$

From gas law we can relate pressure and volume of gas as

$$P = \frac{nRT}{V}$$

From equation-(3.74)

$$P = \frac{nRT_0}{V} e^{\alpha V} \quad \dots (3.80)$$

Differentiating this equation, we get

$$P dV + V dP = nRT_0 \alpha e^{\alpha V} dV \quad \dots (3.81)$$

From equation-(3.79), (3.80) and (3.81), we get

$$C = C_V + \frac{R \left(\frac{nRT_0}{V} e^{\alpha V} \right) dV}{nRT_0 \alpha e^{\alpha V} dV}$$

$$\text{or} \quad C = C_V + \frac{R}{\alpha V} \quad \dots (3.82)$$

Equation-(3.82) gives the molar specific heat of the gas undergoing the given process and students should note that this molar specific heat is given a function of volume of gas thus this process is a nonpolytropic process.

(b) In this case the process equation is given as

$$P = P_0 e^{\alpha V} \quad \dots (3.83)$$

Differentiating, we get

$$dP = P_0 \alpha e^{\alpha V} dV \quad \dots (3.84)$$

If C is the molar specific heat of the gas in this process then from equation-(3.79), (3.83) and (3.84) we have

$$C = C_V + \frac{R(P_0 e^{\alpha V}) dV}{(P_0 e^{\alpha V}) dV + V(P_0 \alpha e^{\alpha V} dV)}$$

$$\text{or} \quad C = C_V + \frac{R}{1 + \alpha V} \quad \dots (3.85)$$

Equation-(3.85) gives the molar heat capacity of the gas in the given process and again we can say that this is also not a polytropic process.

Illustrative Example 3.36

An ideal gas is taken through a process in which the process equation is given as $P = kV^\alpha$, where k and α are positive constants. Find the value of α for which in this process molar heat capacity becomes zero.

Solution

For a given polytropic process $PV^n = \text{constant}$, we know the molar heat capacity is given as

$$C = \frac{R}{\gamma - 1} + \frac{R}{1 - n}$$

In this process equation can be rewritten as

$$PV^{-\alpha} = k \text{ (a constant)}$$

Thus in this process value of polytropic constant n is $-\alpha$. So molar heat capacity of gas in this process is directly given as

$$C = \frac{R}{\gamma - 1} + \frac{R}{1 + \alpha}$$

Given that $C = 0$

or on solving we get

$$\alpha = -\gamma.$$

Illustrative Example 3.37

n moles of a monoatomic ideal gas undergone in a thermodynamic process along the path shown in figure-3.44 from state-1 to state-2. The gas pressure in state-1 is P_0 . Find the amount of heat supplied to the gas in this process and work done by the gas in the process.

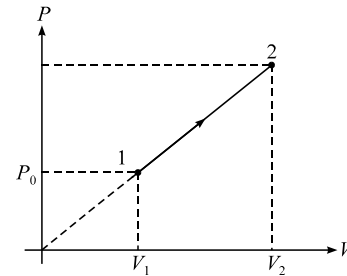


Figure 3.44

Solution

From indicator diagram shown in figure-3.34 it is clear that the process equation of this process can be written as

$$P = kV \quad \text{or} \quad PV^{-1} = k \text{ (constant)} \quad \dots (3.86)$$

[As PV curve is a straight line passing through origin]

Thus here from equation-(3.86), we can say that the process is polytropic with the value of polytropic constant $n = -1$. So the molar heat capacity of the gas can be given as

$$C = C_V + \frac{R}{1 - n}$$

or

$$C = \frac{3R}{2} + \frac{R}{2} = 2R$$

[As for a monoatomic gas $C_V = \frac{3R}{2}$]

If in the process gas temperatures at state-1 and state-2 are T_1 and T_2 then these can be obtained by gas law as

$$PV = nRT$$

For state-1 $T_1 = \frac{P_0 V_1}{nR}$

For this process from equation-(3.86), we have

$$\frac{P_1}{V_1} = \frac{P_2}{V_2}$$

or $P_2 = \frac{V_2}{V_1} P_0$

Thus for state-2 from gas law

$$T_2 = \frac{P_2 V_2}{nR} = \frac{V_2^2 P_0}{nR V_1}$$

If heat supplied in the process in changing the gas state from 1 to 2 is Q , then it is given as

$$Q = n C (T_2 - T_1)$$

or $Q = n(2R) \left[\frac{V_2^2 P_0}{nR V_1} - \frac{P_0 V_1}{nR} \right]$

or $Q = 2P_0 \left(\frac{V_2^2 - V_1^2}{V_1} \right)$

Work done in a polytropic process is given by

$$W = n (C - C_V) (T_2 - T_1)$$

or $W = n \left(2R - \frac{3R}{2} \right) \left(\frac{V_2^2 P_0}{nR V_1} - \frac{P_0 V_1}{nR} \right)$

or $W = \frac{P_0}{2} \left(\frac{V_2^2 - V_1^2}{V_1} \right)$

Work done in the process can also be obtained by the area below the PV curve (in figure-3.45, shaded area), can be given as

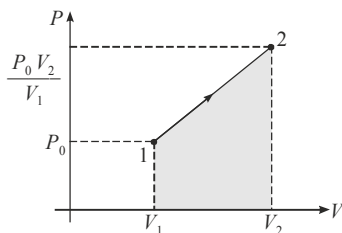


Figure 3.45

$$W = \frac{1}{2} \left(P_0 + P_0 \frac{V_2}{V_1} \right) (V_2 - V_1)$$

or

$$= \frac{P_0}{2} \left(\frac{V_2^2 - V_1^2}{V_1} \right)$$

Illustrative Example 3.38

One mole of an ideal gas with heat capacity at constant pressure C_p undergoes the process $T = T_0 + \alpha V$, where T_0 and α are constant. Find

- heat capacity of the gas as a function of its volume,
- the amount of heat transferred to the gas, if its volume increased from V_1 to V_2 .

Solution

The process equation for the thermodynamic process is given as

$$T = T_0 + \alpha V \quad \dots (3.87)$$

From gas law we have

$$P = \frac{RT}{V} = \frac{R}{V} (T_0 + \alpha V) \quad \dots (3.88)$$

[As $n = 1$ mole]

(a) The heat capacity of a gas in a thermodynamic process is given as

$$C = C_V + \frac{R P dV}{P dV + V dP} \quad \dots (3.89)$$

Differentiating equation-(3.88), we get

$$P dV + V dP = \alpha R dV \quad \dots (3.90)$$

Now from equation-(3.88), (3.89) and (3.90), we have

$$C = C_V + \frac{R \left[\left(\frac{R}{V} \right) (T_0 + \alpha V) \right] dV}{\alpha R dV}$$

or $C = C_V + \frac{R}{\alpha V} (T_0 + \alpha V)$

or $C = C_V + \frac{RT_0}{\alpha V} + R$

or $C = C_p + \frac{RT_0}{\alpha V} \quad [\text{As } C_p = C_V + R]$

(b) As volume of gas increases from V_1 to V_2 , the corresponding temperatures of the gas are

$$T_1 = T_0 + \alpha V_1$$

and $T_2 = T_0 + \alpha V_2$

The amount of heat supplied to the gas can be given as

$$Q = \int dQ = \int_{T_1}^{T_2} C dT \quad [\text{As } n = 1 \text{ mole}]$$

or
$$Q = \int_{T_1}^{T_2} \left[C_P + \frac{RT_0}{\alpha V} \right] dT$$

From the given relation we have

$$V = \frac{T - T_0}{\alpha}$$

Thus
$$Q = \int_{T_1}^{T_2} \left[C_P + \frac{RT_0}{T - T_0} \right] dT$$

$$= C_P (T_2 - T_1) + RT_0 \ln \left(\frac{T_2 - T_0}{T_1 - T_0} \right)$$

$$= C_P \alpha (V_2 - V_1) + RT_0 \ln \left(\frac{V_2}{V_1} \right)$$

Illustrative Example 3.39

One mole of an ideal gas with adiabatic exponent γ whose pressure changes with volume as $P = \alpha V$, where α is a constant, is expanded so that its volume increases η times. Find the change in internal energy and heat capacity of the gas. Initial volume of gas is V_0 .

Solution

The process equation of the thermodynamic process in which gas is undergoing is given as

$$P = \alpha V \quad \dots (3.91)$$

Rearranging we can write

$$PV^{-1} = \alpha \text{ (a constant)} \quad \dots (3.92)$$

Equation-(3.92) shows that this is a polytropic process with the value of polytropic constant $n = -1$. Thus the molar heat capacity of gas in this process can be given as

$$C = \frac{R}{\gamma - 1} + \frac{R}{1 - n}$$

or
$$= \frac{R}{\gamma - 1} + \frac{R}{2} \quad [n = -1]$$

$$= \frac{R}{2} \left[\frac{\gamma + 1}{\gamma - 1} \right]$$

If during the process, temperature of gas changes from T_1 to T_2 then change in internal energy of gas is given as

$$\Delta U = C_V \Delta T \quad [\text{As } n = 1 \text{ mole}]$$

$$= \frac{R}{\gamma - 1} (T_2 - T_1) \quad [\text{As } C_V = \frac{R}{\gamma - 1}] \quad \dots (3.93)$$

Here it is given that initial and final volumes of the gas are V_0 and ηV_0 . Thus the respective pressures in initial and final states are

and
$$P = \alpha V_0$$

$$P = \alpha \eta V_0$$

Thus from gas law we can find initial and final temperatures as

$$T_1 = \frac{P_1 V_1}{R} = \frac{\alpha V_0 \cdot V_0}{R}$$

and
$$T_2 = \frac{P_2 V_2}{R} = \frac{\alpha \eta V_0 \cdot \eta V_0}{R} \quad [\text{As } n = 1 \text{ mole}]$$

Now from equation-(3.93), change in internal energy of gas is given as

$$\Delta U = \frac{R}{\gamma - 1} \left[\frac{\alpha \eta^2 V_0^2}{R} - \frac{\alpha V_0^2}{R} \right]$$

$$= \frac{\alpha V_0^2}{\gamma - 1} [\eta^2 - 1]$$

Illustrative Example 3.40

One mole of an ideal gas, whose adiabatic exponent equal to γ , is expanded so that the amount of heat transferred to the gas is equal to the decrease in internal energy. Find :

- the molar heat capacity of the gas in this process,
- the equation of the process in the variables T, V ;
- the work performed by one mole of the gas when its volume increases η times if the initial temperature of gas is T_0 .

Solution

(a) It is given that in the thermodynamic process amount of heat supplied is equal to the decrease in internal energy of gas thus we have in the process

$$dU = -dQ$$

or
$$nC_V dT = -nC dT$$

or
$$C = -C_V = -\frac{R}{\gamma - 1} \quad \dots (3.94)$$

(b) Now from first law of thermodynamics, we have in this process

$$dQ = dU + dW$$

or $2dQ = dW$

or $-\frac{2nR}{\gamma-1} dT = PdV \quad \dots (3.95)$

We have differential form of gas law

$$PdV + VdP = \left(\frac{1-\gamma}{2}\right) PdV$$

or $\left(\frac{1+\gamma}{2}\right) PdV = -VdP$

or $\left(\frac{1+\gamma}{2}\right) \frac{dV}{V} = -\frac{dP}{P}$

Integrating this equation we get

$$\left(\frac{1+\gamma}{2}\right) \int \frac{dV}{V} = - \int \frac{dP}{P}$$

$$\ln V^{\left(\frac{1+\gamma}{2}\right)} = - \ln P + C$$

or $PV^{\left(\frac{1+\gamma}{2}\right)} = \text{constant} \quad \dots (3.96)$

As we require process equation in T and V , from gas law

$$P = \frac{nRT}{V}$$

Now from equation-(3)

$$\left(\frac{nRT}{V}\right) V^{\left(\frac{1+\gamma}{2}\right)} = \text{constant}$$

or $TV^{\left(\frac{1-\gamma}{2}\right)} = \text{constant} \quad \dots (3.97)$
[As $nR = \text{constant}$]

Process equation in P and V which is given in equation-(3.96) can be directly obtained by the molar heat capacity as from equation-(3.94). We can see that the value of C is a constant not depending on pressure, volume or temperature of gas thus we can say that this process is a polytropic process whose molar specific heat can be given as

$$C = \frac{R}{\gamma-1} + \frac{R}{1-n} = -\frac{R}{\gamma-1}$$

Where n is the polytropic constant of the process

or $\frac{R}{1-n} = -\frac{2R}{\gamma-1}$

or $(1-n) = -\frac{1}{2}(\gamma-1)$

or $2-2n = 1-\gamma$

or $n = \left(\frac{1+\gamma}{2}\right)$

Thus now it is obvious that the process equation of this thermodynamic process in P and V can be simply given as

$$PV^n = \text{constant}$$

or $PV^{\left(\frac{1+\gamma}{2}\right)} = \text{constant}$

Which is same as equation-(3.92)

(c) The work done by a gas in a polytropic process can be given as

$$W = n(C - C_V)(T_2 - T_1) \quad \dots (3.98)$$

Here initial temperature of gas is given as T_0 and the final temperature can be obtained from equation-(3.97) for initial and final state of gas as

$$T_1 V_1^{\left(\frac{\gamma-1}{2}\right)} = T_2 V_2^{\left(\frac{\gamma-1}{2}\right)}$$

Here $T_1 = T_0$ and $V_2 = \eta V_1$, thus

$$T_2 = T_0 \left(\frac{V_1}{V_2}\right)^{\frac{\gamma-1}{2}} = T_0 \eta^{\left(\frac{1-\gamma}{2}\right)}$$

Now from equation-(3.98), work done is

$$W = n(-C_V - C_V)(T_2 - T_1)$$

or $W = -\frac{2R}{\gamma-1} [T_0 (\eta)^{\frac{1-\gamma}{2}} - T_0] \quad [\text{As } n = 1 \text{ mole}]$

or $W = \frac{2RT_0}{\gamma-1} [1 - (\eta)^{\frac{1-\gamma}{2}}]$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Laws of Thermodynamics

Module Number - 29 to 34

Practice Exercise 3.4

(i) One mole of an ideal gas whose adiabatic exponent is γ undergoes a process in which the volume changes with temperatures as $V = \frac{\alpha}{T}$, where α is constant. Find the amount of heat required to raise its temperature by ΔT .

$$\left[\frac{2-\gamma}{\gamma-1} R \Delta T \right]$$

(ii) For the case of an ideal gas find the equation of the process (in the variables T , V) in which the molar heat capacity varies as :

(a) $C = C_V + \alpha T$;

(b) $C = C_V + \beta V$;

(c) $C = C_V + aP$, where α , β , and a are constants.

[(a) $Ve^{-\alpha T/R} = \text{const}$; (b) $Te^{R/\beta V} = \text{const}$; (c) $V - aT = \text{const}$.]

(iii) A gas consisting of monatomic molecules (degrees of freedom = 3) was expanded in a polytropic process so that the rate of collisions of the molecules against the vessel's wall did not change. Find the molar heat capacity of the gas in the process.

[2 R]

(iv) A gas consisting of rigid diatomic molecules was expanded in a polytropic process so that the rate of collisions of the molecules against the vessel's wall did not change. Find the molar heat capacity of the gas in this process.

[3 R]

(v) An ideal gas has an adiabatic exponent γ . In some process its molar heat capacity varies as $C = \alpha/T$, where α is a constant. Find :

(a) the work performed by one mole of the gas during its hating from the temperature T_0 to the temperature η times higher;

(b) the equation of the process in the variables p , V .

[(a) $\alpha \ln \eta - RT_0 (\eta - 1) / (\gamma - 1)$; (b) $pV^\gamma e^{\alpha(\gamma-1)/pV} = \text{const}$.]

(vi) The volume of a gas containing diatomic molecules was increased $\eta = 2$ times in a polytropic process with molar heat capacity $C = R$. How many times will the rate of collision of molecules against the wall of vessel be reduced as a result of this process ?

[$2^{4/3}$ times]

(vii) One mole of an ideal gas whose adiabatic exponent equals γ undergoes a process in which the gas pressure relates to the temperature as $p = aT^\alpha$, where a and α are constants. Find :

(a) the work performed by the gas if its temperature gets an increment ΔT ;

(b) the molar heat capacity of the gas in this process; at what value of α will the heat capacity be negative ?

[(a) $W = (1 - \alpha) R \Delta T$; (b) $C = \frac{R}{(\gamma-1)} + R(1 - \alpha)$; $C < 0$ for $\alpha > \frac{\gamma}{(\gamma-1)}$]

(viii) A gas is taken from state-1 to state-2 from two different paths A and B as shown in figure-3.46. If molar heat capacities of the gas in the two paths are C_A and C_B respectively then which molar heat capacity is greater.

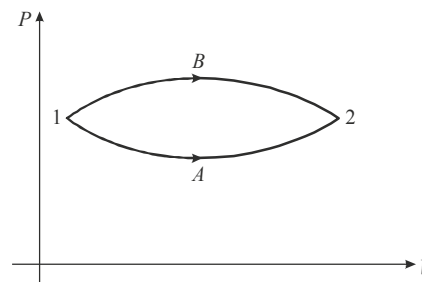


Figure 3.46

[C_B]

3.8 Second Law of Thermodynamics

Most of the thermodynamic processes in nature proceed in one direction but not in opposite direction, for example, we know heat always flows from a hot body to a colder body, never the reverse. There are so many examples in which mechanical energy is completely converted into heat and also so many devices exist which convert heat partially into mechanical energy. But till now there is no machine which can convert heat completely into mechanical energy. Why ? The answer to this question is the second law of thermodynamics. But before proceeding to second law of thermodynamics, we first discuss some basic things related to that.

3.8.1 Reversible Processes

Thermodynamic processes that occur in nature are all irreversible processes. This implies that if a gas is taken from its initial state to final state along a specified path, in practical nature it is not possible to carry the gas back to its initial state along the same path in reverse direction. Reversible processes are idealized in nature and can occur only when the system is in thermal equilibrium within itself and with its surroundings. When two systems are in thermal equilibrium, any change of state that take place can be reversed by making a very elemental change in the conditions of these systems. For example, heat flow between two bodies whose temperature differ by a very small amount can be reversed by making a very small variation

in their temperatures. This we'll discuss in details later. Thus those processes which are reversible are also called equilibrium processes. We know when a system is in thermal equilibrium, no change of state takes place. As discussed, reversible process is an idealization that can never be possible in practical world.

But during a thermodynamic process if the process is carried out very slowly so that it can be considered to be in thermal equilibrium always or if it is of a quasi-static nature we can make such processes approximately reversible and these processes are called quasistatic processes. On the other hand all relatively fast processes or like heat flow with a temperature difference, free expansion of gas or conversion of work to heat are all irreversible processes, these are called, non equilibrium processes.

Now we'll discuss the second law of thermodynamics by considering two broad classes of devices. Heat engines and Refrigerators. Heat engines are partly successful in converting heat into work, and refrigerators are partly successful in transporting heat from colder to hotter bodies.

Earlier while discussing cyclic processes we've learned that the clockwise cycles are called the heat engine cycles and anticlockwise cycles are the refrigeration cycles. Now we discuss both in details.

3.8.2 Heat Engines

Any device that transforms heat partly into work or mechanical energy is called a heat engine. In a heat engine some quantity of a substance undergoes inflow and outflow of heat, expansion and compression and sometimes phase changes. This substance used in heat engine is called working substance or fuel.

The simplest heat engine is a thermodynamic system undergoes a clockwise cyclic process. If the cycle is repeated periodically it leaves the substance in the same state in which it started and during the process some heat is supplied in the expansion process and gas does some work and during compression part of cycle, some work (lesser than first part) is done on the gas and it rejects some heat to the surrounding, lesser than the amount absorbed during expansion. Thus according to first law of thermodynamics the total work done by the gas is equal to the amount of heat supplied to the gas minus the heat which is rejected by the gas during its compression to the initial state.

A heat engine does not use the same substance and over the cycles but it is changed after every cycle by some mechanism. All heat engine absorb heat from a source at a relatively high

temperature, perform some mechanical work and reject some heat at relatively lower temperature. As engine is concerned, the rejected heat is wasted.

When we analyse the heat engine, it helps to think of two bodies with which the working substance of engine interact. One is the heat source and other is the heat sink. Heat source is the one which can supply large amount of heat to the working substance during first part of its cycle and heat sink is the one to which the working substance can reject heat after performing mechanical work during the second part of its cycle. As discussed heat source is at higher temperature and heat sink is at lower temperature.

Figure-3.47 shows the basic block diagram of a heat engine also called energy flow diagram of heat engine. In this diagram engine is represented by the circular block. During the cycle it takes a quantity of heat Q_1 from the heat source at high temperature T_1 and does mechanical work W and rejects a heat Q_2 to the heat sink at lower temperature T_2 .

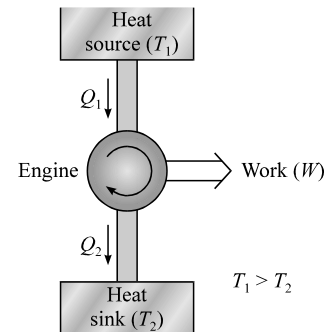


Figure 3.47

Here W is called useful power output of engine and Q_2 is the energy which is wasted. In a heat engine we would like to convert all input heat Q_1 into work but experiments show that it is impossible to have $Q_2 = 0$. According to conservation of energy we have

$$Q_1 = Q_2 + W \quad \dots (3.99)$$

3.8.3 Thermal Efficiency of a Heat Engine

Thermal efficiency of an engine gives the fraction of heat supplied to the engine which is transformed into the useful mechanical work. Thus thermal efficiency of a heat engine is given as

$$\eta = \frac{W}{Q_1} \quad \dots (3.100)$$

From equation-(3.99) we have

$$W = Q_1 - Q_2$$

Thus
$$\eta = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1} \quad \dots (3.101)$$

or
$$\eta = 1 - \left| \frac{Q_{\text{out}}}{Q_{\text{in}}} \right| \quad \dots (3.102)$$

At the time of designing of the engines, designers always try to develop a mechanism in which the amount of useful mechanical work is as wide as possible and to keep the rejected heat as narrow as possible. As Q_2 can never be zero and no engine is having efficiency equal to unity.

3.8.4 Refrigerators

Refrigerator is a device just like a heat engine operating in reverse fashion. A heat engine takes heat from a hot place and gives off heat to a colder place. A refrigerator does the opposite of this. It takes heat from a place at lower temperature and gives it off to a place at relatively higher temperature. A heat engine has a net output of mechanical work where as a refrigerator requires a net input of mechanical work.

Figure-3.48 shows a basic energy flow diagram of a refrigerator. Here the refrigerator draws an amount of heat Q_1 from the colder region which is inside of refrigerator at temperature T_1 . An external mechanical work W is done on refrigerator due to which it rejects some amount of heat Q_2 to a region at higher temperature T_2 , which is generally the outside air. According to energy conservation, for a refrigeration cycle we can write

$$Q_1 + W = Q_2 \quad \dots (3.103)$$

or
$$W = Q_2 - Q_1 \quad \dots (3.104)$$

The best refrigeration cycle is one that removes the greatest amount of heat Q_1 from the inside of refrigerator for the least amount of mechanical work W . The ratio (Q_1/W) , we call this ratio the coefficient of performance of a refrigerator, and it gives the quality of refrigeration cycle. The larger this ratio, the better the refrigerator.

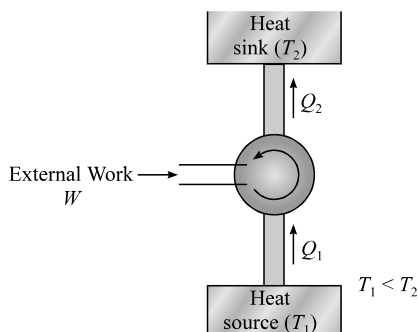


Figure 3.48

The coefficient of performance of a refrigeration cycle is

$$K = \frac{Q_1}{W} = \frac{Q_1}{Q_2 - Q_1} \quad \dots (3.105)$$

3.8.5 Statement of Second Law of Thermodynamics

We've discussed that experiments showed that it is impossible to build a heat engine that converts heat completely to work, that is an engine with 100% thermal efficiency. This leads to the basic statement of second law of thermodynamics. Stated as

"It is impossible for any system to absorb heat and convert it completely into mechanical work leaving the system in its initial state from which the process was started."

The analysis of refrigerator we've discussed in previous section forms the basis for an alternative statement of the second law of thermodynamics. We know heat always flows spontaneously from hotter to colder bodies, never the reverse. A refrigerator does take heat from a colder to a hotter region, but its operation requires an input of mechanical energy or work. Thus we state.

"It is impossible for any process to have only the heat flow from a colder to a hotter body without any involvement of an external agency."

3.8.6 Carnot Cycle

According to second law of thermodynamics, no heat engine can have 100% efficiency. Efficiency of a heat engine depends on the amount of heat taken in and outflow from the heat engine. In 1824, a French scientist Carnot developed a hypothetical, idealized heat engine that has maximum possible efficiency consistent with the second law. The cycle of this engine is called the Carnot Cycle. Now we'll discuss the basic theme of this cycle.

We know conversion of work to heat is an irreversible process and the purpose of a heat engine is a partial reversal of this process, the conversion of heat to work with a given efficiency. The higher the efficiency is, the better the engine will be. For maximum efficiency we must avoid the irreversible processes. This forms the basis of Carnot cycle.

Heat flow through a finite temperature drop is an irreversible process. In carnot cycle when engine takes heat from heat source at temperature T_1 , the engine temperature or the temperature of working substance is also at same temperature T_1 . Similarly when engine rejects heat to the sink at the lower temperature T_2 then during this rejection of heat, engine is also at the same temperature T_2 . The cycle will be ideal if engine

does not exchange any heat during the process when its temperature is changing thus in a single line we can state that every process in the idealized carnot cycle must be either isothermal or adiabatic.

An ideal Carnot cycle is accomplished in four steps, there are:

- (1) The gas or working substance expands isothermally at temperature T_1 and absorbs an amount of heat Q_1 from heat source at temperature T_1 .
- (2) It expands adiabatically until its temperature drops to T_2 where T_2 is the temperature of heat sink.
- (3) It is compressed isothermally at temperature T_2 and rejects an amount of heat Q_2 to the sink.
- (4) Now it is compressed adiabatically back to its initial state at temperature T_1 .

Figure-3.49 shows an indicator diagram for an ideal gas undergoing in carnot cycle.

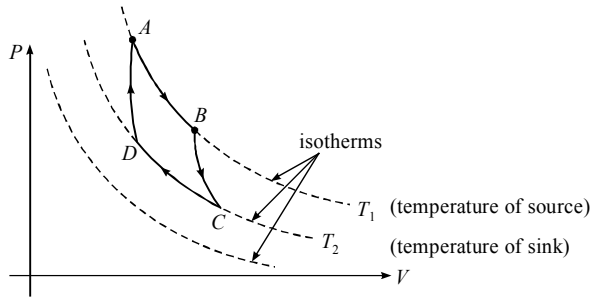


Figure 3.49

As shown in figure-3.49 in a carnot cycle, first gas starts from state-A and in its isothermal expansion upto state-B heat absorbed by the gas is

$$Q_1 = W_{ab} = n R T_1 \ln \left(\frac{V_B}{V_A} \right) \quad \dots (3.106)$$

Here V_B and V_A are the volumes of the gas at state (A) and state (B) respectively. Similarly in isothermal compression of gas from state (C) to state (D), heat rejected by the gas is

$$Q_2 = W_{cd} = n R T_2 \ln \left(\frac{V_C}{V_D} \right) \quad \dots (3.107)$$

In the adiabatic processes from state (B) to state (C) and from state (D) to state (A), we can relate temperatures and volumes of gas as

$$T_1 V_B^{\gamma-1} = T_2 V_C^{\gamma-1} \quad \dots (3.108)$$

$$\text{and} \quad T_1 V_A^{\gamma-1} = T_2 V_D^{\gamma-1} \quad \dots (3.109)$$

From equation-(3.108) and equation-(3.109), we get

$$\frac{V_B}{V_A} = \frac{V_C}{V_D} \quad \dots (3.110)$$

From equation-(3.106) and (3.107), we get

$$\frac{Q_2}{Q_1} = \frac{n R T_2 \ln \left(\frac{V_C}{V_D} \right)}{-n R T_1 \ln \left(\frac{V_B}{V_A} \right)} = -\frac{T_2}{T_1} \quad \dots (3.111)$$

$$\text{or} \quad \left| \frac{Q_2}{Q_1} \right| = \frac{T_2}{T_1}$$

Thus the efficiency of a carnot cycle can be given as

$$\eta = 1 - \left| \frac{Q_2}{Q_1} \right|$$

$$\eta = 1 - \frac{T_2}{T_1} \quad \dots (3.112)$$

This shows that efficiency of a carnot cycle depends only on the temperatures of heat source and heat sink and is large when temperature difference of the two is large and it is very small when the temperatures are very close to each other.

Illustrative Example 3.41

One mole of oxygen undergoes a cyclic process in which volume of the gas changes 10 times within the cycle as shown in figure-3.50. The processes AB and CD are adiabatic while processes BC and DA are isochoric. What is the efficiency of the process ?

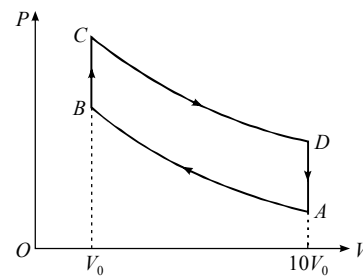


Figure 3.50

Solution

In the cyclic process the efficiency of cycle can be given as

$$\eta = 1 - \left| \frac{Q_{\text{out}}}{Q_{\text{in}}} \right| \quad \dots (3.113)$$

As the process CD and AB are adiabatic, no heat exchange takes place in these processes. In isochoric process BC as

temperature increases, the heat is taken in which is given as

$$\begin{aligned} Q_{\text{in}} &= n C_V (T_C - T_B) \\ &= \frac{5}{2} R (T_C - T_B) \quad \dots (3.114) \end{aligned}$$

$$[\text{As } n = 1 \text{ mole and for } O_2 C_V = \frac{5}{2}R]$$

Similarly in isochoric process A as temperature decreases heat is rejected by the gas, which is given as

$$\begin{aligned} Q_{\text{out}} &= n C_V (T_D - T_A) \\ &\quad [\text{Magnitude of heat rejected}] \\ &= \frac{5}{2} R (T_D - T_A) \quad \dots (3.115) \end{aligned}$$

From equation-(3.113), (3.114) and (3.115), we have

$$\text{Efficiency} \quad \eta = 1 - \frac{T_D - T_A}{T_C - T_B} \quad \dots (3.116)$$

As we know in an adiabatic process the volume and temperature of gas in initial and final states are related as

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

Here we use the same relation for process CD and AB as

$$\begin{aligned} T_C V_0^{\gamma-1} &= T_D (10 V_0)^{\gamma-1} \\ \text{or} \quad T_C &= (10^{\gamma-1}) T_D \quad \dots (3.117) \end{aligned}$$

$$\begin{aligned} \text{and} \quad T_B V_0^{\gamma-1} &= T_A (10 V_0)^{\gamma-1} \\ \text{or} \quad T_B &= (10^{\gamma-1}) T_A \quad \dots (3.118) \end{aligned}$$

Now from equation-(3.116), (3.117) and (3.118), we have

$$\begin{aligned} \text{Efficiency} \quad \eta &= 1 - \frac{(T_D - T_A)}{(10)^{\gamma-1} T_D - (10)^{\gamma-1} T_A} \\ \text{or} \quad &= \left(1 - \frac{1}{10^{\gamma-1}} \right) \\ &= 1 - \frac{1}{10^{0.4}} \quad [\text{As for } O_2, \gamma = 1.4] \\ &= 0.6 = 60\% \end{aligned}$$

Illustrative Example 3.42

Three moles of an ideal gas ($C_P = \frac{7}{2}R$) at pressure P_A and temperature T_A is isothermally expanded to twice its initial volume. It is then compressed at constant pressure to its original volume. Finally the gas is compressed at constant volume to its original pressure P_A .

- Sketch P - V and P - T diagrams for the complete process.
- Calculate the new work done by the gas, and net heat supplied to the gas during the complete process.

Solution

- Sketch of P - V and P - T are shown in figure-3.51(a) and (b).

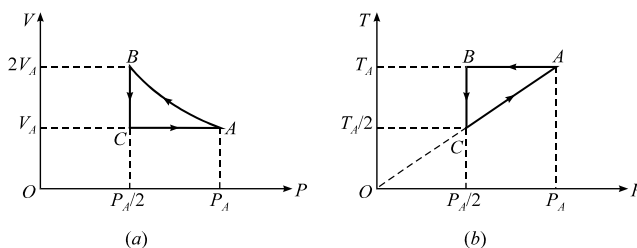


Figure 3.51

The initial state of isothermal expansion is represented by A where pressure is P_A and volume is V_A . Let the final state be represented by B where volume V_B is twice of V_A . Let the pressure be P_B . Then

$$P_A V_A = P_B V_B = P_B (2 V_A)$$

or

$$P_B = P_A \left(\frac{V_A}{2 V_A} \right) = \frac{P_A}{2}$$

When the molecule is compressed to initial volume, the process is represented by BC . Finally the gas is compressed at constant volume to its original pressure. The process is shown by curve CA .

Similarly, P - T diagram can be drawn.

- Workdone in the process AB is given by

$$\begin{aligned} W_1 &= nR T \ln (V_B/V_A) \\ &= 3 \times 8.314 \times T_A \times \ln 2 \\ &= 3 \times 8.314 \times T_A \times 0.693 = 17.29 T_A \end{aligned}$$

Workdone in the process BC is given by

$$\begin{aligned} W_2 &= P \Delta V = P_B \times (V_C - V_B) \\ &= \frac{P_A}{2} \times (V_A - 2V_A) \\ &= -P_A V_A/2 = -n R T_A/2 \\ &= -3 R T_A/2 = -3 \times 8.314 T_A/2 = -12.471 T_A \end{aligned}$$

Workdone during process CA is given by

$$W_3 = P \Delta V = 0 \quad [\text{As } \Delta V = 0]$$

Network

$$\begin{aligned} W &= W_1 + W_2 + W_3 \\ &= 17.26 T_A - 12.45 T_A = 4.81 T_A. \end{aligned}$$

As initial and final states of the gas are same, hence

$$\Delta U = U_A - U_A = 0$$

From first law of thermodynamics

$$\Delta Q = \Delta U + \Delta W = 0 + \Delta W$$

$$Q = W \text{ joule.}$$

Illustrative Example 3.43

One mole of a diatomic ideal gas ($\gamma = 1.4$) is taken through a cyclic process as shown in figure-3.52 starting from point A . In this cycle the process AB is an adiabatic compression. BC is isobaric, CD an adiabatic expansion and DA is isochoric. The volume ratios are $V_A/V_B = 16$ and $V_C/V_B = 2$ and the temperature at A to $T_A = 300^\circ\text{K}$. Calculate the temperature of the gas at the point B and D and find the efficiency of the cycle.

Solution

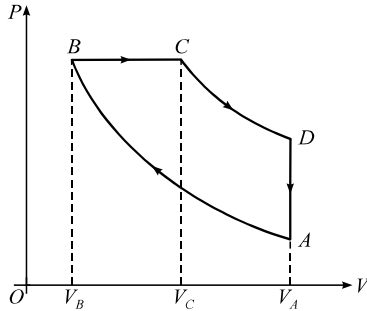


Figure 3.52

The respective cyclic process is shown in figure-3.52. The expansion and compression ratio are given as

$$\frac{V_A}{V_B} = 16 \quad \text{and} \quad \frac{V_C}{V_B} = 2$$

In adiabatic process AB , we have

$$T_A V_A^{\gamma-1} = T_B V_B^{\gamma-1}$$

or

$$\begin{aligned} T_B &= T_A \left(\frac{V_A}{V_B} \right)^{\gamma-1} \\ &= 300 \times (16)^{0.4} \\ &= 909 \text{ K} \end{aligned}$$

Similarly for isobaric process BC , we have

$$\frac{T_C}{V_C} = \frac{T_B}{V_B}$$

or

$$\begin{aligned} T_C &= T_B \left(\frac{V_C}{V_B} \right) \quad \dots (3.119) \\ &= 909 \times 2 = 1818 \text{ K} \end{aligned}$$

Similarly for adiabatic process CD we have

$$T_C V_C^{\gamma-1} = T_D V_D^{\gamma-1}$$

or

$$T_D = T_C \left(\frac{V_C}{V_D} \right)^{\gamma-1}$$

As we have $V_D = V_A$, thus

$$\begin{aligned} \frac{V_C}{V_D} &= \frac{V_C}{V_A} = \frac{V_C}{V_B} \times \frac{V_B}{V_A} \\ &= 2 \times \frac{1}{16} = \frac{1}{8} \end{aligned}$$

Thus from equation-(3.119)

$$\begin{aligned} T_D &= 1818 \times \left(\frac{1}{8} \right)^{0.4} \\ &= 791.3 \text{ K} \end{aligned}$$

The efficiency of cycle can be given as

$$\eta = 1 - \left| \frac{Q_{out}}{Q_{in}} \right|$$

Here we know in process AB and CD , as being adiabatic processes no heat exchange takes place and in isobaric process BC as temperature of gas increases, heat is taken in (say Q_1) and it is given as

$$Q_1 = nC_P(T_C - T_B) \quad \dots (3.120)$$

Similarly in isochoric process DA as temperature of gas decreases heat is rejected (say Q_2) and it can be given as

$$Q_2 = nC_V(T_D - T_A) \quad [\text{A magnitude of rejected heat}]$$

Now efficiency of this cycle can be given as

$$\begin{aligned} \eta &= 1 - \left| \frac{Q_{out}}{Q_{in}} \right| \\ &= 1 - \frac{Q_2}{Q_1} \\ &= 1 - \frac{C_V(T_D - T_A)}{C_P(T_C - T_B)} \\ &= \frac{T_D - T_A}{\gamma(T_C - T_B)} \\ &= \frac{(791.3 - 300)}{(1.4)(1818 - 909)} \\ &= 1 - \frac{491.3}{1272.6} = 0.614 = 61.4\% \end{aligned}$$

NOTE : While solving the problems related to finding the efficiency of a cycle students are advised to first check in which of the path of cycle heat is taken in or heat is rejected and mention it by arrows as for above problem Q_1 and Q_2 can be mentioned as shown in figure-3.53.

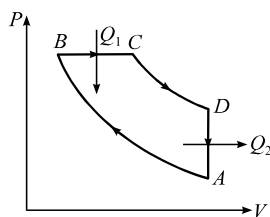


Figure 3.53

Illustrative Example 3.44

One mole of a monatomic ideal gas is taken through the cycle shown in figure-3.54.

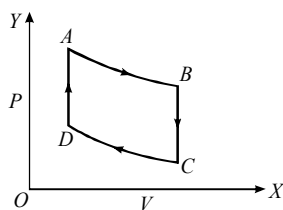


Figure 3.54

$A \rightarrow B$ adiabatic, expansion
 $B \rightarrow C$ cooling at constant volume
 $C \rightarrow D$ adiabatic compression
 $D \rightarrow A$ heating at constant volume

The pressure and temperature at A, B etc. are denoted by $P_A, P_B, \dots; T_A, T_B, \dots$, etc. respectively. Given that

$$T_A = 1000 \text{ K}, P_B = (2/3) P_A \text{ and } P_C = (1/3) P_A.$$

Calculate the following quantities :

- The work done by the gas in process $A \rightarrow B$
- The heat lost by the gas in process $B \rightarrow C$
- The temperature T_D

Given that $(2/3)^{2/5} = 0.85$

Solution

- The workdone in adiabatic process AB is given by

$$W_1 = \frac{R(T_A - T_B)}{\gamma - 1} = \frac{R(T_A - T_B)}{(5/3) - 1}$$

[As for monatomic gas $\gamma = 5/3$]

$$\text{or } W_1 = \frac{3}{2} R(T_A - T_B) \quad \dots (3.121)$$

For adiabatic change

$$P_A^{\gamma-1} T_A^{-\gamma} = P_B^{\gamma-1} T_B^{-\gamma}$$

$$\text{or } \left(\frac{P_A}{P_B}\right)^{2/3} = \left(\frac{T_A}{T_B}\right)^{5/3} \quad \text{or } \left(\frac{T_A}{T_B}\right) = \left(\frac{P_A}{P_B}\right)^{2/5}$$

$$\text{or } \left(\frac{T_A}{T_B}\right) = \left(\frac{3}{2}\right)^{2/5} \quad [\text{As } P_B = (2/3) P_A]$$

$$\begin{aligned} \text{or } T_B &= T_A \left(\frac{2}{3}\right)^{2/5} \\ &= 1000 \times 0.85 = 850 \text{ K} \end{aligned} \quad \dots (3.122)$$

From equation-(3.121)

$$W_1 = \frac{3}{2} \times 8.31 \times (1000 - 850) = 1869.83 \text{ J}$$

- Heat lost by the gas in process BC is given by

$$\begin{aligned} &= C_V(T_B - T_C) = \frac{R}{\gamma - 1} (T_B - T_C) \\ &= \frac{3}{2} R(T_B - T_C) \end{aligned} \quad \dots (3.123)$$

Process BC is under constant volume, hence

$$\frac{P_B}{T_B} = \frac{P_C}{T_C} \quad \text{or } T_C = \left(\frac{P_C}{P_B}\right) \times T_B$$

$$\text{or } T_C = \frac{(P_A/3)}{(2P_A/3)} T_B = \frac{T_B}{2} = 425 \text{ K} \quad \dots (3.124)$$

From equation-(3.123)

$$\text{Heat lost} = \frac{3}{2} \times 8.31 \times (850 - 425) = 5297.63 \text{ J}$$

- For path AB

$$P_A^{\gamma-1} T_A^{-\gamma} = P_B^{\gamma-1} T_B^{-\gamma}$$

$$\text{or } \left(\frac{P_A}{P_B}\right)^{\gamma-1} = \left(\frac{T_A}{T_B}\right)^{\gamma} \quad \dots (3.125)$$

For path BC

$$\frac{P_B}{T_B} = \frac{P_C}{T_C} \quad \text{or } \left(\frac{P_B}{P_C}\right) = \left(\frac{T_B}{T_C}\right) \quad \dots (3.126)$$

For path CD

$$\left(\frac{P_D}{P_C}\right)^{\gamma-1} = \left(\frac{T_D}{T_C}\right)^{\gamma} \quad \dots (3.127)$$

For path AD

$$\left(\frac{P_A}{P_D}\right) = \left(\frac{T_A}{T_D}\right) \quad \dots (3.128)$$

Dividing equation-(3.125) by (3.127), we get

$$\left(\frac{P_A}{P_B} \times \frac{P_C}{P_D}\right)^{\gamma-1} = \left(\frac{T_A}{T_B} \times \frac{T_C}{T_D}\right)^{\gamma} \quad \dots (3.129)$$

Dividing equation-(3.128) by equation-(3.126) we get

$$\left(\frac{P_A}{P_B} \times \frac{P_C}{P_D}\right) = \left(\frac{T_A}{T_B} \times \frac{T_C}{T_D}\right)$$

or
$$\left(\frac{P_A}{P_B} \times \frac{P_C}{P_D}\right)^{\gamma-1} = \left(\frac{T_A}{T_B} \times \frac{T_C}{T_D}\right)^{\gamma-1} \quad \dots (3.130)$$

From equation-(3.129) and (3.130)

$$\left(\frac{T_A}{T_B} \times \frac{T_C}{T_D}\right)^{\gamma} = \left(\frac{T_A}{T_B} \times \frac{T_C}{T_D}\right)^{\gamma-1}$$

$$T_A T_C = T_D T_B$$

$$1000 \times 425 = T_D \times 850$$

$$T_D = 500 \text{ K}$$

Illustrative Example 3.45

Figure-3.55 shows three isothermals at temperature $T_1 = 4000 \text{ K}$, $T_2 = 2000 \text{ K}$, $T_3 = 1000 \text{ K}$. When one mole of an ideal monatomic gas is taken through the paths AB , BC , CD and DA , find the change in internal energy ΔU , the workdone by the gas W and the heat Q absorbed by the gas in each path. Also find these quantities for complete cycle $ABCD$.

Given $V_A = 1 \text{ m}^3$ and $V_B = 2 \text{ m}^3$.

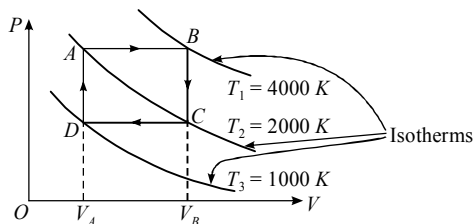


Figure 3.55

Solution

(i) When one mole of an ideal gas undergoes a change in temperature ΔT , the change in its internal energy equals $C_V \Delta T$, where C_V is the molar specific heat of the gas at constant volume. For monoatomic gas $C_V = (3/2)R$. Thus the change of internal energy is given by

$$\begin{aligned} \Delta U_{AB} &= C_V (T_B - T_A) \\ &= (3/2)R (4000 - 2000) = 3000R \\ &= 3000 \times 8.31 = 24.93 \times 10^3 \text{ J} \end{aligned}$$

$$\begin{aligned} \Delta U_{BC} &= C_V (T_C - T_B) \\ &= (3/2)R (2000 - 4000) = -3000R \\ &= -24.93 \times 10^3 \text{ J} \end{aligned}$$

$$\begin{aligned} \Delta U_{CD} &= C_V (T_D - T_C) \\ &= (3/2)R (1000 - 2000) = -1500R \\ &= -12.465 \times 10^3 \text{ J} \end{aligned}$$

$$\begin{aligned} \Delta U_{DA} &= C_V (T_A - T_D) \\ &= (3/2)R (2000 - 1000) = 1500R \\ &= +12.465 \times 10^3 \text{ J} \end{aligned}$$

Total $\Delta U = 0$

(ii) Workdone by gas during path AB under constant pressure

$$\begin{aligned} &= P(V_2 - V_1) = R(T_1 - T_2) = 8.314 \times 2000 \\ &= 16.62 \times 10^3 \text{ J} \end{aligned}$$

No work is done during BC and DA

Workdone on the gas during CD

$$= R(T_2 - T_1) = 8.314 \times 1000 = 8.314 \times 10^3 \text{ J}$$

Net workdone by the gas during the cycle

$$= (16.62 \times 10^3 - 8.314 \times 10^3) = 8.314 \times 10^3 \text{ J}$$

(iii) Heat absorbed during

$$\begin{aligned} AB &= C_P (T_1 - T_2) \\ &= \frac{R\gamma}{\gamma-1} \times 2000 = \frac{8.314 \times (5/3)}{(5/3)-1} \times 2000 \\ &\quad [\text{As } \gamma = 5/3 \text{ for monoatomic gas}] \\ &= 8.314 \times \frac{3}{2} \times 2000 = 41.55 \times 10^3 \text{ J} \end{aligned}$$

Heat released during

$$\begin{aligned} BC &= C_V (T_1 - T_2) \\ &= \frac{R}{\gamma-1} \times 2000 \\ &= 8.314 \times \frac{3}{2} \times 2000 = 24.93 \times 10^3 \text{ J} \end{aligned}$$

Heat released during

$$\begin{aligned} CD &= \frac{R\gamma}{\gamma-1} \times 1000 \\ &= 8.314 \times \frac{5}{2} \times 1000 \\ &= 20.775 \times 10^3 \text{ J} \end{aligned}$$

Heat absorbed during

$$\begin{aligned}
 DA &= \frac{R}{\gamma-1} \times 1000 \\
 &= 8.314 \times \frac{3}{2} \times 10^3 \text{ J} \\
 &= 12.465 \times 10^3 \text{ J}
 \end{aligned}$$

Net heat absorbed by the gas during the cycle

$$\begin{aligned}
 &= (12.465 + 41.55 - 20.775 - 24.93) \times 10^3 \text{ J} \\
 &= 8.314 \times 10^3 \text{ J}
 \end{aligned}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Laws of Thermodynamics

Module Number - 35 to 38

Practice Exercise 3.5

(i) An ideal gas undergoes in a thermodynamic cyclic process shown in figure-3.56. Find the work done by the gas in complete cycle.

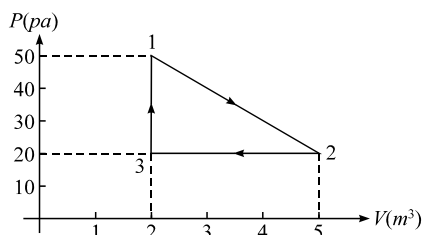


Figure 3.56

[45 J]

(ii) Figure-3.57 shows the VT graph of a thermodynamic process in which gas temperature changes from 300K to 500K during isochoric part of the cycle and volume of gas is doubled during isothermal part of the cycle. This graph is plotted for 2 mole of a gas. Find the total heat rejected by the gas in the complete cycle.

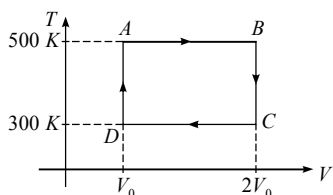


Figure 3.57

[- 2304.64 J]

(iii) Two moles of an ideal monoatomic gas is taken through a cycle $ABCA$ as shown in the PT diagram in figure-3.58. During the process AB , pressure and temperature of the gas vary such that $PT = \text{constant}$. If $T_0 = 300\text{K}$, calculate

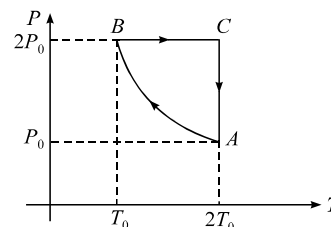


Figure 3.58

(a) The work done by the gas in the process AB

(b) The heat absorbed or released by the gas in each of the process.

[(a) - 1200 R; (b) - 2100 R, 1500 R, 831.6 R]

(iv) One mole of an ideal monoatomic gas is taken round the cyclic process $ABCA$ as shown in figure-3.59. Calculate

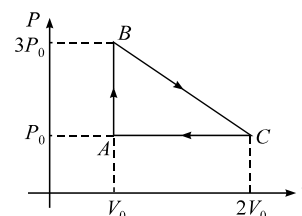


Figure 3.59

(a) The work done by the gas.

(b) The heat rejected by the gas in the path CA and the heat absorbed by the gas in the path AB .

(c) The net heat absorbed by the gas in the path BC .

(d) The maximum temperature attained by the gas during the cycle.

[(a) $P_0 V_0$; (b) $2.5 P_0 V_0$, $3 P_0 V_0$; (c) $0.5 P_0 V_0$; (d) $\frac{25P_0 V_0}{8R}$]

(v) Find the work done by an ideal gas during a closed cycle $1-4-3-2-1$ as shown if $P_1 = 10^5 \text{ Pa}$, $P_0 = 3 \times 10^5 \text{ Pa}$, $P_2 = 4 \times 10^5 \text{ Pa}$, $V_2 - V_1 = 100 \text{ litre}$ and segments $4-3$ and $2-1$ of the cycle are parallel to the V -axis.

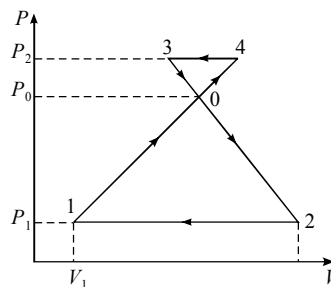


Figure 3.60

[7500 J]

(vi) Three moles of an ideal monoatomic gas perform a cyclic process as shown in the figure-3.61. The gas temperature in different states are $T_1 = 400\text{ K}$, $T_2 = 800\text{ K}$, $T_3 = 2400\text{ K}$, and $T_4 = 1200\text{ K}$. Find the work done during the cycle.

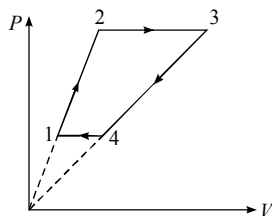


Figure 3.61

[9976.8 J]

(vii) An ideal gas with the adiabatic exponent γ goes a cycle (figure-3.62) within which the absolute temperature varies n -fold. Find the efficiency of this cycle.

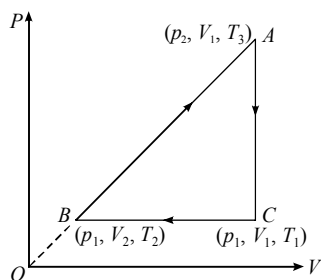


Figure 3.62

$$\left[\left(\frac{\gamma-1}{\gamma+1} \right) \left(\frac{n-1}{n+1} \right) \right]$$

(viii) An ideal gas with the adiabatic exponent γ goes through a direct (clockwise) cycle consisting of adiabatic, isobaric and

isochoric lines. Find the efficiency of the cycle if in the adiabatic process the volume of the ideal gas increases n -fold.

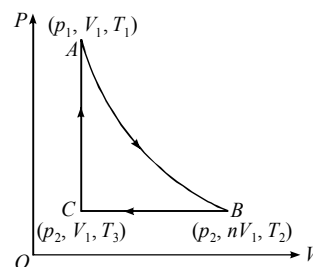


Figure 3.63

$$\left[1 - \gamma \frac{n-1}{n^\gamma - 1} \right]$$

(ix) An ideal gas whose adiabatic exponent is equal to γ goes through a cycle consisting of two isochoric and two isobaric lines as shown in figure-3.64. Find the efficiency of such a cycle, if the absolute temperature of the gas rises n times both in isochoric heating and in isobaric expansion.

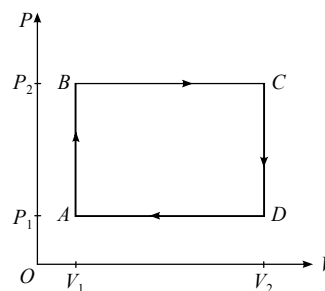


Figure 3.64

$$\left[\frac{(\gamma-1)(n-1)}{1+n\gamma} \right]$$

* * * * *

Discussion Question

Q3-1 When a thermos flask with some water in it is vigorously shaken. Does its temperature rise. Has some heat added to water during the process.

Q3-2 Some gas is enclosed in a piston-cylinder system. It is expanded to double its volume by isobaric or isothermal process. In which process more work is done by the gas.

Q3-3 In a room if door of refrigerator is kept open, will the room temperature decrease.

Q3-4 When a block moves in a straight line on a rough surface. Some heat is dissipated. Is this process reversible.

Q3-5 Can we convert mechanical work completely into heat.

Q3-6 Can we convert heat completely into mechanical work in a cyclic process.

Q3-7 Draw the PV diagram of the process shown in figure-3.65 in VT diagrams.

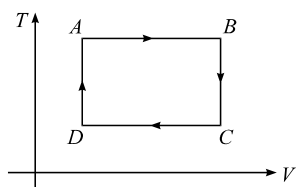


Figure 3.65

Q3-8 When a gas is compressed, it becomes more elastic. Explain.

Q3-9 When we whistle out air on to palm held close to mouth, the air feels cold, but when we blow air out from mouth, keeping it wide open, the air feels hot. Explain.

Q3-10 One mole of an ideal gas changes its state from state-1 to state-2 along the two paths I and II as shown in figure-3.66. In which of the process is the amount of heat absorbed by the gas more. Explain.

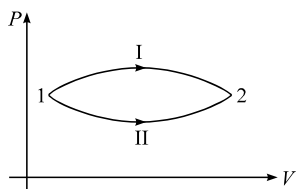


Figure 3.66

Q3-11 Plot approximate graphs between pressure and internal energy of a sample of gas for these processes, (a) isobaric, (b) isochoric, (c) isothermal and (d) adiabatic.

Q3-12 Represent the Carnot cycle on a VT diagram.

Q3-13 The molar heat capacity at constant pressure of all diatomic gases is always same.

Q3-14 A gas is first compressed adiabatically and then isothermally. In both cases, the initial state of the gas is the same. Find in which case more work is done on the gas.

Q3-15 In the polytropic process $PV^2 = \text{constant}$, is the gas cooled or heated with increase in volume.

Q3-16 In the following PT -graph shown in figure-3.67, find what happens to the volume of gas on increasing temperature of the gas.

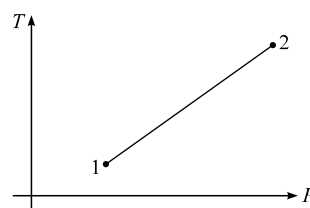


Figure 3.67

Q3-17 In the following VT -graph shown in figure-3.68, find what happens to the pressure of the gas on increasing temperature of the system.

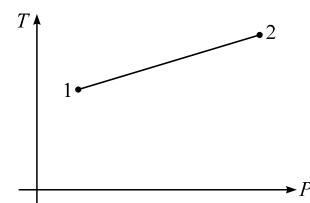


Figure 3.68

Q3-18 What happens to the internal energy of water vapour in the air that condenses on the outside of a cold glass of water ? Is work done or heat exchanged ?

Q3-19 When a mountaineer who eats food, gets warm a lot during a climb and does a lot of mechanical work in raising himself upwards. When he descends, again he also gets warm during descent. Is the source of this energy the same as during the ascent.

Q3-20 The temperature of a gas is increasing, hence its internal energy also increases. Just by observing initial and final states, can we determine whether the internal energy increment was due to work or by heat transfer.

Q3-21 When a hand pump is used to inflate the tires of a bicycle, the pump gets warm after a while. Why ?

Q3-22 There are few materials that contracts when their temperature is increased, such as water between 0°C and 4°C . Would you expect C_p for such materials to be greater or less than C_v ?

Q3-23 A disc rotated about its central axis and gently placed on another coaxial disc at rest. Due to friction between the two first disc retards and the second one starts rotating and after some time both with rotate with a common angular speed. In this process the total internal energy of the system of two discs does not change. State and justify whether this statement is true or false.

Q3-24 “When heat is added to a system, the internal energy of the system must increase”. Justify this statement.

Q3-25 When in a thermally insulated container gas is filled at some temperature above room temperature on ground floor and it is taken to second floor of a building, will the temperature of the gas in it change. Is some work is done by or on the gas in the process.

Q3-26 The outer surface of a metallic cylinder is rubbed by a rough duster, after some. Due to this gas temperature increases. Is some work is done on the gas in the process or some heat is transferred to the gas.

Q3-27 When a tyre bursts, it cools down. Why ?

Q3-28 Consider two thermodynamic processes shown in figure-3.69. In both the processes volume of the gas in initial and final states are same. In which path work done by the gas is more.

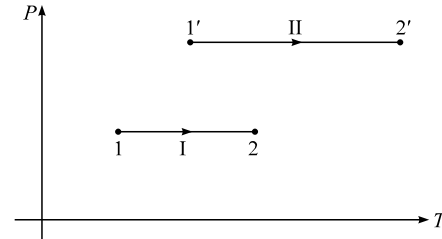


Figure 3.69

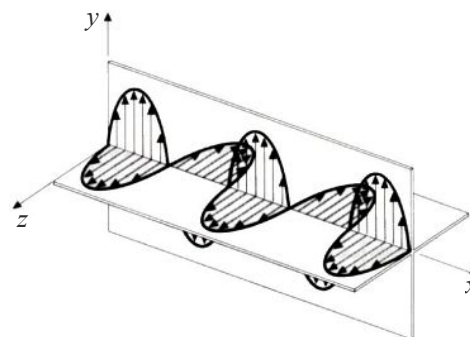
* * * * *

Heat Transfer

FEW WORDS FOR STUDENTS

Upto now we've discussed about the effects of heat supply on different substances solids, liquids and gaseous systems. We've learnt how substances respond on absorbing thermal energy and the basic laws of conservation of energy conserved with thermal energy.

Now in this chapter we'll discuss the different ways how the transfer of energy takes places from one point (place) to another. This section is most essential for complete knowledge of thermal physics. In different situations the ways by which thermal energy transfer takes place depends on so many factors including the surrounding environment. Mainly these ways are classified in three broad categories conduction, convection and radiation. In previous chapters it is learnt that whenever heat is supplied to a body or a system, this is accomplished by either of these three ways.



CHAPTER CONTENTS

- 4.1 Conduction of Heat
- 4.2 Convection of Heat
- 4.3 Radiation of Heat
- 4.4 Newton's Law of Cooling
- 4.5 Black Body Radiation Spectrum

COVER APPLICATION



Figure-(a)



Figure-(b)

Convection can only happen in fluids. This includes liquids and gases and is because the molecules have to be free to move. Heat energy can transfer by convection when there is a significant difference in temperature between two parts of a fluid. When this temperature difference exists, hot fluids rise and cold fluids sink, and then currents, or movements, are created in the fluid. Figure-(a) Shows a pressure cooker inside of which when its bottom is heated, due to convection hot water flows up and heated the vegetables even in upper part of the cooker so it cooks uniformly. Figure-(b) shows a camp fire in which due to fire, hot air is lifted up and cold air from side is pulled and convection currents setup in the surrounding of fire that's why the region above fire is extremely hot compared to side region.

Heat energy can be transferred in three ways from one place to another, Conduction, Convection and Radiation. When one end of a metal rod is heated, the other end gets warm. This is an example of conduction, in which thermal energy is transferred without any net movement of the material itself. It is accomplished due to vibration and collisions of material particles. Conduction is a relatively slow process. A more rapid process of heat transfer is accomplished through the mass-motion or flow of some fluid, such as air or water and is called convection. This transfer takes place when warm air flows through a room and when hot and cold liquids are poured together. A more rapid transfer of thermal energy is accomplished by radiation a process that requires neither contact nor mass flow like the energy from sun comes to us by radiation. We can also feel radiation when standing near by a fire, a room heater or radiators.

4.1 Conduction of Heat

If one end of a metal rod is placed in a flame while the other is held in the hand, that part of the rod one is holding becomes hotter and hotter, although it is not itself in direct contact with the flame. Heat is said to reach the cooler end of the rod by “*thermal conduction*” through the material of the rod. The molecules at the hot end of the rod increase the energy of their vibration as the temperature increases. Then, as they collide with their more slowly moving neighbours on the rod toward colder end, some of their energy is shared with these neighbours, and they in turn pass it along those still farther from the flame. Hence energy of thermal motion is passed along from one molecule to the next, while each individual molecule remains at its original position.

Most of the metals are good conductors of electricity and also of heat. Conduction of electricity is by the free electrons in metal lattice which are detached from their parent molecules and are free to move in the lattice region called conduction band which is evenly spreaded in the whole lattice of the metal. The free electrons also play a part in conduction of heat, and the reason metals are such good heat conductors is that the free electrons provide an effective mechanism for carrying thermal energy from the hotter to the colder portions of the metal.

Conduction of heat can take place in a body only when different parts of body are at different temperatures, and the direction of heat flow is always from points of higher temperature to those at lower temperature. Figure-4.1 shows a metal bar connecting two bodies at temperature T_1 and T_2 ($T_1 > T_2$). The sides of the rod are covered by an insulating material, so that no heat flow takes place to the sides (surroundings). Practically even the best heat insulators also conduct heat to some extent. Perfect heat insulator is just an idealized concept.

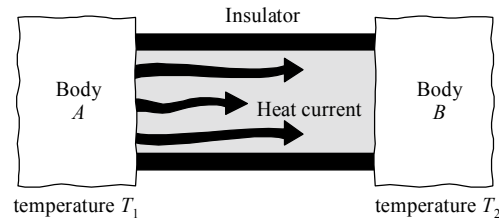


Figure 4.1

In the above example of heat conduction through a bar, after sufficiently long time, the temperature within the bar is found to decrease uniformly with distance from the hot to the cold face. When this uniform temperature gradient is established, at each point within the rod, the temperature remains constant with time. This condition is called steady state heat flow. In next section we'll discuss steady state conduction of heat in more details.

An objects usefulness as a thermal conductor depends in a number of things including its length, thickness and the material from which it is made. Initially we have discussed that when two bodies at different temperatures are connected by a thermally conducting material, heat flows from high temperature body to that which is at a lower temperature. The process by which this thermal energy is transferred through the material is called conduction of heat. Lets discuss conduction in detail step by step.

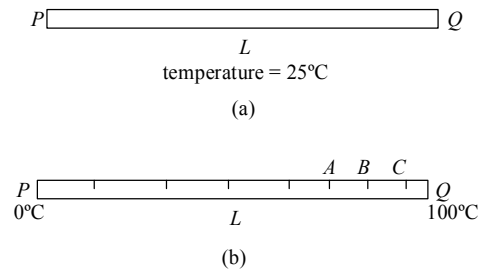


Figure 4.2

Consider an iron rod PQ of length L shown in figure-4.2(a). It is initially at room temperature 25°C . If at an instant left end P of the rod is submerged in an ice bath at 0°C and the other end Q is submerged in boiling water at 100°C . Just after this the temperature of different parts of rod will start changing with time. The part of rod near to end P will have its temperature falling with time and the part near to end Q will have its temperature rising with time. This happens due to heat flow through the rod from high temperature side to lower temperature side. Lets discuss is depth. Consider a part of rod near to end Q (marked as AB in figure-4.2b) as shown in figure-4.3. If temperature to the right of point B is 90°C and that of segment AB and to the left of A are 82°C and 76°C respectively then due to temperature difference some heat say ΔQ_1 flows from end C

to end B then this heat increases the agitation of the molecules of the material in segment BC then due to the vibration and collision of these molecules with their adjacent atoms or molecules, some heat say ΔQ_2 is transferred to segment AB . Some of this ΔQ_1 was absorbed by segment BC , that will raise the temperature of BC by some value, hence ΔQ_2 will be less than ΔQ_1 . Similarly when ΔQ_2 heat comes to segment AB , some of this increases the temperature of segment AB and rest is conducted to the next segments of the rod and in this manner heat is conducted to the other end of the rod for some time.

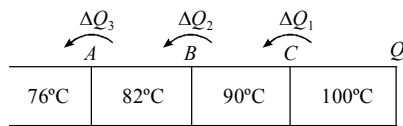


Figure 4.3

After some time we can observe that as the temperature of individual segments of the rod is increasing, the temperature difference between adjacent segments is approaching a constant value as in figure-4.3 the difference of temperature between BC and the end Q was 10°C and that between segments BC and AB was 8°C and after some time as shown in figure-4.4 temperature of the segments CQ , BC and AB are 95°C , 90°C and 86°C respectively the temperature difference is decreasing and hence as temperature difference decreases rate of heat flow becomes slower and ΔQ_1 and ΔQ_2 will be approximately same thus the amount of heat absorbed decreases with time.

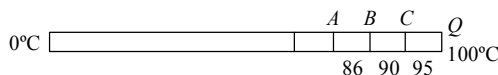


Figure 4.4

Now after a long time when temperature difference between different equally spaced segment of rod becomes equal the rate of heat taken by a segment will be equal to the rate of heat given to the next segment and at this stage no abortion of heat takes place by any segment of rod and temperature of every part of rod becomes constant. This is called steady state of thermal conduction. In steady state throughout the medium rate of flow of heat becomes constant and the amount of heat enters from one end is equal to the amount of heat leaving from the other end. Thus we state in steady state of thermal conduction, in the medium a constant temperature gradient is established and temperature of every part of the medium becomes constant as shown in figure-4.5 for the example we've discussed.

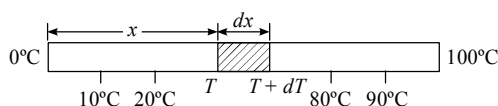


Figure 4.5

Thus in steady state if at a distance x from one end of the rod temperature is T and at a distance $x + dx$, temperature is $T + dT$ than the temperature gradient of the rod at a distance x is given as

$$\frac{dT}{dx} = \text{constant throughout the rod in steady state}$$

This shows that in steady state, in the medium the temperature of medium varies linearly from one end to the other.

4.1.1 Analysis of Heat Conduction in Steady State

In steady state conduction of heat through a medium is uniform and very easy to understand compared to the case before steady state when continuous absorption of heat takes place by every part of the medium during conduction.

To understand the thermal conduction mathematically, we consider two bodies maintained at different temperature T_1 and T_2 respectively as shown in figure-4.6. The two bodies are connected by the thermally conducting rod of length l and area of cross-section A as shown. After connecting the two bodies, it will take some time to be in steady state. When steady flow of heat from one body to another starts. This heat flow rate dQ/dt is observed as

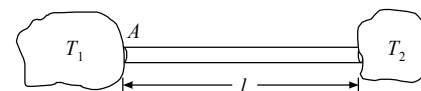


Figure 4.6

- (i) Directly proportional to temperature difference

$$\frac{dQ}{dt} \propto (T_1 - T_2) \quad \dots (4.1)$$

- (ii) Directly proportional to area of cross-section, through which heat is flowing

$$\frac{dQ}{dt} \propto A \quad \dots (4.2)$$

- (iii) Inversely proportional to the length of the medium (rod) through which heat is flowing

$$\frac{dQ}{dt} \propto \frac{1}{l} \quad \dots (4.3)$$

From the equations-(4.1), (4.2) and (4.3), we have

$$\frac{dQ}{dt} = k \frac{A(T_1 - T_2)}{l} \quad \dots (4.4)$$

Here k is the proportionality constant and depends on material of the medium through which heat is flowing and it is termed as thermal conductivity of the medium. The SI units used for heat flow are J/sec or W . So the SI units of k are $\text{J/s-m}^\circ\text{C}$ or $\text{W/m}^\circ\text{C}$. A high thermal conductivity indicates a good heat conductor

and a low thermal conductivity indicates a good heat insulator. When we design a good insulator, the first requirement is to choose a material with a small thermal conductivity so from equation-(4.4) the heat flow is small. In addition by minimizing area of contact A and making the path length l as long as possible, we can further reduce the heat flow.

4.1.2 Heat Transfer Before Steady State

In previous section we've discussed that in steady state heat conduction through the material of a medium takes place in such a manner that no absorption of heat takes place and throughout the material the rate of flow of heat remains constant and a uniform temperature gradient exist in the medium.

As discussed earlier, before steady state the amount of heat transferred in the medium to further sections decreases as some amount of heat is absorbed by the intermediate sections of the material, this is explained in figure-4.7. This figure shows a section of material through which heat is conducted from high temperature end to low temperature end of the medium. Consider two cross-section A and B in the medium. Let at an instant the rate of flow of heat through the cross-section A is $\frac{dQ_1}{dt}$ and that through the cross-section B is $\frac{dQ_2}{dt}$. Then these flow rates are given as

$$\frac{dQ_1}{dt} = kA \left. \frac{dT}{dx} \right|_A$$

and

$$\frac{dQ_2}{dt} = kA \left. \frac{dT}{dx} \right|_B$$

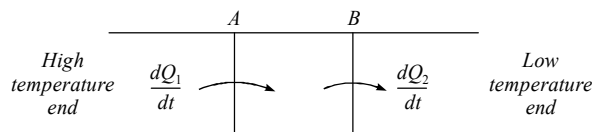


Figure 4.7

Here $\left. \frac{dT}{dx} \right|_A$ and $\left. \frac{dT}{dx} \right|_B$ are the temperature gradients at the location of cross sections A and B respectively.

Here it is obvious that if the medium steady state is not achieved $\frac{dQ_2}{dt}$ must be less than $\frac{dQ_1}{dt}$ as some amount of heat is absorbed by the section AB . Thus here we can say that the rate at which the section AB absorbs heat at this instant is given as

$$\left. \frac{dQ}{dt} \right|_{\text{absorbed by } AB} = \frac{dQ_1}{dt} - \frac{dQ_2}{dt}$$

or

$$= kA \left[\left. \frac{dT}{dx} \right|_A - \left. \frac{dT}{dx} \right|_B \right]$$

If the mass of section AB is m and s is the specific heat of the material of section AB , the rate at which temperature of section

AB rises $\left. \frac{dT}{dt} \right|_{AB}$ can be given as

$$ms \left. \frac{dT}{dt} \right|_{AB} = kA \left[\left. \frac{dT}{dx} \right|_A - \left. \frac{dT}{dx} \right|_B \right]$$

As we know that when steady state is achieved, the temperature gradient at all cross-section of the medium becomes equal. Thus we can say that as the medium is approaching towards steady state the difference in temperature gradients at cross-section A and B decreases and hence the rate at which the temperature of segment AB is increasing also decreases and in steady state it becomes zero. This is the reason why we say that in steady state the temperature of all segments of the medium becomes constant at a steady value (linearly decreasing from one end to another) and no absorption of heat takes place when heat conduction takes place in steady state.

4.1.3 Thermal Resistance and Ohm's Law in Thermal Conduction

One more parameter can be defined to indicate the effectiveness of insulation for a material called thermal resistance of the material. For a given material, thermal resistance is given as

$$R_{th} = \frac{1}{k} \frac{l}{A} \quad \dots (4.5)$$

Here l is the path length for heat flow and A is the area of cross-section of material through which heat is flowing. Students should note that equation-(4.5) can only be used to find thermal resistance of those materials in which cross-sectional area A through which heat is conducted is uniform throughout the path length.

In thermal conduction ohm's law can be stated as "the heat current across a given thermal resistance is directly proportional to the thermal potential (temperature difference) difference across it." Analytically it is given as

$$(T_1 - T_2) = \left(\frac{dQ}{dt} \right) R_{th}$$

[As in current electricity $V_1 - V_2 = IR$]

Here
$$\frac{dQ}{dt} = \frac{kA}{L} (T_1 - T_2) \quad \dots (4.6)$$

Equation- (4.6) is exactly same as equation-(4.5). We use this analogy of current electricity in numerical calculations of thermal conduction. We take few examples to understand this analogy.

Illustrative Example 4.1

Consider two rods of equal cross-sectional area A , one of Aluminium and other of Iron joined end to end as shown in figure-4.8. Length of the two rods and their thermal conductivities are l_1, k_1 and l_2, k_2 respectively. If the ends of the rods are maintained at temperature T_1 and T_2 , ($T_1 > T_2$) find the temperature of the junction in steady state.

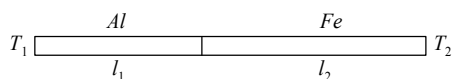


Figure 4.8

Solution

We know in steady state the rate of heat flow or the thermal current throughout the medium remains constant thus we have

$$\left(\frac{dQ}{dt}\right)_{Al} = \left(\frac{dQ}{dt}\right)_{Fe}$$

If junction temperature is assumed T_x then we have

$$\frac{k_1 A (T_1 - T_x)}{l_1} = \frac{k_2 A (T_x - T_2)}{l_2}$$

$$\text{or } T_x = \frac{\frac{k_1 T_1}{l_1} + \frac{k_2 T_2}{l_2}}{\frac{k_1}{l_1} + \frac{k_2}{l_2}} = \frac{k_1 l_2 T_1 + k_2 l_1 T_2}{k_1 l_2 + k_2 l_1}$$

Illustrative Example 4.2

A bar of copper of length 75 cm and a bar of steel of length 125 cm are joined together end to end. Both are of circular cross-section with diameter 2 cm. The free ends of the copper and steel bars are maintained at 100°C and 0°C respectively. The surfaces of the bars are thermally insulated. What is the temperature of the copper-steel junction? What is the heat transmitted per unit time across the junction? Thermal conductivity of copper $9.2 \times 10^{-2} \text{ kcal m}^{-1} \text{ }^\circ\text{C}^{-1} \text{ s}^{-1}$ and that of steel is $1.1 \times 10^{-2} \text{ kcal m}^{-1} \text{ }^\circ\text{C}^{-1} \text{ s}^{-1}$.

Solution

Let T_x be the temperature of the junction of the copper and steel bars (Figure-4.9). In the steady state, the rate of flow of heat in the copper bar must be the same as that in the steel bar,

i.e. (since their cross-sectional areas are equal)

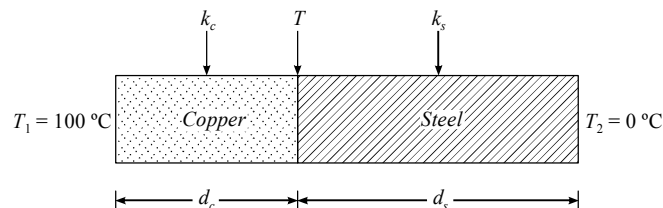


Figure 4.9

$$\frac{dQ}{dt} = \frac{k_c A (T_1 - T_x)}{d_c} = \frac{k_s A (T_x - T_2)}{d_s}$$

$$\text{or } \frac{k_c}{k_s} \times \frac{d_s}{d_c} = \frac{T_x - T_2}{T_1 - T_x} \quad \dots (4.7)$$

Substituting the given values in equation (4.7), we have

$$\frac{9.2 \times 10^{-2}}{1.1 \times 10^{-2}} \times \frac{125}{75} = \frac{T_x - 0}{100 - T_x}$$

$$\text{or } T_x = 93.3^\circ\text{C}$$

or Thus the temperature of the junction is 93.3 °C. The heat flowing through the junction per second is

$$\frac{dQ}{dt} = \frac{k_c A (T_1 - T_x)}{d_c} \quad \dots (4.8)$$

Given $k_c = 9.2 \times 10^{-2} \text{ kcal m}^{-1} \text{ }^\circ\text{C}^{-1} \text{ s}^{-1}$, $d_c = 75 \text{ cm} = 0.75 \text{ m}$, and $A = \pi r^2 = \pi \times (10^{-2} \text{ m})^2 = 3.14 \times 10^{-4} \text{ m}^2$. Also $T_1 = 100^\circ\text{C}$ and $T = 93.3^\circ\text{C}$. Using these values in (4.8) we have

$$\begin{aligned} \frac{dQ}{dt} &= \frac{9.2 \times 10^{-2} \times 3.14 \times 10^{-4} \times (100 - 93.3)}{0.75} \\ &= 2.58 \times 10^{-4} \text{ J/s.} \end{aligned}$$

Illustrative Example 4.3

A cylindrical brass boiler of radius 15 cm and thickness 1.0 cm is filled with water and placed on an electric heater. If the water boils at the rate of 200 gm/s, estimate the temperature of the heater filament. Thermal conductivity of brass $= 109 \text{ J s}^{-1} \text{ m}^{-1} \text{ }^\circ\text{C}^{-1}$ and heat of vaporisation of water $= 2.256 \times 10^3 \text{ J g}^{-1}$.

Solution

Rate of boiling of water = 200 gm/s. Since the heat of vaporisation of water is $2.256 \times 10^3 \text{ J/gm}$, the amount of heat energy required to boil 200 g of water is

$$2.256 \times 10^3 \text{ J g}^{-1} \times 200 \text{ g} = 4.512 \times 10^5 \text{ J}$$

Since water is boiling at the rate of 200 gm/s, the rate at which heat energy is supplied by the heater to water is

$$\frac{Q}{t} = 4.512 \times 10^5 \text{ J s}^{-1} \quad \dots (4.9)$$

Now, Radius of the boiler

$$(r) = 15 \text{ cm} = 0.15 \text{ m}$$

Base area of the boiler

$$(A) = \pi r^2 = 3.142 \times (0.15)^2 = 0.0707 \text{ m}^2$$

Thickness of brass

$$(d) = 1.0 \text{ cm} = 1.0 \times 10^{-2} \text{ m}$$

Thermal conductivity of brass

$$(k) = 109 \text{ J s}^{-1} \text{ m}^{-1} \text{ }^\circ\text{C}^{-1}$$

Temperature of boiling water

$$(T_w) = 100^\circ\text{C}$$

If T_f is the temperature of the filament, the rate at which heat energy is transmitted through the base is given by

$$\frac{Q}{t} = \frac{kA(T_f - T_w)}{d} \quad \dots (4.10)$$

Substituting the values of k , A , T_w and d in (4.10) and equating with (4.9), we get

$$\frac{109 \times 0.0707 (T_f - 100)}{1.0 \times 10^{-2}} = 4.512 \times 10^5$$

$$\text{or} \quad T_f - 100 = 585.5$$

$$\text{or} \quad T_f = 685.5^\circ\text{C}$$

Illustrative Example 4.4

A slab of stone of area 3600 cm^2 and thickness 10 cm is exposed on the lower surface to steam at 100°C . A block of ice at 0°C rests on the upper surface of the slab. If in one hour 4.8 kg of ice is melted, calculate the thermal conductivity of the stone.

Solution

Assuming that heat loss from the sides of the slab is negligible, the amount of heat flowing through the slab is :

$$Q = \frac{kA(T_1 - T_2)t}{d} \quad \dots (4.11)$$

If m is the mass of ice and L the latent heat of fusion, then

$$Q = mL \quad \dots (4.12)$$

Equating (4.11) and (4.12), we have

$$mL = \frac{kA(T_1 - T_2)t}{d}$$

or

$$k = \frac{mLd}{A(T_1 - T_2)t} \quad \dots (4.13)$$

Given

$$m = 4.8 \text{ kg},$$

$$d = 10 \text{ cm} = 0.1 \text{ m},$$

$$A = 3600 \text{ cm}^2 = 0.36 \text{ m}^2$$

$$T_1 = 100^\circ\text{C}, T_2 = 0^\circ\text{C} \text{ and } t = 1 \text{ hour} = (60 \times 60) \text{ s}.$$

We know that

$$L = 80 \text{ cal g}^{-1} = 80,000 \text{ cal kg}^{-1}$$

$$= 80,000 \times 4.2 \text{ J kg}^{-1} = 3.36 \times 10^5 \text{ J kg}^{-1}.$$

Substituting these values in equation (4.13) and solving, we get

$$k = 1.24 \text{ J s}^{-1} \text{ m}^{-1} \text{ }^\circ\text{C}^{-1} \text{ or } 1.24 \text{ W m}^{-1} \text{ K}^{-1}$$

Illustrative Example 4.5

Few rods of material X and Y are connected as shown in figure-4.10. The cross sectional area of all the rods are same. If the end A is maintained at 80°C and the end F is maintained at 10°C . Calculate the temperatures of junctions B and E in steady state. Given that thermal conductivity of material X is double that of Y .

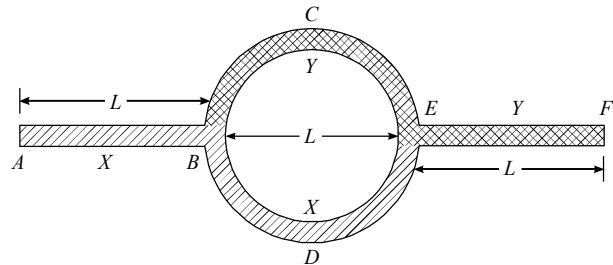


Figure 4.10

Solution

We first find the thermal resistances of the different rods shown in figure-4.10 these are given as

$$R_{AB} = \frac{1}{k_X} \cdot \frac{L}{A}$$

$$R_{BCE} = \frac{1}{k_Y} \cdot \frac{(\pi L/2)}{A}$$

$$R_{BDE} = \frac{1}{k_X} \cdot \frac{(\pi L/2)}{A}$$

$$R_{EF} = \frac{1}{k_Y} \cdot \frac{L}{A}$$

Now in steady state the amount of heat flow from end E to F remains constant as there is no absorption of heat. Then we must have that the amount of heat coming at junction B is equal to the amount of heat having B and same statement can be given for junction E . If temperature of junction B and E are taken as T_B and T_E then we have for junction B .

$$\frac{T_A - T_B}{R_{AB}} = \frac{T_B - T_E}{R_{BCE}} + \frac{T_B - T_E}{R_{BDE}}$$

$$\text{or } \frac{80 - T_B}{\frac{1}{k_X} \cdot \frac{L}{A}} = \frac{T_B - T_E}{\frac{1}{k_Y} \cdot \frac{\pi L/2}{A}} + \frac{T_B - T_E}{\frac{1}{k_X} \cdot \frac{\pi L/2}{A}}$$

$$\text{or } 80 - T_B = \left(\frac{T_B - T_E}{\pi} \right) + \left(\frac{T_B - T_E}{2\pi} \right) \quad [\text{As } k_X = 2k_Y]$$

$$\text{or } 80 - T_B = \frac{3}{2\pi} (T_B - T_E) \quad \dots (4.14)$$

Similarly for junction E , we can write

$$\frac{T_B - T_E}{R_{BCE}} + \frac{T_B - T_E}{R_{BDE}} = \frac{T_E - 10}{R_{EF}}$$

$$\text{or } \frac{T_B - T_E}{\frac{1}{k_Y} \cdot \frac{\pi L/2}{A}} + \frac{T_B - T_E}{\frac{1}{k_X} \cdot \frac{\pi L/2}{A}} = \frac{T_E - 10}{\frac{1}{k_Y} \cdot \frac{L}{A}}$$

$$\text{or } \frac{2(T_B - T_E)}{\pi} + \frac{4(T_B - T_E)}{\pi} = T_E - 10$$

$$\text{or } \frac{3}{4\pi} (T_B - T_E) = T_E - 10 \quad \dots (4.15)$$

Solving equation-(4.14) and (4.15), we get

$$T_E = 19.74^\circ\text{C}$$

$$\text{and } T_B = 60.52^\circ\text{C}$$

Illustrative Example 4.6

The space between two thin concentric metallic spherical shells of radii a and b is filled with a thermal conducting medium of conductivity k . The inner shell is maintained at temperature T_1 and outer is maintained at a lower temperature T_2 . Calculate the rate of flow of heat in radially outward direction through the medium.

Solution

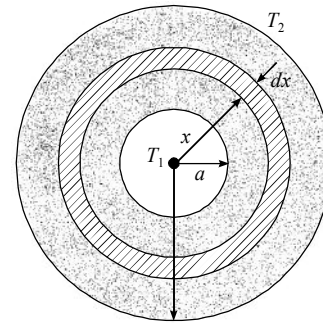


Figure 4.11

Figure-4.11 shows the situation. As we know according to Ohm's Law in thermal conduction, the rate of heat flow or thermal current is given as

$$\frac{dQ}{dt} = \frac{T_1 - T_2}{R_{th}}$$

Where R_{th} is the thermal resistance of the medium through which heat is conducted, which is given as

$$R_{th} = \frac{1}{k} \cdot \frac{L}{A}$$

Where L is the path length and A is the cross-sectional area of the medium through which heat is conducted. But this relation can be used to find thermal resistance only when throughout the path length, area of cross-section A is uniform. In this case in radial direction area is increasing so we can not use the above relation to find thermal resistance.

Here we consider an elemental shell of radius x and width dx in the medium as shown in figure-4.11. If dR be the thermal resistance of this shell then, we have

$$dR = \frac{1}{k} \cdot \frac{dx}{4\pi x^2}$$

Here between the two inner and outer shells, all such small dR resistances can be considered in series combination thus the net thermal resistance of the medium between inner and outer shells is given as

$$R = \int dR = \int_a^b \frac{1}{k} \cdot \frac{dx}{4\pi x^2}$$

or

$$R = \frac{1}{4\pi k} \left[-\frac{1}{x} \right]_a^b = \frac{b-a}{4\pi kab}$$

Now rate of flow of heat from inner to outer shell in radial direction is given as

$$\frac{dQ}{dt} = \frac{T_1 - T_2}{R} = \frac{4\pi kab(T_1 - T_2)}{b-a}$$

Illustrative Example 4.7

A closed cubical box made of a perfectly insulating material has walls of thickness 8 cm and the only way for the heat to enter or leave the box is through two solid cylindrical metal plugs, each of cross-sectional area 12 cm^2 and length 8 cm fixed in the opposite walls of the box. The outer surface A of one plug is kept at a temperature of 100°C while the outer surface of the other plug is maintained at a temperature of 4°C . The thermal conductivity of the material of the plug is $0.5 \text{ cal/cm sec } ^\circ\text{C}$. A source of energy generating 36 cal s^{-1} is enclosed inside the box. Find the equilibrium temperature of the inner surface of the box assuming that it is the same at all points on the inner surface.

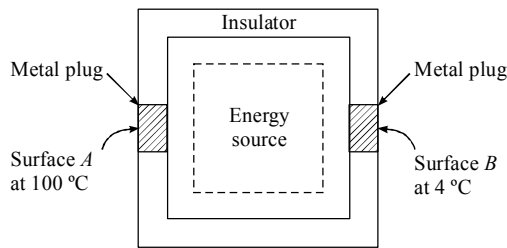
Solution

Figure 4.12

At equilibrium, the total energy generated by the source per second is equal to the heat leaving per second through the two metal plugs (Figure-4.12). Let $T^\circ\text{C}$ be the equilibrium temperature. Then heat leaving the box per second through surface A

$$= \frac{k(T-100) \times 12}{8} \text{ cal s}^{-1}$$

Heat leaving the box per second through the surface B

$$= \frac{k(T-4) \times 12}{8} \text{ cal s}^{-1}$$

Hence
$$\frac{12k}{8} (T-100 + T-4) = 36$$

or
$$2T-104 = \frac{36 \times 8}{12k} = \frac{36 \times 8}{12 \times 0.5} = 48 \text{ or } T = 76^\circ\text{C}$$

Illustrative Example 4.8

Figure-4.13 shows a water tank at a constant temperature. T_0 and a small body of mass m , and specific heat s at a temperature T_1 . Given that $T_1 < T_0$. A metal rod of length L , cross-sectional area A whose thermal conductivity is K is placed between the tank and the body to connect them. Find the temperature of

body as a function of time. Given that the heat capacity of rod is negligible.

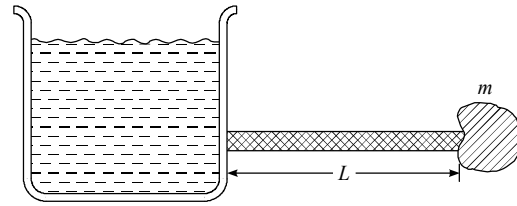


Figure 4.13

Solution

As temperature of water tank is constant at T_0 as it is very large, heat is conducted through the rod to the small body. If at an instant $t = t$, the temperature of small body is T , then the rate of heat flow through the rod can be written as

$$\frac{dQ}{dt} = \frac{KA(T_0 - T)}{L} \quad \dots (4.16)$$

Equation-(4.16) is used for heat conduction is steady state and here as the heat capacity of the rod through which heat is conducted is negligible, it does not absorb any heat thus we can assume that it is always in steady state.

If on dQ amount of heat absorption, let the temperature of small body rises by dT , then we have

$$dQ = msdT$$

Now from equation (4.16)

$$ms \frac{dT}{dt} = \frac{KA}{L} (T_0 - T)$$

or
$$\frac{dT}{T_0 - T} = \frac{KA}{msL} dt$$

Now integrating the above expression in proper limits, we get

$$\int_{T_1}^T \frac{dT}{T_0 - T} = \frac{KA}{msL} \int_{t=0}^t dt$$

$$- \ln \left(\frac{T_0 - T}{T_0 - T_1} \right) = \frac{KA}{msL} t$$

or
$$\frac{T_0 - T}{T_0 - T_1} = e^{-\frac{KA t}{msL}}$$

or
$$T = T_0 - (T_0 - T_1) e^{-\frac{KA t}{msL}}$$

Illustrative Example 4.9

When two bodies of masses m_1 and m_2 with specific heats s_1 and s_2 at absolute temperatures T_{10} and T_{20} ($T_{10} > T_{20}$) are connected by a rod of length l and cross sectional area A with thermal conductivity k . Find the temperature difference of the bodies after time t . Neglect any heat loss due to radiation at any surface.

Solution

Let at time t after connecting the two bodies, they are at temperature, T_1 and T_2 respectively ($T_1 > T_2$). At this time, heat must be flowing from m_1 to m_2 as m_1 is at higher temperature if dQ is the amount of heat flow through the rod from m_1 to m_2 and due to this if dT_1 is fall in temperature of m_1 and dT_2 is rise in temperature of m_2 then we have

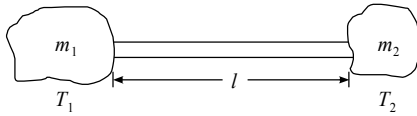


Figure 4.14

$$dQ = -m_1 s_1 dT_1 \quad \dots (4.17)$$

Also $dQ = m_2 s_2 dT_2 \quad \dots (4.18)$

As this heat dQ is conducted through the rod, we have

$$\frac{dQ}{dt} = \frac{kA(T_1 - T_2)}{l}$$

or $dQ = \frac{kA}{l} (T_1 - T_2) dt \quad \dots (4.19)$

As in this expression dQ is given as a function of $(T_1 - T_2)$. Thus from (4.17) and (4.18), we can get

$$dQ \left[\frac{1}{m_1 s_1} + \frac{1}{m_2 s_2} \right] = -d(T_1 - T_2) \quad \dots (4.20)$$

Now from (4.19) and (4.20)

$$\frac{kA(T_1 - T_2)}{l} \left[\frac{1}{m_1 s_1} + \frac{1}{m_2 s_2} \right] dt = -d(T_1 - T_2)$$

or $\frac{d(T_1 - T_2)}{(T_1 - T_2)} = -\frac{kA}{l} \left[\frac{1}{m_1 s_1} + \frac{1}{m_2 s_2} \right] dt$

Integrating the expression within proper limits, we get

$$\int_{T_{10}-T_{20}}^{T_1-T_2} \frac{d(T_1 - T_2)}{(T_1 - T_2)} = -\int_0^t \frac{kA}{l} \left(\frac{1}{m_1 s_1} + \frac{1}{m_2 s_2} \right) dt$$

$$\ln \frac{(T_1 - T_2)}{(T_{10} - T_{20})} = -\frac{kA}{l} \left(\frac{1}{m_1 s_1} + \frac{1}{m_2 s_2} \right) dt$$

or $(T_1 - T_2) = (T_{10} - T_{20}) e^{-\frac{kA}{l} \left(\frac{1}{m_1 s_1} + \frac{1}{m_2 s_2} \right) t}$

Illustrative Example 4.10

A cubical container of side a and wall thickness x ($x \ll a$) is suspended in air and filled n moles of diatomic gas (adiabatic exponent $= \gamma$) in a room where room temperature is T_0 . If at $t = 0$ gas temperature is T_1 ($T_1 > T_0$), find the gas temperature as a function of time t . Assume the heat is conducted through all the walls of container.

Solution

As shown in figure-4.15 if at time $t = 0$, gas temperature is T_1 and after time t , its temperature is T , heat will be conducted through the walls of container, the rate of heat conducted to outside surrounding can be written as

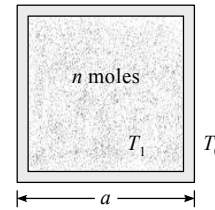


Figure 4.15

$$\frac{dQ}{dt} = \frac{k(6a^2)}{x} [T - T_0] \quad \dots (4.21)$$

If dQ heat is conducted outside in a time dt and due to this if the temperature gas falls by dT , we have

$$dQ = -nC_v dt \quad [\text{As volume of container is constant}]$$

or $dQ = \frac{nR}{\gamma - 1} dT \quad [\text{as } C_v = \frac{R}{\gamma - 1}] \quad \dots (4.22)$

From equation (4.21) and (4.22) we have

$$\frac{nR}{\gamma - 1} \frac{dT}{dt} = -\frac{6ka^2}{x} (T - T_0)$$

or $\frac{dT}{T - T_0} = -\frac{6ka^2(\gamma - 1)}{nRx} dt$

Integrating the above expression within proper limits from beginning ($t = 0$)

$$\int_{T_1}^T \frac{dT}{T - T_0} = - \int_{t=0}^t \frac{6ka^2(\gamma - 1)}{nRx} dt$$

$$\ln \left(\frac{T - T_0}{T_1 - T_0} \right) = - \frac{6ka^2(\gamma - 1)}{nRx} t$$

or

$$T = T_0 + (T_1 - T_0) e^{-\frac{6ka^2(\gamma - 1)}{nRx} t}$$

Illustrative Example 4.11

A layer of ice at 0°C of thickness x_1 is floating on a pond. If the atmospheric temperature is $-T^\circ\text{C}$, show that the time taken for thickness of the layer of ice to increase from x_1 to x_2 is given by

$$t = \frac{\rho L}{2kT} (x_2^2 - x_1^2)$$

where ρ is the density of ice, k its thermal conductivity and L is the latent heat of fusion of ice.

Solution

When the temperature of the air is less than 0°C , the cold air near the surface of the pond takes heat (latent) from the water which freezes in the forms of layers. Figure-4.16. Consequently, the thickness of the ice layer keeps increasing with time. Let x be the thickness of the ice layer at a certain time. If the thickness is increased by dx in time dt , then the amount of heat flowing through the slab in time dt is given by (see figure-4.16)

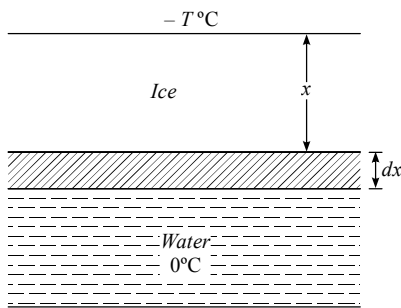


Figure 4.16

$$Q = \frac{kA[0 - (-T)]dt}{x} = \frac{kATdt}{x} \dots (4.23)$$

Where A is the area of the layer of ice and $-T^\circ\text{C}$ is the temperature of the surrounding air. If dm is the mass of water frozen into ice, then $Q = dm \times L$. But $dm = A\rho dx$, where ρ is the density of ice. Hence

$$Q = A\rho Ldx \dots (4.24)$$

Equating (4.23) and (4.24), we have

$$\frac{kATdt}{x} = A\rho Ldx \quad \text{or} \quad dt = \frac{\rho L}{kT} \cdot x dx$$

Integrating, we have

$$\int_0^t dt = \frac{\rho L}{kT} \int_{x_1}^{x_2} x dx \quad \text{or} \quad t = \frac{\rho L}{kT} \left[\frac{x^2}{2} \right]_{x_1}^{x_2} = \frac{\rho L}{2kT} (x_2^2 - x_1^2)$$

Illustrative Example 4.12

Three rods of material X and three rods of material Y are connected as shown in figure-4.17. All the rods are of identical lengths and cross-sectional areas. If the end A is maintained at 60°C and the junction E at 10°C , calculate the temperature of junctions B , C and D . The thermal conductivity of X is $9.2 \times 10^{-2} \text{ kcal m}^{-1} \text{ s}^{-1} \text{ }^\circ\text{C}^{-1}$ and that of Y is $4.6 \times 10^{-2} \text{ kcal m}^{-1} \text{ s}^{-1} \text{ }^\circ\text{C}^{-1}$.

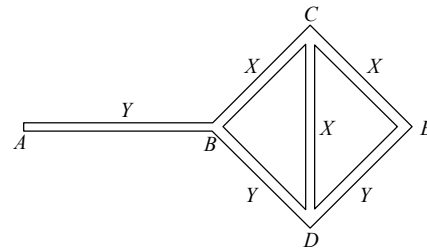


Figure 4.17

Solution

Let k_x and k_y be the thermal conductivities of X and Y respectively and let T_B , T_C and T_D be the temperatures of junctions B , C and D respectively. Given $T_A = 60^\circ\text{C}$ and $T_E = 10^\circ\text{C}$.

In order to solve this problem, we will apply the principle that, in the steady state, the rate at which heat enters a junction is equal to the rate at which heat leaves that junction. In Figure-4.18 the direction of flow of heat is given by arrows.

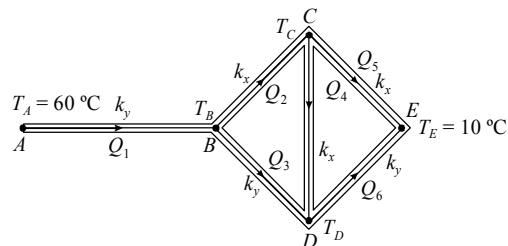


Figure 4.18

For junction B , we have

$$\frac{k_y A (T_A - T_B)}{d} = \frac{k_x A (T_B - T_C)}{d} + \frac{k_y A (T_B - T_D)}{d}$$

or $k_y (T_A - T_B) = k_x (T_B - T_C) + k_y (T_B - T_D)$

or $(T_A - T_B) = \frac{k_x}{k_y} (T_B - T_C) + (T_B - T_D)$

Given $\frac{k_x}{k_y} = \frac{9.2 \times 10^{-2}}{4.6 \times 10^{-2}} = 2$

and $T_A = 60^\circ\text{C}$.

Therefore, $(60 - T_B) = 2(T_B - T_C) + (T_B - T_D)$

or $4T_B - 2T_C - T_D = 60 \quad \dots (4.25)$

For junction C , we have

$$\frac{k_x A (T_B - T_C)}{d} = \frac{k_x A (T_C - T_D)}{d} + \frac{k_x A (T_C - T_E)}{d}$$

Solving we get

$$-T_B + 3T_C - T_D = 10 \quad [\text{At } T_E = 10^\circ\text{C}] \dots (4.26)$$

For junction D , we have

or $\frac{k_y A (T_B - T_D)}{d} + \frac{k_x A (T_C - T_D)}{d} = \frac{k_y A (T_D - T_E)}{d}$

or $(T_B - T_D) + \frac{k_x}{k_y} (T_C - T_D) = (T_D - T_E)$

Putting $\frac{k_x}{k_y} = 2$ and $T_E = 10^\circ\text{C}$ in this equation, we have

$$(T_B - T_D) + 2(T_C - T_D) = (T_D - 10)$$

$$T_B + 2T_C - 4T_D = -10 \quad \dots (4.27)$$

Solving equation (4.25), (4.26) and (4.27), we get

$$T_B = 30^\circ\text{C}$$

and $T_C = T_D = 20^\circ\text{C}$

Practice Exercise 4.1

(i) A wall has two layers A and B each made of different materials. Both the layers have the same thickness. Thermal conductivity of material A is twice that of B . Under thermal equilibrium the temperature difference across the wall is 36°C . What will be the temperature difference across the layer A ?

[12°C]

(ii) A 100 W heater is placed in a cubical container of edge length $6 \times 10^{-2}\text{m}$. The wall thickness of the container is 1 mm. If inside and outside temperature in steady state are 30°C and 25°C , find the thermal conductivity of the material of the box.

[$0.926 \text{ W/m}^\circ\text{C}$]

(iii) One end of metal rod of 1 m length and cross sectional area 10 cm^2 is immersed in boiling water and other end in an ice chamber. If the thermal conductivity of the metal is $92 \text{ cal/ms}^\circ\text{C}$ and the latent heat of fusion of ice is 80 cal/gm , find the amount of ice which will melt in one minute.

[6.9 g]

(iv) Water is filled in a closed cylindrical vessel of 10 cm height and base radius $\sqrt{10/\pi} \text{ cm}$. The open ends of the cylinder are closed by two metal discs made of a material whose thickness is 10^{-3} m and having thermal conductivity $200 \text{ W/m}^\circ\text{C}$. If water temperature inside the cylinder is 50°C and surrounding temperature is 20°C , find the time taken for the temperature to fall by 1°C . Given that the specific heat of water is $4200 \text{ J/kg}^\circ\text{C}$ and heat loss from water only takes place by the discs as the walls are made up of a thermally insulating material.

[0.035 s]

(v) A uniform steel rod of length 50 cm is insulated on its sides. There is a layer of water of thickness 0.2 mm at each end of the rod. If the ends of the rod are exposed to ice at 0°C and steam at 100°C , calculate the temperature gradient in the rod. Thermal conductivities of steel and water are $0.11 \text{ cal cm}^{-1} \text{ s}^{-1} \text{ }^\circ\text{C}^{-1}$ and $1.5 \times 10^{-3} \text{ cal cm}^{-1} \text{ s}^{-1} \text{ }^\circ\text{C}^{-1}$.

[$0.945^\circ\text{C per cm}$]

(vi) Two plates of the same area and the same thickness having thermal conductivities k_1 and k_2 are placed one on top of the other. Show that the thermal conductivity of the composite plate for conduction of heat is given by

$$k = \frac{2k_1 k_2}{(k_1 + k_2)}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Heat Transfer

Module Number - 1 to 11

(vii) Figure-4.19 shows a paddle wheel coupled with an agitator submerged in a water tank placed in an ice bath at 0°C . The thickness of the tank walls is 2 mm and with thermal conductivity $0.5 \text{ W/m}^\circ\text{C}$. The surface area of tank in contact with water is 0.05 m^2 . As the block of mass M attached to wheel goes down agitator rotates. It is found that in steady state the block goes down with a constant speed 0.1 m/s and the temperature of water in the tank remains constant at 1°C . Find the mass of the block. Take $g = 10 \text{ m/s}^2$.

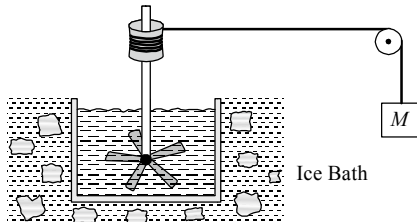


Figure 4.19

[12.5 kg]

(viii) Find the heat current through the frustum of a cone shown in figure-4.20. Temperature of its two ends are maintained at T_1 and T_2 ($T_2 > T_1$) respectively and the thermal conductivity of the material is k .

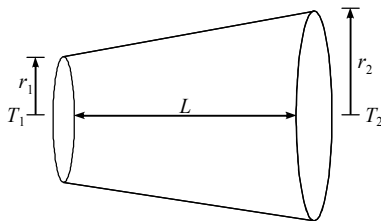


Figure 4.20

$$\left[\frac{k\pi r_1 r_2 (T_2 - T_1)}{L} \right]$$

(ix) Three rods of identical cross sectional area and made from the same metal form the sides of an isosceles triangle ABC right angled at B . The points A and B are maintained at temperatures T and $\sqrt{2} T$ respectively in the steady state. Assume heat flow only by conduction, find the temperature of point C .

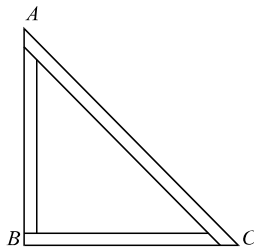


Figure 4.21

$$\left[\frac{3T}{\sqrt{2} + 1} \right]$$

(x) Two identical rods AB and CD , each of length L are connected as shown in figure-4.22. Their cross-sectional area is A and their thermal conductivity is k . Ends A , C and D are maintained at temperatures T_1 , T_2 and T_3 respectively. Neglecting heat loss to the surroundings, find the temperature at B .

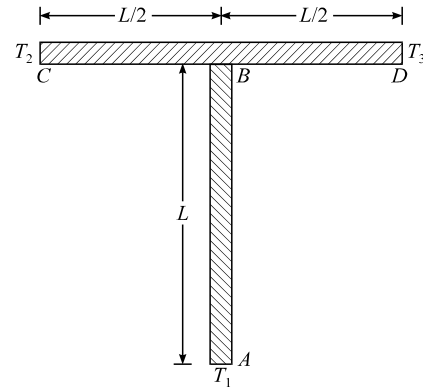


Figure 4.22

$$\left[\frac{T_1 + 2T_2 + 2T_3}{5} \right]$$

(xi) A room is maintained at 20°C by a heater of resistance 20 ohms connected to 200 V mains. The temperature is uniform throughout the room and the heat is transmitted through glass window of area 1 m^2 and thickness 0.2 cm . Calculate the temperature of outside. Thermal conductivity of glass is $0.2 \text{ cal/s m}^\circ\text{C}$ and mechanical equivalent of heat is 4.2 J/cal .

[15.24°C]

4.2 Convection of Heat

Convection is transfer of heat by actual motion of material. Hot air furnace, hot water heating system, and the flow of blood in the body are the examples. If the material is forced to move by a blower or pump, the process is called forced convection. If the material flows due to difference in density, is called natural or free convection.

Free convection in the atmosphere plays a dominant role to determine the daily weather and convection in oceans is also an important heat transfer mechanisms. To understand in a better way we discuss an example, consider the U-tube shown in figure-4.23. In this U-tube its left end A is open to atmosphere and at the other end C a stop X is placed. Initially some water is filled in the tube and temperature of both the arms are equal thus stands equally in both arms. If right arm of U-tube is heated, the water in this arm expands and its density will decrease, thus a larger column of water is needed to balance the pressure produced by the cold water in left column. When stop valve is opened, water starts flowing from the warmer column into the colder column. This increases the pressure at the bottom of U-tube produced by the cold column, and

decreases the pressure at this point due to the hot column. Hence at the bottom of U-tube, water is forced from the cold to the hot side. If heat is continuously supplied to the hot side and removed from the cold side, the circulation continuous indefinitely. The net result is a continuous transfer of heat from hot to cold side of the column.

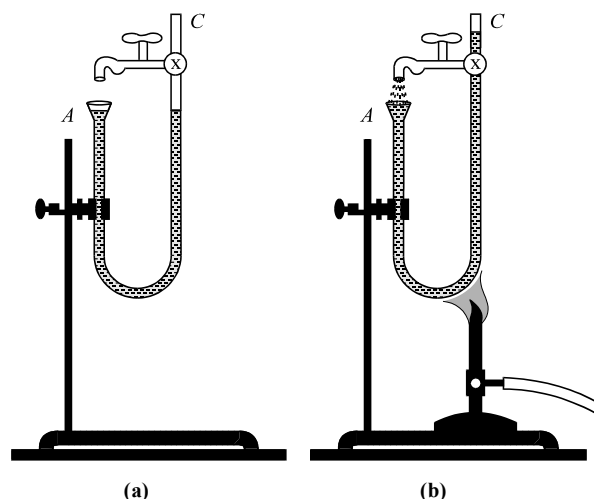


Figure 4.23

There is no simple equation for convective heat transfer as there is for conduction. The heat lost or gained by a surface at one temperature in contact with a fluid at another temperature depends on many factors, such as the shape and orientation of the surface, the mechanical and thermal properties of fluid, and the nature of the fluid flow, laminar or turbulent. Some experimental observations are taken for convection of heat, these are

- (i) The heat current due to convection is directly proportional to the surface area of the main body of fluid in contact with the surface. This is the reason for the large surface areas of heat radiators and cooling fins used in transformers and heat sinks used in power electronic devices.
- (ii) The heat current due to convection is found to be approximately proportional to the temperature difference between the surface and main body of fluid.

Thus we define the heat current due to a surface in contact with a fluid as :

$$\frac{dQ}{dt} = hA\Delta T \quad \dots (4.28)$$

Here A is the surface area in contact and ΔT is the temperature difference between the surface and main body of the fluid and h is a constant called convection coefficient. Values of h are determined experimentally, actually it is found not to be constant but depend on several factors, these are :

- (i) Whether the surface in contact with fluid is flat or curved.
- (ii) Whether the surface is horizontal or vertical.
- (iii) Whether the fluid in contact is liquid or a gas.

- (iv) The density, viscosity, specific heat and thermal conductivity of the fluid.
- (v) Whether the speed of the fluid is small enough to give rise to laminar flow or large enough to cause turbulent flow.
- (vi) Whether conduction or evaporation takes place during flow.

Another example of convection is our human body. The human body produces a great deal of thermal energy, of the food energy transformed within the body, at best 20 percent is used to do work, so over 80 percent appears as thermal energy. During light activity, for example, if this thermal energy were not dissipated, the body temperature would rise about 3°C per hour. Thus we can say that the heat generated by body must be transferred to the outside surrounding. Is this heat transferred by conduction ? The temperature of the skin at normal room temperature is 30 to 35°C where as the interior of body is 37°C . It can be easily proved that this small temperature difference along with low thermal conductivity of tissue, by conduction only a very small amount of heat can be dissipated. Instead the heat is carried to the surface by the blood. In addition to other important responsibilities, blood acts as a convective fluid to transfer heat to just beneath the surface of the skin. It is then conducted through the very small thickness of skin to the surface. Once at the surface, the heat is transferred to the environment by convection, evaporation and radiation.

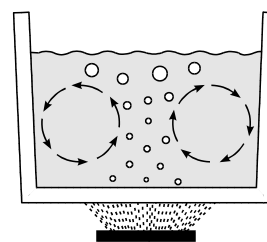


Figure 4.24

Another example of convection is heating water in a pot on a gas burner. When a pot of water is heated on a gas burner as shown in figure-4.24, convection currents are set up as the heated water at the bottom of pot rises because of its reduced density and is replaced by cooler water from above. In figure convection currents are shown in arrows. This principle is used in many heating systems, such as hot water radiator system in houses. Figure-4.25 shows internal diagram of a house having installed such a system. Cold water is heated in furnace, its temperature rises, it expands and start rising up to room-1 as shown. Hot water then enters in the radiators and heat is transferred to the surrounding air by conduction through the walls of radiator and cooled water returns to the furnace. Thus the water circulates because of convection, sometimes pumps are used to make the circulation fast. The air in the room also becomes uniformly heated as a result of convection. The air heated by radiator rises and is replaced by cooled air, resulting the convection air currents as shown in figure-4.25.

Some type of furnaces also depend on convection. Hot air furnaces have openings (called registers) near the floor often do not have fans but depend on natural convection.

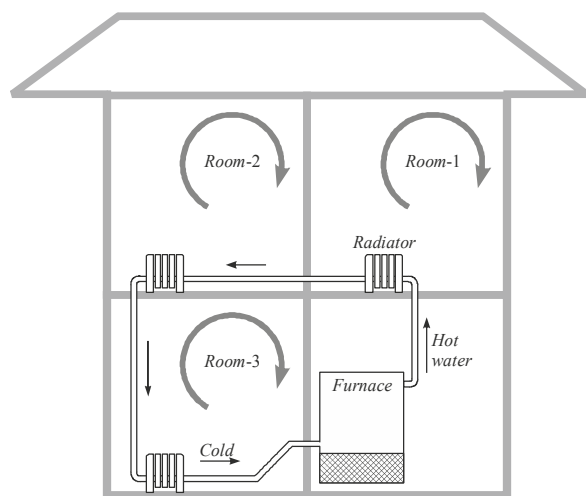


Figure 4.25

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Heat Transfer

Module Number - 12

4.3 Radiation of Heat

Convection and conduction require the presence of matter. Whole life on earth depends on the transfer of energy from the sun, and this energy is transferred to the earth over empty space. This form of energy transfer is referred as radiation. Similarly when a fire is lit and if we stand near by, feel the warmth called radiant energy. Most of the air heated by a fire rises up by convection in the surrounding of fire.

Thermal radiation are electromagnetic waves of several wavelengths (polychromatic radiation) which carry thermal energy. Radiation from the Sun to Earth consists of visible light plus many other wavelengths that the eye is not sensitive to. This includes infrared (IR) radiation which is mainly responsible for heating the earth.

4.3.1 Prevost Theory

In early ages it was assumed that only hot body emits radiation but later a theory of heat radiation was given by Prevost that every body which is at temperature above 0K temperature radiate thermal energy in the form of electromagnetic waves of several wavelengths. Simultaneously the body also absorbs thermal radiation from its surroundings. When a body at higher temperature is kept near one at lower temperature, it radiates

energy more than what it absorbs from the one at lower temperature similarly the colder one absorbs more than what it radiates thus thermal energy is transferred from hot body to the cold body by means of thermal radiations. If two bodies are at same temperature kept close to each other, they radiate and absorbs heat at same rate, so we say that the bodies are in thermal equilibrium. Thus in thermal equilibrium we can not say that no heat flow takes place between bodies. Actually thermal equilibrium between two bodies is a dynamic equilibrium in which when bodies are at same temperature, they radiate and absorb thermal energy at same rate thus no net heat transfer takes place.

4.3.2 Kirchoff's Law

When thermal radiation falls on a material surface three things may happen to it.

- A certain amount of heat E_R will be reflected,
- A certain amount of heat E_A will be absorbed,
- A certain amount of heat E_T will be transmitted

Thus the total amount of incident energy E_I is divided between these three parameters, therefore we can write

$$E_I = E_R + E_A + E_T \quad \dots (4.29)$$

For a shiny surface such as silver, E_R is large and both E_T and E_A are small. For a black surface such as charcoal E_R is small, E_A is large and E_T is small. For a transparent medium like glass E_R is small, E_A small and E_T large.

The energy absorbed by a body can be emitted later or simultaneously along with absorption and clearly if a surface can not absorb radiation strongly it will be unable to emit strongly. Kirchoff Law states the same that “a good emitter is also a good absorber”.

When thermal radiation incident upon a body or emitted by a body equally in all directions, the radiation is said to be isotropic. Some of the radiation incident on a body may be absorbed, reflected and transmitted. In general, the incident radiation of all wavelengths that is absorbed depends on the temperature and the nature of the surface of the absorbing body. The fraction of the total incident power that is absorbed by a body is called its absorptivity. It can be given as

$$\text{absorptivity} = \frac{\text{Total power absorbed by a body}}{\text{Total power incident on the body}}$$

When a body is in thermal equilibrium, the process of absorption and emission of radiant power are equal and opposite. So the total emissivity is equal to total absorptivity. Total emissivity is defined as the fraction of power provided to a general body that is emitted through a material surface as thermal radiation, here the “total” includes all the wavelengths of electromagnetic radiation from the body. Practically it is easier to measure emissivity than absorptivity.

The emissivity of a body depends on both the temperature and the nature of emitting surface. Bodies of same temperature and size but different material emit different amount of total thermal radiation. According to Kirchoff's Law good absorbers are good emitters. For Theoretical purpose we define an ideal substance capable either of absorbing all the thermal radiation falling on it or emitting all the energy provided to it in the form of thermal radiation. Such a substance is called Black Body.

Thus for a black body we can say that its emissivity is 1, as well as its absorptivity. We can compare different substances with black body and obtain the emissivity of respective substances. For example a polished shining steel piece emits only 9% of the thermal radiation as compared to a black body of same size and shape of same temperature, hence its emissivity is 0.09. Similarly emissivity of a rough oxidized steel surface is 0.81 and that of ocean water is 0.96 and some substances like lamp black, coal are there whose emissivity is very nearly unity that is, almost an ideal emitter. The emissivity of a general body is denoted by e and mathematically defined as

$$e = \frac{\text{Radiant power emitted by body at a given temperature}}{\text{Radiant power emitted by a Black Body of same geometry at same temperature}}$$

Thus total thermal radiation power emitted by a General Body is given as

$$\left(\begin{array}{c} \text{Radiation Power} \\ \text{from a} \\ \text{General Body} \\ \text{at a given temperature} \end{array} \right) = e \left(\begin{array}{c} \text{Radiation Power} \\ \text{from a} \\ \text{Black Body} \\ \text{at same temperature} \end{array} \right) \quad \dots (4.30)$$

Experimentally a very good approximation to a black body is provided by a cavity enclosed by high temperature opaque walls regardless of the composition of the material of its interior walls.

4.3.3 Stefan Boltzmann Law

On the basis of experiments, Stefan concluded in 1879 that the total amount of thermal radiation power from a body is directly proportional to the fourth power of its absolute temperature. The mathematical result was derived later by Boltzmann that the total radiation power per unit surface area of a body is given as

$$P_R = \sigma T^4 \text{ J/s.m}^2 \quad \dots (4.31)$$

If a body of surface area A is at temperature T , then total thermal power radiated by the body is

$$P_R = \sigma A T^4 \text{ J/s} \quad \dots (4.32)$$

Later experimentally it is observed that equation-(4.32) does not give correct results in many cases, it gives approximately true results for substances with high emissivities and it was accepted that equation-(4.32) is only valid for black bodies or those having emissivity very close to unity. Thus for general

bodies equation-(4.32) can be generalized as power radiated by a general body of surface area A , at temperature T and with emissivity e , can be given by equation-(4.30) as

$$P_R = e A \sigma T^4 \text{ J/s} \quad \dots (4.33)$$

Any object not only emits energy by radiation, but it also absorbs energy radiated by other bodies in its surrounding. If an object of emissivity e and area A is at a temperature T , it radiates energy at a rate $e A \sigma T^4$. If the object is surrounded by an environment at temperature T_s . The rate at which the surrounding in immediate contact of the body surface radiate energy at a rate $\sigma A T_s^4$ if it is assumed to be a black body and if absorptivity of the body material is a , the amount of energy absorbed by the body is at a rate $a \sigma A T_s^4$. Thus the net heat flow from the object can be given as

$$\frac{dQ}{dt} = e A \sigma T^4 - a A \sigma T_s^4 \quad \dots (4.34)$$

As good absorbers are good emitters we can approximately use $e \approx a$ thus we have

$$\frac{dQ}{dt} = e A \sigma (T^4 - T_s^4) \quad \dots (4.35)$$

When body and surrounding both are at same temperature equation-(4.35) shows that there is no net flow of heat takes place with the body. When $T > T_s$, net flow of heat takes place from body to surrounding and temperature of body decreases. Similarly if $T < T_s$, net flow of heat takes place from surrounding to body and temperature of body increases with time.

If mass of body is m and s is the specific heat of the material of body then we have

$$\frac{dQ}{dt} = ms \frac{dT}{dt} \quad \dots (4.36)$$

Where $\frac{dT}{dt}$ is rate of change of temperature of body. If $\frac{dQ}{dt}$ is rate of heat loss from a body than $\frac{dT}{dt}$ will be the rate at which temperature of body falls and is termed as rate of cooling. Thus if a body of mass m and specific heat s at temperature T is placed in surrounding of temperature T_s then the rate of cooling of body can be given as

$$\frac{dT}{dt} = \frac{e A \sigma}{ms} (T^4 - T_s^4) \text{ K/s} \quad \dots (4.37)$$

4.3.4 Solar Constant

It is defined as "solar energy received per unit surface area of earth surface per unit time, when sun is overhead". If radius of sun is R_s and its temperature T_s , the average rate at which it radiates energy is given as

$$E = \sigma (4\pi R_s^2) T_s^4 \quad \dots (4.38)$$

This radiant power given by equation-(4.38) is isotropically radiated by sun in all direction. If d is the mean distance between earth and sun, the average energy intensity on earth surface for normal incidence of sun rays that is solar constant can be given as

$$s = \frac{E}{4\pi d^2}$$

$$\text{or} \quad s = \frac{\sigma R_s^2 T_s^4}{d^2} \quad \dots (4.39)$$

Now we take few examples to understand the applications of above concepts in details.

Illustrative Example 4.13

An electric heater of power 1 kW emits thermal radiations the surface area of heating element of heater is 200 cm². If this heating element is treated like a black body find the temperature at its surface. Assume its temperature is very much higher than its surroundings.

Solution

As it is given that the temperature of surrounding is very low as compared to that of heating element, we can ignore the amount of heat absorbed by the filament from its surroundings. Thus the power radiated by the heating element is given as

$$P = \sigma A T^4$$

$$\text{or} \quad 1000 = 5.67 \times 10^{-8} \times 0.02 \times T^4$$

$$\text{or} \quad T^4 = 8.82 \times 10^{11}$$

$$\text{or} \quad T = 969.3 \text{ K.}$$

Illustrative Example 4.14

A cube of mass 1 kg and volume 125 cm³ is placed in an evacuated chamber at 27°C. Initially temperature of block is 227°C. Assume block behaves like a black body, find the rate of cooling of block if specific heat of the material of block is 400 J/kg-k.

Solution

In this case the rate of loss of heat by the block is given as

$$\begin{aligned} \frac{dQ}{dt} &= \sigma A (T^4 - T_s^4) \\ &= 5.67 \times 10^{-8} \times 150 \times 10^{-4} \times [(500)^4 - (300)^4] \\ &\quad [\text{surface area of cube is } 6a^2 = 150 \text{ cm}^2] \end{aligned}$$

If $\frac{dT}{dt}$ is rate of cooling of block then we have

$$\frac{dQ}{dt} = ms \frac{dT}{dt}$$

$$\begin{aligned} \text{or} \quad \frac{dT}{dt} &= \frac{1}{ms} \cdot \frac{dQ}{dt} \\ &= \frac{1}{1 \times 400} \times 38.556 \\ &= 0.115 \text{ } ^\circ\text{C/S} \end{aligned}$$

Illustrative Example 4.15

One end A of a metallic rod of length 10 cm is inserted in a furnace whose temperature is 827°C. The curved surface of the rod is insulated. The room temperature is 27°C. When the steady state is attained, the temperature of the other end B of the rod is 702°C. Find the thermal conductivity of the metal. Stefan's constant = $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.

Solution

Since the curved surface of the rod is insulated, heat is lost from the end B of the rod is by radiation. If T_B is the absolute temperature of end B , then the energy radiated per unit area per time from end B is, from Stefan's law,

$$E_1 = \sigma (T_B^4 - T_0^4) \quad \dots (4.40)$$

Where T_0 is the room temperature, σ is Stefan's constant. Also, energy received at end B by conduction through the rod per unit area unit time is

$$E_2 = \frac{k (T_A - T_B)}{l} \quad \dots (4.41)$$

Where T_A = temperature of end A of the rod, l is the length of the rod and k its thermal conductivity. In the steady state $E_1 = E_2$. Equating equation-(4.40) and (4.41) we have

$$\frac{k (T_A - T_B)}{l} = \sigma (T_B^4 - T_0^4)$$

$$\text{or} \quad k = \frac{\sigma l (T_B^4 - T_0^4)}{(T_A - T_B)} \quad \dots (4.42)$$

Given $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$, $l = 0.1 \text{ m}$, $T_B = 702^\circ\text{C} = 975 \text{ K}$, $T_0 = 27^\circ\text{C} = 300 \text{ K}$ and $T_A = 827^\circ\text{C} = 1100 \text{ K}$. Using these values in equation-(4.42), we get

$$k = 36.6 \text{ J s}^{-1} \text{ m}^{-1} \text{ K}^{-4}$$

Illustrative Example 4.16

A solid metallic sphere of diameter 20 cm and mass 10 kg is heated to a temperature of 327°C and suspended in a box in which a constant temperature of 27°C is maintained. Find the rate at which the temperature of the sphere will fall with time. Stefan's constant = $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ and specific heat of metal = $420 \text{ J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$.

Solution

The rate of loss of heat by the sphere is given by

$$\frac{dQ}{dt} = \sigma A (T^4 - T_0^4)$$

Where A is the surface area of the sphere $= 4\pi r^2$, with $r = 10$ cm $= 0.1$ m, $T = 327^\circ\text{C} = 600$ K and $T_0 = 27^\circ\text{C} = 300$ K.

$$\text{Thus } \frac{dQ}{dt} = 5.67 \times 10^{-8} \times 4\pi \times (0.1)^2 \times \{(600)^4 - (300)^4\}$$

$$= 866 \text{ J s}^{-1}$$

Now $dQ = msdT$, where dT is the fall in temperature in time dt .

$$\frac{dQ}{dt} = ms \frac{dT}{dt}$$

$$\text{or } 866 = 10 \times 420 \times \frac{dT}{dt}$$

$$\text{or } \frac{dT}{dt} = \frac{866}{4200} = 0.206^\circ\text{C/s}$$

Illustrative Example 4.17

A cylindrical rod of 50 cm length and having 1 cm^2 cross sectional area is used as a conducting material between an ice bath at 0°C and a vacuum chamber at 27°C as shown in figure-4.26. The end of rod which is inside the vacuum chamber behaves like a black body and is at temperature 17°C in steady state. Find the thermal conductivity of the material of rod and rate at which ice is melting in the ice bath. Given that latent heat of fusion of ice is $3.35 \times 10^5 \text{ J/kg}$.

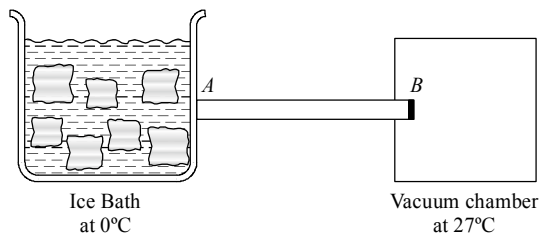


Figure 4.26

Solution

It is given that the system is in steady state. This means that any part of rod is not absorbing any heat. So heat absorbed by the end of the rod which is in vacuum chamber by radiation is fully conducted to the ice bath through the rod. Thus we have

Rate of that conduction through the rod =

Rate of heat absorption by radiation for the vacuum chamber

$$\text{or } \frac{kA(T_B - T_A)}{L}$$

$$= \sigma A (T_{vc}^4 - T_B^4)$$

or

$$\frac{k(17^\circ\text{C} - 0^\circ\text{C})}{0.5}$$

$$= 5.67 \times 10^{-8} [(300)^4 - (290)^4]$$

or

$$k = \frac{5.67 \times 10^{-8} [(300)^4 - (290)^4] \times 0.5}{17}$$

$$= 1.713 \text{ W/m}^\circ\text{C}$$

Using this value of k we can find the rate of heat obtained by the ice bath as

$$\frac{dQ}{dt} = \frac{kA(T_A - T_B)}{l}$$

$$= \frac{1.713 \times 1 \times 10^{-4} \times 17}{0.5}$$

$$= 5.82 \times 10^{-3} \text{ J/s}$$

This heat is used to melt the ice in ice bath. If m mass of ice is being melted per second, then we have

$$\frac{dQ}{dt} = mL$$

$$\text{or } 5.82 \times 10^{-3} = m \times 3.35 \times 10^5$$

$$\text{or } m = 1.74 \times 10^{-8} \text{ kg/s.}$$

Illustrative Example 4.18

The earth receives solar energy at the rate of 2 cal cm^{-2} per minute. Assuming the radiation to be black body in character, estimate the surface temperature of the sun. Given that $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ and angular diameter of the sun = 32 minute of arc.

Solution

Let surface temperature of Sun is T_s then total energy radiated by Sun per second is given as

$$E = \sigma T_s^4 \cdot (4\pi R_s^2)$$

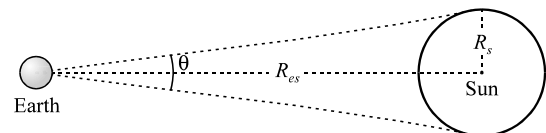


Figure 4.27

Energy received by earth per second per meter square is give as

$$\frac{dQ}{dt} = \frac{E}{4\pi R_{es}^2} = \sigma T_s^4 \left(\frac{R_s}{R_{es}} \right)^2 \quad \dots (4.43)$$

It is given that angular diameter of Sun as observed from earth is 32 min of arc thus, we have

$$\frac{R_s}{R_{es}} = \frac{\theta}{2} = \frac{1}{2} \times \frac{32}{60} = 4.655 \times 10^{-3}$$

and it is given that

$$\frac{dQ}{dt} = \frac{2 \times 4.2 \times 10^4}{60} \text{ J/s-m}^2 \quad \dots (4.44)$$

Now from equation-(4.43) and (4.44) we have

$$5.67 \times 10^{-8} \times T_s^4 \times [4.655 \times 10^{-3}]^2 = \frac{2 \times 4.2 \times 10^4}{60}$$

$$\text{or } T_s^4 = \frac{2 \times 4.2 \times 10^4}{60 \times 5.67 \times 10^{-8} \times (4.655 \times 10^{-3})^2}$$

$$\text{or } T_s^4 = 1.14 \times 10^{15}$$

$$\text{or } T_s = 5810.67 \text{ K}$$

Illustrative Example 4.19

A spherical ball of radius 1 cm coated with a material having emissivity 0.3 is maintained at 1000 K temperature and suspended in a vacuum chamber whose walls are maintained at 300 K temperature. Find rate at which electrical energy is to be supplied to the ball to keep its temperature constant.

Solution

It is given that the temperature of ball is constant. This means that the rate at which it is losing heat by radiation must be equal to the rate at which heat is supplied to this ball externally to keep its temperature constant.

The rate of heat loss by the ball is given as

$$\begin{aligned} \frac{dQ}{dt} &= eA\sigma(T^4 - T_s^4) \\ &= 0.3 \times 4\pi(0.01)^2 \times 5.67 \times 10^{-8} \times [(1000)^4 - (300)^4] \\ &= 21.1 \text{ W} \end{aligned}$$

Thus electrical energy must be supplied to the ball at a rate of 21.1 W.

Illustrative Example 4.20

There are two concentric spherical shells A and B of surface area 20 cm^2 and 80 cm^2 . Surfaces of both the shells behave like black bodies. It is given that the thermal conductivity of material of B is very low and that of A is very high. Initially the temperature of A is 400 K and that of B is 300 K. Find the rate of change of temperature of A and B . Given that the heat capacities of A and B are $50 \text{ J/}^\circ\text{C}$ and $80 \text{ J/}^\circ\text{C}$ respectively.

Solution

It is given that the thermal conductivity of material of A is very high so we can assume that it will not absorb any heat, thus the rate of heat loss by A is given as

$$\frac{dQ}{dt} = A_A \sigma T_A^4$$

$$= 20 \times 10^{-4} \times 5.67 \times 10^{-8} \times (400)^4$$

$$= 2.9 \text{ J/s}$$

If $\frac{dT}{dt}$ is rate of cooling of A then we have

$$\frac{dQ}{dt} = (ms)_A \frac{dT}{dt} \Big|_A$$

$$\text{or } \frac{dT}{dt} \Big|_A = \frac{1}{(ms)_A} \frac{dQ}{dt}$$

$$\text{or } \frac{dT}{dt} \Big|_A = \frac{1}{50} \times 2.9$$

[As for shell A $(ms)_A = 50 \text{ J/s}$]

$$= 0.058 \text{ }^\circ\text{C/s}$$

Now for B it is given that its thermal conductivity is very poor so it absorbs all the heat incident on it (as behaving like a black body). Thus rate of loss of heat from B can be written as

$$\frac{dQ}{dt} = A_B \sigma T_B^4 - A_A \sigma T_A^4$$

$$\begin{aligned} \text{or } \frac{dQ}{dt} &= 80 \times 10^{-4} \times 5.67 \times 10^{-8} \times (300)^4 - 2.9 \\ &= 3.67 - 2.9 = 0.77 \text{ J/s} \end{aligned}$$

If $\frac{dT}{dt} \Big|_B$ is the rate of cooling for shell B , we have

$$\frac{dQ}{dt} = (ms)_B \frac{dT}{dt} \Big|_B$$

$$\text{or } \frac{dT}{dt} \Big|_B = \frac{1}{(ms)_B} \frac{dQ}{dt}$$

$$= \frac{1}{80} \times 0.77$$

[As for shell B $(ms)_B = 80 \text{ J/}^\circ\text{C}$]

$$= 0.00963 \text{ }^\circ\text{C/s}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Heat Transfer

Module Number - 13 to 23

Practice Exercise 4.2

(i) A 500 W lamp loses all of its energy by emission of radiation from the surface. If the area of the surface of the filament is 2.0 cm^2 and its emissivity is 0.5, estimate the temperature of its filament. Given that Stefan constant, $\sigma = 5.7 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$. Neglect radiation received by lamp from surrounding.

[3060.37 K]

(ii) A white-hot metal wire at 3000 K has a radius of 0.075 cm. Calculate the rate per unit length at which it emits radiation if its emissivity is 0.35. Ignore the radiation it receives from the surroundings. Take Stefan's constant $\sigma = 5.7 \times 10^{-8} \text{ W/m}^2\text{K}$

[7611.12 W/m]

(iii) The temperature of the tungsten filament of a 60 W electric bulb is $T = 2000 \text{ K}$. Find the surface area of the filament. The emissivity of the surface is $e = 0.30$. Neglect radiation received from surrounding. Take $\sigma = 5.7 \times 10^{-8} \text{ W/m}^2\text{K}$

[2.192 $\times 10^{-4} \text{ m}^2$]

(iv) A spherical black body with a radius of 12 cm placed in space radiates 450 W power at 500 K. If the radius were halved and the temperature doubled, find the power radiated.

[1800 W]

(v) A block having some emissivity is maintained at 500 K temperature in a surrounding of 300 K temperature. It is observed that, to maintain the temperature of the block, 210 W external power is required to be supplied to it. If instead of this block a black body of same geometry and size is used, 700 W external power is needed for the same. Find the emissivity of the material of the block.

[0.3]

(vi) A solid sphere of copper of radius R and a hollow sphere of the same material of inner radius r and outer radius R are heated to the same temperature and allowed to cool in the same environment. Which of the sphere will cool faster?

[Hollow Sphere]

(vii) The shell of a space station is a blackened sphere in which a temperature $T = 500 \text{ K}$ is maintained due to the operation of appliances of the station. Find the temperature of the shell if the station is enveloped by a thin spherical black screen of nearly the same radius as the radius of the shell.

[594.6 K]

(viii) A copper ball of diameter d was placed in an evacuated vessel whose walls are kept at the absolute zero temperature. The initial temperature of the ball is T_0 . Assuming the surface of the ball to be absolutely black, find how soon its temperature decreases η times. Take specific heat of copper c , density of copper ρ and emissivity e .

$\left[t = \frac{(\eta^3 - 1) c \rho d}{18 \sigma T_0^3 e} \right]$

4.4 Newton's Law of Cooling

When a body radiates thermal energy in an environment at lower temperature, then equation-(4.37) can be used to find

temperature of body as a function of time by integrating the expression

$$\int_{T_0}^T \frac{dT}{T^4 - T_s^4} = - \int_0^t \frac{e A \sigma}{ms} dt \quad \dots (4.45)$$

In equation-(4.45) limits of integration are applied for temperature of body at respective time instants as if at $t = 0$, temperature of body was T_0 and after time $t = t$, its temperature

drops to T . Negative sign in equation-(4.45) shows that $\frac{dT}{dt}$ is negative and with time temperature is falling. In equation-(4.37), for using this equation for analytical purpose a negative sign must also be included.

Expression in equation-(4.45) is not easy to integrate and very bulky and time consuming for even problems of simple time calculations. The expression in equation-(4.37) can be approximated to a simple level for bodies at temperature not very high compared to their surroundings. Equation-(4.37) can be rewritten as

$$\frac{dT}{dt} = \frac{e A \sigma}{ms} (T^2 + T_s^2) (T + T_s) (T - T_s) \quad \dots (4.46)$$

Here if temperature of body is slightly higher than that of its surrounding then we can use $T \approx T_s$ but not $T - T_s = 0$, thus equation-(4.46) becomes

$$\frac{dT}{dt} = - \frac{e A \sigma}{ms} (2 T_s^2) (2 T_s) (T - T_s)$$

$$\text{or} \quad \frac{dT}{dt} = - \frac{4 e A \sigma T_s^3}{ms} (T - T_s)$$

$$\text{or} \quad \frac{dT}{dt} = -k (T - T_s) \quad \dots (4.47)$$

Here $k = \frac{4 e A \sigma T_s^3}{ms}$ is a constant for bodies with small temperature

difference with the surroundings. Equation-(4.47) can be stated as "The rate of cooling for bodies having small temperature difference with the surrounding is directly proportional to the temperature difference of body with its surroundings." This is Newton's Law of cooling. From equation-(4.47) we can use this expression for further calculations of temperature as

$$\frac{dT}{T - T_s} = -k dt$$

Integrating within proper limits we have

$$\int_{T_0}^T \frac{dT}{T - T_s} = - \int_0^t k dt$$

$$\text{or} \quad \ln \left(\frac{T - T_s}{T_0 - T_s} \right) = -kt$$

$$\text{or} \quad T = T_s + (T_0 - T_s) e^{-kt} \quad \dots (4.48)$$

The above expression in equation-(4.48) gives the temperature of body after time t if at $t = 0$ body temperature was T_0 . But this expression will be valid only if T_0 is very close to the surrounding temperature T_s .

According to Newton's Law of cooling. The rate of fall of temperature of a body is directly proportional to the temperature

4.4.1 Average Form of Newton's Law of Cooling

In previous article we've discussed that for bodies with small temperature difference with the surrounding the rate of cooling is directly proportional to the temperature difference of body with surrounding. As temperature difference is small, average rate of cooling of body $\left(\frac{\Delta T}{\Delta t}\right)$ can be taken approximately equal to the instantaneous rate of cooling $\left(\frac{dT}{dt}\right)$. This can be written according to Newton's Law of cooling as

$$\frac{dT}{dt} \approx \frac{\Delta T}{\Delta t} = k(T - T_s)$$

Here T is the average temperature of body during cooling and T_s is the surrounding temperature. For example if a body is at temperature T_1 at $t = 0$ and after time t its temperature drops to T_2 then during cooling average temperature of body can be taken as

$$T = \frac{T_1 + T_2}{2} \quad \dots (4.49)$$

Now according to Newton's law of cooling, average rate at which body cools can be written as

$$\frac{\Delta T}{\Delta t} = \frac{T_2 - T_1}{t} = k \left(\frac{T_1 + T_2}{2} - T_s \right) \quad \dots (4.50)$$

Equation-(4.50) is known as average form of Newton's Law of cooling but is only applicable when T_1 and T_2 are very close to each other, otherwise the average temperature of body can not be given by equation-(4.49). Many times Newton's Law of colling is used to compare the rate of two different substances. In such cases we slightly modify equation-(1.50) with the heat capacity of substance.

By Stefan's law rate of cooling is given as

$$\frac{dT}{dt} = - \frac{4eA\sigma T_s^3}{ms} (T - T_s)$$

Here if substance is changed the parameters e , A , m and s may change so for same emissivity and surface area it can be written as

$$\frac{dT}{dt} = - \frac{k}{ms} (T - T_s)$$

So equation-1.50 can be written as

$$\frac{T_2 - T_1}{t} = - \frac{k}{ms} \left(\frac{T_1 + T_2}{2} - T_s \right)$$

And with variation of e or A you can modify this equation yourself. Lets take some examples to understand the applications of Newton's Law of Cooling.

Illustrative Example 4.21

The temperature of a body in a surrounding of temperature 16°C falls from 40°C to 36°C in 5 mins. Assume Newtons law of cooling to be valid and find the time taken by the body to reach temperature 32°C .

Solution

If we use average form of Newton's Law of cooling, we have

$$\left(\frac{T_2 - T_1}{t} \right) = k \left(\frac{T_2 + T_1}{2} - T_s \right)$$

Initially it is given that $T_2 = 40^\circ\text{C}$, $T_1 = 36^\circ\text{C}$, $t = 5$ min and $T_s = 16^\circ\text{C}$

Thus we have

$$\frac{40 - 36}{5} = k \left[\frac{40 + 36}{2} - 16 \right]$$

or

$$k = \frac{4}{5 \times 22} = 0.036 \text{ min}^{-1}$$

Now if body takes a time t' to come down to 32°C then again using the same relation we have

$$T_2 = 36^\circ\text{C}, T_1 = 32^\circ\text{C}, t = t', T_s = 16^\circ\text{C} \text{ and } k = 0.036 \text{ min}^{-1}$$

Thus we have

$$\frac{36 - 32}{t'} = (0.036) \left[\frac{36 + 32}{2} - 16 \right]$$

or

$$t' = \frac{4}{0.036 \times 18} = 6.173 \text{ min}$$

Illustrative Example 4.22

In a container some water is filled at temperature 50°C . It cools to 45°C in 5 minutes and to 40°C in next 8 minutes. If we assume Newton's law of cooling to be valid in this case, find the surrounding temperature.

Solution

According to average form of Newton's Law of cooling, initially we have

$T_2 = 50^\circ\text{C}$, $T_1 = 45^\circ\text{C}$, $t = 5$ min and let surrounding temperature be T_s , then we have

$$\left(\frac{T_2 - T_1}{t} \right) = k \left(\frac{T_2 + T_1}{2} - T_s \right)$$

$$\text{or } \left(\frac{50-45}{5} \right) = k \left(\frac{50+45}{2} - T_s \right)$$

$$\text{or } 1 = k(47.5 - T_s) \quad \dots (4.51)$$

Again when water cools, we have $T_2 = 45^\circ\text{C}$, $T_1 = 40^\circ\text{C}$, $t = 8 \text{ min}$ and same surrounding temperature T_s , we have

$$\text{or } \left(\frac{45-40}{8} \right) = k \left(\frac{45+40}{2} - T_s \right)$$

$$\text{or } \frac{5}{8} = k(42.5 - T_s) \quad \dots (4.52)$$

Dividing equation-(4.51) by (4.52), we get

$$\frac{8}{5} = \frac{47.5 - T_s}{42.5 - T_s}$$

$$\text{or } 8(42.5 - T_s) = 5(47.5 - T_s)$$

$$\text{or } 3T_s = 340 - 237.5$$

$$\text{or } T_s = \frac{102.5}{3} = 34.17^\circ\text{C}$$

Illustrative Example 4.23

A solid body X of thermal capacity C is kept in an atmosphere whose temperature is $T_A = 300 \text{ K}$. At time $t = 0$, the temperature of X is $T_0 = 400 \text{ K}$. It cools according to Newton's law of cooling. At time t , the temperature is 350 K . At this time ($t = t_1$), the body X is connected to a large box Y at atmospheric temperature T_A , through a conducting rod of length L , cross-sectional area A and thermal conductivity K . The thermal capacity of Y is so large that any variation in its temperature may be neglected. The cross-sectional area A of the connecting rod is small compared to the surface area of X . Find the temperature of X at time $t = 3t_1$.

Solution

If T is the temperature of body X at an instant of time t , then according to Newton's law of cooling, the rate of loss of heat by the body is given by

$$\frac{dQ}{dt} = -(T - T_A)$$

Where T_A is the temperature of the surrounding atmosphere and α is a constant which depends on the emissivity and the surface of the body. If C is the thermal capacity of body, we have

$$\frac{dQ}{dt} = C \frac{dT}{dt}$$

$$\text{Thus } C \frac{dT}{dt} = -\alpha(T - T_A) \quad \dots (4.53)$$

$$\text{or } \frac{dT}{dt} = -k(T - T_A)$$

Where $k = \frac{\alpha}{C}$, is a constant of the body. Therefore,

$$\frac{dT}{(T - T_A)} = -k dt$$

Integrating we have

$$\int \frac{dT}{(T - T_A)} = -k \int dt$$

$$\text{or } \ln(T - T_A) = -kt + c \quad \dots (4.54)$$

Where c is a constant of integration. Given that at $t = 0$, $T = T_0$ ($= 400 \text{ K}$). Using this in equation-(4.54) we have

$$\ln(T_0 - T_A) = -k \times 0 + c = c$$

$$\text{Thus } c = \ln(T_0 - T_A)$$

Using this value of c in equation-(4.54), we get

$$\ln(T - T_A) = -kt + \ln(T_0 - T_A)$$

$$\text{or } \ln \left(\frac{T - T_A}{T_0 - T_A} \right) = -kt$$

$$\left(\frac{T - T_A}{T_0 - T_A} \right) = e^{-kt}$$

Given $T_0 = 400 \text{ K}$ and $T_A = 300 \text{ K}$. Therefore,

$$\frac{T - 300}{(400 - 300)} = e^{-kt} \text{ or } T - 300 = 100 e^{-kt}$$

$$\text{or } T = 100(3 + e^{-kt}) \quad \dots (4.55)$$

Given that at $t = t_1$, $T = 350 \text{ K}$. Using this in equation-(4.55) we have

$$350 = 100(3 + e^{-kt_1})$$

$$\text{or } e^{-kt_1} = \frac{1}{2} \text{ or } -kt_1 = -\ln(2) \text{ or } k = \frac{\ln(2)}{t_1}$$

But $k = \frac{\alpha}{C}$. Therefore,

$$\alpha = \frac{C \ln(2)}{t_1} \quad \dots (4.56)$$

Using (4.56) in (4.53), the rate of loss of heat due to radiation is given by

$$\left(\frac{dQ}{dt} \right)_r = C \left(\frac{dT}{dt} \right)_r = -\frac{C \ln(2)}{t_1} \times (T - T_A) \quad \dots (4.57)$$

When body X is connected to a box Y through a conducting rod, body X will lose heat also by conduction through the rod.

The rate of loss of heat by conduction through a rod of thermal conductivity K , area A and length L is given by

$$\left(\frac{dQ}{dt}\right)_c = -\frac{K A (T - T_A)}{L} \quad \dots (4.58)$$

Thus the total rate of loss of heat by body X at temperature T is

$$\left(\frac{dQ}{dt}\right)_t = \left(\frac{dQ}{dt}\right)_r + \left(\frac{dQ}{dt}\right)_c$$

As C is the thermal capacity of X , hence, using equation-(4.57) and (4.58) we get

$$C \frac{dT}{dt} = -\frac{C \ln(2)}{t_1} \times (T - T_A) - \frac{K A (T - T_A)}{L}$$

$$\text{or} \quad \frac{dT}{(T - T_A)} = -\frac{1}{C} \left[\frac{C \ln(2)}{t_1} dt + \frac{K A}{L} dt \right]$$

Integrating we have

$$\int_{350}^T \frac{dT}{(T - T_A)} = -\frac{1}{C} \left[\frac{C \ln(2)}{t_1} + \frac{K A}{L} \right] \int_{t_1}^{3t_1} dt$$

$$\text{or} \quad \ln(T - T_A) \Big|_{350}^T = -\frac{1}{C} \left[\frac{C \ln(2)}{t_1} + \frac{K A}{L} \right] \times \left[t \right]_{t_1}^{3t_1}$$

$$\text{or} \quad \ln(T - T_A) - \ln(350 - T_A) = -\left[\frac{\ln(2)}{t_1} + \frac{K A}{L C} \right] (3t_1 - t_1)$$

$$\text{or} \quad \ln \left(\frac{T - T_A}{350 - T_A} \right) = -\left[\frac{\ln(2)}{t_1} + \frac{K A}{L C} \right] (2t_1)$$

Given $T_A = 300$ K. Therefore,

$$\ln \left(\frac{T - 300}{350 - 300} \right) = -\left[\frac{\ln(2)}{t_1} + \frac{K A}{L C} \right] (2t_1)$$

$$\text{or} \quad \left(\frac{T - 300}{50} \right) = e \left[-2t_1 \left\{ \frac{\ln(2)}{t_1} + \frac{K A}{L C} \right\} \right]$$

$$= e \left[-2 \ln(2) - \frac{2 K A t_1}{L C} \right]$$

$$\text{or} \quad T = 300 + 50 \times \frac{1}{4} e \left(\frac{-2 K A t_1}{L C} \right)$$

$$\text{or} \quad T = 300 + 12.5 e \left(\frac{-2 K A t_1}{L C} \right)$$

This is the temperature of body X at $t = 3t_1$.

Illustrative Example 4.24

A black walled metal container of negligible heat capacity is filled with water. The container has sides of length 10 cm. It is placed in an evacuated chamber at 27°C . How long will it take for the temperature of water to change from 30°C to 29°C .

Solution

As in this case temperature difference of water with its surrounding is not large, we can use Newton's Law of cooling. If t is the time taken in cooling water from 30°C to 29°C , we have the average from of Newton's law of cooling as

$$\frac{\Delta T}{\Delta t} = \left(\frac{T_2 + T_1}{2} - T_s \right)$$

$$\begin{aligned} \text{Where } k &= \frac{4A\sigma T_s^3}{ms} \\ &= \frac{4 \times 5.67 \times 10^{-8} \times 6 \times 100 \times 10^{-4} \times (300)^3}{1 \times 4200} \\ &= 8.75 \times 10^{-5} \end{aligned}$$

$$\text{or} \quad \frac{30 - 29}{\Delta t} = 8.75 \times 10^{-5} [29.6 - 27]$$

$$\begin{aligned} \text{or} \quad \Delta t &= \frac{1}{8.75 \times 10^{-5} \times 2.5} \\ &= 4571.43 \text{ s} \\ &= 76.2 \text{ min} \\ &= 1 \text{ hr } 16.2 \text{ min.} \end{aligned}$$

Illustrative Example 4.25

A calorimeter of water equivalent 100 gm cools in air in 18 minutes from 60°C to 40°C . When a block of metal of mass 60 gm is heated to 60°C and placed inside the calorimeter. Assume heat loss only by radiation and Newton's Law of cooling to be valid. Find the specific heat of metal if now the system cools from 60°C to 40°C in 20 minutes.

Solution

From average form of Newton's Law of cooling if a body cools from temperature T_2 to T_1 in time t , then we have

$$\frac{T_2 - T_1}{t} = \frac{k}{ms} \left(\frac{T_2 + T_1}{2} - T_s \right)$$

Where ms is the heat capacity of the body and k is the constant which depends on surrounding temperature and surface of body exposed to surrounding. Now in this case it is given that calorimeter alone takes 18 minutes to cool down from 60° to 40°C thus we have

$$\left(\frac{60-40}{18} \right) = \frac{k}{100} \left(\frac{60+40}{2} - T_s \right)$$

[As for calorimeter it is given that $ms = 100 \times 1 = 100$]

$$\text{or} \quad \frac{20}{18} = k(50 - T_s)$$

$$\text{or} \quad \frac{10}{9} = \frac{k}{100} (50 - T_s) \quad \dots (4.59)$$

Now when metal block is placed in it, it takes 20 minutes to cool down from 60°C to 40°C , thus we have

$$\frac{60-40}{20} = \frac{k}{100+60 \times s} \left(\frac{60+40}{2} - T_s \right)$$

$$\text{or} \quad 1 = \frac{k}{100+60s} (50 - T_s) \quad \dots (4.60)$$

Dividing equation-(4.59) by (4.60)

$$\frac{10}{9} = \frac{100+60s}{100}$$

$$\text{or} \quad 1000 = 900 + 540s$$

$$\text{or} \quad s = \frac{100}{540} = 0.185 \text{ cal/gmK.}$$

Illustrative Example 4.26

A metal ball of 1 kg mass is heated by a 20 W heater in a room at 20°C . After some time temperature of ball becomes steady at 50°C . Find the rate of loss of heat by the ball to surrounding when its temperature becomes 50°C . Also find the rate at which it loses heat to the surrounding when its temperature was 30°C .

Solution

It is given that when ball is at 50°C , its temperature becomes steady. This means that rate at which heat is being supplied to the ball, at the same rate, it is losing heat. Thus when ball is at 50°C , the rate of loss of heat to the surrounding by ball is

$$\begin{aligned} \frac{dQ}{dt} &= \text{Power of Heater} \\ &= 20 \text{ W} \end{aligned}$$

When ball was at 30°C , using Newton's Law of cooling we can state that the rate of heat loss by ball is directly proportional to its temperature difference with surrounding, thus we have

$$\begin{aligned} \frac{dQ}{dt} &= k(30 - 20) \\ \frac{dQ}{dt} &= 10k \quad \dots (4.61) \end{aligned}$$

From the state when ball was at 50°C , we can write from Newton's Law of cooling

$$20 = k(50 - 20)$$

$$\text{or} \quad k = \frac{2}{3}$$

Thus from equation-(4.61), we can write

$$\frac{dQ}{dt} = 10 \times \frac{2}{3} = \frac{20}{3} \text{ W.}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Heat Transfer

Module Number - 24 & 25

Practice Exercise 4.3

(i) The temperature of a body falls from 40°C to 36°C in 5 minutes. Find the time after which the temperature of the body will become 32°C . Take surrounding temperature to be 16°C .

[6.11 min]

(ii) A body initially at 80°C cools to 64°C in 5 minutes and to 52°C next 5 minutes. What will be its temperature after next 5 minutes and what is the temperature of the surroundings?

[16°C , 43°C]

(iii) The excess temperature of a hot body above its surroundings is halved in $\tau = 10$ minutes. In what time will it be $\frac{1}{10}$ of its initial value. Assume Newton's law of cooling.

[33.23 min]

(iv) A body cools in a surrounding which is at a constant temperature T_0 . Its temperature T is plotted with time t as shown in figure-4.28. Two tangents are drawn to the curve at the points A and B which meet the time axis at angles of θ_2 and θ_1 as shown. Assuming Newton's law of cooling to be valid show that

$$\frac{\tan \theta_2}{\tan \theta_1} = \frac{T_2 - T_0}{T_1 - T_0}$$

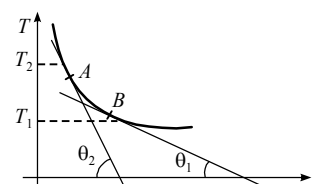


Figure 4.28

(v) A calorimeter of negligible heat capacity contains 100 gm water at 40°C. The water cools to 35°C in 5 minutes. If the water is now replaced by a liquid of same volume as that of water at same initial temperature, it cools to 35°C in 2 minutes. Given specific heats of water and that liquid are 4200 J/kg°C and 2100 J/kg°C respectively. Find the density of the liquid.

[800 kg/m³]

4.5 Black Body Radiation Spectrum

Generally gaseous state spectrum of a substance is discrete lines, it is called line spectrum. In gases atoms are so far apart that interaction between them are negligible and each atom behaves as an isolated system. Hot matter in condensed state in form of solid or liquid always emits radiation with a continuous distribution of wavelengths rather than few particular wavelengths like a line spectrum. We have discussed that a black body is one which absorb all the wavelengths of electromagnetic radiation incident upon it. This is also the best possible emitter of electromagnetic radiation at any wavelength. The continuous spectrum radiation that it emits is called blackbody radiation spectrum.

Stefan-Boltzmann's Law states that the radiation energy emitted per unit surface area or average thermal power radiated per unit area of a black body is given as

$$E = \sigma T^4 \text{ W/m}^2$$

The above formula gives the total amount of energy radiated by the body in the form of electromagnetic radiation but it does not tell any thing about the wavelengths of emitted radiations. As we've discussed earlier that thermal radiation is a polychromatic radiation where total intensity of emitted radiation is distributed among a wide range of wavelengths. Here also this total energy $\sigma T^4 \text{ W/m}^2$ is not uniformly distributed over all the wavelengths. Its distribution can be measured and described by the intensity per wavelength interval denoted by E_λ , called spectral emittance. Here E_λ is defined as $E_\lambda d\lambda$ is the intensity of radiation emitted in the wavelength range from λ to $\lambda + d\lambda$. Thus the total intensity which is given by Stefan-Boltzmann's Law can be written as

$$E = \int_0^\infty E_\lambda d\lambda = \sigma T^4$$

If a variation graph is plotted for spectral emittance E_λ with λ , it is shown in figure-4.29. Experiments show that each graph plotted at a temperature has a peak value of spectral emittance at a wavelength λ_m at which these intensity per wavelength interval is maximum. It is also observed that as temperature of

black body increases the wavelength at which spectral emittance is maximum decreases and experimental data shows that λ_m is inversely proportional to the absolute temperature of body given as

$$\lambda_m \propto \frac{1}{T}$$

or

$$\lambda_m = \frac{b}{T}$$

or

$$\lambda_m T = b \quad \dots (4.62)$$

The above equation-(4.62) is called *Wein's Displacement Law* and here the proportionality constant b is called Wein's constant whose numerical value is

$$b = 2.89 \times 10^{-3} \text{ m-K}$$

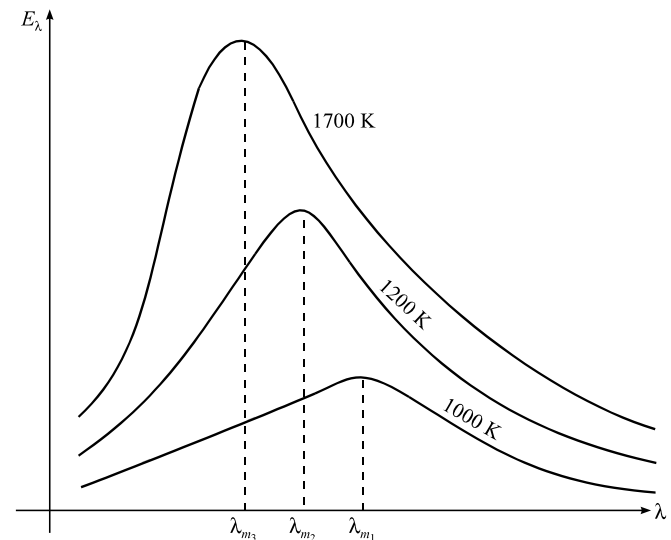


Figure 4.29

Thus as the temperature rises, the peak of E_λ becomes higher and shifts to shorter wavelengths. A body that glows yellow is hotter and brighter than one that glows red. Yellow light has shorter wavelengths than red light. Experiments also show that the shape of the distribution function is same for all the temperature, we can make a curve for one temperature fit any other temperature by only changing the scale on graph.

Several experiments were done by different scientists to develop some empirical expression for calculation of E_λ . The first best appropriate was found by Rayleigh. This result was

$$E_\lambda = \frac{2\pi ckT}{\lambda^4} \quad \dots (4.63)$$

This formula given in equation-(4.63) was found valid only at high wavelengths but it is not valid for small wavelengths. Figure-4.29 shows that at low wavelengths curves fall toward zero but this expression is inversely proportional to λ^4 rises to infinite intensity at lower wavelengths approaching zero.

One more result can be obtained from expression in equation-(4.63). We've read that at maximum value of E_λ , wavelength is λ_m , thus

$$E_\lambda(\max) = \frac{2\pi ckT}{\lambda_m^4}$$

From Wein's Displacement Law, we have

$$\lambda_m = b/T^5, \text{ thus}$$

$$E_\lambda(\max) = \frac{2\pi ck}{b^4} T^5$$

$$\text{or } E_\lambda(\max) \propto T^5 \quad \dots (4.64)$$

Equation-(4.61) states that the maximum spectral emittance from a black body at a given temperature is directly proportional to fifth power of absolute temperature of the body. This is called "*Wein's fifth power Law*". This was initially observed by Wein experimentally which is obtained by a partial correct Raylength relation. Later it was mathematically proven by Plank's Radiation theory.

4.5.1 Plank's Radiation Theory as a Proof of Wein's Law and Stefan Boltzmann's Law

In early years of twentieth century plank derived some mathematical expressions for his experiments on radiation and derived a formula for intensity distribution called Plank's Radiation Law. The final result is

$$E_\lambda = \frac{2\pi hc^2}{\lambda^5 (e^{hc/\lambda kT} - 1)} \text{ W/m}^3 \quad \dots (4.65)$$

Where h is plank's constant, c is speed of light, k is Boltzmann's constant, T is the absolute temperature and λ is the wavelength. This relation also agree well with the experimental Wein's curves shown in figure-4.29.

The Plank's Radiation Law also verifies Wein's displacement law and Stefan-Boltzmann's Law as consequences.

Proof of Wein's Displacement Law

From equation-(4.65) differentiating the expression and equating it to zero we can find the value of λ_m at which E_λ is maximum as

$$\frac{dE_\lambda}{d\lambda} = 0$$

Solving the above equation we get the value of λ_m as

$$\lambda_m = \frac{hc}{4.965 \text{ kT}} \quad \dots (4.66)$$

$$\lambda_m T = 2.89 \times 10^{-3}$$

Which agrees with the Wein's displacement law.

Proof of Wein's Fifth Power Law

Plank's Radiation Law gives the maximum value of E_λ as

$$E_\lambda(\max) = \frac{2\pi hc^2}{\lambda_m^5 (e^{hc/\lambda_m kT} - 1)}$$

From Wein's displacement law we have $\lambda_m = \frac{b}{T}$, thus

$$E_\lambda(\max) = \frac{2\pi hc^2 T^5}{b^5 (e^{hc/kb} - 1)}$$

$$\text{or } E_\lambda(\max) \propto T^5$$

Proof of Stefan-Boltzmann's Law

Stefan-Boltzmann's Law gives the total amount of energy radiated by a black body at a given temperature which can be obtained by finding the area under the Wein's curve distributed from wavelength zero to infinity. This can be given as

$$E = \int_0^\infty E_\lambda d\lambda = \int_0^\infty \frac{2\pi hc^2}{\lambda^5 (e^{hc/\lambda kT} - 1)} d\lambda$$

On solving the above integral we get

$$E = \frac{2\pi^5 k^4}{15c^2 h^3} T^4$$

This agrees with Stefan-Boltzmann Law and also verifies the numerical value of

$$\sigma = \frac{2\pi^5 k^4}{15c^2 h^3} = 5.67 \times 10^{-8} \text{ w/m}^2 \text{K}^4$$

Similarity with Rayleigh Expression

We've already discussed that Rayleigh's expression only agrees with the experimental curves at high wavelength. From Plank's Radiation law if we simplify the expression for high values of λ we get

$$E_\lambda = \frac{2\pi hc^2}{\lambda^5 (e^{hc/\lambda kT} - 1)}$$

For $x \ll 1$ we can use $e^x \simeq 1 + x$. Thus here for high values of λ

we have $e^{hc/\lambda kT} = 1 + \frac{hc}{\lambda kT}$ which result

$$E_\lambda = \frac{2\pi ckT}{\lambda^4}$$

Which is Rayleigh Expression. Thus at high values of wavelengths both Plank's formula and Rayleigh expression approach to same value.

4.5.2 Practical Applications of Kirchoff's Law

For deserts it is commonly heard that days are hot and nights are cold. This can be simply explained as sand is rough and can be approximated as a black body, so it is a good absorber and hence in deserts in day time sand absorbs radiation from sun and so day will be very hot. Now in accordance with Kirchoff's Law, good absorber is a good emitter, so at night when sand emits radiation, will be cold.

Another good example of kirchoff's law in practice can be seen when a metal ball with some black spots on its surface is heated to a high temperature and is taken in dark. In dark surroundings, the black spots shine brilliantly and the shining surfaces of the ball becomes dark. We can simply explain this as during heating black spots absorb radiation and so emit in dark surrounding while the shining parts reflects radiations and absorbs negligible radiation during heating and so does not emit radiation and becomes dull in dark surroundings.

In spectrum analysis kirchoff's law is useful in so many applications. When absorption spectrum of sodium vapours is formed, two dark lines are seen in yellow region in transmitted light due to absorption of these two wavelengths by sodium vapours. When emission spectrum of sodium vapours is obtained the above two bright lines are observed as these vapours emit those radiations which are absorbed initially. This is in accordance with Kirchoff's Law i.e., a good emitter is a good absorber. When spectrum of sun is formed, Dark Fraunhofer lines are found which can be explained on the basis of Kirchoff's Law. When white light is emitted from the central part of the sun (photosphere) passes through its atmosphere (chromosphere). Radiations of those wavelengths will be absorbed by the gases present there which they usually emit, as good emitter is a good absorber, which results dark lines in the spectrum of sun.

Another common example which can be realized in common practice is heating of a coloured glass. At normal temperature a piece of red glass appears red because from all the light falling it reflects red and absorbs other colours, when this glass is heated, it absorbs several wavelengths except red and when it is heated to a very high temperature and taken in dark it emits wavelengths which it was absorbing during heating thus at very high temperature red glass emits radiation other than red so it glows with colour complementary to red such as green or blue. Similarly when a blue glass is heated, it absorbs

all radiation wavelengths other than blue and when at very high temperature it is taken in dark it emits all wavelengths other than blue thus it glows with a colour complementary to blue such as red or yellowish.

Illustrative Example 4.27

A black body at 1500 K emits maximum energy of wavelength 20000 Å. If sun emits maximum energy of wavelength 5500 Å, what would be the temperature of sun.

Solution

According to Wein's displacement law, we have

$$\lambda_m T = \text{constant}$$

$$\text{or} \quad \lambda_{m_1} T_1 = \lambda_{m_2} T_2$$

$$\text{Thus} \quad 20000 \times 10^{-10} \times 1500 = 5500 \times 10^{-10} \times T_2$$

$$\text{or} \quad T_2 = \frac{200}{55} \times 1500 = 5454.54 \text{ K}$$

Illustrative Example 4.28

If the filament of a 100 W bulb has an area 0.25 cm² and behaves as a perfect black body. Find the wavelength corresponding to the maximum in its energy distribution. Given that stefan's constant is $\sigma = 5.67 \times 10^{-8} \text{ J/m}^2\text{s K}^4$, $b = 2.89 \times 10^{-3} \text{ mk}$.

Solution

In the bulb filament given, the energy radiated per sec per m² of its surface area is given as

$$E = \frac{P}{A} = \frac{100}{0.25 \times 10^{-4}} = 4 \times 10^6 \text{ J/s m}^2$$

If T is the temperature of the filament then according to stefan's law, we have

$$E = \sigma T^4$$

$$\text{or} \quad 4 \times 10^6 = 5.67 \times 10^{-8} \times T^4$$

$$\text{or} \quad T^4 = \frac{4 \times 10^6}{5.67 \times 10^{-8}} = 7.055 \times 10^{13}$$

$$\text{or} \quad T = [7.055 \times 10^{13}]^{1/4} = 2898.14 \text{ K}$$

If the filament radiates the maximum energy at a wavelength λ_m , from Wein's displacement law, we have

$$\lambda_m T = b$$

$$\begin{aligned} \text{or} \quad \lambda_m &= \frac{b}{T} \\ &= \frac{2.89 \times 10^{-3}}{2898.14} = 9971.9 \text{ \AA} \end{aligned}$$

Illustrative Example 4.29

A body emits maximum energy at 4253 Å and the same body at some other temperature emits maximum energy at 2342 Å. Find the ratio of the maximum energy radiated by the body in a short wavelength range.

Solution

If E_1 and E_2 are the maximum energy radiated by the body in short wavelength interval at temperature T_1 and T_2 then according to Wein's fifth power law, we have

$$\frac{E_1}{E_2} = \left(\frac{T_1}{T_2} \right)^5 \quad \dots (4.67)$$

According to Wein's displacement law, we have

$$\begin{aligned} \lambda_{m1} T_1 &= \lambda_{m2} T_2 \\ \text{or} \quad \frac{T_1}{T_2} &= \frac{\lambda_{m2}}{\lambda_{m1}} \quad \dots (4.68) \end{aligned}$$

Here λ_{m1} and λ_{m2} are the wavelength at temperatures T_1 and T_2 at which the energy radiated in a small interval of wavelength is maximum.

Now from equation-(4.67) and (4.68), we have

$$\begin{aligned} \frac{E_1}{E_2} &= \left(\frac{\lambda_{m2}}{\lambda_{m1}} \right)^5 \\ \text{or} \quad \frac{E_1}{E_2} &= \left(\frac{234 \times 10^{-10}}{4253 \times 10^{-10}} \right)^5 = 0.0506 \end{aligned}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - HEAT & THERMODYNAMICS

Topic - Heat Transfer

Module Number - 26 to 29

Practice Exercise 4.4

(i) The maximum in the energy distribution spectrum of the sun is at wavelength 4753 Å and its temperature is 6050 K. What will be the temperature of the star whose energy distribution shows a maximum at wavelength 9506 Å.

[3025 K]

(ii) The power radiated by a black body is P , and it radiates maximum energy around the wavelength λ . If the temperature of the black body is now changed so that it radiates maximum energy around a wavelength $3\lambda/4$. By what factor the power radiated by it will increase.

[256/81]

(iii) A furnace is at a temperature of 2000 K. At what wavelength does it emit most intensively?

[14450 Å]

(iv) A black body is at a temperature of 2880 K. The energy of radiation emitted by this object with wavelength between 4990 Å and 5000 Å is E_1 and that between 9990 Å and 10000 Å is E_2 . Find the ratio of E_2 and E_1 .

[4.67]

(v) The radiant emittance of a black body is $R = 250 \text{ kW/m}^2$. At what wavelength will the emissivity of this black body be maximum? ($b = 2.9 \times 10^{-3} \text{ m} \cdot \text{K}$ and $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$)

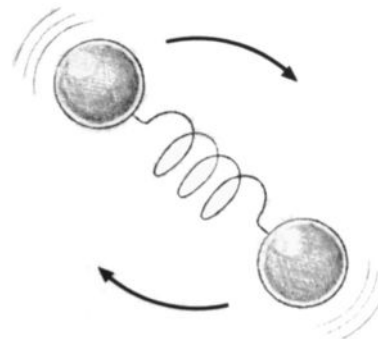
[19944.7 Å]

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Oscillations and Simple Harmonic Motion

FEW WORDS FOR STUDENTS

As motion is concerned we've studied translational and rotational motion and we've applied them in different sections of Physics. There is one more important kind of motion, we introduce now and in further sections of this book we use it. This is oscillatory motion or vibrations. In such a motion it repeats itself and is also termed as periodic motion. The simplest type of oscillatory motion is Simple Harmonic Motion abbreviated as SHM on which we'll focus now in this chapter.



To describe SHM we'll use the basic laws of mechanics you've already covered. Further you'll see that the analysis of SHM is very useful in understanding the concepts of light and electronics.

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COVER APPLICATION

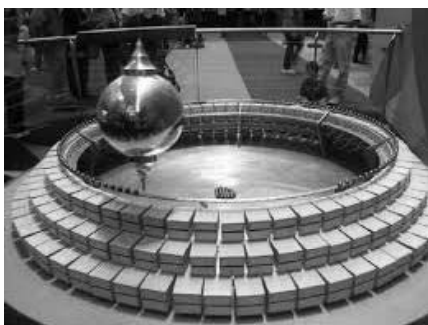


Figure-(a)

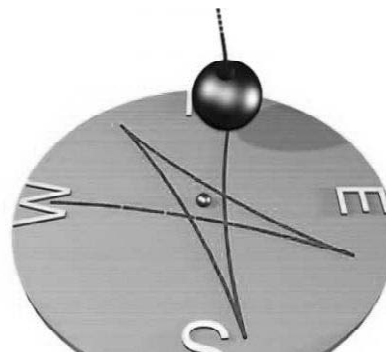


Figure-(b)

Foucault's pendulum is an easy experiment which demonstrates the Earth's rotation. The concept that the Earth revolves was nothing new or radical; the pendulum's accomplishment was to provide a proof that did not require minute observations of the stars or other objects far away from Earth. Foucault's pendulum is a highly localized, easily prepared experiment whose result is clear, powerful, and accessible even to the non-scientist. Figure-(a) shows a Foucault's pendulum of which the plane of oscillations rotates with Earth's rotation with respect to a point relative the Earth and Figure-(b) shows the trajectory of pendulum bob on a flat surface above which it oscillates.

What is periodic motion, we all know the motion of a body or a particle which is repeated after a given period, is called periodic motion. The motion of a particle in a circle, all types of vibrations and any motion in which a particle repeatedly retraces its path of motion.

If T is the period of motion after which it repeats itself then the frequency ν of the periodic motion is the number of cycles performed in one second and it is given as

$$\nu = \frac{1}{T}$$

Units of ν are s^{-1} or per second. A special name is given to the unit of frequency, hertz (Hz) after the discoverer of radio waves.

$$1 \text{ Hz} \equiv 1 \text{ cycle per second}$$

5.1 Periodic Motion and Oscillations

An oscillation is a special type of periodic motion in which a particle moves to and fro about a fixed point called mean position of particle. Oscillations are commonly seen in general life in our surrounding. As discussed, in all type of oscillations, there is always a mean position about which particle can oscillate. This is the position where particle is in equilibrium that is net forces on particle at this position is zero. If particle is displaced from mean position and due to this displacement some forces appear on it which act on particle in a direction directed toward its equilibrium position, these forces are called restoring forces as these forces tend particle to move towards its equilibrium position. Due to restoring forces, particle starts moving toward mean position and when it reaches mean position, it gains some KE due to work done by restoring forces and it will overshoot from this point with some velocity in other direction, again restoring forces appear on particle toward mean position and now particle is retarded and will stop after travelling some distance and will return toward mean position and starts accelerating & in such a way motion is continued which we call oscillation. The maximum displacement of particle from mean position where it will come to rest or from where it was started with zero initial speed is called as Amplitude of oscillations, some different types of oscillations are shown in figure-5.1.

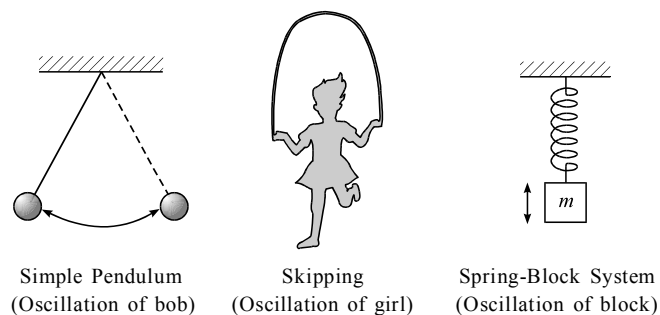


Figure 5.1

5.1.1 Types of Oscillations

There are several different types of oscillations depending on the nature of restoring forces used in oscillations but oscillations are classified in three major categories

- (a) Free oscillations
- (b) Damped oscillations and
- (c) Forced oscillations

Free oscillations are those in which once oscillation of a particle starts, it continues indefinitely till some external damping force appears in opposition to the restoring forces on it. Damped oscillations are those in which when a particle starts its oscillatory motion, due to friction or some other naturally acting opposing forces, the oscillation amplitude of particle start decreasing with time, this is called damping. Forced oscillations are those in which damping is not allowed by applying an external time varying force on particle which compensates the effect of the damping force acting on it, thus it is similar to free oscillations but naturally the oscillations are damped and by applying an external force these are made like free oscillations.

5.2 Simple Harmonic Motion

In this section we discuss a special type of oscillation called simple harmonic motion, abbreviated as SHM. A general oscillation can be regarded as SHM if it satisfies these basic conditions stated as

- (i) The oscillation amplitude of particle must be very small compared to its surrounding dimensions (dimensions of bodies with which it can interact.)
- (ii) During oscillation the acceleration of particle toward mean position due to net restoring forces must be directly proportional to its displacement from mean position.

5.2.1 Representation of SHM

We have discussed that an oscillation can be regarded as SHM if it satisfies the basic requirements to be SHM. Every SHM can be best represented as a projection of a particle in circular motion on its diameter. In fact the motion of projection of a uniform circular motion on its diameter satisfies both the conditions required to be SHM.

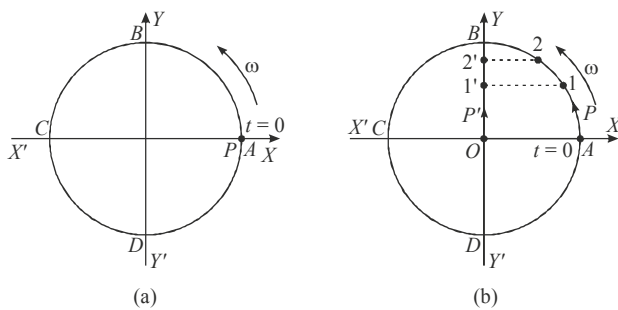


Figure 5.2

Let us analyze it from the figure-5.2 shown. A particle P is executing a uniform circular motion on the circle with a uniform angular velocity ω as shown. The particle is shown at its initial position A at $t = 0$. It starts in anticlockwise direction with uniform angular velocity ω . If we follow the motion of the projection of particle on its vertical diameter in figure-6.2 then we can see, along with P , its projection P' starts from mid point (centre of circle) in upward direction and the respective position of P' for position of P at point 1 and 2 are $1'$ and $2'$ as shown in figure. When P reaches the topmost position, P' will also be at this point and when P starts tracing the second quadrant of circle, P' starts coming down towards point O . Thus we can see, as point P traces its circular path $ABCD$, its projection on diameter YY' follows oscillatory motion along $OBODO \dots$ and so on. This oscillatory motion of P' can be taken as SHM as it follows the conditions to be a SHM. We'll prove it in next section.

Here motion of particle is in a straight line with amplitude equal to the radius of circle.

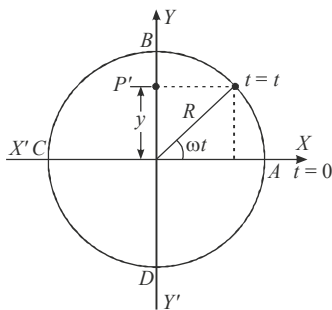


Figure 5.3

As shown in figure-5.3, the displacement of P' from point O as a function of time can be given as

$$y = R \sin \omega t \quad \dots (5.1)$$

As P' is going up, its velocity can be given as

$$v = \frac{dy}{dt} = R\omega \cos \omega t \quad \dots (5.2)$$

Its acceleration is

$$a = \frac{dv}{dt} = -R\omega^2 \sin \omega t$$

From equation (5.1), we have

$$a = -\omega^2 y \quad \dots (5.3)$$

Here ω is a constant thus the acceleration of P' is directly proportional to the displacement from its mean position O and negative sign in equation-(5.3) shows that direction of acceleration is opposite to y that is towards mean position O . Hence the motion of projection P' can be regarded as SHM.

5.2.2 Equation of SHM

Equation of an oscillation is the mathematical expression giving the displacement of oscillating particle from its mean position as a function of time.

If a particle is executing SHM with amplitude A , it can be regarded as the projection of a circular motion of radius A as shown in figure-5.4 if circular motion of point P is at a constant angular velocity ω then this is termed as angular frequency of the point P' which is in SHM.

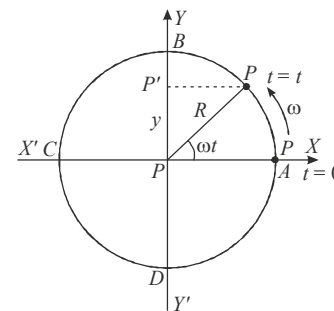


Figure 5.4

Let us consider at $t=0$, P was at point A and starts in anticlockwise direction as shown in figure-5.4. Now in time t , point P traversed by an angle ωt (as shown) and P' reaches to a displacement y as shown, can be given as

$$y = A \sin \omega t \quad \dots (5.4)$$

This equation-(5.4) in this case is called as equation of SHM of point P' which is in SHM. Here ωt which is the angular displacement of point P (in circular motion) is called phase angle of point P' in SHM.

5.2.3 General Equation of SHM

In previous article we've discussed the SHM of point P' which was at its mean position at $t = 0$. But it is not necessary that particle starts its SHM from its mean position. It can start from any point on its path, thus equation-(5.4) can not be accepted as a general equation of SHM, this being the equation of those all SHMs where particle starts (at $t = 0$) their SHM from their mean position.

Now consider figure-5.5 where a particle P' starts its SHM with a point having initial phase angle α anticlockwise on the circle of point P of which it is the projection. Let at $t = 0$, point P was at an angular displacement α from its reference point (point A). Thus the point from which P' will start its SHM is shown in figure-5.5. At this position (projection of P at $t = 0$ on YY') α is called as initial phase of point P' in SHM, as α is the initial angular displacement of particle P which is in circular motion. Now after time t , the angular displacement of P is $(\alpha + \omega t)$ which is called the *instantaneous phase* of point P' at time $t = t$ and at this instant the displacement of point P' from mean position is

$$y = A \sin(\omega t + \alpha) \quad \dots (5.5)$$

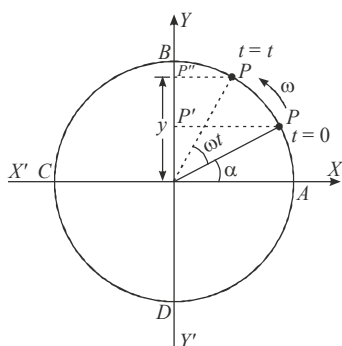


Figure 5.5

This equation-(5.5) is called as general equation of SHM of a particle which starts its SHM with initial phase α . Here initial phase of SHM implies it is the initial angular displacement of the particle which is in circular motion of which the projection is executing SHM.

As equation-(5.4) gives the equation of SHM of those particles which start their SHM from mean position. Similarly we can define an equation of SHM of those particles which start their SHM from their extreme position by substituting $\alpha = \pi/2$ in equation-(5.5). As if a particle starts from its extreme position, we can take its initial phase $\pi/2$ thus its equation of SHM from equation-(5.5) can be given as

$$y = A \cos \omega t \quad \dots (5.6)$$

In all type of problems in which a body or a particle execute SHM, we assume that this is the projection of an another particle who is in circular motion and its projection is executing SHM (that body or particle which is in SHM). Further we will take several examples to analyse this concept.

5.2.4 Velocity and Acceleration of a Particle in SHM

Equation-(5.5) gives the general expression for displacement from mean position of a particle executing SHM as a function of time. Thus velocity of this particle as a function of time can be given as

$$v = \frac{dy}{dt} = A\omega \cos(\omega t + \alpha) \quad \dots (5.7)$$

To convert it in displacement function, we can write

$$\begin{aligned} v &= A\omega \sqrt{1 - \sin^2(\omega t + \alpha)} \\ &= A\omega \sqrt{1 - \frac{y^2}{A^2}} \quad [\text{As } y = A \sin \omega t] \\ v &= \omega \sqrt{A^2 - y^2} \quad \dots (5.8) \end{aligned}$$

Equation-(5.8) gives the velocity of a particle in SHM with amplitude A , and angular frequency ω as a function of its displacement from mean position. From equation-(5.8) we can state that in SHM, particle's velocity is maximum when $y = 0$ i.e. at its mean position and is given as

$$\text{at } y = 0, v_{\max} = A\omega$$

At extreme positions of particle when $y = \pm A$, its velocity is zero where it returns towards its mean position. From equations-(5.5) and (5.8) we can plot the graphs of displacement and velocity as a function of time as shown in figure-5.6.

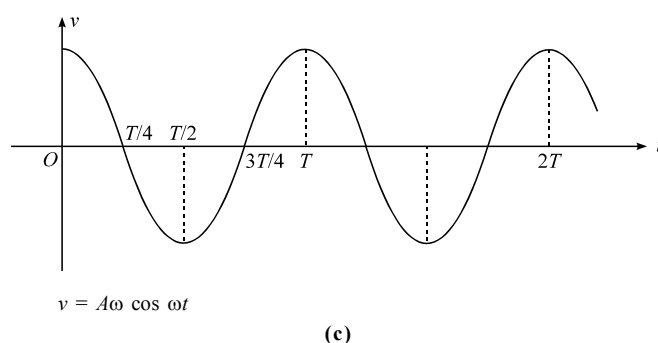
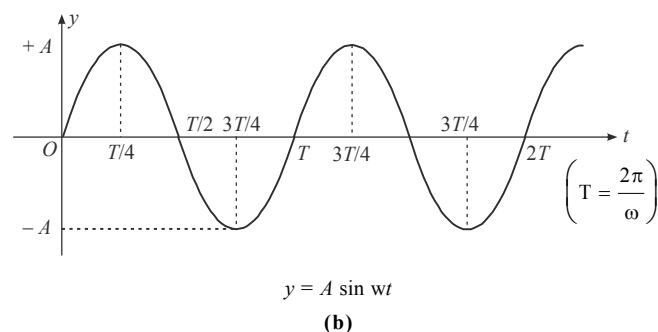
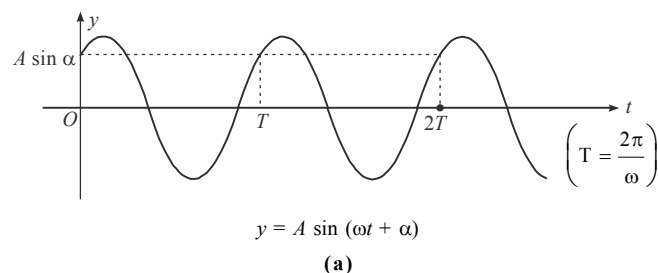


Figure 5.6

Similarly acceleration of particle in SHM can be given as

$$a = \frac{dv}{dt} = -A\omega^2 \sin(\omega t + \alpha) \quad \dots (5.9)$$

$$\text{or} \quad a = -\omega^2 y \quad [\text{As } y = A \sin(\omega t + \alpha)] \quad \dots (5.10)$$

From equation-(5.10) we can see that at mean position ($y = 0$) when velocity of particle is maximum, its acceleration is zero and at extremities where $y = \pm A$, acceleration of particle is maximum and its magnitude is given as

$$a_{\max} = \omega^2 A \quad [\text{Towards mean position}]$$

It also shows that as particle moves away from mean position, its acceleration continuously increases till it reaches its extreme position (at amplitude) when its velocity becomes zero and it returns.

Equation-(5.10) can be rewritten in differential form as

$$\frac{d^2 y}{dt^2} + \omega^2 y = 0 \quad \dots (5.11)$$

This equation-(5.11) is called “Basic Differential Equation” of motion of a particle in SHM. Every expression of displacement y as a function of time which satisfies this equation can also be regarded as an equation of SHM.

Illustrative Example 5.1

Prove that $y = Ae^{i\omega t}$ is an equation of SHM.

Solution

According to given equation in problem, differentiating with respect to time we get,

$$\frac{dy}{dt} = iA\omega e^{i\omega t}$$

Differentiating again with respect to time, we get

$$\frac{d^2 y}{dt^2} = -\omega^2 A e^{i\omega t} = -\omega^2 y \quad [\text{As } y = A e^{i\omega t}]$$

$$\text{Thus we have } \frac{d^2 y}{dt^2} + \omega^2 y = 0$$

This is the basic differential equation of SHM hence $y = Ae^{i\omega t}$ is an equation of SHM.

Illustrative Example 5.2

Find the amplitude and initial phase of a particle in SHM, whose motion equation is given as

$$y = A \sin \omega t + B \cos \omega t$$

Solution

Here in the given equation we can write

$$A = R \cos \phi \quad \dots (5.12)$$

$$\text{and} \quad B = R \sin \phi \quad \dots (5.13)$$

Thus the given equation transforms to

$$y = R \sin(\omega t + \phi) \quad \dots (5.14)$$

Equation-(5.14) is a general equation of SHM and here R is the amplitude of given SHM and ϕ is the initial phase of the oscillating particle at $t = 0$.

Here R is given by squaring and adding equation-(5.12) & (5.13)

$$R = \sqrt{A^2 + B^2}$$

Initial phase ϕ can be given by dividing-(5.13) & (5.12) as

$$\tan \phi = \frac{B}{A}$$

$$\phi = \tan^{-1} \left(\frac{B}{A} \right)$$

NOTE : Equations $y = Ae^{i(\omega t + \phi)}$ and $y = A \sin \omega t + B \cos \omega t$, we can also represent the general equation of SHM.

Illustrative Example 5.3

A body of mass 1 kg is executing simple harmonic motion which is given by $y = 6.0 \cos(100t + 4\pi)$ cm. What is the (i) amplitude of displacement, (ii) Angular frequency, (iii) initial phase, (iv) velocity, (v) acceleration, (vi) maximum kinetic energy?

Solution

The given equation of SHM is

$$y = 6.0 \cos(100t + \pi/4) \text{ cm.}$$

Comparing it with the standard equation of SHM, $y = A \sin(\omega t + \phi)$, we have

(i) Amplitude $A = 6.0$ cm.

(ii) Angular frequency $\omega = 100 \text{ s}^{-1}$

(iii) Initial phase $\phi = \pi/4$

(iv) Velocity $v = \omega \sqrt{A^2 - y^2} = 100 \sqrt{(36 - y^2)} \text{ cm/s}$

(v) Acceleration $= -\omega^2 y = -(100)^2 y = -10^4 y$

(vi) Kinetic energy $= \frac{1}{2} mv^2 = \frac{1}{2} m\omega^2 (A^2 - y^2)$

The kinetic energy of a particle in SHM is maximum, when it passes its mean position i.e. at $y = 0$

$$\begin{aligned} (\text{K.E.})_{\max} &= \frac{1}{2} m v_{\max}^2 = \frac{1}{2} m A^2 \omega^2 \\ &= \frac{1}{2} \times 1 \times 10^4 \times (0.06)^2 \\ &= 18 \text{ J} \end{aligned}$$

Illustrative Example 5.4

A particle of mass 0.8 kg is executing simple harmonic motion with an amplitude of 1.0 metre and periodic time 11/7 sec. Calculate the velocity and the kinetic energy of the particle at the moment when its displacement is 0.6 metre.

Solution

We know that, at a displacement y from mean position particle's velocity is given as

$$v = \omega \sqrt{(A^2 - y^2)}$$

or

$$\begin{aligned} v &= \frac{2\pi}{T} \sqrt{(A^2 - y^2)} \\ &= \frac{2 \times 3.14}{(11/7)} \sqrt{[(1.0)^2 - (0.6)^2]} \\ &= 3.2 \text{ m/s} \end{aligned}$$

Kinetic energy at this displacement is given by

$$\begin{aligned} K &= \frac{1}{2} m v^2 \\ &= \frac{1}{2} \times 0.8 \times (3.2)^2 \\ &= 4.1 \text{ J} \end{aligned}$$

Illustrative Example 5.5

A person normally weighing 60 kg stands on a platform which oscillates up and down harmonically at a frequency 2.0 s^{-1} and an amplitude 5.0 cm. If a machine on the platform gives the person's weight against time, deduce the maximum and minimum reading it will show, take $g = 10 \text{ m/s}^2$.

Solution

As shown in figure-5.7, platform is executing SHM with amplitude and angular frequency given as

$$A = 5.0 \text{ cm}$$

and

$$\omega = 2\pi n = 4\pi \text{ rad/s} \quad [\text{As } n = 2 \text{ s}^{-1}]$$

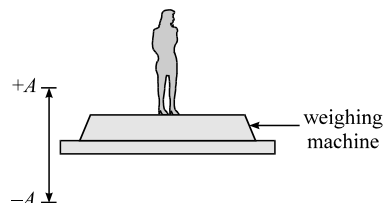


Figure 5.7

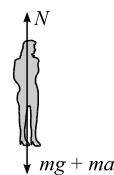


Figure 5.8

Here weighing machine will show weight more than that of man when it is below its equilibrium position when the acceleration of platform is in upward direction. In this situation the free body diagram of man relative to platform is shown in figure-5.8. Here ma is the pseudo force on man in downward direction relative to platform (or weighing machine). As weighing machine will read the normal reaction on it thus for equilibrium of man relative to platform, we have

$$N = mg + ma$$

or

$$N = mg + m(\omega^2 y) \quad [\text{As } |a| = \omega^2 y]$$

Where y is the displacement of platform from its mean position. We wish to find the maximum weight shown by the weighing machine, which is possible when platform is at its lowest extreme position as shown in figure-5.8, thus maximum reading of weighing machine will be

$$\begin{aligned} N &= mg + m\omega^2 A \\ &= 60 \times 10 + 60 \times (4\pi)^2 \times 0.05 \\ &= 600 + 480 \\ &= 1080 \text{ N} = 108 \text{ kg wt} \end{aligned}$$

Similarly the machine will show minimum reading when it is at its upper extreme position when pseudo force on man will be in upward direction, thus minimum reading of weighing machine will be

$$\begin{aligned} N &= mg - m\omega^2 A \\ &= 600 - 480 = 120 \text{ N} = 12 \text{ kg wt} \end{aligned}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Simple Harmonic Motion

Module Number - 1 to 9

Practice Exercise 5.1

(i) The maximum velocity of a body undergoing simple harmonic motion is 0.04 m s^{-1} and its acceleration at 0.02 m from the mean position is 0.06 m s^{-2} . Find its amplitude and period of vibration.

[$2.31 \times 10^{-2} \text{ m}$, 3.625 s]

(ii) A man of mass 60 kg standing on a platform executing SHM in vertical direction. The displacement from the mean of platform varies as

$$y = 0.5 \sin(2\pi ft)$$

Find the minimum value of f , for which the man will feel weightless at the highest point. Take $g = 10 \text{ m/s}^2$.

[0.712 Hz]

(iii) A point moves along the x axis according to the law $x = a \sin^2(\omega t - \pi/4)$. Find :

- the amplitude and period of oscillations; draw the plot $x(t)$;
- the velocity projection v_x as a function of the coordinate x ; draw the plot $v_x(x)$.

[(a) The amplitude is equal to $a/2$, and the period is $T = \pi/\omega$,

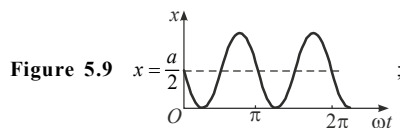
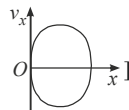


Figure 5.9 $x = \frac{a}{2}$;

(b) $v_x^2 = 4\omega^2 x(a - x)$, see **Figure 5.10**



(iv) (a) When a particle is executing SHM., what is the ratio of mean velocity during motion from one end of the path to the other and the maximum velocity ? (b) What is the ratio the average acceleration during motion from one extremity to the centre to the maximum acceleration ?

[(a) $\frac{2}{\pi}$; (b) $\frac{2}{\pi}$]

(v) A particle executes SHM in a straight line. The maximum speed of the particle during its motion is v_m . Find the average speed of the particle during its SHM.

[$\frac{2v_m}{\pi}$]

(vi) Equation of a particle in SHM is given as

$$x = 4 \sin \omega t + 3 \sin(\omega t + 63^\circ)$$

Here x is in centimeters and t is in seconds. Find the amplitude of oscillation of the particle.

[6.277 cm]

(vii) Figure-5.11 shows the acceleration-displacement graph of a particle in SHM. Find the time period of its SHM.

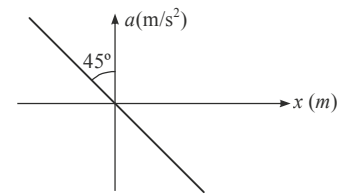


Figure 5.11

[6.28 s]

5.3 Superpositions of Simple Harmonic Motions

When a particle oscillates under the influence of two or more SHMs, its motion equation is not the simple one as discussed earlier for a common SHM equation. The equation of motion of a particle on which two or more than two SHMs are imposed, is given by the principal of superposition which is stated as

“When a particle oscillates under the influence of two or more SHMs then the resultant displacement of the particle is given by the vector sum of the individual displacements produced by the independent SHMs, which are being superposed on the particle.”

If $y_1, y_2, y_3, \dots, y_N$ are the displacements of the particle when it is oscillating only under the N independent SHMs, then on superposing all these SHMs on same particle, its resultant displacement is given as

$$y_R = y_1 + y_2 + \dots + y_N \quad \dots (5.15)$$

These superposing SHMs may be of same or different frequencies. We will discuss this in detail.

5.3.1 Superposition of Same Frequency SHMs

First we discuss superposition of two SHMs, then we generalize the obtained result for more number of SHMs. Let two SHM equations be with amplitude A_1 and A_2 differing in phase by ϕ are producing independent displacements at a particle when acting independently, y_1 and y_2 , which are given as

$$y_1 = A_1 \sin \omega t \quad \dots (5.16)$$

$$y_2 = A_2 \sin(\omega t + \phi) \quad \dots (5.17)$$

When both of these SHMs are superposed on the particle, the resultant displacement of the particle is given by principal of superposition as

$$y_R = y_1 + y_2$$

$$\text{or} \quad = A_1 \sin \omega t + A_2 \sin (\omega t + \phi)$$

$$\text{or} \quad = A_1 \sin \omega t + A_2 \sin \omega t \cos \phi + A_2 \cos \omega t \sin \phi$$

$$\text{or} \quad = (A_1 + A_2 \cos \phi) \sin \omega t + (A_2 \sin \phi) \cos \omega t$$

Here substituting

$$A_1 + A_2 \cos \phi = R \cos \theta \quad \dots (5.18)$$

$$\text{and} \quad A_2 \sin \phi = R \sin \theta \quad \dots (5.19)$$

$$\text{Now we get} \quad y_R = R \sin (\omega t + \theta) \quad \dots (5.20)$$

We can see that equation-(5.20) is also an equation of SHM with amplitude R and initial phase θ . Here R and θ can be obtained from equations-(5.18) and (5.19) as

$$R = \sqrt{(A_1 + A_2 \cos \phi)^2 + (A_2 \sin \phi)^2}$$

$$\text{or} \quad = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi} \quad \dots (5.21)$$

$$\text{and} \quad \tan \theta = \frac{A_2 \sin \phi}{A_1 + A_2 \cos \phi}$$

$$\text{or} \quad \theta = \tan^{-1} \left(\frac{A_2 \sin \phi}{A_1 + A_2 \cos \phi} \right) \quad \dots (5.22)$$

Thus we can say that when two SHMs of same frequency are superposed on a particle then also it executes SHM whose amplitude and initial phase are given by equation-(5.21) and (5.22) respectively.

By observing equation-(5.21) we can say that this result is similar to that used for resultant of two vectors of magnitudes A_1 and A_2 and having an angle ϕ between them. Similarly we can generalize this result for finding the amplitude of SHM of a particle which oscillates under the influence of more than two SHMs of same frequency.

5.3.2 Superposition of Different Frequency SHMs

If two SHMs of different frequencies are superposed on a particle which produces displacements y_1 and y_2 independently, where y_1 and y_2 are given as

$$y_1 = A \sin \omega_1 t \quad \dots (5.23)$$

$$y_2 = A \sin \omega_2 t \quad \dots (5.24)$$

Here we ignore the initial phase difference in the two SHMs as due to different frequencies, the phase difference between the two SHMs continuously change with time. Now according to principle of superposition, the resultant displacement of particle is given as

$$y_R = y_1 + y_2$$

$$= A \sin \omega_1 t + A \sin \omega_2 t$$

$$= 2A \sin \left(\frac{\omega_1 + \omega_2}{2} t \right) \cos \left(\frac{\omega_1 - \omega_2}{2} t \right)$$

In the resultant displacement function, here we can see that there are two sinusoidal functions of time with different arguments which on splitting gives the original SHM equations, superposition of which results this equation. Thus if displacement equation of a particle consists of two or more sinusoidal functions, the same number of independent SHMs are superposed for producing such equation. To understand this, look at the following equation

$$y = 25 \sin (4t) \cos^2 (5t) \quad \dots (5.25)$$

This equation-(5.25) gives the displacement of an oscillating particle and we have to find the numbers of independent SHMs and their frequencies, superposition of which give rise to this displacement. For it we can split the given equation in the manner explained below.

$$y = 25 \sin 4t \left(\frac{1 + \cos(10t)}{2} \right)$$

$$\text{or} \quad y = \frac{25}{2} \sin 4t + \frac{25}{2} \sin (4t) \cos (10t)$$

$$\text{or} \quad y = \frac{25}{2} \sin 4t + \frac{25}{2} [\sin (14t) - \sin (6t)]$$

$$\text{or} \quad y = \frac{25}{2} \sin (4t) + \frac{25}{2} \sin (14t) - \frac{25}{2} \sin (6t)$$

$$\text{or} \quad y = \frac{25}{2} \sin (4t) + \frac{25}{2} \sin (14t) + \frac{25}{2} \sin (6t - \pi) \dots (5.26)$$

From equation-(5.26) it is clear that equation-(5.25) is the resultant of superposition of three independent SHMs, given as

$$y_1 = \frac{25}{2} \sin (4t)$$

$$y_2 = \frac{25}{2} \sin (14t)$$

$$y_3 = \frac{25}{4} \sin (6t - \pi)$$

We can also observe in equation-(5.25) that there are three sinusoidal functions in equation-(5.25) so we can directly state that this equation is the superposition of three SHMs of different frequencies.

5.4 Analysis of Angular SHM

We have discussed the case of SHM of a simple pendulum when its bob oscillates with angular frequency $\omega = \sqrt{\frac{g}{l}}$.

Again consider the same case shown in figure-5.12. If the bob of a simple pendulum is thrown from its bottom most position at velocity v_0 . If amplitude of its oscillation is A , we have

$$v_0 = A\omega = A\sqrt{\frac{g}{l}}$$

[As at mean position velocity of particle is $v = A\omega$]

or

$$A = v_0 \sqrt{\frac{l}{g}}$$

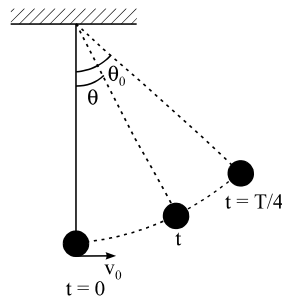


Figure 5.12

When ball is at its extreme position, if its angular displacement is θ_0 , this can be regarded as the angular amplitude of oscillations. As the displacement of bob from its mean position is given as

$$x = A \sin(\omega t + \alpha) \quad \dots (5.27)$$

[General equation of SHM]

If θ and θ_0 are angular displacement and angular amplitude of bob, we have

$$\theta = \frac{x}{l} \quad \text{and} \quad \theta_0 = \frac{A}{l}$$

Thus general equation of SHM of bob in angular form can be given by substituting values of x and A in equation-(5.27) as

$$\theta = \theta_0 \sin(\omega t + \alpha) \quad \dots (5.28)$$

Using above equation we can find the angular velocity of the body which is in angular SHM as

$$\dot{\theta} = \frac{d\theta}{dt} = \theta_0 \omega \cos(\omega t + \alpha) \quad \dots (5.29)$$

or

$$\dot{\theta} = \omega \sqrt{\theta_0^2 - \theta^2} \quad \dots (5.30)$$

NOTE : Here we represent $\frac{d\theta}{dt}$ by $\dot{\theta}$ not ω as the notation ω is already being used for angular frequency of body in SHM.

Similarly angular acceleration of body is given as

$$\ddot{\theta} = \frac{d^2\theta}{dt^2} = -\theta_0 \omega^2 \sin(\omega t + \alpha) \quad \dots (5.31)$$

$$\ddot{\theta} = -\omega^2 \theta \quad \dots (5.32)$$

Thus restoring torque on body is given as

$$\tau_R = -I\ddot{\theta} = -I\omega^2 \theta \quad \dots (5.33)$$

Thus we can state in angular SHM, angular acceleration of body and the restoring torque on body is directly proportional to the angular displacement of body from its mean position and directed toward mean position. Similarly basic differential equation for angular SHM can be written as

$$\frac{d^2\theta}{dt^2} + \omega^2 \theta = 0 \quad \dots (5.34)$$

5.5 How to Find Time Period of SHM

We've discussed that SHM is a special type of oscillation in which the restoring force on the oscillating particle is directly proportional to its displacement from mean position and can be given as

$$F_R = ma = -m\omega^2 y \quad \dots (5.35)$$

Where ω is the angular frequency of the particle in SHM. When a physical situation is given in which a particle can execute SHM, these are the steps to find the time period of particle in its SHM.

5.5.1 Steps to Find Angular Frequency or Time Period of SHM

Step I : When particle is in its equilibrium position, balance all forces acting on it and locate the equilibrium position mathematically. For rotational equilibrium balance all torques acting on body.

Step II : From the equilibrium position, displace the particle slightly by a displacement x and find the expression of net restoring force on it. For angular motion displace the body by a small angle θ and find expression of net restoring torque on it.

Step III : Try to express the net restoring force acting on particle as a proportional function of its displacement from mean position. The final expression should be obtained in the form.

$$F_R = -ky$$

For angular SHM it will be

$$\tau_R = -k\theta$$

Here we put $-ve$ sign as direction of F_R is opposite to the displacement y or τ_R is opposite to θ . If a be the acceleration of particle at this displacement, we have

$$a = -\left(\frac{k}{m}\right)y \quad \dots (5.36)$$

Comparing this equation-(5.36) with the basic differential equation of SHM we get

$$\omega^2 = \frac{k}{m}$$

or

$$\omega = \sqrt{\frac{k}{m}}$$

Similarly for angular SHM, angular acceleration of particle is given as

$$\beta = -\frac{\tau_R}{I} = -\left(\frac{k}{I}\right)\theta$$

Comparing with

$$\beta = -\omega^2\theta$$

we get

$$\omega = \sqrt{\frac{k}{I}}$$

As ω is the angular frequency of the particle in SHM, its time period of oscillation can be given as

$$T = \frac{2\pi}{\omega} \quad \dots (5.37)$$

Using the above steps we can find the oscillation period of particle in SHM in different physical situations.

To understand the concept in detail we first explain some different types of pendulums then we take some examples to discuss in details.

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Simple Harmonic Motion

Module Number - 12 & 15

5.6 Different Types of Pendulums

There are four different types of pendulums are used in general practice. These are :

- (1) Simple Pendulum
- (2) Spring Block Pendulum
- (3) Physical Pendulum or Compound Pendulum
- (4) Torsional Pendulum

5.6.1 Simple Pendulum

This is the most fundamental oscillatory system as shown in figure-5.13(a) a bob of mass m is hanging from a string of

length l . At this position as bob is in equilibrium so tension in string will be equal to weight of the bob, thus

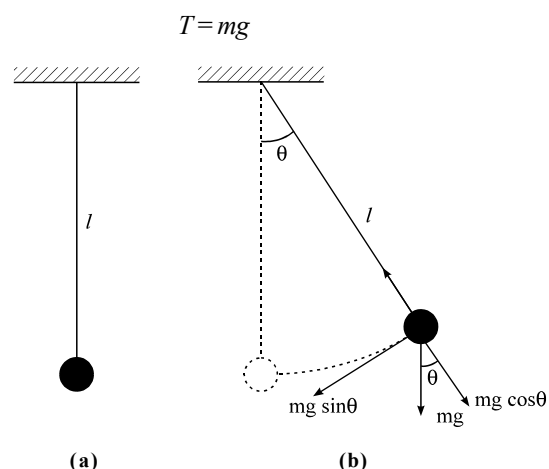


Figure 5.13

If bob is slightly displaced from its equilibrium position and released as shown in figure-5.13(b) by a small angular displacement θ . As still it is in equilibrium along radial direction, at this position, tension in string is

$$T = mg \cos \theta$$

Thus due to $mg \sin \theta$, bob will accelerate towards the equilibrium position thus we can say that in this case $mg \sin \theta$ is behaving as restoring force and for motion of bob, so we have

$$F_R = -mg \sin \theta \quad [-ve \text{ sign for restoring nature}]$$

$$\text{or} \quad F_R = -mg \theta \quad [\text{As for small } \theta, \sin \theta \simeq \theta]$$

If bob is displaced by a distance x from mean position, we have

$$\theta = \frac{x}{l}$$

Now if acceleration of bob toward mean position is a , we have

$$F_R = ma = -mg \frac{x}{l}$$

$$\text{or} \quad a = -\left(\frac{g}{l}\right)x \quad \dots (5.38)$$

Comparing this equation with differential equation of SHM, we get

$$\omega = \sqrt{\frac{g}{l}} \quad \dots (5.39)$$

Thus time period of oscillation of a simple pendulum is

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{l}{g}} \quad \dots (5.40)$$

5.6.2 Spring Block Pendulum

This is another very simple and fundamental oscillating system. Figure-5.14 shows a spring block system in equilibrium. At equilibrium due to the weight of block spring is stretched from its natural length say by a distance h , such that

$$mg = kh \quad \dots (5.41)$$

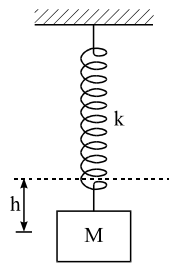


Figure 5.14

Now from this equilibrium position, block is slightly displaced down, say by a distance x and released, then due to spring force it will have a tendency to come back to its equilibrium position and due to this stable equilibrium position it will start oscillating.

When block is stretched down from its equilibrium position by a distance x , the restoring force on block can be written as

$$F_R = - \left[\underbrace{k(x+h)}_{\substack{\text{upward force} \\ \text{(towards mean position)}}} - \underbrace{mg}_{\substack{\text{downward force} \\ \text{(away from mean position)}}} \right]$$

or
$$F_R = -kx$$

[As from equation-(5.41) $mg = kh$]

If acceleration of block toward mean position is a , we have

$$a = -\left(\frac{k}{m}\right)x \quad \dots (5.42)$$

Comparing this equation-(5.42) with standard differential equation of SHM, we get

$$\omega = \sqrt{\frac{k}{m}} \quad \dots (5.43)$$

Thus time period of oscillation is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$$

It shows that when a mass hanging from free end of a spring oscillates it executes SHM with angular frequency $\sqrt{k/m}$.

5.6.3 Compound Pendulum

This is made whenever a rigid body is hanging freely from a horizontally pivoted axis, as shown in figure-5.15. A body of mass m is pivoted at point O through a horizontal axis AA' . The body is hanging freely under gravity and in equilibrium position its centre of mass C is vertically below the suspension point O , at a distance l from O .

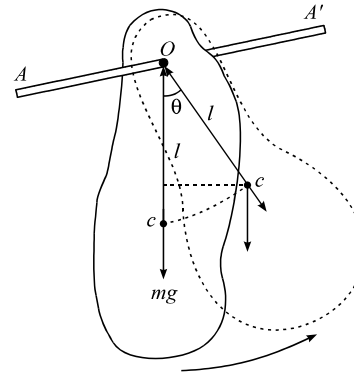


Figure 5.15

If the body is slightly tilted from its equilibrium position by an angle θ , mg will exert a restoring torque on it in opposite direction to restore the equilibrium position. Thus restoring torque on body in dotted position after tilting is

$$\begin{aligned} \tau_R &= -mg \cdot l \sin \theta \\ &\quad \text{[- ve sign for restoring nature]} \\ &= -mgl \theta \quad \text{[For small } \theta, \sin \theta \approx \theta] \end{aligned}$$

If its angular acceleration is α , we have

$$I\alpha = -mgl \theta$$

[Here I is the momentum of inertia of body about axis AA']

or
$$\alpha = -\frac{mgl}{I} \theta \quad \dots (5.44)$$

Comparing equation-(5.44) with standard differential equation of angular SHM we get

$$\begin{aligned} d &= -\cos \theta \\ \omega &= \sqrt{\frac{mgl}{I}} \end{aligned}$$

Thus its time period of oscillation is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I}{mgl}} \quad \dots (5.45)$$

5.6.4 Torisional Pendulum

In a Torisional Pendulum an object is suspended from a wire with rigidity coefficient C . If such a wire is twisted by an angle θ ,

due to its elasticity it exerts a restoring torque $\tau = C\theta$ on the twisted object attached to it.

A general torisional pendulum is shown in figure-5.16. Here a disc D of radius R and mass M is attached to a stiff wire whose other end is suspended from ceiling as shown.

From the equilibrium position of this disc if it is twisted by an angle θ as shown, the wire applies a restoring torque on it, which is given as

$$\tau_R = -C\theta \text{ [-ve sign, for restoring nature]}$$

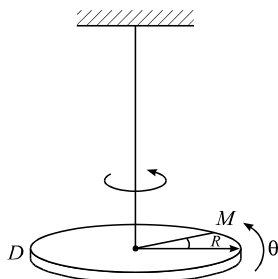


Figure 5.16

If during restoring motion the angular acceleration of disc is α , we can write $\tau = I\alpha$ where I is the moment of inertia of disc about its central axis, thus we have

$$I\alpha = -C\theta$$

$$\text{or} \quad \alpha = -\frac{C}{I} \theta \quad \dots (5.46)$$

Equation-(5.46) resembles with the basic differential equation of SHM in angular form thus we can state the angular frequency of this SHM is

$$\omega = \sqrt{\frac{C}{I}} \quad \dots (5.47)$$

Thus the period of SHM is

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{C}}$$

In the above cases of some pendulums we've discussed, how to find the angular frequency and time period of a body in SHM. Now we take some similar examples of physical situations in which an object is in SHM.

5.6.5 Equivalent Length of a Simple Pendulum

We know that the time period of oscillations of a simple pendulum is given by

$$2\pi \sqrt{\frac{l}{g}} \quad \dots (5.48)$$

Similarly we've also discussed that the time period of a compound pendulum is given as

$$2\pi \sqrt{\frac{I}{Mgl_c}} \quad \dots (5.49)$$

Where I is the moment of inertia of the rigid body about the axis of rotation and l_c is the distance of centre of gravity of body from the suspension point (Axis of rotation). If we consider a simple pendulum of length l_{eq} which has a bob of same mass M as that of rigid body of compound pendulum and has the time period same as that of the compound pendulum then this length of simple pendulum l_{eq} is called "Equivalent length of simple pendulum for the given compound pendulum".

For the time periods of the two pendulums to be equal, we have

$$2\pi \sqrt{\frac{l}{g}} = 2\pi \sqrt{\frac{I}{Mgl_c}} \quad \dots (5.50)$$

If k be the radius of gyration of the rigid body about an axis passing through its centre of mass then the moment of inertia of the rigid body about point of suspension is given as

$$I = Mk^2 + Ml_c^2 \quad \dots (5.51)$$

Now from equation-(5.50), we have

$$\sqrt{\frac{l_{eq}}{g}} = \sqrt{\frac{Mk^2 + Ml_c^2}{Mgl_c}}$$

$$\text{or} \quad l_{eq} = l_c + \frac{k^2}{l_c} \quad \dots (5.52)$$

Equation-(5.52) gives the equivalent length of simple pendulum for the given compound pendulum. One important point to be noted here is if in equation-(5.52) we replace l_c by k^2/l_c , we get

$$l_{eq} = \frac{k^2}{l_c} + \frac{k^2}{k^2/l_c}$$

$$\text{or} \quad = \frac{k^2}{l_c} + l_c \quad \dots (5.53)$$

Which is same as that of equation-(5.52). Thus we can say that if the same rigid body, which is suspended from a point, situated at a distance l_c from centre of gravity of body, we suspend it from a point at a distance k^2/l_c from centre of gravity of the body and oscillate like a compound pendulum, its equivalent length of simple pendulum remains same or the time period of oscillation of body remains same. Consider figure-5.17, a rigid body is suspended at a point O through a horizontal axis AA' .

Here C is the centre of gravity of the body. If it oscillates then the time period of small oscillations can be given as

$$T = 2\pi \sqrt{\frac{l_c + \frac{k^2}{l_c}}{g}} = 2\pi \sqrt{\frac{l_{eq}}{g}} \quad \dots (5.54)$$

As discussed in last section we can say that if the same body is suspended either from any point on circular arcs PQ or RS (as shown in figure-5.17(b)) of radius l_c with centre at C (circle M_1) or any point on the circle of radius k^2/l_c with centre at C (circle M_2), the time period of small oscillations of the body will remain same.

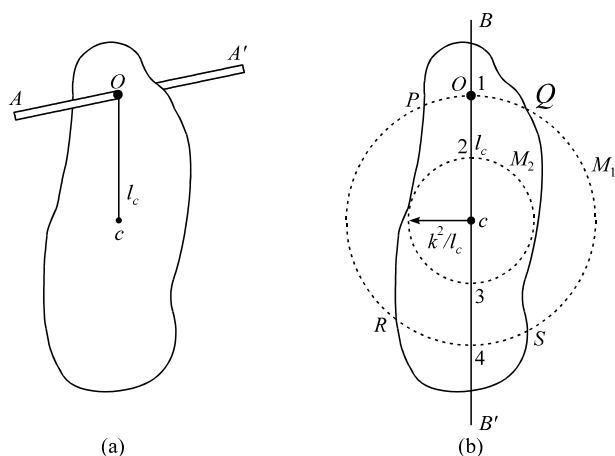


Figure 5.17

Here we can develop one property of the body when it is used as a compound pendulum. This is, when a straight line is drawn passing through the centre of gravity of body as in figure-5.17(b) a line BB' is drawn. There exist four points on this line (as here points 1, 2, 3 and 4) about which if body is suspended, the time period of small oscillation of body remains same. This we can also prove graphically as if we plot the time period of oscillation, the curve looks like as shown in figure-5.18

As shown in figure-5.18, if the body is suspended from C , from equation-(5.54), we can see that if $l_c = 0$, time period becomes ∞ , and at $l_c = k$ i.e. if the body is suspended from a point at a distance equal to radius of gyration of body from C , the time period of oscillation is minimum and at all other suspension points the time period is higher and if we draw a horizontal line in graph at time period T_1 which is more than minimum period, it cuts the graph at four points as shown, which verifies the statement we've discussed earlier.

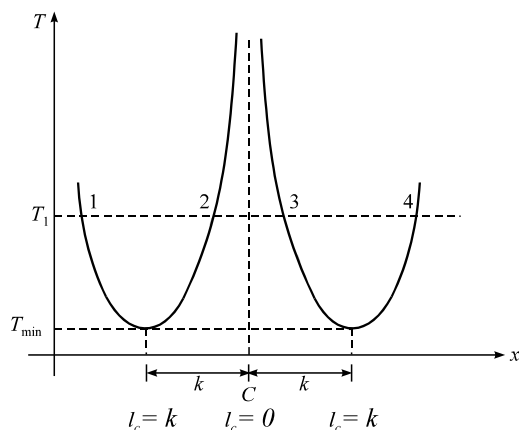


Figure 5.18

From equation-(5.54) we can also find the value of l_c at which this time period has a minimum value by equating $\frac{dT}{dl_c} = 0$, as

$$\frac{dT}{dl_c} = \frac{2\pi}{\sqrt{g}} \cdot \frac{1}{2\sqrt{l_c + \frac{k^2}{l_c}}} \left[1 - \frac{k^2}{l_c^2} \right] = 0$$

or

$$l_c = \pm k$$

Which also verifies the experimental result obtained by graph shown in figure-5.18.

Illustrative Example 5.6

Find the period of small oscillations in a horizontal plane performed by a ball of mass $m = 40$ g fixed at the middle of a horizontally stretched string $l = 1.0$ m in length. The tension of the string is assumed to be constant and equal to $T = 10$ N.

Solution

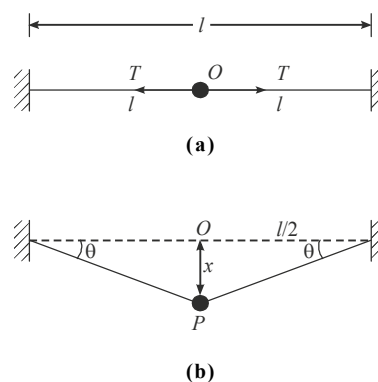


Figure 5.19

The situation is shown in figure-5.19. The ball is in equilibrium at point O as shown in figure-5.19(a). Now it is displaced from O by a distance x in horizontal plane to a position P as shown in figure-5.19(b). The components of tensions on ball toward O are $T \sin \theta$ and $T \sin \theta$, hence the restoring force on ball toward mean position is

$$F_R = -2T \sin \theta$$

[–ve sign for restoring tendency]

or

$$= -\frac{2Tx}{\sqrt{x^2 + l^2/4}}$$

As x is very small compared to l , we can neglect its square compared to $l^2/4$, thus

$$F_R = -\frac{4T}{l}x$$

If a is the acceleration of ball toward mean position, we have

$$a = -\left(\frac{4T}{ml}\right)x$$

Here acceleration is directly proportional to x thus motion of ball is simple harmonic with angular frequency ω , given as

$$\omega = \sqrt{\frac{4T}{ml}}$$

Thus time period of its SHM be given as

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{ml}{4T}} = 3.14 \sqrt{\frac{0.04 \times 1}{10}} = 0.2 \text{ s}$$

Illustrative Example 5.7

A simple pendulum consists of a small sphere of mass m suspended by a thread of length l . The sphere carries a positive charge q . The pendulum is placed in a uniform electric field of strength E directed vertically upwards. With what period will the pendulum oscillate, if the electrostatic force acting on the sphere is less than the gravitational force? (Assume that the oscillations are small).

Solution

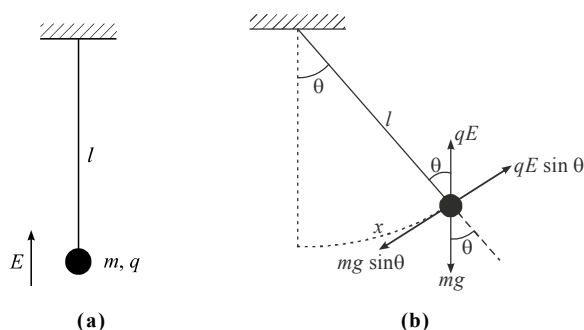


Figure 5.20

Figure-5.20(a) shows the simple pendulum with bob of mass m and charge q with electric field E in the region directed vertically upward.

When bob is in equilibrium, the tension in the string is

$$T = mg - qE \quad \dots (5.55)$$

If we displace this bob by a distance x from its mean position as shown in figure-5.20(b), the restoring force on it can now be written as

$$F_R = -(mg \sin \theta - qE \sin \theta)$$

[If θ is the angular displacement shown in figure]

$$\text{or} \quad F_R = -[mg - qE] \theta \quad [\text{For small } \theta, \sin \theta \simeq \theta]$$

$$\text{or} \quad F_R = (mg - qE) \frac{x}{l} \quad [\text{As } \theta = \frac{x}{l}]$$

If a is the acceleration of bob in the situation shown in figure 5.20(b), we have

$$a = -\frac{\left(g - \frac{qE}{m}\right)}{l} x \quad \dots (5.56)$$

Comparing equation-(5.56) with differential equation of SHM we get, the angular frequency of the SHM of this pendulum bob as

$$\omega = \sqrt{\frac{g - \frac{qE}{m}}{l}}$$

Thus oscillation period is

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{l}{g - \frac{qE}{m}}} \quad \dots (5.57)$$

The result in equation-(5.57) can also be directly obtained as we know the time period of a simple pendulum is given as $T = 2\pi \sqrt{\frac{l}{g}}$ and in a region where an electric field E exist in upward direction, the effective value of acceleration due to gravity for a positively charge particle with a charge q can be used as

$$g_{\text{eff}} = g - \frac{qE}{m}$$

Thus the time period of above pendulum can be directly written as

$$T = 2\pi \sqrt{\frac{l}{g_{\text{eff}}}} = 2\pi \sqrt{\frac{l}{g - \frac{qE}{m}}}$$

If in this problem electric field E is reversed i.e. in downward direction then net effective force on bob in downward direction is increased and effective gravity can be written as

$$g_{\text{eff}} = g + \frac{qE}{m}$$

Thus in this case, the time period of a pendulum becomes

$$T = 2\pi \sqrt{\frac{l}{g + \frac{qE}{m}}}$$

Illustrative Example 5.8

Calculate the period of small oscillations of a floating box as shown in figure-5.21 which was slightly pushed down in vertical direction. The mass of box is m , area of its base is A and the density of liquid is ρ . The resistance of the liquid is assumed to be negligible.

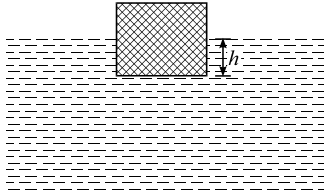


Figure 5.21

Solution

Initially when box is floating in liquid, if its h depth is submerged in liquid then buoyancy force on it is

$$\begin{aligned} F_B &= \text{weight of liquid displaced} \\ &= Ah\rho g \end{aligned}$$

As the box is in equilibrium, we have

$$Ah\rho g = mg \quad \dots (5.58)$$

Now if box is further pushed down by a distance x , net restoring force on it in upward (toward mean position) direction is

$$\begin{aligned} F_R &= -[A(h+x)\rho g - mg] \\ &= -Ax\rho g \quad [\text{As } mg = Ah\rho g] \end{aligned}$$

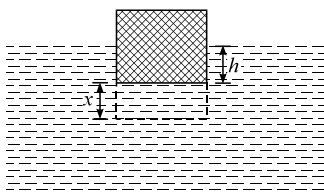


Figure 5.22

If a is the acceleration of box in upward direction we have

$$a = -\left(\frac{A\rho g}{m}\right)x \quad \dots (5.59)$$

Equation-(5.59) shows that the box executes SHM with angular frequency ω given as

$$\omega = \sqrt{\frac{A\rho g}{m}}$$

Thus time period of its oscillation can be given as

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{A\rho g}}$$

Illustrative Example 5.9

A mass M attached to a spring oscillates with a period of 2 s. If the mass is increased by 2 kg, the period increases by one second. Find the initial mass M .

Solution

We know that for a spring block system

$$T = 2\pi \sqrt{\left(\frac{M}{k}\right)}$$

Here k = spring constant.

$$\text{In first case,} \quad 2 = 2\pi \sqrt{\left(\frac{M}{k}\right)} \quad \dots (5.60)$$

$$\text{In second case,} \quad 3 = 2\pi \sqrt{\left(\frac{M+2}{k}\right)} \quad \dots (5.61)$$

Squaring of equation-(5.60) and (5.61) and then dividing (5.61) by (5.60), we have

$$\frac{9}{4} = \frac{M+2}{M} = 1 + \frac{2}{M}$$

Solving we get $M = 1.6 \text{ kg}$.

Illustrative Example 5.10

Two masses m_1 and m_2 are suspended together by a massless spring of spring constant k as shown in figure-5.23. When the masses are in equilibrium, m_1 is removed without disturbing the system. Find the angular frequency and amplitude of oscillation.

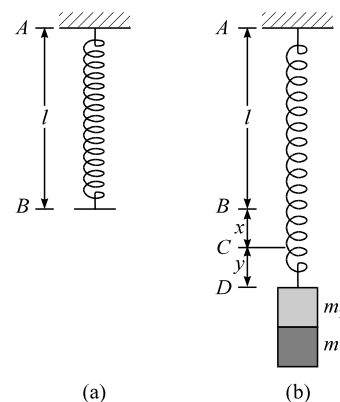


Figure 5.23

Solution

Let AB be the natural length l of the spring as shown in figure-5.23(a). When a mass m_2 is suspended, the spring is stretched to C i.e., $BC = x$. Now on further loading by a mass m_1 ,

let the spring is stretched to a point D i.e., $CD = y$, then we have

$$kx = m_2g \quad \dots (5.62)$$

$$ky = m_1g \quad \dots (5.63)$$

When mass m_1 is removed, the spring block system of mass m_2 starts oscillating about the point C . It's motion is simple harmonic motion. The mass m_2 is executing simple harmonic motion with the angular frequency given as

$$\omega = \sqrt{\left(\frac{m_2}{k}\right)}$$

Thus its time period is

$$T = 2\pi \sqrt{\left(\frac{m_2}{k}\right)}$$

Amplitude of oscillation is given as

$$A = CD = y = \frac{m_1g}{k}$$

Illustrative Example 5.11

Find the frequency of small oscillations of a thin uniform vertical rod of mass m and length l hinged at the point O (figure-5.24). The combined stiffness of each of the spring is equal to k . The mass of the spring is negligible.

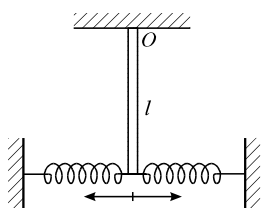


Figure 5.24

Solution

Figure-5.25 shows the equilibrium position of rod. If rod is displaced by a small angle θ as shown in figure-5.25, both the springs are deformed by a distance $x = l\theta$. One is stretched and other is compressed so that both will exert a torque on rod in same direction as restoring torque. Thus net restoring torque on rod is written as

$$\tau_R = -[2kx \times l + mg \times \frac{l}{2} \sin \theta] \\ [-ve \text{ sign for restoring nature}]$$

$$\tau_R = -[2kx \times l + mg \times \frac{l}{2} \theta] \\ [\text{As for small } \theta, \sin \theta \approx \theta]$$

$$\tau_R = -(2kl^2 + \frac{mgl}{2}) \theta \quad [x = l\theta]$$

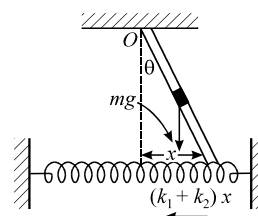


Figure 5.25

If rod has an angular acceleration α , we have, restoring torque

$$\tau_R = \left(\frac{ml}{3}\right)^2 \alpha$$

Where $\frac{ml^2}{3}$ is the moment of inertia of rod about an axis passing through one of its end as in this case. Thus we have

$$\frac{ml^2}{2} \alpha = -\left(2kl^2 + \frac{mgl}{2}\right) \theta$$

or

$$\alpha = -\left(\frac{2kl^2 + \frac{mgl}{2}}{\frac{ml^2}{3}}\right) \theta$$

$$\alpha = -\left(\frac{12kl + 3mg}{2ml}\right) \theta \quad \dots (5.64)$$

Comparing equation-(5.64) with the basic differential equation of SHM, we get

$$\omega = \sqrt{\frac{12kl + 3mg}{2ml}}$$

Thus the oscillation frequency of rod for its small oscillations can be given as

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{12kl + 3mg}{2ml}}$$

Illustrative Example 5.12

A vertical U -tube of uniform cross-section contains water upto a height of 30 cm. Show that if the water on one side is depressed and then released, its motion up and down the two sides of the tube is simple harmonic. Calculate its period.

Solution

Figure-5.26 shows a U -tube of uniform cross-sectional area A . Let the liquid be depressed through the distance y in a limb,

the difference of levels between two limbs will be $2y$ as shown in figure-2.26.

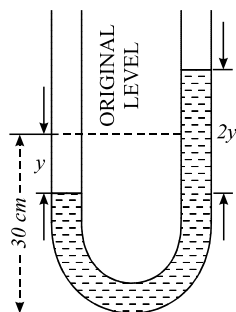


Figure 5.26

The liquid now oscillates about the initial positions.

Excess pressure on whole liquid = (excess height of the liquid column) (density) (g)

$$= 2y \times 1 \times g \quad [\text{As density of water} = 1]$$

Restoring Force on the liquid = Pressure $\times$ area of cross section

$$= 2y g A$$

Due to this force the liquid accelerates and if its acceleration is a , we have

$$ma = -2y g A$$

$$\text{or} \quad (2 \times 30 \times A) a = -2y g A$$

$$\text{or} \quad a = -\frac{g}{30} y$$

Hence acceleration is directly proportional to displacement, so the motion is simple harmonic motion. Thus the time period T is given by

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\left(\frac{30}{g}\right)} = 2\pi \sqrt{\left(\frac{30}{980}\right)} \quad [\text{As } \omega = \sqrt{\frac{g}{30}}]$$

$$= 1.098 \text{ s}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Simple Harmonic Motion

Module Number - 13, 14, 16, 17, 18, 19, 22, 23, 24, 25

Practice Exercise 5.2

(i) A vertical spring block system executes SHM at a frequency of 10 osc/s. At the upper extreme position of the block the spring is unstretched. Find the maximum speed of the particle. Take $g = 10 \text{ m/s}^2$.

$$\left[\frac{1}{2\pi} \text{ m/s} \right]$$

(ii) In the figure-5.27, the spring has a force constant 5000 N m^{-1} . The pulley is light and smooth. The spring and the string are light. The suspended block A is of mass 1 kg. If the block is slightly displaced vertically down from its equilibrium position and released, find the period of its vertical oscillations.

$$[0.1776 \text{ s}]$$

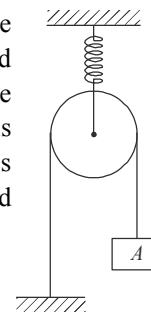


Figure 5.27

(iii) The bob of a simple pendulum is displaced by a small angle θ_0 from the vertical and released. If during its oscillation it is given that the maximum tension in the string is two times the minimum tension, find the value of θ_0 .

$$\left[\cos^{-1} \left(\frac{3}{4} \right) \right]$$

(iv) Determine the period of small longitudinal oscillations of a body with mass m in the system shown in figure-5.28. The stiffness values of the springs are k_1 and k_2 . The friction and the masses of the springs are negligible.

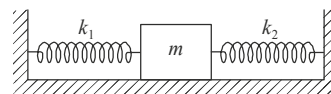


Figure 5.28

$$\left[2\pi \sqrt{\frac{m}{k_1 + k_2}} \right]$$

(v) Find the period of small vertical oscillations of a body with mass m in the system illustrated in figure-5.29. The stiffness values of the springs are k_1 and k_2 , their masses are negligible.

$$\left[2\pi \sqrt{\frac{m(k_1 + k_2)}{k_1 k_2}} \right]$$

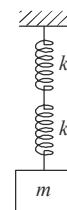


Figure 5.29

(vi) A T-bar of uniform cross section and mass M is supported in a vertical plane by a hinge O and a spring of force constant k at A . Develop a formula for the period for small amplitude rotational oscillations in the plane of the figure-5.30.

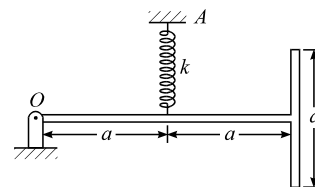


Figure 5.30

$$\left[3\pi \sqrt{\frac{M}{k}} \right]$$

(vii) A cylinder of mass M , radius r and height h is suspended by a spring where the upper end which is fixed is partly submerged in water. In equilibrium the cylinder sinks to half its height. At a certain moment the cylinder was submerged to $2/3$ of the height and then with no initial velocity started to move vertically. If the stiffness constant is k and density of water ρ , find the period.

$$[T = 2\pi \sqrt{\frac{M}{k + \pi r^2 \rho g}}]$$

(viii) A point mass m is suspended at the end of a massless wire of length l and cross-section A . If Y is the Young's modulus of elasticity for the wire, obtain the frequency of oscillation for the simple harmonic motion along the vertical line.

$$[\frac{1}{2\pi} \sqrt{\frac{YA}{ml}}]$$

(ix) An ideal gas whose adiabatic exponents is γ , is enclosed in a vertical cylindrical container and supports a freely moving piston of mass M . The piston and the cylinder have equal cross-sectional area A . Atmospheric pressure is P_0 and when the piston is in equilibrium, the volume of the gas is V_0 . The piston is now displaced slightly from the equilibrium position. Assuming that the system is completely isolated from its surrounding, show that the piston executes simple harmonic motion and find the frequency of oscillation.

$$[\frac{1}{2\pi} \sqrt{\frac{\gamma (P_0 + Mg/A) A^2}{M V_0}}]$$

(x) A light wooden rod fixed at one end is kept horizontal. A load of $m = 0.4$ kg tied to the free end of the rod causes that end to be depressed by $\delta = 2.8$ cm. If this load is set into up and down vibration will it oscillate? Find its frequency of oscillation. Take $g = 10$ m/s².

[3 Hz]

(xi) Determine the period of oscillations of mercury of mass m poured into a bent tube (Figure-5.31) whose right arm forms an angle θ with the vertical. The cross-sectional area of the tube is S . The viscosity of mercury is to be neglected.

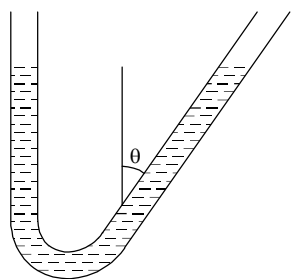


Figure 5.31

$$[T = 2\pi \sqrt{m / S \rho g (1 + \cos \theta)}]$$

5.7 Phase Analysis of a Particle in SHM

We've already discussed about phase angle of a particle in SHM. It is actually the angular displacement of that particle who is in circular motion and whose projection is in SHM. At a general time t , the instantaneous phase of a particle in SHM can be written as $(\omega t + \alpha)$ if α is its initial phase (already discussed).

5.7.1 Phase Difference in Two SHM

Case I: When two SHMs are of same angular frequency

Figure-5.32 shows two particles P' and Q' in SHM with same angular frequency ω . P and Q are the corresponding particles in circular motion for SHM of P' and Q' .

Let P and Q both starts their circular motion at the same time at $t = 0$ then at the same instant P' and Q' starts their SHM in upward direction as shown. As frequency of both are equal, both will reach their extreme position (topmost point) at the same time and will again reach their mean position simultaneously at time $t = T/2$. [T = Time period of SHM = $2\pi/\omega$] and move in downward direction together or we can state that the oscillations of P' and Q' are exactly parallel and at every instant the phase of both P' and Q' are equal, thus phase difference in these two SHMs is zero. These SHMs are called same phase SHMs.

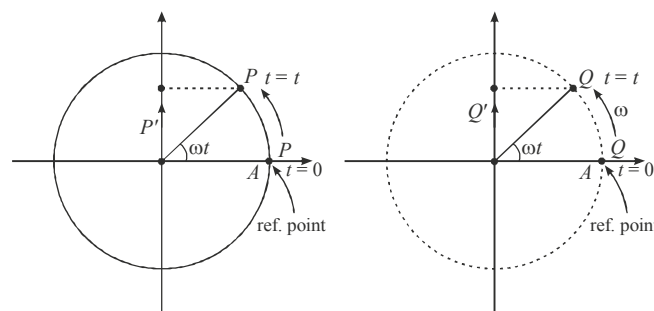


Figure 5.32

Now consider figure-5.32, where we assume if P' starts its SHM at $t = 0$ but Q' will start at time t_1 . In this duration from $t = 0$ to $t = t_1$, P' will move ahead in phase by ωt_1 radians while Q' was at rest. Now Q' starts at time t_1 and move with same angular frequency ω . It can never catch P' as both are oscillating at same angular frequency. Thus here Q' will always lag in phase by ωt_1 then P' or we say P' is leading in phase by ωt_1 then Q' and as ω of both are constant their phase difference will also remains constant so keep it in mind that in two SHMs of same angular frequency, if they have some phase difference, it always remains constant.

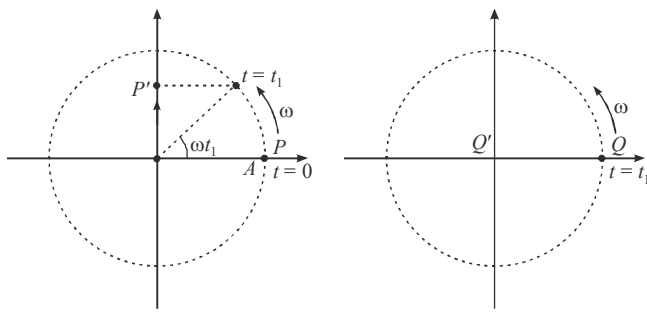


Figure 5.33

Now we consider a special case when time lag between the starting of two SHMs is $T/2$ i.e. half of oscillation period. Consider figure-5.34, here if we assume, particle P' starts at $t = 0$ and Q' at $t = T/2$ when P' completes its half oscillation. Here we can see that the phase difference in the two SHM is π by which Q' is lagging. Here when Q' starts its oscillation in upward direction P' moves in downward direction. As angular velocity of P and Q are same, both complete their quarter revolution in same time. Thus when P' reaches its bottom extreme position, Q' will reach its upper extreme position and then after Q' starts moving downward, P' starts moving upward and both of these will reach their mean position simultaneously but in opposite directions, P' has completed its one oscillation where as Q' is at half of its oscillation due to a phase lag of π .

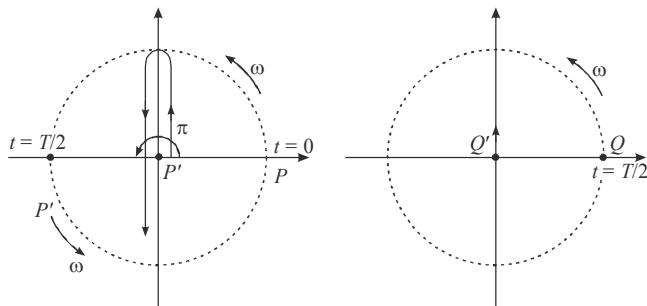


Figure 5.34

Thus if we observe both oscillations simultaneously we can see that oscillations of the two particles P' and Q' are exactly antiparallel i.e. when P' goes up, Q' comes down and at all instants of time their displacements from mean position are equal but in opposite direction if their amplitudes are equal. Such SHMs are called opposite phase SHMs.

Case II: When two SHMs are of different angular frequency

We've discussed that when two SHMs are of same angular frequency, their phase difference does not change with time. Consider two particles in SHM as shown in figure-5.35. Their corresponding particles for circular motion are A and B respectively as shown. If both A' and B' starts their SHM from

mean position at $t = 0$ with angular frequencies ω_1 and ω_2 , then we say at $t = 0$ their phase difference is zero but after time t , their respective phase are $\omega_1 t$ and $\omega_2 t$. Thus after time t , the phase difference in the two SHMs is

$$\phi = (\omega_1 - \omega_2) t \quad \dots (5.65)$$

Thus equation-(5.65) shows that if angular frequencies of the two SHMs are different their phase difference continuously changes with time.

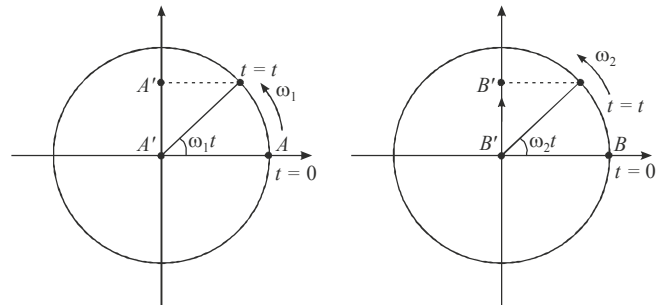


Figure 5.35

5.7.2 Same Phase and Opposite Phase SHMs

Same Phase SHMs

As discussed above two particles execute SHM in such a way that their oscillations are exactly parallel to each other or their phase difference during oscillation is zero, they are said to be in same phase we've seen that this happens when both SHMs are started at same time with same angular frequency. This can happen also when the time lag between starting of the two SHM is T or an integral multiple of time period of SHM. Because if one starts at $t = 0$ and other starts at $t = T$, in this duration first particle will complete its first oscillation and is going to start its second oscillations and the second particle will start in synchronization with the first. Hence the two oscillation will still be parallel or in same phase. Their phase difference in the two SHMs will be 2π . Not only this even if the time lag in starting of the two SHMs is $2T, 3T \dots nT$ or the phase difference in the two SHMs is $4\pi, 6\pi, 8\pi \dots 2n\pi$, then also these SHMs can be treated in same phase.

Thus phase difference in two SHMs of same phase is

$$\phi = 2\pi, 4\pi, 6\pi \dots 2n\pi \quad \dots (5.66)$$

Opposite Phase SHMs

As discussed in previous article, two SHMs are said to be in opposite phase when their oscillations are antiparallel this happens when two SHMs of same angular frequency start with a time lag of $T/2$ and phase difference among the two SHMs is π . By analyzing the situation it can also be stated that the

same thing also happens when the time lag in starting of the two SHMs is $\frac{3T}{2}, \frac{5T}{2}, \dots, (2n+1)\frac{T}{2}$ or the phase difference between the two is $3\pi, 5\pi, \dots, (2n+1)\pi$.

Thus phase difference in two SHMs of opposite phase is

$$\phi = \pi, 3\pi, 5\pi, \dots, (2n+1)\pi \quad \dots (5.67)$$

To understand the concept of phase and phase difference in SHM, we take few examples.

Illustrative Example 5.13

Two particles execute SHM with same amplitude A and same angular frequency ω on same straight line with same mean position. Given that during oscillation they cross each other in opposite direction when at a distance $A/2$ from mean position. Find phase difference in the two SHMs.

Solution

Figure-5.36 shows that two respective particles P' and Q' in SHM along with their corresponding particles in circular motion. Let P' moves in upward direction when crossing Q' at $A/2$ as shown in figure-5.36(a), at this instant phase of P' is

$$\phi_1 = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6} \quad \dots (5.68)$$

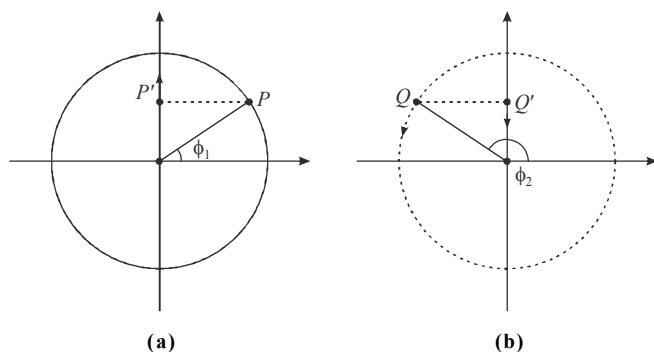


Figure 5.36

Similarly as shown in figure-5.36(b) we can take particle Q' is moving in downward direction (opposite P') at $A/2$, this implies its circular motion particle is in second quadrant thus its phase angle is

$$\phi_2 = \pi - \sin^{-1}\left(\frac{1}{2}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6} \quad \dots (5.69)$$

As both are oscillating at same angular frequency their phase diff. remains constant which can be given from equation-(5.68) and (5.69), as

$$\Delta\phi = \phi_2 - \phi_1 = \frac{5\pi}{6} - \frac{\pi}{6} = \frac{2\pi}{3}$$

Illustrative Example 5.14

A particle executes SHM with amplitude A and angular frequency ω . At an instant when particle is at a distance $A/5$ from mean position and moving away from it. Find the time after which it will come back to this position again and also find the time after which it will pass through mean position.

Solution

Figure-5.37 shows the circular motion representation for the particle P' given in problem. The initial situation of particle is shown in figure. As P moves, its projection P' will go up and then come back to its initial position when P reaches to the corresponding position in second quadrant as shown. In this process P traversed an angular displacement θ with angular velocity ω , thus time taken in the process is

$$t = \frac{\theta}{\omega} = \frac{1}{\omega} [2 \cos^{-1}(\frac{1}{5})] = \frac{2 \cos^{-1}(\frac{1}{5})}{\omega}$$

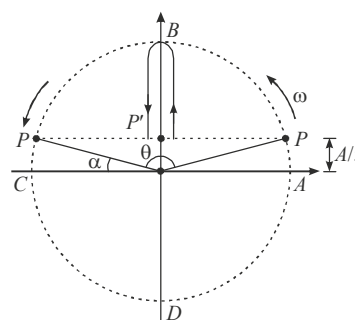


Figure 5.37

Now when P reaches point C , P' will reach its mean position. Thus time taken by P from its initial position to point C is

$$t = \frac{(\pi - \alpha)}{\omega} = \frac{\pi - \sin^{-1}(\frac{1}{5})}{\omega}$$

The same time P' will take from $A/5$ position to mean position through its extreme position.

Illustrative Example 5.15

Two particles executing SHM with same angular frequency and amplitudes A and $2A$ on same straight line with same mean position cross each other in opposite direction at a distance $A/3$ from mean position. Find the phase difference in the two SHMs.

Solution

Figure-5.38 shows the two corresponding particles of circular motion for the two mentioned particles in SHM. Let particle P

is going up and particle Q is going down. From the figure shown, the respective phase differences of particles P' and Q' are

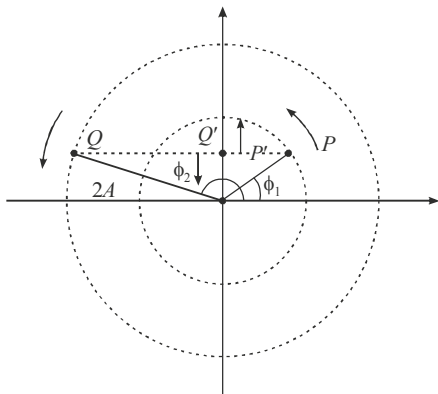


Figure 5.38

$$\phi_1 = \sin^{-1} \left(\frac{1}{3} \right) \quad [\text{Phase angle of } P']$$

and
$$\phi_2 = \pi - \sin^{-1} \left(\frac{1}{6} \right) \quad [\text{Phase angle of } Q']$$

Thus phase difference in the two SHMs is

$$\Delta\phi = \phi_2 - \phi_1 = \pi - \sin^{-1} \left(\frac{1}{6} \right) - \sin^{-1} \left(\frac{1}{3} \right)$$

Illustrative Example 5.16

A particle starts its SHM from mean position at $t = 0$. If its time period is T and amplitude A . Find the distance travelled by the particle in the time from $t = 0$ to $t = \frac{5T}{4}$.

Solution

We know in one complete oscillation i.e. in period T , a particle covers a distance $4A$ and in first one quarter of its period it goes from its mean position to its extreme position as it starts from mean position thus the distance travelled by the particle in time $\frac{5T}{4}$ is $5A$.

Illustrative Example 5.17

Figure-5.39 shows two identical simple pendulums of length l . One is tilted at an angle α and imparted an initial velocity v_1 toward mean position and at the same time other one is projected away from mean position at a velocity v_2 at an initial angular displacement β . Find the phase difference in oscillations of these two pendulums.

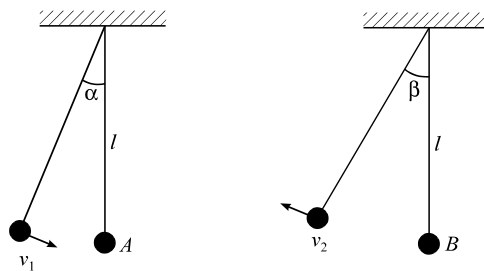


Figure 5.39

Solution

It is given that first pendulum bob is given a velocity v_1 at a displacement $l\alpha$ from mean position, using the formula for velocity we can find the amplitude of its oscillations as

$$V_1 = \omega \sqrt{A_1^2 - (l\alpha)^2}$$

[If A_1 is the amplitude of SHM of this bob.]

As for simple pendulum $\omega = \sqrt{\frac{g}{l}}$ we have

$$V_1^2 = \frac{g}{l} [A_1^2 - (l\alpha)^2]$$

$\Rightarrow$

$$A_1 = \sqrt{l^2\alpha^2 + \frac{v_1^2 l}{g}} \quad \dots (5.70)$$

Similarly if A_2 is the amplitude of SHM of second pendulum bob we have

$$V_2 = \omega \sqrt{A_2^2 - (l\beta)^2}$$

or

$$A_2 = \sqrt{l^2\beta^2 + \frac{v_2^2 l}{g}} \quad \dots (5.71)$$

Now we represent the two SHMs by circular motion representation as shown in figure-5.40.

In first pendulum at $t = 0$ the bob is thrown from a displacement $l\alpha$ from mean position with a velocity v_1 toward mean position. As it is moving toward mean position, in figure-5.40(a), we consider the corresponding circular motion particle A_0 of the bob A is second quadrant, as the reference direction of ω , we consider anticlockwise. As shown in figure initial phase of bob A is given as

$$\phi_1 = \pi - \sin^{-1} \left(\frac{l\alpha}{A_1} \right) \quad \dots (5.72)$$

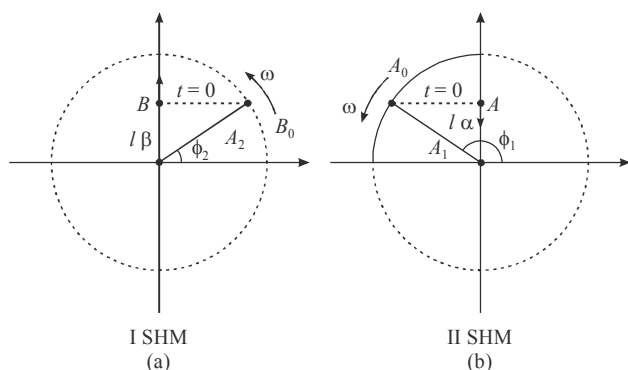


Figure 5.40

Similarly for second pendulum bob B , its corresponding circular motion particle B_0 at $t=0$ is considered as shown in figure-5.40 (b) its initial base is given as

$$\phi_2 = \sin^{-1} \left(\frac{l\beta}{A_2} \right) \quad \dots (5.73)$$

As both the pendulums are identical, their angular frequency for SHM must be same, so their phase difference will not change with time, hence their phase difference can be given as

$$\Delta\phi = \phi_2 - \phi_1 = \sin^{-1} \left(\frac{l\beta}{A_2} \right) + \sin^{-1} \left(\frac{l\alpha}{A_1} \right) - \pi$$

Illustrative Example 5.18

In previous question if second pendulum bob is thrown at velocity v_2 at an angle β from mean position but on other side of mean position. Find the phase difference in the two SHMs now as shown in figure-5.41.

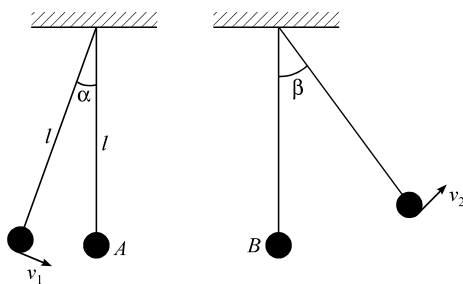


Figure 5.41

Solution

In this case still amplitudes of the two SHMs will remain same and are given by equation-(5.70) & (5.71) but when we represent the two SHMs on their corresponding circular motions, the position of the particle in circular motion in second pendulum is now different as shown in figure-5.42.

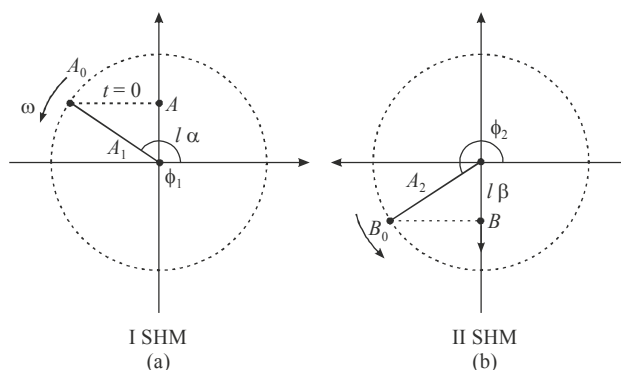


Figure 5.42

As shown in figure-5.42 (a) & (b) the initial phases of the two pendulum bobs are

$$\phi_1 = \pi - \sin^{-1}$$

and

$$\phi_2 = \pi + \sin^{-1} \left(\frac{l\beta}{A_2} \right)$$

As ω for both SHM are same, their phase difference remains constant so it is given as

$$\Delta\phi = \phi_2 - \phi_1 = \sin^{-1} \left(\frac{l\beta}{A_2} \right) + \sin^{-1} \left(\frac{l\alpha}{A_1} \right)$$

Illustrative Example 5.19

A spring block pendulum is shown in figure-5.43. The system is hanging in equilibrium. A bullet of mass $m/2$ moving at a speed u hits the block from downward direction and gets embedded in it. Find the amplitude of oscillation of the block now. Also find the time taken by the block to reach its upper extreme position after hit by bullet.

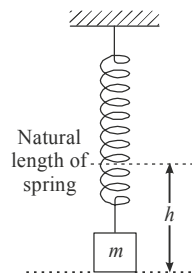


Figure 5.43

Solution

If block is in equilibrium then spring must be at some stretch if it is h , we have

$$mg = kh.$$

If a bullet of mass $m/2$ gets embedded in the block, due to this inelastic impact its new mass becomes $3m/2$ and now the new

mean position of the block will be say at a dept. h_1 from old mean position then, we must have

$$\frac{3}{2} mg = k(h + h_1)$$

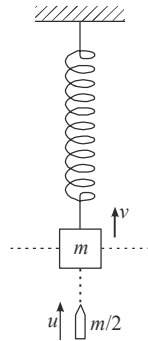


Figure 5.44

$$\frac{3m}{2} g = k(h + h_1)$$

$$\Rightarrow h_1 = \frac{mg}{2k} \quad [\text{As } mg = kh]$$

Just after impact due to inelastic collision if the velocity of block becomes v , we have according to momentum conservation

$$\frac{m}{2} u = \frac{3m}{2} v$$

$$\text{or } v = \frac{u}{3}$$

Now the block executes SHM and at $t = 0$ block is at a distance $h_1 = \frac{mg}{2k}$ above its mean position and having a velocity $\frac{u}{3}$. If amplitude of oscillation is A , we have

$$\frac{u}{3} = \omega \sqrt{A^2 - \left(\frac{mg}{2k}\right)^2}$$

$$\text{or } \frac{u^2}{9} = \frac{2k}{3m} \left[A^2 - \left(\frac{mg}{2k}\right)^2 \right]$$

$$[\text{As for this spring block system } \omega = \sqrt{\frac{k}{(3m/2)}}]$$

$$\Rightarrow A = \sqrt{\frac{mu^2}{6k} + \left(\frac{mg}{2k}\right)^2} \quad \dots (5.74)$$

Now time taken by particle to reach the topmost point can be obtained by circular motion representation as shown in figure-5.45. This figure shows the position of block P and its corresponding circular motion particle P_0 at the $t = 0$. Block P will reach its upper extreme position when particle P_0 will traverse the angle θ and reach the topmost point. As P_0 moves at constant angular velocity ω , it will take a time given as

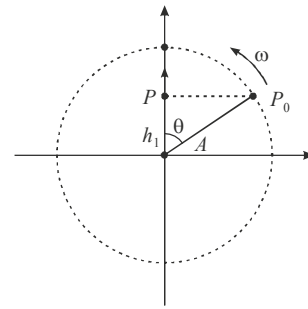


Figure 5.45

$$t = \frac{\theta}{\omega} = \frac{\cos^{-1}(h_1 / A)}{\sqrt{\frac{k}{3m/2}}} = \sqrt{\frac{3m}{2k}} \cos^{-1} \left(\frac{mg}{2kA} \right)$$

Illustrative Example 5.20

Figure-5.46(a) shows a spring block system, hanging in equilibrium. The block of system is pulled down by a distance x and imparted a velocity v in downward direction as shown in figure-5.46(b). Find the time it will take to reach its mean position. Also find the maximum distance to which it will move before returning back towards mean position.

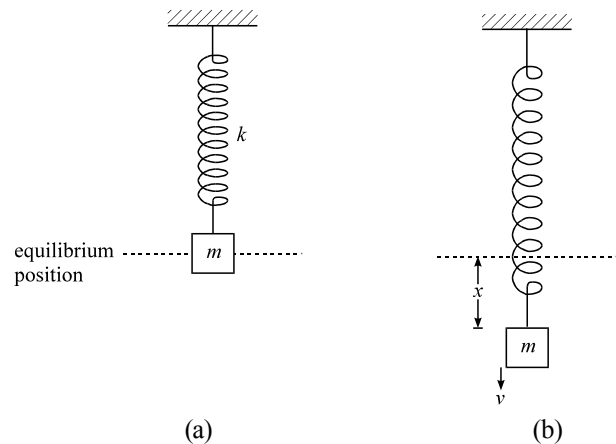


Figure 5.46

Solution

As shown in figure-5.46(b), when the block is pulled down by a distance x and thrown downward, it will start executing SHM. It will go further to a distance A (amplitude) from mean position before returning back which can be found by using the velocity of block v at a displacement x from its mean position as

$$v = \omega \sqrt{A^2 - x^2}$$

or

$$v^2 = \frac{k}{m} (A^2 - x^2)$$

$$[\text{As for spring block system } \omega = \sqrt{\frac{k}{m}}]$$

or

$$A = \sqrt{\frac{mv^2}{k} + x^2}$$

Now time of motion of bob can be obtained by circular motion representation of the respective SHM. Corresponding circular motion representation for this SHM is shown in figure-5.47 at $t = 0$. At $t = 0$, block P is at a distance x from its mean position in downward direction and it is moving downward so we consider its corresponding circular motion particle in III quadrant as shown in reference angular velocity we consider anticlockwise. Now block P will reach its mean position when particle P_0 reaches position A by traversing an angle θ . Shown in figure-5.47. Thus it will take a time given as

$$t = \frac{\theta}{\omega} = \frac{\pi - \sin^{-1}(x/A)}{\sqrt{\frac{k}{m}}}$$

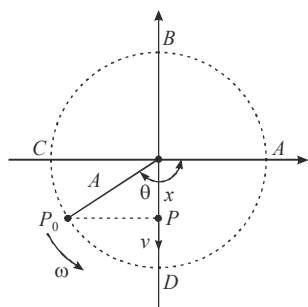


Figure 5.47

or

$$t = \sqrt{\frac{m}{k}} \left(\pi - \sin^{-1} \frac{x}{\sqrt{\frac{mv^2}{k} + x^2}} \right)$$

The maximum distance to which block will move from its initial position is $A - x$ as it goes up to its lower extreme at a distance equal to its amplitude A from to mean position.

Illustrative Example 5.21

Figure-5.48 shows a block of mass m resting on a smooth horizontal ground attached to one end of a spring of force constant k in natural length. If another block of same mass and moving with a velocity u toward right is placed on the block which stick to it due to friction. Find the time it will take to reach its extreme position. Also find the amplitude of oscillations of the combined mass $2m$.

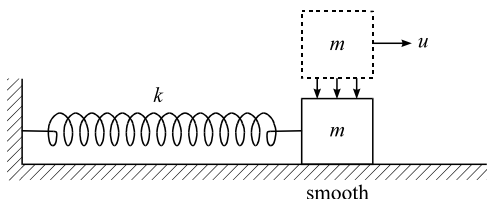


Figure 5.48

Solution

When second mass sticks to the lower mass, due to such an inelastic collision, the velocity of combined block is reduced by half that is $u/2$ to conserve momentum. Now at mean position the velocity of block can be written as

$$\frac{u}{2} = A\omega$$

[If A is the amplitude of oscillation]

$$\text{or} \quad \frac{u}{2} = A \sqrt{\frac{k}{2m}}$$

[As here for combined block new angular frequency of SHM is $\omega = \sqrt{k/2m}$]

$$\text{or} \quad A = \frac{u}{2} \sqrt{\frac{2m}{k}} = u \sqrt{\frac{m}{2k}}$$

As oscillation starts from mean position, in reaching its extreme position, particle has to cover a phase of $\pi/2$ radians, thus time taken by particle to reach its extreme position is

$$t = \frac{\pi/2}{\omega} = \frac{\pi}{2} \sqrt{\frac{k}{2m}}$$

Illustrative Example 5.22

In previous example if block is pulled toward right by a distance x_0 and released, when the block passes through a point at a displacement $x_0/2$ from mean position, another block of same mass is gently placed on it which sticks to it due to friction. Find the new amplitude of oscillation and find the time now it will take in reaching its mean position and extreme position on left side.

Solution

When block was released at x_0 from mean position, this will be the amplitude of oscillation and when it is passing through the position of half amplitude $\frac{x_0}{2}$, its velocity can be given as

$$v = \omega \sqrt{A^2 - \left(\frac{x_0}{2}\right)^2}$$

$$\begin{aligned} \text{or} \quad v &= \sqrt{\frac{k}{m}} \sqrt{x_0^2 - \frac{x_0^2}{4}} \\ &= \frac{\sqrt{3}}{2} x_0 \sqrt{\frac{k}{m}} \end{aligned}$$

When another block of same mass is added to it, due to momentum conservation its velocity becomes half and ω of oscillation will, also change from $\sqrt{k/m}$ to $\sqrt{k/2m}$. Again using

the formula for velocity of SHM at a distance $x_0/2$ from mean position, we get

$$v' = \frac{v}{2} = \omega \sqrt{A'^2 - \left(\frac{x_0}{2}\right)^2}$$

$$\text{or } \frac{\sqrt{3}}{4} x_0 \sqrt{\frac{k}{m}} = \sqrt{\frac{k}{2m}} \sqrt{A'^2 - \frac{x_0^2}{4}}$$

[If A' is the new amplitude of oscillation]

$$\Rightarrow A'^2 = \frac{3}{8} x_0^2 + \frac{x_0^2}{4} = \frac{5}{8} x_0^2$$

$$\text{or } A' = \frac{\sqrt{5}}{2\sqrt{2}} x_0$$

To find time of motion in SHM we use circular motion representation of the respective SHM. The figure-5.49 shows the corresponding circular motion. If at $t = 0$, the second mass is added to the oscillating block, it was at a position $x_0/2$ from mean position and moving towards it, and after adding the mass the new amplitude of oscillation changes to A' and ω changes from $\sqrt{k/m}$ to $\sqrt{k/2m}$. Figure-5.49 shows the corresponding position of particle in circular motion at $t = 0$ in II quadrant. When this particle P_0 will reach the position C after traversing the angle θ , particle P in SHM will reach its mean position and similarly when P_0 will reach position D , P will reach the extreme position on other side. Thus the time taken by P to reach mean position from a position of $x_0/2$ from mean position is given as

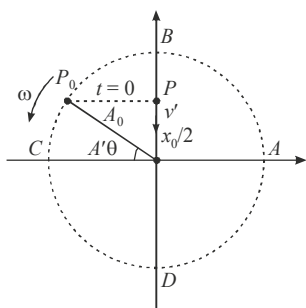


Figure 5.49

$$t_1 = \frac{\theta}{\omega'} = \frac{\sin^{-1}\left(\frac{x_0}{2A'}\right)}{\sqrt{\frac{k}{2m}}}$$

$$\text{or } t_1 = \frac{\sin^{-1}\left(\sqrt{\frac{2}{5}}\right)}{\sqrt{\frac{k}{2m}}} = \sqrt{\frac{2m}{k}} \sin^{-1}\left(\sqrt{\frac{2}{5}}\right)$$

Similarly time taken by P to reach the left end extreme position is

$$t_2 = \frac{\pi/2 + \theta}{\omega'} = \sqrt{\frac{2m}{k}} \left[\frac{\pi}{2} + \sin^{-1}\left(\sqrt{\frac{2}{5}}\right) \right]$$

Illustrative Example 5.23

Figure-5.50 shows a block P of mass m resting on a smooth horizontal surface, attached to a spring of force constant k which is rigidly fixed on the wall on left side, shown in figure-5.34. At a distance l to the right of block there is a rigid wall. If block is pushed toward left so that spring is compressed by a distance $5l/3$ and released, it will start its oscillations. If collision of block with the wall is considered to be perfectly elastic. Find the time period of oscillations of the block.

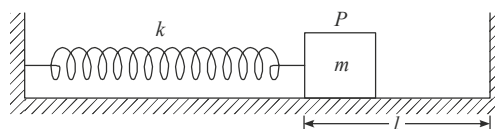


Figure 5.50

Solution

As shown in figure-5.50, as the block is released from rest at a distance $5l/3$ from its mean position, this will be the amplitude of oscillation. But on other side of mean position block can move only upto a distance l from mean position and then it will return from this point with equal velocity due to elastic collision. Consider figure-5.51. If no right wall is present during oscillation block P will execute complete SHM on right side of mean position also up to its amplitude $5l/3$. Thus we can observe the block P at a point X at a distance l from mean position (where in our case wall is present), if block passes this position at speed v (which is the speed with which block hits the wall in our case), after reaching its extreme position Y , it will return and during return path it will cross the position X with the same speed v (as displacement from mean position is same).

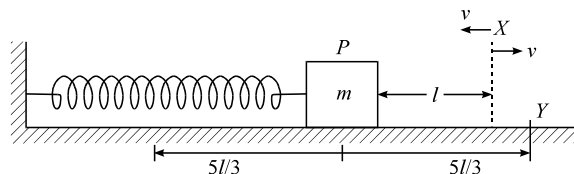


Figure 5.51

Thus we can state that in our case of given problem, block P is executing SHM but it skips a part XYX on right half of its motion due to elastic collision of the block with the wall. If T_0 is the time period of this oscillation we look at figure-5.52, which shows the corresponding circular motion representation.

Here during oscillation of point P , particle P_0 covers its circular motion along $ECDAF$, and from F it instantly jumps to E (due to elastic collision of P with wall at X) and again carry out $ECDAF$ and so on, thus we can find the time of this total motion as

$$t = \frac{\pi + 2\theta}{\omega} = \frac{\pi + 2\sin^{-1}(\frac{3}{5})}{\sqrt{\frac{k}{m}}}$$

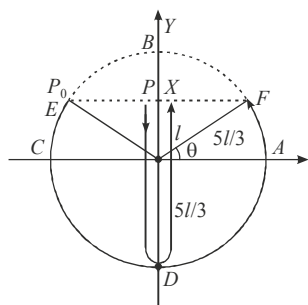


Figure 5.52

$$t = \sqrt{\frac{m}{k}} [\pi + 2\sin^{-1}(\frac{3}{5})] \quad \dots (5.75)$$

We can also find the time period of this motion by subtracting the time of FYE from the total time period as

$$t = \frac{2\pi - 2\cos^{-1}(\frac{3}{5})}{\omega} = \sqrt{\frac{m}{k}} [2\pi - 2\cos^{-1}(\frac{3}{5})] \quad \dots (5.76)$$

These equation-(5.56) & (5.57) will result same numerical value.

Illustrative Example 5.24

Figure-5.53 shows a spring block system hanging in equilibrium. If a velocity v_0 is imparted to the block in downward direction. Find the amplitude of SHM of the block and the time after which it will reach a point at half of the amplitude of block.

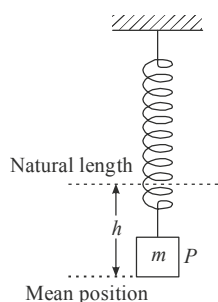


Figure 5.53

Solution

Initially in equilibrium if block is at a depth h below the natural

length of spring then we have

$$mg = kh$$

If at mean position block is imparted a velocity v_0 , this would be the maximum velocity of block during its oscillation. If its amplitude of oscillation is A , then it is given as

$$v_0 = A\omega \quad [\text{Where } \omega = \sqrt{\frac{k}{m}}]$$

or

$$A = \frac{v_0}{\omega} = v_0 \sqrt{\frac{m}{k}}$$

Now to find the time taken by block to reach its half of amplitude point we consider the corresponding circular motion of the SHM as shown in figure-5.54.

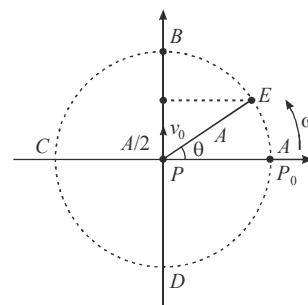


Figure 5.54

Here block P will reach to half of its amplitude when particle P_0 will reach point E shown in figure-5.54 at an angular displacement θ from mean position relative to point A , thus time taken by it is

$$t = \frac{\theta}{\omega} = \frac{\sin^{-1}(\frac{1}{2})}{\sqrt{\frac{k}{m}}} = \frac{\pi}{6} \sqrt{\frac{m}{k}}$$

Illustrative Example 5.25

Figure-5.55 shows a block P of mass M resting on a horizontal smooth floor at a distance l from a rigid wall. Block is pushed toward right by a distance $3l/2$ and released, when block passes from its mean position another block of mass m_1 is placed on it which sticks to it due to friction. Find the value of m_1 so that the combined block just collides with the left wall.

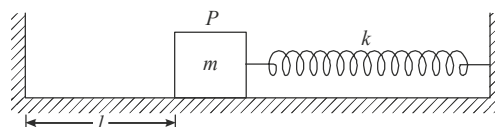


Figure 5.55

Solution

When block P is released from rest from a distance $3l/2$ toward right from mean position, this will be the amplitude of oscillation,

so velocity of block when passing from its mean position is given as

$$v = A\omega = \frac{3l}{2} \sqrt{\frac{k}{m}} \quad \left[\text{As } \omega = \sqrt{\frac{k}{m}} \right]$$

If mass m_1 is added to it and just after if velocity of combined block becomes v_1 , from momentum conservation we have

$$mv = (m + m_1)v_1$$

or

$$v_1 = \frac{m}{(m + m_1)} \left(\frac{3l}{2} \sqrt{\frac{k}{m}} \right)$$

If this is the velocity of combined block at mean position, it must be given as

$$v_1 = A\omega_1 \quad \left[\text{Now } \omega_1 = \sqrt{\frac{k}{m + m_1}} \right]$$

Where A_1 and ω_1 are the new amplitude and angular frequency of SHM of the block. It is given that combined block just reaches the left wall thus the new amplitude of oscillation must be l so we have

$$\frac{m}{(m + m_1)} \cdot \frac{3l}{2} \sqrt{\frac{k}{m}} = l_1 \sqrt{\frac{k}{m + m_1}}$$

or

$$\frac{3\sqrt{m}}{2\sqrt{m + m_1}} = 1$$

or

$$9m = 4m + 4m_1$$

or

$$m_1 = \frac{5}{4}m$$

Practice Exercise 5.3

(i) A particle performing SHM is found at its equilibrium at $t = 1$ second and it is found to have a speed of 0.25 m s^{-1} at $t = 2$ seconds. If the period of oscillation is 6 seconds, calculate.

- (a) amplitude of oscillation and initial phase
(b) velocity of particle at 6 seconds.

[(a) $\frac{3}{2\pi} \text{ m}; -\frac{\pi}{3}$ (b) 0.25 m/s]

(ii) Two light springs of force constant k_1 and k_2 and a block of mass m are in one line AB on a smooth horizontal table such that one end of each spring is fixed on rigid supports and the other end is free as shown in the figure-5.56. The distance between the free ends of the springs is d . If the block moves along AB with a velocity v in between springs, calculate the period of oscillation of the block.

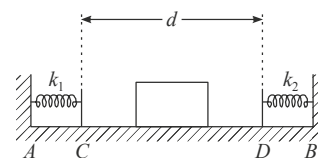


Figure 5.56

$$\left[\pi \left\{ \sqrt{\frac{m}{k_1}} + \sqrt{\frac{m}{k_2}} \right\} + \frac{2d}{v} \right]$$

(iii) Find the phase difference between two particles executing simple harmonic motion with the same frequency if they are found in the states shown in the figure-5.57 at four different points of time.

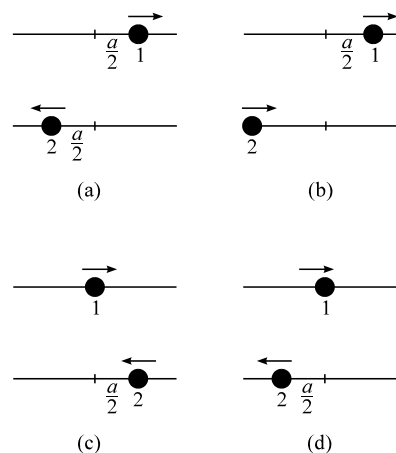


Figure 5.57

[(a) π ; (b) $\frac{2\pi}{3}$; (c) $\frac{5\pi}{6}$; (d) $\frac{7\pi}{6}$]

(iv) A ball is suspended by a thread of length l at the point O on the wall, forming a small angle α with the vertical as shown in figure-5.58. Then the thread with the ball was deviated through a small angle β ($\beta > \alpha$) and set free. Assuming the collision of the ball against the wall to be perfectly elastic, find the oscillation period of such a pendulum.

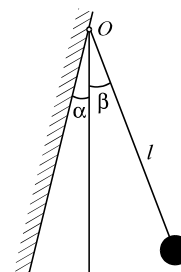


Figure 5.58

$$[T = 2\sqrt{l/g} [\pi/2 + \sin^{-1}(\alpha/\beta)]]$$

(v) A heavy particle is attached to one end of an elastic string, the other end of which is fixed. The modulus of elasticity of the string is such that in equilibrium string length is double its natural length. The string is drawn vertically down till it is four times its natural length and then let go. Show that the particle will return to this point in time $\sqrt{\frac{l}{g}} \left(2\sqrt{3} + \frac{4\pi}{3} \right)$ where l is the natural length of the string.

(vi) Two particles are in SHM with the same amplitude and frequency along the same line and about the same point. If the maximum separation between them is $\sqrt{3}$ times their amplitude, what is the phase difference between them?

[$2\pi/3$]

(vii) A block of mass $m = 1$ kg is attached to a free end of a spring whose other end is fixed with a wall performing simple harmonic motion as shown in figure-5.59. The position of the block from O is given as; $x = 2 + (1/\sqrt{2}) \sin 2t$ where x in meter and t is in second. A shell of same mass is released from smooth the circular path at a height $h = 80$ cm. The shell collide elastically with the block performing SHM and finally reaches upto height 5 cm along circular path. Neglecting friction, find where the collision take place.

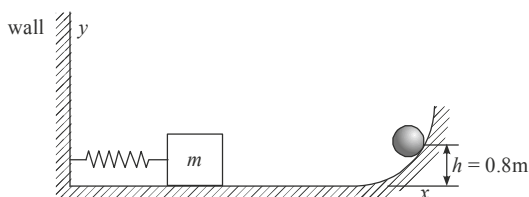


Figure 5.59

[At $x = 1.5$ m or 2.5 m]

(viii) In a SHM the distances of a particle from the middle point of its path at three consecutive seconds are observed to be x , y

and z . Show that the period of oscillation is $\frac{2\pi}{\cos^{-1}\left(\frac{x+z}{2y}\right)}$

(ix) A particle moves with simple harmonic motion in a straight line. In the first second after starting from rest, it travels a distance x_1 , and in the next second it travels a distance x_2 in the same direction. Find the amplitude of the motion.

[$\frac{2x_1^2}{3x_1 - x_2}$]

(x) A particle moves in simple harmonic motion. If the velocities at distances of 4 cm and 5 cm from the equilibrium position are 13 cm per second and 5 cm per second respectively, find the period and amplitude.

[1.6s, 5.02 cm]

(xi) At the moment $t = 0$ a particle starts moving along the x axis so that its velocity projection varies as $v_x = 35 \cos \pi t$ cm/s, where t is expressed in seconds. Find the distance that this particle covers during $t = 2.80$ s after the start.

[60.29 cm]

5.8 Energy of a Particle in SHM

When a particle is in SHM, we know at its extreme position, KE of particle is zero and at mean position, particle's speed is maximum i.e., its KE is maximum. We know at mean position particle is in stable equilibrium where potential energy of particle is minimum about which particle oscillates. During oscillation the total energy of a particle remains constant.

Figure-5.60 shows the graph which represents the variation of potential energy relative to the position of a particle in a force field. At position x_0 potential energy of particle is minimum and in the neighbourhood of x_0 its potential energy is more, thus force on particle is always toward mean position $x = x_0$ and is given by

$$\vec{F} = -\frac{dU}{dx} \hat{i} \quad \dots (5.77)$$

Figure-5.60 shows that at $x = x_0$, dU/dx is zero thus no force acts on particle and at $x > x_0$ from graph, dU/dx is +ve thus $\vec{F}$ is -ve according to equation-(5.77) and force on particle is toward x_0 and at points $x < x_0$, dU/dx is -ve and thus from equation-(5.77) $\vec{F}$ is +ve or towards x_0 thus in such a potential field, a particle can oscillate about $x = x_0$. The region of curve shown in figure is called potential well, in which if a particle is trapped, it can oscillate about the point of lowest potential energy i.e. the point of stable equilibrium.

If particle oscillates about $x = x_0$ with amplitude A , its kinetic energy is maximum at x_0 and as it moves away from x_0 , its PE increases and kinetic energy decreases at extreme position the potential energy of particle becomes U_0 (as shown in graph) and obviously kinetic energy zero. Thus if at mean position its kinetic energy is K_0 we have

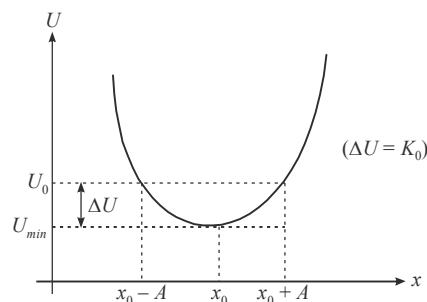


Figure 5.60

Total energy of SHM of particle is

$$E_T = U_{min} + K_0 = U_0$$

Which is also the maximum potential energy of particle when it is at its extreme position.

If particle is executing SHM with amplitude A and angular frequency ω , its velocity at mean position is given as

$$v_{max} = A\omega$$

Thus KE (maximum) of particle at mean position is

$$K_0 = \frac{1}{2} m v_{max}^2 = \frac{1}{2} m \omega^2 A^2 \quad \dots (5.78)$$

If at mean position potential energy of particle is zero then equation-(5.78) represents the total energy of oscillations otherwise total energy of oscillation can be given as

$$E_T = U_{min} + \frac{1}{2} m \omega^2 A^2 \quad \dots (5.79)$$

If $U_{min} = 0$ then $E_T = \frac{1}{2} m \omega^2 A^2$

When particle is in SHM its displacement from mean position can be given as

$$x = A \sin(\omega t + \alpha)$$

At a displacement x from mean position velocity is given as

$$v_x = \omega \sqrt{A^2 - x^2}$$

Thus kinetic energy of particle during SHM at a displacement x from mean position is given as

$$K_x = \frac{1}{2} m v_x^2 = \frac{1}{2} m \omega^2 (A^2 - x^2) \quad \dots (5.80)$$

Now using equation-(5.79), the potential energy of particle at a displacement x from mean position is given as

$$U_x = E_T - K_x \quad [\text{As total energy of particle } E_T \text{ is constant}]$$

$$U_x = U_{min} + \frac{1}{2} m \omega^2 x^2 \quad \dots (5.81)$$

If potential energy of particle at mean position is taken as zero ($U_{min} = 0$). The value of U_x is

$$U_x = \frac{1}{2} m \omega^2 x^2 \quad \dots (5.82)$$

From equation-(5.80) and (5.82) we can write the kinetic and potential energy of a particle in SHM as function of time as

$$U_t = \frac{1}{2} m \omega^2 A^2 \sin^2(\omega t + \alpha) \quad [\text{As } x = A \sin(\omega t + \alpha)]$$

and

$$K_t = \frac{1}{2} m \omega^2 A^2 \cos^2(\omega t + \alpha)$$

Thus at any instant total energy during oscillations is

$$E_T = K_t + U_t = \frac{1}{2} m \omega^2 A^2$$

As we've discussed, the above result is valid if at mean position potential energy is zero ($U_{min} = 0$).

5.8.1 Energy of Oscillation in Angular SHM

When a body is in angular SHM its angular displacement is written as

$$\theta = \theta_0 \sin(\omega t + \alpha)$$

There θ_0 is the angular amplitude and α is the initial phase of particle in SHM.

Similar to previous article here we can directly state that the total energy of oscillations can be given as

$$E_T = \frac{1}{2} I \omega^2 \theta_0^2 \quad \dots (5.83)$$

In equation-(5.83) instead of mass, we use moment of inertia of body and instead of amplitude angular amplitude is used. If at mean position some potential energy U_{min} exist then total energy of body in angular SHM is given as

$$E_T = U_{min} + \frac{1}{2} I \omega^2 \theta_0^2 \quad \dots (5.84)$$

Similarly kinetic and potential energies of the body at an angular displacement θ can be given as

$$K_\theta = \frac{1}{2} I \omega^2 (\theta_0^2 - \theta^2) \quad \dots (5.85)$$

and $U_\theta = \frac{1}{2} I \omega^2 \theta^2 \quad \dots (5.86)$

Similar to previous article, equation-(5.85) and (5.86) are valid if at mean position $U_{min} = 0$.

5.9 Energy Method to Find Frequency of SHM

In previous articles we've read how to find angular frequency of a particle in SHM under some external forces. We first find the equilibrium position of particle then we displace it slightly from its equilibrium position and find the force acting on particle toward its equilibrium position i.e. the restoring force and we try to express this restoring force as a proportional function of displacement and then we compare the acceleration of particle during oscillation with the standard differential equation of SHM.

The above method is appropriate but using energy of oscillations also we can find the angular frequency of SHM of a particle. Some time this method is more useful in solving problem. To explain this method we take few example.

Illustrative Example 5.26

Find angular frequency of a spring block pendulum using energy of oscillations.

Solution

Figure-5.61 shows a spring block system in equilibrium which is at a depth h below the natural length of spring hence we have

$$mg = kh$$

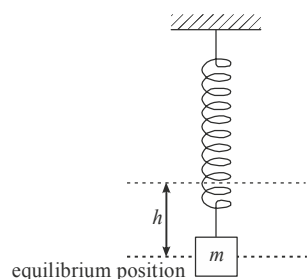


Figure 5.61

Now if the block is displaced down by a distance A below the equilibrium position and released. It starts its oscillation with amplitude A about the mean position. Now during oscillation, we consider the block at position P at a general displacement x from its mean position when it is moving at a speed v as shown in figure-5.62.

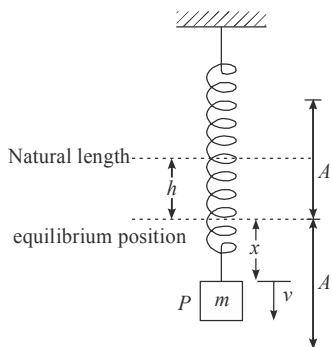


Figure 5.62

If we write the total energy of the oscillating system shown in figure-5.62 at this instant. This is given as

$$E_T = \frac{1}{2}mv^2 + \frac{1}{2}k(x+h)^2 - mgx$$

[Here gravitation work we've considered relative to mean position]

Here we know during oscillation the kinetic and potential energy changes into each other and the total energy of oscillation

remains constant with time, thus we have

$$\frac{dE_T}{dt} = 0$$

$$\text{or } \frac{1}{2}m\left(2v\frac{dv}{dt}\right) + \frac{1}{2}k\left[2(x+h)\frac{dx}{dt}\right] - mg\left(\frac{dx}{dt}\right) = 0$$

Here we can write $v = \frac{dx}{dt}$ and $a = \frac{dv}{dt}$, thus

$$mva + k(x+h)v - mgv = 0$$

$$\text{or } ma + kx + kh - mg = 0$$

$$\text{or } a + \frac{k}{m}x = 0$$

[As from condition of equilibrium $mg = kh$]

Comparing this with standard differential equation of SHM, we have

$$\omega = \sqrt{\frac{k}{m}}$$

Above example explains the way how energy of oscillation can be used to determine the angular frequency of SHM. These are the steps to be followed for this :

Step I : Find equilibrium position of particle by making net force on it equal to zero.

Step II : From equilibrium position displace the particle by a distance A (amplitude of oscillation) and release and let it starts oscillation with amplitude A .

Step III : Consider the particle during oscillations at an intermediate position when its displacement from mean position is x and write the total energy E_T of oscillating system at this instant.

Step IV : Now differentiate this energy of oscillations with respect to time and equate it to zero as this energy does not vary with time.

In step IV when we differentiate this energy expression, we finally get the basic differential equation of SHM as we've seen in example 5.26. In general problems of SHM, if the physical situation involves the concept of mechanical forces, it is easier sometimes to deal the problem using energy expression other wise the conventional method of restoring forces are used. Now we take few more examples using the concept of energy.

Illustrative Example 5.27

Figure-5.63 shows a pulley block system in equilibrium. If the block is displaced down slightly from its equilibrium position and released. Find the time period of oscillation of the system. Assume there is sufficient friction present between pulley and string so that string will not slip over pulley surface.

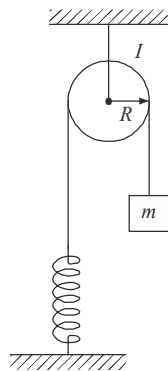


Figure 5.63

Solution

If m is in equilibrium, tension in string must be mg and spring is stretched by h so that $mg = kh$. If we displace the block downward by a distance A and released, it starts executing SHM with amplitude A . During its oscillation we consider the block at a displacement x below the equilibrium position, if it is moving at a speed v at this position, the pulley will be rotating at an angular speed ω given as

$$\omega = \frac{v}{r}$$

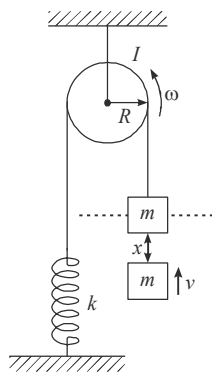


Figure 5.64

Thus at this position the total energy of oscillating system is

$$E_T = \frac{1}{2} mv^2 + \frac{1}{2} I\omega^2 + \frac{1}{2} k(x+h)^2 - mgx$$

Differentiating with respect to time, we get

$$\frac{dE_T}{dt} = \frac{1}{2} m \left(2v \frac{dv}{dt} \right) + \frac{1}{2} I \left(\frac{1}{r^2} \right) \left(2v \frac{dv}{dt} \right) + \frac{1}{2} k$$

$$\left[2(x+h) \frac{dx}{dt} \right] - mg \left(\frac{dx}{dt} \right) = 0$$

$$\text{or } mva + \frac{I}{r^2} va + k(x+h)v - mgv = 0$$

$$\text{or } a + \left(\frac{k}{m + I/r^2} \right) x = 0 \quad \dots (5.87)$$

$$[\text{As } mg = kh]$$

Comparing above equation with standard differential equation of SHM we get

$$\omega = \sqrt{\left(\frac{k}{m + I/r^2} \right)}$$

Thus time period of oscillation is

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m + I/r^2}{k}}$$

Illustrative Example 5.28

Figure-5.65 shows a pulley block system in which a block A is hanging on one side of pulley and on other side a small bead B of mass m is welded on pulley. The moment of inertia of pulley is I and the system is in equilibrium when bead is at an angle α from the vertical. If the system is slightly disturbed from its equilibrium position, find the time period of its oscillations.

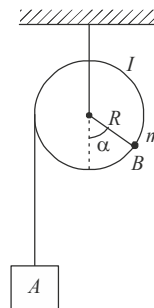


Figure 5.65

Solution

In equilibrium the net torque on pulley must be zero, thus we have

$$MgR = mgR \sin \alpha$$

$$[\text{If mass of block } A \text{ is assumed to be } M]$$

$$\text{or } M = m \sin \alpha \quad \dots (5.88)$$

Now if block is displaced down by distance A and released, it starts oscillating with amplitude A . Now consider the block at

a distance x below the equilibrium position when it is going down at speed v . Figure-5.66 shows the corresponding situation at this instant and the total energy of oscillating system can be written as

$$E_T = \frac{1}{2} Mv^2 + \frac{1}{2} mv^2 + \frac{1}{2} I \left(\frac{v}{R} \right)^2 - Mgx + mgR [\cos \alpha - \cos (\theta + \alpha)]$$

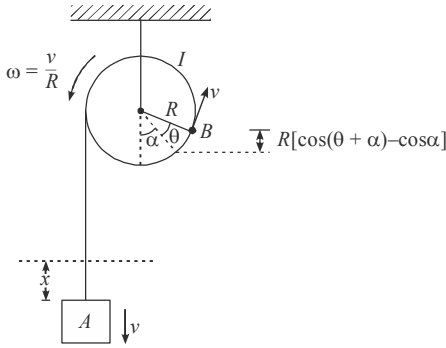


Figure 5.66

Differentiating the above equation w.r. to time, we get

$$\frac{dE_T}{dt} = \frac{1}{2} M \left(2v \frac{dv}{dt} \right) + \frac{1}{2} m \left(2v \frac{dv}{dt} \right) + \frac{1}{2} \frac{I}{R^2} \left(2v \frac{dv}{dt} \right)$$

$$-Mg \left(\frac{dx}{dt} \right) + mgR \left[-\sin(\theta + \alpha) \frac{d\theta}{dt} \right] = 0$$

$$\text{or } Mva + mva + \frac{I}{R^2} va - Mgv + mgR \sin(\theta + \alpha) \left(\frac{v}{R} \right) = 0$$

$$[\text{As } \frac{d\theta}{dt} = \omega = \frac{v}{R}]$$

$$\left(M + m + \frac{I}{R^2} \right) a - Mg - mg [\theta \cos \alpha + \sin \alpha] = 0$$

$$\text{or } \left(M + m + \frac{I}{R^2} \right) a + mg \cos \alpha \cdot \frac{x}{R} = 0$$

$$[\text{As } M = m \sin \alpha \text{ and } \theta = \frac{x}{R}]$$

$$\text{or } a = - \frac{mg \cos \alpha}{R(M + m + I/R^2)} \cdot x \quad \dots (5.89)$$

Comparing equation-(5.89) with basic differential equation of SHM, we get the angular frequency of SHM of system as

$$\omega = \sqrt{\frac{mg \cos \alpha}{R(M + m + I/R^2)}}$$

Thus its time period of oscillations is given as

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{R(M + m + I/R^2)}{mg \cos \alpha}}$$

Illustrative Example 5.29

A particle of mass m is located in a unidimensional potential field where the potential energy of the particle depends on the coordinate x as $U(x) = U_0 (1 - \cos Cx)$; U_0 and C are constants. Find the period of small oscillations that the particle performs about the equilibrium position.

Solution

$$\text{Given that } U(x) = U_0 (1 - \cos Cx)$$

$$\text{We know that } F = m a = - \frac{dU(x)}{dx}$$

$$\text{or } a = \frac{1}{m} \left[- \frac{dU(x)}{dx} \right] = \frac{1}{m} [-U_0 C \sin Cx]$$

$$\text{or } a = - \frac{U_0 C}{m} [Cx] = - \frac{U_0 C^2}{m} x$$

[For small x , we can take $\sin Cx \approx Cx$]

Here acceleration is directly proportional to the negative of displacement. So, the motion is SHM and its time period T is given by

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{(U_0 C^2 / m)}} = 2\pi \sqrt{\left(\frac{m}{U_0 C^2} \right)}$$

Illustrative Example 5.30

The pulley shown in figure-5.67 has a moment of inertia I about its axis and mass m . Find the time period of vertical oscillation of its centre of mass. The spring has spring constant k and the string does not slip over the pulley.

Solution

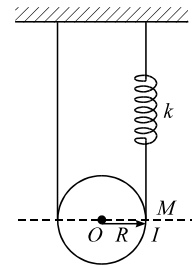


Figure 5.67

Figure-5.67 shows the system in equilibrium. As it is in equilibrium, if mass of pulley is m and tension in string is T , we have

$$T = kx_0$$

Thus $2kx_0 = mg$... (5.90)

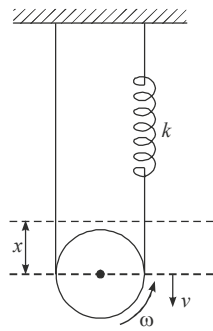


Figure 5.68

Now if the pulley is slightly displaced from its equilibrium position and released, it starts oscillating about the equilibrium position O . During its oscillations we consider the pulley at a depth x below the equilibrium position O as shown in figure-5.68. Due to a further pull by a distance x , total extension in the spring becomes $2(x + x_0)$. If at this instant it is going down with the velocity v and rotating at angular speed ω which is given as v/R , it is given that string does not slip over pulley surface. At this position we write the total energy of this oscillating system as

$$E_T = \frac{1}{2} Mv^2 + \frac{1}{2} I\omega^2 + \frac{1}{2} k [2(x_0 + 2x)]^2 - Mgx$$

differentiating with respect to time, we get

$$\frac{dE_T}{dt} = \frac{1}{2} M \left(2v \frac{dv}{dt} \right) + \frac{1}{2} (I) \left(\frac{2v}{R^2} \frac{dv}{dt} \right) + \frac{1}{2} k$$

$$[2(x_0 + 2x)^2 \cdot 2 \frac{dx}{dt}] - Mg \frac{dx}{dt} = 0$$

$$\text{or } Mva + \frac{Iv}{R^2} a + 2k(x_0 + 2x)v - Mgv = 0$$

$$\text{or } Ma + \frac{Iv}{R^2} a + 2kx_0 + 4kx - Mg = 0 \quad [\text{As } Mg = 2kx_0]$$

$$\text{or } \left(M + \frac{I}{R^2} \right) a + 4kx = 0$$

$$\text{or } a = - \left(\frac{4k}{M + I/R^2} \right) x \quad \dots (5.91)$$

Comparing equation-(5.91) with basic differential equation of SHM, we can say that angular frequency of SHM of pulley is

$$\omega = \sqrt{\frac{4k}{M + I/R^2}}$$

Thus its time period of SHM is given as

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{M + I/R^2}{4k}}$$

Illustrative Example 5.31

A solid uniform cylinder of mass M performs small oscillations in horizontal plane if slightly displaced from its mean position shown in figure-5.69. If it is given that initially springs are in natural lengths and cylinder does not slip on ground during oscillations due to friction between ground and cylinder. Force constant of each spring is k .

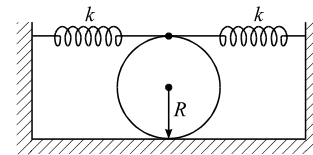


Figure 5.69

Solution

In the situation given in problem, the cylinder is in its equilibrium position when springs are unstrained. When it is slightly rolled and released, it starts executing SHM and due to friction, the cylinder is in pure rolling motion. Now during oscillations we consider the cylinder when it is at a distance x from the mean position and moving with a speed v as shown in figure-5.70. As cylinder is in pure rolling, its angular speed of rotation can be given as

$$\omega = \frac{v}{R}$$

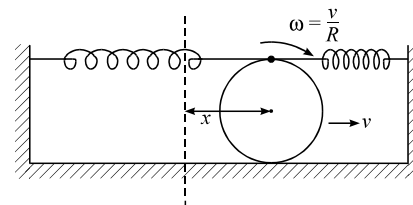


Figure 5.70

As centre of cylinder is at a distance x from the initial position, the springs which connect at a point on its rim must be compressed and stretched by a distance $2x$. Thus at this intermediate position total energy of the oscillating system can be given as

$$E_T = \frac{1}{2} Mv^2 + \frac{1}{2} \left(\frac{1}{2} MR^2 \right) \omega^2 + \frac{1}{2} k (2x)^2 \times 2$$

Differentiating with respect to time, we get

$$\frac{dE_T}{dt} = \frac{1}{2} M \left(2v \frac{dv}{dt} \right) + \frac{1}{2} \left(\frac{1}{2} MR^2 \right) \frac{1}{R^2} \left(2v \frac{dv}{dt} \right) + 4k \left(2x \frac{dx}{dt} \right) = 0$$

$$\text{or } Mva + \frac{1}{2} Mva + 8kxv = 0$$

$$\text{or } a = -\frac{16}{3} \frac{k}{M} x \quad \dots (5.92)$$

Comparing equation-(5.92) with basic differential equation of SHM, we get, the angular frequency of SHM as

$$\omega = \sqrt{\frac{16k}{3M}}$$

Thus time period of these oscillation is

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{3M}{16k}} = \frac{\pi}{2} \sqrt{\frac{3M}{k}}$$

Illustrative Example 5.32

A uniform rod of mass m and length L performs small oscillations about a horizontal axis passing through its upper end. Find the mean kinetic energy of the rod during its oscillation period if at $t = 0$ it is deflected from vertical by an angle θ_0 and imparted an angular velocity ω_0 .

Solution

Here the rod oscillates like a physical pendulum whose angular frequency can be directly given as

$$\omega = \sqrt{\frac{Mgl}{I}}$$

$$\text{Where } I = \frac{ML^2}{3} \quad \text{and} \quad l = \frac{L}{2}$$

$$\text{Thus } \omega = \sqrt{\frac{Mg(L/2)}{ML^2/3}} = \sqrt{\frac{3g}{2L}}$$

It is given that at $t = 0$, rod is at an angular displacement θ_0 and has angular speed ω_0 . If angular amplitude of rod is β then we have

$$\omega_0 = \sqrt{\frac{3g}{2L}} \sqrt{\beta^2 - \theta_0^2}$$

$$\text{Thus } \beta = \sqrt{\frac{2L\omega_0^2}{3g} + \theta_0^2}$$

We know the average kinetic energy of a body in angular SHM over its oscillation period is given by

$$\langle k_t \rangle = \frac{1}{4} I \omega^2 \beta^2$$

$$\text{or } = \frac{1}{4} \left(\frac{ML^2}{3} \right) \left(\frac{3g}{2L} \right) \sqrt{\frac{2L\omega_0^2}{3g} + \theta_0^2}$$

$$\text{or } = \frac{1}{8} MgL \sqrt{\frac{2L\omega_0^2}{3g} + \theta_0^2}$$

Illustrative Example 5.33

Figure-5.71 shows a torisitional pendulum consists of a uniform disc D of mass M and radius R attached to a this rod of torisitional constant C . Find the amplitude and the energy of small torsional oscillations of the disc, if initially the disc was imparted angular speed ω_0 .

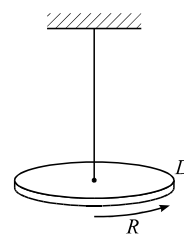


Figure 5.71

Solution

We know for a torisitional pendulum angular frequency of small oscillations is given as

$$\omega = \sqrt{\frac{C}{I}}$$

Where $I = \frac{1}{2} MR^2$, thus

$$\omega = \sqrt{\frac{2C}{MR^2}}$$

As it is given that from mean position the disc is imparted an angular speed ω_0 , if the angular amplitude of oscillations of disc is β , we have

$$\omega_0 = \beta \omega = \beta \sqrt{\frac{2C}{MR^2}}$$

Thus angular amplitude is given as

$$\beta = \sqrt{\frac{MR^2}{2C}} \omega_0$$

For angular SHM the total oscillation energy is given as

$$E_T = \frac{1}{2} I \omega^2 \beta^2 \quad [\text{If } U_{\min} = 0]$$

Thus here oscillation energy of the disc is given as

$$E_T = \frac{1}{2} \left(\frac{1}{2} MR^2 \right) \left(\frac{2C}{MR^2} \right) \left[\frac{MR^2}{2C} \omega_0^2 \right]$$

or

$$E_T = \frac{1}{4} MR^2 \omega_0^2$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Simple Harmonic Motion

Module Number - 10, 11, 20 & 26 to 31

Practice Exercise 5.4

(i) In an SHM, at the initial moment of time, the particle's displacement is 4.3 cm and its velocity is -3.2 m s^{-1} . The particle's mass is 4 kg and its total energy 79.5 J. Write down the equation of the SHM and find the distance travelled by the particle in 0.4 s from the start.

$$[x = 0.05 \sin(126.33 t + 2.1); 1.6 \text{ m}]$$

(ii) A uniform rod of mass m and length l performs small oscillations about the horizontal axis passing through its upper end. Find the mean kinetic energy of the rod averaged over one oscillation period if at the initial moment it was deflected from the vertical by an angle θ_0 and then imparted an angular velocity $\dot{\theta}_0$.

$$[\langle KE \rangle = \frac{1}{8} mgl\theta_0^2 + \frac{1}{12} ml^2 \dot{\theta}_0^2]$$

(iii) A point particle of mass 0.1 kg is executing SHM of amplitude of 0.1 m. When the particle passes through the mean position, its kinetic energy is $8 \times 10^{-3} \text{ J}$. Obtain the equation of motion of this particle if this initial phase of oscillation is 45° .

$$[x = 0.1 \sin(4 t + \pi/4) \text{ m}]$$

(iv) Equal charges $+e$ are fixed at the four corners of a square of side $a\sqrt{2}$. A fifth charge $+e$, whose mass is m , is placed at the centre of the square and is free to move. Show that it is in equilibrium at this point that the equilibrium is stable for all small displacements in the plane of the charges. Find the period of small oscillations along the diagonals of square.

$$[T = 2\pi \sqrt{\frac{2\pi\epsilon_0 ma^3}{e^2}}]$$

(v) Find the time period of oscillations of the system shown in the figure-5.72. The bar is rigid and light. Initially in equilibrium bar is horizontal.

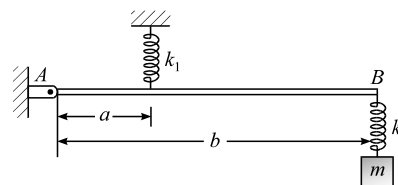


Figure 5.72

$$\left[2\pi \sqrt{\frac{m}{k_1 k_2} \left[k_1 + k_2 \left(\frac{b}{a} \right)^2 \right]} \right]$$

(vi) Find the time period of small oscillations of the spring loaded pendulum. The equilibrium position is vertical as shown. The mass of the rod is negligible and treat mass as a particle.

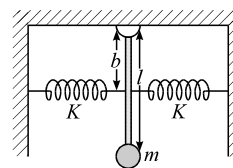


Figure 5.73

$$[T = 2\pi \sqrt{\frac{ml^2}{mgl + 2Kb^2}}]$$

(vii) A massless rod rigidly fixed at O . A string carrying a mass m at one end is attached to point A on the rod so that $OA = a$. At another point B ($OB = b$) of the rod, a horizontal spring of force constant k is attached as shown in figure-5.74. Find the period of small vertical oscillations of mass m around its equilibrium position. Consider rod in vertical position in equilibrium initially.

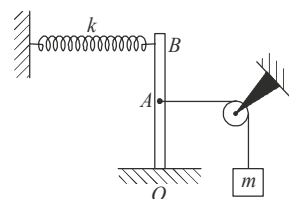


Figure 5.74

$$\left[\frac{2\pi a}{b} \sqrt{\frac{m}{k}} \right]$$

(viii) A particle of mass m is oscillating in SHM along X-axis about its mean position O with angular frequency ω and amplitude A . At the instant when the particle is passing the position $x = \sqrt{3} (A/2)$ and going away from O , an impulsive blow is given to it in the direction of motion. The impulse of this blow is of magnitude $J = m\omega A$. Calculate the new amplitude of vibration in terms of A .

$$[A' = \sqrt{3}A]$$

5.10 Taylor's Method to Find Angular Frequency of a Particle in SHM

For expression of a given mathematical function $y=f(x)$ Taylor's theorem is defined as

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots \quad \dots (5.93)$$

If x is taken as the displacement of a particle from its mean position and the restoring force on particle depends on this x by the function $F_R = f(x)$ then it can be given as

Restoring force on particle at a distance x from mean position is

$$F_R = f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots \quad \dots (5.94)$$

Here $f(0) = 0$ as at $x = 0$ or mean position restoring force is zero and for small displacements of particle higher powers of x can be neglected so restoring force can be given as

$$F_R = -xf'(0)$$

[–ve sign shown the restoring nature]

Acceleration of particle during oscillation is

$$a = \frac{F_R}{m} = -\left(\frac{f'(0)}{m}\right)x \quad \dots (5.95)$$

Comparing this equation with general differential equation of SHM, we get

$$\omega = \sqrt{\frac{f'(0)}{m}} \quad \dots (5.96)$$

Hence time period of this SHM is $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{f'(0)}}$

Illustrative Example 5.34

Two point charges with charge $+Q$ are fixed at pts $(0, r)$ and $(0, -r)$ on $-y$ axis of a coordinate system as shown in figure-5.75. Another small particle of mass m and charge $-q$ is placed at origin of system, where it stays in equilibrium. If this mass m is slightly displaced along $+x$ direction by a small distance x and released, show that it executes SHM and find its time period of oscillations.

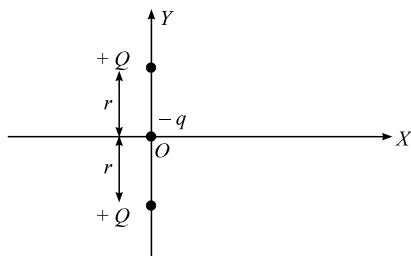


Figure 5.75

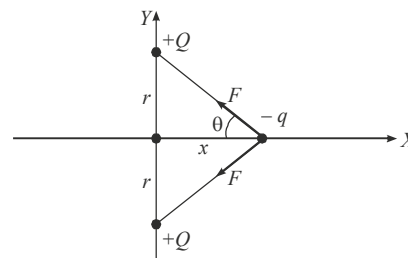


Figure 5.76

If we calculate the net force on particle after its displacement as shown in figure-5.76. The force due to each $+Q$ charge on $-q$ is given by coulomb's law as

$$F = \frac{kqQ}{(r^2 + x^2)} \quad \dots (5.97)$$

Thus net restoring force on it is

$$F_R = -2F \cos \theta = -\frac{2kqQx}{(r^2 + x^2)^{3/2}}$$

[as $\cos \theta = \frac{x}{\sqrt{r^2 + x^2}}$]

As here $x \ll r$, we can neglect x^2 , thus

$$F_R = -\frac{2kqQ}{r^3}x$$

If a is the acceleration of particle, we have

$$a = -\frac{2kqQ}{mr^3}x \quad \dots (5.98)$$

Comparing this equation with basic differential equation of SHM we get angular frequency of SHM of particle as

$$\omega = \sqrt{\frac{2kqQ}{mr^3}} \quad \dots (5.99)$$

Thus its time period is given as

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{mr^3}{2kqQ}}$$

Alternative method by Taylor's theorem :

As we've get the restoring force on particle as

$$F_R = -\frac{2kqQx}{(r^2 + x^2)^{3/2}}$$

We can directly write the angular frequency of SHM as

$$\omega = \sqrt{\frac{f'(0)}{m}} \quad \dots (5.100)$$

Where

$$f'(0) = \left. \frac{dF_R}{dx} \right|_{x=0}$$

$$\text{or} \quad = 2kqQ \left[\frac{(r^2 + x^2)^{3/2} \cdot (1) - x(3/2)[r^2 + x^2]^{1/2}}{(r^2 + x^2)^3} \right] \Big|_{x=0}$$

$$\text{or} \quad = 2kqQ \left[\frac{r^3}{r^6} \right] = \frac{2kqQ}{r^3}$$

Thus from equation-(5.100)

$$\omega = \sqrt{\frac{2kqQ}{mr^3}}$$

Which is same as that of equation-(5.99)

Illustrative Example 5.35

Solve example(5.29) using Taylor's Method

Solution

In question it is given that potential energy of the oscillating particle is given as

$$U = U_0 [1 - \cos(Cx)]$$

Thus force on particle can be given as

$$F = \left| \frac{dU}{dx} \right|$$

$$F = U_0 C \sin(Cx)$$

This force is given as a function of displacement of particle x from its mean position, thus angular frequency of its SHM can be directly given as

$$\omega = \sqrt{\frac{f'(0)}{m}} \quad \dots (5.101)$$

Where

$$f'(0) = \left. \frac{dF}{dx} \right|_{x=0} = U_0 C^2 \cdot \cos(Cx) \Big|_{x=0} = U_0 C^2$$

Thus from equation-(5.101)

$$\omega = \sqrt{\frac{U_0 C^2}{m}}$$

Illustrative Example 5.36

In a given force field, the potential energy of a particle is given as a function of its x -coordinates as

$$U_{(x)} = \frac{p}{x^2} - \frac{q}{x}$$

where a and b are positive constants. Find the period of small oscillations of the particle about its equilibrium position in the field.

Solution

As potential energy of particle is given as

$$U_{(x)} = \frac{p}{x^2} - \frac{q}{x}$$

The force on particle can be given as

$$F = \left| \frac{dU}{dx} \right| = \left| -\frac{2p}{x^3} + \frac{q}{x^2} \right|$$

The equilibrium position of particle can be given for $F=0$, as

$$-\frac{2p}{x^3} + \frac{q}{x^2} = 0$$

$$\text{or} \quad x = \frac{2p}{q}$$

Thus at $x = \frac{2p}{q}$ position particle is in equilibrium. If we find restoring force on particle if it is displaced slightly by a distance z along $+x$ direction then it is given as

$$F_R = - \left[-\frac{2p}{(2p/q + 2)^3} + \frac{q}{(2p/q + 2)^2} \right]$$

[– ve sign for restoring nature]

For small z we have

$$F_R = - \left[-\frac{2p}{(2p/q)^3} \left(1 + \frac{qz}{2p} \right) + \frac{q}{(2p/q)^2} \left(1 + \frac{qz}{2p} \right)^{-2} \right]$$

$$\text{or} \quad F_R = - \left[-\frac{q^3}{4p^2} \left(1 - \frac{3qz}{2p} \right) + \frac{q^3}{4p^2} \left(1 - \frac{bz}{p} \right) \right]$$

$$F_R = - \left[\frac{3q}{2p} - \frac{q}{p} \right] \frac{q^3}{4p^2} z$$

$$F_R = - \left(\frac{q^4}{8p^3} \right) z$$

If a is the acceleration of particle, we have

$$a = - \left(\frac{q^4}{8mp^3} \right) z \quad \dots (5.102)$$

Comparing equation-(5.102) with basic differential equation of SHM, we get the angular frequency of its oscillations as

$$\omega = \sqrt{\frac{q^4}{8mp^3}} = \frac{q^2}{2\sqrt{2m} p^{3/2}} \quad \dots (5.103)$$

This time period of its oscillations is

$$T = \frac{2\pi}{\omega} = 2\pi \left(\frac{2\sqrt{2m} p^{3/2}}{q^2} \right)$$

Alternative method by Taylor's theorem :

As we've got the restoring force on particle is given as

$$F_R = - \left(-\frac{2p}{x^3} + \frac{q}{x^2} \right)$$

[– ve sign for restoring nature]

Then the angular frequency of SHM of particle is given as

$$\omega = \sqrt{\frac{f' \left(\frac{2p}{q} \right)}{m}}$$

[Because now mean position of particle is at $x = \frac{2p}{q}$]

Here $f' \left(\frac{2p}{q} \right) = \left. \frac{dF_R}{dx} \right|_{x=2p/q}$

$$= \frac{6p}{x^4} - \frac{2q}{x^3} \Big|_{x=2p/q}$$

$$= \frac{6p}{(2p/q)^4} - \frac{2q}{(2p/q)^3}$$

$$= \frac{3q^4}{8p^3} - \frac{q^4}{4p^3} = \frac{q^4}{8p^3}$$

Thus angular frequency is given as

$$\omega = \sqrt{\frac{f'(2p/q)}{m}} = \sqrt{\frac{q^4}{8mp^3}} \quad \dots (5.104)$$

Which is same as that of equation-(5.103)

Illustrative Example 5.37

Figure-5.77 shows two identical balls of mass m and charge $+q$. Connected by an ideal spring of constant k . Ball-1 is fixed on a smooth surface as shown in figure. When balls are uncharged, the spring was in its natural length and as both the balls are charged, the spring length gets doubled in equilibrium position of the balls. Find the time period of small oscillations of the ball-2 about the new equilibrium position after charging the two balls.

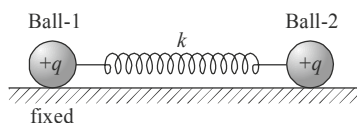


Figure 5.77

Solution

If initial length of spring is l and after charging it becomes $2l$, from coulomb's law we have for equilibrium of ball-2 as

$$\frac{kq^2}{(2l)^2} = kl$$

or $k = \frac{kq^2}{4l^3} \quad \dots (5.105)$

Now if ball is slightly displaced away from ball-1, forms equilibrium position such that the separation between two balls becomes x , then we have restoring force on ball-2 as

$$F_R = \left[k(x-l) - \frac{kq^2}{x^2} \right]$$

[As spring is stretched by $x=l$] ... (5.106)

Equation-(5.106) gives the restoring force as a function of distance x , here we can directly find the angular frequency of SHM of ball-2 using Taylor's Theorem as

$$\omega = \sqrt{\frac{f'(2l)}{m}}$$

[Because equilibrium position of ball-2 is at $x=2l$]

Here $f'(2l) = \left. \frac{dF_R}{dx} \right|_{x=2l}$

$$= \left[k + \frac{2kq^2}{x^3} \right]_{x=2l}$$

$$= k + \frac{2kq^2}{8l^3} = 2k \quad \text{[As } k = \frac{kq^2}{4l^3} \text{]}$$

Thus angular frequency of SHM of ball-2 can be given as

$$\omega = \sqrt{\frac{2k}{m}}$$

Thus the time period of SHM of ball-2 is

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{2k}} \quad \dots (5.107)$$

NOTE : This problem can also be solved by the conventional method of restoring force. We leave this as an exercise for students to get equation-(5.107) by the conventional method.

5.11 Equation of SHM With Shifting of Origin

In previous articles we've discussed that a general equation of SHM of particle is given as

$$y = A \sin(\omega t + \alpha) \quad \dots (5.108)$$

Here mean position of particle is at $y=0$ and it oscillates between points $y=\pm A$. In all cases, it is not necessary that mean position of particle is at origin ($y=0$). If in some case mean position of particle is at $y=y_m$ and it executes SHM with amplitude A then the extreme position of the particle in SHM are $y_m + A$ and $y_m - A$. In such a case the SHM equation can be given as

$$y = y_m + A \sin(\omega t + \alpha) \quad \dots (5.109)$$

The differential equation for SHM can be written by differentiating equation-(5.109) twice as

$$\frac{d^2 y}{dt^2} = -A\omega^2 \sin(\omega t + \alpha)$$

$$\text{or} \quad \frac{d^2 y}{dt^2} = -\omega^2 (y - y_m)$$

$$\text{or} \quad \frac{d^2 y}{dt^2} + \omega^2 y = \omega^2 y_m$$

$$\text{or} \quad \frac{d^2 y}{dt^2} + \omega^2 y = C \quad \dots (5.110)$$

Where $C = \omega^2 y_m$ is a numerical constant, where mean position of particle can be given as

$$y_m = C/\omega^2 \quad \dots (5.111)$$

In some cases of SHM, while solving using energy expression some time equation-(5.110) appears where instead of zero a constant appears on right side of equality, which implies that mean position of oscillation is not at $y=0$ but at $y=C/\omega^2$. Now we take few examples to understand the concept in a better way.

Illustrative Example 5.38

Figure-5.78 shows block of mass m resting on a smooth horizontal plane attached with a spring of natural length l and force constant k . If at $t=0$ an external force F is applied on the block toward right, it starts executing oscillations. Analyse the motion of the block and write the equation of SHM of block by taking origin O at the left wall where spring is connected.

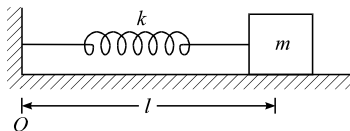


Figure 5.78

Solution

As F force is applied on block, it starts accelerating toward right and its acceleration during motion is given as

$$a = \frac{F - kx}{m}$$

[If x is the displacement of block from its initial position]

After some displacement at $x = F/k$, the block is in equilibrium but gained some velocity so it will move further but now $kx > F$ and it retards and say at $x=l$, it stops thus from work-energy theorem, we have

$$F l - \frac{1}{2} k l^2 = 0$$

$$\text{or} \quad l = \frac{2F}{k}$$

So it is clear that after equilibrium ($x = F/k$) position it travels a further distance F/k which can be regarded as the amplitude of the SHM of block and we can say that at $t=0$ block was started from its left extreme position i.e. at a distance F/k left of the equilibrium position. About the equilibrium position the SHM equation of block can be written as

$$y = \left(\frac{F}{k} \right) \cos \omega t$$

[As at $t=0$, particle starts from its left extreme position

thus its initial phase is $\alpha = 3\pi/2$]

If origin is taken at the point O then equation of SHM i.e. the displacement of block as time function about this origin can be written as

$$y = y_0 - \frac{F}{k} \cos \omega t$$

Here y_0 is the distance co-ordinate of mean position of block from origin which is given by

$$y_0 = l + \frac{F}{k}$$

Thus equation of SHM for origin O is

$$y = l + \frac{F}{k} (1 - \cos \omega t)$$

Here it is important to note that angular frequency of SHM ω is still $\sqrt{\frac{k}{m}}$. There is no effect on the angular frequency due to the presence of external force F , because it is constant.

Always remember that on applying a constant external force on an oscillating system, there is no change in its oscillation frequency. The external force can only change the equilibrium position of oscillations.

Illustrative Example 5.39

Figure-5.79 shows, two balls having charges q_1 connected by a spring of force constant k first ball is fixed in the ceiling and second is hanging vertically in equilibrium. Taking origin at ceiling write the equation of SHM for the motion of hanging ball if at $t = 0$ is given a velocity v_0 in downward direction from equilibrium position. Given that in equilibrium ball with charge q_2 is at a depth y_0 below the ceiling.

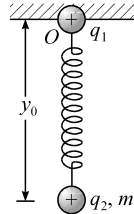


Figure 5.79

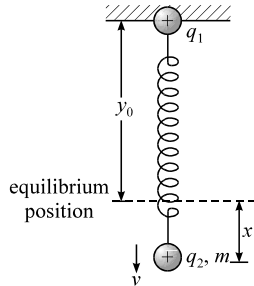
Solution

Figure 5.80

Here during oscillations of second ball, we consider it at an intermediate position at a depth x below the equilibrium position and moving downward at speed v , then the total energy of this oscillating system can be written as

$$E_T = \frac{1}{2}mv^2 + \frac{1}{2}k(x + y_0 - l)^2 + \frac{kq_1q_2}{(y_0 + x)} - mg(x + y_0)$$

[If l is the natural length of spring]

Differentiating with respect to the, we get

$$\frac{dE_T}{dt} = \frac{1}{2}m \left[2v \frac{dv}{dt} \right]$$

$$+ \frac{1}{2}k \left(2(x + y_0 - l) \frac{dx}{dt} \right) - \frac{kq_1q_2}{(y_0 + x)^2} \frac{dx}{dt} - mg \frac{dx}{dt} = 0$$

$$\text{or } mva + k(x + y_0 - l)v - \frac{kq_1q_2}{(y_0 + x)^2}v - mgv = 0$$

$$\text{or } ma + k(x + y_0 - l) - \frac{kq_1q_2}{(y_0 + x)^2} - mg = 0 \quad \dots (5.112)$$

We know at equilibrium position we have

$$k(y_0 - l) = mg + \frac{kq_1q_2}{y_0^2} \quad \dots (5.113)$$

In equation-(5.112) for small x , we have

$$ma + kx + k(y_0 - l) - \frac{kq_1q_2}{y_0^2} \left(1 - \frac{2x}{y_0} \right) - mg = 0 \quad \dots (5.114)$$

From equation-(5.114) and (5.115)

$$ma + kx + \frac{2kq_1q_2}{y_0^3}x = 0$$

$$\text{or } a = - \left(\frac{k}{m} + \frac{2kq_1q_2}{my_0^3} \right) x$$

Thus angular frequency of oscillations of second ball is given as

$$\omega = \sqrt{\frac{k}{m} + \frac{2kq_1q_2}{my_0^3}}$$

If amplitude of SHM is A then we have at mean position velocity of second ball is V_0 , thus

$$V_0 = A\omega \text{ or } A = \frac{V_0}{\omega} = \frac{V_0}{\sqrt{\frac{k}{m} + \frac{2kq_1q_2}{my_0^3}}}$$

As ball starts its SHM at $t = 0$ from its mean position, its equation can be given as

$$y = y_0 + A \sin \omega t$$

5.12 SHM of Free Bodies in Absence of External Forces

We know for oscillations of a body restoring force must be there due to which the body oscillates about its mean position. In some special cases, it is possible that a system oscillates due to only internal forces and internal forces of system provide the required centripetal force for oscillations. We take an Illustrative example to explain such situation.

Illustrative Example 5.40

Figure-5.81 shows two masses m_1 and m_2 connected by a spring of force constant k and the system is placed on a smooth horizontal surface. If the block are compressed slightly and released. Find the time period of such oscillations.

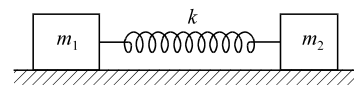
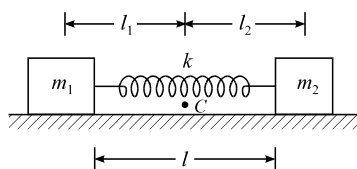


Figure 5.81

Solution**Figure 5.82**

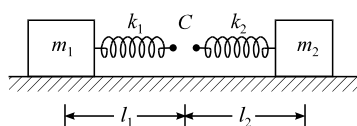
As during oscillations of the two blocks, there is no external force present, thus we can state that the centre of mass of the two block system remain at rest during oscillations. In figure-5.82 if C is the centre of mass of the blocks and l is the natural length of the system, then we can say that in this situation point C will remain at rest and with respect to this point m_1 and m_2 will oscillate independently. We can also split the spring in two parts of lengths l_1 and l_2 which are given by

$$l = \frac{m_2 l}{m_1 + m_2} \quad \text{and} \quad l_2 = \frac{m_1 l}{m_1 + m_2}$$

Now if these two springs are assumed to be fixed at point C separately, the case still remains same as in absence of external forces centre of mass of system C remains at rest as shown in figure-5.83. The respective force constants can be given as

$$k_1 = \frac{kl}{l_1} = \frac{k(m_1 + m_2)}{m_2}$$

and
$$k_2 = \frac{kl}{l_2} = \frac{k(m_1 + m_2)}{m_1}$$

**Figure 5.83**

Now the angular frequency and time period of oscillations can be directly given from that of a spring block system with one end of the spring fixed as

$$\omega = \sqrt{\frac{k}{m}}$$

Here for mass m_1 , we have

$$\omega = \sqrt{\frac{k_1}{m_1}} = \sqrt{\frac{(m_1 + m_2)k}{m_1 m_2}} \quad \dots (5.115)$$

For m_2 , we have

$$\omega = \sqrt{\frac{k_2}{m_2}} = \sqrt{\frac{(m_1 + m_2)k}{m_1 m_2}} \quad \dots (5.116)$$

Both are same thus both the blocks oscillates with same angular frequency given by equation-(5.115) or (5.116). Thus time period of oscillations of the system is given as

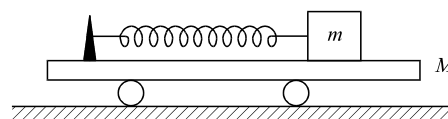
$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m_1 m_2}{(m_1 + m_2)k}}$$

$$\text{or} \quad T = 2\pi \sqrt{\frac{\mu}{k}} \quad \dots (5.117)$$

In equation-(5.117) $\mu = \frac{m_1 m_2}{m_1 + m_2}$ is called reduced mass of the system. Reduced mass can also be regarded as mass of one body relative to the second body (assumed to be fixed) in absence of external forces.

Illustrative Example 5.41

Figure-5.84 shows a cart of mass M on which another mass m is placed and attached to a pole on cart by a spring of force constant k . If the friction between m and surface of cart as well as between cart and floor is neglected find the time period of small oscillation of cart-block system.

**Figure 5.84****Solution**

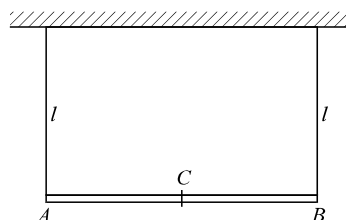
As in problem it is given that there is no friction present anywhere, this implies to external forces are acting on this system. If we assume cart is fixed at its position, we can consider only the block (with reduced mass μ) is oscillating on its surface, thus time period of oscillations of block can be given as

$$T = 2\pi \sqrt{\frac{\mu}{k}} \quad \text{where} \quad \mu = \frac{mM}{m + M}$$

Now we take some more miscellaneous example to understand basic Simple Harmonic Motion in detail.

Illustrative Example 5.42

A uniform rod AB of mass m is suspended by two identical strings of length l as shown in figure-5.85. If this rod is turned by a small angle in horizontal plane about. The vertical axis passing through its centre C . In this process the strings are deviated by a small angle θ_0 from vertical. Then the rod is released to start performing angular oscillations. Find the period of oscillations and the rod's oscillation energy.

**Figure 5.85**

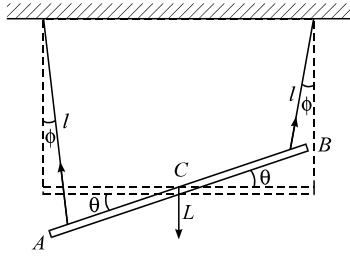
Solution**Figure 5.86**

Figure-86 shows the situation when the rod is in twisted situation if rod is at an angular displacement θ , let string are at deviation ϕ , if length of rod is l , we have from figure-5.86

$$\frac{L}{2} \theta = l \phi \quad \dots (5.118)$$

As the rod vertical is in equilibrium thus we have

$$2T \cos \phi = mg \quad [\text{If } T \text{ is the tension in strings}]$$

For small ϕ , we have

$$T = \frac{1}{2} mg$$

From figure-5.86, we can see that $T \sin \phi$ on both the ends of rod produces a restoring torque on it which is given as

$$\tau_R = -2T \sin \phi \times \frac{L}{2}$$

[– ve sign for restoring nature]

$$\text{or} \quad \tau_R = -\frac{1}{2} mgL \phi \quad [\text{As for small } \phi, \sin \phi \approx \phi]$$

Now from equation-(5.118)

$$\tau_R = -\frac{1}{2} mgL \left[\frac{L}{2l} \theta \right]$$

If α be the angular acceleration of rod during oscillations, we have

$$\tau_R = I \alpha = -\frac{MgL^2}{4l} \theta$$

$$\text{or} \quad \frac{ML^2}{12} \cdot \alpha = -\frac{MgL^2}{4l} \theta$$

$$[\text{As about central axis, for rod, } I = \frac{ML^2}{12}]$$

$$\text{or} \quad \alpha = -\left(\frac{3g}{l}\right) \theta \quad \dots (5.119)$$

Comparing equation-(5.119) with basic differential equation of SHM, we get the angular frequency of oscillations as

$$\omega = \sqrt{\frac{3g}{l}} \quad \dots (5.120)$$

Thus time period of its oscillations can be given as

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{l}{3g}} \quad \dots (5.121)$$

If is given in the question that initially strings are deflected by an angle θ_0 . If β is the angular amplitude of oscillations of rod then from equation-(5.119) we have

$$\frac{L}{2} \beta = l \theta_0$$

or

$$\beta = \frac{2l}{L} \theta_0$$

Now we know that the total energy of oscillation of rod can be given as

$$\begin{aligned} E_T &= \frac{1}{2} I \omega^2 \beta^2 \\ &= \frac{1}{2} \left(\frac{ML^2}{12} \right) \left(\sqrt{\frac{3g}{l}} \right)^2 \left(\frac{2l}{L} \theta_0 \right)^2 \\ &= \frac{1}{2} Mg l \theta_0^2 \end{aligned}$$

Illustrative Example 5.43

Suppose a tunnel is dug through the earth from one side to the other side along a diameter. Show that the motion of a particle dropped into the tunnel is simple harmonic motion. Find the time period. Neglect all the frictional forces and assume that the earth has a uniform density.

$$G = 6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}; \text{ density of earth} = 5.51 \times 10^3 \text{ kg m}^{-3}$$

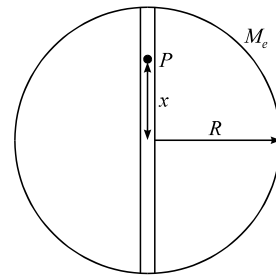
Solution**Figure 5.87**

Figure-5.87 shows the particle P of mass m , dropped in the turned along diameter of earth at a distance x from the centre of earth.

We know intensity gravitational field inside the earth is given as

$$g_{in} = \frac{GM_e}{R_e^3} \cdot x \quad \dots (5.122)$$

Thus restoring force on particle P towards earth centre can be written as

$$F_R = -mg_{in} = -\frac{GM_em}{R_e^3}x \quad \dots (5.123)$$

[– ve sign for restoring nature]

In equation-(5.123), restoring force on particle is always towards centre of earth and it is directly proportional to x , thus it shows that motion of particle is simple harmonic and during motion if its acceleration is a , we have

$$a = -\frac{GM_e}{R_e^3} \cdot x$$

Practice Exercise 5.5

(i) Find the angular frequency of SHM of block of mass m for small oscillation of light rod BD . Consider rod is vertical and springs relaxed in initial state.

$$[\omega = \sqrt{\frac{k_1 k_2 c^2}{m\{k_1 c^2 + k_2 (b+c)^2\}}}]$$

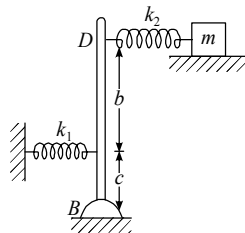


Figure 5.88

(ii) A particle of mass m is suspended at the lower end of a thin rod of negligible mass. The upper end of the rod is free to rotate in the plane of the page about a horizontal axis through the point O . The spring is undeformed when the rod is vertical as in figure-5.89. Show that the motion of the particle is SHM if displaced from mean position and hence find the period of oscillation.

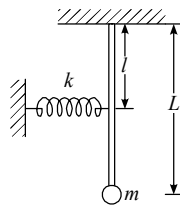


Figure 5.89

$$[2\pi \sqrt{\frac{mL^2}{mgL + kl^2}}]$$

(iii) The cord is light and inextensible in the spring-mass pulley system as in figure-5.90. Find the frequency of vibration if the mass m is displaced slightly and released. Assume that cord does not slip over pulley.

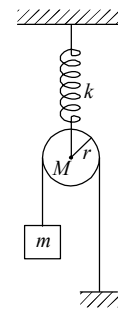


Figure 5.90

$$[\omega = \sqrt{\frac{k}{4m + \frac{3M}{2}}}]$$

(iv) A uniform semicircular cylinder of radius R and weight W is displaced through a small angle θ from its equilibrium position as shown in the figure-5.91. Find the period of small oscillations if it rolls without slipping when displaced slightly from its equilibrium position.

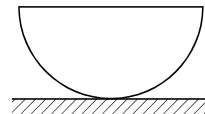


Figure 5.91

$$[T = \pi \sqrt{\frac{(9\pi - 16)R}{2g}}]$$

(v) In the figure-5.92 shown, the springs are unstretched. The left spring is compressed by $2A$ and then released with mass m which is always attached to it. Find the time to touch right spring. Find maximum compression in right spring and equilibrium position of mass m .

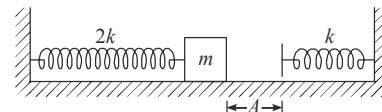


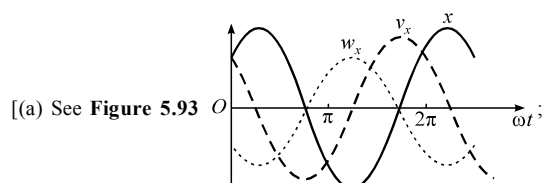
Figure 5.92

$$[\frac{\pi}{3} \sqrt{\frac{m}{2k}}, \frac{\sqrt{22} + 2}{3} A; \text{Equilibrium position is at } \frac{2A}{3} \text{ compression of right spring}]$$

(vi) A point oscillates along the x -axis according to the law $x = a \cos(\omega t - \pi/4)$. Draw the approximate plots

(a) of displacement x , velocity projection v_x , and acceleration projection w_x as functions of time t ;

(b) velocity projection v_x and acceleration projection w_x as function of the coordinate x .



(b) $(v_x/a\omega)^2 + (x/a)^2 = 1$ and $w_x = -\omega^2 x$.]

(vii) A smooth horizontal disc rotates about the vertical axis O (Figure-5.94) with a constant angular velocity ω . A thin uniform rod AB of length l performs small oscillations about the vertical axis A fitted to the disc at a distance a from the axis of the disc. Find the angular frequency of these oscillations.

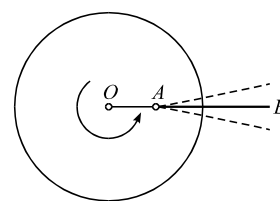


Figure 5.94

$$\left[\sqrt{\frac{3\omega^2 a}{2l}} \right]$$

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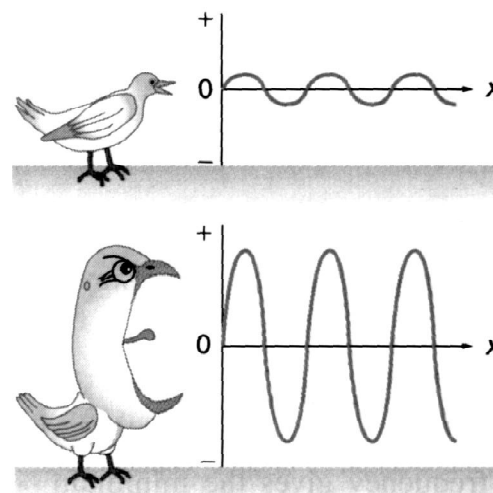
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Waves

FEW WORDS FOR STUDENTS

In mechanical oscillations we have studied the detailed analysis of motion of oscillating particles. In this chapter we use our knowledge of oscillatory motion to explain the behaviour of mechanical waves. A mechanical wave is propagation of energy through a medium due to oscillating medium particles. The basic principles of wave motion are essential for fundamental understanding the behaviour of matter at atomic and subatomic level.



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| 6.2 | <i>Equation of a Simple Harmonic Wave</i> | 6.11 | <i>Stationary Waves</i> |
| 6.3 | <i>Sound Waves</i> | 6.12 | <i>Standing Waves on Clamped String</i> |
| 6.4 | <i>Velocity of a Wave</i> | 6.13 | <i>Wave Resonance</i> |
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COVER APPLICATION



Figure-(a)

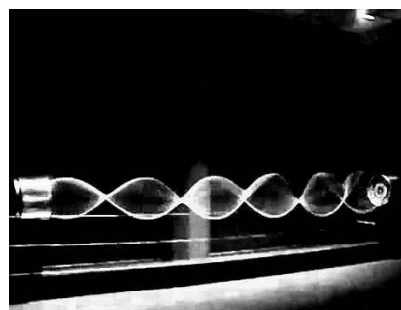


Figure-(b)

Figure-(a) shows a Rubens' tube or flame tube experiment which is an apparatus for demonstrating acoustic standing waves in a tube. It graphically shows the relationship between sound waves and pressure. Above figure shows a tube in which propane gas is passed through a Bunsen burner and inside the tube due to variation of pressure by formation of standing waves the flame length changes with distance. At the position of constructive interference, antinodes are formed and at the points of destructive interference nodes are formed. In above figure it is very important to understand which points are node and which are antinodes in respect of flame length. Figure-(b) shows Melde's experiment setup demonstrating stationary wave in a stretched string.

In simple words a wave is the transfer of energy. We also experience the energy transferred by waves in many situations. We feel how much force an ocean wave can exert, our skin is warmed by the light waves from the sun, we hear sound waves and many more examples are there in our surroundings. Most of the information that we receive comes to us by waves. In this chapter we'll discuss in detail how energy (or information) is transmitted by a wave.

In a wave energy and momentum is transmitted by oscillations. A wave can be classified in two broad categories. Mechanical Waves and Non Mechanical or Electromagnetic waves. In mechanical waves, energy is transferred due to oscillations of medium particles and in non mechanical waves energy transfer takes place by the oscillations of electric and magnetic field vectors. In this chapter we'll mainly discuss all about mechanical waves which requires a physical medium for its propagation.

6.1 Properties of a Mechanical Wave

A wave has three basic properties which decide the characteristic behaviour of wave and these are

- (i) Amplitude of wave (ii) Frequency of wave
- (iii) Velocity of wave

(i) Amplitude : It is the maximum displacement of oscillating medium particle from their mean position. It depends on the source of oscillation or the wave generator. It may vary with the distance of medium particle from the oscillating source.

(ii) Frequency : It is the number of oscillations made by the medium particles per unit time. It also depends on the source of oscillations. Once a wave is produced by a source it is carried by the medium, no matters what the medium may be, the frequency of oscillation remains same wherever the wave will go.

Whenever a wave is produce by a source, it propagates in a medium and during propagations when it encounters another medium in its path then at boundary of the medium a part of wave is reflected into the same medium and a part is transmitted to the other medium. It is obvious that as the reflected and transmitted waves are produced by the initial propagating wave, the frequency of both of these waves must be same as that of the oscillating source of initial wave.

(iii) Velocity of wave : It is the speed with which energy transfers through a medium and it depends mainly on the physical properties of the medium. When a wave is transmitted through a medium, its frequency and amplitude depends on the source of oscillation which produces the wave but its speed is decided by the medium which carries energy.

6.1.1 Different Types of Mechanical Waves

We've discussed that mechanical waves carry energy through a physical medium by oscillations of the medium particles. Depending on the way how these medium particle oscillates and the nature of oscillations, there are several types of mechanical waves. Mainly mechanical waves are classified in two broad categories :

- (i) Transverse waves, (ii) Longitudinal waves.

(i) Transverse Wave : During such type of wave propagation, medium particles oscillate in a plane normal to the direction of propagation of the wave energy as shown in figure-6.1.

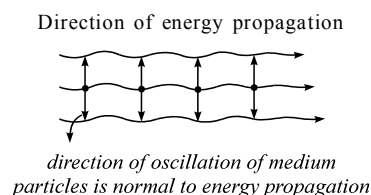


Figure 6.1

(ii) Longitudinal Waves : During such type of wave propagation, medium particles oscillate along the direction of propagation of the wave energy as shown in figure-6.2.

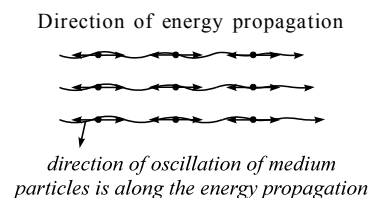


Figure 6.2

The most simple type of a wave is simple harmonic wave which exist in both longitudinal and transverse form. It is the one in which the medium particle oscillates simple harmonically. Now we'll discuss about simple harmonic waves in detail.

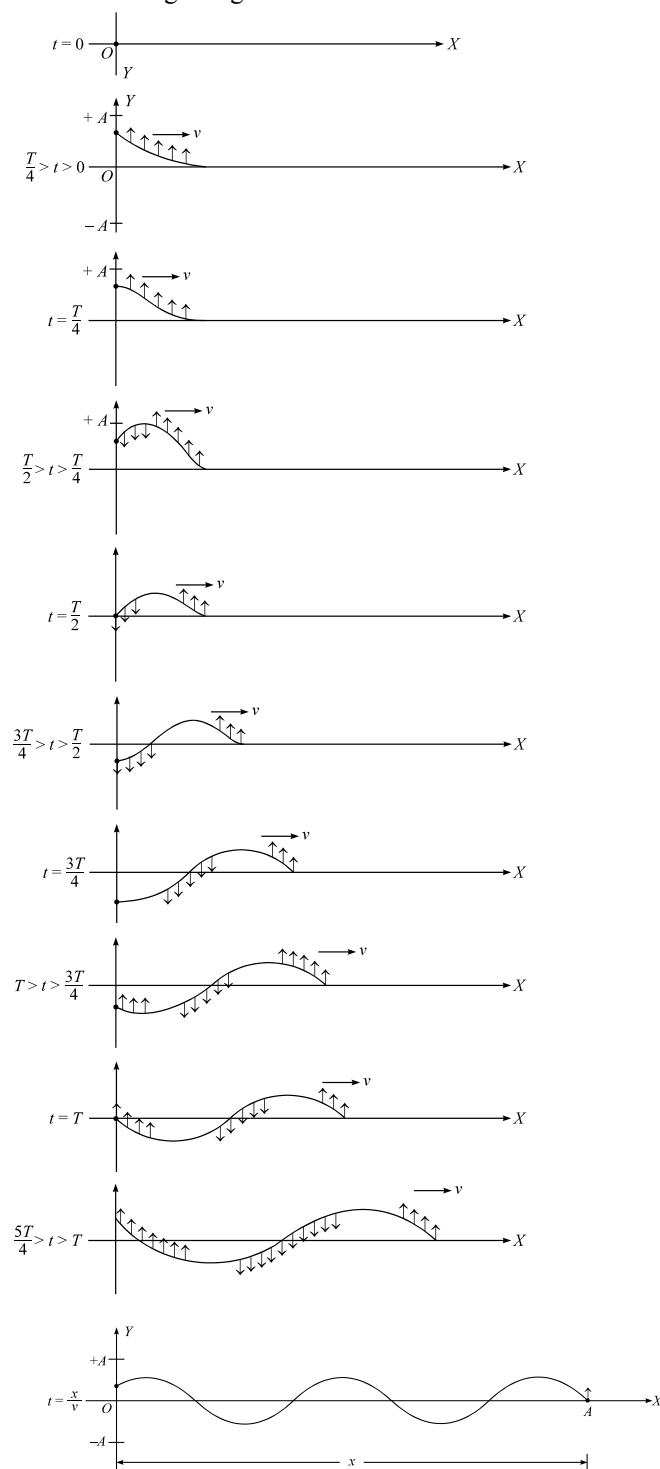
6.2 Equation of a Simple Harmonic Wave

We've discussed that in a simple harmonic wave, all medium particles execute SHM along or perpendicular to the direction of propagation of wave depending on whether wave is longitudinal or transverse. In the chapter of simple harmonic motion we've discussed that the equation of SHM of a particle gives its displacement from mean position as a function of time and can be written in general as

$$y = A \sin(\omega t + \phi) \quad \dots (6.1)$$

Here A is the amplitude of oscillations of the particle with angular frequency ω and ϕ gives the phase angle of particle (on circular projection) at time $t = 0$ when particle starts its SHM with respect to its reference position.

In a simple harmonic wave, all medium particles executes SHM and transfers energy from one particle to another and due to this the wave propagates. Equation of a simple harmonic wave is a mathematical expression similar to equation-(6.1) which gives the displacement of all the medium particles from their mean position as a function of time. Consider the situation shown in figure-6.3. A transverse wave is travelling on a stretched string along +X-direction.



Velocity of medium particle $\longrightarrow$

Figure 6.3

The oscillating source of the wave is at origin O , it starts its oscillations with amplitude A and angular frequency ω at $t = 0$ and the oscillations produced by the source is carried in +X direction by the medium with the wave speed v , as shown. If we consider a medium particle A at a distance x from the origin, will start its oscillations at time $t = x/v$ when wave will first reach this point and its oscillations are started from the mean position with amplitude A and same angular frequency ω . Here we can observe that if origin starts oscillations at $t = 0$ from its mean position, its SHM equation can be given as

$$y = A \sin \omega t \quad \dots (6.2)$$

Medium particle at point A also starts its SHM from its mean position but not at $t = 0$. It will start at time $t = x/v$ where x is the distance, wave covers from origin to point A . In this duration the source of wave or the medium particle at origin will move ahead in phase by an angle ϕ , which is given as

$$\begin{aligned} \phi &= \omega t = \omega \frac{x}{v} \\ &= \frac{2\pi n}{n\lambda} x \\ &= \frac{2\pi}{\lambda} x \\ &= kx \end{aligned} \quad \dots (6.3)$$

Here $k = \frac{2\pi}{\lambda}$ is called angular wave number of the wave. In equation-(6.3) ϕ is the phase difference between point A and origin or this is the phase angle by which SHM of point A lagging with respect to SHM of particle at origin. Thus SHM equation of medium particle at point A can be written as

$$y = A \sin (\omega t - \phi)$$

$$\text{or} \quad y = A \sin (\omega t - kx) \quad \dots (6.4)$$

The expression in equation-(6.4) gives the displacement y of a medium particle from its mean position as a function of time, which is situated at a distance x from the source of oscillations of the wave. This equation-(6.4) is termed as the general equation of a simple harmonic wave propagating in positive x direction.

Equation-(6.4) can be written in different ways as

$$y = A \sin (\omega t - kx) \quad \dots (6.4a)$$

$$y = A \sin \omega \left(t - \frac{x}{v} \right) \quad [\text{As } v = \frac{\omega}{k}] \quad \dots (6.4b)$$

$$y = A \sin \frac{2\pi}{\lambda} (vt - x) \quad [\text{As } \frac{\omega}{v} = \frac{2\pi}{\lambda}] \quad \dots (6.4c)$$

$$y = A \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) \quad [\text{As } \frac{v}{\lambda} = \frac{1}{T}] \quad \dots (6.4d)$$

Above expressions in equations-(6.4a) to (6.4d) represent different forms of equations of a plane propagating wave in positive direction of X -axis. Students are advised to remember all these forms of wave equation in their mind for numerical applications.

6.2.1 Velocity and Acceleration of a Medium Particle in Wave Propagation

When a plane progressive wave propagates along positive x -axis with its source located at $x = 0$ and started at $t = 0$, the equation of motion of a particle at position x is given as

$$y = A \sin (\omega t - kx) \quad \dots (6.5)$$

Equation-(6.5) is also termed as wave equation for the respective wave. Here the velocity of medium particle at position x is given as

$$v_p = \frac{\partial y}{\partial t} = A\omega \cos (\omega t - kx) \quad \dots (6.6)$$

$$= A\omega \sqrt{A^2 - y^2} \quad \dots (6.7)$$

Equation-(6.7) already we've covered in SHM. Here we can also see that during wave propagation the maximum speed of medium particles is given as

$$v_{p(\max)} = A\omega$$

When they cross their mean position, i.e. at $y = 0$.

Further from equation-(6.5), if we differentiate it w.r. to displacement of wave x then we get the slope of displacement curve of wave as

$$\frac{\partial y}{\partial x} = -AK \cos (\omega t - kx) \quad \dots (6.8)$$

From equation-(6.6) and (6.8) we have

$$\begin{aligned} \frac{\partial y}{\partial x} &= -\frac{k}{\omega} \frac{\partial y}{\partial t} \\ &= -\frac{1}{v} \frac{\partial y}{\partial t} \quad [\text{As wave velocity } v = \frac{\omega}{k}] \end{aligned}$$

$$\text{or} \quad v_p = -v \times (\text{slope}) \quad \dots (6.9)$$

Thus in a propagating wave the speed of medium particle at a given position and at a given instant is the negative of the product of wave velocity and the slope of displacement curve of wave at that point and at that instant.

Similarly acceleration of a medium particle is given as

$$\frac{\partial^2 y}{\partial t^2} = -A\omega^2 \sin (\omega t - kx) \quad \dots (6.10)$$

Now differentiating equation-(6.8) w.r. to x again, we get

$$\frac{\partial^2 y}{\partial x^2} = -Ak^2 \sin (\omega t - kx) \quad \dots (6.11)$$

From equation-(6.10) and (6.11) we get

$$\begin{aligned} \frac{\partial^2 y}{\partial t^2} &= \frac{\omega^2}{k^2} \frac{\partial^2 y}{\partial x^2} \\ \frac{\partial^2 y}{\partial t^2} &= v^2 \frac{\partial^2 y}{\partial x^2} \quad \dots (6.12) \end{aligned}$$

There $\frac{\partial^2 y}{\partial x^2}$ is the measure of curvature of the displacement curve, thus we can write that

Acceleration of a medium particle = $v^2 \times$ the curvature of the displacement curve

Here equation-(6.12) is termed as differential form of equation of a propagating wave.

6.2.2 Phase Difference and Path Difference of Medium Particles in Simple Harmonic Wave

We've discussed that when a wave propagates, only energy is transferred through the medium particles and these particles execute SHM at their position. From figure-6.3 it is also clear that as wave propagates, every medium particle follows the phase of its previous particle or in other words the instantaneous phase of a medium particle is same which was that of its previous particle just a moment before. Thus we can say that during wave propagation, phase also travels, that's why some times wave velocity is also termed as phase velocity.

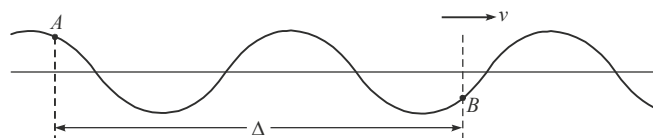


Figure 6.4

Here we can also mention that as phase travels at some speed as that of wave, we can find the phase difference between SHMs of any two medium particles if their path difference is known. Consider figure-6.4 It shows a part of a stretched string carrying a transverse wave travelling at a speed v toward right. Consider two medium particles A and B at a path difference Δ apart. Here we can say that if at an instant A has some instantaneous phase, then the same phase B will be having after a time $t = \Delta/v$ as wave has to travel a distance Δ from A to B . In this duration A will move forward in phase by an angle

$\omega \Delta / v$, which is now the phase lead of A with respect to B . Thus phase difference between two oscillating medium particles having a path difference Δ is given as

$$\phi = \frac{\omega}{v} \Delta = \frac{2\pi}{\lambda} \Delta = k\Delta \quad \dots (6.13)$$

Thus the relation in phase difference between the two points in a propagating wave is related to their path difference is given as

$$\text{Phase difference} = \frac{2\pi}{\lambda} \times \text{path difference} \quad \dots (6.14)$$

The above relation is extremely useful in solving problems of wave motion as well as in problems concerned with principle of superposition, we'll discuss in next section.

6.3 Sound Waves

Sound is type of longitudinal wave. In general majority of longitudinal waves are termed as sound waves. Sound is produced by a vibrating source, like when a gong of a bell is struck with a hammer, sound is produced. The vibrations produced by gong are propagated through air. Through air these vibrations reach to the ear and ear drum is set into vibrations and these vibrations are communicated to human brain. By touching the gong of bell by hand, we can feel the vibrations.

Prongs of tuning fork when hit on a rubber pad vibrates and it is used as the most common source of sound in laboratory experiments. Similarly vibrating string, air column, and any vibrating body produces sound.

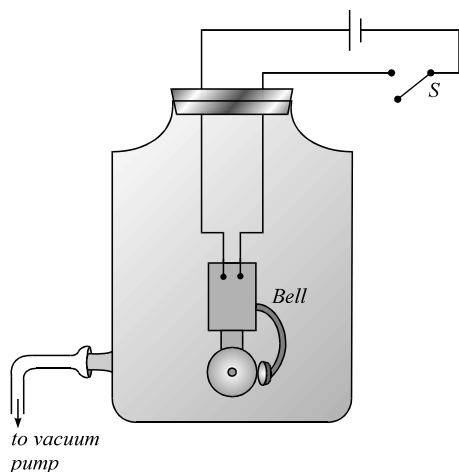


Figure 6.5

As being a mechanical wave sound requires material medium for its propagation. To understand this we discuss a simple experiment shown in figure-6.5. We take a jar and fix an electric

bell in it. The jar is closed by an air tight cork and a small opening at the bottom of jar is connected to a vacuum pump. When we close the switch S , the circuit is completed the bell rings and the sound is heard. Now the air in jar is removed gradually by the vacuum pump. We can listen that sound becomes fainter slowly. When pressure in jar becomes very low, almost no sound is heard, even the bell is vibrating at same pace. Hence we can say without material medium sound propagation is not possible. At normal atmospheric pressure and temperature it is observed that sound travels at speed 332 m/s.

Illustrative Example 6.1

A harmonic oscillation is represented by

$$y = 0.34 \cos (3000t + 0.74)$$

where y and t are in mm and second respectively. Deduce (i) amplitude, (ii) frequency and angular frequency, (iii) time period and (iv) initial phase.

Solution

We know that a simple harmonic oscillation can be represented by

$$y = A \cos (\omega t + \phi)$$

Where A is amplitude ω is angular frequency and ϕ is the initial phase. Comparing this equation with given equation, we have

(i) amplitude $A = 0.34 \text{ mm}$

(ii) angular frequency $\omega = 3000 \text{ Hz}$

Since $\omega = 2\pi n$ where n is the frequency

And thus we get $n = \frac{\omega}{2\pi} = \frac{3000}{2\pi} = \frac{1500}{\pi} \text{ Hz}$

(iii) We know that $T = \frac{1}{n} = \frac{1 \times \pi}{1500} = \frac{\pi}{1500} \text{ sec}$

(iv) Initial phase $\phi = 0.74 \text{ radian}$

Illustrative Example 6.2

An observer standing at sea coast observes 54 waves reaching the coast per minute. If the wavelength of the waves is 10 m, find the velocity. What type of waves did he observe ?

Solution

As 54 waves reach the shore per minute,

$$f = \frac{54}{60} = \frac{9}{10} \text{ Hz}$$

And as wavelength of waves is 10 m

$$v = f\lambda = \frac{9}{10} \times 10 = 9 \text{ m/s}$$

The waves on the surface of water are combined transverse and longitudinal called 'ripples'. In case of surface waves the particles of the medium move in elliptical paths in a vertical plane so that the vibrations are simultaneously back and forth and up and down.

Illustrative Example 6.3

A progressive wave of frequency 500 Hz is travelling with a velocity of 360 m/s. How far apart are two points 60° out of phase ?

Solution

We know that for a wave $v = f\lambda$,

$$\text{So } \lambda = \frac{v}{f} = \frac{360}{500} = 0.72 \text{ m}$$

Now as in a wave path difference is related to phase difference by the relation.

phase difference

$$\Delta\phi = \frac{2\pi}{\lambda} (\text{path difference } \Delta x)$$

Here, phase difference

$$\Delta\phi = 60^\circ = (\pi/180) \times 60 = (\pi/3) \text{ rad}$$

So path difference

$$\Delta x = \frac{\lambda}{2\pi} (\Delta\phi) = \frac{0.72}{2\pi} \times \frac{\pi}{3} = 0.12 \text{ m}$$

Illustrative Example 6.4

A person, standing between two parallel hills, fires a gun. He hears the first echo after 1.5 s and the second after 2.5 s. If the speed of sound is 332 m/s. Calculate the distance between the hills. When will he hear the third echo ?

Solution

Let the person P be at a distance x from hill H_1 and y from H_2 as shown in figure-6.6. The time interval between the original sound and echoes from H_1 and H_2 will be respectively

$$t_1 = \frac{2x}{v} \quad \text{and} \quad t_2 = \frac{2y}{v}$$

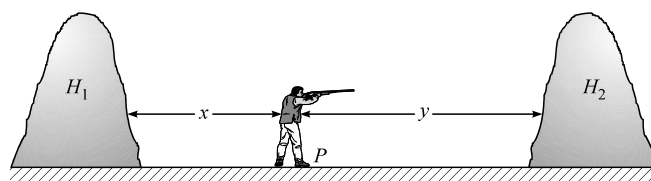


Figure 6.6

So the distance between the hills,

$$x + y = \frac{v}{2} [t_1 + t_2] = \frac{332}{2} [1.5 + 2.5] = 664 \text{ m}$$

Now as I echo will be from H_1 after time t_1 while II echo from H_2 after time t_2 , III echo will be produced due to reflection of sound of I echo from H_2 or of II echo from H_1 , thus we have

$$t_3 = t_1 + t_2 = 1.5 + 2.5 = 4 \text{ s}$$

Thus we can state that III echo will be produced after 4s and in it sound from both I and II echoes will reach simultaneously.

Illustrative Example 6.5

An aeroplane is going towards east at a speed of 510 km/h at a height of 2000 m. At a certain instant, the sound of the plane heard by a ground observer appears to be coming from a point vertically above him. Where is the plane at this instant ? Speed of sound in air = 340 m/s.

Solution

The situation is shown in figure-6.7. The sound reaching the ground observer P , was emitted by the plane when it was at the point Q vertically above his head. The time taken by the sound to reach the observer is

$$t = \frac{2000}{340} = \frac{100}{17} \text{ s}$$

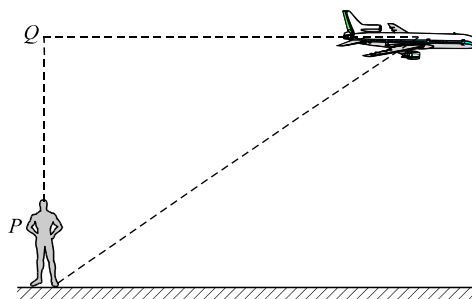


Figure 6.7

The distance moved by the plane during this period is

or

$$\begin{aligned} d &= vt \\ &= \frac{510 \times 5}{18} \times \frac{100}{17} \\ &= 833.33 \text{ m} \end{aligned}$$

Thus, the plane will be 833.33 m ahead of the observer on its line of motion when he hears the sound coming vertically to him.

Illustrative Example 6.6

A light pointer fixed to one prong of a tuning fork touches a vertical plate. The fork is set vibrating and the plate is allowed to fall freely. Eight complete oscillations are counted when the plate falls through 10 cm. What is the frequency of the fork ?

Solution

During the time the plate falls through 10 cm from rest, the tuning fork makes 8 vibrations.

From speed equation we get

$$h = \frac{1}{2} g t^2$$

$$0.1 = \frac{1}{2} \times 9.8 \times t^2 \quad \text{or} \quad t = \frac{1}{7} \text{ s}$$

In $\frac{1}{7}$ s the fork makes 8 vibrations. Therefore in 1s, the number of vibrations fork makes is

$$n = 8 \times 7 = 56$$

Thus frequency of fork is

$$n = 56 \text{ Hz}$$

Illustrative Example 6.7

(a) An ultrasonic transducer used in sonar produces a frequency of 40 kHz. If the velocity of the sound wave in seawater is 5050 ft/s, what is the wavelength ? (b) The transducer is made to emit a short burst of sound and is then turned off. The receiver is turned on. The pulse is reflected from a lurking submarine and received 5.0 s after it was first emitted. How far away is the submarine ?

Solution

$$(a) \quad v = n\lambda$$

$$\text{or} \quad 5050 = 40000 \lambda$$

$$\text{or} \quad \lambda = 0.126 \text{ ft}$$

$$(b) \quad v = \frac{2d}{t}$$

$$\text{or} \quad 5050 = \frac{2d}{5.0}$$

$$\text{or} \quad d = 12600 \text{ ft}$$

Illustrative Example 6.8

When a train of plane wave traverses a medium, individual particle execute periodic motion given by the equation

$$y = 4 \sin \frac{\pi}{2} \left(2t + \frac{x}{8} \right)$$

where the lengths are expressed in cm and time in second. Calculate the amplitude, wavelength, (i) the phase difference for two positions of the same particle which are occupied at time intervals 0.4 second apart and (ii) the phase difference at any given instant of two particles 12 cm apart.

Solution

The equation of a wave motion is given by

$$y = A \sin \frac{2\pi}{\lambda} (vt + x) \quad \dots (6.15)$$

$$\text{Here} \quad y = 4 \sin \frac{\pi}{2} \left(2t + \frac{x}{8} \right)$$

This equation can be written as

$$y = 4 \sin \frac{2\pi}{32} (16t + x) \quad \dots (6.16)$$

Comparing equation-(6.16) with equation-(6.15), we get amplitude $A = 4$ cm; wavelength $\lambda = 32$ cm; wave velocity $v = 16$ cm/s.

Here frequency is given as

$$n = \frac{v}{\lambda} = \frac{16}{32} = \frac{1}{2} = 0.5 \text{ Hz.}$$

(i) Phase of a particle at instant t_1 is given by

$$\phi_1 = \frac{\pi}{2} \left(2t_1 + \frac{x}{8} \right)$$

The phase at instant t_2 is given by

$$\phi_2 = \frac{\pi}{2} \left(2t_2 + \frac{x}{8} \right)$$

The phase difference is given as

$$\begin{aligned} \phi_1 - \phi_2 &= \frac{\pi}{2} \left[\left(2t_1 + \frac{x}{8} \right) - \left(2t_2 + \frac{x}{8} \right) \right] \\ &= \pi(t_1 - t_2) = \pi(0.4) \quad [\text{As } t_1 - t_2 = 0.4] \\ &= 180 \times 0.4 = 72^\circ \quad [\pi \text{ rad} = 180^\circ] \end{aligned}$$

(ii) Phase different at an instant between two particles with path difference Δ is

$$\begin{aligned} \phi &= \frac{2\pi}{\lambda} \times \Delta \\ &= \frac{2\pi}{32} \times 12 \quad [\text{As } \Delta = 12 \text{ cm}] \\ &= \frac{3\pi}{4} \end{aligned}$$

Illustrative Example 6.9

A rod runs midway between two parallel rows of buildings. A motorist moving with a speed of 36 kilometers per hour sounds the horn. He hears the echo one second after he has sounded the horn. Find the distance between the two rows of buildings. When will he hear the echo, a second time? Velocity of sound in air is 330 m/s.

Solution

The situation is shown in figure-6.8.

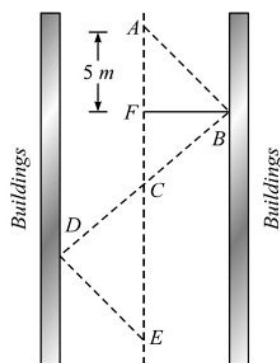


Figure 6.8

Given the velocity of motorist = 36 km/hour = 10 m/s.

So he travels a distance of 10 m in one second. He will hear the first echo after the sound had travelled 330 m through the least path i.e., reflected at point B.

From figure-6.8, we have

$$AB = \sqrt{(AF^2 + BF^2)} = \sqrt{(5^2 + x^2)}$$

$$\text{And } AB + BC = 2\sqrt{(5^2 + x^2)}$$

But we also have $AB + BC = 330$ m

$$\text{Thus } 2\sqrt{(5^2 + x^2)} = 330$$

$$\text{Solving we get } x = 164.9 \text{ m}$$

So the distance between the two rows of building $g = 2x = 329.8$ m. He will hear the second echo at E after the sound had travelled a further distance of 330 m, reflecting from the other row i.e., 2 seconds after the horn is sounded.

Illustrative Example 6.10

An engine approaches a hill with a constant speed. When it is at a distance of 0.9 km it blows a whistle, whose echo is heard by the driver after 5 s. If the speed of sound in air is 330 m/s, calculate the speed of the engine.

Solution

The situation is shown in figure-6.9.

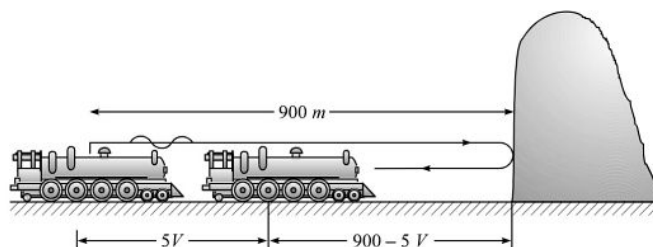


Figure 6.9

If the speed of the engine is v , the distance travelled by the engine in 5 s. will be $5v$. And hence, the distance travelled by sound in reaching the hill and coming back to the moving driver is given as

$$S = 900 + (900 - 5v) = (1800 - 5v)$$

Thus time interval between the original sound and its echo is

$$t = \frac{S}{v_{\text{sound}}}$$

$$5 = \frac{1800 - 5v}{330}$$

On solving we get $v = 30$ m/s

Illustrative Example 6.11

Given the equation for a wave in a string

$$y = 0.03 \sin(3x - 2t)$$

where y and x are in metres and t is in seconds, answer the following:

- At $t = 0$, what is the displacement at 0, 0.1 m.
- At $x = 0.1$ m, what is the displacement at $t = 0$ and $t = 0.2$ s.
- What is the equation for the velocity of oscillation of the particles of the string? What is the maximum velocity of oscillation?
- What is the velocity of propagation of the wave?

Solution

$$\text{Given that } y = 0.03 \sin(3x - 2t)$$

$$\text{(a) At } t = 0, y = 0.03 \sin 3x$$

$$\text{When } x = 0, y = 0.03 \sin 0 = 0$$

$$\text{When } x = 0.1, y = 0.03 \sin(0.3 \times \frac{180}{\pi})$$

or $y = 0.03 \sin \left(\frac{54}{\pi} \right)$

$$= 0.03 \times 0.2954$$

$$= 0.00886 \text{ m} = 8.86 \times 10^{-3} \text{ m}$$

(b) At $x = 0.1 \text{ m}, y = 0.03 \sin (0.3 - 2t)$

When $t = 0, y = 0.03 \sin (0.3)$

$$= 8.86 \times 10^{-3} \text{ m}$$

When $t = 0.2, y = 0.03 \sin (0.3 - 0.4)$

$$= -0.03 \sin 0.1$$

$$= -0.03 \sin (18/\pi)$$

$$= -0.03 \times 0.0999$$

$$= -2.997 \times 10^{-3} \text{ m}$$

(c) Velocity of particle

$$v_p = \frac{dy}{dt}$$

$$= -2 \times 0.03 \cos (3x - 2t)$$

$$= -0.06 \cos (3x - 2t)$$

This is maximum when $\cos (3x - 2t) = 1$

or maximum velocity $= 0.06 \text{ m/s} = 6 \times 10^{-2} \text{ m/s}$

(d) We know that the equation of a progressive wave is given by

$$y = a \sin 2\pi \left(\frac{x}{\lambda} - \frac{t}{T} \right)$$

Comparing the given equation, with this equation, we have

$$\frac{2\pi}{\lambda} = 3 \quad \text{and} \quad \frac{2\pi}{T} = 2$$

or $\lambda = \frac{2\pi}{3} \quad \text{and} \quad 2\pi n = 2 \quad \text{or} \quad n = \frac{1}{\pi}$

Now $v = n\lambda = \frac{1}{\pi} \times \frac{2\pi}{3} = 0.667 \text{ m/s}$

Practice Exercise 6.1

(i) The displacement of a wave is represented by

$$y = 0.25 \times 10^{-3} \sin (500t - 0.25x)$$

where y , t and x are in cm, sec and metre respectively. Deduce

(a) amplitude (b) time period (c) angular frequency (d) wavelength (e) amplitudes of particle velocity and (f) particle acceleration.

[(a) $0.25 \times 10^{-3} \text{ cm}$ (b) $\frac{\pi}{250} \text{ s}$ (c) 500 rad/s

(d) $8\pi \text{ cm}$ (e) 0.125 cm/s (f) 6.25 cm/s^2]

(ii) From a cloud at an angle of 30° to the horizontal we hear the thunder clap 8 s after seeing the lighting flash. What is the height of the cloud above the ground if the velocity of sound in air is 330 m/s ?

[1.320 km]

(iii) A wave is expressed by the equation $y = 0.5 \sin \pi (0.01x - 3t)$, where y and x are in m and t in s . Find the speed of propagation.

[300 m/s]

(iv) A wave propagates on a string in the positive x -direction at a velocity v . The shape of the string at $t = t_0$ is given by $g(x, t_0) = A \sin (x/a)$. Write the wave equation for a general time t .

$$[f(x, t) = A \sin \left(\frac{x - v(t - t_0)}{a} \right)]$$

(v) A man seeing a lightning starts counting seconds, until he hears thunder. He then claims to have found an approximate but simple rule that if the count of second is divided by an integer, the result directly gives, in km, the distance of the lightning source. What is the integer? (Velocity of sound in air $= 330 \text{ m/s}$)

[3]

(vi) A man stands before a large wall at a distance of 50.0 m and claps his hands at regular intervals. Initially, the interval is large. He gradually reduces the interval and fixes it at a value when the echo of a clap merges with the next clap. If he has to clap 10 times during every 3 seconds, find the velocity of sound in air.

[333 m/s]

(vii) A simple harmonic wave has the equation $y = 0.30 \sin (314t - 1.57x)$ where t , x and a are in seconds, metres, and centimeters respectively. Find the frequency and wavelength of this wave. Another wave has the equation $y' = 0.10 \sin (314t - 1.57x + 1.57)$. Deduce phase difference and ratio of intensities of the two wave.

[90° , 9/1]

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Wave Motion

Module Number - 1 to 13

(viii) A plane elastic wave $y = A \cos (\omega t - k x)$ propagates in a medium K . Find the equation of this wave in a reference frame K' moving in the positive direction of x -axis with a constant velocity V relative to the medium K . Wave speed in medium K is v .

$$\left[A \cos \left[\omega \left(1 - \frac{V}{v} \right) t - kx' \right] \right]$$

(ix) The equation of a travelling sound wave is $y = 6.0 \sin (600 t - 1.8 x)$ where y is measured in 10^{-5} m, t in second and x in metre. (a) Find the ratio of the displacement amplitude of the particles to the wavelength of the wave. (b) Find the ratio of the velocity amplitude of the particles to the wave speed.

$$[(a) 1.71 \times 10^{-5}, (b) 1.1 \times 10^{-4}]$$

(x) The equation of a travelling plane sound wave has the form $y = 50 \cos (1800 t - 5.3 x)$, where y is expressed in micrometers, t in seconds, and x in meters. Find :

- the ratio of the displacement amplitude, with which the particles of medium oscillate, to the wavelength;
- the velocity oscillation amplitude of particles of the medium and its ratio to the wave propagation velocity;
- the ratio of oscillation amplitude of relative deformation of the medium to the velocity oscillation amplitude of particles of the medium.

$$[(a) 4.22 \times 10^{-5}, (b) 0.09 \text{ m/s}, 2.65 \times 10^{-4}, (c) 2.94 \times 10^{-3} \text{ m}]$$

(xi) A man standing in front of a mountain at a certain distance beats a drum at regular intervals. The drumming rate is gradually increased and he finds that the echo is not heard distinctly when the rate becomes 40 per minute. He then moves nearer to the mountain by 90 metres and finds that the echo is again not heard when the drumming rate becomes 60 per minute. Calculate

- the distance between the mountain and the initial position of the man,
- the velocity of sound.

$$[270 \text{ m and } 360 \text{ m/s}]$$

(xii) A sound wave of frequency 100 Hz is travelling in air. The speed of sound in air is 350 m/s. (a) By how much is the phase changed at a given point in 2.5 ms ? (b) What is the phase difference at a given instant between two points separated by a distance of 10.0 cm along the direction of propagation ?

$$[(a) \pi/2, (b) 2\pi/35]$$

6.4 Velocity of a Wave

We have discussed that wave velocity is the velocity of energy propagation in the direction of wave motion. Wave velocity of a wave is also termed as its phase velocity as the phase of

medium particles also travels along propagation direction with the wave with this speed. In different type of waves its velocity depends on medium properties in different manner, we will now discuss in detail.

6.4.1 Wave Speed of Transverse Waves on a Stretched String

When on a stretched string shown in figure-6.10, a transverse jerk is given, a pulse is created as shown in figure-6.10(b) and (c) which travels toward right with a wave speed v as shown. We start our analysis by looking at the pulse carefully as shown in enlarged view of figure-6.10(d). For convenience of our analysis we chose a reference frame in which the pulse remains stationary or we assume that our frame is moving along with the pulse at speed v .

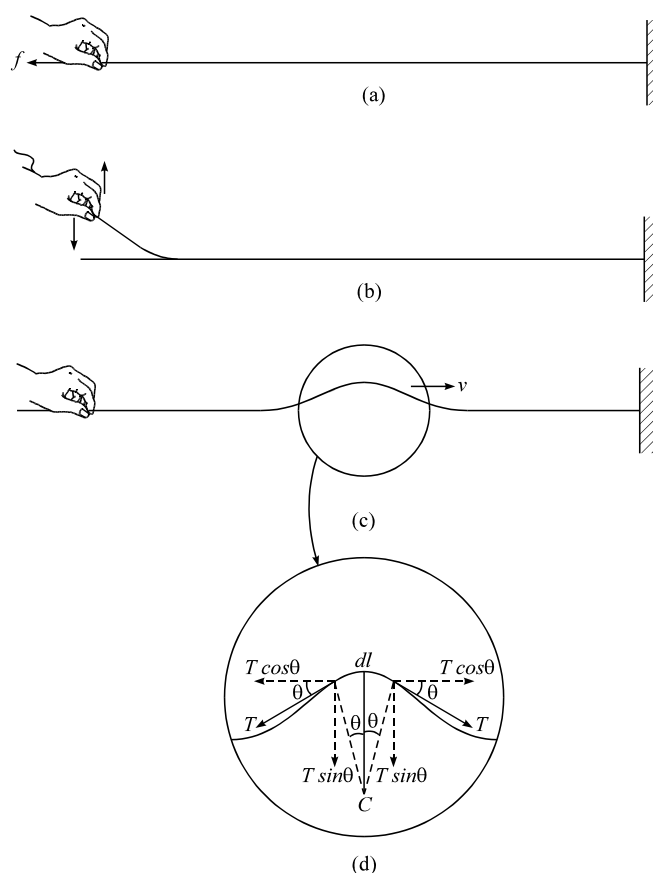


Figure 6.10

Now consider a small element of length dl on this pulse as shown. This element is forming an arc, say of radius R with centre at C and subtending an angle 2θ at C . We can see that two tensions T are acting on the edges of dl along tangential directions as shown. The horizontal components of these tensions cancels each other, but the vertical components add to form a radial restoring force in downward direction, which is given as

$$F_R = 2T \sin \theta$$

$$\begin{aligned} &\simeq 2T\theta \quad [\text{As } \sin \theta \simeq \theta] \\ &= T \frac{dl}{R} \quad [\text{As } 2\theta = \frac{dl}{R}] \quad \dots (6.17) \end{aligned}$$

If μ be the mass per unit length of the string, the mass of this element is given as

$$dm = \mu dl \quad \dots (6.18)$$

In our reference frame if we look at this element, it appears to be moving toward left with speed v then we can say that the acceleration of this element in our reference frame is

$$a = \frac{v^2}{R} \quad \dots (6.19)$$

Now from equation-(6.17), (6.18) and (6.19) we have

$$F_R = \frac{dmv^2}{R}$$

$$\text{or} \quad T \frac{dl}{R} = \frac{(\mu dl)v^2}{R}$$

$$\text{or} \quad v = \sqrt{\frac{T}{\mu}} \quad \dots (6.20)$$

Thus the speed of a transverse wave along a stretched string depends only on the tension and the linear mass density of the string and not on the wave characteristics.

Sometimes when wave amplitude becomes too large, the wave velocity depends on the amplitude also. This discussion in detail is beyond the scope of this book so students are advised, not to use this fact while studying waves on a stretched string.

The above expression of velocity of transverse wave on a stretched string given in equation-(6.20) can also be deduced from Newton's Second Law as discussed in next section.

6.4.2 Wave Speed on a Stretched String from Newton's Second Law

Newton's second law predicts that waves can occur in a medium due to a linear elastic restoring force. Consider an element of length dx of string in which a wave is propagating in positive x -direction in the figure-6.11 shown.

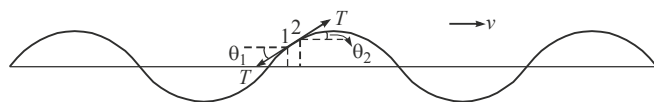


Figure 6.11

As tension in the string can be considered uniform, the restoring force on this element dx toward its mean position for its oscillation is given as

$$F_R = T \sin \theta_1 - T \sin \theta_2 \quad \dots (6.21)$$

As here θ_1 and θ_2 are very small, we can use

$$\sin \theta_1 \simeq \tan \theta_1 = \left. \frac{\partial y}{\partial x} \right|_1$$

[Slope of displacement curve at point 1]

$$\sin \theta_2 \simeq \tan \theta_2 = \left. \frac{\partial y}{\partial x} \right|_2$$

[Slope of displacement curve at point 2]

From equation-(6.21) we have

$$F_R = +T \left[\left. \frac{\partial y}{\partial x} \right|_1 - \left. \frac{\partial y}{\partial x} \right|_2 \right] \quad \dots (6.22)$$

Here $\left. \frac{\partial y}{\partial x} \right|_2 - \left. \frac{\partial y}{\partial x} \right|_1$ is the change in slopes of displacement curve for a differential change in x by dx between points 1 and 2. Thus we can write

$$\left. \frac{\partial y}{\partial x} \right|_2 - \left. \frac{\partial y}{\partial x} \right|_1 = \frac{\partial^2 y}{\partial x^2} dx \quad \dots (6.23)$$

Thus from equation-(6.22) and (6.23)

$$F_R = +T \frac{\partial^2 y}{\partial x^2} dx \quad \dots (6.24)$$

If the element of string has mass dm , we have

$$dm = \mu dx \quad \dots (6.25)$$

Where μ is the mass per unit length of string. Now from Newton's Second Law for this element, we can write

$$F_R = dm \frac{\partial^2 y}{\partial t^2} \quad \dots (6.26)$$

From equation-(6.24), (6.25) and (6.26) we have

$$T \frac{\partial^2 y}{\partial x^2} dx = + (\mu dx) \frac{\partial^2 y}{\partial t^2}$$

$$\text{or} \quad \frac{\partial^2 y}{\partial x^2} = \frac{\mu}{T} \frac{\partial^2 y}{\partial t^2} \quad \dots (6.27)$$

Equation-(6.27) is the wave equation, we have already discussed in previous sections of the chapter. Comparing it with equation-(6.12), we get the wave velocity as

$$v = \sqrt{\frac{T}{\mu}}$$

6.4.3 Velocity of Sound/Longitudinal Waves in Solids

Consider a section AB of medium as shown in figure-6.12(a) of cross-sectional area S . Let A and B be two cross section as shown. Let in this medium sound propagation is from left to right. If wave source is at origin O and when it oscillates, the oscillations at that point propagate along the rod.

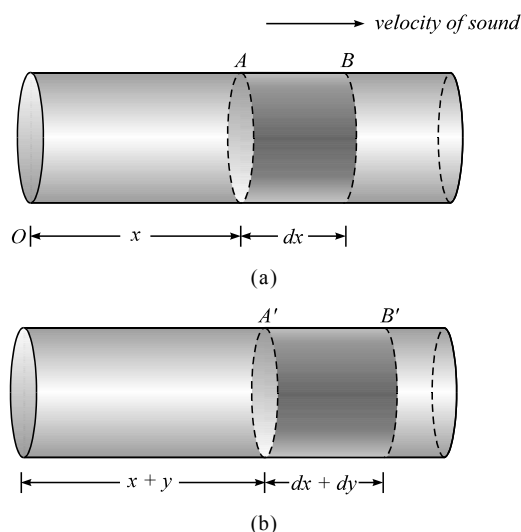


Figure 6.12

Here we say an elastic wave has propagated along the rod with a velocity determined by the physical properties of the medium. Due to oscillations say a force F is developed at every point of medium which produces a stress in rod and is the cause of strain or propagation of disturbance along the rod. This stress at any cross-sectional area can be given as

$$\text{Stress } \mathcal{F}_t = \frac{F}{S} \quad \dots (6.28)$$

If we consider the section AB of medium at a general instant of time t . The end A is at a distance x from O and B is at a distance $x + dx$ from O . Let in time dt due to oscillations, medium particles at A are displaced along the length of medium by y and those at B by $y + dy$. The resulting positions of section are A' and B' shown in figure-6.12(b). Here we can say that the section AB is deformed (elongated) by a length dy . Thus strain produced in it is

Strain in section AB

$$E = \frac{dy}{dx} \quad \dots (6.29)$$

If Young's modulus of the material of medium is Y , we have Young's Modulus

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{\mathcal{F}_t}{E}$$

From equation-(6.28) and (6.29), we have $Y = \frac{F/S}{dy/dx}$

$$\text{or} \quad F = YS \frac{dy}{dx} \quad \dots (6.30)$$

If net force acting of section AB is dF then it is given as

$$dF = dma \quad \dots (6.31)$$

Where dm is the mass of section AB and a be its acceleration, which can be given as for a medium of density ρ .

$$dm = \rho S dx \quad \text{and} \quad a = \frac{d^2 y}{dt^2}$$

From equation-(6.31), we have

$$dF = (\rho S dx) \frac{d^2 y}{dt^2}$$

$$\text{or} \quad \frac{dF}{dx} = \rho S \frac{d^2 y}{dt^2} \quad \dots (6.32)$$

From equation-(6.30) on differentiating w.r. to x , we can write

$$\frac{dF}{dx} = YS \frac{d^2 y}{dx^2} \quad \dots (6.33)$$

From equation-(6.32) and (6.33) we get

$$\frac{d^2 y}{dx^2} = \left(\frac{Y}{\rho} \right) \frac{d^2 y}{dx^2} \quad \dots (6.34)$$

Equation-(6.34) is the different form of wave equation, comparing it with equation-(6.12) we get the wave velocity in the medium can be given as

$$v = \sqrt{\frac{Y}{\rho}} \quad \dots (6.35)$$

6.4.4 Wave Velocity of Longitudinal Waves in Fluid/Gas

Similar to the case of a solid in fluid, instead of Young's Modulus we use Bulk modulus of the medium hence the velocity of longitudinal waves in a fluid medium is given as

$$v = \sqrt{\frac{B}{\rho}} \quad \dots (6.36)$$

Where B is the Bulk modulus of medium.

For a gaseous medium bulk modulus is defined as

$$B = \frac{dP}{(-dV/V)}$$

$$\text{or} \quad B = -V \frac{dP}{dV} \quad \dots (6.37)$$

6.4.5 Newton's Formula for Velocity of Sound in Gases

Newton assumed that during sound propagation temperature of medium remains same hence he stated that propagation of sound in a gaseous medium is an isothermal phenomenon, thus Boyal's law can be applied in the process. So for a section of medium we use

$$PV = \text{constant}$$

Differentiating we get

$$P dV + V dP = 0$$

$$\text{or} \quad -V \frac{dP}{dV} = P$$

or bulk modulus of medium can be given as

$$B = P \quad (\text{Pressure of medium})$$

Newton found that during isothermal propagation of sound in a gaseous medium, bulk modulus of medium is equal to the pressure of the medium, hence sound velocity in a gaseous medium can be given as

$$v = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{P}{\rho}} \quad \dots (6.38)$$

From gas law we have

$$\frac{P}{\rho} = \frac{RT}{M} \quad \dots (6.39)$$

From (6.38) & (6.39) we have

$$v = \sqrt{\frac{RT}{M}} \quad \dots (6.40)$$

From the expression in equation-(6.38) if we find the sound velocity in air at normal temperature and atmospheric pressure we have

Normal atmospheric pressure is

$$P = 1.01 \times 10^5 \text{ Pa}$$

Density of air at NTP is

$$\rho = 1.293 \text{ kg/m}^3$$

Now from equation-(6.38)

$$\begin{aligned} v &= \sqrt{\frac{P}{\rho}} \\ v &= \sqrt{\frac{1.01 \times 10^5}{1.293}} \\ &= 279.48 \text{ m/s} \end{aligned}$$

But the experimental value of velocity of sound determined from various experiments gives the velocity of sound at NTP, 332 m/s. Therefore there is a difference of about 52 m/s between the theoretical and experimental values. This large difference can not be attributed to the experimental errors. Newton was unable to explain error in his formula. This correction was explained by a French Scientist Laplace.

6.4.6 Laplace Correction

Laplace explained that when sound waves propagate in a gaseous medium. There is condensation and rarefactions in the particles of medium. Where there is condensation, particles come near to each other and are heated up, where there is rarefaction, medium expands and there is fall of temperature. Therefore, the temperature of medium at every point does not

remain constant so the process of sound propagation is not isothermal. The total quantity of heat of the system as a whole remains constant. Medium does not gain or lose any heat to the surrounding. Thus in a gaseous medium sound propagation is an adiabatic process. For adiabatic process the relation in pressure and volume of a section of medium can be given as

$$PV^\gamma = \text{constant} \quad \dots (6.41)$$

Here $\rho = \frac{C_p}{C_v}$, ratio of specific heats of the medium.

Differentiating equation-(6.41) we get,

$$dPV^\gamma + \gamma V^{\gamma-1} dV P = 0$$

$$\text{or} \quad dP + \gamma \frac{P dV}{V} = 0$$

$$\text{or} \quad -V \frac{dP}{dV} = \gamma P$$

Bulk modulus of medium

$$B = \gamma P \quad \dots (6.42)$$

Thus Laplace found that during adiabatic propagation of sound, the Bulk modulus of gaseous mediums is equal to the product of ratio of specific heats and the pressure of medium. Thus velocity of sound propagation can be given as

$$v = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{\gamma P}{\rho}} \quad \dots (6.43)$$

$$\text{From gas law} \quad v = \sqrt{\frac{\gamma RT}{M}} \quad \dots (6.44)$$

From equation-(6.43) if we find sound velocity in air at NTP, we have

$$\text{Normal atmospheric pressure } P = 1.01 \times 10^5 \text{ Pa}$$

$$\text{Density of air at NTP } \rho = 1.293 \text{ kg/m}^3$$

$$\text{Ratio of specific heats of air } \gamma = \frac{C_p}{C_v} = 1.42$$

Using equation-(6.43), we get

$$\begin{aligned} v &= \sqrt{\frac{\gamma P}{\rho}} \\ &= \sqrt{\frac{1.42 \times 1.01 \times 10^5}{1.293}} = 333.04 \text{ m/s} \end{aligned}$$

This value is in agreement with experimental value.

6.4.7 Effect of Temperature on Velocity of Sound

From equation-(6.43) we have velocity of sound propagation in a gaseous medium as

$$v = \sqrt{\frac{\gamma RT}{M}}$$

For a given gaseous medium γ , R and M remains constant, thus velocity of sound is directly proportional to square root of absolute temperature of the medium. Thus

$$v \propto \sqrt{T} \quad \dots (6.45)$$

If at two different temperatures T_1 and T_2 , sound velocities in medium are v_1 and v_2 then from equation-(6.45), we have

$$\frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} \quad \dots (6.46)$$

6.4.8 Effect of Pressure on Velocity of Sound

We know from gas law

$$\frac{P}{\rho} = \frac{RT}{M}$$

If temperature of a medium remains constant then on changing pressure, density of medium proportionally changes so that the ratio $\frac{P}{\rho}$ remains constant.

Hence if in a medium, $T = \text{constant}$

Then, $\frac{P}{\rho} = \text{constant}$

Thus velocity of sound, $v = \sqrt{\frac{\gamma P}{\rho}} = \text{constant}$

Therefore, the velocity of sound in air or in a gas is independent of change in pressure.

6.4.9 Effect of Humidity on Velocity of Sound

The density of water vapour at NTP is 0.8 kg/m^3 whereas the density of dry air at NTP is 1.293 kg/m^3 . Therefore water vapour has a density less than the density of dry air. As atmospheric pressure remains approximately same, the velocity of sound is more in moist air than the velocity of sound in dry air

$$v_{\text{moist air}} > v_{\text{dry air}} \quad \dots (6.47)$$

6.4.10 Effect of Wind on Velocity of Sound

If wind is blowing in the direction of propagation of sound, it will increase the velocity of sound. On the other hand if wave propagation is opposite to the direction of propagation of wind,

wave velocity is decreased. If wind blows at speed v_w then sound velocity in the medium can be given as

$$\vec{v} = \vec{v}_s + \vec{v}_w \quad \dots (6.48)$$

Where $\vec{v}_s$ is the velocity of sound in still air.

6.4.11 Velocity of Longitudinal Waves in a Slinky Spring

Suppose a periodic disturbance is produced along a stretched slinky spring as shown in figure-6.13(a) and (b). The resulting wave consists in a series of compressions and expansions that propagate along the spring. In such a case the velocity of these longitudinal wave along the length of spring is given as

$$v = \sqrt{\frac{kL}{\mu}} \quad \dots (6.49)$$

Figure-3.16 shows how by holding a spring when hand oscillates forward and backward due to inertia of spring wave propagation take place.

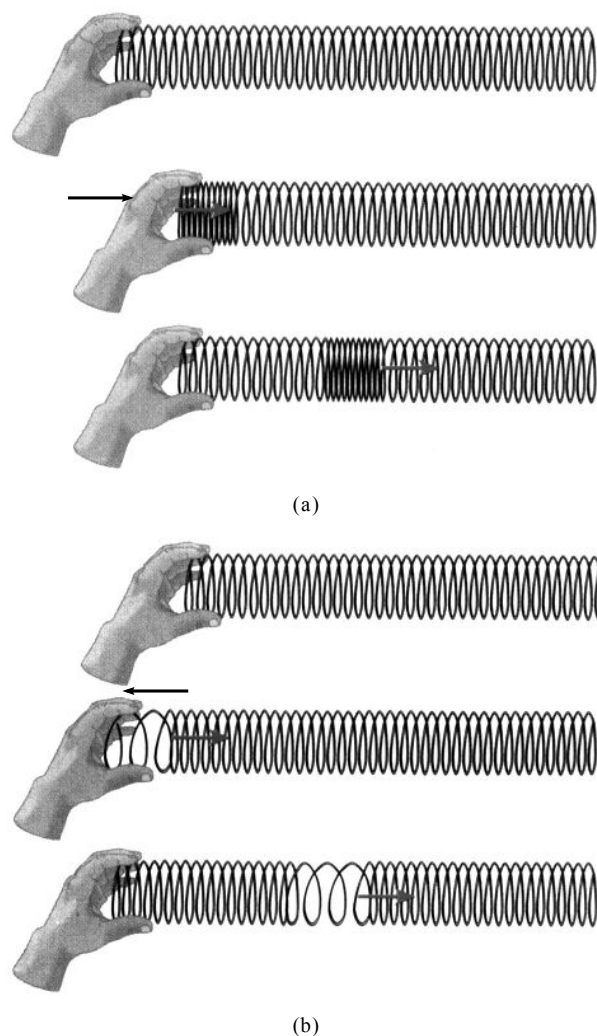


Figure 6.13

Here k is the force constant of the spring.

L is the length of the unstratched spring.

and μ is the linear density of the unstratched spring.

6.4.12 Velocity of Torsional Waves in a Rod

If a rod is damped at one end and other end is periodically twisted in angular oscillations, torsional waves are produced in the rod as shown in figure-6.14.

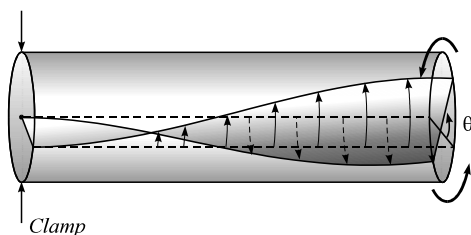


Figure 6.14

The velocity of torsional waves are given as

$$v = \sqrt{\frac{G}{\rho}} \quad \dots (6.50)$$

Where G is the shear modulus of the material of rod. Detailed analysis of this section is beyond the scope of this book, so students should keep the expression in equation-(6.50) as it is in their mind.

6.4.13 Surface Waves in a Liquid

Waves on the surface of a liquid are the most familiar kinds of waves, they are the waves we observe in the ocean and lakes, or simply when we drop a stone into a pond. Compared to previous cases here analytical aspect is more complex and will be omitted here. We will consider here only a descriptive discussion.

The undisturbed surface of a liquid is plane and horizontal. A disturbance on the surface produces a displacement of all molecules directly underneath the surface as a result of intermolecular forces as shown in figure-6.15. The particles moves horizontally as well as vertically and the resulting motion of the particles is elliptical or approximately circular as shown in figure. The amplitude of horizontal and vertical displacements of a volume element of a fluid varies, in general, with depth. The molecules at the bottom do not suffer any vertical displacement due to the pressure of weight of liquid body.

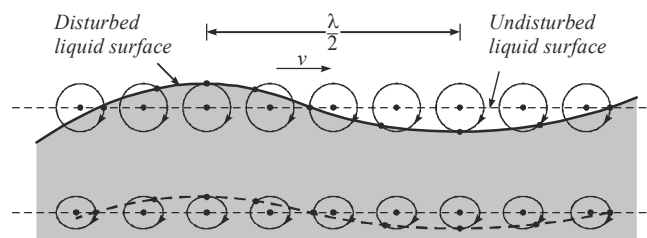


Figure 6.15

At the surface of liquid other forces are playing the role in oscillations. One force is the surface tension of the liquid which gives an upward or downward force on an element of the surface similar to the case of a string. Another force is the weight of the liquid above the undisturbed level of the liquids surface. The waves produced on the surface of liquid propagates with a velocity v given as

$$v = \sqrt{\frac{g\lambda}{2\pi} + \frac{2\pi\sigma}{\rho\pi}} \quad \dots (6.51)$$

Where $g \rightarrow$ acceleration due to gravity

$\sigma \rightarrow$ surface tension of the liquid

$\rho \rightarrow$ density of the liquid

$\lambda \rightarrow$ wavelength of wave.

Here we can see that the velocity of propagation of surface wave depends on the wavelength λ of the waves. Students should note that this expression in equation-(6.51) is valid only when the depth of liquid is large compared to the wavelength λ .

If in a case wavelength λ is large enough so the second term in the expression of equation-(6.51) can be neglected then we have

$$v = \sqrt{\frac{g\lambda}{2\pi}} \quad \dots (6.52)$$

The waves in this case are called gravity waves. In this approximation we can see in equation-(6.52), velocity of propagation is independent from the nature of liquid, since no factor related to the liquid appears in expression of equation-(6.52). Here we can say the longer the wavelength is, faster is the wave propagation. Such types of waves are produced by strong winds over a liquid surface, generally in oceans.

When the wavelength λ is very small so that the first term in equation-(6.51) can be neglected, the wave velocity of surface waves can be given as

$$v = \sqrt{\frac{2\pi\sigma}{\rho\pi}} \quad \dots (6.53)$$

These waves are called ripples or capillary waves. These are the waves observed when a very gentle wind blows, over the surface of water or when a container is subject to vibrations of high frequency. In this case we can see from equation-(6.53), the longer the wavelength slower the wave propagation.

Now we take some examples on wave velocity in different mediums.

Illustrative Example 6.12

Calculate the increase in velocity of sound for 1°C rise of temperature, if the velocity of sound at 0°C is 332 m/s.

Solution

We know that
$$\frac{v_t}{v_0} = \sqrt{\left(\frac{T_t}{T_0}\right)} = \sqrt{\left(\frac{273+t}{273}\right)}$$

or
$$v_t = v_0 \left(1 + \frac{t}{273}\right)^{1/2}$$

$$= v_0 \left(1 + \frac{t}{2 \times 273}\right) = v_0 \left(1 + \frac{t}{546}\right)$$

So,
$$v_t - v_0 = v_0 \times \frac{t}{546}$$

Thus increase in velocity per °C is

$$= \frac{v_0 \times 1}{546}$$

$$= \frac{332}{546} = 0.61 \text{ m/s}$$

Illustrative Example 6.13

A 4.0 kg block is suspended from the ceiling of an elevator through a string having a linear mass density of $19.2 \times 10^{-3} \text{ kg/m}$. Find the speed (with respect to the string) with which a wave pulse can proceed on the string if the elevator accelerate up at the rate of 2.0 m/s^2 . Take $g = 10 \text{ m/s}^2$.

Solution

As elevator accelerates up at 2 m/s^2 , tension in the string is

$$T = m(g + a)$$

$$= 4(10 + 2) = 48 \text{ N}$$

Linear mass density of string is

$$\mu = 19.2 \times 10^{-3} \text{ kg/m}$$

Speed of transverse waves on string is

$$v = \sqrt{\frac{T}{\mu}}$$

$$= \sqrt{\frac{48}{19.2 \times 10^{-3}}}$$

$$= 50 \text{ m/s}$$

Illustrative Example 6.14

At what temperature will the speed of sound in air becomes double of its value at 0°C ?

Solution

We know that
$$\frac{v_t}{v_0} = \sqrt{\left(\frac{T_t}{T_0}\right)}$$

Here
$$v_t = 2 v_0$$

or
$$\frac{2 v_0}{v_0} = \sqrt{\left(\frac{T_t}{273}\right)}$$

or
$$4 = \frac{T_t}{273}$$

or
$$T_t = 4 \times 273 = 1092 \text{ K}$$

$$t = 1092 - 273 = 819^\circ\text{C}$$

Illustrative Example 6.15

The planet Jupiter has an atmosphere mainly of methane at a temperature – 130°C. Calculate the velocity of sound on this planet assuming γ for the mixture to be 1.3 (Gas constant $R = 8.3 \text{ joules/mol } ^\circ\text{C}$).

Solution

We know that
$$v = \sqrt{\left(\frac{\gamma P}{d}\right)} = \sqrt{\left(\frac{\gamma R T}{M}\right)}$$

According to the given problem,

$$\gamma = 1.30, R = 8.3 \times 10^3 \text{ J/kg mol K,}$$

and
$$T = -130^\circ\text{C} = -130 + 273 = 143 \text{ K}$$

$$M = \text{Molecular weight of methane (CH}_4\text{)}$$

$$= 12 + 4 = 16$$

Hence
$$v = \sqrt{\left(\frac{1.3 \times 8.3 \times 10^3 \times 143}{16}\right)}$$

$$= 311 \text{ m/s}$$

Illustrative Example 6.16

How long will it take sound waves to travel the distance l between the points A and B if the air temperature between them varies linearly from T_1 to T_2 ? The velocity of sound propagation in air is equal to $v = \alpha\sqrt{T}$, where α is a constant.

Solution

For linear variation of temperature, we can write temperature at a distance x from point A is

$$T_x = T_1 + \frac{T_2 - T_1}{l}x$$

Thus velocity of sound at this point is given as

$$v = \alpha \sqrt{T_1 + \left(\frac{T_2 - T_1}{l}\right)x}$$

$$\text{or } \frac{dx}{dt} = \alpha \sqrt{T_1 + \left(\frac{T_2 - T_1}{l}\right)x}$$

$$\text{or } \alpha \frac{dx}{\sqrt{T_1 + \left(\frac{T_2 - T_1}{l}\right)x}} = dt$$

Integrating the above expression within proper limits, we get

$$\text{or } \int_0^l \frac{dx}{\alpha \sqrt{T_1 + \left(\frac{T_2 - T_1}{l}\right)x}} = \int_0^t dt$$

$$\frac{2l}{\alpha(T_2 - T_1)} \left[\sqrt{T_1 + \frac{T_2 - T_1}{l}x} \right]_0^l = t$$

$$\text{or } t = \frac{2l}{\alpha(T_2 - T_1)} [\sqrt{T_2} - \sqrt{T_1}]$$

$$\text{or } t = \frac{2l}{\alpha[\sqrt{T_2} + \sqrt{T_1}]}$$

Alternative solution :

$$\text{Given that } v = \alpha\sqrt{T}$$

$$\text{or velocity at point } A \text{ is } v_1 = \alpha\sqrt{T_1}$$

$$\text{Velocity at point } B \text{ is } v_2 = \alpha\sqrt{T_2}$$

$$\text{The average velocity } \bar{v} = \frac{v_1 + v_2}{2}$$

$$\text{or } \bar{v} = \frac{\alpha(\sqrt{T_1} + \sqrt{T_2})}{2}$$

Distance between A and $B = l$

Thus time from A to B is

$$t = \frac{\text{distance}}{\text{velocity}} = \frac{l}{[\alpha(\sqrt{T_1} + \sqrt{T_2})/2]} \\ = \frac{2l}{\alpha(\sqrt{T_1} + \sqrt{T_2})}$$

Illustrative Example 6.17

The speed of a transverse wave, going on a wire having a length 50 cm and mass 5.0 kg, is 80 m/s. The area of cross-section of the wire is 1.0 mm^2 and its Young's modulus is $16 \times 10^{11} \text{ N/m}^2$. Find the extension of the wire over its natural length.

Solution

The linear mass density is

$$\mu = \frac{5 \times 10^{-3}}{50 \times 10^{-2}} = 1.0 \times 10^{-2} \text{ kg/m}$$

The wave speed is

$$v = \sqrt{\frac{T}{\mu}}$$

Thus, the tension is

$$T = \mu v^2 \\ = (1.0 \times 10^{-2}) \times 6400 = 64 \text{ N}$$

The Young's modulus is given by

$$Y = \frac{T/A}{\Delta L/L}$$

The extension is, therefore,

$$\Delta L = \frac{TL}{AY} \\ = \frac{64 \times 0.50}{1.0 \times 10^{-6} \times 16 \times 10^{11}} = 0.02 \text{ mm}$$

Illustrative Example 6.18

Two blocks each having a mass of 3.2 kg are connected by a wire CD and the system is suspended from the ceiling by another wire AB (figure-6.16). The linear mass density of the wire AB is 10 g/m and that of CD is 8 g/m. Find the speed of a transverse wave pulse produced in AB and in CD .

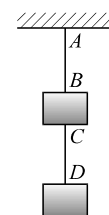


Figure 6.16

Solution

Tension in string AB is

$$T_{AB} = 6.4 \text{ g} = 64 \text{ N}$$

Thus speed of transverse waves in string AB is

$$v_{AB} = \sqrt{\frac{T_{AB}}{\mu_{AB}}} = \sqrt{\frac{64}{10 \times 10^{-3}}} \\ = \sqrt{6400} = 80 \text{ m/s}$$

Tension in strong CD is

$$T = 3.2 \text{ g} = 32 \text{ N}$$

Thus speed of transverse waves in string CD is

$$v_{CD} = \sqrt{\frac{T_{CD}}{\mu_{DC}}} = \sqrt{\frac{32}{8 \times 10^{-3}}} = \sqrt{4000} \\ = 63.24 \text{ m/s}$$

Illustrative Example 6.19

A heavy but uniform rope of length L is suspended from a ceiling. (a) Write the velocity of a transverse wave travelling on the string as a function of the distance from the lower end. (b) If the rope is given a sudden sideways jerk at the bottom, how long will it take for the pulse to reach the ceiling? (c) A particle is dropped from the ceiling at the instant the bottom end is given the jerk. Where will the particle meet the pulse?

Solution

Let m be the mass of the hanging rope, then its linear mass density will be

$$\mu = \frac{M}{L} \quad \dots (6.54)$$

(a) At a distance x above the lower end if we consider a cross section A then tension at point A will be due to the weight of the lower part and it is given as

$$\text{Tension at } A \text{ is, } T = \frac{M}{L} xg \quad \dots (6.55)$$

Now velocity of transverse waves at point A is given as

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{\frac{M}{L} xg}{M/L}} = \sqrt{xg} \quad \dots (6.56)$$

(b) If a jerk is given at the lower end of rope, it propagation in upward direction and its velocity at a distance x from lower end is given by equation-(6.56). We can find the time taken by pulse of jerk to reach the top by integration expression in equation-(6.56) as

$$\frac{dx}{dt} = \sqrt{xg}$$

$$\text{or } \frac{dx}{\sqrt{x}} = \sqrt{g} dt$$

Integrating this expression in proper limits we get

$$\int_0^L \frac{dx}{\sqrt{x}} = \int_0^t \sqrt{g} dt$$

$$\text{or } [2\sqrt{x}]_0^L = \sqrt{g} t$$

$$\text{or } t = 2\sqrt{\frac{L}{g}}$$

(c) When a particle is dropped from the top it falls by a distance $(L - x)$ in time t . When it will meet the pulse and if pulse has

travelled a distance x , Thus time taken by pulse to travel a distance x from bottom is

$$t = 2\sqrt{\frac{x}{g}}$$

In this time the distance fallen by particle in its free fall motion is

$$(L - x) = \frac{1}{2} gt^2$$

$$\text{or } L - x = \frac{1}{2} g \left(2\sqrt{\frac{x}{g}} \right)^2$$

$$\text{or } L - x = 2x$$

$$\text{or } x = \frac{L}{3}$$

Thus particle and the pulse meet at a distance $\frac{L}{3}$ from the bottom.

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Wave Motion

Module Number - 14 to 25

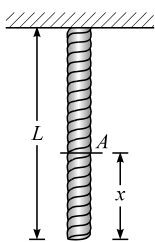


Figure 6.17

Practice Exercise 6.2

(i) A steel wire of length 64 cm weighs 5 g. If it is stretched by a force of 8 N, what would be the speed of a transverse wave passing on it?

[32 m/s]

(ii) A travelling wave is produced on a long horizontal string by vibrating an end up and down sinusoidally. The amplitude of vibration is 1.0 cm and the displacement becomes zero 200 times per second. The linear mass density of the string is 0.10 kg/m and it kept under a tension of 90 N. (a) Find the speed and the wavelength of the wave. (b) assume that the wave moves in the positive x -direction and at $t = 0$, the end $x = 0$ is at its positive extreme position. Write the wave equation. (c) Find the velocity and acceleration of the particle at $x = 50$ cm at time $t = 10$ ms.

$$[(a) 30 \text{ m/s, } 30 \text{ cm} \quad (b) y = (1.0 \text{ cm}) \cos 2\pi \left[100t - \frac{x}{0.3} \right]]$$

$$(c) - 5.45 \text{ m/s, } 2.0 \text{ km/s}]$$

(iii) Two wires of different densities but same area of cross section are soldered together at one end and are stretched to a tension T . The velocity of a transverse wave in one wire is double of that in the second wire. Find the ratio of the density of the first wire to that of the second wire.

[0.25]

(iv) A particle on a stretched string supporting a travelling wave, takes 5.0 ms to move from its mean position to the extreme position. The distance between two consecutive particles, which are at their mean positions, is 2.0 cm. Find the frequency, the wavelength and the wave speed.

[50 Hz, 4.0 cm, 2.0 m/s]

(v) The speed of sound as measured by a student in the laboratory on a winter day is 340 m/s when the room temperature is 17°C. What speed will be measured by another student repeating the experiment on a day when the room temperature is 32°C?

[348.68 m/s]

(vi) A string of length 40 cm and weighing 10 g is attached to a spring at one end and to a fixed wall at the other end. The spring has a spring constant of 160 N/m and is stretched by 1.0 cm. If a wave pulse is produced on the string near the wall, how much time will it take to reach the spring?

[0.05 s]

(vii) Find the change in the volume of 1.0 litre kerosene when it is subjected to an extra pressure of $2.0 \times 10^5 \text{ N/m}^2$ from the following data. Density of kerosene = 800 kg/m^3 and speed of sound in kerosene = 1330 m/s.

[0.141 cm³]

(viii) The constant γ for oxygen as well as for hydrogen is 1.40. If the speed of sound in oxygen is 470 m/s, what will be the speed in hydrogen at the same temperature and pressure?

[1880 m/s]

(ix) A tuning fork of frequency 440 Hz is attached to a long string of linear mass density 0.01 kg/m kept under a tension of 49 N. The fork produces transverse waves of amplitude 0.50 mm on the string. (a) Find the wave speed and the wavelength of the waves. (b) Find the maximum speed and acceleration of a particle of the string. (c) At what average rate is the tuning fork transmitting energy to the string?

[(a) 70 m/s, 15.9 cm, (b) 1.381 m/s, 3.872 km/s, (c) 0.667 W]

(x) Velocity of sound in a tube containing air at 27°C and at a pressure of 76 cm of Hg is 300 m/s. What will its velocity be when the pressure is increased to 100 cm of Hg and the temperature is kept constant?

[Same 300 m/s]

(ix) A blast gives a sound of intensity 0.80 W/m^2 and frequency 1 kHz. If the density of air is 1.3 kg/m^3 and speed of sound in air is 330 m/s find the amplitude of the sound wave.

[$9.656 \times 10^{-6} \text{ m}$]

6.5 Principle of Superposition

This principle defines the displacement of a medium particle which is oscillating under the influence of two or more than two waves. The principle of superposition is stated as :

“When two or more waves superpose on a medium particle than the resultant displacement of that medium particle is given by the vector sum of the individual displacements produced by the component waves at that medium particle independently.”

Let $\vec{y}_1, \vec{y}_2, \dots, \vec{y}_N$ are the displacements produced by N independent waves at a medium particle in absence of others then the displacement of that medium, when all the waves are superposed at that point, is given as

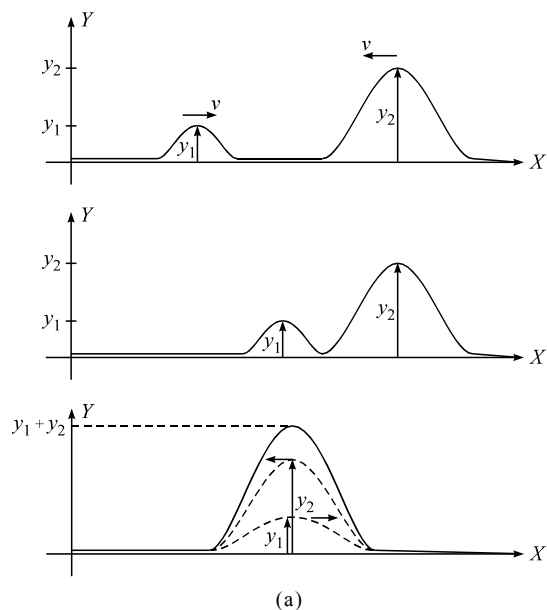
$$\vec{y} = \vec{y}_1 + \vec{y}_2 + \vec{y}_3 + \dots + \vec{y}_N \dots (6.57)$$

If all the waves are producing oscillations at that point are collinear then the displacement of the medium particle where superposition is taking place can be simply given by the algebraic sum of the individual displacements. Thus we have

$$y = y_1 + y_2 + \dots + y_N \dots (6.58)$$

The above equation is valid only if all individual displacements $y_1, y_2, \dots, y_N$ are along same straight line. In this book mainly we'll deal with the problems of superposition of this type.

A simple example of superposition can be understood by figure-6.18. Suppose two wave pulses are travelling simultaneously in opposite directions as shown. When they overlap each other the displacement of particle on string is the algebraic sum of the two displacements as the displacements of the two pulses are in same direction. Figure-6.18(b) also shows the similar situation when the wave pulses are in opposite sides.



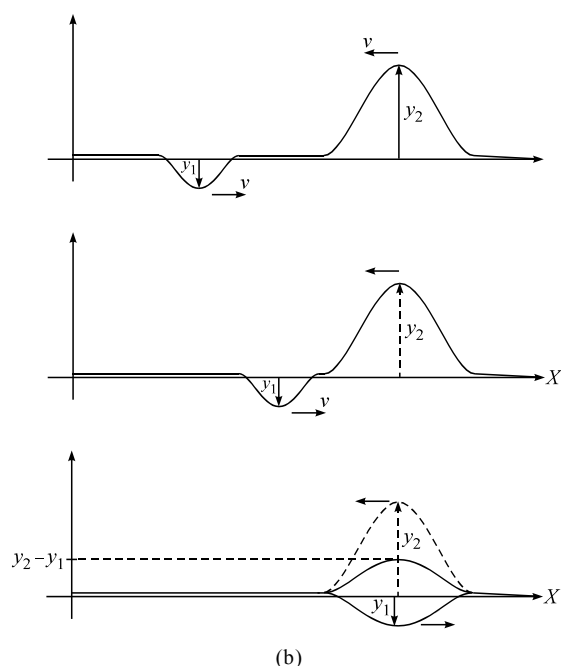


Figure 6.18

6.5.1 Application of Principle of Superposition of Waves

There are several different phenomenon which takes place during superposition of two or more waves depending on the wave characteristics which are being superposed. We'll discuss some standard phenomena, and these are :

- (1) Interference of Wave
- (2) Stationary Waves
- (3) Beats
- (4) Lissajou's Figures (Not discussed here in detail.)

Lets discuss these in detail.

6.6 Interference of Waves

Suppose two sinusoidal waves of same wavelength and amplitude travel in same direction along the same straight line (may be on a stretched string) then superposition principle can be used to define the resultant displacement of every medium particle. The resultant wave in the medium depends on the extent to which the waves are in phase with respect to each other, that is, how much one waveform is shifted from the other waveform. If the two waves are exactly in same phase, that is the shape of one wave exactly fits on to the other wave then they combine to double the displacement of every medium particle as shown in figure-6.19(a). This phenomenon we call as constructive interference. If the superposing waves are exactly out of phase or in opposite phase then they combine to cancel all the displacements at every medium particle and medium remains in the form of a straight line as shown in figure-6.19(b).

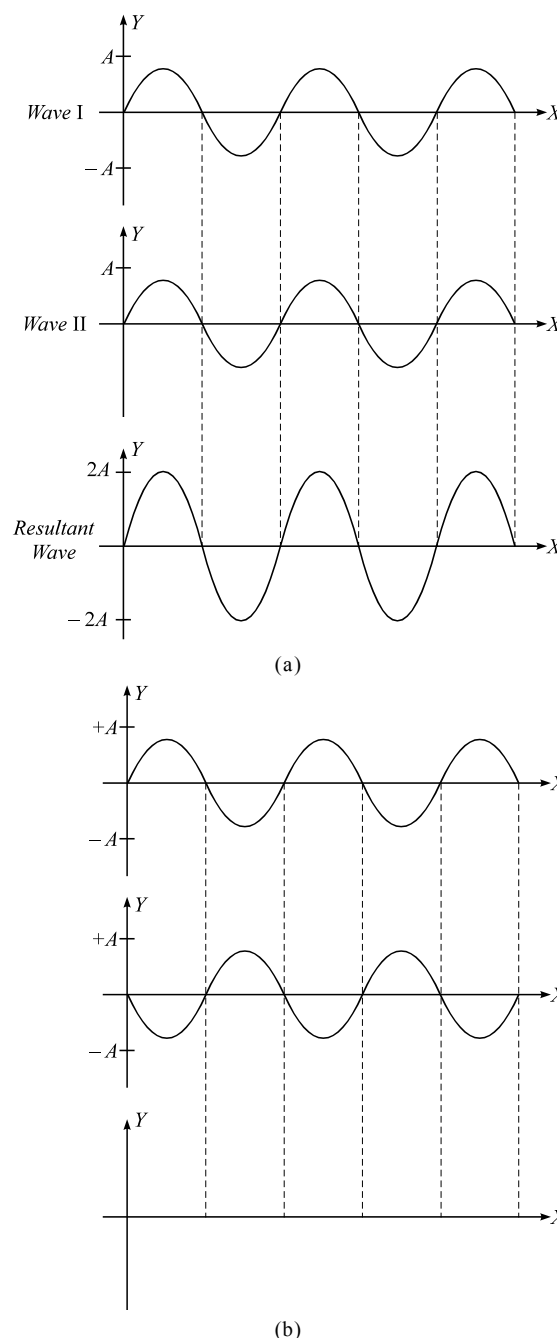


Figure 6.19

This phenomenon we call destructive interference. Thus we can state that when two waves meet, they interfere constructively if they meet in same phase and destructively if they meet in opposite phase. In either case the wave patterns do not shift relative to each other as they propagate. Such superposing waves which have same form and wavelength and have a fixed phase relation to each other, are called coherent waves. Sources of coherent waves are called coherent sources. Two independent sources can never be coherent in nature due to practical limitations of manufacturing process. Generally all coherent sources are made either by splitting of the waveforms of a single source or the different sources are fed by a single main energy source.

In simple words interference is the phenomenon of superposition of two coherent waves travelling in same direction.

We've discussed that the resultant displacement of a medium particle when two coherent waves interfere at that point, as sum or difference of the individual displacements by the two waves if they are in same phase (phase difference = 0, 2π , ...) or opposite phase (phase difference = π , 3π , ...) respectively. But the two waves can also meet at a medium particle with phase difference other than 0 or 2π , say if phase difference ϕ is such that $0 < \phi < 2\pi$, then how is the displacement of the point of superposition given? Now we discuss the interference of waves in details analytically.

6.6.1 Analytical Treatment of Interference of Waves

Now we discuss mathematically, how the displacement of the point of superposition varies with time. Let displacement of a medium particle due to a propagating wave be

$$y_1 = A_1 \sin(\omega t) \quad \dots (6.59)$$

At the same medium particle if another wave of same angular frequency ω and different amplitude A_2 arrives with a phase shift ϕ then the displacement of the medium particle from mean position due to this second wave can be given as

$$y_2 = A_2 \sin(\omega t - \phi) \quad \dots (6.60)$$

Now the resultant displacement of this medium particle can be given by the principle of superposition. If the displacements produced by two waves at that point are y_1 and y_2 along same line of motion then the net displacement of this medium particle is given as

$$\begin{aligned} y &= y_1 + y_2 \\ &= A_1 \sin \omega t + A_2 \sin(\omega t - \phi) \\ &= A_1 \sin \omega t + A_2 \sin \omega t \cos \phi - A_2 \cos \omega t \sin \phi \end{aligned}$$

Rearranging the terms of $\sin \omega t$ and $\cos \omega t$ separately as

$$y = (A_1 + A_2 \cos \phi) \sin \omega t - (A_2 \sin \phi) \cos \omega t$$

Now substituting

$$R \cos \theta = A_1 + A_2 \cos \phi \quad \dots (6.61)$$

$$\text{and} \quad R \sin \theta = A_2 \sin \phi \quad \dots (6.62)$$

$$\text{We get} \quad y = R \cos \theta \sin \omega t - R \sin \theta \cos \omega t$$

$$\text{or} \quad y = R \sin(\omega t - \theta) \quad \dots (6.63)$$

Equation-(6.63) is an equation of SHM, thus we can state that after superposition of the two waves, this medium particle executes SHM with amplitude R and initial phase lag θ with respect to the SHM produced at the point by the first wave.

Here R and θ can be given by equation-(6.61) and (6.62). Squaring and adding the two equations, we get

$$\begin{aligned} R &= \sqrt{(A_1 + A_2 \cos \phi)^2 + (A_2 \sin \phi)^2} \\ R &= \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi} \quad \dots (6.64) \end{aligned}$$

Dividing equation-(6.62) and (6.61) gives

$$\begin{aligned} \tan \theta &= \frac{A_2 \sin \phi}{A_1 + A_2 \cos \phi} \\ \text{or} \quad \theta &= \tan^{-1} \left(\frac{A_2 \sin \phi}{A_1 + A_2 \cos \phi} \right) \quad \dots (6.65) \end{aligned}$$

Equation-(6.64) and (6.65) are the results similar to those obtained by parallelogram rule of vector addition. If A_1 and A_2 are two vectors and ϕ is the angle between their directions then the resultant vector of the two is given by equation-(6.63) and the direction of resultant with the first vector is given by equation-(6.64). Thus we can conclude that when two or more waves of same frequency which differ in phase, superpose on a medium particle then the resulting motion of that medium particle is also SHM with same frequency. Its amplitude can be given by treating the individual amplitudes as vectors with their phase differences as the angles between them and finding the resultant of these vectors.

6.6.2 Interference of two Coherent Waves of Same Amplitude

We've discussed in previous article that when two coherent waves of different amplitudes A_1 and A_2 with a phase difference ϕ superpose on a medium particle, the resulting amplitude of that medium particle is given as

$$R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$$

If the two waves are of equal amplitude $A_1 = A_2 = A$, then R is given as

$$\begin{aligned} R &= \sqrt{A^2 + A^2 + 2A^2 \cos \phi} \\ &= \sqrt{2A^2(1 + \cos \phi)} \\ &= 2A \cos \phi/2 \quad \dots (6.66) \end{aligned}$$

Here we can see that the resultants amplitude R of the medium particle after superposition, depends on the amplitudes of component waves and on the phase difference ϕ between the two component waves. Thus if the phase difference between the two waves changes at the point of superposition, the resulting amplitude of that medium particle also changes. From equation-(6.64). We can see that the amplitude at the point of interference is maximum when

$$\cos \phi = +1$$

$$\text{or when} \quad \phi = 2N\pi \quad [N \in I] \quad \dots (6.67)$$

Then the maximum value of R is given as

$$\begin{aligned} R_{max} &= \sqrt{A_1^2 + A_2^2 + 2A_1A_2} \\ R_{max} &= A_1 + A_2 \quad \dots (6.68) \end{aligned}$$

If $A_1 = A_2 = A$ then $R_{max} = 2A$

Thus when the phase difference between two superposition waves is an integral multiple of 2π i.e. when the two waves superpose on a medium particle in same phase then the resultant amplitude of that medium particle will be maximum given by equation-(6.68) and this situation is called constructive interference of waves at the medium particle.

Similarly from equation-(6.64) we can see that the amplitude at the point of superposition is minimum when

$$\begin{aligned} \cos \phi &= -1 \\ \text{or when } \phi &= (2N+1)\pi \quad [N \in I] \quad \dots (6.69) \end{aligned}$$

Then the minimum amplitude is given as

$$\begin{aligned} R_{min} &= \sqrt{A_1^2 + A_2^2 - 2A_1A_2} \\ &= A_1 - A_2 \quad \dots (6.70) \end{aligned}$$

If $A_1 = A_2 = A$ then $R_{min} = 0$

Thus when the phase difference between two superposing waves is an odd multiple of π i.e., when the two waves superpose on a medium particle in opposite phase, then the resultant amplitude of that medium particle will be minimum and is given by equation-(6.70) and this situation is called destructive interference of the waves at the medium particle.

6.6.3 Condition on path difference at the point of interference

As we've already studied that two waves interfering at a point having their path difference Δ and phase difference ϕ are related as

$$\phi = \frac{2\pi}{\lambda} \cdot \Delta$$

For the case of constructive interference, waves must interfere in same phase for which $\phi = 2N\pi$ so for constructive interference the condition of path difference between waves at the point of interference is

$$\Delta = \frac{\lambda}{2\pi} \cdot 2N\pi = N\lambda$$

Thus if at a point path difference between interfering waves is an integral multiple of wavelength then waves superpose in same phase and constructive interference take place.

Similarly for the case of destructive interference, waves must interfere in opposite phase for which $\phi = (2N+1)\pi$ so the

condition of path difference between waves at the point of interference for destructive interference is

$$\Delta = \frac{\lambda}{2\pi} \cdot (2N+1)\pi = (2N+1) \frac{\lambda}{2}$$

Thus if at a point path difference between interfering waves is odd multiple of half wavelength then the waves superpose at the point in opposite phase and destructive interference take place.

6.7 Intensity of Wave

When a wave travels through a medium, energy is transferred from one part of medium to another part. The wave intensity is defined as the average amount of energy flow in the medium per unit time and per unit of its cross-sectional area. Thus intensity is measured in units of watt/m². Intensity of a wave can also be given as average power per unit cross-sectional area. Let us find mathematically the average power and intensity of a wave.

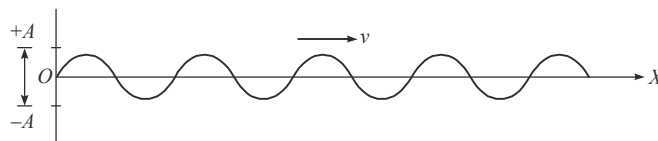


Figure 6.20

Consider a wave propagating along positive x -direction having angular frequency ω and amplitude A moving at speed v . We know wave speed is defined as the distance traveled by energy in one second. In one second wave imparts oscillation energy to all the medium particles in a length v meters. If ρ be the density of medium and S be the area of cross section through which energy flows, then the total oscillation energy of all the medium particles, oscillating with amplitude A and angular frequency ω is given as

$$\begin{aligned} E &= \frac{1}{2} m \omega^2 A^2 \\ &= \frac{1}{2} (\rho S v) \omega^2 A^2 \quad [\text{As } m = \rho S v] \end{aligned}$$

$$P = E = 2\pi^2 n^2 A^2 \rho v S \quad \dots (6.71)$$

Equation-(6.71) gives the oscillation energy of all the medium particles in a length v meter, which is the energy crossing a given cross section per unit time hence it can be regarded as the average power of wave. Now we can simply calculate the intensity of wave as

$$I = \frac{P}{S} = 2\pi^2 n^2 A^2 \rho v \quad \dots (6.72)$$

If a wave is propagating in a medium then in the above equation the parameters frequency n , density ρ and velocity v remains constant and we can state that the intensity of a wave is directly

proportional to the square of amplitude of the wave is

$$I \propto A^2$$

or
$$I = k A^2 \quad \dots (6.73)$$

Here k is a constant whose value can be obtained from equation-(6.72) as

$$k = 2\pi^2 n^2 \rho v \quad \dots (6.74)$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Wave Motion

Module Number - 26 to 29

6.7.1 Wave Intensity in Interference

We've discussed that when two coherent waves superpose each other on a medium particle interference takes place. The resultant amplitude at the point of interference depends on the phase difference of the two waves at the point of interference. If the amplitudes of two interfering waves are A_1 and A_2 then the resulting amplitudes of two interfering waves are A_1 and A_2 then the resulting amplitude R at the point of interference depends on their phase difference ϕ as

$$R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi} \quad \dots (6.75)$$

If I_R be the intensity at the point of interference then it can be given as

$$I_R \propto R^2$$

or
$$I_R = k R^2$$

or
$$I_R = k (A_1^2 + A_2^2 + 2A_1A_2 \cos \phi)$$

or
$$I_R = k A_1^2 + k A_2^2 + 2k A_1 A_2 \cos \phi$$

or
$$I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi \quad \dots (6.76)$$

Here $I_1 = k A_1^2$ and $I_2 = k A_2^2$ are the intensities of first and second wave respectively. The expression in equation-(6.76) gives the resultant intensity at the point of interference due to superposition of two coherent waves having independent intensities I_1 and I_2 respectively. Equation-(6.76) shows that the resultant intensity at the point of interference depends on the individual intensities I_1 , I_2 and the phase difference ϕ between the two waves at that point. If the waves are of equal intensities $I_1 = I_2 = I_0$ then after interference the intensity at the point of interference is given by equation-(6.76) as

$$I_R = I_0 + I_0 + 2 I_0 \cos \phi$$

or
$$I_R = 2 I_0 (1 + \cos \phi)$$

or
$$I_R = 4 I_0 \cos^2 \frac{\phi}{2} \quad \dots (6.77)$$

We know when two coherent waves interfere constructively, phase difference between the two is zero or multiple of 2π . Thus from equation-(6.77) $\cos \phi = +1$ and maximum intensity at the point of constructive interference can be given as

$$I_{max} = I_1 + I_2 + \sqrt{I_1 I_2} \quad \dots (6.78)$$

$$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2 \quad \dots (6.79)$$

For waves of equal intensities if $I_1 = I_2 = I_0$ then we have

$$I_{max} = 4 I_0 \quad \dots (6.80)$$

Similarly for destructive interference as the phase difference between waves should be an odd multiple of π , we have

$$\cos \phi = -1$$

Thus from equation-(6.76), the maximum intensity at the point of destructive interference can be given as

$$I_{min} = I_1 + I_2 - 2\sqrt{I_1 I_2} \quad \dots (6.81)$$

or
$$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2 \quad \dots (6.82)$$

For waves of equal intensities of $I_1 = I_2 = I_0$, then we have

$$I_{min} = 0 \quad \dots (6.83)$$

From equation-(6.78) and (6.81) we can see that in constructive interference the resultant intensity is more than the sum of individual intensities of the component waves and in destructive interference the resultant intensity is less than the sum of individual intensities of the component waves.

6.7.2 Power of a Wave and Wave Energy Density of a Transverse Wave

As a transverse wave move along a string, it carries energy in the direction of wave travel. Consider an element of string of width dx as shown in figure-6.21, it oscillates in a direction perpendicular to wave propagation. During oscillation this element has both kinetic and potential energies. Kinetic energy due to its motion and potential energy due to the amount it is stretched. Its kinetic energy is given as

$$dk = \frac{1}{2} (\mu dx) \left(\frac{\partial y}{\partial t} \right)^2 \quad \dots (6.84)$$

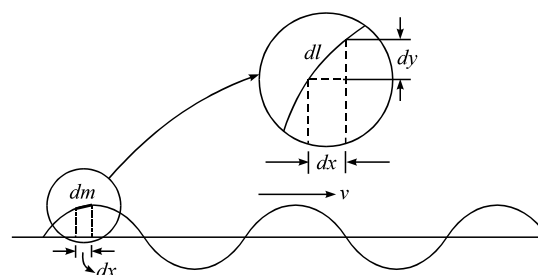


Figure 6.21

Here μ is the linear density of string and $\frac{\partial y}{\partial t}$ is the instantaneous velocity of the element. Therefore the kinetic energy per unit length of string can be given as

$$\frac{dK}{dx} = \frac{1}{2} \mu \left(\frac{\partial y}{\partial t} \right)^2 \quad \dots (6.85)$$

When the element goes from its mean position to a height y , its length changes to dl , which is given as

$$dl = \sqrt{\partial x^2 + \partial y^2} \quad \dots (6.86)$$

In stretching the element from dx to dl , work done by the tension T of the string will be stored in the form of potential energy in it, which is given as

$$dU = T(\partial l - \partial x)$$

or
$$dU = T dx \left(\sqrt{1 + \left(\frac{\partial y}{\partial x} \right)^2} - 1 \right)$$

or
$$dU = T dx \left(1 + \frac{1}{2} \left(\frac{\partial y}{\partial x} \right)^2 - 1 \right)$$

[Using binomial approximation]

or
$$dU = \frac{1}{2} T dx \left(\frac{\partial y}{\partial x} \right)^2 \quad \dots (6.87)$$

Thus potential energy per unit length can be given as

$$\frac{dU}{dx} = \frac{1}{2} T \left(\frac{\partial y}{\partial x} \right)^2 \quad \dots (6.88)$$

Total energy of string carrying a wave per unit length is called its energy density and is given as

$$E = \frac{dK}{dx} + \frac{dU}{dx}$$

or
$$E = \frac{1}{2} \mu \left(\frac{\partial y}{\partial t} \right)^2 + \frac{1}{2} T \left(\frac{\partial y}{\partial x} \right)^2 \quad \dots (6.89)$$

If we consider a wave propagating in positive x direction, its displacement equation can be given as

$$y = A \sin(\omega t - kx) \quad \dots (6.90)$$

From equation-(6.90) we get

$$\frac{\partial y}{\partial t} = A \omega \cos(\omega t - kx) \quad \dots (6.91)$$

And
$$\frac{\partial y}{\partial x} = -Ak \cos(\omega t - kx) \quad \dots (6.92)$$

From equation-(6.89), (6.91) and (6.92) we have

$$E = \frac{1}{2} \mu A^2 \omega^2 \cos^2(\omega t - kx) + \frac{1}{2} T A^2 k^2 \cos^2(\omega t - kx) \quad \dots (6.93)$$

We have wave velocity

$$v = \sqrt{\frac{T}{\mu}} \quad \text{or} \quad T = \mu v^2$$

And we also have
$$v = \frac{\omega}{k}$$

or
$$T = \frac{\mu \omega^2}{k^2} \quad \dots (6.94)$$

From equation-(6.93) and (6.94) we have

$$E = \mu A^2 \omega^2 \cos^2(\omega t - kx) \quad \dots (6.95)$$

Equation-(6.95) gives the total energy of string in a wave is propagating. The wave travels a distance v meter in one second thus power of a wave on string can be given as

$$P = Ev$$

$$= \mu A^2 \omega^2 v \cos^2(\omega t - kx) \quad \dots (6.96)$$

For a string if ρ be the density and S be its cross sectional area, then we have

$$\mu = S\rho$$

Thus power of a wave can be given from equation-(6.96)

$$P = 4\pi^2 n^2 A^2 \rho v S \cos^2(\omega t - kx) \quad \dots (6.97)$$

Equation-(6.97) gives the instantaneous power transmitted by a wave during its propagation. Here average power transmitted can be obtained by substituting average value of time function $\cos^2(\omega t - kx)$ which is $1/2$. Thus average power transmitted by a wave can be given as

$$P_{av} = 4\pi^2 n^2 A^2 \rho v S \left(\frac{1}{2} \right)$$

or
$$P_{av} = 2\pi^2 n^2 A^2 \rho v S \quad \dots (6.98)$$

Equation-(6.98) is same as that of equation-(6.71) which we have already derived in section-ABC.

6.7.3 Quink's Tube

This is an apparatus used to demonstrate the phenomenon of interference and also used to measure velocity of sound in air. This is made up of two U-tubes A and B as shown in figure-6.22. Here the tube B can slide in and out from the tube A . There are two openings P and Q in the tube A . At opening P , a tuning fork or a sound source of known frequency n_0 is placed and at the other opening a detector is placed to detect the resultant sound of interference occurred due to superposition of two sound waves coming from the tubes A and B .

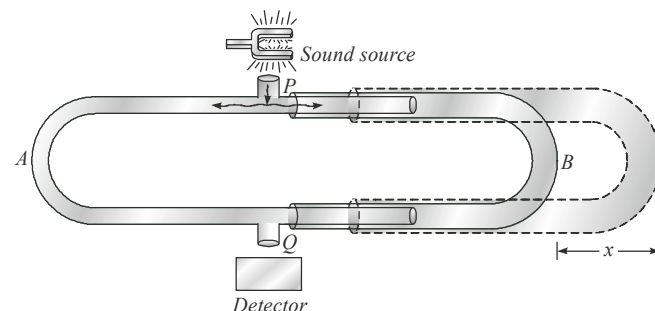


Figure 6.22

Initially tube B is adjusted so that detector detects a maximum. At this instant if length of paths covered by the two waves

from P to Q from the side of A and side of B are l_1 and l_2 respectively then for constructive interference we must have

$$l_2 - l_1 = N\lambda \quad \dots (6.99)$$

If now tube B is further pulled out by a distance x so that next maximum is obtained and the length of path from the side of B is l'_2 then we have

$$l'_2 = l_2 + 2x \quad \dots (6.100)$$

Where x is the displacement of the tube. For next constructive interference of sound at point Q , we have

$$l'_2 - l_1 = (N+1)\lambda \quad \dots (6.101)$$

From equation-(6.99), (6.100) and (6.101), we get

$$l'_2 - l_2 = 2 \times Q = \lambda$$

or
$$x = \frac{\lambda}{2} \quad \dots (6.102)$$

Thus by experiment we get the wavelength of sound as for two successive points of constructive interference, the path difference must be λ . As the tube B is pulled out by x , this introduces a path difference $2x$ in the path of sound wave through tube B . If the frequency of the source is known, n_0 , the velocity of sound in the air filled in tube can be given as

$$\begin{aligned} v &= n_0 \lambda \\ &= 2n_0 x \end{aligned} \quad \dots (6.103)$$

6.7.4 Seebeck's Tube

This is also an apparatus used to demonstrate phenomenon of interference and it can also be used to find velocity of sound in air. This is a hollow tube with a freely moving piston as shown in figure-6.23.

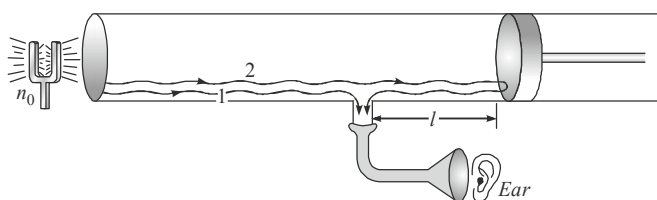


Figure 6.23

There is an opening in the tube which can be used as a point where sound can be detected directly by human ear or a detector as shown in figure. When a source of sound oscillates near the mouth of the tube (tuning fork as shown). At the opening two sound waves will interfere. One direct from the source and second, which is reflected from the piston. Here we can see that the path difference between direct wave and reflected wave at opening can be given as

$$\Delta = 2l \quad \dots (6.104)$$

Where l is the distance between opening and piston shown in figure-6.23.

In the experiment first the piston is adjusted so that a maximum sound is detected at opening. This implies that the path difference between the two waves must be an integral, multiple of wavelength λ .

Thus we have,
$$\Delta = 2l = N\lambda \quad (N \in I) \quad \dots (6.105)$$

Now the piston is slowly moved out so that the intensity of sound detected changed and again becomes maximum after some time. At this instant say if the distance of piston from opening becomes l' , then we can say for next successive constructive interference the path difference of the two waves is

$$\Delta' = 2l' = (N+1)\lambda \quad \dots (6.106)$$

Here if displacement of piston is x , we have

$$x = l' - l$$

From equation-(6.105) & (6.106)

$$x = \frac{\lambda}{2}$$

or
$$\lambda = 2x \quad \dots (6.107)$$

If the source frequency is known, n_0 , then velocity of sound in air can be written as

$$\begin{aligned} v &= n_0 \lambda \\ &= 2xn_0 \end{aligned} \quad \dots (6.108)$$

Illustrative Example 6.20

In a large room a person receives direct sound waves from a source 120 m away from him. He also receives waves from the same source which reach him, being reflected from the 25 m high ceiling at a point halfway between them. For which wavelength will these two sound waves interfere constructively?

Solution

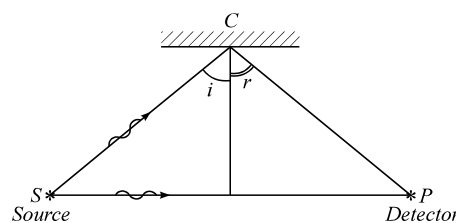


Figure 6.24

As shown in figure-6.24 for reflection from the ceiling

Path
$$SCP = SC + CP = 2SC$$

[As $\angle i = \angle r$, $SC = CP$]

or $\text{Path } SCP = 2\sqrt{60^2 + 25^2} = 130 \text{ m}$

So path difference between interfering waves along paths SCP , and SP ,

$$\Delta x = 130 - 120 = 10 \text{ m}$$

Now for constructive interference at P ,

$$\Delta x = n\lambda, \text{ i.e. } 10 = n\lambda$$

or $\lambda = \frac{10}{n} \text{ with } n = 1, 2, 3 \dots$

i.e., $\lambda = 10 \text{ m}, 5 \text{ m}, (10/3) \text{ m}$ and so on

Illustrative Example 6.21

Figure-6.25 shows a tube structure in which a sound signal is sent from one end and is received at the other end. The semicircular part has a radius of 20.0 cm. The frequency of the sound source can be varied electronically between 1000 and 4000 Hz. Find the frequencies at which maxima of intensity are detected. The speed of sound in air = 340 m/s.

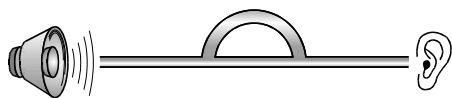


Figure 6.25

Solution

The sound wave reaches detector by two paths simultaneously by straight as well as semicircular track. The wave through the straight part travels a distance $l_1 = 2 \times 20 \text{ cm}$ and the wave through the curved part travels a distance $l_2 = \pi (20 \text{ cm}) = 62.8 \text{ cm}$ before they meet again and travel to the receiver. The path difference between the two waves received is, therefore,

$$\begin{aligned} \Delta l &= l_2 - l_1 = 62.8 \text{ cm} - 40 \text{ cm} \\ &= 22.8 \text{ cm} = 0.228 \text{ m} \end{aligned}$$

The wavelength of either wave is $\frac{v}{n} = \frac{340}{n}$. For constructive interference, $\Delta l = N\lambda$, where N is an integer.

or, $0.228 = N \left(\frac{340}{n} \right)$

or, $n = N \left(\frac{340}{0.228} \right) = N(1491.2) \text{ Hz}$
 $= N(1490) \text{ Hz}.$

Thus, the frequencies within the specified range which cause maxima of intensity are 1490 Hz and 2980 Hz.

Illustrative Example 6.22

Two sources S_1 and S_2 , separated by 2.0 m, vibrate according to equation $y_1 = 0.03 \sin \pi t$ and $y_2 = 0.02 \sin \pi t$ where y_1, y_2 and t are in M.K.S. units. They send out waves of velocity 1.5 m/s. Calculate the amplitude of the resultant motion of the particle co-linear with S_1 and S_2 and located at a point (a) to the right of S_2 (b) to the left of S_2 and (c) in the middle of S_1 and S_2 .

Solution

The situation is shown in figure-6.26.

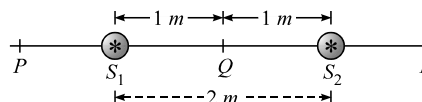


Figure 6.26

The oscillations y_1 and y_2 have amplitudes $A_1 = 0.03 \text{ m}$ and $A_2 = 0.02 \text{ m}$ respectively.

The frequency of both sources is $n = \frac{\omega}{2\pi} = \frac{1}{2} = 0.5 \text{ Hz}$

Now wavelength of each wave $\lambda = \frac{v}{n} = \frac{1.5}{0.5} = 3.0 \text{ m}$

(a) The path difference for all points P_2 to the right of S_2 is

$$\Delta = (S_1 P_2 - S_2 P_2) = S_1 S_2 = 2 \text{ m}$$

Phase difference $\phi = \frac{2\pi}{\lambda} \times \text{Path difference}$

$$= \frac{2\pi}{3} \times 2.0 = \frac{4\pi}{3}$$

The resultant amplitude for this point is given by

$$\begin{aligned} R &= \sqrt{A_1^2 + A_2^2 + 2A_1 A_2 \cos \phi} \\ &= \sqrt{(0.03)^2 + (0.02)^2 + 2 \times 0.03 \times 0.02 \times \cos (4\pi/3)} \end{aligned}$$

Solving we get

$$R = 0.0265 \text{ m}$$

(b) The path difference for all points P , to the left of S_1

$$\Delta = (S_2 P - S_1 P) = S_1 S_2 = 2.0 \text{ m}$$

Hence the resultant amplitude for all points to the left of S_1 is also 0.0265 m

(c) For a point Q , midway between S_1 and S_2 , the path difference is zero i.e., $\phi = 0$. Hence constructive interference takes place at Q , thus amplitude at this point is maximum and given as

$$R = \sqrt{A_1^2 + A_2^2 + 2A_1 A_2}$$

$$\begin{aligned}
 &= A_1 + A_2 \\
 &= 0.03 + 0.02 \\
 &= 0.05 \text{ m}
 \end{aligned}$$

Illustrative Example 6.23

Two point sources of sound are placed at a distance d and a detector moves on a straight line parallel to the line joining the sources as shown in figure-6.27 at a distance D away from sources. Initially Detector is situated on the line so that it is equidistant from both the sources. Find the displacement of detector when it detects n^{th} maximum sound and also find its displacement when it detects n^{th} minimum sound.

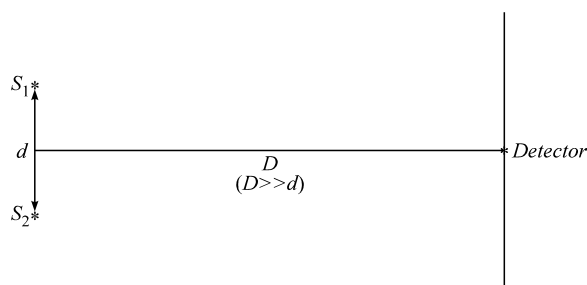


Figure 6.27

Solution

The situation is shown in figure-6.28.

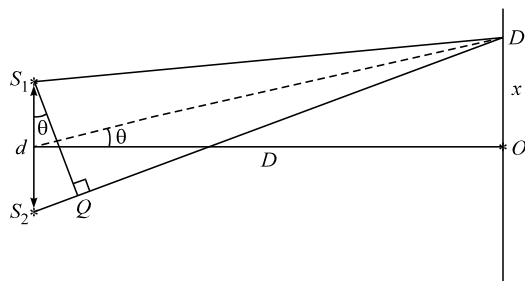


Figure 6.28

Let us consider the situation when detector move by a distance x as shown. Let at this position the path difference between the waves from S_1 and S_2 to detector be Δ then we have

$$\begin{aligned}
 \Delta &= S_2D - S_1D \\
 &\approx S_2Q
 \end{aligned}$$

[Where S_1Q is perpendicular on line S_2D]

Here if θ is small angle as $D \gg d$, we have

$$\begin{aligned}
 S_2Q &= d \sin \theta \approx d\theta \\
 &= d \frac{x}{D}
 \end{aligned}$$

Thus at the position of detector, path difference is

$$\Delta = \frac{dx}{D} \quad \dots (6.109)$$

The expression for path difference in equation-(6.109) is an important formula for such problems. Students are advised to keep this formula in mind for future use.

When detector was at point O , path difference was zero and it detects a maxima, now if detector detects n^{th} maximum then its path difference at a distance x from O can be given as

$$\begin{aligned}
 \Delta &= n\lambda \\
 \text{or} \quad \frac{dx}{D} &= n\lambda \\
 \text{or} \quad x &= \frac{n\lambda D}{d}
 \end{aligned}$$

Similarly if detector detects n^{th} minima then the path difference between two waves at detector can be given as

$$\begin{aligned}
 \Delta &= (2n-1) \frac{\lambda}{2} \\
 \text{or} \quad \frac{dx}{D} &= (2n-1) \frac{\lambda}{2} \\
 \text{or} \quad x &= \frac{(2n-1)\lambda D}{2d}
 \end{aligned}$$

Illustrative Example 6.24

Two small loudspeakers A, B (1 m apart) are connected to the same oscillator so that both emit coherent sound waves of frequency 1700 Hz in phase. A sensitive detector, moving parallel to the line AB along LQ , 2.4 m away, detects a maximum wave at P on the perpendicular bisector MP of AB and another maximum wave when it first reaches a point Q directly opposite to B as shown in figure-6.29. Calculate the speed of the sound waves in air.

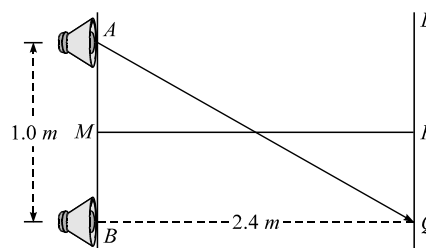


Figure 6.29

Solution

$AQ - BQ = \lambda$ for constructive interference of waves at Q

$$\text{or} \quad \sqrt{1.00^2 + 2.40^2} - 2.40 = \lambda \quad \text{or} \quad \lambda = 0.2 \text{ m}$$

Thus we get $v = n\lambda$

$$\text{or} \quad v = 1700 \times 0.2 = 340 \text{ m/s}$$

Illustrative Example 6.25

A source emitting sound of frequency 180 Hz is placed in front of a wall at distance of 2 m from it. A detector is also placed in front of the wall at the same distance from it. Find the minimum distance between the source and the detector for which the detector detects a maximum of sound. Speed of sound in air = 360 m/s.

Solution

The situation is shown in figure-6.30.

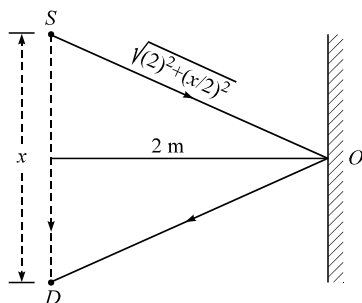


Figure 6.30

Suppose the detector is placed at a distance of x meter from the source. The direct wave received from the source travels a distance x meter. The wave reaching the detector after reflection from the wall has travelled a distance of $2[\sqrt{(2)^2 + x^2/4}]$ meter. Thus the path difference between the two waves at detector is

$$\Delta = \left\{ 2\sqrt{(2)^2 + \frac{x^2}{4}} - x \right\} \quad \dots (6.110)$$

Constructive interference will take place when $\Delta = \lambda, 2\lambda, \dots$. Thus the minimum distance x for which a maximum occurs at detector, the path difference will be

$$\Delta = \lambda \quad \dots (6.111)$$

The wavelength is $\lambda = \frac{v}{n} = \frac{360}{180} = 2 \text{ m}$

From equation-(6.110) and (6.111), we have

$$\Delta = 2\sqrt{2^2 + \frac{x^2}{4}} - x = 2$$

or, $\sqrt{4 + \frac{x^2}{4}} = 1 + \frac{x}{2}$

or, $4 + \frac{x^2}{4} = 1 + \frac{x^2}{4} + x$

or, $x = 3 \text{ m}$

Thus, the detector should be placed at a distance of 3 m from the source to detect a maximum sound

Illustrative Example 6.26

Three plane sources of sound of frequency $n_1 = 400 \text{ Hz}$, 401 Hz and $n_3 = 402 \text{ Hz}$ of equal amplitude a each, are sounded together. A detector receives waves from all the three sources simultaneously. It can detect signals of amplitude $\geq A$. Calculate (a) period of one complete cycle of intensity received by the detector and (b) time for which the detector remains idle in each cycle of intensity.

Solution

(a) We know that $y = A \sin \omega t = A \sin (2\pi n t)$

The displacements of the medium particles caused by these waves are given as

$$y_1 = A \sin (800 \pi t) \quad \dots (6.112)$$

$$y_2 = A \sin (802 \pi t) \quad \dots (6.113)$$

and

$$y_3 = A \sin (804 \pi t) \quad \dots (6.114)$$

The resultant displacement of medium particle at time t is given by

$$\begin{aligned} y &= y_1 + y_2 + y_3 \\ &= A [\sin (800 \pi t) + \sin (802 \pi t) + \sin (804 \pi t)] \\ &= A [\{\sin (800 \pi t) + \sin (804 \pi t)\} + \sin (802 \pi t)] \\ &= A [1 + \cos 2 \pi t] \sin 802 \pi t \end{aligned}$$

The resultant is also a plane wave. Let its amplitude be R . Then, $y = R \sin \omega t$

Here $R = A (1 + \cos 2 \pi t) \quad \dots (6.115)$

Equation-(6.115) shows that the resultant amplitude (or resultant intensity, $I \propto R^2$) varies with time.

The resultant amplitude is maximum, when

$$2\pi t = 0, 2\pi, 4\pi, \dots$$

or

$$t = 0, 1, 2, 3, \dots$$

Hence period of one complete cycle of intensity is one second.

(b) Given that signal is detected when $A \geq a$

Thus $\cos 2 \pi t \geq 0$

Thus $\cos 2 \pi t$ should lie either in first quadrant or in fourth quadrant i.e., between either 0 to $\pi/2$ or $3\pi/2$ and 2π .

So, during first cycle of intensity, the signal is detected when

$$0 \leq t \leq \frac{1}{4} \quad \text{and} \quad \frac{3}{4} \leq t \leq 1$$

This shows that detector remains ideal from

$$t = \frac{1}{4} \text{ s to } t = \frac{3}{4} \text{ s}$$

Therefore, in each cycle of intensity, the detector remains ideal

$$\text{for } \frac{3}{4} - \frac{1}{4} = \frac{2}{4} = 0.5 \text{ s}$$

Illustrative Example 6.27

Two coherent narrow slits emitting of wavelength λ in the same phase are placed parallel to each other at a small separation of 2λ . the sound is detected by moving a detector on the screen S at a distance D ($\gg \lambda$) from the slit S_1 as shown in figure-6.31. Find the distance x such that the intensity at P is equal to the intensity at O .

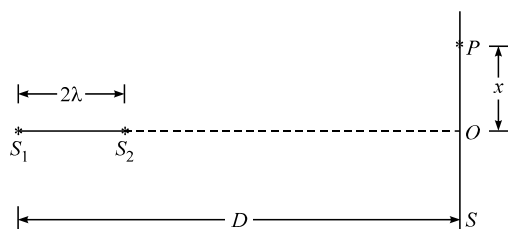


Figure 6.31

Solution

When detector is at O , we can see that the path difference in the two waves reaching O is $d = 2\lambda$ thus at O detector receives a maximum sound. When it reaches P and again there is a maximum sound detected at P the path difference between two waves must be $\Delta = \lambda$. Thus from figure-6.32 the path difference at P can be given as

$$\begin{aligned} \Delta &= S_1P - S_2P \simeq S_1Q \\ &= d \cos \theta \\ &= 2\lambda \cos \theta \end{aligned}$$

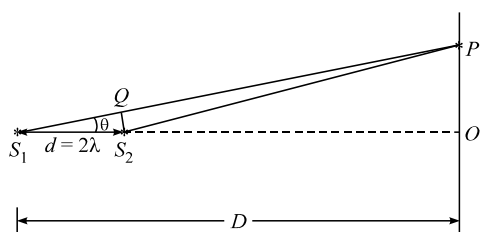


Figure 6.32

And we have at point P , path difference $\Delta = \lambda$, Thus

$$\Delta = 2\lambda \cos \theta = \lambda$$

$$\text{or, } \cos \theta = \frac{1}{2}$$

$$\text{or, } \theta = \frac{\pi}{3}$$

Thus the value of x can be written as

$$x = D \tan \theta = D \tan \left(\frac{\pi}{3} \right) = \sqrt{3} D$$

Illustrative Example 6.28

Figure-6.33 shows two coherent sources S_1 and S_2 which emit sound of wavelength λ in phase. The separation between the sources is 3λ . A circular wire of large radius is placed in such a way that S_1S_2 lies in its plane and the middle point of S_1S_2 is at the centre of the wire. Find the angular positions θ on the wire for which constructive interference takes place.

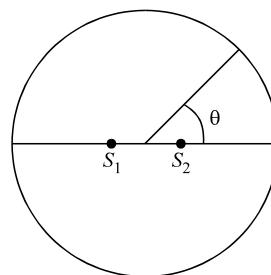


Figure 6.33

Solution

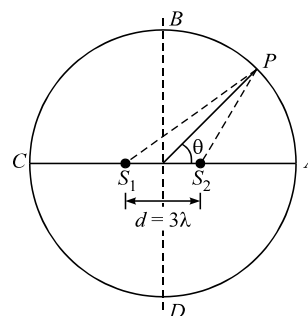


Figure 6.34

From previous question, we can say that for a point P on the circle shown in figure-6.34. The path difference in the two waves at P is

$$\begin{aligned} \Delta &= S_1P - S_2P = d \cos \theta \\ &= 3\lambda \cos \theta \end{aligned}$$

We know for constructive interference at P . The path difference must be an integral multiple of wavelength λ . Thus for a maxima at P , we have

$$3\lambda \cos \theta = 0; \quad 3\lambda \cos \theta = \lambda; \quad 3\lambda \cos \theta = 2\lambda; \quad 3\lambda \cos \theta = 3\lambda$$

$$\text{or, } \theta = \frac{\pi}{2} \quad \text{or } \theta = \cos^{-1} \frac{1}{3} \quad \text{or } \theta = \cos^{-1} \frac{2}{3} \quad \text{or } \theta = 0$$

There are four points A, B, C and D on circle at which $\theta = 0$ or $\frac{\pi}{2}$ and there are two points in each quadrant at $\theta = \cos^{-1} \frac{1}{3}$ and $\theta = \cos^{-1} \frac{2}{3}$ at which constructive interference takes place. Thus there are total twelve points on circle at which maxima occurs.

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Interference of Waves

Module Number - 1 to 17

Practice Exercise 6.3

(i) A source emitting sound of frequency 180 Hz is placed in front of an obstacle at a distance of 2 m from it. A detector is also placed at the position of source. (a) find the minimum distance between by which source and the detector are separated normal to the line passing through obstacle for which the detector detects a maximum of sound. (b) How much farther to the right must the obstacle be moved if the two waves are to be out of phase by 180° . (Speed of sound in air = 360 m/s).

[(a) 3 m; (b) 0.598 m]

(ii) Two audio speakers are kept some distance apart and are driven by the same amplifier system. A person is sitting at a place 6.0 m from one of the speakers and 6.4 m from the other. If the sound signal is continuously varied from 500 Hz to 5000 Hz, what are the frequencies for which there is a destructive interference at the place of the listener? Speed of sound in air = 320 m/s.

[1200 Hz, 2000 Hz, 2800 Hz, 3600 Hz and 4400 Hz]

(iii) Two points sources of sound are kept at a separation of 10 cm. They vibrate in phase to produce waves of wavelength 5.0 cm. What would be the phase difference between the two waves arriving a point 20 cm from one source

(a) on the line joining the source
(b) on the perpendicular bisector of the line joining the sources ?

[(a) zero, (b) zero]

(iv) Two sources of sound S_1 and S_2 vibrate at same frequency and are in phase (figure-6.35). The intensity of sound detected at a point P as shown in the figure-6.35 is I_0 . If θ equals 45° , what will be the intensity of sound detected at this point if one of the sources is switched off ?

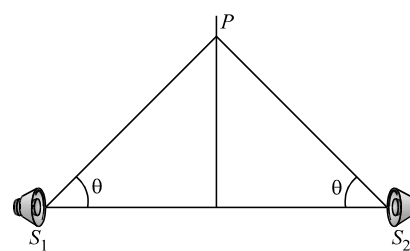


Figure 6.35

$[I_0/4]$

(v) Sound waves from a tuning fork A reach a point P by two separate paths ABP and ACP . When ACP is greater than ABP by 11.5 cm, there is silence at P . When the difference is increased to 23 cm the sound becomes loudest at P and again when it increases to 34.5 cm there is silence again and so on. Calculate the minimum frequency of the fork if the velocity of sound is taken to be 331.2 m/s.

[1440 Hz]

(vi) Three sources of sound S_1, S_2 and S_3 of equal intensity are placed in a straight line with $S_1S_2 = S_2S_3$ (figure-6.36). At a point P , far away from the sources, the wave coming from S_2 is 120° ahead in phase of that from S_1 . Also, the wave coming from S_3 is 120° ahead of that from S_2 . what would be the resultant intensity of sound at P ?

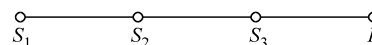


Figure 6.36

[zero]

(vii) Sounds from two identical sources S_1 and S_2 reach a point P . When the sounds reach directly, and in the same phase, the intensity at P is I_0 . The power of S_1 is now reduced by 64%, and the phase difference between S_1 and S_2 is varied continuously. The maximum and minimum intensities recorded at P are now I_{max} and I_{min} . Find I_{max}/I_{min} .

$[I_{max}/I_{min} = 16]$

(viii) At two points S_1 and S_2 on a liquid surface two coherent wave sources are set in motion at $t = 0$ with the same phase. The speed of the waves in the liquid $v = 0.5$ m/s, the frequency of vibration $\eta = 5$ Hz and the amplitude $A = 0.04$ m.

At a point P of the liquid surface which is at a distance $x_1 = 0.30$ m from S_1 and $x_2 = 0.34$ m from S_2 a piece of cork floats :

(a) Find the displacement of the cork at $t = 3$ s.
(b) Find the time t_0 that elapse from the moment the wave sources were set in motion until the moment that the cork passes through the equilibrium position for the first time.

[(a) $y = -0.02344$ m (b) $t_0 = 0.74$ s]

(ix) Two point sound sources A and B each of power $25\pi\text{W}$ and frequency 850Hz are 1m apart. (a) determine the phase difference between the waves emitting from A and B received by detector D as in figure-6.37 (b) also determine the intensity of the resultant sound wave as recorded by detector D . Velocity of sound = 340m/s .

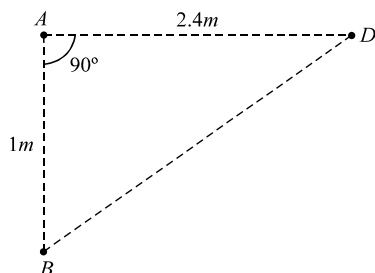


Figure 6.37

[(a) π , (b) 0.0064 W/m^2]

(x) Two sources of sound of the same frequency produce sound intensities I and $4I$ at a point P when used individually. If they are used together such that the sounds from them reach a point P with a phase difference of $2\pi/3$, find the intensity at point P .

[$3I$]

(xi) A source of sound S and a detector D are placed at some distance from one another. A big cardboard is placed near the detector and perpendicular to the line SD as shown in figure-6.38. It is gradually moved away and it is found that the intensity changes from a maximum to a minimum as the board is moved through a distance of 20 cm . Find the frequency of the sound emitted. Velocity of sound in air is 336 m/s .

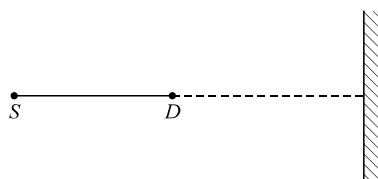


Figure 6.38

[420 Hz]

6.8 Compression Waves

When a longitudinal wave propagates in a gaseous medium, it produces compression and rarefaction in the medium periodically. The region where compression occurs, the pressure is more than the normal pressure of the medium and the region where rarefaction occurs, the pressure is lesser than the normal pressure of the medium. Thus we can also describe longitudinal waves in a gaseous medium as pressure waves and these are also termed as compression waves in which the pressure at different points of medium also varies periodically with their

displacements. Let us discuss the propagation of excess pressure in a medium in longitudinal wave analytically.

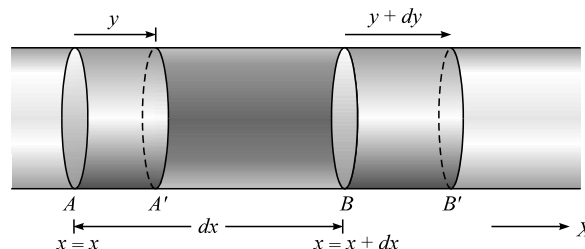


Figure 6.39

Consider a longitudinal wave propagating in positive x -direction as shown in figure-6.39. Figure shows a segment AB of the medium of width dx . In this medium let a longitudinal wave is propagating whose equation is given as

$$y = A \sin(\omega t - kx) \quad \dots (6.116)$$

Where y is the displacement of a medium particle situated at a distance x from the origin, along the direction of propagation of wave. In figure-6.39 AB is the medium segment whose a medium particle is at position $x = x$ and B is at $x = x + dx$ at an instant. If after some time t medium particle at A reaches to a point A' which is displaced by y and the medium particle at B reaches to point B' which is at a displacement $y + dy$ from B . Here dy is given by equation-(6.116) as

$$dy = -Ak \cos(\omega t - kx) dx$$

Here due to displacement of section AB to $A'B'$ the change in volume of it's section is given as

$$dV = -S dy \quad [S \rightarrow \text{Area of cross-section}]$$

$$= SA k \cos(\omega t - kx) dx$$

The volume of section AB is

$$V = S dx$$

Thus volume strain in section AB is

$$\frac{dV}{V} = \frac{-SAk \cos(\omega t - kx) dx}{S dx}$$

or

$$\frac{dV}{V} = -Ax \cos(\omega t - kx)$$

If B is the bulk modulus of the medium, then the excess pressure in the section AB can be given as

$$\Delta P = B \left(\frac{dV}{V} \right) \quad \dots (6.117)$$

$$\Delta P = B A k \cos(\omega t - kx)$$

or

$$\Delta P = \Delta P_0 \cos(\omega t - kx) \quad \dots (6.118)$$

Here ΔP_0 is the pressure amplitude at a medium particle at position x from origin and ΔP is the excess pressure at that

point. Equation-(6.118) shows that excess pressure varies periodically at every point of the medium with pressure amplitude ΔP_0 , which is given as

$$\begin{aligned}\Delta P_0 &= B A k \\ &= \frac{2\pi}{\lambda} A B \quad \dots (6.119)\end{aligned}$$

Equation-(6.118) is also termed as the equation of pressure wave in a gaseous medium. We can also see that the pressure wave differs in phase by $\pi/2$ from the displacement wave and displacement maxima occur where the pressure is at its normal level. Remember that pressure maxima implies that the pressure at a point is pressure amplitude times more or less than the normal pressure level of the medium.

Wave intensity for longitudinal waves at a point can also be expressed in terms of its pressure amplitude as from equation-(6.118)

$$I = 2\pi^2 n^2 A^2 \rho v$$

$$\text{or} \quad I = \frac{1}{2} \frac{4\pi^2}{\lambda^2} A^2 \rho v^3$$

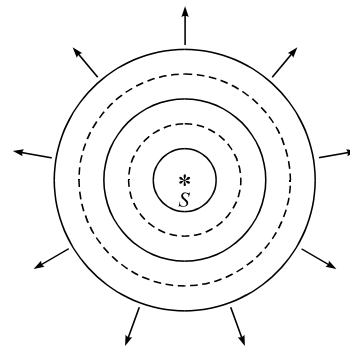
$$\text{or} \quad I = \frac{1}{2} \left(\frac{\Delta P_0}{B} \right)^2 \rho v^3$$

$$\text{or} \quad I = \frac{\Delta P_0^2}{2\rho v} \quad \left[\text{As } v = \sqrt{\frac{B}{\rho}} \right] \quad \dots (6.120)$$

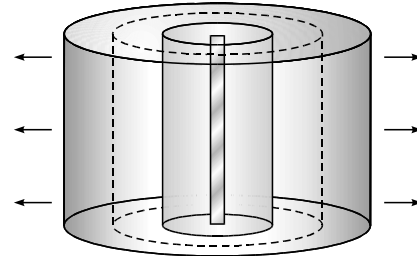
Expression in equation-(6.120) relates the wave intensity at a point and the pressure amplitude of the wave at that point for longitudinal waves.

6.9 Spherical and Cylindrical Waves

Till now we've discussed the longitudinal and transverse waves travelling in one direction only. Consider a source of sound which emits sound waves in all directions as shown in figure-6.40(a). When source vibrates, it emits oscillations in air in its surrounding space. It emits waves in the three dimensional spherical region in its surrounding and the oscillations produced by source in spherical region propagate radially with speed of sound in the medium (air). These waves which travel in three dimensional space in spherical form are called spherical waves. All the particles in the surrounding of the source where it produce oscillations on a spherical surface, we call spherical wave front. Once a wavefront is created, it expands with speed of sound and all wavefronts are in the form of concentric spherical shells.



(a) Waves emitted by a point source



(b) Waves emitted by a line source

Figure 6.40

Similarly we can define cylindrical waves as shown in figure-6.40(b). As spherical waves are produced by a point source, cylindrical waves are produced by a line source. The best example to understand a cylindrical wave is a tube light, which is a source of cylindrical light waves. The wavefront produced by a line source is cylindrical and here also the radius of wavefront increases with speed of waves.

6.9.1 Intensity of Spherical and Cylindrical Waves

If a point source of power P watt produces a spherical wave then the power emitted by the source in its wavefronts is continuously spreaded in all directions. If we find the wave intensity at point A at a distance x from the source S , as shown in figure-6.41 then it can be directly given as

$$I_A = \frac{P}{4\pi x^2} \text{ W/m}^2 \quad \dots (6.121)$$

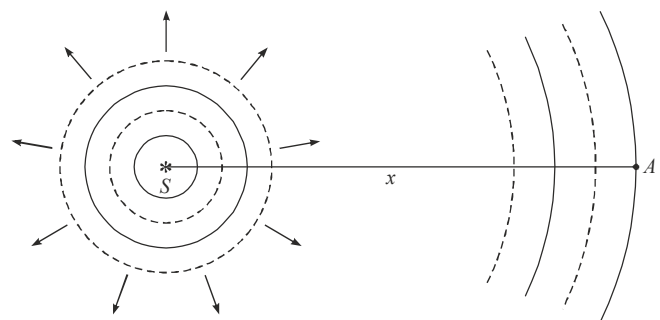


Figure 6.41

As at point A the total area of wavefront in which the power P is distributed is $4\pi x^2$ thus the intensity can be given by the expression in equation-(6.121).

Similar to this is case of a line source of power P and length l , the intensity at a distance x from the source can be given as

$$I = \frac{P}{2\pi x l} \quad \dots (6.122)$$

6.9.2 Equation of Spherical and Cylindrical Waves

In general a wave equation is given as

$$y = A \sin(\omega t - kx)$$

Where A is the amplitude of oscillations of medium particles. In case of a spherical wave we know the intensity of wave depends on the distance from the source. We've read that if P is the power of the source the intensity at a distance x can be given as

$$I = \frac{P}{4\pi x^2} \quad \dots (6.123)$$

We also know that the wave intensity is directly proportional to the square of amplitude of oscillations of medium particles thus in case spherical waves, we can say that the wave amplitude is inversely proportional to the distance from the source, as

$$A \propto \sqrt{I} \quad \dots (6.124)$$

or
$$A \propto \frac{1}{x}$$

or
$$A = \frac{A_0}{x} \quad \dots (6.125)$$

Here A_0 is a proportionality constant. Thus the wave equation of a spherical wave can be given as

$$y = \frac{A_0}{x} \sin(\omega t - kx) \quad \dots (6.126)$$

Similarly in case of cylindrical wave the intensity is inversely proportional to the distance from the source so we can say that here the wave amplitude is inversely proportional to the square root of the distance from the source these the amplitude of a cylindrical wave can be given as

$$A = \frac{A_0}{\sqrt{x}} \quad \dots (6.127)$$

Thus the equation of a cylindrical wave can be given as

$$y = \frac{A_0}{\sqrt{x}} \sin(\omega t - kx) \quad \dots (6.128)$$

6.10 Measurements of Sound Levels

The human ear is an extremely sensitive detector, capable of hearing sounds over an extremely large range of intensities. For example the sound produced by a thunder storm is 10^5 to

10^7 times greater than the sound intensity due to a buzzing mosquito, yet we can hear both sounds clearly. Our ears are sensitive to this enormous range of sound intensities. During observation people perceive a sound to be about twice as loud as a reference sound when its intensity is ten times as large as the reference sound. A sound perceived to be four times as loud as a reference requires an increase in sound intensity by a factor of 100. This relationship is approximately logarithmic, or we can state that loudness is proportional to the logarithm of the sound intensity.

The loudness of a sound is measured in units of decibel, dB . The decibel unit is one tenth of the size of the bel (a unit named in honor of Alexander Graham Bell). The intensity level in decibel is defined to be ten times the logarithm of the ratio of two intensities. If two sound intensities I_1 and I_2 are in W/m^2 then loudness level of I_2 with respect to I_1 can be given as

$$L \text{ (in dB)} = 10 \log_{10} \frac{I_2}{I_1} \quad \dots (6.129)$$

Here we say second sound is L dB louder than first sound. If we fix a standard reference level I_0 then we can measure the loudness level of any sound with respect to the reference intensity I_0 as

$$L \text{ (in dB)} = 10 \log_{10} \frac{I}{I_0} \quad \dots (6.130)$$

For example if a sound intensity exceeds the reference intensity I_0 by a factor of 4, then we have

$$L = 10 \log 4 = 10 (0.6) = 6 \text{ dB}$$

Thus here we can say that intensity level of the sound is 6 dB above the reference sound level. We can see that the measure of sound intensity level in decibel is actually a comparison of loudness of two different sound levels. So a given sound level cannot be expressed in units of decibels unless the reference level is fixed. The standard reference level is taken $I_0 = 10^{-12} W/m^2$, which is the sound intensity that can just be heard by a person with good hearing. This intensity $10^{-12} W/m^2$ is called the threshold of hearing. Figure-6.42 shows the comparison of different loudness levels of different commonly heard sounds in day to day life.

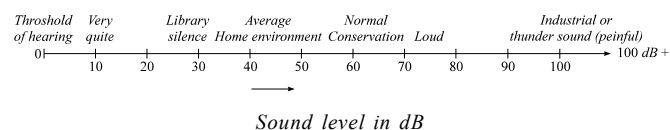


Figure 6.42

6.10.1 Comparison of Loudness Levels

Loudness of two sound waves are compared by their loudness levels as sound intensity is not a direct measure for perception

of loudness of sound wave so comparison of two sound waves is done by measuring the difference in loudness levels of the two sounds.

If there are two waves having intensities I_1 and I_2 , their loudness levels are given as

$$L_1 = 10 \log \frac{I_1}{I_0} \text{ dB}$$

and
$$L_2 = 10 \log \frac{I_2}{I_0} \text{ dB}$$

Difference in loudness level (in dB) is given as

$$\begin{aligned} \Delta L \text{ (in dB)} &= L_2 - L_1 = 10 \log \frac{I_2}{I_0} - 10 \log \frac{I_1}{I_0} \\ &= 10 \log \frac{I_2}{I_1} \end{aligned}$$

Here we say sound of wave-2 is ΔL dB louder than sound of wave-1.

Illustrative Example 6.29

(a) The power of sound from the speaker of a radio is 20 mW. By turning the knob of volume control the power of sound is increased to 400 mW. What is the power increase in dB as compared to original power?

(b) How much more intense is an 80 dB sound than a 20 dB whisper?

Solution

(a) As intensity is power per unit area, for a given source $P \propto I$, if L_1 and L_2 are the initial and final loudness levels then we have

$$L_2 - L_1 = 10 \log (I_2/I_1)$$

The increase in loudness level is

or
$$\Delta L = 10 \log \frac{P_2}{P_1} = 10 \log \frac{400}{20}$$

or
$$\Delta L = 10 [\log 20] \approx 13 \text{ dB}$$

(b) Increase in loudness level of sound is

$$L_2 - L_1 = 10 \log (I_2/I_1)$$

So,
$$80 - 20 = 10 \log (I_2/I_1)$$

or
$$6 = \log (I_2/I_1)$$

or
$$(I_2/I_1) = 10^6$$

Illustrative Example 6.30

A dog while barking delivers about 1 mW of power. If this power is uniformly distributed over a hemispherical area, what is the sound level at a distance of 5 m? What would the sound level be if instead of 1 dog, 5 dogs start barking at the same time delivering 1 mW of power?

Solution

As power is distributed uniformly in a hemisphere, intensity at a distance of 5 m from the source will be

$$I = \frac{P}{S} = \frac{P}{(1/2)4\pi r^2} = \frac{10^{-3}}{2 \times \pi \times 5^2} = 6.37 \mu\text{W/m}^2$$

Thus loudness level is

$$L = 10 \log \frac{I}{I_0} = 10 \log \frac{6.37 \times 10^{-6}}{(10^{-12})}$$

or
$$L = 10 [\log 6.37 + 6 \log 10] = 10 [0.80 + 6]$$

or
$$L = 68 \text{ dB}$$

If there are 5 dogs barking at the same time and same level, $I_2 = 5I_1$. So

$$L_2 - L_1 = 10 \log \frac{I_2}{I_1} = 10 \log \frac{5I_1}{I_1}$$

or
$$L_2 = L_1 + 10 \log 5$$

or
$$L_2 = 68 + 10 \times 0.7 = 75 \text{ dB}$$

Illustrative Example 6.31

The sound level at a point is increased by 30 dB. By what factor is the pressure amplitude increased?

Solution

The sound level in dB is

$$L = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

If L_1 and L_2 are the sound levels and I_1 and I_2 are the corresponding intensities in the two cases,

$$L_2 - L_1 = 10 \left[\log_{10} \left(\frac{I_2}{I_0} \right) - \log_{10} \left(\frac{I_1}{I_0} \right) \right]$$

or,
$$30 = 10 \log_{10} \left(\frac{I_2}{I_1} \right)$$

or,
$$\frac{I_2}{I_1} = 10^3$$

As the intensity is proportional to the square of the pressure amplitude, thus we have

$$\frac{\Delta p_2}{\Delta p_1} = \sqrt{\frac{I_2}{I_1}} = \sqrt{1000} \approx 32$$

Illustrative Example 6.32

What is the maximum possible sound level in dB of sound waves in air? Given that density of air = 1.3 kg/m^3 , $v = 332 \text{ m/s}$ and atmospheric pressure $P = 1.01 \times 10^5 \text{ N/m}^2$.

Solution

For maximum possible sound intensity, maximum pressure amplitude of wave can be at most equal to atmospheric pressure, so we have

$$\Delta p_0 = P = 1.01 \times 10^5 \text{ N/m}^2$$

$$\text{So } I = \frac{\Delta p_0^2}{2\rho v} = \frac{(1.01 \times 10^5)^2}{2 \times 1.3 \times 332} = 1.18 \times 10^7 \text{ W/m}^2$$

Thus loudness level is

$$L = 10 \log \frac{I}{I_0} \approx 10 \log \frac{10^7}{10^{-12}} = 190 \text{ dB}$$

Illustrative Example 6.33

A window whose area is 2 m^2 opens on street where the street noise result in an intensity level at the window of 60 dB. How much 'acoustic power' enters the window via sound waves. Now if an acoustic absorber is fitted at the window, how much energy from street will it collect in five hours?

Solution

Loudness level is given as

$$L = 10 \log \left(\frac{I}{I_0} \right)$$

$$\text{Thus } 10 \log \frac{I}{I_0} = 60$$

$$\text{or } \frac{I}{I_0} = 10^6$$

$$\text{or } I = (10^{-12} \times 10^6) = 10^{-6} \text{ W/m}^2 = 1 \mu\text{W/m}^2$$

And we have power of sound is

$$\begin{aligned} P &= IS \\ &= 1 \times 10^{-6} \times 2 = 2 \mu\text{W} \end{aligned}$$

Thus energy is given as

$$\begin{aligned} E &= P \times t \\ &= 2 \times 10^{-6} \times 5 \times 60 \times 60 = 36 \times 10^{-3} \text{ J} \end{aligned}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Interference of Waves

Module Number - 30 to 42

Practice Exercise 6.4

(i) If the intensity of sound is doubled, by how many decibels does the sound level increase?

[3.01 dB]

(ii) A certain sound level is increased by an additional 30 dB. Find the factor by which (a) its intensity increases and (b) its pressure amplitude increases.

[(a) 10^3 , (b) 31.62]

(iii) If the sound level in a room is increased from 50 dB to 60 dB, by what factor is the pressure amplitude increased?

[$\sqrt{10}$]

(iv) A typical loud sound wave with a frequency of 1 kHz has a pressure amplitude of about 10^{-4} atm . (a) At $t = 0$, the pressure is maximum at some point x_1 . What is the displacement at that point at $t = 0$? (b) What is the maximum value of the displacement at any time and place? Consider air as diatomic ideal gas.

[(a) 0, (b) $3.753 \times 10^{-6} \text{ m}$]

(v) A source of sound operates at 2.0 kHz, 20 W emitting sound uniformly in all directions. The speed of sound in air is 340 m/s and the density of air is 1.2 kg/m^3 . (a) What is the intensity at a distance of 6.0 m from the source? (b) What will be the pressure amplitude at this point? (c) What will be the displacement amplitude at this point?

[(a) 44.2 mW/m^2 , (b) 6.0 Pa, (c) $1.16 \times 10^{-6} \text{ m}$]

(vi) The intensity of sound from a point source is $1.0 \times 10^{-8} \text{ W/m}^2$ at a distance of 5.0 m from the source. What will be the intensity at a distance of 25 m from the source?

[$4.0 \times 10^{-10} \text{ W/m}^2$]

(vii) The noise level in a class-room in absence of the teacher is 50 dB when 50 students are present. Assuming that on the average each student outputs same sound energy per second, what will be the noise level if the number of students is increased to 100?

[53.01 dB]

(viii) At a distance $r_0 = 20.0$ m from a point isotropic source of sound the loudness level $L_0 = 30.0$ dB. Neglecting the damping of the sound wave, find :

- (a) the loudness level at a distance $r = 10.0$ m from the source;
 (b) the distance from the source at which the sound is not heard.

[(a) 36.02 dB; (b) 632.45 m]

(ix) In a good FM radio receiver, the radio signal detected may be as much as 65 dB greater than the noise signal. What is the ratio of signal intensity to noise intensity ?

$[3.16 \times 10^6]$

(x) Two sound waves move in the same direction in a medium. If the average powers transmitted across a cross section by them are equal while their wavelength are in the ratio 1: 2. Find the ratio of their pressure amplitudes.

[1]

6.11 Stationary Waves

In previous sections we've discussed that when two coherent waves superpose on a medium particle, phenomenon of interference takes place. Similarly when two coherent waves travelling in opposite direction superpose then simultaneous interference at all the medium particles takes place. These waves interfere to produce a pattern on all the medium particles what we call, a stationary waves. If the two interfering waves which travel in opposite direction carry equal energies then no net flow of energy takes place in the region of superposition. Within this region redistribution of energy takes place between medium particles. There are some medium particles where constructive interference takes place and hence energy increases and on the other hand there are some medium particles where destructive interference takes place and energy decreases. Now we'll discuss the stationary waves in analytically.

Let two waves of equal amplitude travelling in opposite direction along x -axis. The wave equation of the two waves can be given as

$$y_1 = A \sin(\omega t - kx)$$

[Wave travelling in $+x$ direction]... (6.131)

and

$$y_2 = A \sin(\omega t + kx)$$

[Wave travelling in $-x$ direction]... (6.132)

When the two waves superpose on medium particles, the resultant displacement of the medium particles can be given as

$$y = y_1 + y_2$$

or

$$y = A \sin(\omega t - kx) + A \sin(\omega t + kx)$$

or

$$y = A$$

$$[\sin \omega t \cos kx - \cos \omega t \sin kx + \sin \omega t \cos kx + \cos \omega t \sin kx]$$

or

$$y = 2A \cos kx \sin \omega t \quad \dots (6.133)$$

Equation-(6.133) can be rewritten as

$$y = R \sin \omega t \quad \dots (6.134)$$

Where

$$R = 2A \cos kx \quad \dots (6.135)$$

Here equation-(6.134) is an equation of SHM. It implies that after superposition of the two waves the medium particles executes SHM with same frequency ω and amplitude R which is given by equation-(6.135). Here we can see that the oscillation amplitude of medium particles depends on x i.e. the position of medium particles. Thus on superposition of two coherent waves travelling in opposite direction the resulting interference pattern, we call stationary waves, the oscillation amplitude of the medium particle at different positions is different.

At some points in medium the resultant amplitude is maximum which are given as

R is maximum when

$$\cos kx = \pm 1$$

$$\text{or} \quad \frac{2\pi}{\lambda} x = N\pi \quad [N \in \mathbb{I}]$$

$$\text{or} \quad x = \frac{N\lambda}{2}$$

$$\text{or} \quad x = 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, \dots$$

and the maximum value of R is given as

$$R_{\max} = \pm 2A \quad \dots (6.136)$$

Thus in the medium at positions $x = 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, \dots$ the waves interfere constructively and the amplitude of oscillations becomes $2A$. Similarly at some points of the medium, the waves interfere destructively, the oscillation amplitude becomes minimum i.e. zero in this case. These are the points where R is minimum, when

$$\cos kx = 0$$

$$\text{or} \quad \frac{2kx}{\lambda} = (2N+1) \frac{\pi}{2}$$

$$\text{or} \quad x = (2N+1) \frac{\lambda}{4} \quad [N \in \mathbb{I}]$$

$$\text{or} \quad x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \dots$$

and the minimum value of R is given as

$$R_{\min} = 0 \quad \dots (6.137)$$

Thus in the medium at positions $x = \lambda/4, 3\lambda/4, 5\lambda/4, \dots$ the waves interfere destructively and the amplitude of oscillation becomes zero. These points always remain at rest. Figure-6.43 shows the oscillation amplitude of different medium particles in a stationary wave.

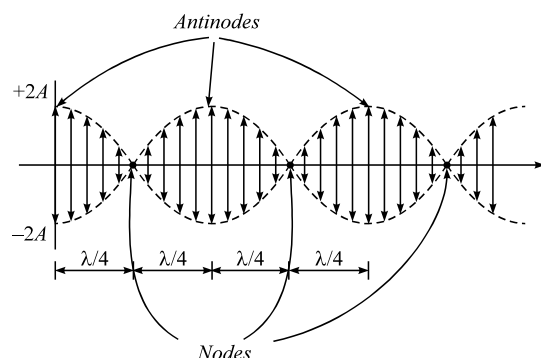


Figure 6.43

In figure we can see that the medium particles at which constructive interference takes place are called antinodes of stationary wave and the points of destructive interference are called nodes of stationary wave which always remain at rest.

Figure-6.44 explains the movement of medium particles with time in the region where stationary waves are formed. Let us assume that at an instant $t = 0$ all the medium particles are at their extreme positions as shown in figure-6.44(a). Here points A, B, C, D are the nodes of stationary waves where medium particles remain at rest. All other particles start moving toward their mean positions and at $t = T/4$ all particles cross their mean position as shown in figure-6.44(c), you can see in the figure that the particles at nodes are not moving. Now the medium crosses their mean position and starts moving on other side of mean position toward the other extreme position. At time $t = T/2$, all the particles reach their other extreme position as shown in figure-6.44(e) and at time $t = 3T/4$ again all these particles cross their mean position in opposite direction as shown in figure-6.44(g). Finally at $t = T$, after completing oscillation all the medium particles are in their initial position as shown in figure-6.44(i).

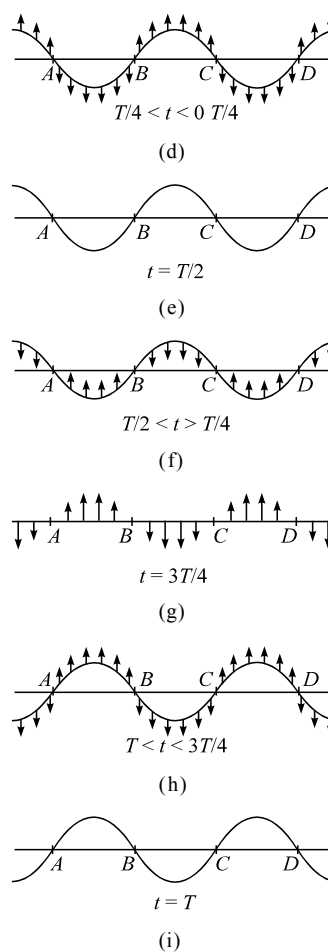
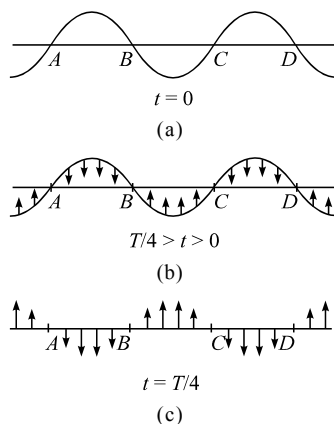


Figure 6.44

Based on the above analysis of one complete oscillations of the medium particles, we can make some inferences for a stationary wave. These are :

- (i) In oscillations of a stationary wave in a region, some points are always at rest (nodes) and some oscillate with maximum amplitudes (antinodes). All other medium particles oscillate with amplitudes less than those of antinodes.
- (ii) All medium particles between two successive nodes oscillate in same phase and all medium particles on one side of a node oscillate in opposite phase with those on the other side of the same node.
- (iii) In the region of a stationary wave during one complete oscillation all the medium particles come in the form of a straight line twice.
- (iv) If the component wave amplitudes are equal, then in the region where stationary wave is formed, no net flow of energy takes place, only redistribution of energy takes place in the medium.

When amplitudes of the two component waves are equal then the resultant amplitude at node is zero and these particles always remain at rest, which does not allow energy to propagate in any direction. But when the component waves have unequal amplitude say A_1 and A_2 then the medium particles situated at nodes also oscillate with amplitude $|A_1 - A_2|$ and in this case the amplitude of oscillations of medium particles at antinodes will be $|A_1 + A_2|$. Such an interference pattern is called partial stationary wave. Because in this case as the two component waves have unequal, amplitudes, their powers will also be different in opposite directions. Due to this after superposition there must be some flow of energy in the direction of propagation of wave having higher amplitude. The oscillation amplitude pattern of different medium particles in a pure stationary wave and a particle stationary wave is shown in figure-6.45

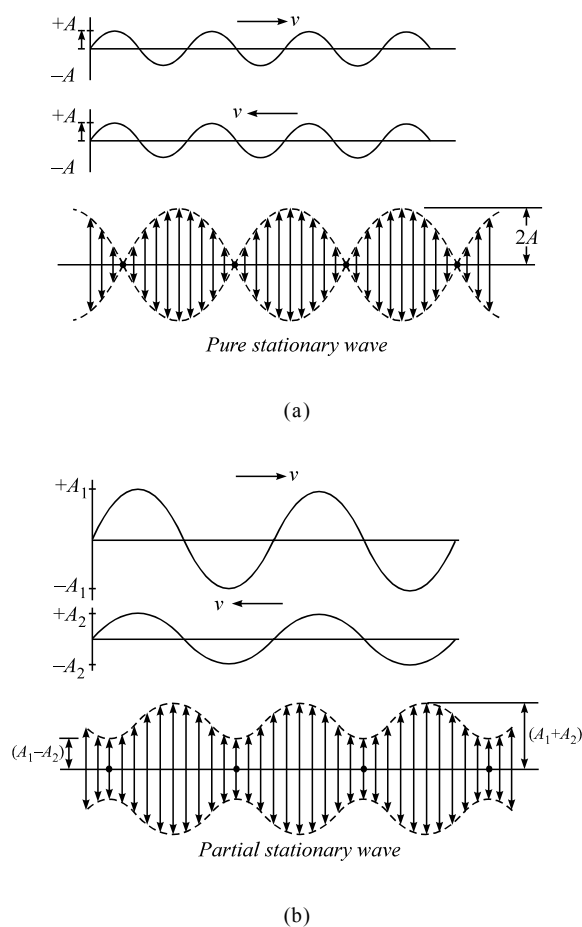


Figure 6.45

A partial stationary wave can also be regarded as a superposition of a pure stationary wave with a propagating wave. Analytically we can discuss this. Let the two different amplitude coherent waves travelling in opposite direction be

$$y_1 = A_1 \sin(\omega t - kx) \quad \dots (6.138)$$

[Travelling in $+x$ direction]

and

$$y_2 = A_2 \sin(\omega t + kx) \quad [A_1 > A_2] \quad \dots (6.139)$$

[Travelling in x -direction]

The region in which these waves will superpose the resultant displacement of medium particles can be given as

$$y = y_1 + y_2$$

$$= A_1 \sin(\omega t - kx) + A_2 \sin(\omega t + kx)$$

First term of the above equation can be split as

$$y = A_2 \sin(\omega t - kx) + (A_1 - A_2) \sin(\omega t - kx) + A_2 \sin(\omega t + kx)$$

$$y = 2A_2 \cos kx \sin \omega t + (A_1 - A_2) \sin(\omega t - kx) \quad \dots (6.140)$$

The first term in the expression in equation-(6.140) is the equation of a pure stationary wave with antinode amplitude $2A_2$ and node amplitude zero and second term in equation-(6.140) is a propagating wave travelling in $+x$ direction with amplitude $A_1 - A_2$. Here we can state that net energy flow takes place in the medium due to the propagating wave in $+x$ direction as $A_1 > A_2$.

6.11.1 Different Equations for a Stationary Wave

Consider two equal amplitude waves travelling in opposite direction as

$$y_1 = A \sin(\omega t - kx) \quad \dots (6.141)$$

and

$$y_2 = A \sin(\omega t + kx) \quad \dots (6.142)$$

The result of superposition of these two waves is

$$y = 2A \cos kx \sin \omega t \quad \dots (6.143)$$

Which is the equation of stationary wave where $2A \cos kx$ represents the amplitude of medium particle situated at position x and $\sin \omega t$ is the time sinusoidal factor. This equation-(6.143) can be written in several ways depending on initial phase differences in the component waves given by equation-(6.141) and (6.142). If the superposing waves are having an initial phase difference π , then the component waves can be expressed as

$$y_1 = A \sin(\omega t - kx) \quad \dots (6.144)$$

$$y_2 = -A \sin(\omega t - kx) \quad \dots (6.145)$$

Superposition of the above two waves will result

$$y = 2A \sin kx \cos \omega t \quad \dots (6.146)$$

Equation-(6.146) is also an equation of stationary wave but here amplitudes of different medium particles in the region of interference is given by

$$R = 2A \sin kx \quad \dots (6.147)$$

Similarly the possible equations of a stationary wave can be written as

$$y = A_0 \sin kx \cos (\omega t + \phi) \quad \dots (6.148)$$

$$y = A_0 \cos kx \sin (\omega t + \phi) \quad \dots (6.149)$$

$$y = A_0 \sin kx \sin (\omega t + \phi) \quad \dots (6.150)$$

$$y = A_0 \cos kx \cos (\omega t + \phi) \quad \dots (6.151)$$

Here A_0 is the amplitude of antinodes. In a pure stationary wave it is given as

$$A_0 = 2A$$

Where A is the amplitude of component waves. If we care fully look at equation-(6.148) to (6.151), we can see that in equation-(6.148) and (6.150), the particle amplitude is given by

$$R = A_0 \sin kx \quad \dots (6.152)$$

Here at $x = 0$, there is a node as $R = 0$ and in equation-(6.149) and (6.151) the particle amplitude is given as

$$R = A_0 \cos kx \quad \dots (6.153)$$

Here at $x = 0$, there is an antinode as $R = A_0$. Thus we can state that in a given system of co-ordinates when origin of system is at a node we use either equation-(6.148) or (6.150) for analytical representation of a stationary wave and we use equation-(6.149) or (6.151) for the same when an antinode is located at the origin of system.

6.11.2 Velocity and Acceleration of Medium Partical in Stationary Wave

In a stationary wave we have discussed that the displacement of a medium particle from its mean position is given by either of equation (6.148) to (6.151) we take any one of these to represent a stationary wave in a medium as

$$y = A_0 \sin kx \cos \omega t \quad \dots (6.154)$$

Here velocity of a medium particle at a position x can be given by

$$v_p = \frac{\partial y}{\partial t} = \omega A_0 \sin kx \cos \omega t \quad \dots (6.155)$$

Differentiating again equation-(6.155) only w.r. to time gives the acceleration of the medium particle at a position x as

$$a_p = \frac{\partial^2 y}{\partial t^2} = -\omega^2 A_0 \sin kx \sin \omega t \quad \dots (6.156)$$

Similarly we can find the slope of displacement curve of a stationary wave at a particular instant of time from equation-(6.154) as

$$\frac{\partial y}{\partial x} = k A_0 \cos kx \sin \omega t \quad \dots (6.157)$$

Illustrative Example 6.34

The following equation represents standing wave set up in a medium,

$$y = 4 \cos \frac{\pi x}{3} \sin 40 \pi t,$$

where x and y are in cm and t in sec. Find out the amplitude and the velocity of the two component waves and calculate the distance between adjacent nodes. What is the velocity of a medium particle at $x = 3$ cm at time $1/8$ s?

Solution

The given equation of stationary wave is

$$y = 4 \cos \frac{\pi x}{3} \sin 40 \pi t$$

$$\text{or } y = 2 \times 2 \cos \frac{2\pi x}{6} \sin \frac{2\pi (120)t}{6} \quad \dots (6.158)$$

We know that

$$y = 2 A \cos \frac{2\pi x}{\lambda} \sin \frac{2\pi v t}{\lambda} \quad \dots (6.159)$$

Comparing the equations-(6.158) and (6.159), we get

$$A = 2 \text{ cm}, \lambda = 6 \text{ cm} \quad \text{and} \quad v = 120 \text{ cm/s}$$

The component waves are

$$y_1 = A \sin \frac{2\pi}{\lambda} (v t - x)$$

$$\text{and } y_2 = A \sin \frac{2\pi}{\lambda} (v t + x)$$

Distance between two adjacent nodes

$$= \frac{\lambda}{2} = \frac{6}{2} = 3 \text{ cm}$$

Particle velocity

$$\frac{dy}{dt} = 4 \cos \frac{\pi x}{3} \cos (40 \pi t) \times 40 \pi$$

$$= 160 \pi \cos \left(\frac{\pi x}{3} \right) \cos 40 \pi t$$

At $x = 3$, $t = 1/8$, the particle velocity is given by

$$= 160 \pi \cos \pi \cos \left(40\pi \times \frac{1}{8} \right) = 160 \pi \text{ cm/s}$$

Illustrative Example 6.35

A wave is given by the equation

$$y = 10 \sin 2 \pi (100 t - 0.02 x) + 10 \sin 2 \pi (100 t - 0.02 x)$$

Find the loop length, frequency, velocity and maximum amplitude of the stationary wave produced.

Solution

We know the equation of a stationary wave is given by

$$\begin{aligned}
 y &= A \sin \left[\frac{2\pi}{\lambda} (vt - x) \right] + A \sin \left[\frac{2\pi}{\lambda} (vt + x) \right] \\
 &= 2A \cos \frac{2\pi x}{\lambda} \sin \frac{2\pi vt}{\lambda} \\
 &= R \sin \frac{2\pi}{\lambda} vt
 \end{aligned}$$

Here $R = 2A \cos \frac{2\pi x}{\lambda}$ is the amplitude of medium particle situated at a distance x .

The given equation can be expressed as

$$\begin{aligned}
 y &= 10 \sin \left[\frac{2\pi}{50} (5000t - x) \right] + 10 \sin \left[\frac{2\pi}{50} (5000t + x) \right] \\
 &= 2 \times 10 \cos \left(\frac{2\pi x}{50} \right) \sin \left(\frac{2\pi}{50} 5000t \right)
 \end{aligned}$$

Comparing it with standard equation of stationary wave, we get wavelength

$$\lambda = 50 \text{ units}$$

$$\text{Wave velocity } v = 5000 \text{ units}$$

$$\text{Thus amplitude } R = 2 \times 10 \cos \frac{2\pi x}{50}$$

and Maximum amplitude

$$R_{\max} = 2 \times 10 = 20 \text{ units}$$

As wave velocity is

$$v = 5000 \text{ units,}$$

$$\text{Frequency } = \frac{v}{\lambda} = \frac{5000}{5} = 100 \text{ units}$$

and loop length is

$$\frac{\lambda}{2} = \frac{50}{2} = 25 \text{ units.}$$

Illustrative Example 6.36

A progressive and a stationary simple harmonic wave each has the same frequency of 250 Hz, and the same velocity of 30 m/s. Calculate

- the phase difference between two vibrating points on the progressive wave which are 100 cm apart,
- the equation of motion of the progressive wave if its amplitude is 0.03 m,
- the distance between nodes in the stationary wave,
- the equation of motion, the stationary wave if its amplitude is 0.01 m.

Solution

$$\text{Given, } n = 250 \text{ Hz, } v = 30 \text{ m/s}$$

$$\text{or } \lambda = \frac{v}{n} = \frac{3}{25} \text{ m} = 12 \text{ cm.}$$

(i) Phase difference between two points at a distance $\lambda = 2\pi$

Phase difference between two points, unit distance apart = $\frac{2\pi}{\lambda}$

Thus phase difference for a distance of 10 cm

$$= \frac{2\pi}{\lambda} \times 10 = \frac{2\pi}{12} \times 10 = \frac{5}{3} \pi$$

(ii) Now $A = 0.03, \lambda = (3/25) \text{ m}$

$$\text{and } \frac{1}{T} = n = 250 \text{ Hz}$$

The general equation of a plane progressive wave is given by

$$y = A \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$$

$$\text{Here } y = 0.03 \sin 2\pi (250t - 25x/3)$$

(iii) The distance between nodes in stationary wave

$$= \frac{\lambda}{2} = \frac{12}{6} = 2 \text{ cm}$$

(iv) Equation of stationary wave is given by

$$y = 2A \cos \frac{2\pi x}{\lambda} \sin \frac{2\pi vt}{\lambda}$$

$$= 2A \cos \frac{2\pi x}{\lambda} \sin \frac{2\pi t}{T}$$

$$\text{Here } 2A = 0.01, \lambda = \frac{3}{25} \text{ m and } \frac{1}{T} = 250 \text{ Hz}$$

$$\text{or } y = 0.01 \cos \frac{5\pi x}{3} \sin 500\pi t$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Stationary Waves and Beats

Module Number - 1 to 8

6.12 Standing Waves on Clamped String

In previous sections we've discussed about the reflection of a wave pulse on a string when it arrives at a boundary point

which may be a fixed or a free end. Now we'll discuss what happens when a propagating simple harmonic wave is reflected by a fixed end of a string. Consider the situation shown in figure-6.46.

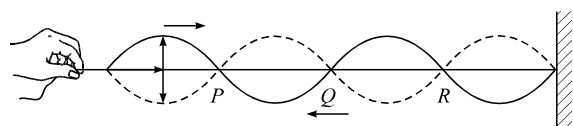


Figure 6.46

A stretched string is clamped at a point on a rigid vertical wall. The other end of string is given oscillations manually which creates a continuous propagating wave toward the wall from which it is reflected with a phase difference of π radians which start propagating in direction opposite to incident wave. If we assume ideal reflection from the wall then the reflected wave will also have the same amplitude as that of the incident wave. As here two coherent waves (as during reflection frequency does not change) travelling in opposite direction superpose each other, stationary waves are formed on the string. As we can see in figure-6.46. At points P , Q , R nodes are formed and between every two adjacent node an antinode is there. But this is not so simple that you move one end of a clamped string up and down and you'll see nodes and antinodes. This will happen only at some particular oscillation frequencies which depend on several factors like length of string, tension in string, linear mass density of string etc. Thus if a clamped string is oscillated like the way as shown in figure-6.46 stationary waves will be formed or not it depends on the frequency of source of oscillations (in our case it is hand) but theoretically when incident and reflected waves superpose, stationary waves must be formed then why does the practical situation differ. This we'll discuss now in detail.

First we discuss one more example of oscillations of a string clamped at both the ends as shown in figure-6.47. If we pluck the string at its mid point and release it will start oscillating as shown in figure-6.47(b). Actually when we pluck the string at its mid point and release, two transverse waves are generated and start moving toward the clamps C_1 and C_2 from middle portion of string. These are reflected from the clamps and superpose on the length of the string and stationary wave is formed between C_1 and C_2 . As being rigid clamps the string particles are not allowed to oscillate at these points. Here at C_1 and C_2 , nodes must be formed and there is one antinode at the mid point of the string. Similarly if at the mid point of the string we put a support and we pluck the left portion of the string a loop is formed and the wave of same frequency transmits to the right portion and another loop is formed there as shown in figure-6.47(c). Now if the support at the mid point is removed

then also string oscillations remains same, it continues to oscillates in two loops as already there was a node at the point of support so it does not make any difference whether there is any support present or not. The support was required only to initiate the oscillations. In the same fashion we can make string to oscillate in three, four or more loops by placing a support at $\frac{1}{3}$ rd, $\frac{1}{4}$ th length from one clamp to initiate the oscillations in the beginning.

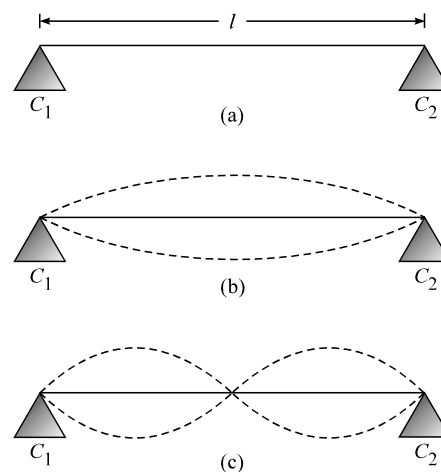


Figure 6.47

Here we can see that when there is no support, the string oscillates full length as one loop. In this case the separation between two nodes is the string length l i.e. $\frac{\lambda}{2}$. In second case we have $\frac{l}{2} = \frac{\lambda}{2}$ and so on. This implies that as number of loops in the string increases the wavelength decreases and hence frequency of oscillations increases.

Thus in a clamped string oscillations, stationary waves are formed only when the string length is equally divided into loops otherwise there will not be a node at one of the clamps. So for the formation of stationary wave in a clamped string the string length must be an integral multiple of $\frac{\lambda}{2}$, where λ is the wave length of the wave on string.

6.12.1 Normal Modes of Oscillations of a Clamped String

We've discussed that stationary waves can be established in a clamped string and it can oscillate at different frequencies. These frequencies at which a stable stationary wave can be produced in a clamped string are called different modes of oscillations. Now we discuss these modes in detail.

When a string clamped at both the ends oscillates. The minimum frequency at which it can oscillate is called its fundamental frequency or the fundamental mode of its oscillations. This

case is shown in figure-6.48 when only one antinode is formed between two nodes (clamps). In this case the wavelength is at its maximum possible value.

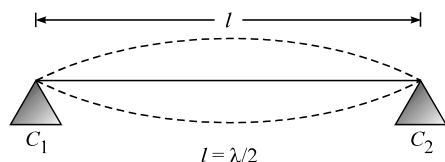


Figure 6.48

In this case oscillation frequency of string can be given as

$$n = \frac{v}{\lambda} = \frac{v}{2l} \quad [\text{As here } \frac{\lambda}{2} = l]$$

$$= \frac{1}{2l} \sqrt{\frac{T}{\mu}} \quad \dots (6.160)$$

Here T is the tension at which string is stretched and μ is the linear mass density of the string. The next higher frequency at which string oscillates with stationary waves is the one when two loops are formed as shown in figure-6.49

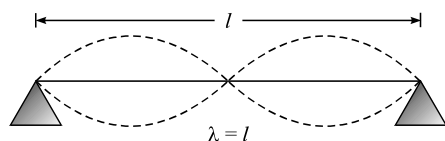


Figure 6.49

In this case oscillation frequency of string can be given as

$$n_2 = \frac{v}{\lambda} = \frac{v}{l} \quad [\text{As here } \lambda = l]$$

$$= \frac{2}{2l} \sqrt{\frac{T}{\mu}} = 2n_1 \quad \dots (6.161)$$

This frequency n_2 is the next higher frequency after fundamental frequency at which stable stationary waves are formed in the string. This mode of oscillations is called first overtone. As this frequency is double that of fundamental frequency this is termed as second harmonic frequency of string oscillations. All the frequencies which are integral multiples of fundamental frequency are called harmonic frequencies.

If we come to the next mode of oscillations of this string it is the case when it vibrates in three loops as shown in figure-6.50.

Here $l = \frac{3\lambda}{2}$

or $\lambda = \frac{2l}{3}$

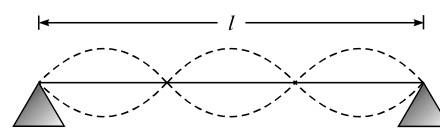


Figure 6.50

In this case the oscillation frequency of the string is given as

$$n_3 = \frac{v}{\lambda} = \frac{3v}{2l}$$

$$= \frac{3}{2l} \sqrt{\frac{T}{\mu}} = 3n_1 \quad \dots (6.162)$$

This mode of oscillation is called second overtone of the string oscillations and as the frequency is three times the fundamental frequency, it is also termed as third harmonic. Similarly if a string oscillates in P loops, its frequency can be written as

$$n_p = \frac{pv}{2l} = \frac{p}{2l} \sqrt{\frac{T}{\mu}} = pn_1$$

Where $p = 1, 2, 3, \dots$... (6.163)

The above frequencies n_p at which stationary waves can be formed are the harmonic frequencies and this series of these frequencies is called harmonic series.

6.13 Wave Resonance

When a pendulum or a block at the end of a spring is set into motion by a periodic force, the system moves with largest amplitude or with maximum energy when the frequency of force equals the frequency of natural vibrations of the free system. We can take a simple example to explain resonance. If we push a child on a swing periodically then the oscillations of swing become stronger if the pushing frequency is equal to the frequency of natural oscillations of the swing. Thus the concept of resonance is that the oscillations of a free system are most strong when the frequency of driving force is equal to the frequency of the system's free vibrations.

6.13.1 Resonance in Clamped String Oscillations

Resonance can also be observed in case of a clamped string oscillations consider a string of length l clamped between the two clamps C_1 and C_2 . The length and tension in the string adjusted so that here we have the fundamental frequency of this string is 10 Hz, thus we have

$$n_1 = \frac{v}{2l} = 10 \text{ Hz} \quad \dots (6.164)$$

This implies that this string can sustain stationary waves at all frequencies which are multiple of 10 Hz (Harmonic frequencies). At frequencies other than 10 Hz the oscillation energy of string will be damped as its length will not be integral multiple of $\frac{\lambda}{2}$.

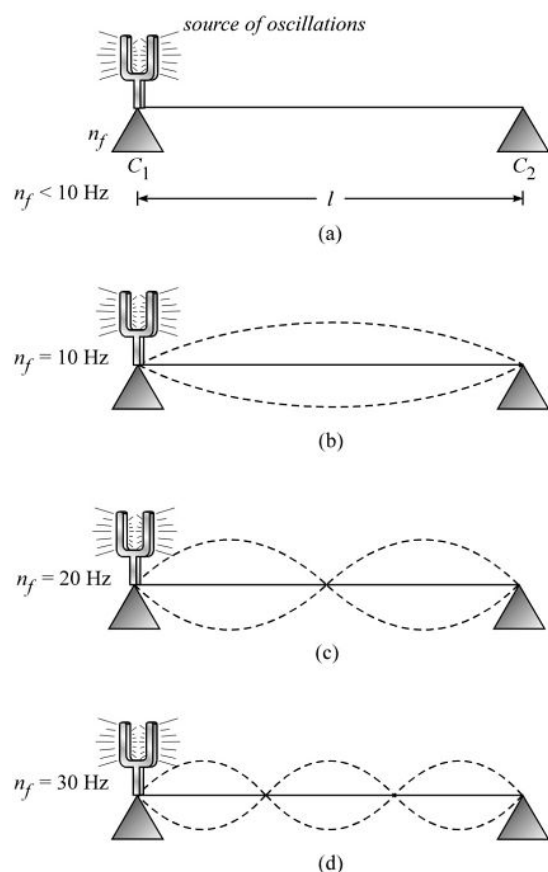


Figure 6.51

Let us use a tuning fork of frequency $n_f < 10$ Hz to establish oscillations in the string as shown in figure-6.51(a). As at frequencies less than 10 Hz, stationary waves can not be established in the string thus the energy supplied by the tuning fork will damp and string will oscillate very little or it appears motionless. But if the frequency of the tuning fork (driving agent) matches with any of the harmonic frequencies of the string, it starts oscillating widely and the amplitude of its oscillations is also so significant that it can be seen as shown in figure-6.51(b), (c) and (d). This is a common method of establishing stationary waves in a clamped string. Thus a string resonates whenever the external oscillating source frequency matches with any of the harmonic frequency of the string. In previous example if a tuning fork of frequency 43.2 Hz is used to setup oscillations in the string it will oscillate at 43.2 Hz but will appear almost straight as shown in figure-6.51(a) as the whole of supplied energy will clamp.

6.13.2 Sonometer

It is a device used to measure velocity of transverse mechanical wave in a stretched metal wire. The principle of sonometer is based on resonance of string vibrations. Working oscillations

are induced in a clamped string by an external source like a tuning fork or an oscillator and the corresponding oscillations in string will become stronger when resonance takes place i.e. the frequency of oscillation of source matches with any of the harmonic of the string vibration.

Figure-6.52 shows basic structure and of a sonometer setup arrangement. It consists of a wooden box M on which a wire AB is stretched, by a hanging weight as shown in figure-6.52. On sonometer box there are two clamps C_1 and C_2 placed which can slide under the wire to change the length of wire between the clamps.

When an oscillating tuning fork is placed in contact with the sonometer wire as shown in figure-6.52. Some oscillations are transferred to the wire. If tension in wire is T and n_0 be the frequency of tuning fork, the wavelength of wave in wire is

$$\lambda = \frac{v}{n_0} = \frac{1}{n_0} \cdot \sqrt{\frac{T}{\mu}} \quad \dots (6.165)$$

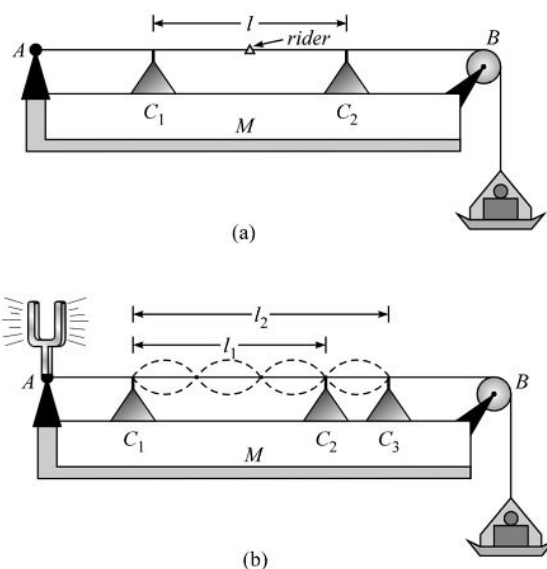


Figure 6.52

When the length between clamps is an integral multiple of $\lambda/2$ then stationary waves are established in the portion of wire between C_1 and C_2 . To adjust this, clamp C_1 is fixed and C_2 is displaced so that a small rider (a piece of paper) on wire starts jumping violently on wire and falls indicating that the oscillation amplitude of wire is increasing and stationary waves are established. Let in this situation the length between clamps is l_1 . Now again C_2 is displaced away from C_1 so that again resonance is obtained. This will happen again when the clamp reaches the position C_3 and when next node of stationary waves is present as shown in figure-6.52(b). Let this length be l_2 .

So we can say that if l_1 and l_2 are the two successive resonance lengths then we have

$$l_2 - l_1 = \frac{\lambda}{2}$$

or wavelength of wave is

$$\lambda = 2(l_2 - l_1)$$

As frequency n_0 of oscillating source is known, we can find the velocity of wave in wire as

$$\begin{aligned} v &= n_0 \lambda \\ &= 2n_0(l_2 - l_1) \end{aligned} \quad \dots (6.166)$$

Equation-(6.166) gives the practically measured value of velocity of transverse waves in stretched wire. This can be compared with the theoretical value of v given by $\sqrt{\frac{T}{\mu}}$.

6.13.3 Vibrations of Composite Strings

We have discussed the vibration of a clamped string in previous section, now we discuss the stationary waves in a string composed of two different strings as shown in figure-6.53. Here two strings S_1 and S_2 of different material and lengths are joined end to end and tied between two clamps as shown. Now when we induce oscillations in this composite string, stationary waves are established only at those frequencies which matches with any one harmonic of both the independent string S_1 and S_2 .

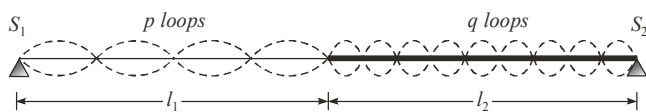


Figure 6.53

Let at a frequency stationary waves are established in this string so that string S_1 oscillates in p loops and strings S_2 oscillates in q loops as frequency of both the string is equal, we have

$$n_{ext} = n_{S_1} = n_{S_2}$$

$$\text{or} \quad n_{ext} = \frac{p}{2l_1} \sqrt{\frac{T}{\mu_1}} = \frac{q}{2l_2} \sqrt{\frac{T}{\mu_2}} \quad \dots (6.167)$$

[As T = same in both the string]

$$\text{or} \quad \frac{p}{q} = \frac{l_1}{l_2} \sqrt{\frac{\mu_1}{\mu_2}} \quad \dots (6.168)$$

In equation-(6.168) p and q must be integers, thus the minimum values of p and q which satisfies equation-(6.168) will decide fundamental frequency or the least frequency at which stationary waves can exist in this composite string and the oscillation frequency can be obtained from equation-(6.167).

6.13.4 Melde's Experiment

This is an experiment for demonstration of transverse stationary wave in a stretched string.

In Melde's experiment, one end of the string is connected to the prong of an electrically oscillated tuning fork. The other end of the string is connected to the scale pan. The string passes over a smooth friction less pulley. The distance between tuning fork and pulley can be adjusted. There are two different ways in which oscillations can be established in the string

Case-I: Transverse Mode of vibration

As shown in figure-6.54, tuning fork vibrates right angle to the length of the string. In this case frequency of vibration of string, is equal to the frequency of the tuning fork. First we adjust the length of string so that stationary waves are formed in string. In this case the vibrations of string are strong enough so that the loops can be seen in the string. If string vibrates in p loops as shown in figure, the frequency of string oscillations can be given as

$$n_s = \frac{p}{2l} \sqrt{\frac{T}{\mu}} \quad \dots (6.169)$$

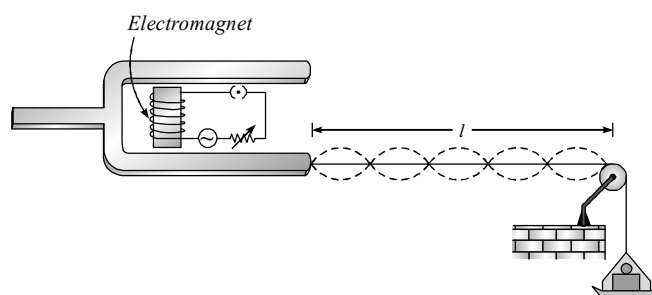


Figure 6.54

If n_f be the frequency of oscillation of tuning fork then we have

$$n_f = n_s = \frac{p}{2l} \sqrt{\frac{T}{\mu}} \quad \dots (6.170)$$

(2) Longitudinal Mode of vibrations

In this case the vibrations of tuning fork are along the length of the string. The orientation of tuning fork is shown in figure-6.55. In this case for one complete vibrations of the tuning fork, the string completes only half of its vibration so the frequency of vibration of string is half of that of the oscillation frequency of tuning fork. If string vibrates in p loops then we have,

$$\frac{n_f}{2} = n_s = \frac{p}{2l} \sqrt{\frac{T}{\mu}} \quad \dots (6.171)$$

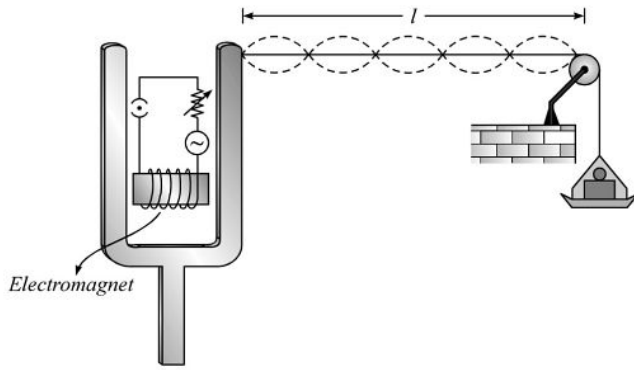


Figure 6.55

We can see if the frequency of tuning fork remains constant from equation-(6.170) and (6.171) we can write

$$p\sqrt{T} = \text{constant} \quad [\text{As } \mu = \text{constant}] \quad \dots (6.172)$$

If the tension in the string is changed then number of loops in the stationary wave formed varies according to equation-(6.172). This is called as Melde's law.

Thus if in Melde's experiment stationary waves are formed at two different values of tension T_1 and T_2 at which p_1 and p_2 loops are formed in the string then we can write as

$$\frac{p_1}{p_2} = \sqrt{\frac{T_2}{T_1}} \quad \dots (6.173)$$

6.13.5 Energy Associated with the Medium Particles in a Stationary Wave on a String

Consider a stationary wave on a stretched string of length l as shown in figure-6.56. The maximum amplitude of medium particles in this stationary wave at antinodes is A_0 . Then the amplitude of medium particles at a distance x from one end can be given as

$$R = A_0 \sin kx$$

Now consider an element of length dx at a distance x from one end as shown. If frequency of oscillation of string is ω then the total energy contained in this element of length dx can be given as

$$dE = \frac{1}{2} dm \omega^2 R^2$$

$$\text{or} \quad dE = \frac{1}{2} (\mu dx) \omega^2 A_0^2 \sin^2 kx \quad \dots (6.174)$$

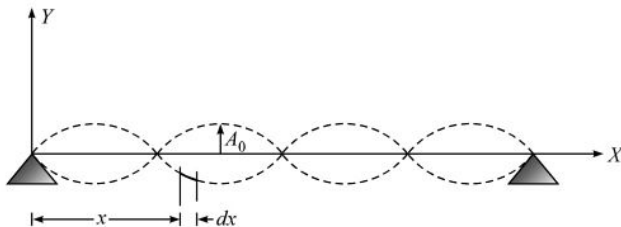


Figure 6.56

Now total energy contained in the string can be obtained by integrating the expression in equation-(6.174) for whole length of string as

$$\begin{aligned} E &= \int dE = \int_0^L \frac{1}{2} \mu \omega^2 A_0^2 \sin^2 kx dx \\ &= \frac{1}{2} \mu \omega^2 A_0^2 \int_0^L \left(\frac{1 - \cos 2kx}{2} \right) dx \\ &= \frac{1}{4} \mu \omega^2 A_0^2 \left[x - \frac{\sin 2kx}{2k} \right]_0^L \\ &= \frac{1}{4} \mu \omega^2 A_0^2 [L] \\ &= \frac{1}{4} \mu \omega^2 A_0^2 L \quad \dots (6.175) \end{aligned}$$

6.14 Vibrations of Clamped Rod

We have discussed the resonant vibrations of a string clamped at two ends. Now we discuss the oscillations of a rod clamped at a point on its length as shown in figure-6.57. Figure shows a rod AB clamped at its middle point. If we gently hit the rod at its one end, it begins to oscillate and in the natural oscillations the rod vibrates at its lowest frequency and maximum wavelength, which we call fundamental mode of oscillations. With maximum wavelength when transverse stationary waves setup in the rod, the free ends vibrate as antinodes and the clamped end as a node as shown in figure. Here if λ be the wavelength of the wave, we have

$$l = \frac{\lambda}{2}$$

$$\text{or} \quad \lambda = 2l \quad \dots (6.176)$$

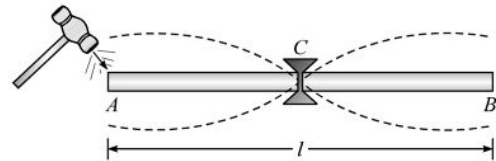


Figure 6.57

Thus the frequency of fundamental oscillations of a rod clamped at mid point can be given as

$$n_0 = \frac{v}{\lambda} = \frac{1}{2l} \sqrt{\frac{Y}{\rho}} \quad \dots (6.177)$$

Where Y is the Young's modulus of the material of rod and ρ is the density of the material of rod.

Next higher frequency at which rod vibrates will be then one when wave length is decreased to a value so that one node is

inserted between mid point and an end of rod as shown in figure-6.58.

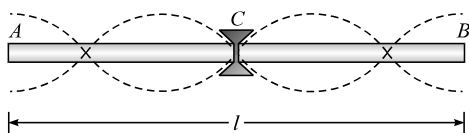


Figure 6.58

In this case if λ be the wavelength of the waves in rod, we have

$$l = \frac{3\lambda}{2}$$

or
$$\lambda = \frac{2l}{3} \quad \dots (6.178)$$

Thus in this case the oscillation frequency of rod can be given as

$$v_1 = \frac{v}{\lambda} = \frac{3}{2l} \sqrt{\frac{Y}{\rho}} = 3n_0 \quad \dots (6.179)$$

This is called first overtone frequency of the damped rod or third harmonic frequency. Similarly, the next higher frequency of oscillation i.e. second overtone of the oscillating rod can be shown in figure-6.59. Here if λ be the wavelength of the wave then it can be given as

$$l = \frac{5\lambda}{2}$$

or
$$\lambda = \frac{2l}{5} \quad \dots (6.180)$$

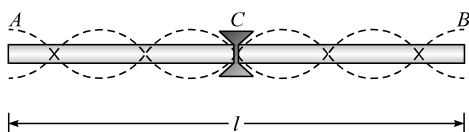


Figure 6.59

Thus the frequency of oscillation of rod can be given as

$$v_2 = \frac{v}{\lambda} = \frac{5}{2l} \sqrt{\frac{Y}{\rho}} = 5n_0 \quad \dots (6.181)$$

Thus the second overtone frequency is the fifth harmonic of the fundamental oscillation frequency of rod. We can also see from the above analysis that the resonant frequencies at which stationary waves are setup in a damped rod are only odd harmonics of fundamental frequency.

Thus when an external source of frequency matching with any of the harmonic of the damped rod then stationary waves are setup in the rod.

Illustrative Example 6.37

A wire of density $9 \times 10^3 \text{ kg/m}^3$ is stretched between two clamps 1 m apart and is subjected to an extension of $4.9 \times 10^{-4} \text{ m}$. What will be the lowest frequency of transverse vibration in the wire ? (Young's modulus of material = $9 \times 10^{10} \text{ N/m}^2$).

Solution

Let A be the area of cross-section of wire and T be the tension applied.

The lowest frequency of transverse vibration is given by

$$n = \frac{1}{2l} \sqrt{\left(\frac{T}{\mu}\right)} \quad \dots (6.182)$$

Where m = mass per unit length of wire

= Volume of unit length $\times$ density

$$= A \times 1 \times \rho$$

or
$$n = \frac{1}{2l} \sqrt{\left(\frac{T}{A\rho}\right)} \quad \dots (6.183)$$

We know that
Young's modulus

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{T/A}{\Delta l/l}$$

or
$$T = YA \left(\frac{\Delta l}{l}\right) \quad \dots (6.184)$$

From equations-(6.183) and (6.184)

$$n = \frac{1}{2l} \sqrt{\left(\frac{YA \Delta l}{l A \rho}\right)} = \frac{1}{2l} \sqrt{\left(\frac{Y \Delta l}{l \rho}\right)}$$

Substituting the given values

$$n = \frac{1}{2 \times 1} \sqrt{\left(\frac{(9 \times 10^{10}) \times (4.9 \times 10^{-4})}{1 \times (9 \times 10^3)}\right)} \\ = 35 \text{ Hz}$$

Illustrative Example 6.38

The fundamental frequency of a 1.5 m long, stretched steel wire is 175 Hz. The density of steel is $7.8 \times 10^3 \text{ kg/m}^3$. (i) Find the speed of transverse waves in the wire. (ii) Calculate the longitudinal stress of the wire. (iii) If the tension in the wire is increased by 3%, calculate the percentage change in the frequency of the wire.

Solution

(i) We know that the fundamental frequency of the wire is given by

$$n = \frac{1}{2l} \sqrt{\left(\frac{T}{\mu}\right)} \quad \dots (6.185)$$

The speed of the transverse wave in the wire is

$$v = \sqrt{\left(\frac{T}{\mu}\right)} = n \times 2l \quad \dots (6.186)$$

Substituting the given values, we have

$$v = 175 \times (2 \times 1.5) = 525 \text{ m/s}$$

(ii) Suppose the weight suspended from the wire be Mg . Then

$$T = Mg \quad \text{and} \quad \mu = \pi r^2 \rho$$

or

$$v = \sqrt{\left(\frac{T}{\mu}\right)} = \sqrt{\left(\frac{Mg}{\pi r^2 \rho}\right)} \quad \dots (6.187)$$

Now

$$525 = \sqrt{\left(\frac{Mg}{\pi r^2 \rho}\right)} \quad [\text{As } v = 525 \text{ m/s}]$$

or

$$\frac{Mg}{\pi r^2} = (525)^2 \times \rho = (525)^2 \times (7.5 \times 10^3)$$

$$= 2.15 \times 10^9 \text{ N/m}^2$$

(iii) When the tension of the wire is increased by 3%, then the new tension becomes

$$T' = T + \frac{3}{100} T = 1.03 T$$

Now the new frequency

$$= n' = \frac{1}{2l} \sqrt{\left(\frac{T'}{\mu}\right)} \quad \dots (6.188)$$

Dividing equation-(6.188) by equation-(6.185), we get

$$\frac{n'}{n} = \sqrt{\left(\frac{T'}{T}\right)} = \sqrt{\left(\frac{1.03T}{T}\right)}$$

$$= \sqrt{1.03}$$

or

$$\frac{n'}{n} = 1.015$$

or

$$n' = 1.015 \times n = n + 0.015 n$$

$$= n + (3n/200)$$

Thus percentage change in frequency

$$\left(\frac{n' - n}{n}\right) \times 100 = \frac{3}{2} = 1.5$$

Illustrative Example 6.39

A wire having a linear density of 0.05 gm/cm^3 is stretched between two rigid supports with a tension of $4.5 \times 10^7 \text{ dynes}$. It is observed that the wire resonates at a frequency of 420 Hz . The next highest frequency at which the same wire resonates is 490 Hz . Find the length of the wire.

Solution

Let the frequency 420 Hz corresponds to p th harmonic. The formula for p th harmonic is given by

$$n_p = \frac{p}{2l} \sqrt{\left(\frac{T}{\mu}\right)}$$

Hence,

$$420 = \frac{p}{2l} \sqrt{\left(\frac{T}{\mu}\right)} \quad \dots (6.189)$$

For the next higher frequency p is $(p + 1)$, hence

$$490 = \frac{p+1}{2l} \sqrt{\left(\frac{T}{\mu}\right)} \quad \dots (6.190)$$

From equations-(6.189) and (6.190), we have

$$\frac{490}{420} = \frac{p+1}{p}$$

Solving we get, $p = 6$

Substituting the value p in equation-(6.189), we get

$$420 = \frac{6}{2l} \left(\frac{4.5 \times 10^7}{0.05}\right)^{1/2}$$

$$l = 214.3 \text{ cm}$$

Illustrative Example 6.40

In Melde's experiment it was found that the string vibrates in 3 loops when 8 gm were placed in the pan. What mass should be placed in the pan to make the string vibrate in 5 loops? (Neglect the mass of string).

Solution

According to Melde's Law

$$P\sqrt{T} = \text{constant}$$

or

$$P_1 \sqrt{T_1} = P_2 \sqrt{T_2}$$

$$3\sqrt{8} = 5\sqrt{T}$$

or

$$T = \frac{9 \times 8}{25}$$

$$= 2.88 \text{ gm}$$

Illustrative Example 6.41

Find the ratio of the fundamental tone frequencies of two identical strings after one of them was stretched by $\eta_1 = 2.0\%$ and the other by $\eta_2 = 4.0\%$. The tension is assumed to be proportional to the elongation.

Solution

In case of a stretched string

$$n = \frac{1}{2l} \sqrt{\left(\frac{T}{\mu}\right)}$$

Where μ is the mass per unit length.

If M be the total mass of the wire, then

$$n = \frac{1}{2l} \sqrt{\left(\frac{TL}{\mu}\right)} \quad \dots (6.191)$$

It should be remembered that when the wire is stretched, the total mass of wire M remains constant.

If original length was l , then after elongation

$$l_1 = 1.02l \quad \text{and} \quad l_2 = 1.04l$$

Given that tension $\propto$ elongation

$$\text{Hence} \quad \frac{T_1}{T_2} = \frac{\eta_1}{\eta_2} = \frac{2}{4} = 0.5 \quad \dots (6.192)$$

The new frequencies are

$$n_1 = \frac{1}{2(1.02l)} \sqrt{\left[\frac{T_1(1.02l)}{M}\right]} \quad \dots (6.193)$$

$$\text{and} \quad n_2 = \frac{1}{2(1.04l)} \sqrt{\left[\frac{T_2(1.04l)}{M}\right]} \quad \dots (6.194)$$

Dividing equation-(6.194) by equation-(6.193), we get

$$\begin{aligned} \frac{n_2}{n_1} &= \sqrt{\left[\left(\frac{1.02}{1.04}\right)\left(\frac{T_2}{T_1}\right)\right]} = \sqrt{\left[\frac{1.02}{1.04} \times \frac{1}{0.5}\right]} \\ &= 1.4 \end{aligned}$$

Illustrative Example 6.42

The fundamental frequency of a sonometer wire increases by 6 Hz if its tension is increased by 44% keeping the length constant. Find the change in the fundamental frequency of the sonometer wire, the length of the wire is increased by 20% keeping the original tension in the wire.

Solution

In case of vibration of a string, fundamental frequency is given by

$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

So if length of given wire is kept constant

$$(f'/f) = (T'/T)^{1/2}$$

and as here $f' = f + 6$ and $T' = T + 0.44T = 1.44T$

$$\frac{(f+6)}{f} = \sqrt{\frac{1.44T}{T}}$$

$$\text{or} \quad f = 30 \text{ Hz}$$

Now if keeping the original tension (T), the length wire is changed, we have

$$\frac{f''}{f} = \frac{l}{l''} = \frac{1}{1.20} \quad [\text{As } l'' = l + 0.20l = 1.20l]$$

$$\text{or} \quad f'' = \frac{30}{1.2} = 25 \text{ Hz}$$

$$\text{Hence} \quad \Delta f = f'' - f = 25 - 30 = -5 \text{ Hz}$$

Thus, fundamental frequency will decrease by 5 Hz.

Illustrative Example 6.43

A sonometer wire fixed at one end has a solid mass M hanging from its other end to produce tension in it. It is found that a 70 cm length of the wire produces a certain fundamental frequency when plucked. When the same mass M is hanging in water, completely submerged in it, is found that the length of the wire has to be changed by 5 cm in order that it will produce the same fundamental frequency. Calculate the density of the material of the mass M hanging from the wire.

Solution

The fundamental frequency n_1 is given by

$$n_1 = \frac{1}{2l} \sqrt{\left(\frac{T}{\mu}\right)}$$

Here $l = 70$ cm, $T = Mg$ and μ , mass per unit length

$$n_1 = \frac{1}{2 \times 70} \sqrt{\left(\frac{Mg}{\mu}\right)} \quad \dots (6.195)$$

When the mass M is submerged in water, the frequency remains the same for a length of 65 cm.

$$\text{Hence we have} \quad n_1 = \frac{1}{2 \times 65} \sqrt{\left(\frac{M'g}{\mu}\right)} \quad \dots (6.196)$$

Where $M'g$ is the effective weight of mass in water.

From equation-(6.195) and (6.196)

$$\frac{1}{2 \times 70} \sqrt{\left(\frac{Mg}{\mu}\right)} = \frac{1}{2 \times 65} \sqrt{\left(\frac{M'g}{\mu}\right)}$$

$$\text{or } \sqrt{\left(\frac{M}{M'}\right)} = \frac{14}{13} \quad \dots (6.197)$$

Now using the Archimedes principle, we shall calculate the value of M' in terms of M . Let ρ be the density of the material of the mass M .

$$\text{Volume of mass} = (M/\rho) \text{ cm}^3$$

$$\text{Volume of water displaced} = (M/\rho) \text{ cm}^3$$

$$\text{or } \text{Upthrust} = (M/\rho) g$$

$$\text{Weight of mass } M \text{ in water } M'g = Mg - \left(\frac{M}{\rho}\right) g$$

$$\text{or } M' = M \left(1 - \frac{1}{\rho}\right) \quad \dots (6.198)$$

Substituting the value of M' in equation-(6.197), we get

$$\sqrt{\left(\frac{M}{M(1-1/\rho)}\right)} = \frac{14}{13}$$

$$\text{Solving we get, } \rho = 7.26 \text{ gm/cm}^3 = 7.26 \times 10^3 \text{ kg/m}^3$$

Illustrative Example 6.44

The length of the sonometer wire between two fixed ends is 100 cm. Where should the two bridges be placed so as to divide the wire into three segments whose fundamental frequencies are in the ratio of 1 : 2 : 3 ?

Solution

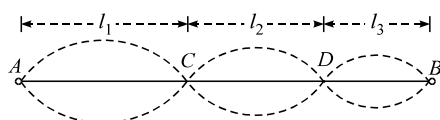


Figure 6.60

Figure-6.60 shows a sonometer wire AB between two fixed ends. Let the two bridges be placed at C and D such that the wire is divided into three segments (AC , CD and DB) of lengths l_1 , l_2 and l_3 respectively whose frequencies are in the ratio of 1 : 2 : 3

For the length of string, we have

$$AC + CD + DB = 100 \text{ cm} \quad \dots (6.199)$$

Let n_1 , n_2 and n_3 be the fundamental frequencies of these segments respectively, then

$$\frac{n_1}{n_2} = \frac{1}{2} \quad \text{and} \quad \frac{n_2}{n_3} = \frac{2}{3} \quad \dots (6.200)$$

We know that frequency is inversely proportional to the length of the segment because when tension remains constant

$$n \propto \frac{1}{l}$$

$$\text{or } n l = \text{constant}$$

$$\text{or } n_1 l_1 = n_2 l_2 = n_3 l_3$$

$$\text{or } l_1 = \frac{n_2}{n_1} l_2 = 2 l_2$$

$$\text{and } l_3 = \frac{n_2}{n_3} l_2 = \frac{2}{3} l_2 \quad \dots (6.201)$$

Substituting these values in equation-(6.199), we have

$$2 l_2 + l_2 + \frac{2}{3} l_2 = 100$$

$$\text{or } l_2 = 27.27 \text{ cm}$$

From equation-(6.201),

$$l_1 = 2 \times 27.27 = 54.54 \text{ cm}$$

$$\text{and } l_3 = \frac{2}{3} \times 27.27 = 18.18 \text{ cm}$$

Illustrative Example 6.45

A uniform rope of length 12 m and mass 6 kg hangs vertically from a rigid support. A block of mass 2 kg is attached to the free end of the rope. A transverse pulse of wavelength 0.06 m is produced at the lower end of the rope. What is the wavelength of the pulse when it reaches the top of the rope?

Solution

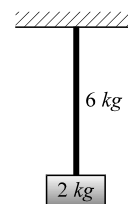


Figure 6.61

As the rope is heavy, its tension will be different at different points. The tension at the free end will be 2 g and that at the upper end it will be 8 g.

$$\text{We have, } v = n\lambda$$

$$\text{or } \sqrt{\frac{T}{\mu}} = n\lambda$$

$$\text{or } \sqrt{\frac{T}{\lambda}} = n\sqrt{\mu} \quad \dots (6.202)$$

The frequency of the wave pulse will be the same everywhere on the rope as it depends only on the frequency of the source. The mass per unit length is also the same throughout the rope as it is uniform. Thus, we have

As $\sqrt{\frac{T}{\lambda}}$ is constant, we have

$$\frac{\sqrt{2g}}{0.06\text{m}} = \frac{\sqrt{8g}}{\lambda_1}$$

Solving we get $\lambda_1 = 0.12\text{ m}$

Where λ_1 is the wavelength at the top of the rope.

Illustrative Example 6.46

A metallic rod of length 1m is rigidly clamped at its mid-point. Longitudinal stationary waves are set up in the rod in such a way that there are two nodes on either side of the mid-point. The amplitude of an antinode is $2 \times 10^{-6}\text{ m}$. Write the equation of motion at a point 2 cm from the mid-point and those of the constituent waves in the rod. (Young's modulus = $2 \times 10^{11}\text{ N/m}^2$, density = 800 kg/m^3).

Solution

The situation is shown in figure-6.62.

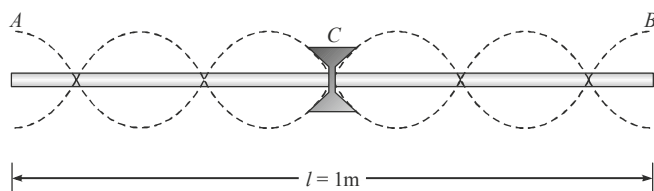


Figure 6.62

From figure, we can calculate the wavelength as

$$\lambda = 0.4\text{ m} = 40\text{ cm}$$

Velocity of longitudinal waves in the rod can be given as

$$v = \sqrt{\left(\frac{Y}{\rho}\right)} = \sqrt{\left(\frac{2 \times 10^{11}}{8000}\right)} = \sqrt{\left(\frac{1 \times 10^8}{4}\right)} = \frac{10^4}{2} = 5000\text{ m/s}$$

$$\text{Now } n = \frac{v}{\lambda} = \frac{5000}{0.4} = 12500\text{ Hz}$$

Assuming left end of the rod as origin, the equation of stationary waves is given by

$$y = 2A \cos \frac{2\pi x}{\lambda} \sin(2\pi n t)$$

$$\text{or } y = 2A \cos \frac{2\pi x}{\lambda} \sin(2\pi \times 12500 t) \quad \dots (6.203)$$

Where amplitude at any instant t is given by

$$R = 2A \cos \left(\frac{2\pi x}{0.4} \right)$$

$$\text{At } x=0, \quad R = 2A = 2 \times 10^{-6}\text{ m}$$

Thus, equation-(6.203) can be written as

$$y = 2 \times 10^{-6} \cos(5\pi x) \sin(25000\pi t) \quad \dots (6.204)$$

At a point 2 cm from mid-point to the right

$$x = 50 + 2 = 52\text{ cm} = 0.52\text{ m}$$

$$y = 2 \times 10^{-6} \cos(5\pi \times 0.52) \sin(25000\pi t)$$

$$\text{or } y = 2 \times 10^{-6} \cos(2.6\pi) \sin(25000\pi t) \quad \dots (6.205)$$

This is the required equation of stationary waves in the rod.

Now we can write the equations of constituents waves in the rod as

$$y_1 = 1 \times 10^{-6} \sin(25000\pi t - 5\pi x)$$

$$\text{and } y_2 = 1 \times 10^{-6} \sin(25000\pi t + 5\pi x)$$

Illustrative Example 6.47

A string 120 cm in length sustains a standing wave, with the points of the string at which the displacement amplitude is equal to 3.5 mm being separated by 15.0 cm. Find the maximum displacement amplitude. To which overtone do these oscillations correspond ?

Solution

We can see from figure-6.63, in a stationary wave two successive of equal amplitude, if separated by equal distances then this distance must be $\frac{\lambda}{4}$. Thus we have

$$\frac{\lambda}{4} = 15\text{ cm}$$

$$\text{or } \lambda = 60\text{ cm}$$

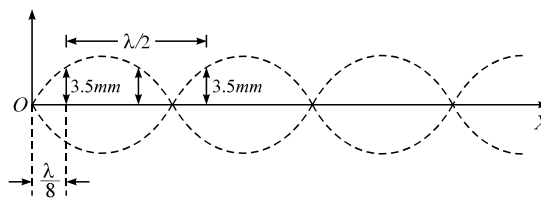


Figure 6.63

Thus there are four loops in 120 cm length of string. This corresponds to 3rd overtone oscillations. As shown in figure if we consider origin O is at a node then the amplitude of a general medium particle, at a distance x from O can be given as

$$R = A_0 \sin kx$$

Where A_0 is the maximum displacement amplitude. First point

from the origin where amplitude is 3.5 mm is at distance $\frac{\lambda}{8} = 7.5$ cm. Thus we have

$$3.5 = A_0 \sin\left(\frac{2\lambda}{60} \times 7.5\right)$$

or
$$A_0 = \frac{3.5}{\sin(\pi/4)} = 3.5\sqrt{2} \text{ mm}$$

Illustrative Example 6.48

A guitar string is 90 cm long and has a fundamental frequency of 124 Hz. Where should it be pressed to produce a fundamental frequency of 186 Hz?

Solution

The fundamental frequency of a string fixed at both ends is given by

$$n = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

As F and μ are fixed,

$$\frac{n_1}{n_2} = \frac{L_2}{L_1}$$

or
$$L_2 = \frac{n_1}{n_2} L_1 = \frac{124}{186} \times 90 = 60 \text{ cm}$$

Thus, the string should be pressed at 60 cm from an end.

Illustrative Example 6.49

In a Melde's experiment when the tension is 100 gm and the tuning fork vibrates at right angles to the direction of the string, the later is thrown into four segments. If now the tuning fork is set to vibrate along the string, find what additional weight which will make the string vibrate in one segment.

Solution

In first case the string vibrates in four loops and if n_0 be the oscillation frequency of tuning fork we have

$$N = \frac{4}{2l} \sqrt{\frac{T_1}{\mu}} \quad \dots (6.206)$$

In second case if string vibrate in one segment, we have

$$\frac{N}{2} = \frac{1}{2l} \sqrt{\frac{T_2}{\mu}} \quad \dots (6.207)$$

From (6.206) and (6.207), we get

$$\begin{aligned} \sqrt{T_2} &= 2\sqrt{T_1} \\ T_2 &= 4T_1 \end{aligned}$$

As $T_1 = 100$ gm, we get

$$T_2 = 400 \text{ gm}$$

This additional weight required is 300 gm.

Illustrative Example 6.50

The vibrations of a string of length 600 cm fixed at both ends are represented by the equation

$$y = 4 \sin\left(\pi \frac{x}{15}\right) \cos(90 \pi t)$$

where x and y are in cm and t in seconds.

- What is the maximum displacement of a point $x = 5$ cm ?
- Where are the nodes located along the string ?
- What is the velocity of the particle at $x = 7.5$ cm at $t = 0.25$ s.
- Write down the equations of the component waves whose superposition gives the above wave

Solution

The given equation is

$$y = 4 \sin\left(\frac{\pi x}{15}\right) \cos(96 \pi t)$$

This can be written as

$$y = 2 \times 2 \sin\left(\frac{2 \pi x}{300}\right) \cos \frac{2 \pi (1440) t}{30} \quad \dots (6.208)$$

Comparing equation-(6.208) with standard equation of stationary wave, we get

This shows that $A = 2$ cm $\lambda = 30$ cm and $v = 1440$ cm/s

- The maximum displacement is given by

$$y_{\max} = 4 \sin \frac{2 \pi x}{30}$$

- For a point at $x = 5$ cm,

$$y_{\max} = 4 \sin \frac{2 \pi 5}{30} = \frac{4\sqrt{3}}{2} 2\sqrt{3} \text{ cm}$$

- As $\lambda = 30$ cm, hence the nodes are located along the string at places

$$0, 15 \text{ cm}, 30 \text{ cm}, 45 \text{ cm}, 60 \text{ cm}$$

The velocity of the particle is given by

$$\frac{dy}{dt} = -4 \sin\left(\frac{\pi x}{15}\right) \sin(96 \pi t) 96 \pi$$

For $x = 7.5$ cm and $t = 0.25$ sec, we have

$$\begin{aligned} \frac{dy}{dt} &= -4 \sin\left(\frac{\pi \cdot 7.5}{15}\right) \sin(96 \pi \times 0.25) 96 \pi \\ &= 0 \end{aligned}$$

(iv) The equations of component waves are

$$y_1 = A \sin \frac{2\pi}{\lambda} (x - vt) = 2 \sin \frac{2\pi}{30} (x - 1440 t)$$

or $y_1 = 2 \sin 2\pi \left(\frac{x}{30} - 48 t \right) \quad \dots (6.209)$

and $y_2 = A \sin \frac{2\pi}{\lambda} (x + vt) = 2 \sin \frac{2\pi}{30} (x + 1440 t)$

or $y_2 = 2 \sin 2\pi \left(\frac{x}{30} + 48 t \right) \quad \dots (6.210)$

Equations-(6.209) and (6.210) represent the equations of component waves superposition of which results the given stationary wave.

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Stationary Waves and Beats

Module Number - 12 to 31

Practice Exercise 6.5

(i) A steel wire of length 1 m and density 8000 kg/m^3 is stretched tightly between two rigid supports. When vibrating in its fundamental mode, its frequency is 200 Hz.

- What is the velocity of transverse waves along this wire?
- What is the longitudinal stress in the wire?
- If the maximum acceleration of the wire is 800 m/s^2 , what is the amplitude of vibration at the mid-point?

[(a) 400 m/s , (b) $1.28 \times 10^9 \text{ N/m}^2$, (c) $5 \times 10^{-4} \text{ m}$]

(ii) A sonometer wire having a length of 1.50 m between the bridges vibrates in its second harmonic in resonance with a tuning fork of frequency 256 Hz. What is the speed of the transverse wave on the wire?

[384 m/s]

(iii) The length of the wire shown in figure-6.64 between the pulleys is 1.5 m and its mass is 12.0 g. Find the frequency of vibration with which the wire vibrates in two loops leaving the middle point of the wire between the pulleys at rest. Take $g = 10 \text{ m/s}^2$.

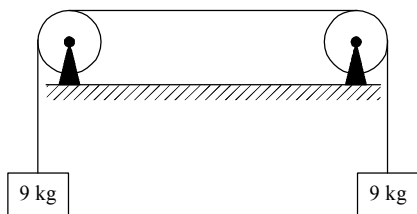


Figure 6.64

[70.71 Hz]

(iv) A steel wire fixed at both ends has a fundamental frequency of 200 Hz. A person can hear sound of maximum frequency 14 kHz. What is the highest harmonic that can be played on this string which is audible to the person?

[70]

(v) Figure-6.65 shows a string stretched by a block going over a pulley. The string vibrates in its tenth harmonic in unison with a particular tuning fork. When a beaker containing water is brought under the block so that the block is completely dipped into the beaker, the string vibrates in its eleventh harmonic. Find the density of the material of the block.

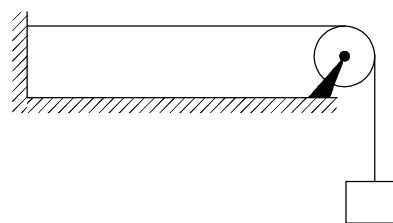


Figure 6.65

[$5.76 \times 10^3 \text{ kg/m}^3$]

(vi) A wire of diameter 0.04 cm and made of steel of density 8000 kg/m^3 is under a tension of 79 N. A fixed length of 50 cm is set into transverse vibrations. How would you cause vibrations of frequency 840 Hz to predominate in intensity? Locate the support and plucking points on wire for this.

[$\frac{1}{6}$ point is to be plucked]

(vii) The displacement of the medium in sound wave is given by the equation

$$= A \cos (ax + bt)$$

where A , a and b are positive constants. The wave is reflected by an obstacle situated at $x = 0$. The intensity of the reflected wave is 0.64 times that of the incident wave.

- What is the wavelength and frequency of the incident wave?
- Write the equation for the reflected wave.
- In the resultant wave formed after reflection, find the maximum and minimum values of the particle speed in the medium.
- Express the resultant waves as a superposition of a standing wave and a travelling wave. What are the positions of the antinodes of the standing wave? What is the direction of propagation of the travelling wave?

[(a) $\frac{2\pi}{a}$, $\frac{b}{a}$, (b) $-0.8 A \cos (bt - ax)$, (c) $1.8 Ab$, $0.2 Ab$,

(d) $2A \sin b t \sin a x + 0.2 A \cos (bt - ax)$]

(viii) A uniform horizontal rod of length 40 cm and mass 1.2 kg is supported by two identical wires as shown in figure-6.66. Where should a mass of 4.8 kg be placed on the rod so that the same tuning fork may excite the wire on left into its fundamental vibrations and that on right into its first overtone? Take $g = 10 \text{ m/s}^2$.

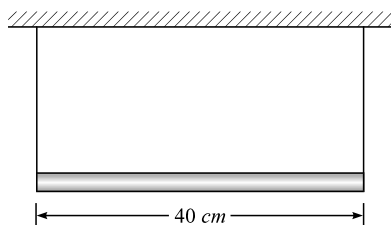


Figure 6.66

[5 cm from the left end]

(ix) A 2 m long string fixed at both ends is set into vibrations in its first overtone. The wave speed on the string is 200 m/s and the amplitude is 0.5 cm. (a) Find the wavelength and the frequency. (b) Write the equation giving the displacement of different points as a function of time. Choose the X-axis along the string with the origin at one end and $t = 0$ at the instant when the point $x = 50 \text{ cm}$ has reached its maximum displacement.

[(a) 2 m, 100 Hz, (b) $(0.5 \text{ cm}) \sin(\pi x) \cos(200 \pi t)$]

(x) Three resonant frequencies of a string are 90, 150 and 210 Hz. (a) Find the highest possible fundamental frequency of vibration of this string. (b) Which harmonics of the fundamental are the given frequencies? (c) Which overtones are these frequencies. (d) If the length of the string is 80 cm, what would be the speed of a transverse wave on this string?

[(a) 30 Hz (b) 3rd, 5th and 7th (c) 2nd, 4th and 6th (d) 48 m/s]

6.15 Waves in a Vibrating air Column

Hollow pipes have long been used for making musical sounds. A hollow pipe we call organ pipe. To understand how these work, first we examine the behaviour of air in a hollow pipe that is open at both ends. If we blow air across one end, the disturbance due to the moving air at that end propagates along the pipe to the far end. When it reaches far end, a part of the wave is reflected, similar in the case when a wave is reflected along a string whose end point is free to move. Since the air particles are free to move at the open end, the end point is an antinode. If one end of the pipe is closed off, the air is not free to move any further in that direction and closed end becomes a node. Now the resonant behaviour of pipe is completely changed. Similar in the case of string, here also all harmonic frequencies are possible and resonance may take place if the frequency of external source matches with any of the one harmonic frequency of pipe. Let us discuss in detail.

6.15.1 Vibration of Air in a Closed Organ Pipe

When a tuning fork is placed near the open end of a pipe. The air in the pipe oscillates with the same frequency as that of tuning fork. Here the open end should be an antinode and closed end should be a node for perfect reflection of waves from either end or for formation of stationary waves. Since one end is a node and other is an antinode, the lowest frequency (largest wavelength) vibration has no other nodes or antinodes between ends as shown in figure-6.67(a). This is the fundamental (minimum) frequency at which stationary waves can be formed in a closed organ pipe. Thus if the wavelength is λ then we can see from figure-6.67(a), which shows the displacement wave of longitudinal waves in the closed organ pipe.

$$l = \frac{\lambda}{4} \quad \dots (6.211)$$

or

$$\lambda = 4l$$

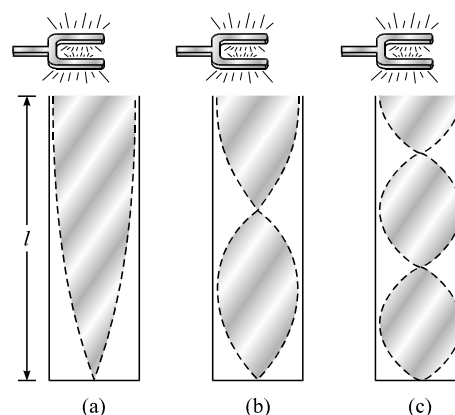


Figure 6.67

Thus fundamental frequency of oscillations of a closed organ pipe of length l be given as

$$n_1 = \frac{v}{\lambda} = \frac{v}{4l} \quad \dots (6.212)$$

Similarly first overtone of closed pipe vibrations is shown in figure-6.67(b) here wavelength λ' and pipe length l are related as

$$l = \frac{3\lambda}{4} \quad \dots (6.213)$$

or

$$\lambda = \frac{4l}{3}$$

Thus frequency of first overtone oscillations of a closed organ pipe of length l can be given as

$$n_2 = \frac{v}{\lambda'} = \frac{3v}{4l} \quad \dots (6.214)$$

$$= 3n_1 \quad \dots (6.215)$$

This is three times the fundamental frequency thus after fundamental only third harmonic frequency exist for a closed organ pipe at which resonance can take place or stationary waves can be formed in it.

Similarly next overtone, second overtone is shown in figure-6.67(c) Here the wavelength λ'' and pipe length l are related as

$$l = \frac{5\lambda}{4} \quad \dots (6.216)$$

or
$$\lambda = \frac{4l}{5}$$

Thus the frequency of second overtone oscillation of a closed organ pipe of length l can be given as

$$n_3 = \frac{v}{\lambda''} = \frac{5v}{4l} = 5 n_1 \quad \dots (6.217)$$

This is fifth harmonic frequency of fundamental oscillations. From above analysis it is clear that the resonant frequencies of the closed organ pipe are only odd harmonics of the fundamental frequency. Thus when a tuning fork is used to oscillate a closed organ pipe, the air in pipe oscillates with the same frequency as that of tuning fork but in pipe stationary waves are formed or resonance take place only when the frequency of fork matches with any of the odd harmonic of its fundamental frequency.

6.15.2 Vibration of Air in Open Organ Pipe

Figure-6.68 shows the resonant oscillations of an open organ pipe. The least frequency at which an open organ pipe resonates is the one with longest wavelength when at both the open ends of pipe antinodes are formed and there is one node is between as shown in figure-6.68(a). In this situation the wavelengths of sound in air λ is related to length of organ pipe as

$$l = \frac{\lambda}{2} \quad \dots (6.218)$$

or
$$\lambda = 2l$$

Thus the fundamental frequency of organ pipe can be given as

$$n_1 = \frac{v}{\lambda} = \frac{v}{2l} \quad \dots (6.219)$$

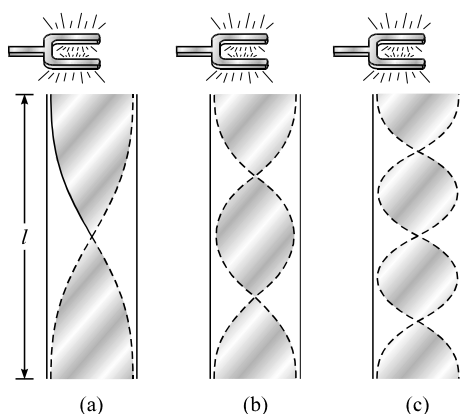


Figure 6.68

Similarly next higher frequency at which the open organ pipe

resonate is shown in figure-6.68(b) which we call first overtone. Here the wavelength λ' is related to the length of pipe as

$$l = \lambda' \quad \dots (6.220)$$

Thus here resonant frequency for first overtone is given as

$$n_2 = \frac{v}{\lambda'} = \frac{v}{l} \quad \dots (6.221)$$

$$= 2n_1 \quad \dots (6.222)$$

Which is second harmonic of fundamental frequency. Similarly as shown in figure-6.68(c), in second overtone oscillations, the wavelength λ'' of sound is related to the length of pipe as

$$l = \frac{3\lambda''}{2} \quad \dots (6.223)$$

or
$$\lambda'' = \frac{2l}{3} \quad \dots (6.224)$$

Thus the frequency of second overtone oscillations of an open organ pipe can be given as

$$n_3 = \frac{v}{\lambda''} = \frac{3v}{2l} \quad \dots (6.225)$$

$$= 3n_1 \quad \dots (6.226)$$

Which is third harmonic of fundamental frequency. The above analysis shows that resonant frequencies for formation of stationary waves includes all the possible harmonic frequencies for an open organ pipe.

6.15.3 Natural Oscillations of Organ Pipes

When we initiate some oscillations in an organ pipe, which harmonics are excited in the pipe depends on how initial disturbance is produced in it. For example, if you gently blow across the top of an organ pipe it resonates softly at its fundamental frequency. But if you blow much harder you hear the higher pitch of an overtone because the faster airstream creates higher frequencies in the exciting disturbances. This sound effect can also be achieved by increasing the air pressure to an organ pipe.

6.15.4 Kundt's Tube

This is an apparatus used to find velocity of sound in a gaseous medium or in different materials. It consists of a glass tube as shown in figure-6.69 one end of which a piston B is fitted which is attached to a wooden handle H and can be moved inside and outside the tube and fixed, the rod M of the required material is fixed at clamp C in which the velocity of sound is required, at on end of rod a disc A is fixed as shown.

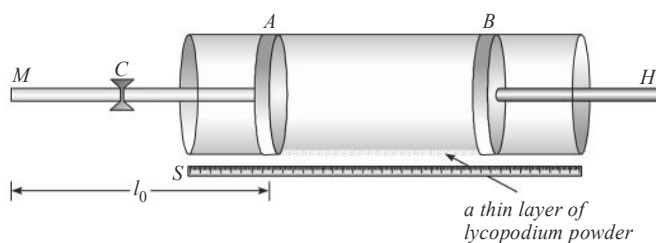


Figure 6.69

In the tube air is filled at room temperature and a thin layer of lycopodium powder is put along the length of the tube. It is a very fine powder particles of which can be displaced by the air particles also.

When rod M is gently rubbed with a resin cloth or hit gently, it starts oscillating in fundamental mode as shown in figure-6.70, frequency of which can be given as

$$n_{\text{rod}} = \frac{v}{\lambda} = \frac{1}{2l_0} \sqrt{\frac{Y}{\rho}} \quad [\text{As } l_0 = \frac{\lambda}{2}] \quad \dots (6.227)$$

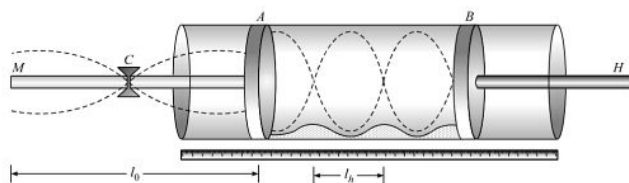


Figure 6.70

In fundamental mode, the free ends of the rod behaves as antinodes and the clamped point C acts as a node, we have also discussed earlier. These vibrations are transferred to the air column in the tube through the disc A and the air column starts oscillating at the same frequency. Now the piston B is adjusted so that the air column resonates with the vibrations of rod. In resonance condition the lycopodium powder sets itself in the form of heaps at the position of nodes as at antinodes the air particles vibrate with maximum amplitude and displaces the powder to the adjacent nodes. We can measure the length between successive heaps of powder, scale S attached with the tube, let it be l_h then we can write

$$l_h = \frac{\lambda_a}{2} \quad [\lambda_a = \text{wavelength of sound in air}]$$

and we know that frequency of sound in air and rod is equal, thus

$$n_{\text{rod}} = n_{\text{air}}$$

$$\text{or} \quad \frac{v_{\text{rod}}}{\lambda_{\text{rod}}} = \frac{v_{\text{air}}}{\lambda_{\text{air}}}$$

$$\begin{aligned} \text{or} \quad v_{\text{rod}} &= \left(\frac{\lambda_{\text{rod}}}{\lambda_{\text{air}}} \right) v_{\text{air}} \\ &= \frac{2l_0}{l_h} \times v_{\text{air}} = \left(\frac{l_0}{l_h} \right) \times v_{\text{air}} \quad \dots (6.228) \end{aligned}$$

If the Young's modulus and density of material of rod is known then using equation-(6.227) and (6.228) we can find velocity of sound in air or in a gas which is filled in the tube, as

$$n_{\text{rod}} = n_{\text{gas}}$$

$$\frac{1}{2l_0} \sqrt{\frac{Y}{\rho}} = \frac{v_{\text{gas}}}{2l_h}$$

or

$$v_{\text{gas}} = \left(\frac{l_h}{l_0} \right) \sqrt{\frac{Y}{\rho}} \quad \dots (6.229)$$

6.15.5 Resonance Tube

This is an apparatus used to determine the velocity of sound in air experimentally and also to compare the frequencies of two tuning forks.

Figure-6.71 shows the setup of a resonance tube experiment. There is a long tube T in which initially water is filled up to the top and the water level can be changed by moving a reservoir R up and down.

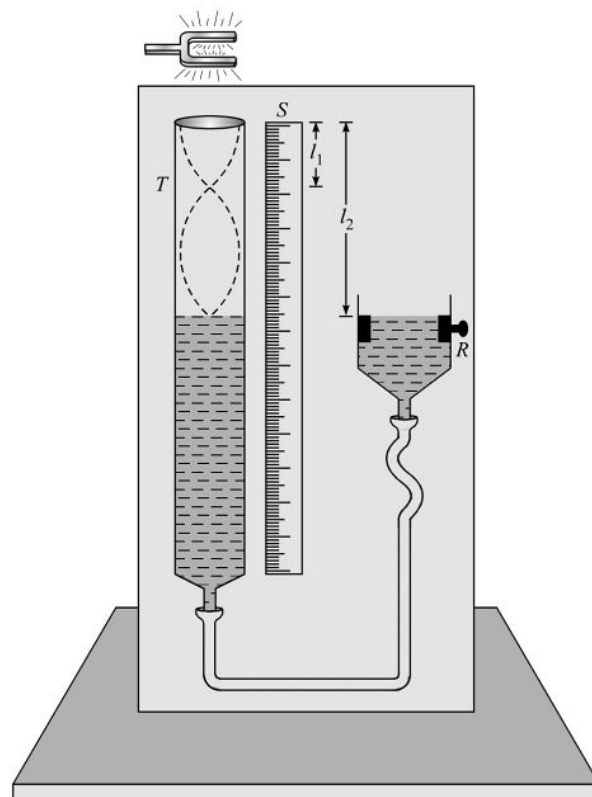


Figure 6.71

A tuning fork of known frequency n_0 is struck gently on a rubber pad and brought near the open end of tube T due to which oscillations are transferred to the air column in the tube above the water level. Now we gradually decrease the water level in the tube. This air column behaves like a closed organ pipe and

the water level as closed end of pipe. As soon as water level reaches a position where there is a node of corresponding stationary wave, in air column, resonance takes place and maximum sound intensity is detected. Let at this position length of air column be l_1 . If water level is further decreased, again maximum sound intensity is observed when water level is at another node i.e. at a length l_2 as shown in figure. Here if we find two successive resonance lengths l_1 and l_2 , we can get the wavelength of the wave as

$$l_2 - l_1 = \frac{\lambda}{2}$$

or $\lambda = 2(l_2 - l_1)$

Thus sound velocity in air can be given as

$$v = n_0 \lambda = 2n_0(l_2 - l_1)$$

6.15.6 End Correction in Organ Pipes

We have discussed, that in an organ pipe the incident and reflected waves superpose and give rise to establishment of stationary waves at harmonic frequencies. First we discuss how a wave is reflected from the open end of an organ pipe.

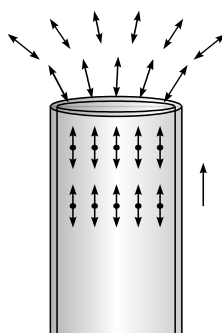


Figure 6.72

Consider an organ pipe shown in figure-6.72. Here we consider a wave is propagating towards its open end. As due to longitudinal wave medium (air) particles oscillates along the length of pipe as shown in figure-6.72. But the oscillations are along the length of the pipe within the boundaries of the pipe. When wave reaches the open end, due to collisions the medium particles outside the pipe scatters in the direction away from pipe and due to this medium (air) density reduces outside the pipe and from the region of this rarer medium the wave is reflected.

Here we can see that when a wave reaches the open end of a pipe it penetrates atmosphere upto to a small depth where the density is decreased and then it is reflected back into the pipe. Thus the wave is not exactly reflected from the open end of the pipe. Hence in the formation of stationary waves in organ pipe we say always there is an antinode at the open end of the pipe but in fact antinode is formed a little above the open end as

shown in figure-6.73. The distance above the open end antinode formed is called end correction and is represented by e .

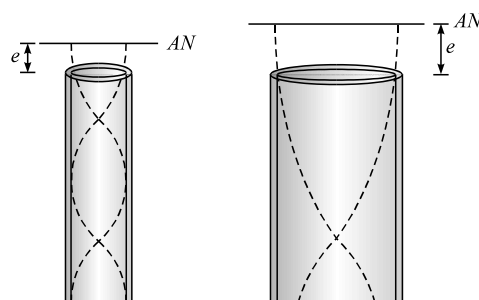


Figure 6.73

It is observed that end correction depends on the radius of organ pipe and is experimentally determined and expressed as

$$e = 0.6r \quad \dots (6.230)$$

Thus for a broad pipe end correction is more than a narrow pipe. When we find the different harmonic frequencies of oscillations of air column in organ pipe, we must account end corrections. Now taking into account end correction the fundamental frequency of a closed pipe of length l is taken as

$$n_0 = \frac{v}{4(l+e)} \quad [\text{One end open}] \quad \dots (6.231)$$

and fundamental frequency of an open pipe of length l is taken as

$$n_0 = \frac{v}{2(l+2e)} \quad [\text{Both ends open}] \quad \dots (6.232)$$

Illustrative Example 6.51

A tube of certain diameter and of length 48 cm is open at both ends. Its fundamental frequency of resonance is found to be 320 Hz. The velocity of sound in air is 320 m/s. Estimate the diameter of the tube. One end of the tube is now closed. Calculate the lowest frequency of resonance for the tube.

Solution

The displacement curves of longitudinal waves in a tube open at both ends is shown in figure-6.74(a) and (b).

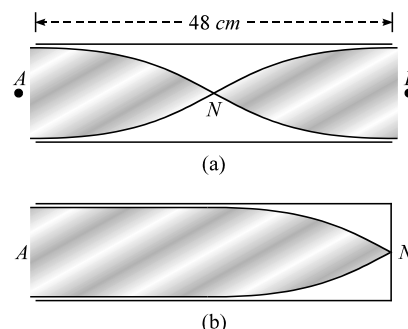


Figure 6.74

Let r be the radius of the tube. We know that antinodes occur slightly outside the tube at a distance $0.6r$ from the tube end.

The distance between two antinodes is given by

$$\frac{\lambda}{2} = 48 + 2 \times 0.6r$$

We have $\lambda = \frac{v}{n} = \frac{32000}{320} = 100 \text{ cm}$

or $50 = 48 + 1.2r$

or $r = \frac{2}{1.2}$
 $= 1.67 \text{ cm}$

Thus diameter of the tube is

$$D = 2r = 3.33 \text{ cm}$$

When one end is closed, then

$$\begin{aligned} \frac{\lambda}{4} &= 48 + 0.6r \\ &= 48 + 0.6 \times 1.67 \\ &= 49 \end{aligned}$$

or $\lambda = 4 \times 49$
 $= 196 \text{ cm}$

Now $n = \frac{v}{\lambda} = \frac{32000}{196} = 163.3 \text{ Hz}$

Illustrative Example 6.52

An open organ pipe filled with air has a fundamental frequency of 500 Hz. The first harmonic of another organ pipe closed at one end and filled with CO_2 has the same frequency as that of the first harmonic of the open organ pipe. Calculate the length of each organ pipe. The velocity of sound in air and CO_2 are 300 m/s and 264 m/s, respectively.

Solution

To solve this problem, we have to remember that the fundamental mode is itself the first harmonic.

$$n = \frac{c}{2l} \text{ for an open pipe}$$

or $500 = \frac{300}{2l} \Rightarrow l = 30 \text{ cm}$

$$n = \frac{c}{4l} \text{ for a closed pipe}$$

or $500 = \frac{264}{4l} \Rightarrow l = 13.2 \text{ cm}$

Illustrative Example 6.53

A tuning fork having a frequency of 340 Hz is vibrated just above a cylindrical tube. The height of the tube is 120 cm. Water is slowly poured in it. What is the minimum height of water required for resonance? Velocity of sound in air = 340 m/s.

Solution

In a closed tube, the first resonance takes place, when the length of air column $= \lambda/4$

The wavelength of sound in air

$$\begin{aligned} \lambda &= \frac{v}{n} = \frac{340}{340} \\ &= 1 \text{ m [As } v = 340 \text{ m/s and } n = 340] \end{aligned}$$

Here length of first resonance can be given as

$$\begin{aligned} \frac{\lambda}{4} &= 0.25 \text{ m} \\ &= 25 \text{ cm} \end{aligned}$$

The length of second resonance

$$\begin{aligned} \frac{3\lambda}{4} &= \frac{3}{4} \text{ m} \\ &= 75 \text{ cm} \end{aligned}$$

The length of third resonance

$$\begin{aligned} \frac{5\lambda}{4} &= \frac{5}{4} \text{ m} \\ &= 125 \text{ cm} \end{aligned}$$

As the length of the tube is only 120 cm, hence third resonance is not possible.

Therefore, the minimum height of water required for resonance
 $= 120 - 75 = 45 \text{ cm}$

Illustrative Example 6.54

The fundamental frequency of a closed organ pipe is equal to the first overtone frequency of an open organ pipe. If the length of the open pipe is 60 cm, what is the length of the closed pipe?

Solution

We know the fundamental frequency of a closed organ pipe is $\frac{v}{4l_1}$ and first overtone frequency of open pipe is $\frac{v}{l_2}$. Thus we have

$$\begin{aligned} \frac{v}{4l_1} &= \frac{v}{60} \\ l_1 &= \frac{1}{4} \times 60 \text{ cm} = 15 \text{ cm} \end{aligned}$$

Illustrative Example 6.55

A pipe of length 1.5 m closed at one end is filled with a gas and it resonates in its fundamental with a tuning fork. Another pipe of the same length but open at both ends is filled with air and it resonates in its fundamental with the same tuning fork. Calculate the velocity of sound at 0°C in the gas, given that the velocity of sound in air is 360 m/s at 30°C, where the experiment is performed.

Solution

The frequency of the fundamental note emitted by an open pipe is given by

$$n = \frac{v}{2l} = \frac{360}{2 \times 1.5}$$

[As $v = 360$ m/s and $l = 1.5$ m]

$$= 120 \text{ Hz}$$

Let the velocity of sound in the gas at 30°C = v' .

Frequency of the fundamental note emitted by closed pipe filled with the gas is given by

$$n_1 = \frac{v'}{4l} = \frac{v'}{4 \times 1.5} = \frac{v'}{6}$$

But

$$n_1 = n,$$

or

$$120 = v'/6 \quad \text{or} \quad v' = 720 \text{ m/s}$$

If the velocity of sound in the gas at 0°C = v_0 , then

$$\frac{v_0}{v'} = \sqrt{\left(\frac{T_0}{T}\right)}$$

Here,

$$T_0 = 0^\circ\text{C} = 273^\circ\text{K},$$

$$T = 30^\circ\text{C} = 30 + 273 = 303^\circ\text{K}$$

and

$$v' = 720 \text{ m/s}$$

$$\frac{v_0}{720} = \sqrt{\left(\frac{273}{300}\right)} \quad \text{or} \quad v_0 = 683.4 \text{ m/s}$$

Illustrative Example 6.56

A certain organ pipe resonates in its fundamental mode at a frequency of 500 Hz in air. What will be the fundamental frequency if the air is replaced, by hydrogen at the same temperature? The density of air is 1.20 kg/m³ and that of hydrogen is 0.089 kg/m³.

Solution

Suppose the speed of sound in hydrogen is v_h and that in air is v_a . The fundamental frequency of an organ pipe is proportional

to the speed of sound in the gas contained in it. If the fundamental frequency with hydrogen in the tube is v , we have

$$\frac{v}{500} = \frac{v_h}{v_a} = \sqrt{\frac{\rho_a}{\rho_h}} = \sqrt{\frac{1.2}{0.089}} = 3.67$$

or

$$v = 3.67 \times 500 \text{ Hz} \approx 1840 \text{ Hz}.$$

Illustrative Example 6.57

Clamped at the middle, a metal rod of length 1 metre and density 7.5 gm/cm³ gives dust heaps at intervals of 8 cm. Calculate Young's modulus of the material of the rod. Velocity of sound in the gas used is 400 m/s.

Solution

The situation is shown in figure-6.75.

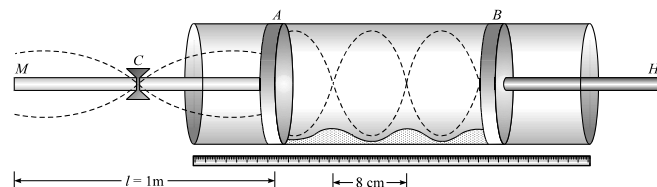


Figure 6.75

Let λ_r and λ_g be the wavelength in the rod and gas column respectively. Since the rod is clamped in the middle, it vibrates with a node in the middle and antinodes at the ends. Therefore

$$\frac{\lambda_r}{2} = 1 \text{ m} = 100 \text{ cm}, \quad [\text{As } \lambda_r = 200 \text{ cm}]$$

and

$$\frac{\lambda_g}{2} = 8 \text{ cm} \quad [\text{As } \lambda_g = 16 \text{ cm}]$$

If v_r and v_g be the sound velocities in the rod and in the gas respectively and n be the frequency of vibration, then frequency of oscillations is given as

$$n = \frac{v_{rod}}{\lambda_{rod}} = \frac{v_{gas}}{\lambda_{gas}}$$

or

$$v_{rod} = \frac{\lambda_{rod}}{\lambda_{gas}} \times v_{gas}$$

or

$$= \frac{200}{16} \times 400$$

or

$$= 5000 \text{ m/s}$$

Now velocity of longitudinal waves in a metal rod is given by

$$v_{rod} = \sqrt{\left(\frac{Y}{d}\right)}$$

or

$$Y = v_{rod}^2 \times d$$

or
$$Y = (500,0)^2 \times 7.5 \times 10^3$$

$$= 1.875 \times 10^{11} \text{ N/m}^2$$

Illustrative Example 6.58

To determine the sound propagation velocity in air by acoustic resonance technique one can use a pipe with a piston and a sonic membrane closing one of its ends. Find the velocity of sound if the distance between the adjacent positions of the piston at which resonance is observed at a frequency $n = 2000$ Hz is equal to $l = 8.5$ cm.

Solution

When the two positions of the resonance are obtained at distances l_1 and l_2 respectively, then the velocity v of sound is given by

$$v = 2n(l_2 - l_1)$$

Where n is the frequency at resonance.

Here $n = 2000$ Hz,

$$(l_2 - l_1) = 8.5 \text{ cm} = 8.5 \times 10^{-2} \text{ m}$$

$$v = 2 \times 2000 \times 8.5 \times 10^{-2}$$

$$= 340 \text{ m/s}$$

Illustrative Example 6.59

AB is a cylinder of length 1.0 m filled with a thin flexible diaphragm C (Figure-6.76) at the middle and two other thin flexible diaphragms A and B at the ends. The portions AC and BC contain hydrogen and oxygen gases respectively. The diaphragms A and B are set into vibrations of same frequency. What is the minimum frequency of these vibrations for which the diaphragm C is a node? Under the conditions of the experiment, the velocity of sound in hydrogen is 1100 m/s and in oxygen is 300 m/s.

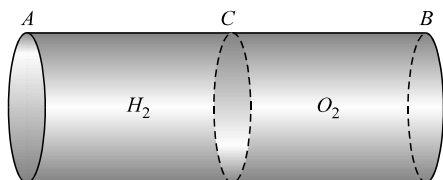


Figure 6.76

Solution

A and B are set in oscillation, hence antinodes are situated at these points. The portions of the cylinders i.e., AC and CB behaves as a close cylinder at the end C . Thus there is immediate node at C .

The fundamental frequency of each pipe corresponds to just one node and one anti-node. Let n_1 and n_2 be the fundamental frequencies of gases in AC and BC respectively, then

$$n_1 = \frac{v_1}{4l} = \frac{1100}{4 \times 0.5} = 550 \text{ Hz}$$

or

$$AC = l = 0.5$$

$$n_2 = \frac{v_2}{4l} = \frac{300}{4 \times 0.5} = 150 \text{ Hz}$$

Here v_1 and v_2 are the velocities of sound in hydrogen and oxygen respectively.

As the two frequencies are different and hence the two gas columns are not vibrating in the fundamental mode. In case of a closed pipe only odd harmonics with frequencies 3, 5, 7, 9, ..., etc. times the fundamental frequency are observed. Now the problem is to find out that which harmonics of n_1 and n_2 have the same frequency. We notice that

$$\frac{n_1}{n_2} = \frac{500}{150} = \frac{11}{3}$$

or

$$3n_1 = 11n_2$$

This shows that third harmonic of n_1 and eleventh harmonic of n_2 have equal frequencies. Similarly, 6th harmonic of n_1 and 22th harmonic of n_2 have equal frequencies and so on.

Thus common minimum frequency is

$$= 3n_1 = 3 \times 500 = 1650 \text{ Hz}$$

$$= 11n_2 = 11 \times 150 = 1650 \text{ Hz.}$$

Illustrative Example 6.60

In Kundt's tube experiment the following observations were made : Length of the brass rod is 100 cm; average length of a loop in air is 10.3 cm and in carbon-di-oxide = 8.0 cm. Calculate the velocity of sound in brass and in CO_2 . What is the frequency of the note ?

(Given the velocity of sound in air at the temperature of the experiment to 350 m/s)

Solution

Let λ_r , λ_a and λ_g be the wavelengths in cm in the rod, in air and in gas respectively. Then

$$\frac{\lambda_r}{2} = 100 \quad \text{or} \quad \lambda_r = 200 \text{ cm,}$$

$$\frac{\lambda_a}{2} = 10.3 \quad \text{or} \quad \lambda_a = 20.6 \text{ cm,}$$

and $\frac{\lambda_g}{2} = 8.0$ or $\lambda_g = 16$ cm,

Let v_r , v_a and v_g be the sound velocities in rod, air and gas respectively. Then

$$\frac{v_r}{v_a} = \frac{\lambda_r}{\lambda_a} \quad \text{or} \quad v_r = \frac{\lambda_r}{\lambda_a} \times v_a$$

$$\begin{aligned} \text{or} \quad v_r &= \frac{200}{20.6} \times 350 \\ &= 3.4 \times 10^3 \text{ m/s} \end{aligned}$$

Similarly we have

$$v_g = 350 \times \frac{16}{20.6} = 271.8 \text{ m/s}$$

If n be frequency of the note, then we have

$$v_a = n \lambda_a$$

$$\begin{aligned} \text{or} \quad n &= \frac{v_a}{\lambda_a} = \frac{350}{0.206} \\ &= 1699.02 \text{ Hz} \end{aligned}$$

Illustrative Example 6.61

The air column in a pipe closed at one end is made to vibrate in its second overtone by a tuning fork of frequency 440 Hz. The speed of sound in air is 330 m/s. End corrections may be neglected. Let P_0 denote the mean pressure at any point in the pipe, and ΔP_0 the maximum amplitude of pressure variation.

- Find the length L of the air column.
- What is the amplitude of pressure variation at the middle of the column?
- What are the maximum and minimum pressure at the open-end of the pipe?
- What are the maximum and minimum pressure at the closed-end of the pipe?

Solution

(a) In case of closed organ pipe as fundamental frequency is $(v/4L)$ and only odd harmonics are present, second overtone will mean fifth harmonic and so

$$f = \frac{5v}{4L} = 440 \text{ Hz}$$

$$\text{and hence} \quad L = \frac{5 \times 330}{4 \times 440} = \frac{15}{16} \text{ m}$$

(b) In terms of pressure as at the position of displacement antinode there is pressure node and vice-versa, the variation of pressure amplitude of standing pressure waves along the length of the column with $x = 0$ at its open end will be

$$p = \Delta p_0 \sin kx = \Delta p_0 \sin \left(\frac{2\pi}{\lambda} x \right) \quad \left[\text{As } k = \frac{2\pi}{\lambda} x \right] \quad \text{or}$$

Now as for second overtone $L = (5/4)\lambda$, so at the middle.

$$x = \frac{L}{2} = \frac{5}{8} \lambda$$

and hence

$$p = \Delta p_0 \sin \frac{2\pi}{\lambda} \left(\frac{5}{8} \lambda \right) = \Delta p_0 \sin \left(\frac{5}{4} \pi \right)$$

$$\text{or} \quad |p| = \Delta p_0 \times \frac{1}{\sqrt{2}} = \frac{\Delta p_0}{\sqrt{2}}$$

(c) For free end as $x = 0$, $p = 0$, i.e., the amplitude of pressure wave is zero (as it is a node), so

$$p_{\max} = p_{\min} = p_0 \pm 0 = p_0$$

(d) For closed end $x = (5/4)\lambda$, so the amplitude of pressure wave

$$|p| = \left| \Delta p_0 \sin \frac{2\pi}{\lambda} \left(\frac{5}{4} \lambda \right) \right| = \Delta p_0$$

Thus maximum and minimum pressures are given as

$$p_{\max} = p_0 + \Delta p_0 \quad \text{and} \quad p_{\min} = p_0 - \Delta p_0$$

Illustrative Example 6.62

A pop-gun consists of a cylindrical barrel 3 cm³ in cross-section closed at one end by a cork and having a well fitting piston at the other. If the piston is pushed slowly in, the cork is finally ejected, giving a pop, the frequency of which is found to be 512 Hz. Assuming that the initial distance between the cork and the piston was 25 cm and that there is no leakage of air, calculate the force required to eject the cork. Atmospheric pressure = 1 kg wt/cm², $v = 340$ m/s.

Solution

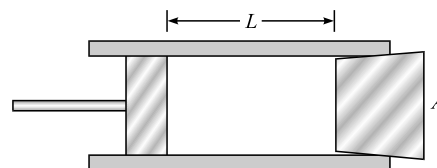


Figure 6.77

When the cork is ejected, the situation is shown in figure-6.77. Let the position of the piston from end A be L cm. So, this forms a closed pipe of length L . It produces a note of frequency 512 Hz. Now

$$n = \frac{v}{4L} \quad \text{or} \quad 512 = \frac{340}{4L}$$

$$L = \frac{340}{512 \times 4} = 0.166 \text{ m} = 16.6 \text{ cm}$$

Before the piston is moved, $P = 1 \text{ kg wt/cm}^2$ and $V = 25 \times 3 \text{ cm}^3$.
When the cork is ejected, let pressure be P' .

The volume of air inside

$$V' = 16.6 \times 3 \text{ cm}^3$$

From Boyle's law,

$$PV = P'V'$$

$$(1)(25 \times 3) = P' \times (16.6 \times 3)$$

$$P' = \left(\frac{25}{16.6} \right) \text{ kg wt/cm}^2$$

$$= 1.5 \text{ kg wt/cm}^2$$

Thus pressure inside the barrel $= 1.5 \text{ kg wt/cm}^2$

Pressure difference = Pressure inside – Pressure outside

$$= (1.5 - 1) \text{ kg wt/cm}^2$$

$$[\text{As Outside pressure} = 1 \text{ kg wt/cm}^2]$$

$$= 0.5 \text{ kg wt/cm}^2$$

Force on piston = Pressure $\times$ area

$$= 0.5 \times 3$$

$$= 1.5 \text{ kg wt}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Stationary Waves and Beats

Module Number - 32 to 46

Practice Exercise 6.6

(i) Find the fundamental frequency and the first four overtones of a 15 cm pipe :

(a) if the pipe is closed at one end, and

(b) if the pipe is open at both ends.

(c) How many overtones may be heard by a person of normal hearing in each of the above cases? Velocity of sound in air $= 330 \text{ m/s}$.

[(a) 550, 1650, 2750, 3850, 4950 Hz (b) 1100, 2200, 3300, 4400, 5500 Hz (c) 17, 17]

(ii) The separation between a node and the next antinode in a vibrating air column is 25 cm. If the speed of sound in air is 340 m/s, find the frequency of vibration of the air column.

[340 Hz]

(iii) For a certain organ pipe, three successive resonance frequencies are observed at 425 Hz, 595 Hz and 765 Hz respectively. Taking the speed of sound in air to be 340 m/s.

(a) Explain whether the pipe is closed at one end or open at both ends

(b) Determine the fundamental frequency and length of the pipe.

[(a) closed-end, (b) 85 Hz, 1 m]

(iv) A pipe is closed at one end by a membrane which may be considered a seat of displacement node and provided with a piston at the other end. The membrane is set to sonic oscillations of frequency 2000 Hz. Find the velocity of sound if on moving the piston, resonance occurs at the interval of 8.5 cm.

[340 m/s]

(v) A 'pop' gun consists of a tube 25 cm long closed at one end by a cork and at the other end by a tightly fitted piston. The piston is pushed slowly in. When the pressure rises to one and half times the atmospheric pressure, the cork is violently blown out. Calculate the frequency of the 'pop' caused by its ejection. Speed of sound in air is 340 m/s.

[510 Hz]

(vi) A cylindrical metal tube has a length of 50 cm and is open at both ends. Find the frequencies between 1000 Hz and 2000 Hz at which the air column in the tube can resonate. Speed of sound in air is 340 m/s.

[1020 Hz, 1360 Hz and 1700 Hz]

(vii) A uniform tube of length 60 cm stands vertically with its lower end dipping into water. When the length above water successively taken at 14.8 cm and 48.0 cm, the tube responds to a vibrating tuning fork of frequency 512 Hz. Find the lowest frequency to which the tube will respond when it is open at both ends.

[283.30 Hz]

(viii) A tuning fork having frequency of 340 Hz is vibrated just above a cylindrical tube. The height of the tube is 120 cm. Water is slowly poured in. What is the minimum height of water required for resonance? (velocity of sound in air is $v = 340 \text{ m/s}$).

[45 cm]

(ix) A tube closed at one end has a vibrating diaphragm at the other end, which may be assumed to be a displacement node. It is found that when the frequency of the diaphragm is 2000 Hz, a stationary wave pattern is set up in which the distance between adjacent nodes is 8 cm. When the frequency is gradually reduced, the stationary wave pattern disappears but another

stationary wave pattern reappears at a frequency of 1600 Hz. Calculate:

- The speed of sound in air,
- The distance between adjacent nodes at a frequency of 1600 Hz,
- The distance between the diaphragm and the closed end,
- The next lower frequencies at which stationary wave patterns will be obtained.

[(a) 320 m/s (b) 10 cm (c) 40 cm (d) 1200 Hz, 800 Hz and 400 Hz]

6.16 Beats

When two sources of sound that have almost the same frequency are sounded together, an interesting phenomenon occurs. A sound with a frequency average of the two is heard and the loudness of sound repeatedly grows and then decays, rather than being constant. Such a repeated variation in amplitude of sound are called “beats”.

If the frequency of one of the wave source is changed, there is a corresponding change in the rate at which the amplitude varies. This rate is called beat frequency. As the frequencies come close together, the beat frequency becomes slower. A musician can tune a guitar to another source by listening for the beats while increasing or decreasing the tension in each string, eventually the beat frequency becomes very low so that effectively no beats are heard, and the two sources are then in tune.

We can also explain the phenomenon of beats mathematically. Let us consider the two superposing waves have frequencies n_1 and n_2 then their respective equations of oscillation are

$$y_1 = A \sin 2\pi n_1 t \quad \dots (6.233)$$

and $y_2 = A \sin 2\pi n_2 t \quad \dots (6.234)$

On superposition at a point, the displacement of the medium particle is given as

$$y = y_1 + y_2$$

$$y = A \sin 2\pi n_1 t + A \sin 2\pi n_2 t$$

$$y = 2A \cos 2\pi \left(\frac{n_1 - n_2}{2} \right) t \sin 2\pi \left(\frac{n_1 + n_2}{2} \right) t \quad \dots (6.235)$$

$$y = R \sin 2\pi \left(\frac{n_1 + n_2}{2} \right) t \quad \dots (6.236)$$

There equation-(6.236) gives the displacement of medium particle where superposition takes place, it shows that the particle executes SHM with frequency $\frac{n_1 + n_2}{2}$, average of the two

superposing frequencies and with amplitude R which varies with time, given as

$$R = 2A \cos 2\pi \left(\frac{n_1 - n_2}{2} \right) t \quad \dots (6.237)$$

Here R becomes maximum when

$$\cos 2\pi \left(\frac{n_1 - n_2}{2} \right) t = \pm 1$$

or $2\pi \left(\frac{n_1 - n_2}{2} \right) t = N\pi \quad [N \in I]$

or $t = \frac{N}{n_1 - n_2} \quad \dots (6.238)$

or at time $t = 0, \frac{1}{n_1 - n_2}, \frac{2}{n_1 - n_2}, \dots$

At all the above time instants the sound of maximum loudness is heard, similarly we can find the time instants when the loudness of sound is minimum, it occurs when

$$\cos 2\pi \left(\frac{n_1 - n_2}{2} \right) t = 0$$

or $2\pi \left(\frac{n_1 - n_2}{2} \right) t = (2N + 1) \frac{\pi}{2} \quad [N \in I]$

or $t = \frac{2N + 1}{2(n_1 - n_2)} \quad \dots (6.239)$

or at time instants $t = \frac{1}{2(n_1 - n_2)}, \frac{3}{2(n_1 - n_2)}, \dots$

Here we can see that these time instants are exactly lying in the middle of the instants when loudest sound is heard. Thus on superposition of the above two frequencies at a medium particle, the sound will be increasing, decreasing, again increasing and decreasing and so on. This effect is called beats. Here the time between two successive maximum or minimum sounds is called beat period, which is given as

Beat Period T_B = time between two successive maxima = time between two successive minima

$$= \frac{1}{n_1 - n_2} \quad \dots (6.240)$$

Thus beat frequency or number of beats heard per second can be given as

$$F_B = \frac{1}{T_B} = n_1 - n_2 \quad \dots (6.241)$$

The superposition of two waves of slightly different frequencies is graphically shown in figure-6.78. The resulting envelope of the wave formed after superposition is also shown in figure-6.78(b). Such a wave when propagates, produces “beat” effect at the medium particles.

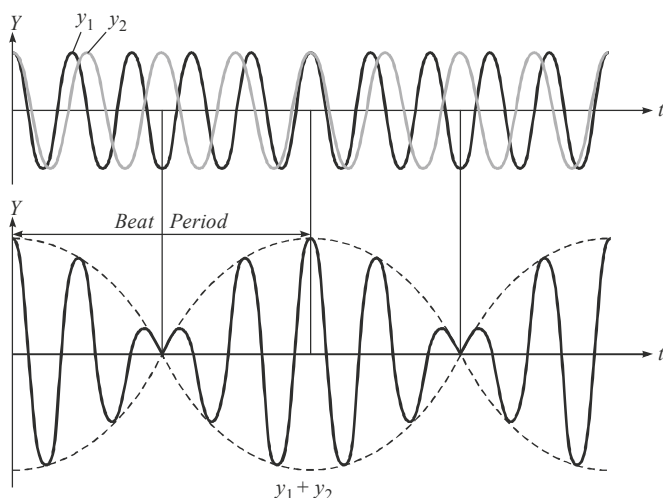


Figure 6.78

6.16.1 Echo

The repetition of sound produced due to reflection by a distant extended surface like a different, hill, well, building etc. is called an echo. The effect of sound on human ear remains for approximately one tenth of a second. If the sound is reflected back in a time less than $1/10$ of a second, no echo is heard. Hence human ears are not able to distinguish a beat frequency of 10 Hz or more than 10 Hz.

Illustrative Example 6.63

Two identical sonometer wires have a fundamental frequency of 500 Hz when kept under the same tension. What fractional increase in the tension of one wire would cause an occurrence of 5 beats per second, when both wires vibrate together?

Solution

The fundamental frequency of wire is given as

$$n_0 = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

On increasing tension by ΔT the new frequency will be

$$\begin{aligned} n'_0 &= \frac{1}{2L} \sqrt{\frac{T + \Delta T}{\mu}} \\ &= n_0 \sqrt{1 + \frac{\Delta T}{T}} \\ &\approx n_0 \left(1 + \frac{\Delta T}{2T} \right) \end{aligned}$$

Now the fractional change in frequency will be

$$\frac{n'_0 - n_0}{n_0} = \frac{1}{2} \frac{\Delta T}{T}$$

or

$$\begin{aligned} \frac{\Delta T}{T} &= 2 \times \frac{5}{500} \\ &= 0.02 \end{aligned}$$

Illustrative Example 6.64

Two wires are fixed on a sonometer wire. Their tensions are in the ratio 8 : 1, the lengths in the ratio 36 : 35, the diameters in the ratio 4 : 1 and the densities in the ratio 1 : 2. Find the frequency of the beats produced if the note of the higher pitch has frequency of 360 Hz.

Solution

For the two wires, we have

$$n_1 = \frac{1}{2l_1} \sqrt{\left(\frac{T_1}{\pi r_1^2 \rho_1} \right)} \sqrt{\left(\frac{T_2}{\pi r_2^2 \rho_2} \right)}$$

and

$$n_2 = \frac{1}{2l_2} \sqrt{\left(\frac{T_2}{\pi r_2^2 \rho_2} \right)}$$

or

$$= \frac{l_2}{l_1} \sqrt{\left[\frac{T_1}{T_2} \left(\frac{r_2}{r_1} \right)^2 \frac{\rho_2}{\rho_1} \right]} \quad \dots (6.242)$$

According to the given problem

$$\frac{l_2}{l_1} = \frac{36}{35}; \quad \frac{T_1}{T_2} = \frac{8}{1}; \quad \frac{r_1}{r_2} =$$

$$= \frac{4}{1} \quad \text{and} \quad \frac{\rho_1}{\rho_2} = \frac{1}{2}$$

Substituting these values in equation-(6.242)

$$\text{or} \quad \frac{n_1}{n_2} = \frac{35}{36} \sqrt{\left[\left(\frac{8}{1} \right) \left(\frac{1}{16} \right) \cdot \left(\frac{2}{1} \right) \right]} = \frac{35}{36} \quad \dots (6.243)$$

It is clear from equation-(6.243) that n_2 has higher pitch so

$$n_2 = 360$$

$$\text{Now} \quad \frac{n_1}{360} = \frac{35}{36}$$

$$\text{or} \quad n_1 = 350 \text{ Hz}$$

Thus beat frequency is given as

$$\begin{aligned} \Delta n &= 360 - 350 \\ &= 10 \text{ Hz} \end{aligned}$$

Illustrative Example 6.65

A tuning fork of frequency 300 Hz resonates with an air column closed at one end at 27°C. How many beats will be heard in the vibrations of the fork and the air column at 0°C? End correction is negligible.

Solution

As the air column at 27°C resonates with a tuning fork of frequency 300 hertz and hence at 27°C its own frequency is also 300 Hz. Let l be the length of air column and v_2 be the speed of sound at 27°C, then frequency of air column at 27°C is

$$n = \frac{v_2}{4l} = 300 \text{ Hz} \quad \dots (6.244)$$

Suppose the speed of sound at 0°C be v_0 . We know that the speed is directly proportional to the square root of the absolute temperature, thus we have

$$\frac{v_0}{v_2} = \sqrt{\left(\frac{273}{27 + 273}\right)} = \sqrt{\left(\frac{273}{300}\right)} = 0.954$$

or $v_0 = 0.954 \times v_2$

Thus frequency of air column at 0°C,

$$n' = \frac{v_0}{4l} = \frac{0.954 \times v_2}{4l} \quad \dots (6.245)$$

Substituting the value of $(v_2/4l)$ from equation-(6.244) in equation-(6.245), we get

$$n' = 0.954 \times 300 = 286 \text{ Hz}$$

As the frequency of tuning fork is 300 Hz, the beat frequency is given as

$$\begin{aligned} \Delta n &= 300 - 286 \\ &= 14 \text{ Hz.} \end{aligned}$$

Illustrative Example 6.66

A string under a tension of 129.6 N produces 10 beats per second when it is vibrated along with a tuning fork. When the tension in the string is increased to 160 N, it sounds in unison with the same tuning fork. Calculate the fundamental frequency of tuning fork.

Solution

Let n be the frequency of tuning fork. The frequency of string will be either $(n + 10)$ or $(n - 10)$. As the tension in the string increases, its frequency increases ($n \propto \sqrt{T}$) and becomes n . This shows that the initial frequency (at $T = 129.6 \text{ N}$) of string will be $(n - 10)$. Hence

$$n - 10 = \frac{1}{2l} \sqrt{\left(\frac{129.6}{\mu}\right)} \quad \dots (6.246)$$

$$\text{Finally, } n = \frac{1}{2l} \sqrt{\left(\frac{160}{\mu}\right)} \quad \dots (6.247)$$

Dividing equation-(6.246) by equation-(6.247), we get

$$\frac{n-10}{n} = \sqrt{\left(\frac{129.6}{160}\right)}$$

Solving we get $n = 100 \text{ Hz}$

Illustrative Example 6.67

Two forks A and B when sounded together produce 4 beats per second. The fork A is in unison with 30 cm length of a sonometer wire, and B is in unison with 25 cm length of the same wire at the same tension. Calculate the frequency of the forks.

Solution

As in case of vibrations of string under specific tension, we know that frequency $n \propto (1/L)$ or $n = (K/L)$. As forks A and B are in unison (*equal frequencies*) with 30 cm and 25 cm length of a given sonometer wire respectively. If n_A and n_B be the frequencies of the two tuning forks, we have,

$$n_A = \frac{K}{30} \quad \text{and} \quad n_B = \frac{K}{25}$$

$$\text{or} \quad \frac{n_A}{n_B} = \frac{25}{30} \quad \dots (6.248)$$

Further as the forks A and B produce 4 beats when sounded together,

$$n_A \sim n_B = 4$$

But from equation-(6.248) it is clear that $n_A < n_B$

$$\text{Thus we have } n_B - n_A = 4 \quad \dots (6.249)$$

Solving equation-(6.248) and (6.249) for n_A and n_B , we get

$$n_A = 20 \text{ Hz} \quad \text{and} \quad n_B = 24 \text{ Hz}$$

Illustrative Example 6.68

A column of air at 51°C and a tuning fork produce 4 beats per second when sounded together. As the temperature of air column is decreased, the number of beats per second tends to decrease and when the temperature is 16°C the two produce one beat per second. Find the frequency of tuning fork.

Solution

The frequency of air column is given by

$$n = \frac{v}{2l}$$

Neglecting end correction, we have

$$\begin{aligned} \frac{n_1}{n_2} &= \frac{v_1}{v_2} = \sqrt{\left(\frac{T_1}{T_2}\right)} \\ &= \sqrt{\left(\frac{273+51}{273+16}\right)} = \sqrt{\left(\frac{324}{289}\right)} \\ &= 1.059 \end{aligned}$$

This shows that $n_2 < n_1$

As the number of beats with n_2 is less than the number of beats with n_1 , hence the frequency of air column must be greater than the frequency of the tuning fork. If n be the frequency of tuning fork, then

$$n_1 = n + 4 \quad \text{and} \quad n_2 = n + 1$$

$$\text{or} \quad \frac{n+4}{n+1} = 1.059$$

This gives $n = 49.8 \text{ Hz}$

Illustrative Example 6.69

The first overtone of an open organ pipe beats with the first overtone of a closed organ pipe with a beat frequency of 2.2 Hz. The fundamental frequency of the closed organ pipe is 110 Hz. Find the lengths of the pipes.

Solution

Let the lengths of open and closed pipes be l_1 and l_2 respectively. We know that

$$\text{first overtone of open organ pipe, } n_1 = \frac{v}{l_1}$$

$$\text{first overtone of closed organ pipe, } n_2 = \frac{3v}{4l_2}$$

$$\text{fundamental frequency of closed organ pipe } n = \frac{v}{4l_2}$$

According to the given question, as $n = 110 \text{ Hz}$, we have

$$110 = \frac{v}{4l_2} = \frac{330}{4l_2}$$

$$\text{or} \quad l_2 = \frac{330}{4 \times 110} = 0.75 \text{ m}$$

Further, it is given that beat frequency is 2.2 Hz, we have

$$f_B = 2.2 = \frac{v}{l_1} - \frac{3v}{4l_2}$$

$$\text{or} \quad 2.2 = \frac{2 \times 330}{2l_1} - \frac{3 \times 330}{4 \times 0.75}$$

$$\text{or} \quad 2.2 = \frac{330}{l_1} - 330$$

$$\text{or} \quad 332.2 = \frac{330}{l_1}$$

$$\text{or} \quad l_1 = \frac{330}{332.2} = 0.993 \text{ m}$$

Again, Beat frequency can also be given as

$$f_B = 2.2 = \frac{3v}{4l_2} - \frac{v}{l_1}$$

$$\text{or} \quad 2.2 = \frac{3 \times 330}{4 \times 0.75} - \frac{2 \times 330}{2l_1}$$

$$2.2 = 330 - \frac{330}{l_1}$$

Solving we get $l_1 = 1.006 \text{ m}$

Illustrative Example 6.70

The vibration portion of a wire which is stretched with a weight of 6.48 kg weighs 0.5 gm. When sounding in fundamental note, it is found to give 20 beats in 5 seconds, with a vibrating tuning fork of frequency 256. If the length of the wire is slightly decreased, the note emitted by it is observed to be in unison with that of the fork. Calculate the original length of the wire.

Solution

When the length of the wire is l metre, the number of beats per second = $20/5 = 4$.

If the length is decreased, the frequency increase and the beats disappear. The original frequency of the wire is less than 256 by 4.

So the frequency of the wire of length l is

$$n = 256 - 4 = 252 \text{ Hz}$$

Tension in wire is

$$T = 6.48 \times 9.8 \text{ N}$$

Linear mass density of wire is

$$\mu = \frac{0.5 \times 10^{-3}}{l} \text{ kg/m}$$

We know natural frequency of oscillations of the wire is given as

$$n_o = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

or

$$252 = \frac{1}{2l} \sqrt{\left(\frac{6.48 \times 9.8 \times L}{0.5 \times 10^{-3}} \right)}$$

or

$$(252)^2 = \frac{1}{4l^2} \times \frac{6.48 \times 9.8 \times L}{0.5 \times 10^{-3}}$$

On solving we get

$$l = 0.5 \text{ m} = 50 \text{ cm}$$

Illustrative Example 6.71

When 0.98 m long metallic wire is stressed, an extension of 0.02 m is produced. An organ pipe 0.5 m long and open at both ends, when sounded with this stressed metallic wire, produces 8 beats in its fundamental mode. By decreasing the stress in the wire, the number of beats are found to decrease. Find the young's modulus of wire. The density of metallic wire is 10^4 kg/m^3 and sound velocity in air is 292 m/s.

Solution

We know the fundamental frequency of open pipe is given as

$$n_1 = \frac{v}{2l} = \frac{292}{2 \times (0.5)} = 292 \text{ Hz} \quad \dots (6.250)$$

Let L' be the stressed length of the wire. Then

$$L' = L + \Delta L, \text{ where } L = \text{initial length}$$

$$= 0.98 + 0.02 = 1.0 \text{ m}$$

The fundamental frequency of stressed wire is

$$n = \frac{1}{2L'} \sqrt{\left(\frac{T}{\pi r^2 \rho} \right)} \quad \dots (6.251)$$

Where r = radius of the wire and ρ = density of wire

Given that the wire produces 8 beats with pipe. Thus we have

$$n = n_1 \pm 8 = 292 \pm 8$$

$$= 300 \quad \text{or} \quad 284$$

As the number of beats decreases by decreasing the stress and hence

$$n = 300$$

From equation-(6.251), we have

$$\frac{T}{\pi r^2} = 4 n^2 \rho (L')^2$$

Now

$$\begin{aligned} Y &= \frac{(T / \pi r^2)}{(\Delta L / L)} = \frac{4 n^2 \rho (L')^2}{(\Delta L / L)} \\ &= \frac{4 \times (300)^2 \times 10^4 \times (1.0)^2}{(0.02 / 0.98)} \\ &= 17.64 \times 10^{10} \text{ N/m}^2 \end{aligned}$$

Illustrative Example 6.72

A string 25 cm long and having a mass of 2.5 gm is under tension. A pipe closed at one end is 40 cm long. When the string is set vibrating in its first overtone and the air in the pipe in its fundamental frequency, 8 beats per second are heard. It is observed that decreasing the tension in the string decreases the beat frequency. If the speed of sound in air is 320 m/s find the tension in the string.

Solution

Linear mass density of string is

$$\mu = \frac{m}{L} = \frac{2.5 \times 10^{-3}}{0.25} = 0.01 \text{ kg/m}$$

For first overtone (i.e., second harmonic) the frequency of the string is

$$n_1 = \frac{2}{2L} \sqrt{\frac{T}{\mu}} = \frac{2}{2 \times 0.25} \sqrt{\frac{T}{0.01}} = 40\sqrt{T}$$

Fundamental frequency of closed organ pipe is given as

$$n_c = \frac{v}{4l} = \frac{320}{4 \times 0.4} = 200 \text{ Hz}$$

Now as the two produces 8 beats per second, we have

$$200 - 40\sqrt{T} = 8 \quad [\text{case-I}]$$

$$\text{or} \quad 40\sqrt{T} - 200 = 8 \quad [\text{case-II}]$$

Now as decreasing the tension decreases the beat frequency, case-I is not permissible. Thus we have

$$40\sqrt{T} = 208,$$

or

$$\begin{aligned} T &= \left[\frac{208}{40} \right]^2 \\ &= 27.04 \text{ N} \end{aligned}$$

Illustrative Example 6.73

A metal wire of diameter 1 mm is held on two knife edges separated by a distance of 50 cm. The tension in the wire is 100 N. The wire, vibrating with its fundamental frequency and a vibrating tuning fork together produces 5 beats per second. The tension in the wire is then reduced to 81 N. When the two are excited, beats are heard at the same rate. Calculate

- the frequency of the fork and
- the density of the material of wire

Solution

Let the frequency of tuning fork be n , then in the first case the fundamental frequency of the wire will be $(n + 5)$, which is given as

$$(n + 5) = \frac{1}{2l} \sqrt{\left(\frac{T}{\mu}\right)}$$

Here $T = 100$ N, $l = 50$ cm = 0.5 m, on substituting values, we get

$$(n + 5) = \frac{1}{2 \times 0.5} \sqrt{\left(\frac{100}{\mu}\right)} = \frac{10}{\sqrt{\mu}} \quad \dots (6.252)$$

In the second case, $T = 81$ N. In this case the frequency of wire will be $(n - 5)$. Thus we have

$$(n - 5) = \frac{1}{2 \times 0.5} \sqrt{\left(\frac{81}{\mu}\right)} = \frac{9}{\sqrt{\mu}} \quad \dots (6.253)$$

Subtracting equation-(6.253) from equation-(6.252), we get

$$10 = \frac{10}{\sqrt{\mu}} - \frac{9}{\sqrt{\mu}} = \frac{1}{\sqrt{\mu}}$$

or

$$\mu = \frac{1}{100} = 0.01 \text{ kg/m}$$

But we know that linear mass density of a wire of cross sectional radius r and density ρ is given as

$$\mu = \pi r^2 \rho$$

or

$$\rho = \frac{m}{\pi r^2} = \frac{0.01}{3.14 (5 \times 10^{-4})^2}$$

$$= 12732.5 \text{ kg/m}^3$$

From equation-(6.252), we have

$$n + 5 = \frac{10}{\sqrt{0.01}} = 100 \text{ Hz}$$

$$n = 100 - 5$$

$$= 95 \text{ Hz}$$

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Stationary Waves and Beats

Module Number - 47 to 54

Practice Exercise 6.7

(i) A closed pipe and an open pipe sounding together produce 5 beats per second. If the length of the open pipe is 30 cm, find by how much the length of the closed pipe should be changed to bring the two pipes in unison. Take speed of sound in air is 330 m/s.

[0.1376 cm]

(ii) A tuning fork produces 4 beats per second with another tuning fork of frequency 256 Hz. The first one is now loaded with a little wax and the beat frequency is found to increase to 6 per second. What was the original frequency of the tuning fork?

[252 Hz]

(iii) If two sound waves, $y_1 = 0.3 \sin 596 \pi [t - x/330]$ and $y_2 = 0.5 \sin 604 \pi [t - x/330]$ are superimposed, what will be the (a) frequency of resultant wave (b) frequency at which beats are produced. (c) The ratio of maximum and minimum intensities of beats.

[(a) 300 Hz, (b) 4 Hz, (c) 16]

(iv) As set of 25 tuning forks is arranged in a series of decreasing frequencies. Each fork gives 3 beats with the succeeding one. The first fork is the octave of the last. Calculate the frequency of the first and the 16th tuning fork.

[144 Hz and 99 Hz]

(v) A piano wire A vibrates at a fundamental frequency of 600 Hz. A second identical wire B produces 6 beats per second with it when the tension in A is slightly increased. Find the ratio of the tension in A to the tension in B .

[1.02]

(vi) You are given four tuning forks; the lowest frequency of the fork is 300 Hz. By striking two tuning forks at a time, 1, 2, 3, 5, 7 and 8 Hz beat frequencies are heard. What are the possible frequencies of the other three forks?

[301, 303 and 308 Hz]

(vii) There are three sources of sound of equal intensities with frequency 400, 401 and 402 Hz. What is the beat frequency heard if all are sounded simultaneously?

[1 Hz]

(viii) A tuning fork of frequency 256 Hz produces 4 beats per second with a wire of length 25 cm vibrating in its fundamental mode. The beat frequency decreases when the length is slightly shortened. What could be the minimum length by which the wire be shortened so that it produces no beats with the tuning fork?

[0.39 cm]

6.17 Doppler's Effect

When a car at rest on a road sounds its high frequency horn and you are also standing on the road near by, you'll hear the sound of same frequency it is sounding but when the car approaches you with its horn sounding, the pitch (frequency) of its sound seems to drop as the car passes. This phenomenon was first described by an Austrian Scientist Christien Doppler, is called the Doppler effect. He explained that when a source of sound and a listener are in motion relative to each other, the frequency of the sound heard by the listener is not the same as the source frequency. Lets discuss the Doppler effect in detail for different cases.

6.17.1 Stationary Source and Stationary Observer

Figure-6.79 shows a stationary sources of frequency n_0 which produces sound waves in air of wavelength λ_0 given as

$$\lambda_0 = \frac{v}{n_0} \quad [v = \text{speed of sound in air}]$$

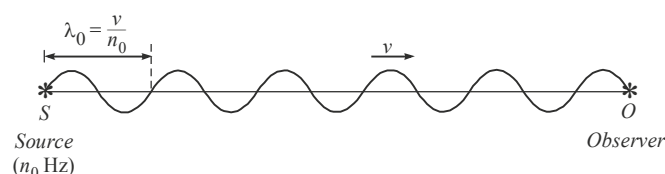


Figure 6.79

Although sound waves are longitudinal, here we represent sound waves by the transverse displacement curve as shown in figure-6.79 to understand the concept in a better way. As source produces waves, these waves travel towards, stationary observer O in the medium (air) with speed v and wavelength λ_0 . As observer is at rest here it will observe the same wavelength

λ_0 is approaching it with speed v so it will listen the frequency n given as

$$n = \frac{v}{\lambda_0} = n_0 \quad \dots (6.254)$$

[Same as that of source]

This is why when a stationary observer listens the sound from a stationary source of sound, it detects the same frequency sound which the source is producing. Thus no Doppler effect takes place if there is no relative motion between source and observer.

6.17.2 Stationary Source and Moving Observer

Figure-6.80 shows the case when a stationary sources of frequency n_0 produces sound waves which have wavelength in air given as

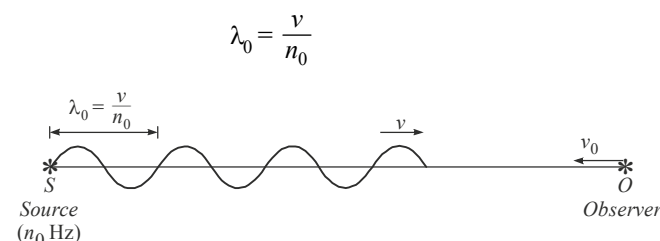


Figure 6.80

These waves travel toward moving observer with velocity v_0 towards, the source. When sound waves approach observer, it will receive the waves of wavelength λ_0 with speed $v + v_0$ (relative speed). Thus the frequency of sound heard by observer can be given as

Apparent frequency

$$\begin{aligned} n_{ap} &= \frac{v + v_0}{\lambda_0} \\ &= \frac{v + v_0}{\left(\frac{v}{n_0}\right)} = n_0 \left(\frac{v + v_0}{v}\right) \quad \dots (6.255) \end{aligned}$$

Similarly we can say that if the observer is receding away from the source the apparent frequency heard by the observer will be given as

$$n_{ap} = n_0 \left(\frac{v - v_0}{v}\right) \quad \dots (6.256)$$

6.17.3 Moving Source and Stationary Observer

Figure-6.81 shows the situation when a moving source S of frequency n_0 produces sound waves in medium (air) and the waves travel toward observer with velocity v .

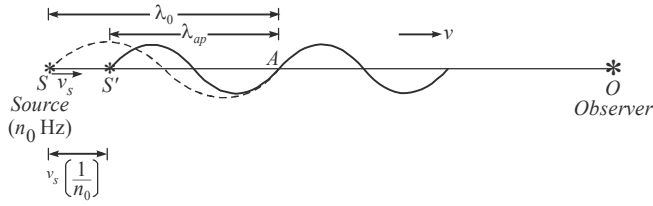


Figure 6.81

Here if we carefully look at the initial situation when source starts moving with velocity v_s as well as it starts producing waves. The period of one oscillation is $\left(\frac{1}{n_0}\right)$ sec and in this duration source emits one wavelength λ_0 in the direction of propagation of waves with speed v , but in this duration the source will also move forward by a distance $v_s \left(\frac{1}{n_0}\right)$. Thus the effective wavelength of emitted sound in air is slightly compressed by this distance as shown in figure-6.81. This we term as apparent wavelength of sound in medium (air) by the moving source. This is given as

Apparent wavelength

$$\begin{aligned}\lambda_{ap} &= \lambda_0 - v_s \left(\frac{1}{n_0}\right) \quad \dots (6.257) \\ &= \frac{\lambda_0 n_0 - v_s}{n_0} = \frac{v - v_s}{n_0}\end{aligned}$$

Now this wavelength will approach observer with speed v (as O is at rest). Thus the frequency of sound heard by observer can be given as

Apparent frequency

$$\begin{aligned}n_{ap} &= \frac{v}{\lambda_{ap}} \\ &= \frac{v}{(v - v_s)/n_0} \\ &= n_0 \left(\frac{v}{v - v_s}\right) \quad \dots (6.258)\end{aligned}$$

Similarly if source is receding away from observer, the apparent wavelength emitted by source in air toward observer will be slightly expanded and the apparent frequency heard by the stationary observer can be given as

$$n_{ap} = n_0 \left(\frac{v}{v + v_s}\right) \quad \dots (6.259)$$

6.17.4 Moving Source and Moving Observer

Let us consider the situation when both source and observer are moving in same direction as shown in figure-6.82 at speeds v_s and v_0 respectively.

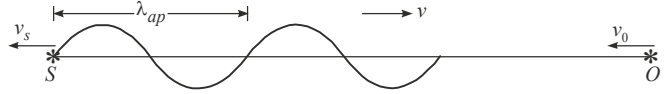


Figure 6.82

In this case the apparent wavelength emitted by the source behind it is given as

$$\lambda_{ap} = \frac{v + v_s}{n_0}$$

Now this wavelength will approach the observer at relative speed $v + v_0$, thus the apparent frequency of sound heard by the observer is given as

$$\begin{aligned}n_{ap} &= \frac{v + v_0}{\lambda_{ap}} \\ &= n_0 \left(\frac{v + v_0}{v + v_s}\right) \quad \dots (6.260)\end{aligned}$$

By looking at the expression of apparent frequency given by equation-(6.260), we can easily develop a general relation for finding the apparent frequency heard by a moving observer due to a moving source as

$$n_{ap} = n_0 \left[\frac{v \pm v_0}{v \pm v_s} \right] \quad \dots (6.261)$$

Here + and – signs are chosen according to the direction of motion of source and observer. The sign convention related to the motion direction can be stated as :

- For both source and observer v_0 and v_s are taken in equation-(6.261) with – ve sign if they are moving in the direction of $\vec{v}$ i.e. the direction of propagation of sound from source to observer.
- For both source and observer v_0 and v_s are taken in equation-(6.261) with + ve sign if they are moving in the direction opposite to $\vec{v}$ i.e. opposite to the direction of propagation of sound from source to observer.

6.17.5 Doppler Effect in Reflected Sound

When a car is moving toward a stationary wall as shown in figure-6.83. If the car sounds a horn, wave travels toward the wall and is reflected from the wall. When the reflected wave is

heard by the driver, it appears to be of relatively high pitch. If we wish to measure the frequency of reflected sound then the problem must be handled in two steps.

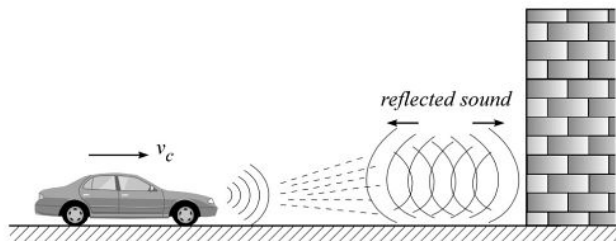


Figure 6.83

First we treat the stationary wall as a stationary observer and car as a moving source of sound of frequency n_0 . In this case the frequency received by the wall is given as

$$n_1 = n_0 \left(\frac{v}{v - v_c} \right) \quad \dots (6.262)$$

Now wall reflects this frequency and behaves like a stationary source of sound of frequency n_1 and car (driver) behave like a moving observer with velocity v_c . Here the apparent frequency heard by the car driver can be given as

$$\begin{aligned} n_{ap} &= n_1 \left(\frac{v + v_c}{v} \right) \\ \text{or} \quad &= n_0 \left(\frac{v}{v - v_c} \right) \times \left(\frac{v + v_c}{v} \right) \\ &= n_0 \left(\frac{v + v_c}{v - v_c} \right) \quad \dots (6.263) \end{aligned}$$

Same problem can also be solved in a different manner by using method of sound images. In this procedure we assume the image of the sound source behind the reflector. In previous example we can explain this by situation shown in figure-6.84

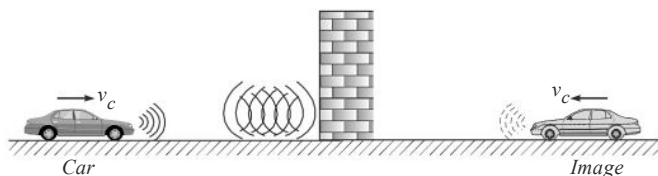


Figure 6.84

Here we assume that the sound which is reflected by the stationary wall is coming from the image of car which is at the back of it and coming toward it with velocity v_c . Now the frequency of sound heard by car driver can directly be given as

$$n_{ap} = n_0 \left[\frac{v + v_c}{v - v_c} \right] \quad \dots (6.264)$$

This method of images for solving problems of Doppler effect

is very convenient but is used only for velocities of source and observer which are very small compared to the speed of sound and it should not be used frequently when the reflector of sound is moving.

6.17.6 Doppler's Effect for Accelerated Motion

For the case of a moving source and a moving observer, we know the apparent frequency observer can be given as

$$n_{ap} = n_0 \left[\frac{v \pm v_0}{v \pm v_s} \right] \quad \dots (6.265)$$

Here v is the velocity of sound and v_0 and v_s are the velocity of observer and source respectively.

When a source or observer has accelerated or retarded motion then in equation-(6.265) we use that value of v_0 at which observer receives the sound and for source, we use that value of v_s at which it has emitted the wave.

The alternative method of solving this case is by the traditional method of compressing or expanding wavelength of sound by motion of source and using relative velocity of sound with respect to observer

6.17.7 Doppler's Effect when Source and Observer are not in Same Line of Motion

Consider the situation shown in figure-6.85. Two cars 1 and 2 are moving along perpendicular roads at speeds v_1 and v_2 . When car-1 sound a horn of frequency n_0 , it emits sound in all directions and say car-2 is at the position, shown in figure-6.85 when it receives the sound. In such cases we use velocity components of the cars along the line joining the source and observer thus the apparent frequency of sound heard by car-2 can be given as

$$n_{ap} = n_0 \left[\frac{v + v_2 \cos \theta_2}{v - v_1 \cos \theta_1} \right] \quad \dots (6.266)$$

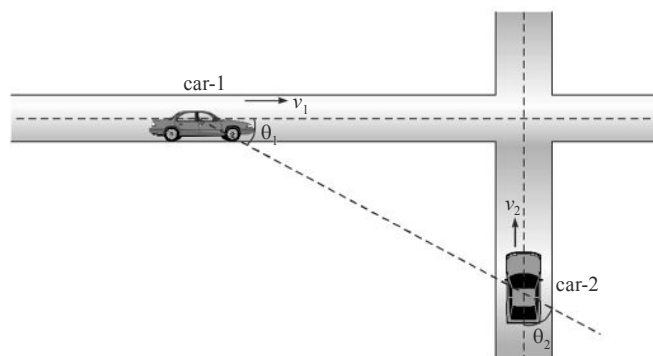


Figure 6.85

6.17.8 Doppler's Effect in Light

We've discussed the Doppler effect in sound. In case of light, the frequency of light radiation changes due to relative motion between light source and observer. According to the theory of electromagnetic waves, material medium is not necessary for the propagation of light, whereas for sound waves, material medium is necessary. In sound waves Doppler's effect is not symmetric i.e. the apparent frequencies are different when the source is moving toward a stationary observer and when the observer is moving toward a stationary source.

But the case is not same in light waves. In light, the Doppler effect is symmetric. The apparent frequency is same when either the source is moving or observer is moving. Let us consider an example, this will explain the concept in detail.

Consider a source of light with frequency ν_0 is moving toward a stationary observer at speed v_s then the frequency observed can be written as

$$\begin{aligned}\nu_{ap} &= \nu_0 \left[\frac{c}{c - v_s} \right] \\ \text{or} \quad \nu_{ap} &= \nu_0 \left[\frac{1}{1 - \frac{v_s}{c}} \right] \\ &= \nu_0 \left(1 - \frac{v_s}{c} \right)^{-1} \approx \nu_0 \left(1 + \frac{v_s}{c} \right) \quad \dots (6.267) \\ &\quad \left[\text{As } \frac{v_s}{c} \ll 1 \right]\end{aligned}$$

Similarly if we consider an observer is moving at speed v_0 toward a stationary light source of frequency ν_0 then the apparent frequency observed by the observer can be given as

$$\nu_{ap} = \nu_0 \left[\frac{c + v_0}{c} \right] = \nu_0 \left(1 + \frac{v_0}{c} \right) \quad \dots (6.268)$$

If $v_s = v_0$ equation-(6.267) and (6.268) gives same results. This happens because the velocity of light is very large compared to that of source and observer.

Similarly in case of light when source and observer both are moving we can consider any one at rest and consider relative velocity of one with respect to other for finding the apparent frequency observed.

In general cases we take observer at rest and consider source moving relative to observer because relative velocity of light in any physical condition can not exceed c . If λ_0 and λ_{ap} be the actual and apparent wavelength of light then for a stationary observer and a moving source, we have

$$\nu_{ap} = \nu_0 \left[\frac{c}{c - v_s} \right]$$

$$\text{or} \quad \frac{c}{\lambda_{ap}} = \frac{c}{\lambda_0} \left[\frac{c}{c - v_s} \right]$$

$$\text{or} \quad \lambda_0 c - \lambda_0 v_s = \lambda_{ap} c$$

$$\text{or} \quad v_s = \frac{(\lambda_0 - \lambda_{ap})}{\lambda_0} c$$

$$v_s = \frac{\Delta \lambda}{\lambda_0} c \quad \dots (6.269)$$

Here $\Delta \lambda$ is the wavelength shift due to motion of source and this equation-(6.269) gives the relative velocity of source toward observer. Whenever we observe wavelength shift, if received wavelength (λ_{ap}) is less than actual wavelength (λ_0) or $\Delta \lambda$ is positive we say source is approaching the observer and when received wavelength (λ_{ap}) is more than the actual wavelength (λ_0) or $\Delta \lambda$ is negative we say source is moving away from observer. In the field of astronomy, Doppler effect is found to be very useful.

6.17.9 Applications of Doppler Effect in Light

(i) Doppler's Shift

When radiation coming from distant galaxies and nebulae are analyzed by radio telescopes and compared with their natural radiation wavelength focussed on mean wavelength on a visible spectrum, it is observed that the coming radiation has a shift toward red end of visible spectrum approximately by 200 Å i.e. The wave length of coming radiation is 200 Å more than their actual wavelength. This gives the idea that universe is expanding. The shifting of analyzed wavelength toward red end of spectrum is called red shift and when some comet or a heavenly body is coming toward earth, the wavelength of the analyzed light decreases and on spectrum it shift toward violet end and is called violet shift or blue shift.

(ii) Velocity and Rotation of Sun

When light coming from eastern and western edges of sun are observed and analyzed, Doppler's shift shows that the shift is due to a velocity of about 2×10^3 m/s. But when light from north and south edges is observed, no such shift is observed. This shows that the sun rotates about the north south axis.

(iii) Discovery of Double Stars

By the constant observation of the sky, it has been found that some of the stars that appear to be single are actually double stars and are known as spectroscopic binaries. These stars

revolve about each other. When one is approaching the earth, and the other is going away from the earth, there is a shift in their spectral lines and a single spectral line is split up into two lines whose separation depends upon time and the time period is equal to the time-period of revolution of the stars. By this method a number of double-star systems have been found.

(iv) Saturn's Rings

The planet Saturn has been found to be surrounded by concentric rings. With the help of Doppler effect it has been found that these rings are not solids but consist of a number of 'satellites' moving around the Saturn in these orbits. If the rings were solids, the outer edge of the ring should have greater velocity than the inner edge. But, according to the principle of a satellite.

$$\frac{mv^2}{R} = \frac{GMm}{R^2}$$

or
$$v^2 = \frac{GM}{R}$$

Thus, the velocity of the satellite in the inner orbit is more than that in the outer orbit. This fact has been established by the Doppler shift. Thus, the rings of the Saturn are not solids but there are a large number of satellites moving in these orbits called the rings of Saturn.

Illustrative Example 6.74

A locomotive whistle 256 Hz is moving towards a stationary observer with a velocity $1/20^{\text{th}}$ that of sound. What will be the frequencies of the notes heard by observer before and after the engine passes it ?

Solution

We know that
$$n' = n \frac{(v - v_0)}{(v - v_s)}$$

(i) When locomotive is approaching the stationary observer

$$n = 256, \quad v_s = \frac{v}{20} \quad \text{and} \quad v_0 = 0$$

or
$$n' = 256 \frac{v - 0}{v - (v/20)} = 256 \times \frac{20}{19}$$

$$= 269.5 \text{ Hz}$$

(ii) When locomotive passes the stationary observers

$$v_s = -v/20$$

or
$$n' = 256 \frac{v - 0}{v - (-v/20)} = 256 \times \frac{20}{21}$$

$$= 243.8 \text{ Hz}$$

Illustrative Example 6.75

Two tuning forks with natural frequencies of 340 Hz each move relative to a stationary observer. One fork moves away from the observer, while the other moves towards him at the same speed. The observer hears beats of frequency 3 Hz. Find the speed of this tuning fork.

Solution

Let the velocity of each tuning fork with respect to stationary observer be v_s . Apparent frequency of the tuning fork coming towards the stationary observer is given by

$$n_1 = \frac{n v}{v - v_s} \quad \dots (6.270)$$

Apparent frequency of the tuning fork moving away from the stationary observer is given by

$$n_2 = \frac{n v}{v + v_s} \quad \dots (6.271)$$

From equations-(6.270) and (6.272)

$$n_1 - n_2 = n v \left[\frac{1}{v - v_s} - \frac{1}{v + v_s} \right]$$

or
$$n_1 - n_2 = n v \left[\frac{2 v_s}{v^2 - v_s^2} \right] \quad \dots (6.272)$$

Substituting the given values, we have

$$3 = 340 \times 340 \left[\frac{2 v_s}{(340)^2 - v_s^2} \right]$$

or
$$3 [(340)^2 - v_s^2] = 2 \times (340)^2 v_s$$

or
$$3 \times (340)^2 - 3 v_s^2 = 2 \times (340)^2 v_s$$

or
$$3 v_s^2 + 2 \times (340)^2 v_s - 3 \times (340)^2 = 0$$

or
$$v_s = \frac{-2 \times (340)^2 \pm \sqrt{4 \times (340)^4 + 4 \times 3 \times 3 (340)^2}}{2 \times 3}$$

Solving we get $v_s = 1.5 \text{ m/s}$

Illustrative Example 6.76

A vibrating tuning fork tied to the end of a string 1.988 metre long is whirled round a circle. If it makes two revolutions in a second, calculate the ratio of the frequencies of the highest and the lowest notes heard by an observer situated in the plane of the tuning fork. Velocity of sound is 350 m/s.

Solution

Number of revolutions per second = 2

Radius of the circle = 1.988 m

As Linear velocity of the tuning fork is

$$v = 2 \times 2 \pi r = 4 \times \frac{22}{7} \times 1.988 = 25 \text{ m/s}$$

(i) Apparent frequency when the tuning fork is approaching the listener

$$n_1 = \frac{v n}{v - v_s} = \frac{350 n}{350 - 25} = \frac{14}{13} n \quad [\text{Highest note}]$$

(ii) Apparent frequency when the tuning fork is moving away from the listener

$$n_2 = \frac{v n}{v + v_s} = \frac{50 n}{(350 + 25)} = \frac{14}{15} n \quad [\text{Lowest note}]$$

The ratio of highest note to the lowest note is given by

$$\frac{n_1}{n_2} = \frac{14 n}{13} \times \frac{15}{14 n} = \frac{15}{13} = 1.154$$

Illustrative Example 6.77

A sources of sonic oscillations with frequency $n = 1700 \text{ Hz}$ and a receiver are located on the same normal to a wall. Both the source and receiver are stationary, and the wall recedes from the source with velocity $u = 6.0 \text{ cm/s}$. Find the beat frequency registered by the receiver. The velocity of sound is equal to $v = 340 \text{ m/s}$.

Solution

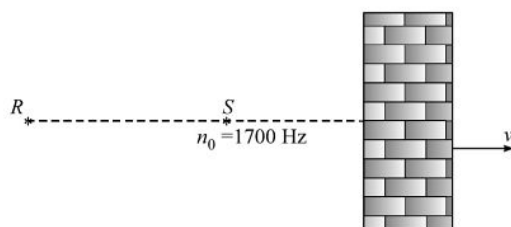


Figure 6.86

As both source and receiver are at rest the frequency of sound waves which directly reach to receiver, will be 1700 Hz and the frequency of sound which wall will receive as a moving observer is

$$n_1 = n_0 \left[\frac{v}{v + u} \right]$$

The wall now behaves as a moving source of frequency n_1 which when received by receiver, the frequency observed is

$$\begin{aligned} n_2 &= n_1 \left[\frac{v - u}{v} \right] \\ &= n_0 \left(\frac{v - u}{v + u} \right) \end{aligned}$$

$$= 1700 \left[\frac{340 - 0.06}{340 + 0.06} \right]$$

$$= 1700 \times \frac{339.94}{340.06}$$

$$= 1699.4 \text{ Hz}$$

Thus beat frequency received by detector is

$$\Delta n = 1700 - 1699.4 = 0.6 \text{ Hz}$$

Illustrative Example 6.78

If the earth is moving towards a stationary star at a speed of $30 \text{ kilometres per second}$, find the apparent wavelength of light emitted from the star. The real wavelength has the value $5875 \text{ \AA}$.

Solution

Here earth (observer) is approaching the star and hence, the mutual distance is decreasing. So, the observer will notice an increase in frequency or decrease in wavelength. If v be the relative velocity of the source and c , the velocity of light, then the change in wavelength is given by

$$\Delta \lambda = \frac{v}{c} \times \lambda.$$

Substituting the given values, we have

$$\Delta \lambda = \frac{30 \times 10^3}{3 \times 10^8} \times 5875 \text{ \AA}$$

$$= 5875 \times 10^{-4} \text{ \AA}.$$

Altered wavelength

$$\lambda' = \lambda - \Delta \lambda$$

$$= 5875 \text{ \AA} - (5875 \times 10^{-4} \text{ \AA})$$

or

$$\lambda' = 0.9999 \times 5875 \text{ \AA}$$

$$= 5874.4125 \text{ \AA}$$

Illustrative Example 6.79

A source of sound of frequency 256 Hz is moving rapidly towards a wall with a velocity of 5 m/s . How many beats per second will be heard if sound travels at a speed of 330 m/s ?

Solution

Case-I : When observer is between wall and the source

In this case, the observer is stationary and the source in motion towards observer. The apparent frequency for stationary observer is given by

$$n' = n \left[\frac{v}{v - v_s} \right]$$

Here $n = 256 \text{ Hz}$, $v = 330 \text{ m/s}$

and $v_s = 5 \text{ m/s}$

$$\text{or } n' = 256 \left(\frac{330}{330 - 5} \right) = \frac{256 \times 330}{325} = 259.93 \text{ Hz}$$

In this case, the apparent frequency received by the observer from source is 259.93. The observer also receives the sound reflected from the wall. As reflection does not cause any change in frequency and hence the frequency of reflected sound is also 259.93 Hz.

Thus beats frequency is

$$\Delta n = 259.93 - 259.93 = 0$$

Case-II : When the sources is between wall and observer

For direct sound, source is moving away from the observer.

$$\text{Hence } n'' = n \left(\frac{v}{v + v_s} \right) = \frac{330}{330 + 5} \times 256 = 252.2 \text{ Hz}$$

For reflected sound, source is moving towards wall. Hence

$$n' = n \left(\frac{v}{v - v_s} \right) = 259.9$$

Thus beat frequency is

$$\begin{aligned} \Delta n &= n' - n'' = 259.9 - 252.2 \\ &= 7.7 \approx 8 \text{ Hz} \end{aligned}$$

Illustrative Example 6.80

A radar wave has a frequency of $7.8 \times 10^9 \text{ s}^{-1}$. The reflected wave from an aeroplane shows a frequency difference of $2.7 \times 10^3 \text{ s}^{-1}$ on the higher side. Deduce the velocity of the aeroplane in the line of sight.

Solution

Suppose the aeroplane is approaching the observer with velocity v_a , then the velocity of image of source which is producing reflected wave will be $v_s = 2v_a$, (As in case of plane mirror, when the mirror moves with a velocity v , then the image moves with a velocity $2v$).

$$\text{In this case, } n' = n \left[\frac{c}{c - v_s} \right] = n \left[1 + \frac{v_s}{c} \right]$$

$$\text{or } n' = n + n \frac{v_s}{c}$$

$$\text{Here } n' - n = 2.7 \times 10^3 \text{ s}^{-1},$$

$$\text{or } n = 7.8 \times 10^9 \text{ s}^{-1} \quad \text{and} \quad c = 3 \times 10^8 \text{ m/s}$$

$$\text{or } 2.7 \times 10^3 = 7.8 \times 10^9 \left(\frac{v_s}{3 \times 10^8} \right) = \frac{78 v_s}{3}$$

$$\text{or } v_s = \frac{3}{78} \times 2.7 \times 10^3 \text{ m/s}$$

$$\text{Now } v_a = \frac{v_s}{2} = \left[\frac{3 \times 2.7 \times 10^3}{2 \times 78} \right] \times \frac{60 \times 60}{1000}$$

$$\text{or } = 1.87 \times 10^3 \text{ km/hr.}$$

Illustrative Example 6.81

Two distant sources situated together emit sound each of frequency 300 cycles per second. If one of them were to approach and the other to recede from a stationary observer each with a velocity of $1/100^{\text{th}}$ the velocity of sound, calculate the number of beats per second heard by the observer.

Solution

The situation is shown in figure-6.87.

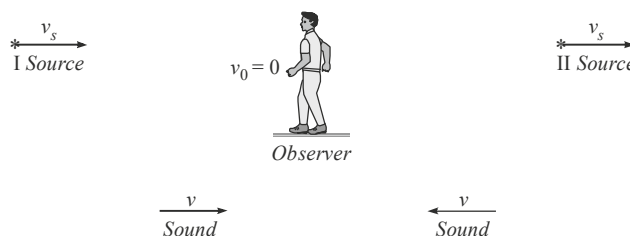


Figure 6.87

The first source is approaching the stationary observer. The frequency of the source n_1 as heard by observer is given by

$$n_1 = n \left(\frac{v}{v - v_s} \right)$$

Where n is the actual frequency of source

$$\begin{aligned} \text{or } n_1 &= 300 \left(\frac{v}{v - v/100} \right) \\ &= \frac{300 \times 100}{99} \\ &= 303.03 \text{ Hz} \end{aligned}$$

The second source is receding from the observer in the direction opposite to the sound. Hence the apparent frequency n_2

$$\begin{aligned} n_2 &= \left(\frac{v}{v + 0.01v} \right) \times 300 \\ &= \frac{300 \times 100}{101} \\ &= 297.03 \text{ Hz} \end{aligned}$$

Hence beat frequency detected by observer is

$$\Delta n = 303.03 - 297.03 = 6 \text{ Hz}$$

Illustrative Example 6.82

A source of sonic oscillations with frequency $n = 1700 \text{ Hz}$ and a receiver are located at the same point. At the moment $t = 0$, the source starts receding from the receiver with constant acceleration $a = 10.0 \text{ m/s}^2$. Assuming the velocity of sound to be equal to $v = 340 \text{ m/s}$, find the oscillation frequency registered by the stationary receiver $t = 10.0 \text{ s}$ after the start of motion.

Solution

In 10 second source will travel a distance

$$S_0 = \frac{1}{2} a (10)^2 = \frac{1}{2} \times 10 \times (10)^2 = 500 \text{ m}$$

At this instant observer will receive that sound which source had emitted at time $t = t_1$. At time t_1 source was at a distance S from the source.

$$S = \frac{1}{2} a t_1^2 = \frac{1}{2} (10) t_1^2 = 5 t_1^2$$

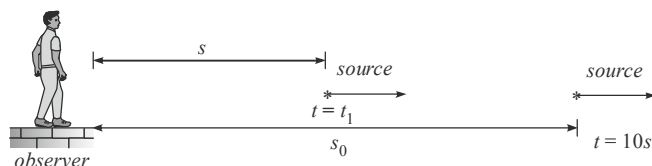


Figure 6.88

Now in a time $(10 - t_1)$ sound travels a distance S from source to observer. Thus, we have

$$S t_1^2 = 340 (10 - t_1)$$

$$\text{or } S t_1^2 + 340 t_1 - 3400 = 0$$

$$\text{or } t_1 = \frac{-340 \pm \sqrt{(340)^2 + 4(5)(3400)}}{2 \times 5}$$

$$= 8.85 \text{ s} \quad [\text{Ignoring negative root}]$$

Thus velocity of source at instant t_1 is

$$v = a t_1 = 10 \times 8.85 = 88.5 \text{ m/s}$$

Thus frequency observed by observer at 10 sec is given as

$$n = n_0 \left[\frac{340}{340 + 88.5} \right]$$

$$= 1700 \times \frac{340}{428.5}$$

$$= 1350 \text{ Hz}$$

Illustrative Example 6.83

A source of sound with natural frequency $n = 1.8 \text{ kHz}$ moves uniformly along a straight line separated from a stationary observer by a distance $l = 250 \text{ m}$. The velocity of the source is equal to $\eta = 0.80$ fraction of the velocity of sound. Find :

- the frequency of sound received by the observer at the moment when the source gets closest to him;
- the distance between the source and the observer at the moment when the observer receives a frequency $n_{ap} = n$.

Solution

The situation is shown in figure-6.89.

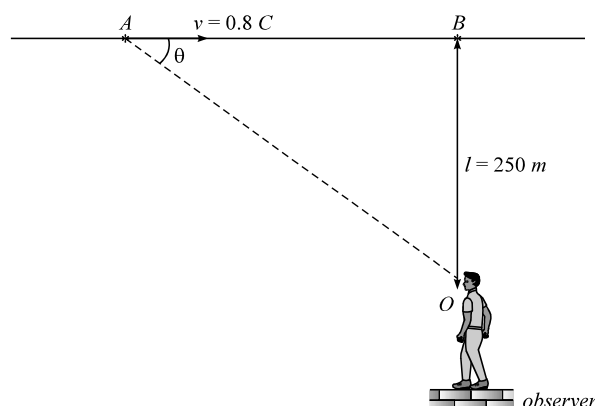


Figure 6.89

- Figure-6.89 shows the corresponding situation. Source gets closest to observer when it is at position B but at this instant observer receives those sound waves which source had emitted from an earlier position A shown in figure-6.89, so that when source reaches A to B , in the same time sound reaches A to O . Thus the apparent frequency heard by observer is

$$n_{ap} = n_0 \left[\frac{v}{v - v_s \cos \theta} \right] \quad \dots (6.273)$$

Here, we have

$$\cos \theta = \frac{AB}{AO} \quad \dots (6.274)$$

If sound takes t_1 time from A to O , we have

$$AB = v_s t_1 = 0.8 t_1$$

and

$$AO = v t_1$$

Now from equation-(6.274)

$$\cos \theta = 0.8$$

Now from equation-(6.273)

$$n_{ap} = n_0 \left[\frac{v}{v - (0.8)v} \right] = n_0 \left[\frac{1}{0.36} \right] = \frac{1.8 \times 10^3}{0.36} = 5 \text{ kHz}$$

(b) Observer receives a frequency equal to original frequency of source corresponding to the waves emitted by source from position B , as in this case velocity component of source is there in the direction of observer.

In this case the time taken by sound to reach observer from B is

$$t_0 = \frac{l}{v}$$

In this direction distance travelled by source is

$$S = v_s t = 0.8 vt = 0.8 l$$

At the instant when observer receives this sound the separation between source and observer can be given as

$$\begin{aligned} x &= \sqrt{l^2 + (0.8l)^2} = \sqrt{1.64} \times l \\ &= \sqrt{1.16} \times 250 \\ &= 320 \text{ m} \end{aligned}$$

Illustrative Example 6.84

A locomotive approaching a crossing at a speed of 80 miles/hr, sounds a whistle of frequency 400 Hz when 1 mile from the crossing. There is no wind, and the speed of sound in air is 0.200 mile/s. What frequency is heard by an observer 0.60 miles from the crossing on the straight road which crosses the railroad at right angles?

Solution

The situation is shown in figure-6.90.

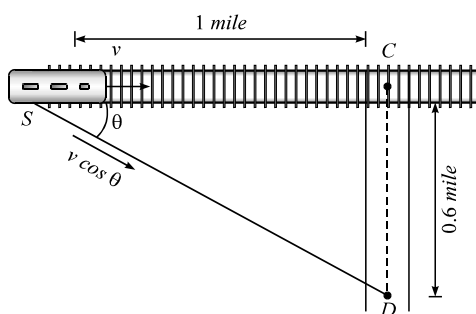


Figure 6.90

Here is $SD = \sqrt{1^2 + 0.6^2} = 1.166 \text{ miles}$

$$\cos \theta = \frac{SC}{SD} = \frac{1}{1.166} = 0.857$$

Thus speed of source along the line of sight

$$\begin{aligned} v_s &= v \cos \theta = \frac{80}{60 \times 60} \times 0.857 \\ &= 0.019 \text{ mile/s} \end{aligned}$$

Thus the apparent frequency observed is

$$n = n_0 \left[\frac{v_{\text{sound}}}{v_{\text{sound}} - v_s} \right] = 400 \left[\frac{0.2}{0.2 - 0.019} \right] = 442 \text{ Hz}$$

6.18 Shock Waves

The general equation, we've developed for finding the apparent frequency by Doppler's effect, is given as

$$n_{ap} = n_0 \left(\frac{v \pm v_0}{v \pm v_s} \right) \quad \dots (6.275)$$

If a source is moving toward an observer at a speed equal to the speed of sound. That is $v_s = v$ then from equation-(6.275) the detected frequency n_{ap} will be infinitely high. This means that the source is moving so fast that the effective wavelength of produced sound in air is compressed to almost zero. This we can understand step by step as shown by figure-6.92.

Figure-6.91(a) shows the sound waves produced by a stationary source in three dimensional, space. The spherical region surrounding the source in which it creates oscillations in phase is called spherical wavefront produced by the source. These wavefronts propagate in radially outward direction with speed of sound as shown. All the spherical wavelengths with increasing size are concentric spherical shells with centre at source. If we look at figure-6.91(b) when source is moving toward right at speed v_s ($v_s < v$), here we can say that if initially a wavefront was emitted by source when it was at position s_1 , then this wavefront will expand (sound propagates) continuously with centre located at point s_1 , but source is now moved to another position s_2, s_3 and so on. In this case we can see that once if a wavefront is created (sound is produced), it is carried by the medium only and this propagation does not depend on source thus the wavefront will expand (sound travels) with its centre located at the same position from where it was created. Thus here it is clear that in front of source the separation between successive wavefronts is decreased (λ is compressed) and in the region behind the source the separation between successive wavefronts is increased (λ is expended). This is the same phenomenon of Doppler effect. Which we've described earlier in spherical waves.

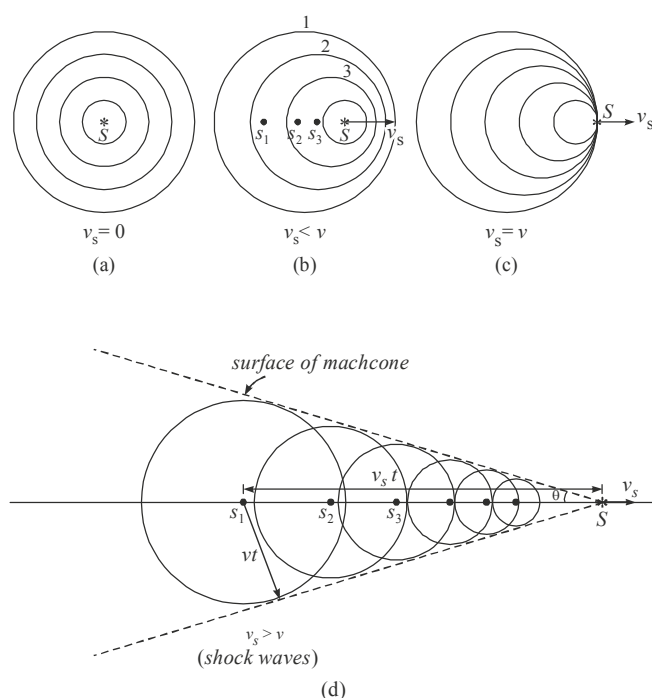


Figure 6.91

As the speed v_s of source approaches the speed of sound v , as shown in figure-6.91(c), then it keeps pace with its own spherical wavefronts, the wavelength approaches zero and the waves produced by the source from its different locations move together and pile up on each other. In such a case source must exert large force on medium particles to produce compressions and rarefaction in front of it. According to Newton's third Law, the medium exerts equally large force back on the source or we can say that when source speed approaches speed of sound then in the medium (air), there is a large increase in air resistance (aerodynamic drag) to its motion. This is called as “*sound barrier*.” This is the reason when a supersonic plane crosses its speed $v_s = v$ or sound barrier, it experiences a very high aerodynamic drag. When source speed v_s is greater in magnitude than v the source of sound is called supersonic. For supersonic sources the Doppler effect is no longer valid.

When sources speed is greater than speed of sound v , the situation is shown in figure-6.91(d). The source comes out of each spherical wavefront it produces as wavefronts expands at speed of sound v . We can see with time that a series of wavefronts are created by source during its motion and each spreads out in a sphere created at the position of source when it emitted the wavefront. After a time t , the wavefront emitted from a position s_1 , has spread to a sphere of radius vt as shown in figure-6.91(c), and the source has moved a greater distance $v_s t$, here we can see that all the spreading wavefronts created from local intermediate position of the source bunch along a

conical envelope as shown by dotted line in figure. At the surface of this cone we can say that all the wavefronts are oscillating in same phase, thus at the medium particles due to constructive interference the amplitude of oscillations will be very large. This very high energy wave with the new conical wavefront (effectively) is called shock wave, the angle θ made by the shock wave cone (called mach cone) with the line of motion of source can be given as

$$\sin \theta = \frac{vt}{v_s t} = \frac{v}{v_s} \quad \dots (6.276)$$

In this relation of equation-(6.276) the ratio $\frac{v_s}{v}$ is called Mach number. It is greater than 1 for all supersonic speeds. Here $\sin \theta$ is reciprocal of Mach number.

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Doppler's Effect

Module Number - 1 to 25

Practice Exercise 6.8

(i) A policeman on duty detects a drop of 10% in the pitch of the horn of motion car as it crosses him. If the velocity of sound is 330 metres per second, calculate the speed of the car.

[17.37 m/s]

(ii) A car travelling at 10 m/s sounds its horn which has a frequency of 500 Hz and this is heard in another car which is travelling behind the first car in the same direction, with a velocity of 20 m/s. The sound can also be heard in the second car by reflection from a bridge head. What frequency will the driver of the second car hear ? ($v = 340$ m/s)

[545.45 Hz]

(iii) The wavelength of light coming from a distant galaxy is found to be 0.5% more than that coming from a source on earth. Calculate the velocity of the galaxy.

[1.5×10^6 m/s]

(iv) A band playing music at a frequency n_0 is moving towards a wall at a speed v_b . A motorist is following the band with a speed v_m . If v is the speed of sound obtain an expression for the beat frequency heard by the motorist.

$$\left[n_0 \left[\frac{(v + v_m)(2v_b)}{(v^2 - v_b^2)} \right] \right]$$

(v) An astronaut is approaching the moon. He sends out a radio signal of frequency 5×10^9 Hz and finds out that the frequency shift in echo received is 10^3 Hz. Find his speed of approach.

[30 m/s]

(vi) If a vibrating fork is rapidly moved towards a wall, beats may be heard between the direct and reflected sounds. Calculate beat frequency if the frequency of fork is 512 Hz and approaches the wall with a velocity of 300 cm/s. The velocity of sound is 330 m/s. Consider observer is behind the fork.

[9.3 Hz]

(vii) A train *A* crosses a station with a speed of 40 m/s and whistles a short pulse of natural frequency 596 Hz. Another train *B* is approaching towards the same station with the same speed along a parallel track. Two tracks are 99 m apart. When train *A* whistles train *B* is 152 m away from the station as shown in figure-6.92. If velocity of sound in air is 330 m/s, calculate the frequency of the pulse heard by driver of train *B*.

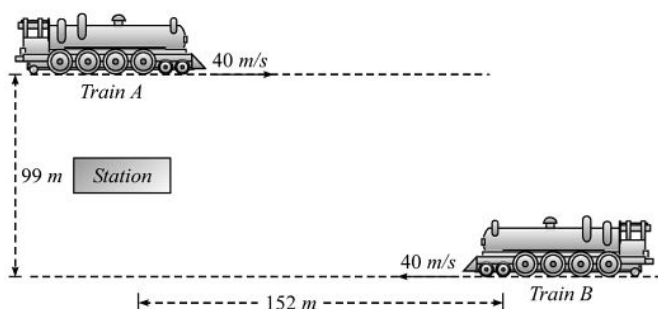


Figure 6.92

[724 Hz]

(viii) How fast would you have to go through a red light to have it appear green if the wavelengths of red and green light are respectively 620 nm and 540 nm? Is it possible to achieve this speed on earth?

[3.87×10^4 km/s, No]

6.19 Reflection and Refraction of Waves

Whenever a wave incident on the boundary of two different media, it splits in three parts (i) A part of incident wave bounces back into the same medium called reflected wave. (ii) A part of incident wave is transmitted into the other medium, called refracted wave (iii) A part of incident wave is absorbed by medium particles of both the medium at the boundary of media, called absorbed wave energy. If the incident wave amplitude is

A ; then the amplitudes of reflected, transmitted and absorbed waves can be given as

$$A_r = a_r A_i$$

$$A_t = a_t A_i$$

and

$$A_a = a_a A_i$$

Where a_r , a_t and a_a are called reflection coefficient, transmission coefficient and absorption coefficient for the respective wave amplitudes and for the three we have

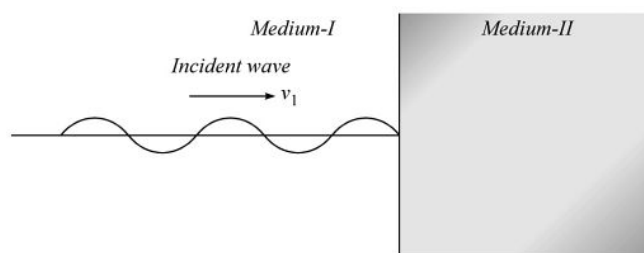
$$a_r + a_t + a_a = 1 \quad \dots (6.277)$$

One more thing students should note that we have already discussed that whenever a wave is emitted from a source its frequency remains constant during propagation, no matters in which medium wave is propagating. Similarly here also we can state that the frequencies of reflected and transmitted waves must be same as that of incident wave.

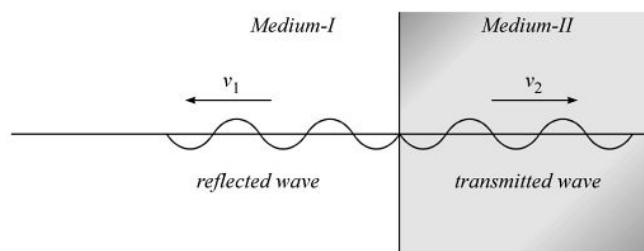
Now we find corresponding reflection and transmission coefficients for the reflected and transmitted wave amplitudes.

Let us consider a wave propagating in medium-I (velocity = v_1) incident on a medium boundary as shown in figure-6.93(a). If displacement amplitude of incident wave is A ; its equation can be written as

$$y_i = A_i \sin(\omega t - k_1 x) \quad [\text{Where } k_1 = \frac{\omega}{v_1}] \quad \dots (6.278)$$



(a)



(b)

Figure 6.93

If A_r and A_t be the amplitudes of respective reflected and transmitted waves as shown in figure-6.93(b), the corresponding equations are

For reflected wave

$$y_r = A_r \sin(\omega t + k_1 x) \quad \dots (6.279)$$

and for transmitted wave

$$y_t = A_t \sin(\omega t - k_2 x) \quad \left[\text{Where } k_2 = \frac{\omega}{v_2} \right] \quad \dots (6.280)$$

As wave is continuous at the boundary i.e. at $x = 0$ the displacement of medium particles in the two mediums must be equal, thus we have

$$\begin{aligned} y_i + y_r &= y_t \\ A_i + A_r &= A_t \end{aligned} \quad \dots (6.281)$$

At the boundary, the slope of displacement curve must also be continuous thus at $x = 0$, we also have

$$\frac{dy_i}{dx} + \frac{dy_r}{dx} = \frac{dy_t}{dx}$$

Now from equation-(6.278), (6.279) and (6.280), we have

$$-A_i k_1 \cos \omega t + A_r k_1 \cos \omega t = -A_t k_2 \cos \omega t$$

$$\text{or} \quad A_i - A_r = A_t \left(\frac{k_2}{k_1} \right) \quad \dots (6.282)$$

Adding equation-(6.281) and (6.282) we get

$$2A_i = A_t \left(\frac{k_2}{k_1} + 1 \right)$$

$$\text{or} \quad A_t = \left(\frac{2k_1}{k_1 + k_2} \right) A_i \quad \dots (6.283)$$

If a_t be the transmission coefficient for the boundary of the two media we have

$$A_t = a_t A_i \quad \dots (6.284)$$

comparing equation-(6.283) and (6.284) we get

Transmission coefficient,

$$a_t = \frac{2k_2}{k_1 + k_2} = \frac{2v_2}{v_1 + v_2} \quad \dots (6.285)$$

Now dividing equation-(6.282) and (6.281) we get

$$\frac{A_i - A_r}{A_i + A_r} = \frac{k_2}{k_1}$$

$$\text{or} \quad \frac{A_i}{A_r} = \frac{k_1 + k_2}{k_1 - k_2}$$

$$\text{or} \quad A_r = \left(\frac{k_1 - k_2}{k_1 + k_2} \right) A_i \quad \dots (6.286)$$

If a_r be the reflection coefficient for the boundary of the two medium we have

$$A_r = a_r A_i \quad \dots (6.287)$$

Comparing equation-(6.286) and (6.287) we get

Reflection coefficient,

$$a_r = \frac{k_1 - k_2}{k_1 + k_2} = \frac{v_2 - v_1}{v_1 + v_2} \quad \dots (6.288)$$

From equation-(6.285) and (6.287) we can see that transmission coefficient is always positive but reflection coefficient can be negative when $v_2 < v_1$ or when a wave propagating in a rarer medium is reflected from the boundary of a denser medium.

The negative sign with reflection coefficient shows that whenever a wave is reflected from the boundary of a denser medium, there is always a phase addition of π radian takes place in reflected wave, due to this the wave is inverted on reflection if it is a transverse wave.

6.19.1 Reflection of a Wave Pulse in a Stretched String

Let us consider a case of reflection of a transverse wave pulse in a stretched string from a wall as shown. Here we can consider that the wave velocity in second medium (wall), $v_2 = 0$ as wall is rigid then we have from equation-(6.285) and (6.286)

$$a_r = -1 \text{ and } a_t = 0$$

This means that incident wave is completely reflected with a phase change of π . As shown in figure-6.94.

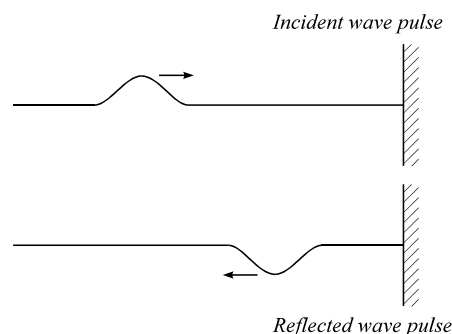


Figure 6.94

Now consider the case shown in figure-6.95. Two different strings are joined end to end stretched by a tension T . String 1 is lighter than string 2 so the wave speed in string 1 is more than that in string 2. When a wave pulse traveling from string 1 toward junction incident on the junction a part of energy is reflected and a part is transmitted to the other string. We can see as shown in figure-6.95(a) that when wave pulse travelling on lighter string (rarer medium) is reflected from heavier string

(denser medium) the reflected pulse suffers a phase change of π radians and inverted. But if a wave pulse travelling from heavier string incident on the junction, during reflection no phase change occurs as we can see from figure-6.95(b)

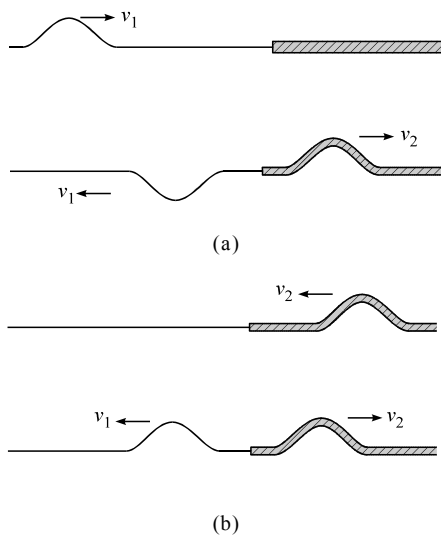


Figure 6.95

Web Reference at www.physicsgalaxy.com

Age Group - High School Physics | Age 17-19 Years

Section - OSCILLATION & WAVES

Topic - Stationary Waves and Beats

Module Number - 9, 10, 11

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